

RESEARCH METHODOLOGY

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RESEARCH METHODOLOGY

To all my teachers

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Preface

The economy of a country is controlled by several factors, namely, population size, industrial activities, agriculture, policies of the government, culture of the people, educational system, infrastructure facilities. The policies and guidelines of both the central and state governments facilitate the integration, coordination and control of all the activities of the nation with the objective of maximizing the growth of the nation. Hence, it is clear that each entity of the nation is forced to have a competitive role to maximize its productivity for its survival. The productivity of different organizations can be improved through several research studies. Each research study is an organized set of activities to study and develop a model or procedure/technique to analyze results of a realistic problem supported by literature and data such that its objectives are optimized and further make recommendations/interferences for implementation.

Teaching industrial engineering students at Anna University, management students at Pondicherry University and the IGNOU study centre, Pondicherry, further the experience of having published four books on management and industrial engineering through Prentice-Hall of India, has given me a balanced mix of expertise to write this book in a clear and accessible style so as to benefit students, teachers and executives in business and public organizations.

This text contains in-depth coverage of concepts and techniques of research methodology. It begins with an overview of research methodology and it is followed by fifteen chapters: data collection and presentation, review of basic statistical measures, design and analysis of experiments, attitude measurement and scales, probability distribution, sampling methods and distributions, tests of hypotheses, nonparametric tests, basic multivariate analysis, advanced multivariate analysis, multidimensional scaling and conjoint analysis, algorithmic research, models for industries and public systems, simulation, and report writing and presentation.

First, comprehensive coverage of research problems, types of research and research process is presented. It is followed by guidelines for data collection and presentation which form the basic foundation for further analyses. The topics on measures of central tendencies, variation and skewness provides continuity to basic statistical methods. Design of experiment is the basic building blocks of any research which helps collect data needed for further analyses. So, an in-depth discussion of different experimental designs and analysis such as completely randomized design, randomized complete block design, latin square design, factorial design, special designs using Yate's algorithm and expected mean square rule is presented next. It is followed by comprehensive coverage of fundamentals of data collection methods and different scales. Next, a brief introduction to probability distribution is presented which will definitely make readers comfortable in understanding following chapters. Sampling is an inevitable process in any research and hence, a detailed presentation on sampling methods, sampling distributions for mean, variance and proportion as well as topics on confidence interval is followed. The most important topic on tests of hypotheses is presented next which gives an in-depth idea of hypotheses concerning means, proportions, variances as well as on chi-square tests. Though the

earlier chapter focussed on parametric tests, in some situations, it may not be possible to estimate its parameters. Hence, nonparametric tests, which play a vital role in decision making which comprises of one-sample tests, two-samples tests and K -samples tests, are presented next. Following two chapters are on multivariate analysis focussing on correlation analysis, forecasting, linear regression, time series, discriminant analysis, factor analysis, and cluster analysis. Further advanced topics like multidimensional scaling and conjoint analysis are presented next.

Many of the business decision-making processes are mainly dependent on the results obtained by algorithms. Hence, in the next chapter, the concept of algorithm, types of algorithm and algorithmic procedure, technique for comparison of algorithms and introduction to meta heuristics are presented. It is followed by an overview of applications of modelling in different functional areas of management and public systems. As some of the decision environments are probabilistic in nature, the topic on simulation which helps researchers deal with such situations is given next. Finally, report writing and presentation are dealt with in a comprehensive manner which will aid researchers to document their research studies and analysis along with findings.

In this book, well-defined algorithms for the different statistical methods, wherever necessary, are provided and the workings of different statistical techniques with fitting examples are demonstrated. The comprehensive chapter on design and analysis of experiments is unique to this book. Throughout the text, a large number of tables and figures are presented to illustrate the concepts and techniques. At the end of each chapter, to test conceptual understanding of the subject and put theory into practice, review questions are included. Also, it has answers to numerical questions, wherever applicable. Throughout the book, simple and easy-to-read style is followed.

This comprehensive textbook is designed for a course on research methodology for the students of MBA, M.Com., MA (Economics) and MA (Sociology). This book further acts as a guide for practicing professionals to design and execute different types of research.

I express my gratefulness and appreciations to my colleagues and academic friends for having shared their experiences and giving constructive suggestions in deciding the structure and contents of the book. I am very much thankful to Prentice-Hall of India, especially its editorial and production team, for their encouraging interactions with me and meticulous processing of the manuscript. Finally, I thank the authorities of Pondicherry University for having sanctioned me sabbatical leave for a full academic year to write this book.

I look forward to suggestions for improving the contents of this book.

R. PANNEERSELVAM

CHAPTER

1

OVERVIEW OF RESEARCH METHODOLOGY

1.1 INTRODUCTION

The economy of a country is controlled by several factors, viz., population size, industrial activities, agriculture, policies of its government, culture of the people, educational system, infrastructure facilities, etc. In the process of satisfying the basic needs (food, shelter and clothing) people engage themselves in various activities such as agriculture, housing and textile industries. Each of the above industries is backed by various other industries. For example, a textile industry is supported by machine tool, agriculture and chemical industries. Major portion of the product of each of the industries is consumed by the public and a portion of it is fed as raw materials/equipments to other industries. Over and above these industrial activities, education and training of required manpower for these industries are taken care by educational institutions. In addition to these activities, there are many service organizations, viz., health, transportation, postal, legal, insurance, banking, etc., catering to the needs of the society.

Policies and guidelines of the Central government and state governments facilitate the integration, coordination and control of all the activities of the nation with the objective of maximizing the growth of the nation. The above discussion makes it clear that each entity of the nation is forced to have competitive role to maximize its productivity for its survival. The productivity of different organizations can be improved through several research studies.

The productivity of a business system is the ratio between its output and input, where the output corresponds to annual total income of the business system and the input corresponds to the annual total cost of different resources consumed and services utilized by the business system to realize the annual production. Then

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}} = \frac{\text{Annual revenue}}{\text{Annual cost}}$$

Take the case of a consumer non-durable, like a toothpaste. The main factors which affect the demand of brand-X toothpaste are:

- Ingredients of the toothpaste
- Taste of the toothpaste

- Selling price per unit
- Size of the population
- Income level of people
- Number of competing brands
- Locations of consumers.

The objective of the company producing brand-X toothpaste is to identify the optimal level of production of the toothpaste by taking the above factors into account such that the overall productivity which is defined as the ratio between the annual revenue and the annual cost of the production and selling of that brand is maximized. The company can achieve the above objectives through research and development activities.

If we carefully examine the factors affecting the sales of the toothpaste, they can be classified into qualitative and quantitative factors. One has to develop a suitable model using the data of these factors augmented with a detailed survey. This is an example of a research study to improve the productivity of an organization.

Take another example of introducing a manufacturing technology and procedure, like just-in-time (JIT) for the assembly line of an automobile company. In this case, the company is left with the following two options:

- Identification and use of the state-of-the-art technique of JIT
- Conducting a company sponsored research project through an educational institution to develop a technique of JIT, which will result in lesser throughput time of the product assembled using the assembly line.

Consider the social problem of a joint family system. In the past, the joint family system was very much in existence in all the families. Though this system has several advantages, over a period of time, due to various valid and invalid reasons, the family system is divided into nucleus type. As a result, old age homes are growing in large number. This is considered as a greater social concern. Further, the per capita savings also reduces under nucleus type because of more overhead expenses. A detailed study can be conducted to find out a suitable system, which will give security to olders and improve the national economic growth.

Research One can visualize the fact that a detailed study is required in each practical situation for better results. Any effort which is directed to such study for better results is known as a *research*. In other words, a research is an organized set of activities to study and develop a model or procedure/technique to find the results of a realistic problem supported by literature and data such that its objective(s) is(are) optimized and further make recommendations/interferences for implementations.

Research methodology A system of models, procedures and techniques used to find the results of a research problem is called a *research methodology*.

1.2 MATHEMATICAL TOOLS FOR ANALYSIS

Mathematical tools for study and analysis of realistic systems are discussed as follows:

Statistical techniques

The different statistical methods, which are used in a research, are listed as follows: (These are explained later in this book.)

1. Methods to find measures of central tendency, viz., mean, median and mode.
2. Methods to find measures of dispersion, viz., variance, coefficient of variance, coefficient of skewness, quartile deviation, etc.
3. Forecasting methods
4. Factorial design
5. Sampling methods
6. Tests of hypotheses
7. Factor analysis
8. Discriminant analysis
9. Conjoint analysis.

Operations research models

Different operations research models are listed as follows: (The concept of operations research models are discussed in Chapter 14.)

1. Linear programming
2. Integer programming
3. Distance related network techniques
4. Project management
5. Inventory control
6. Dynamic programming
7. Queueing theory
8. Replacement analysis
9. Game theory
10. Goal programming
11. Simulation
12. Production scheduling
13. Non-linear programming.

Fundamental research on mathematical tools

Case-study-oriented applications of the existing mathematical tools in different areas are presented earlier. But, to cope with the challenges introduced by the dynamism of evolving real-world business systems and practices, a continuous research should be carried out to establish modified techniques and algorithms of existing tools or newer techniques and algorithms. Some of the examples of conducting a research to modify the existing techniques are as follows:

- The existing transportation technique may be modified for better accuracy of results.
- Developing a technique to find the optimal distribution pattern in a transportation problem with quantity discounts.
- Devising improved techniques for scheduling problems.
- Modified integer programming technique with stronger cuts.

Some of the examples of conducting a research to develop newer techniques and models are listed as follows:

- Model for media-mix of a multinational company
- Manpower planning model for a changing business environment
- Aggregate planning model for a multiplant organization
- Goal programming approach for a portfolio design
- Meta heuristics (simulated annealing, genetic and tabu search) based algorithms for various challenging combinatorial problems in each of the functional areas of management and public systems.

1.3 RESEARCH PROBLEMS IN MANAGEMENT

Business executives in each of the functional areas of management, viz., marketing, production, finance and personnel are confronted with several interacting decision-making problems in their day-to-day operation. Many of these problems demand a systematic research which will aid the executives to take optimal and effective decisions.

1.3.1 Marketing

Different research topics studied under *marketing* management are:

- Product identification
- Demand estimation
- Demand-supply analysis
- Product development
- Market segmentation
- Media mix
- Sales promotion programme
- Product launching
- Design of distribution channel
- Pricing
- Marketing strategy formulation
- Product differentiation and product positioning
- Packaging
- Design of advertisements
- Planning sales force
- Buying behaviour.

1.3.2 Production

Different research topics studied under *production* management are:

- Forecasting
- Capacity planning
- Plant location
- Plant layout
- Line balancing
- Line of balance
- Inventory control
- Aggregate planning
- Master production scheduling
- Materials requirements planning
- Single-machine scheduling
- Flow-shop scheduling
- Job-shop scheduling
- Project management
- Maintenance management
- Quality control
- Work design
- Modern production system design, like JIT, computer integrated manufacturing, flexible manufacturing, agile manufacturing, logistics management, total quality management, etc.

1.3.3 Finance

Different research topics studied under *financial* management are:

- Models for break-even analysis of multi-product
- Economic evaluation of alternatives
- Study of financial parameters of organizations
- Capital budgeting
- Ratio analysis
- Portfolio management
- Primary market study
- Secondary market study
- Balance of payment
- Inflation
- Deflation
- Economic growth models.

1.3.4 Personnel

Different research topics studied under *personnel* management are:

- Manpower planning
- Performance appraisal systems
- Conflict management
- Study of organizational climate
- Design of incentive plans
- Leadership styles
- Training methods
- Change management
- Negotiation and wage settlement
- Labour welfare study.

1.4 TYPES OF RESEARCH

A research can be classified into explanatory research, conclusive research, modelling research and algorithmic research.

1.4.1 Exploratory Research

Exploratory research is an initial research which analyzes the data and explores the possibility of obtaining as many relationships as possible between different variables without knowing their end-applications. This means that a general study will be conducted without having any specific end-objective except to establish as many relationships as possible between the variables of the study. This research provides a basis for general findings. Researchers and practitioners can explore the possibility of using such general findings in future. This type of research lay the foundation for the formulation of different hypotheses of research problems.

Different types of exploratory researches are literature survey, experience survey and study of problems to have an insight.

Literature survey

Literature survey is a study involving a collection of literatures in the selected area of research in which the researcher has limited experience, and critical examination and comparison of them to have better understanding. It also helps the researchers to update with the past data, data sources and results, which will be useful for related researches in future.

Some examples of such researches are review of wholesale price index, demographic analysis, periodic stock exchange index reports, cross comparison of census details of different countries, etc.

Experience survey

Experience survey is a survey of experiences of experts/specialists in a particular field which will act as a database for future research. This survey gathers the experiences of specialists in terms of their skill and knowledge which have been developed over a period of time, or through other studies. The outcome of this survey is to generate ideas which will help carry out future research with minimal data collection. The decision making in each of the functional areas of management under probabilistic situation is a complex process. For such decision,

the study of the experiences of the executives can be carried out using the experience survey. Some of the examples of the experience survey are:

- Bidding of tenders
- Technology forecasting
- Materials planning
- Manpower planning
- Production scheduling
- Union/state government budget exercise
- Investment decisions
- Portfolio management
- Company's annual budget exercise
- Monsoon prediction
- Product pricing.

Study of insight stimulating examples

This is a special kind of study oriented to have insight into research topics. This kind of study is mainly case study oriented. For example, the implementation of just-in-time manufacturing system in a company involves nurturing of supplier firms so that there will be greater reliability of supply at appropriate time to make the JIT system operational. But, there are so many factors like, culture and loyalty of suppliers, government policies, etc., which will influence the success of such nurturing of vendors. In a country, like Japan, it is very much existing in full scale. This is being followed in some industries situated in many countries including India especially after the emergence multinational companies with supply chain management. So, the success of the implementation of the JIT system in Indian industries can be studied through case study approach coupled with a carefully designed questionnaire.

1.4.2 Conclusive Research

As stated earlier, the exploratory research lays the foundation for the formulation of hypothesis (hypotheses). *Conclusive* research tests the hypothesis (hypotheses) of a research problem formulated by exploratory research and draws definite conclusion(s) for implementation. After validating the hypothesis (hypotheses), a decision-making framework can be formulated.

The conclusive research is classified into descriptive research and experimental research. They are explained in the following text:

Descriptive research

As stated earlier, a *descriptive* research is carried out with specific objective(s) and hence it results in definite conclusions. This research tries to describe the characteristics of the respondents in relation to a particular product or a practice/culture of importance.

For example, consider television as a product. The degree of use of the television varies with respect to age, sex, income level and profession of the respondents as well as place and time of use. Hence, the degree of use of television to different types of respondents will be of importance to a researcher.

There are three different players in deciding the usage of television: the manufacturer of television, broadcasting agencies of programmes and the viewers.

The manufacturers can come out with facilities which make the television more user friendly. Some of the facilities are:

- Remote, which helps the viewers to change channel as per their taste. This facility avoids the movement of viewers, which will be of great help for aged people.
- Provision of locking some channel(s) which may cause disturbance in the minds of youths.
- Providing a range of models to suit different income groups.
- Built-in mechanism to receive signals effectively which leads to clear picture.

The broadcasting stations can come out with programmes which suit to different groups based on sex, age, income, place and profession. Further, the programmes should be scheduled such that maximal viewers of these categories are benefited.

The viewers are the ultimate group which puts the television into maximum use. They must be aware of the programmes of different channels and accordingly they can have a tentative plan of using it with minimal disturbance to others in the family or neighbourhood.

Based on the above discussions, a study can be conducted by a manufacturing company to find the degree of use of the facilities provided in its television set. Similar study can be conducted by a broadcasting station to find out the degree of use of its TV programmes. In this study, the following hypotheses may be made about the programmes as well as viewers:

- The programmes of the channel are no way different from that of other channels.
- The programmes of the channel are useful to the viewers.
- The programme timings of the channel are worst fit for viewers.
- The viewers are fully aware of the programmes of the channel.
- TV is an exhibit in many houses
- Viewers feel that TV is a nuisance for school going children.
- Viewers feel that TV is a boon for children to improve their general knowledge.

The researcher can design a suitable questionnaire to capture data from the respondents to test the above hypotheses and come out with conclusions after data analysis. The kind of conclusions which are derived in such studies are descriptive in nature.

Experimental research

An *experimental* research is used to study the effect of a set of factors on the response variable of a system of study. This research is conducted in a controlled environment, and analyzed using ANOVA.

Consider an example of rating of the taste of a soft drink by respondents on 1 to 10 scale. Let the factors which affect the response variable, namely the rating, be the age and profession of the respondents. A sample experimental design is shown in Table 1.1. In Table 1.1, the number of levels of the factor, 'Age' is 4 and that of 'Profession' is also 4. For each experimental combination, three replications are carried out to minimize the error. After collecting the data from the respondents, as per the design shown in Table 1.1, a

factorial analysis can be done to test whether each factor or the combination of factors has a significant effect on the response variable, 'rating'. Each factor and the combination of the factors are known as *components of the ANOVA model*. The conclusion may be that whether or not each component has effect on the response variable at a given significance level.

Table 1.1 Experimental Design

Age (years)	Profession			
	Employees	Unemployed	Business	Housewife
< 25	⋮	⋮	⋮	⋮
25–35	⋮	⋮	⋮	⋮
35–60	⋮	⋮	⋮	⋮
> 60	⋮	⋮	⋮	⋮

1.4.3 Modelling Research

A model is an abstraction of reality. Many of the real-life problems of business situations can be formulated as models. Different types of models applied to business situations are symbolic models, mathematical models and simulation model.

Symbolic model

Symbolic model is a representation of the performance measure of the system of interest in terms of its variables. This means that the attributes of a system are related by an equation. Regression equation is an example of a symbolic model.

Mathematical model

Mathematical models are mainly operations research models which are aimed to solve complex real-life problems arising in the direction and management of large systems of men, machines, materials and money in industry, business and government. Some examples of mathematical models are linear programming models, transportation models, assignment models, inventory models, replacement models, goal programming models, parametric programming models, quadratic programming models, chance constrained programming models, separable programming models, etc.

In this category of research, one can do any one of the following types of research:

- Development of improved method for each of the mathematical modelling techniques
- Use of any one or a combination of existing mathematical models to obtain the results of a complex problem.

Simulation model

Simulation is an experiment conducted over a real-life stochastic system in a scaled time-frame to extract as many average operational statistics as possible to formulate respective decision guidelines at all the levels of management in industry, business and government. The simulation model can be either continuous or discrete. In *continuous* simulation model, the clock unit of the simulation is continuously incremented whereas in *discrete* simulation model, the clock unit of the simulation is incremented in discrete manner. So, in discrete simulation, the time of simulation can be either scaled-up or scaled-down depending on the situation and the required accuracy.

1.4.4 Algorithmic Research

An *algorithm* is a well-defined sequence of steps to solve a problem of interest in industry, business and government. Research problems can be classified into *combinatorial* problems and *polynomial* problems. If a problem takes the exponential form of time complexity and volume complexity, then it is termed as a combinatorial problem; if it takes the polynomial form of time complexity and volume complexity, then it is termed as a polynomial problem. If a problem comes under polynomial category, the job of the researcher is to develop the exact algorithm to obtain its optimal solution; otherwise, the researcher should concentrate in developing efficient heuristic for the problem. A detailed account of this type of research is given in Chapter 13.

1.5 RESEARCH PROCESS

A research process involves the following sequence of steps:

1. Problem definition
2. Objectives of the research
3. Research design
4. Data collection
5. Data analysis
6. Interpretation of results
7. Validation of results.

They are now explained.

1.5.1 Problem Definition

A research problem must be identified and defined without any ambiguity. The productivity is the core objective of any organization for its survival in the prevailing competitive environment. So, each organization is involved in improving its productivity through continuous research in each functional area of management. In any organization, the budget allocation for R&D activities is limited, which forces the research team to select the best combination of research projects out of the competing alternative research projects in different functional areas. So, the research team should identify research projects which will have more value addition to the organization. Also, each selected research project must be clearly defined.

There may be a situation in which the researcher is fully aware of the symptoms relating to certain deficiency in achieving an organizational goal. But, he/she may not be in a position to clearly spell out the problem which is causing such deficiency. Unless it is clearly identified, it will not be possible to proceed further to carry out the project. If a researcher proceeds with ill-defined problems, he/she may end up with misleading conclusions or aborting the research project in the middle due to poor pay-off identified through interim evaluation of the research. Hence, the research problem should be clearly defined.

1.5.2 Objectives of the Research

The objectives of the research must be identified by taking the system requirements into account. In the process of identifying the objectives of the research, the researcher must finalize the following.

- Research questions
- Hypotheses
- Boundary of the study.

Research questions are the problems which are not resolved till date. One should ask questions relating to the purpose of the study, place of the study, present state of the research issue and the possible means of finding solution to the research problem. For example,

Purpose: What is the purpose of the study?

Place: Where does the study take place?

Present state: What is the present state of the research issue?

Means: How is the purpose achieved? Why that particular method?

Answers to these questions will provide a starting point to crystallize the objectives and hypotheses of the research problem.

A *hypothesis* is an assumption about a population of the study. A sample hypothesis is as follows:

Hypothesis: Age has no influence over achieving sales target of a salesperson.

Generally, a hypothesis is formulated for a situation where the inference is not explicit. In the above case, the hypothesis may be true or false. The correct fact can be ascertained only after collecting and analyzing the related data.

In addition to the above two aspects, research questions and hypotheses, the researcher should clearly define the boundary of the study. In an ideal case, consideration of the whole universe of the study may be beneficial for the organization. But, such an assumption will make the study infeasible because of its size, complexities and constraints. Hence, the boundary of the study should be properly defined.

The objectives of the research may be classified into *quantitative* and *qualitative*.

The quantitative objectives aim to optimize certain measure of performances of the system of study. Some of the examples of quantitative objectives are as follows:

- A study on advertising budget aims to plan the allocation of advertising fund in relation to other expenditures such that the incremental sales revenue is maximized.
- A study on training methods aims to evaluate the relative merits and demerits of various training methods applied to a company with a view to maximize the labour productivity.
- A study on layout design aims to minimize the cost of materials handling and maximize the utilization of equipments.
- A study to estimate the demand of a product with a high precision.

The qualitative objectives aim to test the significance of hypotheses of a study of importance. Some of the examples of qualitative objectives are as follows:

- A study on testing the effect of the level of education of sales force of a company on its sales revenue.
- A study on the impulse buying behaviour of customers at a grocery stores.
- A study on testing the effect of cultural change on national economy.
- A study on testing the effect of industrial automation on the level of employment.

1.5.3 Research Design

Once the research project is identified and defined clearly, the next stage is to design the research. The research design provides a complete guidelines for data collection. Following are the essence of a research design:

- Selection of research approach
- Design of sampling plan
- Design of experiment
- Design of questionnaire.

Selection of research approach

As discussed in the previous section, there are two major classification of research approaches, viz., exploratory research and conclusive research for survey based researches. Based on the requirements of the study, the researcher should decide about the type of study to be conducted. The modelling research should be used to find the best result through a model which consists of an objective function and a set of constraints. The algorithmic research should be used to find the optimal or near optimal solution using either exact algorithm or heuristic, respectively. Depending on the type of the reality, the fittest research approach should be selected.

Design of sampling plan

A *sampling plan* is a mechanism by which the sampling units of a study are selected from the sampling frame of the population. The selection of the sampling plan in a study in turn affects the cost and time to conduct the study, and the reliability of inferences of the study. Hence, it should be selected with utmost care. The sampling plan can be classified into *probability* sampling plans and *non-probability* sampling plans. Different sampling plans in each of these categories are listed as follows:

Probability sampling plans

1. Simple random sampling
2. Systematic sampling
3. Stratified random sampling
4. Cluster sampling
5. Multi-stage sampling.

Non-probability sampling plans

1. Convenience sampling
2. Judgment sampling
3. Quota sampling
4. Snowball sampling.

(These sampling plans are explained in detail in Chapter 7.) Depending on the population size, required precision and available time to carry out the research project, a suitable sampling plan is to be selected.

Design of experiment

A study involves different response variables. Each response variable may be affected by several factors. To test the effect of these factors on a response variable, a suitable experiment is to be designed such that the necessary data for testing the significance of the effects of the factors on the response variable are collected and the inferences of the test are highly reliable. There are two main steps of designing the experiment:

- (a) Identify the response variables of the study.
- (b) For each response variable, repeat the following steps.
 - Identify the factors affecting the response variable.
 - Decide on the type of each of the factors (a factor may be either fixed factor or random factor).
 - Fix the number of levels (treatments) of each factor.
 - Form the skeleton of the experiment, as shown in Table 1.1.
 - Write the model of the experiment and define its components.

Design of questionnaire

The data can be classified into *primary* data and *secondary* data. The data which is collected for the first time by direct observation is called primary data. The data which is obtained from existing records, publications, etc., is known as secondary ones.

Different methods of primary data collection are observation method, personal interview, telephone interview and mail survey. The success of survey methods depends on the strength of the questionnaire used. A questionnaire consists of a set of well-formulated questions to probe and obtain responses from respondents.

The generalized steps of designing questionnaire are presented as follows:

1. Identification of research issues and finalization of the set of hypotheses;
2. For each issue, formulation of a set of questions and then deciding about the content and format of each question;
3. Deciding question wording depending upon the types of the question;
4. Arrangement of questions in the questionnaire in appropriate sequence and also deciding the format of the questionnaire;
5. Pre-testing questionnaire;
6. Review of questionnaire for improvements.

The questionnaire must contain provisions to collect all the data items which are required for testing different hypotheses of the experiment as well as for testing the hypotheses of other tests relating to various research issues.

1.5.4 Data Collection

Data are the basic input to any decision-making process in a business. The processing of data gives statistics of importance of the study. As stated earlier, data can be classified into primary data and secondary data.

The different methods which are used for primary data collection are observation method, personal interview, telephone interview and mail survey. In an observation method, the investigator will collect data through personal observations. Personal interview is a survey method of data collection which employs a questionnaire. Personal interview can be classified into door-to-door interview, executive interview, mall intercept surveys, self-administered questionnaires and purchase intercept technique. Telephonic interview is considered to be a cost effective and dominant data collection method. The unique features of telephone interview are selecting telephone numbers, call outcomes, call timing and call reports. Mail survey is a data collection method in which questionnaires are mailed to potential respondents who in turn fill and return them at their convenience.

As stated earlier, the secondary data are collected from sources which have already been created for the purpose of first-time use and future uses. The secondary data can be obtained from internal sources and external sources. The internal sources of secondary data for marketing applications are sales records, marketing activity, cost information, distributor reports and feedback and customer feedback. The different external sources of secondary data are government publications, foreign government publications, journals, publications of trade associations, books, magazines, newspapers, annual reports, research reports in universities, etc.

1.5.5 Data Analysis

After data are collected, proper tools and techniques should be used for classification and analysis of data.

The tools of classification of data are frequency distribution, cumulative frequency distribution, relative frequency distribution and charts. *Charts* are graphical representation of data. Different types of charts are pie chart, bar chart, stacked bar chart, histogram, frequency

polygon and ogive curves. The classification tools serve as data presentation techniques for clear interpretations. The analysis of data involves one or more of the following tasks:

- Computation of statistics, viz., mean, median, mode, standard deviation, coefficient of variation, coefficient of skewness, etc.
- Designing regression equation for estimating response variable as a function of set of independent variables
- Performing correlation analysis
- Testing different hypotheses relating to various issues of the research
- Factor analysis
- Discriminant analysis
- Conjoint analysis.

1.5.6 Interpretation of Results

In a research through modelling, following steps are involved:

- Defining the real-world problem
- Formulating a model of the real-world problem
- Deducing model conclusions
- Interpretation of real-world conclusions
- Validation of results.

These above steps clearly indicate that the real-world problem is translated into a model through which the analysis can be easily done. Later, the researcher should infer the results of the research problem from that of the model by taking the assumptions and the kind of transformation, which took place while modelling the original problem. Take the case of a product-mix problem. In this problem, the objective is to determine the optimal combination of production volumes of the products of a company, which results in maximum profit. This problem can be modelled as a linear programming problem and then using the 'Simplex method' one can find its solution.

From the solution of the linear programming problem, it will be difficult to infer the results of the research problem, because of certain transformation which took place while modelling it as the linear programming problem. So, the researcher must trace back those transformation which have been done while modelling and inferring the results of the product-mix problem accordingly from the results of the linear programming problem.

In a survey based research, many assumptions are made about the population, sample, sampling method and scaling while collecting data. The analysis of data after data collection yields a set of results either in the form of statistics, regression equations, identification of significant factors or in the form of acceptance or rejection of different hypotheses. The researcher must infer the results of the original research issues from the results obtained through data analysis as explained in the case of modelling research by taking various assumptions of the study into account.

The algorithmic type of research gives the optimal or near optimal solution as per the assumed transformation of the real-life problem into a form suitable for the algorithm. This solution should be inferred in such a way that it is possible to implement it to bring about the desired results for the real-life situation.

1.5.7 Validation of Results

In a research study through modelling/algorithm, the results after interpretation must be validated by using past data. The process of validation of results ensures the credibility of the results. If there is any mismatch between the results of the model and that of the real-world problem in the past, then the assumptions and modelling exercise should be revisited till the results are validated.

In the case of testing of hypotheses, the concept of validation has different meaning. The credibility of the inferences of testing of hypotheses are already built-in through a significance level. Inferences from the research can act as guidelines for framing policies or some correction/change in the system of study for further betterment.

QUESTIONS

1. Define research and discuss the need for research.
2. Distinguish between fundamental research and applied research. Give examples.
3. What is research methodology? Give a brief account of mathematical tools for the analysis in research methodology.
4. Distinguish between exploratory research and conclusive research.
5. Discuss the types of exploratory research with examples.
6. Discuss the types of conclusive research with examples.
7. Distinguish between modelling research and algorithmic research.
8. Discuss the research problems in different functional areas of management.
9. What are the types of research? Explain them in brief.
10. List and explain the steps of research process?

CHAPTER

2

DATA COLLECTION AND PRESENTATION

2.1 INTRODUCTION

The reliability of managerial decisions depends on the quality of data. The quality of data can be expressed in terms of its representative feature of the reality which can be ensured by the usage of a fitting data collection method.

As has been discussed earlier, data can be classified into primary data and secondary data. The data which are collected from the field under the control and supervision of an investigator is known as primary data. This type of data are generally afresh and collected for the first time.

Some of the examples of primary data are listed below:

- Data collected using a survey to determine the market segment of the product of a company
- Data of a study to determine the infrastructure facilities of educational institutions in a state
- Data of a study to determine the morale of the employees in a company.

If data are collected from journals, magazines, government publications, annual reports of companies, etc., then such data are called as secondary data. In each of these sources of data, the process of data collection has already been done by the respective organization/agency.

The distinction between primary data and secondary data is only relative. The primary data of one study will serve as secondary data of another study. For example, the census data of a country is an example of primary data to find the size of its population. This in turn will serve as data to plan public distribution system of each taluk of different districts in each state of the country. The same data under such situation is called as secondary data. The industry handbook is an example of primary data established through a survey, which can be used as secondary data by researchers in management and commerce disciplines.

2.2 PRIMARY DATA

As stated earlier, primary data are useful for current studies as well as for future studies. Hence, it should be collected with utmost care. The different methods which are used for primary data collection are: observation method, personal interview, telephone interview and mail survey.

2.2.1 Observation Method

In the *observation method*, the investigator will collect data through personal observations.

Consider an example in which an investigator collects the data about the organizational climate in an organizations through direct observation. In this study, the investigator will speak to the employees at different levels of the organization, observe the behaviour of the employees to assess the organizational climate.

Consider the case of work sampling in a shop floor to determine the standard time of a job which is manufactured by a set of machines. In such study, the investigator observes the state of these machines through random sampling. The state of a machine may be either in working state or idle state. At the end of the study, based on the direct observations, the investigator will determine the standard time of the job that is manufactured using those machines.

Continuous monitoring of stock exchange index and share prices movements through newspaper and magazines is an example of observational method which will help investment companies and individuals effective management of portfolios.

Generally, observation method of data collection deals with the recording of the behaviour of respondents/sampling units. In this method, the investigator will observe the behaviour of the respondents in disguise. Take the case of customers transacting with a bank. Here, the behaviour of the customers like, patience while waiting, way of moving with the bank employees, helping fellow customers in filling different forms, informing the bankers if there is any excess credit in their pass books, returning excess currency to the cashier if given by him, opinion of the customers about the bank through their casual discussions, time spent in reading circulars in the notice board, etc., will be observed by the investigator. The identity of the investigator should not be revealed to the customers. If it is known to the customers, they may change their behaviour.

This method helps capture the behaviour of customers directly. But, it is a time consuming and costly exercise. Also, it suffers from personnel biases of investigators which will distort the findings.

2.2.2 Personal Interview

Personal interview is a survey method of data collection which employs a questionnaire. The components of the personal interview are the researcher, the interviewer, interviewee and the interview environment. Under the guidance of the researcher several interviewers will be sent with questionnaires to meet the respondents/interviewees of the survey for seeking responses to the questions in the questionnaire under the defined interview environment. The environment is defined in terms of place, time and several other factors which will

influence interviewees. Personal interview can be classified into door-to-door interview, executive interview, mall intercept surveys, self-administered questionnaires and purchase intercept technique.

In *door-to-door interviewing*, the interviewer will go to the residence of each selected respondents/customers and obtain responses for the items in the questionnaire by direct interviewing with him/her for consumer goods or any other topic of study. This is a detailed in-depth survey method which seeks responses with better precision. It is a time-consuming and costly process.

Executive interviewing is same as door-to-door interviewing except that the interviewees are industrial executives and the products are industrial goods. This is a costly and time-consuming process. The average waiting time of the interviewers at the offices of the executives before commencing the interview process is too long because of the busy schedule of the executives.

Mall intercept interview is conducted in shopping complexes where a diverse spectrum of customers can be interviewed. The people who pass through that place will be used as respondents for filling the questionnaires. In this method, the interviewers are stationed in a particular place of an area and the people who come to that area will be sampled and served with questionnaires. Hence, it is a less costly and quick process.

In *self-administered interview*, the respondents will be issued with questionnaires and they will be asked to fill them on their own. This means that the interviewers will be absent while filling the questionnaires. This method has the advantage of removing interviewer's bias. But, the absence of the interviewer will make the respondents to assume some irrelevant responses for some open ended questions for which the respondents have limited knowledge/doubt. It is a less costly and quick process.

Purchase intercept technique is a direct interviewing method in which the customers will be intercepted and interviewed after they select their items from the showroom. The main advantage of this method is that it will enable the buyers to have better recall and compare the product of interest with competing brands with greater accuracy since they are yet to complete the buying process. But, in this method, the sampling is restricted to the buyers who completed the selection of some items to buy. Hence, a complete randomized selection of sampling units is prohibited in this method.

2.2.3 Telephone Interview

Telephone interview is considered to be a cost effective and dominant data collection method because of the following reasons:

- The travel time of interviewers is totally eliminated
- The cost of travel of interviewers is also eliminated
- Greater possibility of reaching the customers all over the geography
- Total time of conducting the interview of the sample is least when compared to other methods
- Greater possibility of random selection of respondents among the population having telephone connection

- There is a greater probability of reaching the respondent unlike the nonresponse problems of personal interview.

Though the telephone interview has many advantages, it has the following drawbacks:

- Impossible to employ visual aids
- Exclusion of population who are not having telephone connection
- Interviewee may discontinue conversation in the mid-way unlike the personnel interview in which there is greater probability of completing the interview fully because of the rapport developed between the interviewer and the interviewee through personal contact.

The components of the telephone interview are same as the personal interview, viz., the researcher, the interviewer, interviewee and the interview environment. The unique features of telephone interview are selecting telephone numbers, call outcomes, call timing and call report.

Selection of telephone numbers

Telephone numbers must be selected such that they cover the entire population of interest. The methods for selection of telephone numbers are as follows:

- Telephone directory reference
- Random-digit dialing
- Systematic random-digit dialing
- Plus-one dialing.

The traditional method of selection of telephone numbers is the selection of numbers from telephone directory. The sampling frame is designed by randomly selecting telephone numbers from the respective telephone directories of different regions.

In random-digit dialing, without the help of telephone directory, the telephone numbers are dialed randomly by specifying area code and within each area code a randomly generated telephone number. This process is repeated till the required number of respondents are interviewed.

In systematic random-digit dialing, initially an area code and a telephone number are randomly selected. Then a constant number is added to the initial number to get the second number which is to be dialed. Similarly, the same constant number is to be added to the immediate preceding number generated to find the remaining required number of telephone numbers of the respondents.

The plus-one dialing is a directory assisted telephone interview. In this method of interview, the required number of telephone numbers are generated from one or more telephone directories. Then the constant '1' is added to say, each of the last four digits of each selected telephone number. This generates a new sample of telephone numbers. This sample may also contain the telephone numbers which are not present in telephone directories because the telephone numbers of new subscribers will be absent in telephone directories till they are updated. So, this method can overcome such deficiency.

Call outcomes

In telephone interview, the following outcomes of a call noticed:

1. The telephone is not in service.
2. The number dialed is busy.
3. No one answers the call.
4. The number dialed is a fax number.
5. Answering machine responds.
6. The call is responded by someone else instead of the targetted respondent.
7. The person belonging to the dialed number is not in the sampling frame.
8. The call is responded by the targeted respondent.

Following are important guidelines for handling call outcomes:

1. If the telephone is not in service, then eliminate that number from the sampling frame.
2. If the number dialed is busy, then contact that number later.
3. If the call is not answered by any one, try later.
4. If the number dialed is a fax number, give a fax to that number requesting to provide the respondent's telephone number and possible time of contact.
5. If the call is answered by an answering machine, then leave a message about yourself and the purpose of your call, and dial the same number later.
6. If the call is responded by someone else instead of the targetted respondent, then introduce yourself and brief about the purpose of your study and try that number later.
7. If the person belonging to the dialed number is not in the sampling frame, then eliminate that number once for all.
8. If the call is responded by the targetted respondent and if he/she is free, conduct the interview; otherwise contact him/her again at his/her convenience.

Call timing

The timing of the calls should be such that maximum number of respondents are available for interview on first dialing. The respondents may be classified into working, non-working, business groups.

For consumer survey, it is preferable to call the respondents belonging to all the three groups mentioned above in evening time during weekdays and day time during weekends. For survey relating to industrial marketing, the respondents belonging to a business group and part of working group should be called during 10⁰⁰ A.M. and 4 P.M. only on weekdays. The remaining part of the working group may be called during weekends.

Call report

It is a catalogue of events about telephone interviewing. The interviewer should record telephone numbers called, and for each number, the name of the respondent, call date and

time, duration of the call, any special remark of the respondent, etc., in the call report. This report is a kind of database for the interviewers which is served to manage the respondents who are yet to be interviewed.

2.2.4 Mail Survey

Mail survey is a data collection method in which questionnaires are mailed to potential respondents who in turn fill and return them at their convenience. This method has the following advantages:

- Less cost of data collection
- Less time of data collection
- Wider coverage of population
- Better accuracy of data
- Absence of interviewer's bias.

But it has the following drawbacks.

- The identity of the respondents is not known to the interviewers.
- The questionnaire may be filled in by the assistant/family members of the respondent.
- Some respondents may not return filled-in questionnaires.
- There may be delay from the part of the respondents in returning the filled-in questionnaires.

In mail survey, the researcher selects the required number of potential respondents of the study from mailing list/ mailing panel provided by some organizations. Then a carefully designed questionnaire is despatched to each of the potential respondents.

The mailing of questionnaires involves the following tasks:

- Selecting the type of envelope
- Determining the mode of postage
- Designing covering letter
- Deciding questionnaire length, content, layout and format
- Notification and follow-up details
- Type of incentive, if any, to be given to potential respondents.

In some cases, even before mailing the questionnaires, a notification is sent to the respondents which makes them aware of the purpose of the study before hand. After mailing the questionnaires, reminders are to be sent to the respondents at predetermined intervals to speedup the matter.

If the response rate is very poor, then more reminders should be mailed to the respondents. In spite of this effort, if the response rate is low, to cope up with the deficit number of respondents, either the personal interview or the telephone interview may be used. As an alternative, a kind of extrapolation can be done based on the responses of the respondents who replied very late. Here, these delayed respondents are equated to non-response type respondents for the purpose of extrapolation.

As an alternative to mail survey, with the emergence of communication facilities, one can use either *fax survey* or *Web survey*. In a fax survey, the questionnaire is sent to the respondents through fax. In turn, the respondents are expected to send the filled-in questionnaires through fax or mail. This method quickens the entire process. The success of this method relies on the infrastructural facilities available with the respondents. Hence, this is used only for the organizational respondents for whom such facilities are available. In the spirit of the study, organizations support its employees to use such fax facility. In a Web-based survey, the questionnaire is posted on a secure Web site. The invitation to participate in the Web survey would be posted on a company Web site which receives high traffic from target customers. If a customer visits that particular Web site and clicks that particular invitation banner, then the customer will be connected to the secure Web site, where a detailed interview is conducted. After finishing the interview, the customer is placed back to his/her original point of entry into this system. The moment, a customer finishes the interview in a secure Web site, the researcher can view the results of the interview instantaneously at his/her station. When compared to other methods, this has the provision of even filtering nonsensical responses. It is considered to be the fastest method of data collection. But, the units of the sampling frame are limited to the customers who own computers or visits Web sites very often on rental machines.

2.2.5 Questionnaire Design

As mentioned in the previous section, the different methods of primary data collection, viz., personal interview, telephone interview and mail survey come under survey methods. The success of survey methods depends on the strength of the questionnaire used. A questionnaire consists of a set of well-formulated questions to probe and obtain responses from respondents.

Though there is a variation in the form and content of questionnaires from situation to situation, the generalized steps of designing a questionnaire are discussed now.

Identification of research issues

The definition of research issues and related objectives is an important task while designing questionnaire because the content of the questionnaire mainly depends on these issues based on which measurements are made. Hence, they should be clearly identified and defined properly. If the researcher is not clear about the kind of measurements to be made, then the questionnaire will suffer from ambiguous questions. So the answers to these questions will have no value for the study. This amounts to garbage-in and garbage-out. So, the researcher should define different hypotheses for various issues of the research which in turn help him/her to decide about the measurements to be made.

Formulation of questions and format

If the types of measurements to be made are clearly known, then the researcher can formulate necessary questions relating to each measurement. While formulating a question, its content should be such that each respondent grasps the intention of the question very quickly as originally thought by the researcher. This will minimize the distortion of the research focus. Also, the format of each question in relation to that of the overall questionnaire should be decided.

The format of a question can be of three types: open-ended questions, close-ended questions and structured questions.

Open-ended questions In this type of question, the interviewer writes the answer of the respondent verbatim. In this type, there is no pre-coded classification of answers to the question.

In a survey to find customer opinion on the interest rate of housing loan, a sample open-ended question without classification is as:

What are the reasons for availing housing loan?

The possible answer for this question may be:

Mainly to avail tax benefit and inculcate saving habit in me.

The open-ended questions may bring unconventional answers which are not known to the researcher. Generally, this type of question is used for probing type question, introductory question to get opinion about the components of the system of study, sensitive question, etc. If the number of alternative answers is many, then one can resort to this type of question. This is alternatively known as *unstructured* question.

Though it has certain benefits, the answers to unstructured questions may not be realistic as expected if the respondents are not articulate and educated. In telephone interviews, interviewers find it difficult to write verbatim of the answers given by the respondents to open-ended questions.

Close-ended questions Close-ended questions are structured questions. It has two formats:

1. Questions with multiple responses out of which the respondents have to select one or more choices.
2. Questions with rating scale with discrete responses or continuous range.

A sample format of the first type is as shown below:

What is the number of dependents with you?

- | | |
|---------------------|-----------|
| (a) None | (b) One |
| (c) Two | (d) Three |
| (e) Four and above. | |

A sample format of the second type is demonstrated based on the joint family system as follows:

Question: "Joint family system leads to better GDP of a nation." Please tick one of the following:

- | | |
|---------------------------------|-------------------------|
| (a) Strongly agree | (b) Moderately agree |
| (c) Strongly disagree | (d) Moderately disagree |
| (e) Neither agree nor disagree. | |

An extreme form of close-ended question has only two responses. The respondent has to select one of these two responses which is demonstrated with reference to a study on consumer survey. This question has dichotomous responses. For example,

Have you used any one of our company's products?

(a) Yes

(b) No

Deciding question wording

The wording of each question should fully convey its meaning to the respondents. The degree of the level of understanding varies from person to person. If questions are not easily understood by the respondents, they will respond with random answers to questions, which may distort the research findings.

Guidelines for question wording are as follows:

1. The vocabulary used should be simple, direct and familiar to all respondents.
2. The words of the question should not have vague or ambiguous meaning.
3. As far as possible, each question should have single part. Otherwise, the respondents will have difficulty in answering all the parts of the questions.
4. Questions should not be leading to answers which will suffer from biases.
5. Questions should be loaded with more interrelated items, which bring out the respondents attitude and real opinion in the form of their responses.
6. The instructions of the questions should be clear.
7. The questions must be applicable to all the respondents in the sampling frame.
8. Keep the number of words of each questions as minimal as possible.

Appropriate sequence of questions

After formulating necessary questions, they should be arranged in an appropriate sequence such that respondents are able to have better grasp of them and provide accurate responses.

A preferred sequence of questions is listed here:

1. First few questions relating to personal details of the respondents.
2. Introductory questions to establish a rapport with the respondents.
3. Simple and direct questions to build an opinion that the questionnaire is simple to answer.
4. Focussed questions relating to objectives and issues of the research.
5. Main body of the questionnaire should contain questions to obtain needed information.

Pre-testing of questionnaire

Once the questionnaire is ready, it should be pre-tested through a pilot survey involving the respondents in the proposed sampling frame. This exercise is mainly intended to test the degree of understanding the meaning of the questions, difficulty in understanding the questions by the respondents if the meaning of the questions are conveyed correctly, to check the relevance of the questions, to ascertain the interest of the respondents, etc.

Review of questionnaire for improvements

The purpose of pre-testing of questionnaire is to obtain information to improve its content, format and sequence. Based on the information, the questionnaire should be revised in its format, content and sequence for final use in the survey.

A SAMPLE QUESTIONNAIRE

A study was conducted for a tele-services company to find the expectations of customers using telephone booths at Chennai and their profiles. The format of the questionnaire used in this study is presented below:

QUESTIONNAIRE

STUDY ON CUSTOMER EXPECTATIONS AND PROFILES OF PCO BOOTHS AT CHENNAI

* Address of Telephone Booth:

Customer's personal profile:

1. NAME:

2. AGE:

(a) Up to 18 yrs

(b) 19-24 yrs

(c) 25-35 yrs

(d) 36-45 yrs

(e) 46-55 yrs

(f) More than 55 yrs.

3. GENDER:

(a) Male ()

(b) Female ().

4. MONTHLY HOUSEHOLD INCOME:

(a) Less than Rs. 5000

(b) Rs. 5000-10000

(c) Rs. 10000-15000

(d) Rs. 15000-25000

(e) More than Rs. 25000.

5. OCCUPATION:

(a) Service sector

(b) Government

(c) Public

(d) Private

(e) Business

(f) Student/housewife

(g) Others (specify).

If business, tick one of the following:

(a) Trading

(b) Manufacturing

(c) Retail

(d) Service industry

(e) Others (specify).

Are you operating from Chennai?

Yes ()

No ()

If None, mention your base city: _____

6. QUALIFICATION:

(a) School

(b) Undergraduate

(c) Postgraduate

(d) Doctorate

(e) Others

7. OWNERSHIP DETAILS: (Please specify the brand.)

Two wheeler:

Four wheeler:

Cellphone: Roaming () STD ()

Credit card:

Data profile:

1. How often do you make a call at a PCO? (Please tick the cell.)

		No. of times			
		1	2	3	> 3
Periodicity	(a) Daily				
	(b) Weekly				
	(c) Monthly				

2. What is the type of call made by you?

(a) Personal

(b) Business

(c) Both

3. What are reasons for using PCO?

.....

4. Do you come to this booth all the time?

(a) Yes, why?

(b) No, why?

5. Do you have a telephone/cellphone?

(a) If yes, then go to item 6.

(b) If no, why?

Go to item 7.

6. Do you have an STD/ISD connection at home/office?

(a) If yes, why do you come to PCO booth to make calls?

.....

(b) If no, why?

.....

Go to item 8.

7. Are you planning to get a telephone service along with a STD/ISD facility in the near future?

(a) If yes, why?

.....

(b) If no, why?

.....

8. Are you aware of the charge rates at different timings?

(a) If yes, mention them

.....

(b) No

9. At what time do you make STD calls?

(a) Morning

(b) Afternoon

(c) Evening

(d) Night

10. At what time do you make ISD calls?

(a) Morning

(b) Afternoon

(c) Evening

(d) Night

11. What is your average weekly spending on PCO calls?

(a) Less than Rs. 50

(b) Rs. 50-100

(c) Rs. 100-250

(d) Rs. 250-500

(e) Rs. 500-1000

(f) More than Rs. 1000

12. Are you satisfied with the billings at PCO?

(a) Yes ()

(b) No ()

If No, give reasons:

.....

13. What are your general expectations from PCO billings?

.....

14. Have you encountered any PCO booth with tampered billings?

(a) Yes ()

(b) No ()

15. Rank your order of preference of expectation from 1-6 at PCO booths?

(a) Good aeration ()

(b) Privacy ()

(c) Good seating facility ()

(d) Recreation ()

(e) Accurate billing ()

(f) Exact changes ().

Any other, specify:

16. "Privacy of conversation at PCO booths can be guaranteed."

(a) Strongly agree

(b) Moderately agree

(c) Strongly disagree

(d) Moderately disagree

(e) Neither agree nor disagree

17. "Some may feel that coming to PCO booths to make calls is below the dignity of a person."

(a) Strongly agree

(b) Moderately agree

(c) Strongly disagree

(d) Moderately disagree

(e) Neither agree nor disagree.

18. List the inconveniences faced by you at PCO booths.

.....

THANK FOR YOUR COOPERATION

2.2.6 Editing of Primary Data

After executing the survey, it is highly essential to review the answers of the respondents provided in the questionnaires from the angle of legibility, completeness, consistency and homogeneity.

Legibility To transform the responses into useful information, the responses must be legible. The filling of questionnaires by some of the interviewers may not be legible. Under such situation, the research investigator should ask the respective interviewers to rewrite it or it can be corrected by the investigator with the knowledge of the interviewers.

Completeness All the items in the questionnaire should have respondents' answers. But, some of the items may be unfilled. This may be due to the negligence of the interviewers while recording data or he/she would have forgotten to ask such items. If it is due to the negligence on the part of the interviewers, then he/she may be asked to recall the responses for those items either from his/her memory or from his/her scratch pad. If no attempt has been made about those items, then the respective respondent may be approached again to seek his/her responses for those items.

Consistency The investigator should examine the data for consistency. The responses should be accurate and match with the realistic figures. Sometimes there may be mismatch of responses of related items of the questionnaire. By applying logical analysis, one can easily identify such flaws and rectify them with interpolation or extrapolation of responses based on other answers. As an example, the monthly expenditure may not match with his monthly income. Under such a situation, the researcher should identify the true response out of the two responses—monthly income and monthly expenditure—and estimate the

other responses using either interpolation or extrapolation. There may be repeating pattern of responses in the questionnaire. Such repeating pattern is a sign of manipulation done on the part of the interviewers without really collecting the data from the respondents. Such questionnaires may be omitted or another attempt may be made with other interviewers.

Homogeneity. The investigator should randomly check a sample of questionnaires collected by each interviewer for homogeneity of data. This means that whether the questions have been understood in the same way by all the respondents from the point of view of dimensions like, income per period. The period may be weekly, monthly or yearly. All the answers of the respondents for such items should be based on the same period. Otherwise, it will distort the mean income of the respondents. Similarly, there should be homogeneity in the units of the quantity of purchase/consumption of a product by the respondents. Here, two dimensions are involved, viz., consumption per period and units of measurement of weight/volume. Let the actual consumption of the product is 50 kg per month. This may be responded like 50 grams per month or 50 kg per week which will distort the respective mean value.

2.3 SECONDARY DATA

As stated earlier, secondary data are collected from sources which have been already created for the purpose of first-time use and future uses.

The secondary data collection involves less cost, time and effort. Sometimes more accurate data can be obtained only from secondary data. Such data are collected for some other purpose. So, sometimes, it may not suit for the present purpose. There is no control over secondary data collection and hence it may not be accurate for many applications. Further, it may not be available in the required format. Sometimes, it may be outdated and may not meet the requirements of the present study.

The sources for secondary data can be classified as internal sources and external sources.

2.3.1 Internal Sources of Secondary Data

Internal sources of secondary data for marketing applications are as follows:

- Sales records
- Marketing activity
- Cost information
- Distributor reports and feedback
- Customer feedback.

A sales system consists of a set of integrated elements which also includes customer profile and intermediaries details. So, the different reports starting with invoice to customers-summary report can be used as sources for secondary data for future study.

The extent of marketing efforts in terms of infrastructures, labour force, sales promotions and advertisements will form marketing activity. The record of these activities will provide

information in terms of cost, marketing force profile, feedback of sales promotions and effect of advertisements.

With the emergence of computer technology, it is possible to systematically record customer feedback about product failures, after-sales services, product utility, etc. either directly or through distributors, in addition to, distributors' feedback on timeliness of stock replenishment, quality of product, match of product range with the actual customer inquiries.

Likewise, one can recognize different internal sources of data with respect to studies in other areas like, personnel, finance, production, public systems, etc.

2.3.2 External Sources of Secondary Data

Various external sources of secondary data are government publications, foreign government publications, journals, publications of trade associations, books, magazines, newspapers, annual reports, research reports in universities, industry handbook, publications of statistics department, census data, Indian Agricultural Statistics, index number of wholesale prices in India, Reserve Bank of India Bulletin, etc.

Besides, publications from UNESCO, WHO, ILO, UNO also form sources of secondary data. Case studies and findings of research articles in journals of different disciplines, viz., trade, commerce, economics, engineering, sociology, history, etc., form up-to-date form of secondary data for advanced research studies.

2.4 DATA PRESENTATION

After collecting data, they must be classified and presented in meaningful forms to have better insight of a research problem. Different tools of classification and presentation of research data are listed as follows:

- Frequency distribution
- Cumulative frequency distribution
- Relative frequency distribution
- Charts.

2.4.1 Frequency Distribution

If data are of repeating nature, then they should be presented in forms of the number of occurrences of each value of the data of a particular type.

The frequency distribution can be classified into discrete frequency distribution and continuous frequency distribution which are demonstrated in Tables 2.1 and 2.2, respectively.

Table 2.1 Discrete Frequency Distribution

<i>Income category</i>	<i>Number of respondents (frequency)</i>
Low income	300
Medium income	200
High income	100

Table 2.2 Continuous Frequency Distribution

Monthly income (in rupees) (class interval)	Number of respondents (frequency)
0-5000	20
5000-10000	30
10000-15000	40
15000-20000	60
20000-25000	30
25000-30000	20

2.4.2 Cumulative Frequency Distribution

The cumulative frequency distribution is a modified form of frequency distribution, as shown in Table 2.3. In a given row, the value in the last column is the cumulative value of the frequencies shown in its last but one column up to that value.

Table 2.3 Cumulative Frequency Distribution

Monthly income (in rupees) (class interval)	Number of respondents (frequency)	Cumulative frequency
0-5000	20	20
5000-10000	30	50
10000-15000	40	90
15000-20000	60	150
20000-25000	30	180
25000-30000	20	200

2.4.3 Relative Frequency Distribution

The relative frequency distribution is a modified form of frequency distribution, as shown in Table 2.4. In a given row, the value in the last column is the ratio between the frequency of that row and the total frequency.

Table 2.4 Relative Frequency Distribution

Monthly income (in rupees) (class interval)	Number of respondents (frequency)	Relative frequency
0-5000	20	$20/200 = 0.10$
5000-10000	30	$30/200 = 0.15$
10000-15000	40	$40/200 = 0.20$
15000-20000	60	$60/200 = 0.30$
20000-25000	30	$30/200 = 0.15$
25000-30000	20	$20/200 = 0.10$

2.4.4 Charts

Data can be presented in graphical form (or charts), which are classified as: pie chart, bar chart, stacked bar chart, histogram, frequency polygon and ogive curve.

Pie chart This is a chart in which the area of a circle is divided into different sectors representing different categories such that the angle of each sector is proportional to the frequency of the corresponding category. The data shown in Table 2.1 are shown in the form of a pie chart in Fig. 2.1.

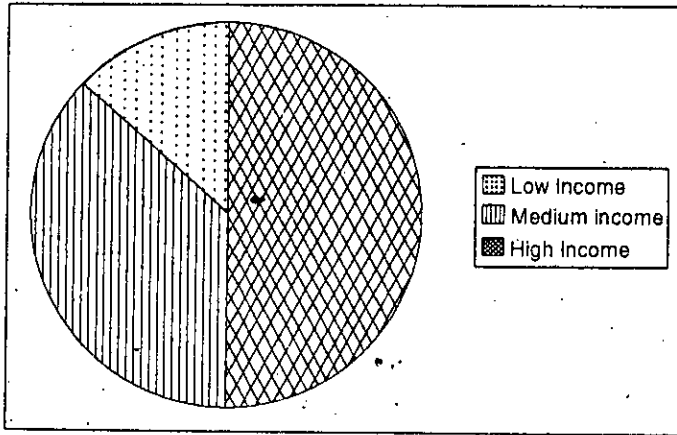


Fig. 2.1 Pie Chart.

Bar chart Bar chart is a graphical view of the given data such that the frequency of each category is shown as a vertical strip against that category in proportion to the heights of other such vertical strips. The data shown in Table 2.1 are shown in the form of a bar chart in Fig. 2.2. In this figure, each category is taken on X-axis and a vertical bar corresponding to its frequency in proportion to other frequencies is marked in Y direction.

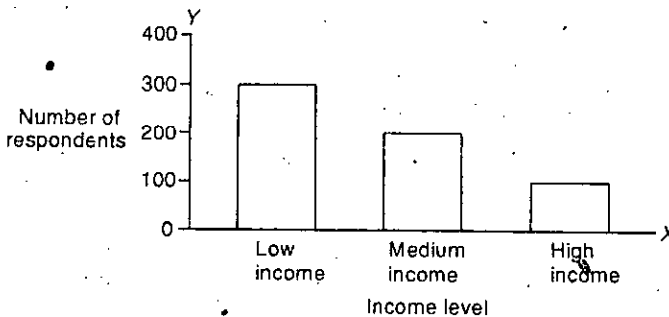


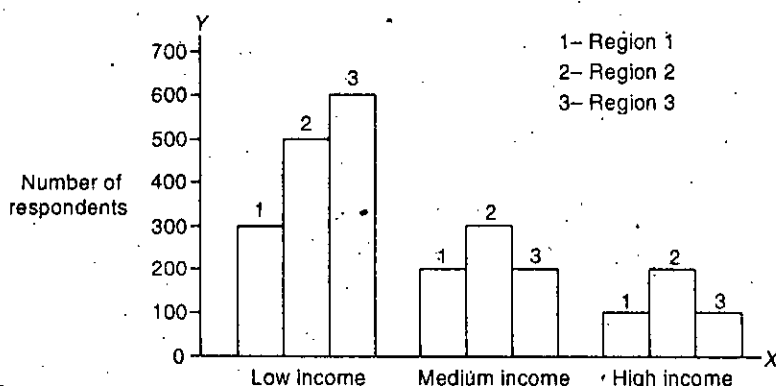
Fig. 2.2 Bar chart of income distribution.

Stacked bar chart Stacked bar chart is a graphical view of the given multiple data such that the frequency of each category of the multiple data is shown as an adjacent vertical strip against that category in proportion to the heights of other such vertical strips within that multiple data at each point on X-axis and across different points on X-axis. Consider the data, as shown in Table 2.5.

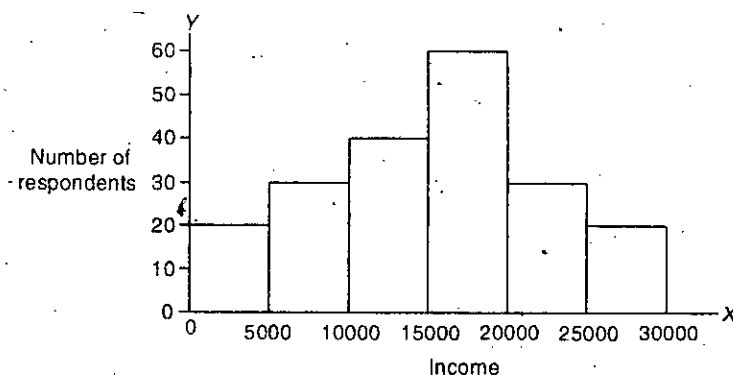
Table 2.5 Discrete Frequency Distribution of Number of Respondents

Income category	Number of respondents (frequency)		
	Region 1	Region 2	Region 3
Low income	300	500	600
Medium income	200	300	200
High income	100	200	100

The data shown in Table 2.5 are shown in the form of a stacked bar chart in Fig. 2.3. In this figure, each category of income is taken on *X*-axis and a vertical bar corresponding to the frequency of number of respondents of each region in each income group is shown as an adjacent vertical strip within that income group in proportion to other such frequencies which are marked in *Y* direction.

**Fig. 2.3** Stacked bar chart of number of respondents.

Histogram *Histogram* is a graphical view of the given data such that the frequency of each range of the variable of interest is shown as adjacent vertical strip against that range of the variable in proportion to the heights of other such vertical strips. The data shown in Table 2.2 is shown in the form of a histogram in Fig. 2.4. In this figure, for each class

**Fig. 2.4** Histogram of income distribution.

interval, the corresponding vertical bar proportionate to its frequency is placed in Y direction over its range on X -axis.

Frequency polygon *Frequency polygon* is a line diagram connecting the mid-points at the top of different vertical strips of the histogram of a given problem. The data shown in Table 2.2 are shown in the form of a frequency polygon in Fig. 2.5.

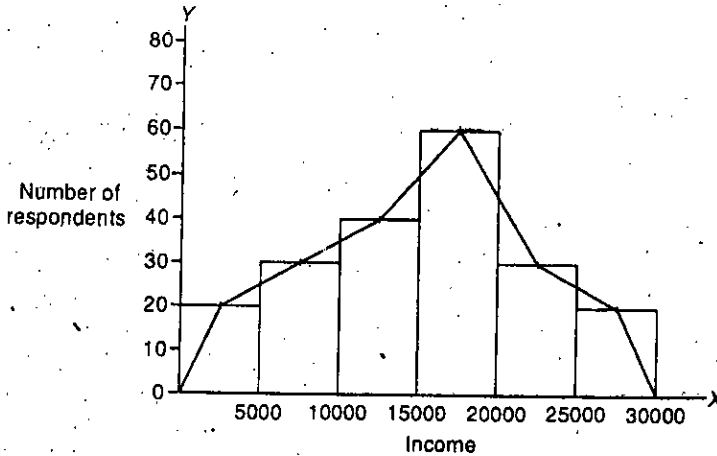


Fig. 2.5 Frequency polygon of income distribution.

Ogive curve *Ogive curve* is a cumulative frequency curve. This can be classified into *less-than-ogive curve* and *more-than-ogive curve*. In *less-than-ogive curve*, the upper limit of each class interval is taken on X -axis in increasing order. For each such upper limit on X -axis, the cumulative frequency of all the class intervals from the first class interval to that class interval is taken on Y -axis. The data shown in Table 2.2 are converted into a form, which is suitable for *less-than-ogive curve*, as shown in Table 2.6. The corresponding *less-than-ogive curve* is shown in Fig. 2.6.

Table 2.6 Cumulative Frequency Distribution for Less-than-ogive Curve

Monthly income (in rupees) (class interval)	Number of respondents (frequency)	Cumulative frequency
Less than 5000	20	20
Less than 10000	30	50
Less than 15000	40	90
Less than 20000	60	150
Less than 25000	30	180
Less than 30000	20	200

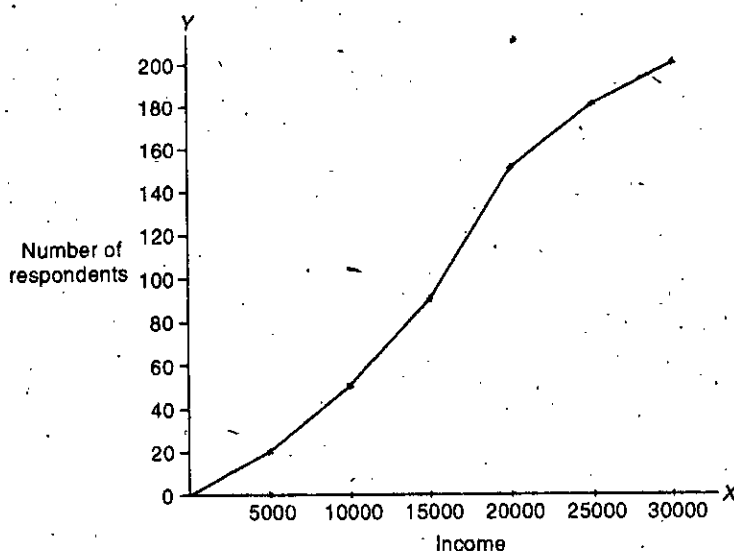


Fig. 2.6 Less-than-ogive curve of income distribution.

In more-than-ogive curve, the lower limit of each class interval is taken on X -axis in increasing order. For each such lower limit on X -axis, the cumulative frequency of all the class intervals from that class interval to the last class interval is taken on Y -axis. The data shown in Table 2.2 are converted into a form which is suitable for more-than-ogive curve, as shown in Table 2.7. The corresponding more-than-ogive curve is shown in Fig. 2.7.

Table 2.7 Cumulative Frequency Distribution for More-than-ogive Curve

Monthly income (in rupees) (class interval)	Number of respondents (frequency)	Cumulative frequency
More than 0	20	200
More than 5000	30	180
More than 10000	40	150
More than 15000	60	110
More than 20000	30	50
More than 25000	20	20

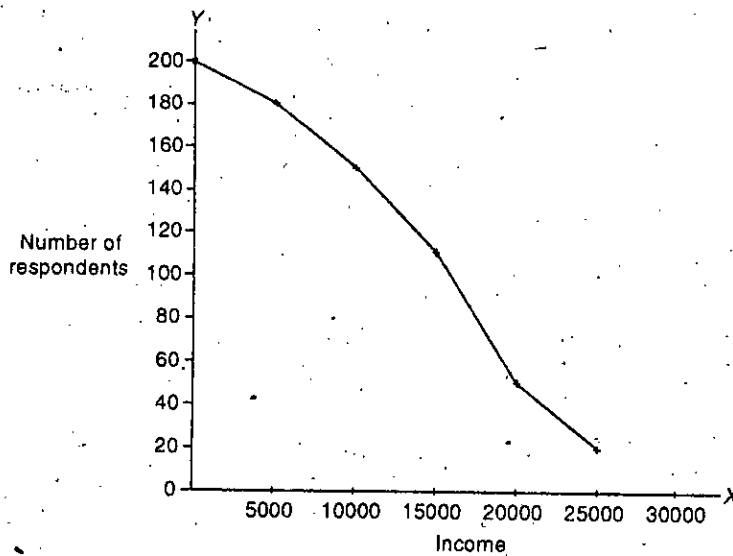


Fig. 2.7 More-than-ogive curve of income distribution.

QUESTIONS

1. Define data and give some examples of data. Distinguish between primary data and secondary data.
2. What are the methods of collecting primary data? Compare and contrast them.
3. What is an observation method? Explain it with examples.
4. Give the classification of a personal interview and explain them.
5. Discuss the features of a telephone interview.
6. What are the methods of selecting telephone number in telephone interview? Explain them.
7. What are the possible outcomes of call outcomes?
8. Write short notes on call timings?
9. What is a mail survey? What are the tasks in mailing questionnaires?
10. Discuss the steps of questionnaire design.
11. Give brief account of editing primary data.
12. What are secondary data? What are the internal and external sources of secondary data?
13. What are the data presentation tools? Explain them in brief.
14. What is frequency distribution? Give an example.

15. What is the relative frequency distribution? Give an example.
16. What are the types of charts? Compare and contrast them.
17. Illustrate each of the following with an example:
 - (a) Pie chart
 - (b) Bar chart
 - (c) Stacked bar chart
 - (d) Histogram
 - (e) Frequency polygon
 - (f) Ogive curves.
18. Design a questionnaire to collect data for analyzing employees morale in a thermal power station.
19. Design a questionnaire to study the impulse buying behaviour of consumers in a grocery stores.
20. Design a questionnaire to study the status of TQM applications in Indian industries.
21. Design a questionnaire to study the scope of non-renewable sources of energy in India.

CHAPTER

3

REVIEW OF BASIC STATISTICAL MEASURES

3.1 INTRODUCTION

The data of a given situation should be characterized by some statistical measures for the purpose of estimation or comparison with similar data or making inference about the sample population to which the data belong.

Statistical measures can be classified into measures of central tendency, measures of variation and measures of skewness.

Measures of central tendency are arithmetic mean, weighted arithmetic mean, median, mode, geometric mean and harmonic mean. Measures of variation are range, quartile deviation, average deviation, standard deviation and coefficient of variation. The measure of skewness is relative skewness.

3.2 MEASURES OF CENTRAL TENDENCY

In this section, the concept of each of the measures of central tendency is demonstrated using suitable examples.

3.2.1 Arithmetic Mean

It is the most widely used measure of central tendency. It is defined as the ratio between the sum of the observations and the number of observations.

Arithmetic mean of ungrouped data

The formula to compute the arithmetic mean of ungrouped data is:

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

where $\bar{X}$ is the arithmetic mean; X_i , the i th observation; and n , the total number of observations.

Example 3.1 Determine the arithmetic mean of salaries of the employees, as shown in Table 3.1.

Table 3.1 Salaries of Employees

Employee no.	1	2	3	4	5	6	7
Monthly salary (Rs.)	12,000	14,500	8,500	13,500	13,500	17,500	11,500

Solution. The number of observations, $n = 7$. Then

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} = \frac{12,000 + 14,500 + 8,500 + 13,500 + 13,500 + 17,500 + 11,500}{7} = \text{Rs. } 13,000$$

Arithmetic mean of grouped data

The formula for arithmetic mean of grouped data is:

$$\bar{X} = \frac{\sum_{i=1}^n f_i X_i}{\sum_{i=1}^n f_i}$$

where $\bar{X}$ is the arithmetic mean of grouped data; X_i , the mid-point of the i th class interval; f_i , the frequency of the i th class interval; and n , the total number of class intervals.

Example 3.2 The minimum and the maximum of the monthly salaries paid to the employees of a computer firm are Rs. 8,000 and Rs. 40,000. The frequency distribution of the monthly salary of the employees of the firm is presented in Table 3.2. Determine the arithmetic mean.

Table 3.2 Frequency Distribution of Monthly Salary

Monthly salary (Rs.)	No. of employees	Monthly salary (Rs.)	No. of employees
8,000–12,000	400	24,000–28,000	200
12,000–16,000	350	28,000–32,000	150
16,000–20,000	325	32,000–36,000	75
20,000–24,000	250	36,000–40,000	25

Solution. The data of Table 3.2 is presented in Table 3.3 with additional data for computing the arithmetic mean.

Table 3.3 Frequency Distribution of Monthly Salaries with Additional Data

Monthly salary (Rs.)	Mid-point of class interval (X_i)	Frequency (f_i)	$f_i X_i$
8,000–12,000	10,000	400	4,000,000
12,000–16,000	14,000	350	4,900,000
16,000–20,000	18,000	325	5,850,000
20,000–24,000	22,000	250	5,500,000
24,000–28,000	26,000	200	5,200,000
28,000–32,000	30,000	150	4,500,000
32,000–36,000	34,000	75	2,550,000
36,000–40,000	38,000	25	950,000
Total		1,775	33,450,000

The total number of employees, $n = 1,775$. Then

$$\bar{X} = \frac{\sum_{i=1}^n f_i X_i}{\sum_{i=1}^n f_i} = \frac{33,450,000}{1,775} = \text{Rs. } 18,845.07$$

The arithmetic mean of monthly salaries = Rs. 18,845.07.

Shortcut method for arithmetic mean of grouped data

The arithmetic mean of the grouped data can be obtained using the shortcut method, which is presented as follows:

$$\bar{X} = A + \frac{\sum_{i=1}^n f_i d_i}{\sum_{i=1}^n f_i} C = A + \frac{\sum_{i=1}^n f_i d_i}{N} C$$

where $\bar{X}$ is the arithmetic mean of grouped data; X_i , the mid-point of the i th class interval; f_i , the frequency of the i th class interval; A , the assumed mean (preferably around the mid of the series); C , the width of class interval; $d_i = (X_i - A)/C$; n , the total number of class intervals; and N , the sum of the frequencies.

Example 3.3 The minimum and the maximum of the ages of the employees of a textile firm are 20 years and 60 years, respectively. The frequency distribution of the age of the employees of the firm is presented in Table 3.4. Determine the arithmetic mean of the age of the employees.

Table 3.4 Frequency Distribution of Ages of Employees

Age (years)	No. of employees	Age (years)	No. of employees
20–24	40	40–44	40
24–28	35	44–48	50
28–32	35	48–52	45
32–36	25	52–56	35
36–40	30	56–60	30

Solution. The data in Table 3.4 is presented in Table 3.5 with additional data to compute the arithmetic mean. For this example, $N = 365$ and $C = 4$. Let the assumed mean (A) be 38. Therefore,

$$d_i = \frac{X_i - A}{C} = \frac{X_i - 38}{4}$$

Table 3.5 Frequency Distribution of Ages of Employees with Additional Data

Age (years)	Mid-point (X_i)	Frequency (f_i)	$d_i = (X_i - 38)/4$	$f_i d_i$
20–24	22	40	-4	-160
24–28	26	35	-3	-105
28–32	30	35	-2	-70
32–36	34	25	-1	-25
36–40	38	30	0	0
40–44	42	40	1	40
44–48	46	50	2	100
48–52	50	45	3	135
52–56	54	35	4	140
56–60	58	30	5	150
Total		365		205

$$\bar{X} = A + \frac{\sum_{i=1}^n f_i d_i}{N} \cdot C = 38 + \frac{205}{365} \times 4 = 40.25$$

The arithmetic mean of the age of the employees = 40.25 years.

3.2.2 Weighted Arithmetic Mean

In some situation, all the observations may not have equal importance. Under such

situations, the observations are given differential weights. The measure of central tendency of such data is known as *weighted arithmetic mean*.

Weighted arithmetic mean of ungrouped data

The formula to compute the weighted arithmetic mean of ungrouped data is:

$$\bar{X}_w = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}$$

where $\bar{X}_w$ is the weighted arithmetic mean; X_i , the i th observation; w_i , the weight of the i th observation; and n is the total number of observations.

Example 3.4 The demand of a product is estimated using the three-year moving average method, which is nothing but the weighted arithmetic mean method. The demand in units of the years 2000, 2001 and 2002 are 400, 500 and 450, respectively. The corresponding weights are 1, 2 and 3, respectively. Find the demand for the year 2003.

Solution. The formula to get the weighted arithmetic mean (three year moving average) is:

$$\bar{X}_w = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i} = \frac{400 \times 1 + 500 \times 2 + 450 \times 3}{1 + 2 + 3} = 458.33 \text{ units}$$

Therefore, the demand for the year 2003 = 458.33 units.

Example 3.5 M.G. School of Management uses the scheme of weights which is shown in Table 3.6 for evaluating the assignments given to students. Each assignment will have five questions. Each question is evaluated in terms of grade A or B or C or D or E depending on the depth of discussion and correctness of its answer. The numerical equivalents of these grades are 5, 4, 3, 2 and 1, respectively. The minimum grade given to any of the questions is E. For a given student, in a given subject, if the grades given to the answers of the questions 1, 2, 3, 4 and 5 are A, C, D, E and A, respectively, find the weighted score and weighted grade of the given subject of that student using the grading scheme shown in Table 3.7.

Table 3.6 Weights of Questions

Question no. (i)	1	2	3	4	5
Weight (w_i)	2	1	3	2	1

Table 3.7 Grading Scheme

Weighted score	<1.5	1.5–2.4	2.5–3.4	3.5–4.4	4.5–5.0
Grade	E	D	C	B	A

Solution. The details of scores of the questions in a given subject of the given student are summarized in Table 3.8.

Table 3.8 Details of Scores of Questions

Question no. (i)	Weight (w_i)	Grade assigned	Numerical of grade (X_i)	Equivalent ($w_i X_i$)
1	2	A	5	10
2	1	C	3	3
3	3	D	2	6
4	2	E	1	2
5	1	A	5	5
Total	9			26

Then,

$$\bar{X}_w = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i} = \frac{26}{9} = 2.89$$

The weighted score in the given subject of the given student is 2.89 and the corresponding grade is C.

3.2.3 Median

The *median* is another measure of central tendency which corresponds to the half of the total frequency. As per this measure, 50 per cent of the observations are above the median value and another 50 per cent of the observations are below the median value. This measure can be computed even for problems with open-ended class interval(s). This is the main advantage of the median when compared to the arithmetic mean.

Median of ungrouped data

If the number of observations is *odd*, then the median is the middle value after the observations are arranged in either ascending or descending order as per reality.

If the number of observations is *even*, then median is the arithmetic mean of the middle most two observations after the observations are arranged in either ascending or descending order as per reality.

Example 3.6 The values of sales orders booked by a salesman in a month are: Rs. 1,00,000, Rs. 1,50,000, Rs. 3,00,000, Rs. 75,000 and Rs. 1,75,000. Find the median value of the sales order.

Solution. The ascending order of the values of the sales orders is: Rs. 75,000, Rs. 1,00,000, Rs. 1,50,000, Rs. 1,75,000, Rs. 3,00,000. The number of sales order in the month is 5, which is odd. Therefore, the middle-most value is Rs. 1,50,000. The median value of the sales = Rs. 1,50,000.

Example 3.7 The values of sales orders booked by a salesman in a month are: Rs. 1,00,000, Rs. 1,50,000, Rs. 3,00,000, Rs. 75,000, Rs. 1,75,000 and Rs. 4,00,000. Find the median value of the sales order.

Solution. The ascending order of the values of the sales orders is: Rs. 75,000, Rs. 1,00,000, Rs. 1,50,000, Rs. 1,75,000, Rs. 3,00,000, Rs. 4,00,000. The number of sales order in the month is 6, which is even. Therefore, the middle-most two values are Rs. 1,50,000 and Rs. 1,75,000.

$$\text{The arithmetic mean of these values} = \frac{150,000 + 175,000}{2}$$

$$\text{The median value of sales} = \text{Rs. } 1,62,500$$

Median of grouped data

The formula to determine the median of grouped data is:

$$\text{Median} = L + \frac{(N/2) - F}{f} C$$

where L is the lower limit of the median class; N , the total of all the frequencies; f , the frequency of the median class; F , the cumulative frequency of the previous class, with reference to the median class; and C is the width of class interval.

Example 3.8 The distribution of monthly travelling allowance (in thousands of rupees) claimed by salesmen of a firm is as presented in Table 3.9. Find the median of the travelling allowance of the salesmen.

Table 3.9 Frequency Distribution of Monthly Travelling Allowance of Salesmen

Allowance (Rupees in thousands)	Number of salesmen	Allowance (Rupees in thousands)	Number of salesmen
Less than 8	5	16-18	40
8-10	10	18-20	45
10-12	15	20-22	20
12-14	20	22-24	15
14-16	35	- More than 24	11

Solution. The data of the given problem is presented along with additional details, as in Table 3.10.

Table 3.10 Data of Example 3.8 with Additional Details

Allowance (Rupees in thousands)	Frequency (f_i)	Cumulative frequency Σf_i
Less than 8	5	5
8-10	10	15
10-12	15	30
12-14	20	50
14-16	35	85
16-18	40	125
18-20	45	170
20-22	20	190
22-24	15	205
More than 24	11	216

The row in bold face is the median class.

The total number of salesmen, $N = \sum_{i=1}^n f_i = 216$.

Therefore,

$$\frac{N}{2} = \frac{216}{2} = 108$$

The class interval for which the cumulative frequency just exceeds the half of the total frequency ($N/2$) is 16-18. Hence, it is treated as the median class. Now, we have the following details:

Lower limit of the median class (L)	= 16
Frequency of the median class (f)	= 40
Cumulative frequency of the previous class (F)	= 85
Width of class interval (C)	= 2

Then,

$$\text{Median} = L + \frac{(N/2) - F}{f} C = 16 + \frac{108 - 85}{40} \times 2 = 17.15$$

The median of the travelling allowance of the salesmen is Rs. 17.15 thousands (i.e. Rs. 17,150).

3.2.4 Mode

It is another measure of central tendency, like median which is applicable for frequency distributions with open ended class interval(s). It is the value of X_i for which the frequency/class frequency is the maximum.

Mode of ungrouped data

The mode of ungrouped data is the value of X_i , which has the maximum frequency.

Example 3.9 The number of defects found in 10 different samples of a finished product are 12, 9, 5, 15, 8, 9, 12, 17, 12 and 10. Find the mode of the number of defects found in the samples.

The frequency distribution of the number of defects is shown in Table 3.11.

Table 3.11 Distribution of Number of Defects

No. of defects	5	8	9	10	12	15	17
No. of samples	1	1	2	1	3	1	1

Solution. From Table 3.11, it is clear that the mode of the number of defects is 12 because it has the maximum frequency of 3.

Mode of grouped data

The formula to find the mode of grouped data is:

$$\text{Mode} = L + \frac{f_1}{f_1 + f_2} C$$

where L is the lower limit of the modal class; f_1 , the absolute difference between the frequency of the modal class and that of its immediately preceding class; f_2 , is the absolute difference between the frequency of the modal class and that of its immediately succeeding class; and C is the width of class interval.

Example 3.10 The distribution of annual revenues of 100 firms in a district is presented in Table 3.12. Find the mode of the annual revenue of the firms.

Table 3.12 Distribution of Annual Revenues

Annual revenue (in crores of rupees)	Number of firms	Annual revenue (in crores of rupees)	Number of firms
Less than 2	10	10-12	5
2-4	15	12-14	4
4-6	20	14-16	3
6-8	30	More than 16	5
8-10	8		

Solution. The frequency distribution of annual revenue of the firms is reproduced as in Table 3.13.

Table 3.13 Distribution of Annual Revenues

Annual revenue (in crores of rupees)	Frequency
Less than 2	10
2-4	15
4-6	20
6-8	30
8-10	8
10-12	5
12-14	4
14-16	3
More than 16	5

The modal class shown in the table is in bold face. The maximum frequency is 30 and the corresponding class interval is 6-8 (fourth class interval). Hence, from the Table 3.13, $L = 6$, $f_1 = 10$, $f_2 = 22$ and $C = 2$. Then,

$$\text{Mode} = L + \frac{f_1}{f_1 + f_2} C = 6 + \frac{10}{10 + 22} \times 2 = 6.625$$

Therefore, the mode of the annual revenue of the firms is 6.625 crores.

Relationship between mean, median and mode

The relationship between mean, median and mode is as:

$$\text{Mode} = 3 (\text{Median}) - 2 (\text{Mean})$$

The relationship can be used to determine the mean of a frequency distribution with open-ended class interval(s), because for such distribution mean cannot be found directly.

Example 3.11 (Modified version of Example 3.10) The distribution of annual revenues of 100 firms in a district is presented in Table 3.14. Find the mean of the annual revenue of the firms.

Table 3.14 Distribution of Annual Revenue

Annual revenue (in crores of rupees)	Number of firms	Annual revenue (in crores of rupees)	Number of firms
Less than 2	10	10-12	5
2-4	15	12-14	4
4-6	20	14-16	3
6-8	30	More than 16	5
8-10	8		

Solution. The frequency distribution of annual revenues of the firms along with cumulative frequencies is given in Table 3.15.

Table 3.15 Distribution of Annual Revenue along with Cumulative Frequencies

Allowance (in crores of rupees)	Frequency (f_i)	Cumulative frequency
Less than 2	10	10
2-4	15	25
4-6	20	45
6-8	30	75
8-10	8	83
10-12	5	88
12-14	4	92
14-16	3	95
More than 16	5	100

The row in bold face is the median class. The total number of firms, $N = \sum_{i=1}^n f_i = 100$. Then

$$\frac{N}{2} = \frac{100}{2} = 50$$

The class interval for which the cumulative frequency just exceeds half of the total frequency ($N/2$) is 6-8. Hence, it is treated as the median class. Now, we get the following details:

Lower limit of the median class (L)	= 6
Frequency of the median class (f)	= 30
Cumulative frequency of the previous class (F)	= 45
Width of class interval (C)	= 2

Then

$$\text{Median} = L + \frac{(N/2) - F}{f} \cdot C = 6 + \frac{50 - 45}{30} \times 2 = 6.33 \text{ crores}$$

The median of the annual revenue of the firms is Rs. 6.33 crores.

From Example 3.10, it is known that the *mode* of the annual revenue of the firms is Rs. 6.625 crores.

The formula linking mean, median and mode is:

$$\text{Mode} = 3 (\text{Median}) - 2 (\text{Mean}) \quad \text{or} \quad \text{Mean} = \frac{3 (\text{Median}) - (\text{Mode})}{2}$$

Therefore,

$$\text{Mean} = \frac{3 \times 6.33 - 6.625}{2} = 6.1825$$

Hence, the mean of the annual revenue of the firms is Rs. 6.1825 crores.

3.2.5 Geometric Mean

Let the number of observations be n . The geometric mean is a measure of central tendency which is equal to the n th root of the product of the observations. The formula for the geometric mean is:

$$GM = (X_1 X_2 X_3 \cdots X_i \cdots X_n)^{1/n}$$

where X_i is the i th observation.

The application areas of the geometric mean are as follows:

- To construct of index number
- To provide large weights for small observations
- To provide small weight for large observations
- To measure the growth of a population.

Example 3.12 The growth rate of the demand of a product during the last four years: 1999, 2000, 2001 and 2002, when compared to the demand of the base year 1998, are 20%, 30%, 35% and 40%, respectively (Table 3.16). Find the average growth rate of the demand of the product.

Table 3.16 Growth Rate and Demand of the Product

Year	Growth rate (%)	Demand of the product (X_i)
1999	20	$100 + 20 = 120$
2000	30	$100 + 30 = 130$
2001	35	$100 + 35 = 135$
2002	40	$100 + 40 = 140$

Solution. Number of observations, $n = 4$. Let the demand of the product during 1998 = 100 units. Now, refer to Table 3.16 for growth rate and demand of the product in the respective years.

$$\begin{aligned} \text{Geometric mean of the demands, } GM &= (X_1 X_2 X_3 X_4)^{1/4} \\ &= (120 \times 130 \times 135 \times 140)^{1/4} \\ &= 131.04 \end{aligned}$$

Now, the average growth rate of demand (R) during the four-year period is computed as follows:

$$(100 + R) = GM = 131.04 \quad \text{or} \quad R = 131.04 - 100 = 31.04\%$$

3.2.6 Harmonic Mean

Harmonic mean is another measure of central tendency which is used to express the average rate like, production volume per day, kilometre per litre of fuel, distance travelled by hour, etc. The formula for the harmonic mean is:

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_i} + \dots + \frac{1}{x_n}}$$

where HM is the harmonic mean; x_i , the i th observation; and, n , the total number of observations.

From the formula, it is clear that the harmonic mean is the reciprocal of the average of the reciprocals of the individual observations.

Example 3.13 A doctor advised Mr. Ram, a diabetic patient, to have early morning tread mill exercise for 30 minutes at a speed convenient to him daily for a week and report to him later about the speed at which he walked on the tread mill on each day. The patient accordingly did the exercise for a week and the speeds recorded were 3 km/hr, 4 km/hr, 4.5 km/hr, 5 km/hr, 3.5 km/hr, 4.75 km/hr and 4.25 km/hr on these days, respectively. Now, assist the doctor for computing the average speed of Mr. Ram during his early morning exercise.

Solution. The number of observations, $n = 7$. Then

$$\begin{aligned} HM &= \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_i} + \dots + \frac{1}{x_n}} \\ &= \frac{7}{\frac{1}{3} + \frac{1}{4} + \frac{1}{4.5} + \frac{1}{5} + \frac{1}{3.5} + \frac{1}{4.75} + \frac{1}{4.25}} \\ &= 4.03 \text{ km/hr} \end{aligned}$$

Therefore, the average speed of Mr. Ram on the tread mill exercise is 4.03 km/hr.

3.3 MEASURES OF VARIATION

The measures of central tendency are the estimates of the average of a set of observations. While, interpreting this average value, we assume that the estimated values are uniform. The average value does not explain the differences between the observations.

Let us take the case of two different salesmen (X and Y) operating in two different regions of a company. Let the monthly sales revenue generated by these two salesmen be shown in Table 3.17.

Table 3.17 Sales Revenue of Salesmen

Year	Salesman-X	Saleman-Y
1	75,000	10,000
2	75,000	15,000
3	75,000	20,000
4	75,000	30,000
5	75,000	40,000
6	75,000	3,35,000
Total sales	4,50,000	4,50,000
Mean sales	75,000	75,000

It is clear from the table that the mean sales of each of the salesmen is Rs. 75,000. So, if the mean alone is taken to judge the performance of the salesmen, it is not possible to discriminate them, though there is a strong evidence of poor performance of the salesman-Y during initial years.

Hence, the differences between sales figures of a salesman should be studied in the form of variation which is alternatively called as spread. In this case, the sales revenues of the salesman-X is consistent when compared to that of the salesman-Y.

The *variation (spread)* is a measure of the extent to which the individual observations vary with reference to the average of the given set of observations.

The different measures of variation, which are discussed in this section are: range, quartile deviation, mean deviation, standard deviation and coefficient of variation.

3.3.1 Range

Range is the difference between the highest value and the lowest value of the given set of observations:

$$R = H - L$$

where R is the range of the observations; H , the highest value of the observations; and L , the lowest value of the observations.

Coefficient of range is another measure of variation which is the ratio between the difference of the extreme values and the sum of the extreme values:

$$\text{Coefficient of range} = \frac{H - L}{H + L}$$

3.3.2 Quartile Deviation

Quartile deviation is the half of the difference between the third quartile and the first quartile.

The *third quartile* is the value of the variable of interest (X) corresponding to the first 75 per cent of the total frequency of grouped data.

The *first quartile* is the value of the variable of interest (X) corresponding to the first 25 per cent of the total frequency of grouped data.

The formula for the quartile deviation is as presented below:

$$QD = \frac{Q_3 - Q_1}{2}$$

where Q_1, Q_3 are the first and third quartile, respectively such that

$$\text{First quartile, } Q_1 = L_1 + \frac{(N/4) - F}{f_1} C$$

where L_1 is the lower limit of the first quartile class; F , the cumulative frequency of the previous class with respect to the first quartile class; f_1 is the frequency of the first quartile class; and, C is the width of class interval. We also have,

$$\text{Third quartile, } Q_3 = L_3 + \frac{(3N/4) - F}{f_3} C$$

where L_3 is the lower limit of the third quartile class; F the cumulative frequency of the previous class with respect to the third quartile class; f_3 is the frequency of the third quartile class; and C is the width of class interval.

The relative measure with respect to the quartile deviation is given by the following formula which is known as *coefficient of deviation*.

$$\text{Coefficient of quartile deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Example 3.14 The distribution of yearly labour welfare expenses (in lakhs of rupees) of 88 firms in an industrial estate is shown in Table 3.18. Find the quartile deviation of the yearly labour welfare expenses.

Table 3.18 Distribution of Labour Welfare Expenses

Labour welfare expenses	No. of firms	Labour welfare expenses	No. of firms
Below 10	5	50-60	10
10-20	8	60-70	7
20-30	10	70-80	4
30-40	15	80-90	3
40-50	20	Above 90	6

Solution. The distribution of labour welfare expenses is shown in Table 3.19 along with cumulative frequencies.

Table 3.19 Distribution of Labour Welfare Expenses with Cumulative Frequencies

Labour welfare expenses	No. of firms (f_i)	Cumulative frequency	Quartile class
Below 10	5	5	
10-20	8	13	
20-30	10	23	First
30-40	15	38	
40-50	20	58	
50-60	10	68	Third
60-70	7	75	
70-80	4	79	
80-90	3	82	
Above 90	6	88	

Here, we have

$$N = \sum_{i=1}^n f_i = 88, \quad C = 10$$

As $N/4 = 88/4 = 22$, the class interval for which the cumulative frequency just exceeds 22 is 20-30. Hence, $L_1 = 20$, $f_1 = 10$ and $F = 13$. Then

$$\text{First quartile, } Q_1 = L_1 + \frac{(N/4) - F}{f_1} C = 20 + \frac{22 - 13}{10} 10 = 29$$

We also have, $3N/4 = 3 \times 88/4 = 66$. Therefore, the class interval for which the cumulative frequency just exceeds 66 is 50-60. Hence, $L_3 = 50$, $f_3 = 10$ and $F = 58$. Then

$$\text{Third quartile, } Q_3 = L_3 + \frac{(3N/4) - F}{f_3} C = 50 + \frac{66 - 58}{10} 10 = 58$$

Also,

$$\text{Quartile deviation, } QD = \frac{Q_3 - Q_1}{2} = \frac{58 - 29}{2} = \text{Rs. 14.5 lakhs}$$

and

$$\text{Coefficient of quartile deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{58 - 29}{58 + 29} = 0.333$$

3.3.3 Average Deviation

Average deviation (AD) is the mean of the absolute deviations of the individual observations from either arithmetic mean or median of the given set of observations.

The formula to compute the average deviation of ungrouped data is as given below:

$$\text{Average deviation, AD} = \frac{\sum |X_i - \bar{X}|}{n}$$

or

$$\text{Average deviation, AD} = \frac{\sum |X_i - \text{Median}|}{n}$$

where X_i is the i th observation; $\bar{X}$, the arithmetic mean; and n , the number of observations. The formula to compute the average deviation of grouped data is given as:

$$\text{Average deviation, AD} = \frac{\sum f_i |X_i - \bar{X}|}{N}$$

or

$$\text{Average deviation, AD} = \frac{\sum f_i |X_i - \text{Median}|}{N}$$

where X_i is the i th observation; f_i , the frequency of the i th observation; $\bar{X}$, the arithmetic mean; and N , the total frequency ($\sum f_i$).

The relative measure with respect to average deviation is defined as coefficient of average deviation:

$$\text{Coefficient of AD} = \frac{\text{AD}}{\text{Median}}$$

or

$$\text{Coefficient of AD} = \frac{\text{AD}}{\text{Mean}}$$

If the average deviation is based on mean, then the coefficient of average deviation is computed by dividing the average deviation by the mean. If the average deviation is based on median, then the coefficient of average deviation is computed by dividing it by the median.

Example 3.15 The distribution of number of shares applied to the recently concluded equity issue of an automobile company is as shown in Table 3.20. Compute the average deviation as well as coefficient of average deviation for the number of shares applied.

Table 3.20 Distribution of Number of Shares Applied

No. of shares applied for	No. of applications	No. of shares applied for	No. of applications
50-100	2500	300-350	750
100-150	1500	350-400	675
150-200	1300	400-450	525
200-250	1100	450-500	450
250-300	900		

Solution. The distribution of the number of shares applied along with cumulative frequencies is given in Table 3.21.

Table 3.21 Distribution of Number of Shares Applied along with Cumulative Frequencies

No. of shares applied for	Mid-point X_i	No. of applications f_i	Cumulative Σf_i frequency	$d_i = (X_i - 275)/50$	$f_i d_i$	$ X_i - \bar{X} $	$f_i X_i - \bar{X} $
50-100	75	2,500	2,500	-4	-10,000	135	337,500
100-150	125	1,500	4,000	-3	-4,500	85	127,500
150-200	175	1,300	5,300	-2	-2,600	35	45,500
200-250	225	1,100	6,400	-1	-1,100	15	16,500
250-300	275	900	7,300	0	0	65	58,500
300-350	325	750	8,050	1	750	115	86,250
350-400	375	675	8,725	2	1,350	165	111,375
400-450	425	525	9,250	3	1,575	215	112,875
450-500	475	450	9,700	4	1,800	265	119,250
Total					-12,725		1,015,250

Here, $N = 9700$. Then

$$\bar{X} = A + \frac{\sum_{i=1}^n f_i d_i}{N} C = 275 - \frac{12725}{9700} \times 50 = 209.41 \approx 210$$

We also have

$$\frac{N}{2} = \frac{9700}{2} = 4850$$

The class interval for which the cumulative frequency just exceeds 4850 is 150-200. Therefore, $L = 150$, $f = 1300$, $F = 4000$, $C = 50$. Then

$$\text{Median} = L + \frac{(N/2) - F}{f} C = 150 + \frac{4850 - 4000}{1300} \times 50 = 182.69 \approx 183$$

and

$$\text{Average deviation, AD} = \frac{\sum f_i |X_i - \bar{X}|}{N} = \frac{1015250}{9700} = 104.67 \approx 105$$

Determination of coefficient of average deviation Since the average deviation is computed based on the mean, the coefficient of average deviation is:

$$\text{Coefficient of AD} = \frac{\text{AD}}{\text{Mean}} = \frac{105}{210} = 0.5$$

3.3.4 Standard Deviation

Standard deviation is the widely used measure of variation. Alternatively, it is called *root mean square deviation* and its notation is σ . It is the root of the variance, σ^2 .

Standard deviation for ungrouped data

The formula to compute the standard deviation of ungrouped data is:

$$\sigma = \left[\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} \right]^{1/2}$$

where X_i is the i th observation; $\bar{X}$, the mean of the observations; n , the number of observations.

Standard deviation for ungrouped data with frequencies

The formula to compute the standard deviation of ungrouped data with frequencies is:

$$\sigma = \left[\frac{\sum_{i=1}^n f_i (X_i - \bar{X})^2}{N} \right]^{1/2}$$

where X_i is the i th observation; $\bar{X}$, the mean of the observations; f_i , the frequency of the i th observation; and N , is the total frequency.

Standard deviation for grouped data

The formula to compute the standard deviation of grouped data, which are small in magnitudes is given as:

$$\sigma = \left[\frac{\sum_{i=1}^n f_i X_i^2}{N} - \left(\frac{\sum_{i=1}^n f_i X_i}{N} \right)^2 \right]^{1/2} = \left[\frac{\sum_{i=1}^n f_i X_i^2}{N} - \bar{X}^2 \right]^{1/2}$$

where X_i is the i th observation; f_i , the frequency of the i th observation and N , the total frequency.

The shortcut formula to compute the standard deviation of grouped data which are large in magnitude is:

$$\sigma = \left[\frac{\sum_{i=1}^n f_i d_i^2}{N} - \left(\frac{\sum_{i=1}^n f_i d_i}{N} \right)^2 \right]^{1/2} \times C$$

where $d_i = (X_i - A)/C$; X_i , the i th observation; f_i , the frequency of the i th observation; N the total frequency; C , the width of class interval; and A , an assumed mean.

Example 3.16 The distribution of the price of the equity share of a company in a stock exchange is shown in Table 3.22. Find the mean and standard deviation of the share price of the company.

Table 3.22 Distribution of Share Price

Share price (Rs.)	50-60	60-70	70-80	80-90	90-100
No. of sessions	20	30	45	55	50

Solution. The distribution of the share price with additional details for calculating the standard deviation is shown in Table 3.23.

Table 3.23 Distribution of Share Price with Additional Details

Share price (Rs.)	Mid-point X_i	No. of sessions f_i	$d_i = (X_i - 75)/10$	$f_i d_i$	$f_i d_i^2$
50-60	55	20	-2	-40	80
60-70	65	30	-1	-30	30
70-80	75	45	0	0	0
80-90	85	55	1	55	55
90-100	95	50	2	100	200
Total		200		85	365

From the table, $\sum f_i = N = 200$, $\sum f_i d_i = 85$, $\sum f_i d_i^2 = 365$ and $C = 10$.

$$\sigma = \left[\frac{\sum_{i=1}^n f_i d_i^2}{N} - \left(\frac{\sum_{i=1}^n f_i d_i}{N} \right)^2 \right]^{1/2} \times C = \left[\frac{365}{200} - \left(\frac{85}{200} \right)^2 \right]^{1/2} \times 10 = 12.82$$

and

$$\bar{X} = A + \frac{\sum_{i=1}^n f_i d_i}{N} C = 75 + \frac{85}{200} \times 10 = 79.25$$

The mean and standard deviation of the share price of the company are Rs. 79.25 and Rs. 12.82, respectively.

3.3.5 Coefficient of Variation

Coefficient of variation is another measure of variation, which is an improved measure over standard deviation. It is the ratio between the standard deviation and mean, and it is usually represented by percentage. This measure indicates the consistency of the observations within a sample and discriminates a given set of samples sharply in terms of consistency.

$$CV = \frac{\sigma}{\bar{X}} \times 100$$

where CV is the coefficient of variation; σ , the standard deviation; and $\bar{X}$, the mean.

Example 3.17 Consider two different businesses with equal initial investments and equal lives whose expected cash flows in crores are as summarized in Table 3.24. Identify the business which is more consistent in terms of the cash flow.

Table 3.24 Cash Flows of Two Businesses

Period	1	2	3	4	5
Business-A	80	85	70	100	115
Business-B	100	70	90	80	150

Solution. For business-A

$$\bar{X}_A = \frac{80 + 85 + 70 + 100 + 115}{5} = \text{Rs. 90 crores}$$

$$\sigma_A^2 = \frac{(80 - 90)^2 + (85 - 90)^2 + (70 - 90)^2 + (100 - 90)^2 + (115 - 90)^2}{5} = 250$$

$$\sigma_A = \text{Rs. 15.81 crore.}$$

$$CV_A = \frac{\sigma}{\bar{X}} \times 100 = \frac{15.81}{90} \times 100 = 17.57\%$$

For business-B,

$$\bar{X}_B = \frac{100 + 70 + 90 + 80 + 150}{5} = \text{Rs. 98 crores}$$

$$\sigma_H^2 = \frac{(100-98)^2 + (70-98)^2 + (90-98)^2 + (80-98)^2 + (150-98)^2}{5} = 776$$

$$\sigma_H = \text{Rs. } 27.86 \text{ crores}$$

$$CV_H = \frac{\sigma}{\bar{X}} \times 100 = \frac{27.86}{98} \times 100 = 28.43\%$$

Inference: Since the coefficient of variation of the business-A is less than that of the business-B, the cash flow of the business-A is more consistent when compared to that of the business-B.

3.4 MEASURE OF SKEWNESS

In some situation, the mean as well as standard deviation may be the same for different samples. But, still there may be some other characteristic on which some more inference can be drawn. The shape of a distribution is one such characteristic which is a measure of concern for many business decisions. The shape may be either symmetrical or asymmetrical. The shape of a distribution explains the nature of the distribution of frequencies of observations, which is defined as *coefficient of skewness* and its range is from -1 to $+1$. If the coefficient of skewness is 0 , then the distribution is symmetrical. If it is positive, the distribution is positively skewed. This means that the distribution will have more thin tail portion on the right-hand side. Under such situation, the relationship between mean, median and mode is as:

$$\text{Mean} > \text{Median} > \text{Mode}$$

If the coefficient is negative, the distribution is negatively skewed. This means that the distribution will have more thin tail portion on the left-hand side of the distribution. Under such situation, the relationship between mean, median and mode is:

$$\text{Mean} < \text{Median} < \text{Mode}$$

There are three different formulas to compute the coefficient of skewness, which are discussed now.

Karl Pearson's coefficient of skewness

If the mean, mode and standard deviation are known, the formula for the coefficient of skewness of a distribution is:

$$CS = \frac{\text{Mean} - \text{Mode}}{\sigma}$$

where CS is the coefficient of skewness and σ , the standard deviation.

Approximate formula for coefficient of skewness

If the mode of a distribution is not known, then the coefficient of skewness can be determined differently. We know the relationship between mean, median and mode as:

$$\text{Mode} = 3 (\text{Median}) - 2 (\text{Mean}).$$

Substitution of the above in this formula for coefficient of skewness developed by Karl Pearson yields the following formula, which will give an approximate value.

$$CS = \frac{3 (\text{Mean} - \text{Median})}{\sigma}$$

Bowley's coefficient of skewness

If the distribution is open ended, then the formula developed by Bowley for the coefficient of skewness can be shown as:

$$CS = \frac{Q_3 + Q_1 - 2(\text{Median})}{Q_3 - Q_1}$$

where CS is the coefficient of skewness; Q_1 , the first quartile of the distribution; and Q_3 , the third quartile of the distribution.

Example 3.18 There are several factors affecting organizational productivity. In this direction, labour absenteeism is a factor of concern for the productivity engineer of an automobile company. He has collected data pertaining to the labour absenteeism during the last year and summarized its distribution as shown in Table 3.25.

Table 3.25 Distribution of Labour Absenteeism

Absenteeism (in days)	No. of employees	Absenteeism (in days)	No. of employees
0-4	10	16-20	24
4-8	76	20-24	36
8-12	100	24-28	14
12-16	150	28-32	2

Determine the Karl Pearson's coefficient of skewness for the labour absenteeism for the above distribution and offer your comment.

Solution. The data on labour absenteeism is reproduced along with additional data, as shown in Table 3.26.

Table 3.26 Data on Labour Absenteeism with Additional Details

Absenteeism (in days)	Mid-point (x_i)	No. of employees (f_i)	$d_i = (x_i - 14)/4$	$f_i d_i$	$f_i d_i^2$
0-4	2	10	-3	-30	90
4-8	6	76	-2	-152	304
8-12	10	100	-1	-100	100

(contd.)

Absenteeism (in days)	Mid-point (X_i)	No. of employees (f_i)	$d_i = (x_i - 14)/4$	$f_i d_i$	$f_i d_i^2$
12-16	14	150	0	0	0
16-20	18	24	1	24	24
20-24	22	36	2	72	144
24-28	26	14	3	42	126
28-32	30	2	4	8	32
Total		412		-136	820

Here, $N = 412$, $A = 14$, and $C = 4$. Then

$$\bar{X} = A + \frac{\sum_{i=1}^n f_i d_i}{N} \cdot C = 14 + \frac{-136}{412} \times 4 = 12.68$$

The maximum frequency is 150 and the corresponding class interval is 12-16. Therefore, $L = 12$, $f_1 = 50$, $f_2 = 126$ and $C = 4$. Then

$$\text{Mode} = L + \frac{f_1}{f_1 + f_2} \cdot C = 12 + \frac{50}{50 + 126} \times 4 = 13.14$$

and

$$\begin{aligned} \text{Standard deviation, } \sigma &= \left[\frac{\sum_{i=1}^n f_i d_i^2}{N} - \left(\frac{\sum_{i=1}^n f_i d_i}{N} \right)^2 \right]^{1/2} \times C \\ &= \left[\frac{820}{412} - \left(\frac{-136}{412} \right)^2 \right]^{1/2} \times 4 \\ &= 5.49 \end{aligned}$$

where the coefficient of skewness is

$$CS = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{12.68 - 13.14}{5.49} = -0.0838$$

Comment: Since the coefficient of skewness is negative, the distribution is skewed to the left. But the extent of distortion in the shape of the distribution is very small because the value of the coefficient is too small.

Example 3.19 There are several factors affecting organizational productivity. In this direction, machine breakdown time is a factor of concern for the productivity engineer of an engineering company. He has collected data pertaining to the machine breakdown time during the last month and summarized its distribution as in Table 3.27.

Table 3.27 Distribution of Machine Breakdown Time

Machine breakdown time (hrs)	No. of machines	Machine breakdown time (hrs)	No. of machines
Less than 4	26	16-20	100
4-8	50	20-24	60
8-12	70	24-28	50
12-16	80	More than 28	24

Determine the coefficient of skewness for the machine breakdown time for the above distribution and offer your comment.

Solution. The data on machine breakdown time is reproduced along with additional data as in Table 3.28.

Table 3.28 Data on Machine Breakdown Time with Additional Details

Machine breakdown time (hrs)	No. of machines (f_i)	Cumulative frequency	Quartile class
Less than 4	26	26	
4-8	50	76	
8-12	70	146	First
12-16	80	226	
16-20	100	326	Median
20-24	60	386	Third
24-28	50	436	
More than 28	24	460	

Since the distribution is open ended, Bowley's coefficient of skewness should be used for this situation.

Here, class interval, $C = 4$ and $N = 460$ or $N/4 = 115$. The class interval for which the cumulative frequency just exceeds 115 is 8-12. Therefore, $L_1 = 8$, $f_1 = 70$, $F = 76$. Then

$$Q_1 = L_1 + \frac{(N/4) - F}{f_1} C = 8 + \frac{115 - 76}{70} \times 4 = 10.23$$

As $N/2 = 230$, the class interval for which the cumulative frequency just exceeds 230 is 16-20.

Therefore, $L_2 = 16$, $f_2 = 100$, $F = 226$. Then

$$\text{Median, } Q_2 = L_2 + \frac{(N/2) - F}{f_2} C = 16 + \frac{230 - 226}{100} \times 4 = 16.16$$

Determination of Q_3 : As $3N/4 = 345$, the class interval for which the cumulative frequency just exceeds 345 is 20-24.

Therefore, $L_3 = 20$, $f_3 = 60$, $F = 326$. Then

$$Q_3 = L_3 + \frac{(3N/4) - F}{f_3} C = 20 + \frac{345 - 326}{60} \times 4 = 21.27$$

Now, the Bowley's coefficient of skewness is

$$CS = \frac{Q_3 + Q_1 - 2(\text{Median})}{Q_3 - Q_1} = \frac{21.27 + 10.23 - 2 \times 16.16}{21.27 - 10.23} = -0.0743$$

Comment: Since the coefficient of skewness is negative, the distribution is skewed to the left. But the extent of distortion in the shape of the distribution is very small because the value of the coefficient is too small.

QUESTIONS

1. Determine the arithmetic mean of the salaries of the employees which are given as:

Employee no.	1	2	3	4	5	6	7
Monthly salary (Rs.)	13,000	16,000	9,500	14,500	20,500	19,500	7,500

2. Determine the arithmetic mean of the annual revenues (in crores of Rs.) of a firm as shown below.

Year	1	2	3	4	5	6	7
Annual revenue	50	70	60	100	250	225	250

3. The minimum and the maximum of the monthly salaries paid to the employees of a computer firm are Rs. 4,000 and Rs. 36,000, respectively. The frequency distribution of the monthly salary of the employees of the firm is given, as in the following table:

Monthly salary (Rs.)	Number of employees	Monthly salary (Rs.)	Number of employees
4000–8000	150	20000–24000	200
8000–12000	250	24000–28000	175
12000–16000	275	28000–32000	100
16000–20000	350	32000–36000	50

Determine the arithmetic mean.

- The mean annual R&D expenditure of 150 firms in an industrial estate is Rs. 60 lakh. The mean annual R&D expenditure of the large-scale industries in that estate is Rs. 70 lakh and that of the medium-scale industries is Rs. 55 lakh. Find the number of large-scale industries and the number of medium-scale industries in that estate.
- The minimum and the maximum of the ages of the employees of a textile firm are 20 years and 60 years, respectively. The frequency distribution of the age of the employees of the firm is presented below:

Age (years)	Number of employees	Age (years)	Number of employees
20–24	20	40–44	50
24–28	30	44–48	40
28–32	45	48–52	35
32–36	50	52–56	35
36–40	55	56–60	20

Determine the arithmetic mean.

- The distribution of monthly income of customers in a locality is given, as in the following table: Find the arithmetic mean of the income of the customers.

Income (in thousands Rs.)	Number of customers	Income (in thousands Rs.)	Number of customers
6–8	10	16–18	80
8–10	25	18–20	40
10–12	40	20–22	20
12–14	60	22–24	15
14–16	75	24–26	10

- The demand of a product is estimated using the four-year moving average method, which is nothing but the weighted arithmetic mean method. The demand in units of the years 1999, 2000, 2001 and 2002 are 300, 350, 400 and 325, respectively. The corresponding weights are 1, 2, 3 and 4, respectively. Find the demand for the year 2003.

8. Beta incorporation has introduced a performance appraisal system in which there are five different factors for evaluating each employee at managerial position. Employees are rated on 0–10 scale for all the factors. The scheme of weights for the factors are as shown below:

Factor	A	B	C	D	E
Weight	3	1	2	4	2

If an employee is rated with 8, 7, 6, 9 and 5 points for the factors A, B, C, D and E, respectively, find the overall weighted rating of the employee.

9. The distribution of monthly travelling allowance (in thousands of rupees) claimed by salesmen of a firm is as presented in the following table:

Allowance (in thousands Rs.)	Number of salesmen	Allowance (in thousands Rs.)	Number of salesmen
Less than 4	10	12–14	50
4–6	20	14–16	60
6–8	15	16–18	40
8–10	25	18–20	30
10–12	30	More than 20	20

Find the median of the travelling allowance of the salesmen.

10. The distribution of the years of experience of employees at a particular cadre in a multinational company is shown as follows:

Experience (years)	Number of employees	Experience (years)	Number of employees
Less than 2	26	10–12	38
2–4	40	12–14	28
4–6	30	14–16	20
6–8	40	16–18	10
8–10	28	More than 18	4

Find the median of the years of experience of the employees in that cadre.

11. The distribution of annual revenues of 100 firms in a district is presented below:

Annual revenue (in crores of rupees)	Number of firms	Annual revenue (in crores of rupees)	Number of firms
Less than 6	10	14–16	4
6–8	12	16–18	5
8–10	25	18–20	3
10–12	28	More than 20	4
12–14	9		

Find the mode of the annual revenue of the firms.

12. The distribution of number of months of holding of the shares of a company by its share holders is presented below:

No. of months of holding	No. of share holders ('000)	No. of months of holding	No. of share holders ('000)
Less than 2	5	10-12	6
2-4	8	12-14	20
4-6	10	14-16	12
6-8	7	16-18	9
8-10	9	More than 18	2

Find the mode of the number of months of holding of the shares by its share holders.

13. The distribution of annual revenues of the firms in a district is presented below:

Annual revenue (in crores of Rs.)	Number of firms	Annual revenue (in crores of Rs.)	Number of firms
Less than 8	20	24-28	30
8-12	35	28-32	24
12-16	40	32-36	24
16-20	50	More than 36	10
20-24	45		

Find the mean of the annual revenue of the firms.

14. Based on a survey, the distribution of the number of years of usage of a particular brand of washing machine by the first buyers is shown as follows:

No. of years of usage	No. of respondents	No. of years of usage	No. of respondents
Less than 2	10	6-8	25
2-4	15	8-10	20
4-6	20	10-12	12
		More than 12	10

Find the mean of the number of years of usage of the washing machine by the first buyers.

15. The growth rate of the demand of a product during the last four years, 1999, 2000, 2001 and 2002 when compared to the demand of the base year 1998 are 15%, 20%, 25% and 30%, respectively. Find the average growth rate of the demand of the product.
16. The kilometre per litre given by a new vehicle on five different occasions are 50 km/hr, 60 km/hr, 55 km/hr, 60 km/hr and 45 km/hr. Find the average kilometre per litre given by the new vehicle.

17. The distribution of yearly labour welfare expenses (in lakhs of rupees) of the firms in an industrial estate is shown as follows:

<i>Labour welfare expenses</i>	<i>No. of firms</i>	<i>Labour welfare expenses</i>	<i>No. of firms</i>
Below 5	8	25-30	50
5-10	12	30-35	40
10-15	20	35-40	28
15-20	32	40-45	12
20-25	46	45-50	4

Find the quartile deviation of the yearly labour welfare expenses.

18. The distribution of loan amount (in lakhs of rupees) sanctioned to industries by a bank is shown as in the following table:

<i>Loan amount</i>	<i>No. of firms</i>	<i>Loan amount</i>	<i>No. of firms</i>
Below 4	20	20-24	25
4-8	16	24-28	23
8-12	20	28-32	28
12-16	28	32-36	6
16-20	40	36-40	2

Find the quartile deviation of the loan amount.

19. The distribution of number of shares applied to the recently concluded equity issue of an automobile company is shown as follows:

<i>No. of shares applied for</i>	<i>No. of applications</i>	<i>No. of shares applied for</i>	<i>No. of applications</i>
50-100	500	300-350	850
100-150	1500	350-400	760
150-200	1800	400-450	600
200-250	1600	450-500	400
250-300	1000		

Compute the average deviation as well as coefficient of average deviation for the number of shares applied.

20. The distribution of monthly inventory in lakhs of rupees in the warehouse of a textile company in the last 5 years is shown in the following table:

<i>Monthly inventory</i>	<i>No. of months</i>	<i>Monthly inventory</i>	<i>No. of months</i>
5-10	5	30-35	5
10-15	6	35-40	7
15-20	8	40-45	6
20-25	9	45-50	4
25-30	10		

Compute the average deviation as well as coefficient of average deviation of the monthly inventory held by the warehouse.

21. The distribution of the price of the equity share of a company in a stock exchange is shown below:

Share price (Rs.)	50-60	60-70	70-80	80-90	90-100
No. of sessions	50	90	96	80	40

Find the mean and standard deviation of the share price of the company.

22. Based on a survey, the distribution of income (in thousands of rupees) of the respondents in a locality is as shown below:

Income (Rs. '000)	4-6	6-8	8-10	10-12	12-14	14-16
No. of respondents	80	90	100	70	60	50

Find the mean and standard deviation of the income of the people in that locality.

23. Consider two different businesses with equal initial investments and equal lives whose expected cash flows in crores are as summarized below:

Period	1	2	3	4	5
Business-A	90	150	100	200	300
Business-B	80	100	200	300	100

Identify the business which is more consistent in terms of the cash flow.

24. Consider two different salesmen of a company whose annual sales figures in crores of rupees for the past 5 years are as shown below:

Year	1	2	3	4	5
Salesman-X	85	90	95	150	200
Salesman-Y	100	120	50	60	125

Identify the salesman who is more consistent in terms of the annual sales made.

25. There are several factors affecting organizational productivity. In this direction, labour absenteeism is a factor of concern for the productivity engineer of an automobile company. He has collected data pertaining to the labour absenteeism during the last year and summarized its distribution as shown below:

Absenteeism (in days)	No. of employees	Absenteeism (in days)	No. of employees
0-4	20	16-20	80
4-8	56	20-24	40
8-12	76	24-28	30
12-16	90	28-32	10

Determine the Karl Pearson's coefficient of skewness for the labour absenteeism and offer your comment.

26. Find the coefficient of skewness of the following data:

Age (in years)	Number of employees	Age (in years)	Number of employees
22-26	30	38-42	70
26-30	50	42-46	60
30-34	70	46-50	40
34-38	80	50-54	30

27. There are several factors affecting organizational productivity. In this direction, machine breakdown time is a factor of concern for the productivity engineer of a textile company. He has collected data pertaining to the machine breakdown time during the last month and summarized its distribution as shown below:

Machine breakdown time (hrs)	Number of machines	Machine breakdown time (hrs)	Number of machines
Less than 4	40	10-12	90
4-6	50	12-14	30
6-8	90	14-16	20
8-10	100	More than 16	40

Determine the coefficient of skewness for the machine breakdown time and offer your comment.

28. The distribution of the cost of machines in a company is as shown below:

Cost of machine (Rs. in lakhs)	Number of machines	Cost of machine (Rs. in lakhs)	Number of machines
Less than 1	20	4-5	40
1-2	25	5-6	20
2-3	45	6-7	6
3-4	50	More than 7	4

Determine the coefficient of skewness of the cost of the machines and offer your comment.

CHAPTER

4

DESIGN AND ANALYSIS OF EXPERIMENTS

4.1 INTRODUCTION

The significance of the difference between the mean of two different samples can be tested using either paired t -test or Z -test depending on the sample size. If the sample size is less than 30, t -test is to be used; otherwise, Z -test is to be used.

As an example, two different samples may represent the sales revenues in four different periods by two different salesmen of a company. Here, the objective is to test the significance of the difference between the mean sales revenue of the two salesmen. Since, the sample size is less than 30, one can use the paired t -test for this purpose.

If the number of salesmen is n which is more than 2, the number of pairs of mean sales-revenues of the salesmen for comparison will be " C_2 " which will be more than 2. Under this situation, it is not possible to compare the " C_2 " pairs using the single paired t -test. The above comparisons may be done by repeating the paired t -test for " C_2 " times. But, it will be tedious.

Similarly, the testing of the significance of the difference between the variances of sales revenues of two different salesmen can be done using F -test. As stated earlier, if the number of salesmen of a company is more than 2, the number of pairs of salesmen will be " C_2 ". Then comparisons of the variances of the sales revenues of " C_2 " pairs of salesmen are not possible with a single F -test. Such limitation necessitates a comprehensive technique like 'Analysis of variance' (ANOVA).

4.2 ANALYSIS OF VARIANCE (ANOVA)

Decompositioning of total variability into its components is called analysis of variance (ANOVA). Consider an example of analyzing sales revenues of different salesmen, as shown in Table 4.1.

In the table, Y_{ij} is the i th sales data (i th replication) of the j th salesman. Salesman is the factor which has effect on the response variable Y_{ij} . Let it be factor A . a is the number of salesmen of the company. This is also known as the *number of levels/treatments* of the factor A . n is the number of sales data under each salesman. This is also known as the *number of replications* under each level of the factor A .

Table 4.1 Sales Data of Salesmen

		Salesman					
		1	2	...	j	...	a
Replication	1	Y_{11}	Y_{12}	...	Y_{1j}	...	Y_{1a}
	2	Y_{21}	Y_{22}	...	Y_{2j}	...	Y_{2a}
	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
	i	Y_{i1}	Y_{i2}	...	Y_{ij}	...	Y_{ia}
	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
	n	Y_{n1}	Y_{n2}	...	Y_{nj}	...	Y_{na}

Types of factor

A factor, which has effect on the response variable of an experiment, can be classified into fixed factor or random factor.

Fixed factor In an experiment, if a specific set of treatments of a factor is selected with certainty, then that factor is termed as a *fixed* factor. Under such situation, the inference of the analysis of the experiment is applied to only the set of treatments of that factor.

For example, if four salesmen (A, B, C and D) are selected specifically for studying their effect on the sales revenue, then the inference of the analysis is applied to only those four salesmen (A, B, C and D). The effect of the treatments of the fixed factor A on the response variable is given by the following formula.

$$\text{Effect of fixed factor } A = \frac{\sum_{j=1}^a A_j^2}{a-1}$$

where $(a-1)$ is the degrees of freedom of the factor A , and A_j is the effect of the j th treatment of the factor A .

Random factor In an experiment, if a set of treatments of a factor is selected randomly from among the available treatments, then that factor is termed as a *random* factor. Under such situation, the inference of the analysis of the experiment can be generalized to all the treatments of that factor.

For example, if four salesmen (A, D, M and X) are selected randomly from among the available salesmen (A, B, C, ..., Z) for studying their effect on the sales revenue, then the inference of the analysis of the experiment can be generalized to all the treatments (A, B, C, ..., Z) of the factor, salesman.

The effect of the treatments of the random factor A on the response variable is given by the following formula:

$$\text{Variance component of the random factor } A = \sigma_A^2$$

where σ_A^2 is the variance of the random factor A . The model of this situation is

$$Y_{ij} = \mu + A_j + e_{ij}$$

where μ is the overall mean of the sales revenue, A_j is the effect of the j th treatment of the factor A (salesman) on the response variable Y_{ij} , e_{ij} is the random error associated with the i th replication of the j th treatment of the factor A.

Partitioning of the total variability is presented as follows:

Total corrected sum of squares is:

$$SS_{\text{total}} = \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{..})^2$$

The equation is rewritten by adding and subtracting the j term $\bar{Y}_{.j}$ as per the following presentation:

$$\sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{..})^2 = \sum_{i=1}^n \sum_{j=1}^a [(\bar{Y}_{.j} - \bar{Y}_{..}) + (Y_{ij} - \bar{Y}_{.j})]^2$$

or

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{..})^2 &= \sum_{i=1}^n \sum_{j=1}^a (Y_{.j} - \bar{Y}_{..})^2 + \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{.j})^2 \\ &\quad + 2 \sum_{i=1}^n \sum_{j=1}^a (\bar{Y}_{.j} - \bar{Y}_{..}) \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{.j}) \end{aligned}$$

The last term on the right-hand side of the equation is equal to 0, because

$$\sum_{i=1}^n (Y_{ij} - \bar{Y}_{.j}) = Y_{.j} - n\bar{Y}_{.j} = Y_{.j} - nY_{.j}/n = 0$$

Therefore, the total sum of squares corrected is reduced to the following:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{..})^2 &= \sum_{i=1}^n \sum_{j=1}^a (Y_{.j} - \bar{Y}_{..})^2 + \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{.j})^2 \\ &= n \sum_{j=1}^a (\bar{Y}_{.j} - \bar{Y}_{..})^2 + \sum_{i=1}^n \sum_{j=1}^a (Y_{ij} - \bar{Y}_{.j})^2 \end{aligned}$$

Now, we get

Total sum of squares = (Sum of squares of treatment) + (Sum of squares of error)

That is;

$$SS_{\text{total}} = SS_A + SS_{\text{error}}$$

$$\text{Mean sum of squares of treatment} = \frac{\text{Sum of squares of treatment}}{\text{Degrees of freedom of treatment}} = \frac{SS_A}{a-1}$$

$$\text{Mean sum of squares of error} = \frac{\text{Sum of squares of error}}{\text{Degrees of freedom of error}}$$

$$= \frac{SS_{\text{error}}}{a(n-1)}$$

$$= \frac{SS_{\text{error}}}{N-a}$$

where N is the total number of observations in the experiment.

Following are the hypotheses of the model:

Null hypothesis, $H_0: \bar{Y}_{.1} = \bar{Y}_{.2} = \bar{Y}_{.3} = \bar{Y}_{.4} = \dots = \bar{Y}_{.a}$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of treatment means.

The generalized results are summarized in Table 4.2.

Table 4.2 Generalized Results with Single Factor

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	$a - 1$	$SS_{\text{treatments}}$	$SS_{\text{treatments}}/(a-1)$	$MSS_{\text{treatments}}/MSS_{\text{error}}$
Within treatments	$a(n-1)$	SS_{error}	$SS_{\text{error}}/[a(n-1)]$	$SS_{\text{error}}/[a(n-1)]$
Total	$N - 1$	SS_{total}		

In Table 4.2, the F ratio is obtained by dividing the mean sum of squares of the treatments by the mean sum of squares of the error by assuming the factor of concern (factor A) as the fixed factor.

In an experiment, if we have mixed factors or random factors, then dividing different mean sum of squares of the components of the model by the error mean sum of squares as in the fixed effect model, is incorrect. Such model requires a special procedure to identify the F ratios to test the effect of the model components on the response variable.¹

Different designs and related techniques discussed in this chapter are:

- Completely randomized design
- Randomized complete block design
- Latin square design
- Duncan's multiple range test
- Factorial design

1. This is explained in Section 4.7, which discusses 'expected mean squares rule'.

- 2^n factorial design
- Expected mean squares rule.

In this chapter, except the last subsection, the factors of the respective model are assumed as fixed factors. Hence, there is no special mention about the type of the factors involved in the examples of these subsections.

4.3 COMPLETELY RANDOMIZED DESIGN

The concept and application of completely randomized design are demonstrated through the following example 4.1:

Example 4.1 Ever Style Incorporation has a showroom at Chennai which deals with garments and health-care products. Recently, it has recruited 10 management graduates from each of the leading four business schools (A, B, C and D). It is keen in analyzing the sales performance of these graduates with respect to the assignment to different sections of the showroom. The products of the showroom are grouped and displayed into four sections, namely S_1 , S_2 , S_3 and S_4 . It is considering a study of sales over a period of 4 weeks (W_1 , W_2 , W_3 and W_4) in all the four sections. Four graduates from each of the four business schools (graduates A_1 , A_2 , A_3 and A_4 from the business school A; graduates B_1 , B_2 , B_3 and B_4 from the business schools B; graduates C_1 , C_2 , C_3 and C_4 from the business school C; graduates D_1 , D_2 , D_3 and D_4 from the business school D) are considered for allotment to different sections during different periods. The sixteen graduates are assigned to different combinations of the section and period, as shown in Table 4.3.

Table 4.3 Experimental Combinations of Completely Randomized Design

		Period			
		W_1	W_2	W_3	W_4
Section	S_1	A_1	C_1	B_4	C_3
	S_2	B_2	A_2	C_2	D_1
	S_3	D_2	B_3	C_4	B_1
	S_4	A_4	D_4	A_3	D_3

The weekly sales as per the completely randomized design, as shown in Table 4.3, are summarized in Table 4.4.

Table 4.4 Weekly Sales in Completely Randomized Design

		Period (Rupees in lakhs)			
		W_1	W_2	W_3	W_4
Section	S_1	23 (A_1)	35 (C_1)	20 (B_4)	25 (C_3)
	S_2	20 (B_2)	34 (A_2)	29 (C_2)	28 (D_1)
	S_3	40 (D_2)	30 (B_3)	45 (C_4)	30 (B_1)
	S_4	34 (A_4)	34 (D_4)	45 (A_3)	40 (D_3)

Rearranged data of Table 4.4 with respect to the graduates of different business schools, as the only factor, are shown in Table 4.5. In this table, the graduates of the business schools is the single factor. Different business schools (A, B, C and D) are different treatments (T_1 , T_2 , T_3 and T_4) of this design. Further, weekly sales of different graduates of a business school are termed as replications of that business school.

Table 4.5 Weekly Sales in Completely Randomized Design with Respect to Graduates of Different Business Schools

	Business school (Rupees in lakhs)			
	A	B	C	D
	23 (A_1)	30 (B_1)	35 (C_1)	28 (D_1)
	34 (A_2)	20 (B_2)	29 (C_2)	40 (D_2)
	45 (A_3)	30 (B_3)	25 (C_3)	40 (D_3)
	34 (A_4)	20 (B_4)	45 (C_4)	34 (D_4)
$Y_{.j}$	136	100	134	142

The model of completely randomized design is:

$$Y_{ij} = \mu + T_j + e_{ij}$$

where μ is the overall mean; Y_{ij} , the i th observation under the j th treatment of the factor; business school; T_j , the effect of the j th treatment of the factor, business school; and e_{ij} , the random error associated with the i th observation under the j th treatment of the factor, business school.

Null hypothesis, $H_0 : T_1 = T_2 = T_3 = T_4$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of the treatment means.

The relationship between different sum of squares of this model is as follows:

$$\begin{aligned} \text{Total sum of squares} &= (\text{Sum of the squares of treatments}) \\ &+ (\text{Sum of squares of errors}) \end{aligned}$$

That is

$$SS_{\text{total}} = SS_{\text{treatments}} + SS_{\text{error}}$$

In this case

$$SS_{\text{total}} = SS_{\text{business school}} + SS_{\text{error}}$$

The generalized shortcut formulas to compute the sum of squares of different components of the model are:

$$\begin{aligned} SS_{\text{total}} &= \sum_{i=1}^n \sum_{j=1}^a Y_{ij}^2 - \frac{Y_{..}^2}{N} \\ SS_{\text{treatments}} &= \sum_{j=1}^a \frac{Y_{.j}^2}{n} - \frac{Y_{..}^2}{N} \end{aligned}$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{treatments}}$$

where a is the number of treatments ($a = 4$); n , the number of replications under each treatment ($n = 4$; 4 graduates from each business school); $Y_{..}$, the sum of Y_{ij} over all values of i and j ; $Y_{.j}$, the sum of Y_{ij} over all values of i for a given j ; and N , the total number of observations in the experiments ($4 \times 4 = 16$).

For the example problems, we have

$$Y_{..} = \sum_{i=1}^4 \sum_{j=1}^4 Y_{ij} = 512$$

$$Y_{.1} = 136, \quad Y_{.2} = 100, \quad Y_{.3} = 134 \quad \text{and} \quad Y_{.4} = 142$$

Then

$$SS_{\text{total}} = \sum_{i=1}^4 \sum_{j=1}^4 Y_{ij}^2 = \frac{Y_{..}^2}{16}$$

$$= (23)^2 + (30)^2 + (35)^2 + (28)^2 + (34)^2 + (20)^2 + \dots + (45)^2 + (34)^2 - \frac{(512)^2}{16}$$

$$= 938$$

$$SS_{\text{treatments}} = \sum_{j=1}^4 \frac{Y_{.j}^2}{4} - \frac{Y_{..}^2}{16}$$

$$= \frac{(136)^2 + (100)^2 + (134)^2 + (142)^2}{4} - \frac{(512)^2}{16}$$

$$= 270$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{treatments}} = 938 - 270 = 668$$

The distribution of the total sum of squares of this design is shown diagrammatically in Fig. 4.1.

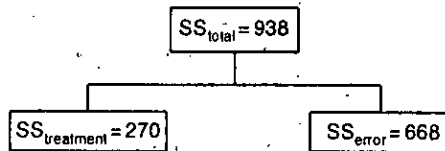


Fig. 4.1 Distribution of total sum of squares of completely randomized design of Example 4.1.

The generalized results and results applied to the example problem of this design are summarized in Tables 4.6 and 4.7, respectively.

Table 4.6 Generalized Results in Completely Randomized Design

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	$a - 1$	$SS_{\text{treatments}}$	$SS_{\text{treatments}} / (a - 1)$	$SS_{\text{treatments}} / MSS_{\text{error}}$
Within treatments	$N - a$	SS_{error}	$SS_{\text{error}} / (N - a)$	
Total	$N - 1$	SS_{total}		

Table 4.7 Results in Completely Randomized Design (Example 4.1)

Source of variation	Degrees of freedom	sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	3	270	$270/3 = 90$	$90/55.67 = 1.617$
Within treatments (error)	12	668	$668/12 = 55.67$	
Total	15	938		

In Table 4.7, the value of the calculated F ratio is 1.617. The value of F ratio from statistical table for a significance level of 0.05 and degrees of freedom (3,12) is 3.49. Also,

$$F_{\text{calculated}} [1.617] < F_{\alpha = 0.05 \text{ and } df = (3,12)} [3.49]$$

Hence, the null hypothesis (H_0) should be accepted.

Inference: This mean that there is no significant difference in terms of weekly sales made between the graduates of different business schools.

Example 4.2 Alpha Industries Limited has deputed four different batches of its employees to four different training organizations (A, B, C and D) for the same training programme, which aims to train them in improving their decision-making skills. Each batch consists of five employees with similar qualifications and work experience. After the training programme, the company conducted a common examination to test their level of additional knowledge gained through the training programme. The percentage scores of the employees of the batches are summarized in Table 4.8.

Table 4.8 Percentage Scores of Employees of Batches in the Examination

Training organization			
A	B	C	D
80	70	65	90
90	60	50	89
96	55	58	85
85	85	55	95
70	90	40	80

Perform ANOVA to check whether there is a significant difference between the training organizations in terms improving the decision-making skills of the employees by assuming a significance level of 0.05.

Solution. The model of this problems is:

$$Y_{ij} = \mu + T_j + e_{ij}$$

where μ is the overall mean; Y_{ij} , the i th observation under the j th treatment of the factor, training organization; T_j , the effect of the j th treatment of the factor, training organization; e_{ij} , the random error associated with the i th observation under the j th treatment of the factor training organization,

Null hypothesis, $H_0: T_1 = T_2 = T_3 = T_4$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of treatment means.

The data of Table 4.8 is reproduced below in Table 4.9 along with column totals.

Table 4.9 Column Totals of Table 4.8

Training organization			
A	B	C	D
80	70	65	90
90	60	50	89
96	55	58	85
85	85	55	95
70	90	40	80
421	360	268	439

We have

$$Y_{..} = \sum_{i=1}^5 \sum_{j=1}^4 Y_{ij} = 1488$$

$$Y_{.1} = 421, Y_{.2} = 360, Y_{.3} = 268 \text{ and } Y_{.4} = 439$$

Then

$$SS_{\text{total}} = \sum_{i=1}^5 \sum_{j=1}^4 Y_{ij}^2 - \frac{Y_{..}^2}{20}$$

$$= (80)^2 + (70)^2 + (65)^2 + (90)^2 + (90)^2 + (60)^2 + \dots + (40)^2 + (80)^2 - \frac{(1488)^2}{20}$$

$$= 5368.8$$

$$SS_{\text{treatments}} = \sum_{j=1}^4 \frac{Y_{.j}^2}{5} - \frac{Y_{..}^2}{20}$$

$$= \frac{(421)^2 + (360)^2 + (268)^2 + (439)^2}{5} - \frac{(1488)^2}{20}$$

$$= 3570$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{treatments}}$$

$$= 5368.8 - 3570$$

$$= 1798.8$$

The results of this problem are summarized in Table 4.10.

Table 4.10 Results in Completely Randomized Design (Example 4.2)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	3	3570.0	$3570/3 = 1190$	$1190/112.43 = 10.58$
Within treatments	16	1798.8	$1798.8/16 = 112.43$	
Total	19	5368.8		

In Table 4.10, the value of the calculated F ratio is 10.58. The value of F ratio from the statistical table for a significance level of 0.05 and degrees of freedom (3,16) is 3.24. Then

$$F_{\text{calculated}} [10.58] > F_{\alpha=0.05 \text{ and } df=(3,16)} [3.24]$$

Hence, the null hypothesis (H_0) should be rejected.

Inference: This means that there is a significant difference in terms of percentage scores between the batches trained by different training organizations.

4.4 RANDOMIZED COMPLETE BLOCK DESIGN

The concept of randomized complete block design is demonstrated through the following example:

Example 4.3 In Table 4.3 of Example 4.1, there is no assignment of graduate of each business school in each of the sections of the showroom except the section, S_2 . Hence, there is no homogeneity in the assignment of graduates to different sections of the showroom. (This is considered to be a major limitation of the completely randomized design.) Hence, in this section, randomized complete block design is presented which will introduce homogeneity in each block (section), as shown in Table 4.11. (In this table, each section of the showroom—each row—is known as a block and assigned with one graduate from each of the business schools which brings homogeneity in it.)

Table 4.11 Experimental Combinations of Randomized Complete Block Design (Example 4.3)

		Period			
		W_1	W_2	W_3	W_4
Section	S_1	A_1	B_1	C_3	D_1
	S_2	B_2	C_2	A_2	D_4
	S_3	A_3	D_3	B_3	C_4
	S_4	C_1	B_4	D_2	A_4

In practice, there will be changes in the values of the weekly sales of the graduates of different business schools if a different experimental design is used. For the purpose of simplicity, as per the design shown in Table 4.11, the data from Table 4.4 is mapped into Table 4.12. The data of Table 4.12 is rearranged and presented, as shown in Table 4.13.

Table 4.12 Weekly Sales Rupees in Lakh in Randomized Complete Block Design (Example 4.3)

		Period			
		W_1	W_2	W_3	W_4
Section	S_1	(A_1) 23	(B_1) 30	(C_3) 25	(D_1) 28
	S_2	(B_2) 20	(C_2) 29	(A_2) 34	(D_4) 34
	S_3	(A_3) 45	(D_3) 40	(B_3) 30	(C_4) 45
	S_4	(C_1) 35	(B_4) 20	(D_2) 40	(A_4) 34

Table 4.13 Rearranged Weekly Sales Rupees in Lakh in Randomized Complete Block Design (Example 4.3)

		Business school				
		A	B	C	D	$Y_{.j}$
Section	S_1	(A ₁) 23	(B ₁) 30	(C ₃) 25	(D ₁) 28	106
	S_2	(A ₂) 34	(B ₂) 20	(C ₂) 29	(D ₄) 34	117
	S_3	(A ₃) 45	(B ₃) 30	(C ₄) 45	(D ₃) 40	160
	S_4	(A ₄) 34	(B ₄) 20	(C ₁) 35	(D ₂) 40	129
$Y_{.j}$		136	100	134	142	

The Model of the randomized complete block design is as follows:

$$Y_{ij} = \mu + B_i + T_j + e_{ij}$$

where μ is the overall mean; Y_{ij} , the observation with respect to the j th treatment of the factor (business school) and i th block (section); B_i , the effect of the i th block (section); T_j , the effect of the j th treatment of the factor (business school) and e_{ij} , the random error associated with the i th block (section) and the j th treatment of the factor (business school).

Hypothesis with respect to treatment (business school):

Null hypothesis, $H_0: T_1 = T_2 = T_3 = T_4$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of treatment means.

Hypothesis with respect to block (section):

Null hypothesis, $H_0: B_1 = B_2 = B_3 = B_4$

Alternate hypothesis, H_1 : Block means are not equal for at least one pair of block means.

The relationship between different sum of squares of this model is shown as follows:

$$\begin{aligned} \text{Total sum of squares} &= (\text{Sum of squares of blocks}) \\ &\quad + (\text{Sum of squares of treatments}) \\ &\quad + (\text{Sum of squares of errors}) \end{aligned}$$

That is

$$SS_{\text{total}} = SS_{\text{blocks}} + SS_{\text{treatments}} + SS_{\text{error}}$$

For the example problem,

$$SS_{\text{total}} = SS_{\text{section}} + SS_{\text{business school}} + SS_{\text{error}}$$

The generalized shortcut formulas to compute the sum of squares of different components of the model are:

$$SS_{\text{total}} = \sum_{i=1}^b \sum_{j=1}^a Y_{ij}^2 - \frac{Y_{..}^2}{N}$$

$$SS_{\text{blocks}} = \sum_{i=1}^b \frac{Y_{i.}^2}{a} - \frac{Y_{..}^2}{N}$$

$$SS_{\text{treatments}} = \sum_{j=1}^a \frac{Y_{.j}^2}{b} - \frac{Y_{..}^2}{N}$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{blocks}} - SS_{\text{treatments}}$$

where a is the number of treatments ($a=4$); b , the number of blocks ($b=4$); $Y_{..}$, the sum of Y_{ij} overall values of i and j ; $Y_{.j}$, the sum of Y_{ij} overall values of i for a given j ; $Y_{i.}$, the sum of Y_{ij} overall values of j , for a given i ; and, N , the total number of observations in the experiments ($4 \times 4 = 16$).

For this design, the following calculations are done based on Table 4.13:

$$Y_{..} = \sum_{i=1}^4 \sum_{j=1}^4 Y_{ij} = 512$$

$$Y_{.1} = 136, Y_{.2} = 100, Y_{.3} = 134 \text{ and } Y_{.4} = 142$$

$$Y_{1.} = 106, Y_{2.} = 117, Y_{3.} = 160 \text{ and } Y_{4.} = 129$$

Therefore,

$$SS_{\text{total}} = \sum_{i=1}^4 \sum_{j=1}^4 Y_{ij}^2 - \frac{Y_{..}^2}{16}$$

$$= (23)^2 + (30)^2 + (25)^2 + (28)^2 + (34)^2 + (20)^2 + \dots + (35)^2 + (40)^2 - \frac{(512)^2}{16}$$

$$= 938$$

$$SS_{\text{treatments}} = \sum_{j=1}^4 \frac{Y_{.j}^2}{4} - \frac{Y_{..}^2}{16}$$

$$= \frac{(136)^2 + (100)^2 + (134)^2 + (142)^2}{4} - \frac{(512)^2}{16}$$

$$= 270$$

$$\begin{aligned}
 SS_{\text{blocks}} &= \sum_{i=1}^4 \frac{Y_i^2}{4} - \frac{Y_{..}^2}{16} \\
 &= \frac{(106)^2 + (117)^2 + (160)^2 + (129)^2}{4} - \frac{(512)^2}{16} \\
 &= 407.5
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{error}} &= SS_{\text{total}} - SS_{\text{blocks}} - SS_{\text{treatments}} \\
 &= 938 - 407.5 - 270 \\
 &= 260.5
 \end{aligned}$$

The distribution of the total sum of squares of this design is shown diagrammatically in Fig. 4.2.

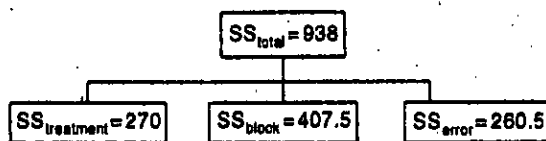


Fig. 4.2 Distribution of total sum of squares of randomized complete block design of Example 4.3.

The generalized results and results applied to the example problem of this design are summarized in Tables 4.14 and 4.15, respectively.

Table 4.14 Generalized Results in Randomized Complete Block Design

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	$a - 1$	$SS_{\text{treatments}}$	$SS_{\text{treatments}}/(a - 1)$	$MSS_{\text{treatments}}/MSS_{\text{error}}$
Between blocks	$b - 1$	SS_{blocks}	$SS_{\text{blocks}}/(b - 1)$	$MSS_{\text{blocks}}/MSS_{\text{error}}$
Error	$N - a - b + 1$	SS_{error}	$SS_{\text{error}}/(N - a - b + 1)$	
Total	$N - 1$	SS_{total}		

Table 4.15 Results in Randomized Complete Blocks Design (Example 4.3)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	3	270.0	$270/3 = 90$	$90/28.94 = 3.11$
Between blocks	3	407.5	$407.5/3 = 135.83$	$135.83/28.94 = 4.69$
Error	9	260.5	$260.5/9 = 28.94$	
Total	15	938.0		

In Table 4.15, the value of the calculated F ratio for the treatment is 3.11, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 9) is 3.86. The calculated F ratio for the block is 4.69, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 9) is 3.86.

Component—treatment (business school): For this component,

$$F_{\text{calculated}} [3.11] < F_{\alpha=0.05 \text{ and } df=(3,9)} [3.86]$$

Hence, the null hypothesis H_0 should be accepted.

Inference: This mean that there is no significant difference in terms of weekly sales made between the graduates of different business schools.

Component—block (section): For this component,

$$F_{\text{calculated}} [4.69] > F_{\alpha=0.05 \text{ and } df=(3,9)} [3.86]$$

Hence, the null hypothesis (H_0) should be rejected.

Inference: This mean that there is significant difference in terms of weekly sales made between different sections of the showroom.

Example 4.4 A business school recently conducted a management meet titled 'Synergy' in which 'stock exchange simulation game' was an event. Students from 15 different business schools participated in the meet. Five different teams (T_1, T_2, T_3, T_4 and T_5) have been formed from these 15 schools to participate in the simulation game. Each team consisted of a team leader (L_1, L_2, L_3, L_4 and L_5) and four members. The details of the game were as follows:

- Each team was given 10 lakhs rupees.
- Daily quotations were displayed in an online network system.
- Rumours, political news and industrial news were broadcasted over a mike.
- Based on these information, the teams effected purchases and sales.
- At the end of the game, the net worth of each team was assessed.
- Then, team leaders were rotated to serve in all other teams as team leaders and the game was repeated four more times.

It is clear that the experimentation is a randomized complete block design. The net worths of the teams (in lakhs of rupees) after the experiment are, as summarized in Table 4.16.

Table 4.16 Results of Simulation Game

		Team Leader				
		L ₁	L ₂	L ₃	L ₄	L ₅
Team	T ₁	40	30	55	25	35
	T ₂	45	60	10	22	33
	T ₃	38	55	40	55	28
	T ₄	30	27	32	56	17
	T ₅	45	34	20	34	37

Perform the randomized complete block design to check whether there is a significant difference between the team leaders (L₁, L₂, L₃, L₄ and L₅) as well as between different teams (T₁, T₂, T₃, T₄ and T₅) in terms of final net worth by assuming a significance level of 0.05.

Solution. In this problem, 'team leader' is considered to be the factor and 'team' is considered to be the block. The model of the randomized complete block design applied to this example is:

$$Y_{ij} = \mu + T_i + L_j + e_{ij}$$

where μ is the overall mean; Y_{ij} , the observation with respect to j th treatment of the factor (team leader) and i th block (team); T_i , the effect of the i th block (team); L_j , the effect of the j th treatment of the factor (team leader); and e_{ij} , the random error associated with the i th block (team) and the j th treatment of the factor (team leader).

Hypothesis with respect to treatment (team leader): In this case,

Null hypothesis, $H_0: L_1 = L_2 = L_3 = L_4 = L_5$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of treatment means.

Hypothesis with respect to block (team): Here,

Null hypothesis, $H_0: T_1 = T_2 = T_3 = T_4 = T_5$

Alternate hypothesis, H_1 : Block means are not equal for at least one pair of block means.

The data of this example is reproduced with row totals and column totals, as shown in Table 4.17.

Table 4.17 Results of Simulation Game

	Team leader					$Y_{i.}$
	L_1	L_2	L_3	L_4	L_5	
T_1	40	30	55	25	35	185
T_2	45	60	10	22	33	170
T_3	38	55	40	55	28	216
T_4	30	27	32	56	17	162
T_5	45	34	20	34	37	170
$Y_{.j}$	198	206	157	192	150	903

For the design, the following calculations are done based on the Table 4.17:

$$Y_{..} = \sum_{i=1}^5 \sum_{j=1}^5 Y_{ij} = 903$$

$$Y_{.1} = 198, Y_{.2} = 206, Y_{.3} = 157, Y_{.4} = 192 \text{ and } Y_{.5} = 150$$

$$Y_{1.} = 185, Y_{2.} = 170, Y_{3.} = 216, Y_{4.} = 162 \text{ and } Y_{5.} = 170$$

Now, we get

$$SS_{\text{total}} = \sum_{i=1}^5 \sum_{j=1}^5 Y_{ij}^2 - \frac{Y_{..}^2}{25}$$

$$= (40)^2 + (30)^2 + (55)^2 + (25)^2 + (35)^2 + (45)^2 + \dots + (34)^2 + (37)^2 - \frac{(903)^2}{25}$$

$$= 4118.64$$

$$SS_{\text{team leader}} = \sum_{j=1}^5 \frac{Y_{.j}^2}{5} - \frac{Y_{..}^2}{25}$$

$$= \frac{(198)^2 + (206)^2 + (157)^2 + (192)^2 + (150)^2}{5} - \frac{(903)^2}{25}$$

$$= 514.24$$

$$SS_{\text{team}} = \sum_{i=1}^5 \frac{Y_{i.}^2}{5} - \frac{Y_{..}^2}{25}$$

$$= \frac{(185)^2 + (170)^2 + (216)^2 + (162)^2 + (170)^2}{5} - \frac{(903)^2}{25}$$

$$= 368.64$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{treatments}} - SS_{\text{blocks}}$$

$$= 4118.64 - 514.24 - 368.64$$

$$= 3235.76$$

The results of this example are summarized in Table 4.18.

Table 4.18 Results in Randomized Complete Block Design (Example 4.4)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	4	514.24	$514.24/4 = 128.56$	$128.56/202.24 = 0.636$
Between blocks	4	368.64	$368.64/4 = 92.16$	$92.16/202.24 = 0.456$
Error	16	3235.76	$3235.76/16 = 202.24$	
Total	24	4118.64		

In Table 4.18, the value of the calculated F ratio of the treatment is 0.636, whereas its table value with the significance level of 0.05 and degrees of freedom (4,16) is 3.01. The calculated F ratio for the block is 0.456, whereas its table value with the significance level of 0.05 and degrees of freedom (4, 16) is 3.01.

Component—treatment (team leader): For this component,

$$F_{\text{calculated}} [0.636] < F_{\alpha = 0.05 \text{ and } df = (4,16)} [3.01]$$

Hence, the null hypothesis, H_0 should be accepted.

Inference: This means that there is no significant difference in terms of net worth between the team leaders.

Component—block (team): For this component,

$$F_{\text{calculated}} [0.456] < F_{\alpha = 0.05 \text{ and } df = (4,16)} [3.01]$$

Hence, the null hypothesis, H_0 should be accepted.

Inference: This means that there is no significant difference in terms of net worth between different teams.

4.5 LATIN SQUARE DESIGN

The concept of Latin square design is demonstrated through Example 4.5, which is an extension of Example 4.3. This design is an improvement over the randomized complete block design.

Example 4.5 Consider Example 4.3. In Table 4.11 of this example, each section (row) has the assignment of one graduate from each of the business schools. But, it has no assignment of graduate of each business school in each of the weeks (columns) of the showroom except the week W_3 . Hence, there is no homogeneity in the assignment of graduates to different weeks of the showroom. This is considered to be a major limitation of the randomized complete block design. Hence, in this section, Latin square design is presented, which will introduce homogeneity in each week, as shown in Table 4.19. In this table, 'period' of the showroom (each column) is known as another type of block in addition to the earlier type of block, 'section'. Each period is assigned with one graduate from each of the business schools, which brings homogeneity in each column.

Table 4.19 Experimental Combinations in Latin Square Design (Example 4.5)

		Period			
		W_1	W_2	W_3	W_4
Section	S_1	A_1	D_2	B_1	C_1
	S_2	B_2	C_4	A_2	D_3
	S_3	C_3	A_3	D_4	B_4
	S_4	D_1	B_3	C_2	A_4

In practice, there will be changes in the values of the weekly sales of the graduates of different business schools if a different experimental design is used. For the purpose of simplicity, as per the design shown in Table 4.19, the data from Table 4.12 is mapped into Table 4.20.

Table 4.20 Weekly Sales in Latin Square Design

		Period (Rupees in lakhs)				$Y_{i..}$
		W_1	W_2	W_3	W_4	
Section	S_1	$(A_1) 23$	$(D_2) 40$	$(B_1) 30$	$(C_1) 35$	128
	S_2	$(B_2) 20$	$(C_4) 45$	$(A_2) 34$	$(D_3) 40$	139
	S_3	$(C_3) 25$	$(A_3) 45$	$(D_4) 34$	$(B_4) 20$	124
	S_4	$(D_1) 28$	$(B_3) 30$	$(C_2) 29$	$(A_4) 34$	121
$Y_{..k}$		96	160	127	129	

The model of the Latin square design is as follows:

$$Y_{ijk} = \mu + B_i + T_j + P_k + e_{ijk}$$

where μ is the overall mean; Y_{ijk} , the observation in the i th row (section) and k th column (period) for the j th treatment of the factor (business school); B_i , the effect of the i th block representing section; T_j , the effect of the j th treatment of the factor; (business school); P_k , the effect of the k th block representing period; and e_{ijk} is the random error associated with the i th row (section) and the k th column (period) for the j th treatment of the factor (business school).

Hypothesis with respect to treatment (business school):

Null hypothesis, $H_0: T_1 = T_2 = T_3 = T_4$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of treatment means.

Hypothesis with respect to rows (section):

Null hypothesis, $H_0: B_1 = B_2 = B_3 = B_4$

Alternate hypothesis, H_1 : Row means are not equal for at least one pair of row means.

Hypothesis with respect to columns (period):

Null hypothesis, $H_0: P_1 = P_2 = P_3 = P_4$

Alternate Hypothesis, H_1 : Column means are not equal for at least one pair of column means.

The relationship between different sum of squares of this model is:

Total sum of squares = (Sum of squares of treatments) + [Sum of squares of rows (section)]
+ [Sum of squares of columns (period)]
+ (Sum of squares of errors)

That is

$$SS_{\text{total}} = SS_{\text{rows}} + SS_{\text{treatments}} + SS_{\text{columns}} + SS_{\text{error}}$$

For the example problem,

$$SS_{\text{total}} = SS_{\text{section}} + SS_{\text{business school}} + SS_{\text{period}} + SS_{\text{error}}$$

The generalized shortcut formulas to compute the sum of squares of different components of the model are:

$$SS_{\text{total}} = \sum_i \sum_j \sum_k Y_{ijk}^2 - \frac{Y^2}{N}$$

$$SS_{\text{rows}} = \sum_{i=1}^b \frac{Y_{i..}^2}{c} - \frac{Y^2}{N}$$

$$SS_{\text{treatments}} = \sum_{j=1}^a \frac{Y_{.j.}^2}{b} - \frac{Y^2}{N}$$

$$SS_{\text{columns}} = \sum_{k=1}^c \frac{Y_{..k}^2}{b} - \frac{Y_{...}^2}{N}$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{rows}} - SS_{\text{treatments}} - SS_{\text{columns}}$$

where a is the number of treatments ($a = 4$); b , the number of rows ($b = 4$); c , the number of columns ($c = 4$); $Y_{...}$, the sum of Y_{ijk} overall values of i, j and k ; $Y_{.j.}$, the sum of Y_{ijk} over all values of i and k for a given j ; $Y_{i..}$, the sum of Y_{ijk} overall values of j and k for a given i ; $Y_{..k}$, the sum of Y_{ijk} overall values of i and j for a given value of k ; and N , the total number of observations in the experiments ($4 \times 4 = 16$).

For the example under consideration:

$$Y_{...} = \sum_i \sum_j \sum_k Y_{ijk} = 512$$

$$Y_{1..} = 128, Y_{2..} = 139, Y_{3..} = 124 \text{ and } Y_{4..} = 121$$

$$Y_{..1} = 96, Y_{..2} = 160, Y_{..3} = 127 \text{ and } Y_{..4} = 129$$

Now, we get

$$\begin{aligned} SS_{\text{total}} &= \sum_i \sum_j \sum_k Y_{ijk}^2 - \frac{Y_{...}^2}{16} \\ &= (23)^2 + (40)^2 + (30)^2 + (35)^2 + (20)^2 + (45)^2 + \dots + (29)^2 + (34)^2 - \frac{(512)^2}{16} \\ &= 938 \text{ (Same as in Examples 4.1 and 4.3)} \end{aligned}$$

$$\begin{aligned} SS_{\text{section}} &= \sum_{i=1}^4 \frac{Y_{i..}^2}{4} - \frac{Y_{...}^2}{16} \\ &= \frac{(128)^2 + (139)^2 + (124)^2 + (121)^2}{4} - \frac{(512)^2}{16} \\ &= 46.5 \end{aligned}$$

$$\begin{aligned} SS_{\text{period}} &= \sum_{k=1}^4 \frac{Y_{..k}^2}{4} - \frac{Y_{...}^2}{16} \\ &= \frac{(96)^2 + (160)^2 + (127)^2 + (129)^2}{4} - \frac{(512)^2}{16} \\ &= 512.5 \end{aligned}$$

By rearranging the data of Table 4.20, as in Table 4.5, the column totals for the treatments are as follows:

$$Y_{.1.} = 136, \quad Y_{.2.} = 100, \quad Y_{.3.} = 134, \quad Y_{.4.} = 142$$

Therefore,

$$\begin{aligned} SS_{\text{treatments}} &= \sum_{j=1}^4 \frac{Y_{.j.}^2}{4} - \frac{Y_{...}^2}{16} \\ &= \frac{(136)^2 + (100)^2 + (134)^2 + (142)^2}{4} - \frac{(512)^2}{16} \\ &= 270 \text{ (Same as in Examples 4.1 and 4.3)} \\ SS_{\text{error}} &= SS_{\text{total}} - SS_{\text{section}} - SS_{\text{treatments}} - SS_{\text{period}} \\ &= 938 - 46.5 - 270 - 512.5 \\ &= 109 \end{aligned}$$

The distribution of sum of squares of this design is shown diagrammatically in Fig. 4.3.

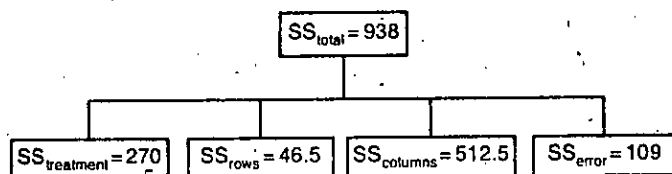


Fig. 4.3 Distribution of sum of squares of Latin square design.

The generalized results and results applied to the example problem of this design are summarized in Tables 4.21 and 4.22, respectively.

Table 4.21 Generalized Results in Latin Square Design

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	$a - 1$	$SS_{\text{treatments}}$	$\frac{SS_{\text{treatments}}}{a - 1}$	$\frac{MSS_{\text{treatments}}}{MSS_{\text{error}}}$
Between rows	$b - 1$	SS_{rows}	$\frac{SS_{\text{rows}}}{b - 1}$	$\frac{MSS_{\text{rows}}}{MSS_{\text{error}}}$
Between columns	$c - 1$	SS_{columns}	$\frac{SS_{\text{blocks}}}{c - 1}$	$\frac{MSS_{\text{columns}}}{MSS_{\text{error}}}$
Error	$N - a - b - c + 2$	SS_{error}	$\frac{SS_{\text{error}}}{N - a - b - c + 2}$	
Total	$N - 1$	SS_{total}		

Table 4.22 Results in Latin Square Design

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between treatments	3	270.0	$270/3 = 90.00$	$90/18.17 = 4.95$
Between rows	3	46.5	$46.5/3 = 15.50$	$15.5/18.17 = 0.85$
Between columns	3	512.5	$512.5/3 = 170.83$	$170.83/18.17 = 9.40$
Error	6	109.0	$109/6 = 18.17$	
Total	15	938.0		

In Table 4.22, the value of the calculated F ratio for the treatment is 4.95 whereas its table value with the significance level of 0.05 and degrees of freedom (3, 6) is 4.76. The calculated F ratio for the rows is 0.85, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 6) is 4.76. The calculated F ratio for the columns is 9.40, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 6) is 4.76.

Component—treatment (business school): For this component,

$$F_{\text{calculated}} [4.95] > F_{\alpha = 0.05 \text{ and } df = (3,6)} [4.76]$$

Hence, the null hypothesis (H_0) should be rejected.

Inference: This means that there is significant difference in terms of weekly sales made between the graduates of different business schools.

Component—rows (section): For this component,

$$F_{\text{calculated}} [0.85] < F_{\alpha = 0.05 \text{ and } df = (3,6)} [4.76]$$

Hence, the null hypothesis H_0 should be accepted.

Inference: This means that there is no significant difference in terms of weekly sales made between different sections of the showroom.

Component—columns (period): For this component,

$$F_{\text{(calculated)}} [9.40] > F_{\alpha = 0.05 \text{ and } df = (3,6)} [4.76]$$

Hence, the null hypothesis, H_0 should be rejected.

Inference: This means that there is significant difference in terms of weekly sales made between different weeks (period) of the showroom.

Example 4.6 The production manager of a firm is keen in analyzing the hourly production volume of a product. He feels that operators (A, B, C, D and E), machines (M_1 , M_2 , M_3 , M_4 and M_5) and suppliers of raw materials (S_1 , S_2 , S_3 , S_4 and S_5) will affect the production volume of the product. Hence, he designed a Latin square design, as shown in Table 4.23. The corresponding observations are shown in Table 4.24.

Table 4.23 Experimental Combinations in Latin Square Design (Example 4.6)

		Supplier				
		S ₁	S ₂	S ₃	S ₄	S ₅
Machine	M ₁	A	B	C	D	E
	M ₂	B	C	D	E	A
	M ₃	C	D	E	A	B
	M ₄	D	E	A	B	C
	M ₅	E	A	B	C	D

Table 4.24 Hourly Production Volume as per Latin Square Design (Example 4.6)

		Supplier				
		S ₁	S ₂	S ₃	S ₄	S ₅
Machine	M ₁	12 (A)	8 (B)	9 (C)	15 (D)	18 (E)
	M ₂	14 (B)	7 (C)	13 (D)	10 (E)	12 (A)
	M ₃	7 (C)	15 (D)	12 (E)	8 (A)	6 (B)
	M ₄	16 (D)	17 (E)	13 (A)	11 (B)	12 (C)
	M ₅	9 (E)	14 (A)	8 (B)	10 (C)	8 (D)

Perform statistical analysis to test the following at a significance level of 0.05:

- Whether there is significant difference between operators in terms of production volume of the product;
- Whether there is significant difference between machines in terms of production volume of the product;
- Whether there is significant difference between suppliers in terms of production volume of the product.

Solution. The model of the Latin square design of this example is as follows:

$$Y_{ijk} = \mu + M_i + O_j + S_k + e_{ijk}$$

where μ is the overall mean; Y_{ijk} , the observation in the i th row (machine) and k th column (supplier) for the j th treatment of the factor (Operator); M_i , the effect of the i th block representing machine; O_j , the effect of the j th treatment of the factor (Operator); S_k , the effect of the k th block representing supplier; and e_{ijk} , the random error associated with the i th row (machine) and the k th column (supplier) for the j th treatment of the factor (operator).

Hypothesis with respect to treatment (operator):

Null hypothesis, $H_0: O_1 = O_2 = O_3 = O_4 = O_5$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of treatment means.

Hypothesis with respect to rows (machine):

Null hypothesis, $H_0: M_1 = M_2 = M_3 = M_4 = M_5$

Alternate Hypothesis, H_1 : Row means are not equal for at least one pair of row means.

Hypothesis with respect to columns (supplier):

Null hypothesis, $H_0: S_1 = S_2 = S_3 = S_4 = S_5$

Alternate hypothesis, H_1 : Column means are not equal for at least one pair of column means.

The data of Table 4.24 is reproduced in Table 4.25 with row totals and column totals.

Table 4.25 Hourly Production Volume in Latin Square Design (Example 4.6)

		Supplier					
		S_1	S_2	S_3	S_4	S_5	$Y_{..}$
Machine	M_1	12 (A)	8 (B)	9 (C)	15 (D)	18 (E)	62
	M_2	14 (B)	7 (C)	13 (D)	10 (E)	12 (A)	56
	M_3	7 (C)	15 (D)	12 (E)	8 (A)	6 (B)	48
	M_4	16 (D)	17 (E)	13 (A)	11 (B)	12 (C)	69
	M_5	9 (E)	14 (A)	8 (B)	10 (C)	8 (D)	49
$Y_{..k}$		58	61	55	54	56	284

For the example under consideration,

$$Y_{...} = \sum_i \sum_j \sum_k Y_{ijk} = 284$$

$$Y_{1..} = 62, Y_{2..} = 56, Y_{3..} = 48, Y_{4..} = 69 \text{ and } Y_{5..} = 49$$

$$Y_{..1} = 58, Y_{..2} = 61, Y_{..3} = 55, Y_{..4} = 54 \text{ and } Y_{..5} = 56$$

Then, we get

$$\begin{aligned}
 SS_{\text{total}} &= \sum_i \sum_j \sum_k Y_{ijk}^2 - \frac{Y_{...}^2}{25} \\
 &= (12)^2 + (8)^2 + (9)^2 + (15)^2 + (18)^2 + (14)^2 + \dots + (10)^2 + (8)^2 - \frac{(284)^2}{25} \\
 &= 271.76
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{machine}} &= \sum_{i=1}^5 \frac{Y_{i..}^2}{5} - \frac{Y_{...}^2}{25} \\
 &= \frac{(62)^2 + (56)^2 + (48)^2 + (69)^2 + (49)^2}{5} - \frac{(284)^2}{25} \\
 &= 62.96
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{supplier}} &= \sum_{k=1}^5 \frac{Y_{..k}^2}{5} - \frac{Y_{...}^2}{25} \\
 &= \frac{(58)^2 + (61)^2 + (55)^2 + (54)^2 + (56)^2}{5} - \frac{(284)^2}{25} \\
 &= 6.16
 \end{aligned}$$

The data of Table 4.24 is rearranged as shown in Table 4.26, to compute the sum of squares of 'operator'.

Table 4.26 Rearranged Hourly Production Volume in Latin Square Design (Example 4.6)

	Operator				
	A	B	C	D	E
$Y_{1.}$	12	8	9	15	18
$Y_{2.}$	12	14	7	13	10
$Y_{3.}$	8	6	7	15	12
$Y_{4.}$	13	11	12	16	17
$Y_{5.}$	14	8	10	8	9
$Y_{.j}$	59	47	45	67	68

we get

$$Y_{1.} = 59, \quad Y_{2.} = 47, \quad Y_{3.} = 45, \quad Y_{4.} = 67, \quad Y_{5.} = 66$$

Then,

$$\begin{aligned}
 SS_{\text{operator}} &= \sum_{j=1}^5 \frac{Y_{.j}^2}{5} - \frac{Y_{...}^2}{25} \\
 &= \frac{(59)^2 + (47)^2 + (45)^2 + (67)^2 + (66)^2}{5} - \frac{(284)^2}{25} \\
 &= 85.76
 \end{aligned}$$

$$\begin{aligned}
 SS_{\text{error}} &= SS_{\text{total}} - SS_{\text{machine}} - SS_{\text{operator}} - SS_{\text{supplier}} \\
 &= 271.76 - 62.96 - 85.76 - 6.16 \\
 &= 116.88
 \end{aligned}$$

The ANOVA results of this problem are summarized in Table 4.27.

Table 4.27 Results in Latin Square Design (Example 4.6)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between operators	4	85.76	$85.76/4 = 21.44$	$21.44/9.74 = 2.20$
Between machines	4	62.96	$62.96/4 = 15.74$	$15.74/9.74 = 1.62$
Between suppliers	4	6.16	$6.16/4 = 1.54$	$1.54/9.74 = 0.16$
Error	12	116.88	$116.88/12 = 9.74$	
Total	24	271.76		

In Table 4.27, the value of the calculated F ratio for 'operator' is 2.2, whereas its table value with the significance level of 0.05 and degrees of freedom (4,12) is 3.26. The calculated F ratio for 'machine' is 1.62; whereas its table value with the significance level of 0.05 and degrees of freedom (4,12) is 3.26. The calculated F ratio for 'supplier' is 0.16, whereas its table value with the significance level of 0.05 and degrees of freedom (4,12) is 3.26.

Component—treatment (operator): For this component,

$$F_{\text{calculated}} [2.2] < F_{\alpha=0.05 \text{ and } df=(4,12)} [3.26]$$

Hence, the null hypothesis, H_0 should be accepted.

Inference: This means that there is no significant difference between operators in terms of hourly production volume.

Component—rows (machines): For this component,

$$F_{\text{calculated}} [1.62] < F_{\alpha=0.05 \text{ and } df=(4,12)} [3.26]$$

Hence, the null hypothesis, (H_0) should be accepted.

Inference: This means that there is no significant difference between machines in terms of hourly production volume.

Component—columns (supplier): For this component,

$$F_{\text{calculated}} [0.16] < F_{\alpha=0.05 \text{ and } df=(4,12)} [3.26]$$

Hence, the null hypothesis, (H_0) should be accepted.

Inference: This means that there is no significant difference between suppliers in terms of hourly production volume.

4.6 DUNCAN'S MULTIPLE RANGE TEST

If the null hypothesis of any component in any of the models discussed earlier in this chapter is rejected (i.e. there is significant difference between the treatment means of the factor with respect to that component in terms of the response variable), one can use Duncan's multiple range test to compare the means of the treatments of that component in that model. This test was developed by Duncan (1955).

The steps of this test are:

Step-1: Arrange the treatment averages (' a ' number of averages) in the ascending order from left to right.

Step-2: Find the standard error of each treatment mean using the following formula.

$$SE_{\bar{y}_j} = \sqrt{\frac{MSS_{\text{error}}}{n}}$$

where n is the number of replications in each treatment of the factor.

Step-3: From Duncan's table of significant ranges, obtain $a-1$ ranges $[r_\alpha(j, df)]$ for $j = 2, 3, 4, \dots, a$ and for given α and df , where α is the level of significance and df is the degrees of freedom of the error term.

The $a-1$ ranges are:

$$r_\alpha(2, df), r_\alpha(3, df), r_\alpha(4, df), \dots, r_\alpha(a, df)$$

Step-4: Obtain $a-1$ least significant ranges by multiplying the $a-1$ significant ranges by the standard error as:

$$r_\alpha(j, df) \times SE_{\bar{y}_j} \quad j = 2, 3, 4, \dots, a$$

Step-5: Place the treatment means in the ascending order on a horizontal line. Compare the difference between the largest treatment mean and the smallest treatment mean with the range $r_\alpha(a, df)$. Then, compare the difference between the largest treatment mean and the second smallest treatment mean with the range $r_\alpha(a-1, df)$. Like this, the comparison should be carried out until all the smaller treatment means are compared with the largest treatment mean.

Compare the difference between the second largest treatment mean and the smallest treatment mean with the range $r_\alpha(a-1, df)$. Then, compare the difference between the second largest treatment mean and the second smallest treatment mean with the range $r_\alpha(a-2, df)$. Like this, the comparison should be carried out until all the smaller treatment means are compared with the second largest treatment mean.

The process of comparison should be repeated until all possible $a(a-1)/2$ pairs of means are checked.

Guidelines for inferences For a particular pair of means, if the difference between them is less than or equal to the corresponding least significant range (LSR), then there is no significant difference between those two treatment means; otherwise, there is significant difference between those two treatment means.

Example 4.7 Consider Example 4.2 again for a quick reference.

- Perform ANOVA to check whether there is significant difference between the training organizations in terms improving the decision-making skills of the employees by assuming a significance level of 0.05.
- If there is significant difference between the treatment means of the factor, Training Organization, then perform Duncan's Multiple Range Test and comment on the results.

Solution. (a) The ANOVA results has already been discussed earlier (refer to Table 4.10).

(b) Since the null hypothesis is rejected, for further interpretation on the difference between different treatment means, Duncan's multiple range test is performed as follows:

Step-1: The ascending order of treatment means are:

$\bar{Y}_3$	$\bar{Y}_2$	$\bar{Y}_1$	$\bar{Y}_4$
53.6	72.0	84.2	87.8

Step-2: The standard error of each treatment mean is

$$SE_{\bar{Y}_j} = \sqrt{\frac{MSS_{\text{error}}}{n}} = \sqrt{\frac{112.43}{5}} = 4.74$$

Step-3: Obtain 3 (i.e. $a - 1$) significant ranges from the Duncan's table of multiple ranges for $j = 4$ and $df = 16$ and with $\alpha = 0.05$ as shown in Table 4.28.

Table 4.28 Significant Ranges

j	Significant range
2	3.00
3	3.15
4	3.23

Step-4: Find the least significant ranges (LSR) by multiplying the significant ranges with the standard error as shown in Table 4.29.

Table 4.29 Least Significant Ranges (LSR)

j	LSR
2	$3.00 \times 4.74 = 14.22$
3	$3.15 \times 4.74 = 14.93$
4	$3.23 \times 4.74 = 15.31$

Step-5: Place the treatment means in the ascending order on the horizontal line, as shown in Fig. 4.4.

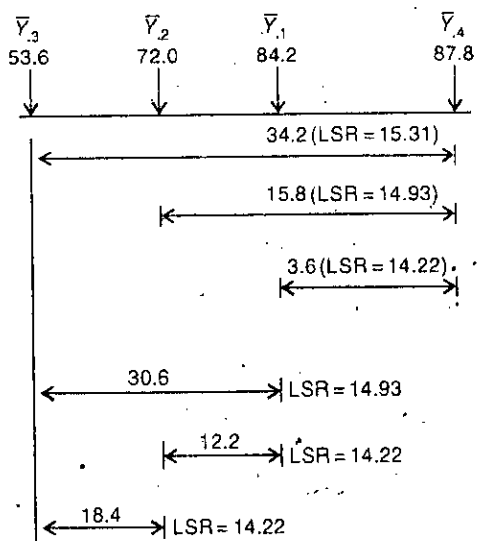


Fig. 4.4 Difference between treatment means and corresponding LSR values.

Table 4.30 Summary of Comparison of Duncan's Multiple Range Test

Treatment pair	Actual difference between means	Least significant range	Remark
3 and 4	34.2	15.31	Significant
2 and 4	15.8	14.93	Significant
1 and 4	3.6	14.22	Not significant
3 and 1	30.6	14.93	Significant
2 and 1	12.2	14.22	Not significant
3 and 2	18.4	14.22	Significant

The comparison of differences between different pairs of treatment means with the corresponding least significant ranges are summarized in Table 4.30. From Table 4.30, it is clear that the actual difference between the means of the treatment pairs (3,4), (2,4), (3,1) and (3,2) are significant. For the remaining pairs of treatments [(1,4) and (2,1)], the actual differences between their means are not significant.

4.7 FACTORIAL DESIGN

If the number of factors is more than one, then there is a need for a generalized design of experiment. Such design is called as *factorial* design. Analysis of many of the real-life situations will require this type of design.

The concept of factorial design is demonstrated with a problem involving two factors as in Example 4.8.

Example 4.8 The sales manager of Krishna Engineering Company wants to study the effect of sales regions as well as salesmen on quarterly sales revenue. So, he has designed a factorial experiment involving two factors as shown in Table 4.31, with two replications for each experimental combination:

Table 4.31 Quarterly Sales of Krishna Engineering Company

		Salesman (B) (Rupees in lakhs)			
		1	2	3	4
Sales region (A)	East	65	30	50	20
		70	35	65	40
	West	30	60	20	50
		35	55	25	40

As per this design each salesman is assigned to each region for two different quarters. Perform ANOVA and check the following at the significance level of 0.05.

- Whether there is a significant difference in terms of sales revenue between sales regions
- Whether there is a significant difference in terms of sales revenue between salesmen.
- Whether there is a significant difference in terms of sales revenue between different combinations of sales region and salesman.

Solution. The data in Table 4.31 are presented in Table 4.32 along with row totals and column totals.

Table 4.32 Quarterly Sales of Krishna Engineering Company

		Salesman (B) (Rupees in lakhs)				
		B_1	B_2	B_3	B_4	
Sales region (A)	A_1	65	30	50	20	375
		70	35	65	40	
	A_2	30	60	20	50	315
		35	55	25	40	
	$Y_{..}$	200	180	160	150	690

This problem has two factors, namely sales region (A) and salesman (B). The number of treatments for the factor A is 2 (A_1 and A_2) and that of the factor B is 4 (B_1 , B_2 , B_3 and B_4). The number of replications for each of the eight experimental combinations (A_1B_1 , A_1B_2 , A_1B_3 , A_1B_4 , A_2B_1 , A_2B_2 , A_2B_3 and A_2B_4) is 2.

The generalized model of the factorial design is as follows:

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}$$

where a is the number of treatments of the factor A ; b , the number of treatments of the factor B ; n , the number of replications under each of the experimental combinations of the factor A and the factor B ; μ , the overall mean; Y_{ijk} the k th replication under the i th treatment of the factor A and the j th treatment of the factor B ; A_i , the effect of the i th treatment of the factor A ; B_j , is the effect of the j th treatment of the factor B . AB_{ij} is the effect of the i th treatment of the factor A and the j th treatment of the factor B (interaction effect); and e_{ijk} , the random error associated with the k th replication under the i th treatment of the factor A and the j th treatment of the factor B .

Hypothesis with respect to treatment of the factor A:

Null hypothesis, $H_0: A_1 = A_2 = A_3 = \dots = A_a$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of the treatment means of the factor A .

Hypothesis with respect to treatment of the factor B:

Null hypothesis, $H_0: B_1 = B_2 = B_3 = \dots = B_b$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of the treatment means of the factor B .

Hypothesis with respect to interaction component, AB:

Null hypothesis, $H_0: A_1B_1 = A_1B_2 = A_1B_3 = \dots = A_2B_3 = A_2B_4$

Alternate hypothesis, H_1 : Interaction means are not equal for at least one pair of interaction means.

The relationship between different sum of squares of this model is:

$$\begin{aligned} \text{Total sum of squares} &= (\text{Sum of squares of the factor } A) \\ &+ (\text{Sum of squares of the factor } B) \\ &+ (\text{Sum of squares of the interaction } AB) \\ &+ (\text{Sum of squares of errors}) \end{aligned}$$

That is,

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_{\text{error}}$$

For the example problem;

$$SS_{\text{total}} = SS_{\text{region}} + SS_{\text{salesman}} + SS_{\text{region} \times \text{salesman}} + SS_{\text{error}}$$

The generalized shortcut formulas to compute the sum of squares of different components of the model are as follows:

$$SS_{\text{total}} = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n Y_{ijk}^2 - \frac{Y_{\dots}^2}{N}$$

$$SS_{\text{rows}} = \sum_{i=1}^a \frac{Y_{i..}^2}{bn} - \frac{Y_{...}^2}{N}$$

$$SS_{\text{columns}} = \sum_{j=1}^b \frac{Y_{.j.}^2}{an} - \frac{Y_{...}^2}{N}$$

$$SS_{\text{subtotals}} = \sum_{i=1}^a \sum_{j=1}^b \frac{Y_{ij.}^2}{n} - \frac{Y_{...}^2}{N}$$

$$SS_{\text{interaction}} = SS_{\text{subtotals}} - SS_{\text{rows}} - SS_{\text{columns}}$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{rows}} - SS_{\text{columns}} - SS_{\text{interaction}}$$

where $Y_{...}$ is the sum of Y_{ij} overall values of i, j and k ; $Y_{.j.}$, the sum of Y_{ijk} overall values of i and k for a given j ; $Y_{i..}$, the sum of Y_{ijk} overall values of j and k for a given i ; $Y_{..k}$, the sum of Y_{ijk} overall values of i and j for a given value of k ; and N is the total number of observations in the experiments ($abn = 2 \times 4 \times 2 = 16$).

For the example under consideration, we have

$$Y_{...} = \sum_{i=1}^2 \sum_{j=1}^4 \sum_{k=1}^2 Y_{ijk} = 690$$

$$Y_{1..} = 375, Y_{2..} = 315$$

$$Y_{..1} = 200, Y_{..2} = 180, Y_{.3.} = 160 \text{ and } Y_{.4.} = 150$$

Then,

$$SS_{\text{total}} = \sum_{i=1}^2 \sum_{j=1}^4 \sum_{k=1}^2 Y_{ijk}^2 - \frac{Y_{...}^2}{16}$$

$$= (65)^2 + (30)^2 + (50)^2 + (20)^2 + (70)^2 + (35)^2 + \dots + (25)^2 + (40)^2 - \frac{(690)^2}{16}$$

$$= 4093.75$$

$$SS_A = \sum_{i=1}^2 \frac{Y_{i..}^2}{8} - \frac{Y_{...}^2}{16}$$

$$= \frac{(375)^2 + (315)^2}{8} - \frac{(690)^2}{16}$$

$$= 225$$

$$SS_B = \sum_{j=1}^4 \frac{Y_{.j}^2}{4} - \frac{Y_{...}^2}{16}$$

$$= \frac{(200)^2 + (180)^2 + (160)^2 + (150)^2}{4} - \frac{(690)^2}{16}$$

$$= 368.75$$

Table 4.32 is rearranged as in Table 4.33 to compute the sum of squares of subtotals.

Table 4.33 Data to Compute Sum of Squares of Subtotals

A_1B_1	A_1B_2	A_1B_3	A_1B_4	A_2B_1	A_2B_2	A_2B_3	A_2B_4
65	30	50	20	30	60	20	50
70	35	65	40	35	55	25	40
$Y_{.1}$ 135	65	115	60	65	115	45	90

Then, we get

$$SS_{\text{subtotals}} = \sum_{i=1}^2 \sum_{j=1}^4 \frac{Y_{ij}^2}{2} - \frac{Y_{...}^2}{16}$$

$$= \frac{(135)^2 + (65)^2 + (115)^2 + (60)^2 + (65)^2 + (115)^2 + (45)^2 + (90)^2}{2} - \frac{(690)^2}{16}$$

$$= 3668.75$$

$$SS_{AB} = SS_{\text{subtotals}} - SS_A - SS_B$$

$$= 3668.75 - 225 - 368.75$$

$$= 3075$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_A - SS_B - SS_{AB}$$

$$= 4093.75 - 225 - 368.75 - 3075$$

$$= 425$$

The generalized results and results applied to the example problem of this design are summarized in Table 4.34 and Table 4.35, respectively.

Table 4.34 Generalized Results in Factorial Design

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
Between rows	$a - 1$	SS_{rows}	$\frac{SS_{\text{rows}}}{a - 1}$	$\frac{MSS_{\text{rows}}}{MSS_{\text{error}}}$
Between columns	$b - 1$	SS_{columns}	$\frac{SS_{\text{columns}}}{b - 1}$	$\frac{MSS_{\text{columns}}}{MSS_{\text{error}}}$
Between Rows $\times$ columns	$(a - 1)(b - 1)$	$SS_{\text{rows} \times \text{columns}}$	$\frac{SS_{\text{rows} \times \text{columns}}}{(a - 1)(b - 1)}$	$\frac{MSS_{\text{rows} \times \text{columns}}}{MSS_{\text{error}}}$
Error	$ab(n - 1)$	SS_{error}	$\frac{SS_{\text{error}}}{ab(n - 1)}$	
Total	$abn - 1$	SS_{total}		

Table 4.35 Results in Factorial Design (Example 4.8)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
SS_A	1	225.00	$225.00/1 = 225.00$	$225.00/53.13 = 4.24$
SS_B	3	368.75	$368.75/3 = 122.92$	$122.92/53.13 = 2.31$
SS_{AB}	3	3075.00	$3075/3 = 1025.00$	$1025.00/53.13 = 19.29$
SS_{error}	8	425.00	$425/8 = 53.13$	
Total	15	4093.75		

In Table 4.35, the value of the calculated F ratio for the factor A is 4.24, whereas its table value with the significance level of 0.05 and degrees of freedom (1, 8) is 5.32. The calculated f ratio for the factor B is 2.31, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 8) is 4.07. The calculated f ratio for the interaction AB is 19.29, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 8) is 4.07.

Component—factor A : For this component,

$$F_{\text{calculated}} [4.24] < F_{\alpha=0.05 \text{ and } df=(1,8)} [5.32]$$

Hence, the null hypothesis (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different sales regions.

Component—factor B : For this component,

$$F_{\text{calculated}} [2.31] < F_{\alpha=0.05 \text{ and } df=(3,8)} [4.07]$$

Hence, the null hypothesis (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different salesmen.

Component—interaction, AXB: For this component,

$$F_{\text{calculated}} [19.29] > F_{\alpha=0.05 \text{ and } df=(3,8)} [4.07]$$

Hence, the null hypothesis (H_0) should be rejected.

Inference: This means that there is a significant difference in quarterly sales because of the interaction between the factor 'sales region' and the factor 'salesman'. Alternatively, this means that the interaction between 'sales region' and 'salesman' has significant effect on quarterly sales.

Example 4.9 The quality control department of a fabric finishing plant is studying the effect of several factors on the dyeing of cotton-synthetic cloth used to manufacture men's shirts. Two operators, three cycle times and two temperatures were selected and two small specimens of the cloth were dyed under each combination of experimental conditions. The finished cloth was compared to a standard and a numerical score was assigned. The results are shown in Table 4.36. Analyze the data and draw conclusions at the significance level of 0.05.

Table 4.36 Numerical Scores of Specimens of Cloth

		Temperature			
		250°		300°	
		Operator-1	Operator-2	Operator-1	Operator-2
40		23	27	24	38
		25	26	28	35
50		36	34	37	34
		36	39	35	36
60		28	35	26	36
		27	34	25	34

Solution. The data in Table 4.36 is presented in Table 4.37 along with row totals and column totals. This problem has three factors: cycle time (A), temperature (B) and operator (C).

Let a be the number of treatments for the factor A ($a=3$); b , the number of treatments for the factor B ($b=2$); c be the number of treatments for the factor C ($c=2$); and n be the number of replications under each experimental combination ($n=2$).

Table 4.37 Numerical Scores of Specimens of Cloth

		Temperature (B)				
		250°		300°		
		Operator (C)		Operator (C)		
		1	2	1	2	$\bar{Y}_{i...}$
Cycle time (A)	40	23	27	24	38	226
		25	26	28	35	
	50	36	34	37	34	287
		36	39	35	36	
	60	28	35	26	36	245
		27	34	25	34	
		370		388		
$Y_{.j}$						
$Y_{.k}$		350		408		

The generalized model of the factorial design with three factors is as follows:

$$Y_{ijkl} = \mu + A_i + B_j + AB_{ij} + C_k + AC_{ik} + BC_{jk} + ABC_{ijk} + e_{ijkl}$$

where μ is the overall mean; Y_{ijkl} , the l th replication for the i th treatment of the factor A , j th treatment of the factor B and k th treatment of the factor C ; A_i , the effect of the i th treatment of the Factor A ; B_j , the effect of the j th treatment of the factor B ; AB_{ij} , the effect of the i th treatment of the factor A and the j th treatment of the factor B (two-way interaction effect); C_k , the effect of the k th treatment of the factor C ; AC_{ik} , the effect of the i th treatment of the factor A and the k th treatment of the factor C (two-way interaction); BC_{jk} , the effect of the j th treatment of the factor B and the k th treatment of the factor C (two-way interaction); ABC_{ijk} , the effect of the i th treatment of the factor A , j th treatment of the factor B and the k th treatment of the factor C (three-way interaction); and e_{ijkl} , the random error associated with the l th replication under the i th treatment of the factor A , j th treatment of the factor B and k th treatment of the factor C .

Hypotheses: As in the previous example, one can define H_0 and H_1 for each of the seven components (A_i , B_j , AB_{ij} , C_k , AC_{ik} , BC_{jk} , ABC_{ijk}) in the model.

The relationship between different sum of squares of this model is:

Total sum of squares = (Sum of squares of the factor A)
 + (Sum of squares of the factor B)
 + (Sum of squares of the interaction AB).
 + (Sum of squares of the factor C)
 + (Sum of squares of the interaction AC)
 + (Sum of squares of the interaction BC)
 + (Sum of squares of the interaction ABC)
 + (Sum of squares of errors)

That is,

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_C + SS_{AC} + SS_{BC} + SS_{ABC} + SS_{\text{error}}$$

The generalized shortcut formulas to compute the sum of squares of different components of the model are as follows:

$$\bar{SS}_{\text{total}} = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c \sum_{l=1}^n Y_{ijkl}^2 - \frac{Y_{....}^2}{N}$$

$$SS_A = \sum_{i=1}^a \frac{Y_{i...}^2}{bcn} - \frac{Y_{....}^2}{N}$$

$$SS_B = \sum_{j=1}^b \frac{Y_{.j..}^2}{acn} - \frac{Y_{....}^2}{N}$$

$$SS_{AB(\text{subtotals})} = \sum_{i=1}^a \sum_{j=1}^b \frac{Y_{ij..}^2}{cn} - \frac{Y_{....}^2}{N}$$

$$SS_{AB} = SS_{AB(\text{subtotals})} - SS_A - SS_B$$

$$SS_C = \sum_{k=1}^c \frac{Y_{...k}^2}{abn} - \frac{Y_{....}^2}{N}$$

$$SS_{AC(\text{subtotals})} = \sum_{i=1}^a \sum_{k=1}^c \frac{Y_{i.k.}^2}{bn} - \frac{Y_{....}^2}{N}$$

$$SS_{AC} = SS_{AC(\text{subtotals})} - SS_A - SS_C$$

$$SS_{BC(\text{subtotals})} = \sum_{j=1}^b \sum_{k=1}^c \frac{Y_{.jk.}^2}{an} - \frac{Y_{....}^2}{N}$$

$$SS_{BC} = SS_{BC \text{ (subtotals)}} - SS_B - SS_C$$

$$SS_{ABC \text{ (subtotals)}} = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^c \frac{Y_{ijk}^2}{an} - \frac{Y_{....}^2}{N}$$

$$SS_{ABC} = SS_{ABC \text{ (subtotals)}} - SS_A - SS_B - SS_{AB} - SS_C - SS_{AC} - SS_{BC}$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_A - SS_B - SS_{AB} - SS_C - SS_{AC} - SS_{BC} - SS_{ABC}$$

where N is the total number of observations in the experiments ($abcn = 3 \times 2 \times 2 \times 2 = 24$).

For the example under consideration, we have

$$Y_{....} = \sum_{i=1}^3 \sum_{j=1}^2 \sum_{k=1}^2 \sum_{l=1}^2 Y_{ijkl} = 758$$

$$Y_{1...} = 226, Y_{2...} = 287, Y_{3...} = 245$$

$$Y_{.1..} = 370, Y_{.2..} = 388$$

$$Y_{..1.} = 350, Y_{..2.} = 408$$

Therefore,

$$\begin{aligned} SS_{\text{total}} &= \sum_{i=1}^3 \sum_{j=1}^2 \sum_{k=1}^2 \sum_{l=1}^2 Y_{ijkl}^2 - \frac{Y_{....}^2}{24} \\ &= (23)^2 + (27)^2 + (24)^2 + (38)^2 + (25)^2 + (26)^2 + \dots + (25)^2 + (34)^2 - \frac{(758)^2}{24} \\ &= 609.834 \end{aligned}$$

$$\begin{aligned} SS_A &= \sum_{i=1}^3 \frac{Y_{i...}^2}{8} - \frac{Y_{....}^2}{24} \\ &= \frac{(226)^2 + (287)^2 + (245)^2}{8} - \frac{(758)^2}{24} \\ &= 243.584 \end{aligned}$$

$$\begin{aligned} SS_B &= \sum_{j=1}^2 \frac{Y_{.j..}^2}{12} - \frac{Y_{....}^2}{24} \\ &= \frac{(370)^2 + (388)^2}{12} - \frac{(758)^2}{24} \\ &= 13.5 \end{aligned}$$

$$\begin{aligned}
 SS_C &= \sum_{k=1}^2 \frac{Y_{..k}^2}{12} - \frac{Y_{...}^2}{24} \\
 &= \frac{(350)^2 + (408)^2}{12} - \frac{(758)^2}{24} \\
 &= 140.166
 \end{aligned}$$

Computation of SS_{AB} : To compute the sum of squares of the interaction AB , Table 4.37 is rearranged, as shown in Table 4.38.

Table 4.38 Rearranged Data to Compute Sum of Squares of AB

		Temperature (B)					
		250°			300°		
Cycle time (A)	40	23	101	27	24	125	38
		25		26	28		35
	50	36	145	34	37	142	34
		36		39	35		36
	60	28	124	35	26	121	36
		27		34	25		34

Note: Subtotals of AB are given in the respective cells.

We also have,

$$\begin{aligned}
 SS_{AB(\text{subtotals})} &= \sum_{i=1}^3 \sum_{j=1}^2 \frac{Y_{ij..}^2}{4} - \frac{Y_{...}^2}{24} \\
 &= \frac{(101)^2 + (125)^2 + (145)^2 + (142)^2 + (124)^2 + (121)^2}{4} - \frac{(758)^2}{24} \\
 &= 317.834 \\
 SS_{AB} &= SS_{AB \text{ subtotals}} - SS_A - SS_B \\
 &= 317.834 - 243.584 - 13.5 \\
 &= 60.75
 \end{aligned}$$

Computation of SS_{AC} : To compute the sum of squares of the interaction AC , Table 4.37 is rearranged, as shown in Table 4.39.

Table 4.39 Rearranged Data to Compute Sum of Squares of AC

		Operator (C)			
		1		2	
Cycle time (A)	40	23	100	27	126
		25		26	38
	50	36	144	34	143
		36		39	34
	60	28	106	35	139
		27		34	36

Note: Subtotals of AC are given in the respective cells.

We have,

$$\begin{aligned}
 SS_{AC(\text{subtotals})} &= \sum_{i=1}^3 \sum_{k=1}^2 \frac{Y_{i.k}^2}{4} - \frac{\bar{Y}^2}{24} \\
 &= \frac{(100)^2 + (126)^2 + (144)^2 + (143)^2 + (106)^2 + (139)^2}{4} - \frac{(758)^2}{24} \\
 &= 464.334 \\
 SS_{AC} &= SS_{AC(\text{subtotals})} - SS_A - SS_C \\
 &= 464.334 - 243.584 - 140.166 \\
 &= 80.584
 \end{aligned}$$

Computation of SS_{BC} : To compute the sum of squares of the interaction BC, Table 4.37 is rearranged, as shown in Table 4.40.

Table 4.40 Rearranged Data to Compute Sum of Squares of BC

Temperature (B)			
250°		300 °	
Operator (C)		Operator (C)	
1	2	1	2
23	27	24	38
25	26	28	35
36	34	37	34
36	39	35	36
28	35	26	36
27	34	25	34
175	195	175	213

Then

$$SS_{BC(\text{subtotals})} = \sum_{j=1}^2 \sum_{k=1}^2 \frac{Y_{jk}^2}{6} - \frac{Y_{...}^2}{24}$$

$$= \frac{(175)^2 + (195)^2 + (175)^2 + (213)^2}{6} - \frac{(758)^2}{24}$$

$$= 167.166$$

$$SS_{BC} = SS_{BC(\text{subtotals})} - SS_B - SS_C$$

$$= 167.166 - 13.5 - 140.166$$

$$= 13.5$$

Computation of SS_{ABC} : To compute the sum of squares of the interaction ABC , Table 4.37 is rearranged, as shown in Table 4.41.

Table 4.41 Rearranged Data to Compute Sum of Squares of ABC

		Temperature			
		250°		300°	
		Operator - 1	Operator - 2	Operator - 1	Operator - 2
Cycle time (A)	40	23 25	27 26	24 28	38 35
		48	53	52	73
	50	36 36	34 39	37 35	34 36
		72	73	72	70
	60	28 27	35 34	26 25	36 34
		55	69	51	70

$$SS_{ABC(\text{subtotals})} = \sum_{i=1}^3 \sum_{j=1}^2 \sum_{k=1}^2 \frac{y_{ijk}^2}{2} - \frac{Y_{...}^2}{24}$$

$$= \frac{(48)^2 + (53)^2 + (52)^2 + (73)^2 + (72)^2 + (73)^2 + \dots + (51)^2 + (70)^2}{2} - \frac{(758)^2}{24}$$

$$= 574.834$$

$$SS_{ABC} = SS_{ABC(\text{subtotals})} - SS_A - SS_B - SS_C - SS_{AB} - SS_{AC} - SS_{BC}$$

$$= 574.834 - 243.584 - 13.5 - 140.166 - 60.75 - 80.584 - 13.5$$

$$= 22.75$$

$$\begin{aligned}
 SS_{\text{error}} &= SS_{\text{total}} - (SS_A + SS_B + SS_{AB} + SS_C + SS_{AC} + SS_{BC} + SS_{ABC}) \\
 &= 609.834 - (243.584 + 13.5 + 60.75 + 140.166 + 80.584 + 13.5 + 22.75) \\
 &= 35
 \end{aligned}$$

The results of ANOVA of this example are summarized in Table 4.42. From this table, it is clear that the components A_i , AB_{ij} , C_k , AC_{ik} and ABC_{ijk} are significant. This means that these components have significant effect on the scores. The remaining components (B_j and BC_{jk}) are not significant.

Table 4.42 Results in Factorial Design (Example 4.9)

Source of Variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio (calculated)	F ratio (at $\alpha = 0.05$)	Comment
SS_A	2	243.584	121.79	41.71	3.89	Significant
SS_B	1	13.500	13.50	4.62	4.75	Not significant
SS_{AB}	2	60.750	30.38	10.40	3.89	Significant
SS_C	1	140.166	140.17	48.00	4.75	Significant
SS_{AC}	2	80.584	40.29	13.80	3.89	Significant
SS_{BC}	1	13.500	13.50	4.62	4.75	Not significant
SS_{ABC}	2	22.750	11.38	3.90	3.89	Significant
SS_{error}	12	35.000	2.92			
Total	23	609.834				

4.8 2ⁿ FACTORIAL EXPERIMENT

If the number of levels, in a factorial experiment, is two for each of the factors in it, then the corresponding design is called 2ⁿ design, where n is the number of factors in the experiment. Yates has developed a generalized algorithm to find the sum of squares of the components of the model of a given 2ⁿ factorial experiment, which will be explained in this section. Before introducing Yates' algorithm, the concept of 2² and 2³ designs are presented in the following sections:

4.8.1 2² Factorial Experiment

In a 2² factorial experiment, the number of factors is 2. Suppose they are A and B . Each factor has only two levels. Let the level 1 be 'low level' and the level 2 be 'high level'. These are shown in Fig. 4.5. The low level and high level of each factor are represented by 0 and 1, respectively.

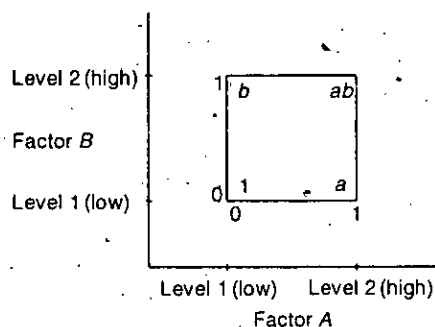


Fig. 4.5 Graphical view of 2^2 design.

The model of this two-factor factorial experiment is:

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}$$

where μ is the overall mean; Y_{ijk} , the k th replication under the i th treatment of factor A and j th treatment of factor B ; A_i , the effect of the i th treatment of the factor A ; B_j , the effect of the j th treatment of the factor B ; AB_{ij} , the effect of the i th treatment of the factor A and the j th treatment of the factor B (interaction effect); e_{ijk} , the random error associated with the k th replication under the i th treatment of the factor A and the j th treatment of the factor B ; and k , the number of replications under each of the experimental combinations of the factor A and the factor B .

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_{\text{error}}$$

Effect of a factor: The average effect of a factor is the average of changes of the responses due to change in the levels of that factor under the levels of all other factors.

The letter combinations 1, a , b and ab represent the totals of replications in the respective treatments (levels) combinations.

At the low level of B , the average change in the response if the level of the factor A is changed from 0 to 1 is $(a - 1)/n$. Similarly, at the high level of B , the average change in the response if the level of the factor A is changed from 0 to 1 is $(ab - b)/n$.

The average effect of the factor A is the average of above two averages as given by the following formula:

$$A = \frac{1}{2n} [(a - 1) + (ab - b)] = \frac{1}{2n} (-1 + a - b + ab)$$

At the low level of A , the average change in the response if the level of the factor B is changed from 0 to 1 is $(b - 1)/n$. Similarly, at the high level of A , the average change in the response if the level of the factor B is changed from 0 to 1 is $(ab - a)/n$.

The average effect of the factor B is the average of the above two averages as given by the following formula:

$$B = \frac{1}{2n} [(b - 1) + (ab - a)] = \frac{1}{2n} (-1 - a + b + ab)$$

The effect of the interaction term AB is the average difference between the effect of A at high level of B and at a low level of B , which is as:

$$AB = \frac{1}{2n} [(ab - b) - (a - 1)] = \frac{1}{2n} (+1 - a - b + ab)$$

The coefficients of different terms of the average effects of the factors A and B , and the interaction term AB are shown in Table 4.43.

Table 4.43 Coefficients of Terms of Effects of Components of 2² Model

Model component	Effect of		
	A	B	AB
1	-1	-1	+
A	+1	-1	-
B	-1	+1	-
AB	+1	+1	+

The sum of squares of each term of the model can be obtained by using the following formula:

$$\text{Sum of squares} = \frac{(\text{Contrast})^2}{2^2 k} = \frac{(\text{Contrast})^2}{4k}$$

where the contrasts of the terms are:

$$A = -1 + a - b + ab$$

$$B = -1 - a + b + ab$$

$$AB = +1 - a - b + ab$$

and k , the number of replications in each combination of treatments.

Example 4.10 The surface finish of products produced in a machine shop is suspected to be affected by a factor 'operator' and another factor 'machine'. The data of this experiment with two replications in different treatment combinations are summarized in Table 4.44. Perform ANOVA and check the significance of the components of the related model when α is 0.05.

Table 4.44 Data (Example 4.10)

		Machine (B)	
		1	2
Operator (A)	1	65 70	20 40
	2	30 35	50 40

Solution. Let, the factor, 'operator' be A and the another factor 'machine' be B . Here, all the factors have been assumed as fixed factors. Number of levels for each of the factors, $n = 2$. Number of replications in each treatment combination, $k = 2$. Significance level, $\alpha = 0.05$.

The model of this problem is as shown below:

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}$$

The definitions of the components of this model are as already explained in this section. The data in Table 4.44 are reproduced in Table 4.45, with necessary notations and the totals of different treatment combinations (1, a , b , ab).

Table 4.45 Data with Necessary Information.

		Machine (B)	
		1	2
Operator (A)	1	65 70 1 135	20 40 b 60
	2	30 35 a 65	50 40 ab 90

Then,

$$\begin{aligned} \text{Sum of squares of } A &= \frac{(\text{Contrast } A)^2}{4k} = \frac{(-1 + a - b + ab)^2}{4 \times 2} \\ &= \frac{(-135 + 65 - 60 + 90)^2}{8} \\ &= 200 \end{aligned}$$

$$\begin{aligned} \text{Sum of squares of } B &= \frac{(\text{Contrast } B)^2}{4k} = \frac{(-1 - a + b + ab)^2}{4 \times 2} \\ &= \frac{(-135 - 65 + 60 + 90)^2}{8} \\ &= 312.5 \end{aligned}$$

$$\begin{aligned} \text{Sum of squares of } AB &= \frac{(\text{Contrast } AB)^2}{4k} \\ &= \frac{(1 - a - b + ab)^2}{4 \times 2} = \frac{(135 - 65 - 60 + 90)^2}{8} \\ &= 1250 \end{aligned}$$

$$\begin{aligned}
 \text{Total sum of squares} &= \sum \sum \sum Y_{ijk}^2 - \frac{Y_{...}^2}{k2^2} \\
 &= 17350 - \frac{(350)^2}{8} \\
 &= 2037.5
 \end{aligned}$$

$$\begin{aligned}
 \text{Sum of squares of error} &= SS_{\text{total}} - SS_A - SS_B - SS_{AB} \\
 &= 2037.5 - 200 - 312.5 - 1250 \\
 &= 275
 \end{aligned}$$

The ANOVA table for this problem is shown in Table 4.46.

Table 4.46 Results of ANOVA (Example 4.10)

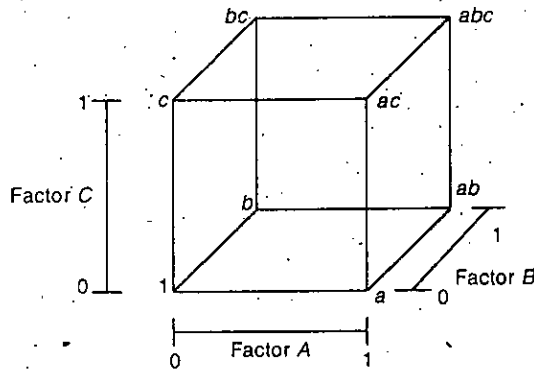
Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
SS _A	1	200.0	200/1 = 200	2.910
SS _B	1	312.5	312.5/1 = 312.5	4.676
SS _{AB}	1	1250.0	1250/1 = 1250	18.182
SS _{error}	4	275.0	275/4 = 68.75	
Total	7	2037.5		

The F value for (1, 4) degrees of freedom at $\alpha = 0.05$ is 7.71. Based on this value, one can conclude that the factor A as well as factor B , which is not having effect on the response variable, (surface finish), but the interaction effect AB has a significant effect on the response variable.

So, the 'operator' as well as 'machine' is not having effect on the surface finish; but the interaction between 'operator and machine' has effect on the surface finish.

4.8.2 2³ Design

In 2³ factorial design, the number of factors is 3. Suppose they are A , B and C . Each factor has only two levels. Let the level 1 be 'low level' and the level 2 be 'high Level'. These are shown in Fig. 4.6. The low level and high level of each factor are represented by 0 and 1, respectively.

Fig. 4.6 Graphical view of 2^3 design.

The model of this three-factor factorial experiment is:

$$Y_{ijkl} = \mu + A_i + B_j + AB_{ij} + C_k + AC_{ik} + BC_{jk} + ABC_{ijk} + e_{ijkl}$$

where μ the overall mean; Y_{ijkl} , the l th replication under the i th treatment of the factor A , j th treatment of the factor B and k th treatment of the factor C ; A_i the effect of the i th treatment of the factor A ; B_j the effect of the j th treatment of the factor B . AB_{ij} the effect of the i th treatment of the factor A and the j th treatment of the factor B (two-way Interaction effect); C_k , the effect of the k th treatment of the factor C ; AC_{ik} , the effect of the i th treatment of the factor A and the k th treatment of the factor C (two-way interaction); BC_{jk} , the effect of the j th treatment of the factor B and the k th treatment of the factor C (two-way interaction); ABC_{ijk} , the effect of the i th treatment of the factor A , j th treatment of the factor B and the k th treatment of the factor C (three-way interaction); e_{ijkl} , the random error associated with the l th replication under the i th treatment of the factor A , j th treatment of the factor B and k th treatment of the factor C .

The relationship between different sum of squares of this model is:

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_C + SS_{AC} + SS_{BC} + SS_{ABC} + SS_{\text{error}}$$

The letter combinations 1, a , b , ab , c , ac , bc and abc represent the totals of replications in the respective treatments (levels) combinations.

Based on the discussions of 2^2 design, the average effects of each of the components of the 2^3 factorial experiment are:

$$A = \frac{1}{4n} (-1 + a - b + ab - c + ac - bc + abc)$$

$$B = \frac{1}{4n} (-1 - a + b + ab - c - ac + bc + abc)$$

$$AB = \frac{1}{4n} (1 - a - b + ab + c - ac - bc + abc)$$

$$C = \frac{1}{4n} (-1 - a - b - ab + c + ac + bc + abc)$$

$$AC = \frac{1}{4n} (1 - a + b - ab - c + ac - bc + abc)$$

$$BC = \frac{1}{4n} (1 + a - b - ab - c - ac + bc + abc)$$

$$ABC = \frac{1}{4n} (-1 + a + b - ab + c - ac - bc + abc)$$

The coefficients of the effects of the above components are summarized in Table 4.47.

Table 4.47 Coefficients of Terms of Effects of Components of 2³ Model

	A	B	AB	C	AC	BC	ABC
1	-	-	+	-	+	+	-
A	+	-	-	-	-	+	+
B	-	+	-	-	+	-	+
AB	+	+	+	-	-	-	-
C	-	-	+	+	-	-	+
AC	+	-	-	+	+	-	-
BC	-	+	-	+	-	+	-
ABC	+	+	+	+	+	+	+

The sum of squares of each term of the model can be obtained by using the following formula:

$$\text{Sum of square} = \frac{(\text{Contrast})^2}{2^3 \times 1} = \frac{(\text{Contrast})^2}{8 \times 1}$$

where the contrasts of the terms are:

$$A = (-1 + a - b + ab - c + ac - bc + abc)$$

$$B = (-1 - a + b + ab - c - ac + bc + abc)$$

$$AB = (1 - a - b + ab + c - ac - bc + abc)$$

$$C = (-1 - a - b - ab + c + ac + bc + abc)$$

$$AC = (1 - a + b - ab - c + ac - bc + abc)$$

$$BC = (1 + a - b - ab - c - ac + bc + abc)$$

$$ABC = (-1 + a + b - ab + c - ac - bc + abc)$$

and 1 is the number of replications in each combination of treatments.

4.8.3 Yates' Algorithm for 2^n Factorial Experiment

Yates' algorithm is a generalized algorithm which gives the sum of squares of different components of a model of 2^n factorial experiment, where n is the total number of factors and each factor has only two levels/treatments.

The steps of this algorithm are described as follows:

Step-1: Arrange the standard order of the model components, columnwise. Let it be column x .

The standard orders of a few designs are as:

2^2 design: 1, a , b , ab

2^3 design: 1, a , b , ab , c , ac , bc , abc

2^4 design: 1, a , b , ab , c , ac , bc , abc , d , ad , bd , abd , cd , acd , bcd , $abcd$

Step-2: Write the response totals of the corresponding model components in the next column. Let it be column y .

Step-3: Obtain the entries of column 1 using the following steps:

3.1: Obtain each of the first half of the entries from top in the column 1 by adding the consecutive pair of entries from the top of the column y .

3.2: Obtain each of the second-half of the entries from $[(2^n/2) + 1]$ position in the column 1 by adding the consecutive pair of entries from the top of the column y by changing the sign of the first entry in that pair.

Step-4: Obtain the entries of column 2 using the results of column 1 and by following the steps as followed for the column 1.

Step-5: Obtain the entries of column 3 using the results of column 2 and by following the steps as followed for the column 2.

Step-6: Obtain the entries of the remaining columns up to column n in the same manner, where n is the total number of factors.

Step-7: Find the sum of squares of each component of the model using the formula:

$$SS = \frac{(\text{Corresponding entry in column } n)^2}{k2^n}$$

where n is the total number of factors and k , the number of replications under each treatment combination of the factorial table.

Step-8: Find the total sum of squares in the usual way.

Step-9: The error sum of squares is obtained using the formula:

$$\begin{aligned} \text{Sum of squares of error} &= (\text{Total sum of squares}) \\ &\quad - (\text{Sum of the sum of squares} \\ &\quad \quad \text{of the model components}) \end{aligned}$$

Step-10: Perform ANOVA and draw conclusions.

Example 4.11 The surface finish of products produced in a machine shop is suspected to be affected by three factors: 'operator', 'machine' and 'depth of cut'. The data of this

experiment with two replications in different treatment combinations are summarized in Table 4.48. Perform ANOVA and check the significance of the components of the related model when α is 0.05.

Table 4.48 Data of Example 4.11

		Machine (B)			
		1		2	
		Depth of cut		Depth of cut	
		0.5 mm	1 mm	0.5 mm	1 mm
Operator (A)	1	65	60	20	60
		70	50	40	55
2		30	40	50	40
		35	55	45	60

Solution. Let the factor, 'operator' be A , the factor 'machine' be B and the factor 'depth of cut' be C . Here, all the factors are assumed as fixed factors.

Here number of levels for each of the factors, $n = 2$, number of replications $l = 2$; and the significance level, $\alpha = 0.05$. The model of this problem is

$$Y_{ijkl} = \mu + A_i + B_j + AB_{ij} + C_k + AC_{ik} + BC_{jk} + ABC_{ijk} + e_{ijkl}$$

The definitions of the components of this model are as explained earlier.

The total of the responses in each experimental combination is shown in Table 4.49 along with its treatment combination.

Table 4.49 Data of Example 4.11 with Additional Data

		Machine (B)			
		1		2	
		Depth of cut (c)		Depth of cut (c)	
		0.5 mm	1 mm	0.5 mm	1 mm
Operator (A)	1	65 1 135	60 c 110	20 b 60	60 bc 115
		70	50	40	55
2		30 a 65	40 ac 95	50 ab 95	40 abc 100
		35	55	45	60

The calculations of Yates' algorithm applied to this problem are summarized in Table 4.50.

Table 4.50 Calculations of Yates' Algorithm for Example 4.11

Component (x)	Response total (y)	Column			Sum of squares
		1	2	3	
1	135	200	355	775	—
a	65	155	420	-65	264.0625
b	60	205	-35	-35	76.5625
ab	95	215	-30	105	689.0625
c	110	-70	-45	65	264.0625
ac	95	35	10	5	1.5625
bc	115	-15	105	55	189.0625
abc	100	-15	0	-105	689.0625

The total sum of squares and error sum of squares are computed as follows:

$$\begin{aligned}
 SS_{\text{total}} &= \sum \sum \sum \sum Y_{ijkl}^2 - \frac{Y_{\dots}^2}{k^2} \\
 &= 40325 - \frac{775^2}{2 \times 2^3} \\
 &= 2785.938
 \end{aligned}$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_A - SS_B - SS_{AB} - SS_C - SS_{AC} - SS_{BC} - SS_{ABC} = 612.5$$

The results of ANOVA of this problem are summarized in Table 4.51.

Table 4.51 Results of ANOVA (Example 4.11)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
SS _A	1	264.0625	264.0625	3.449
SS _B	1	76.5625	76.5625	1.000
SS _{AB}	1	689.0625	689.0625	9.000
SS _C	1	264.0625	264.0625	3.449
SS _{AC}	1	1.5625	1.5625	0.020
SS _{BC}	1	189.0625	189.0625	2.469
SS _{ABC}	1	689.0625	689.0625	9.000
SS _{Error}	8	612.5000	76.5625	
Total	15	2785.9380		

The F value for (1, 8) degrees of freedom and at the significance level of 0.05 is 5.32. Based on this table value, the interaction component AB and ABC are having significant effect on the surface finish. All the factors (A , B and C) and other interaction terms AC and BC are not having significant effect on the surface finish.

So, the interaction terms 'operator $\times$ machine' and 'operator $\times$ machine $\times$ depth of cut' are having significant effect on the surface finish. All the factors, 'machine', 'operator' and 'depth of cut', and the other interaction terms, 'operator $\times$ depth of cut' and machine $\times$ depth of cut' are not having significant effect on the surface finish.

4.9 EXPECTED MEAN SQUARE (EMS) RULE

As stated in Section 4.1, in a given model, if all the factors are fixed factors, then the F statistic to test the effect of each and every component in it is formed by dividing the mean sum of squares of that component with the mean sum of squares of the error component.

If a model consists of a combination of fixed factors and random factors, or purely random factors, then the method of forming the F statistic to test the effect of each and every component of that model by dividing its mean sum of squares with the mean sum of squares of error is not correct. Under such situation, the F statistic to test the effect of each component should be formed differently using Expected Mean Square Rule.

If the factor A is a fixed factor, with ' a ' number of treatments, then its effect is the ratio between the sum of squares of the model component with respect to the factor A and the degrees of freedom of that factor.

$$\text{Effect of the factor } A = \frac{\sum_{i=1}^a A_i^2}{a-1}$$

where A_i is the effect of the i th treatment of the factor A .

If the factor A is a random factor with ' a ' number of treatments, then its variance component is σ_A^2 .

If the factor A is a fixed factor and the factor B is random factor, and these two factors are the factors of a model, then the model component AB is random. Under this situation, the variance component of AB is σ_{AB}^2 .

If the factors A and B are random factors, and these two factors are the factors of a model, then the model component AB is random. Under this situation, the variance component of AB is σ_{AB}^2 .

Replicates of any model is always considered to be random.

Rules (steps) for formation of expected mean squares

The steps for the formation of expected mean squares for different components of a model are discussed here. The steps are simultaneously demonstrated through a model with the factor A and factor B , which are fixed. Let the number of treatments of the factor A and

factor B be a and b , respectively. The number of replications in each experimental combination be n . The corresponding model is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + e_{(ij)k}$$

where Y_{ijk} is the k th replication under the i th treatment of the factor A and j th treatment of the factor B ; μ , the overall mean; A_i , the effect of the i th treatment of the factor A ; B_j , the effect of the j th treatment of the factor B ; $(AB)_{ij}$, the effect of the i th treatment of the factor A and j th treatment of the factor B ; and $e_{(ij)k}$, the random error associated with the k th replication under the i th treatment of the factor A and j th treatment of the factor B .

Note: In the error term, it is essential to place all the subscripts within parentheses, except the subscript with respect to the replication. In this model, the three subscripts (i , j and k) of the error term $e_{(ij)k}$ are classified into two groups, namely *live* subscript and *dead* subscripts. Here, the subscripts, i and j , which are enclosed within parentheses are called 'dead' subscripts. The subscript k , which is outside the parentheses is called 'live' subscript. The subscript(s) of all other components of the model are live subscript(s). Also, note that the interaction term AB is placed within parentheses $[(AB)_{ij}]$.

Now, the steps for formulation of expected mean squares follow:

Step-1: Prepare a table, as shown in Table 4.52, according to the guidelines given as follows:

- Place all the subscripts column-wise.
- Write the number of levels (treatments)/replications of the factor with respect to each of the subscripts just above it.
- Write the type of the factor (fixed/random) indicated by F or R just above the previous row with respect to each of the subscripts.
- Write the components of the model in the standard order row-wise one after another at the extreme left portion of the table.
- In the row corresponding to the error term, write 1 in each column if any one of the dead subscripts of the error term matches with that column subscript.

Table 4.52 Template to Compute Expected Mean Squares

Model component	F	F	R	(Factor type)
	a	b	n	(Levels)
	i	j	k	(Subscript)
A_i				
B_j				
$(AB)_{ij}$				
$e_{(ij)k}$	1	1		

Step-2: In each row, if any one of the live subscripts matches with the subscript of a column, do the following:

- If the column is headed by fixed factor, write 0 in the corresponding row-column combination.
- If the column is headed by random factor, write 1 in the corresponding row-column combination.

The corresponding incorporations of 0's and 1's are shown in Table 4.53.

Table 4.53 Modified Table with Respect to Step-2

Model component	F a i	F b j	R n k
A_i	0		
B_j		0	
$(AB)_{ij}$	0	0	
$\sigma_{(ijk)}$	1	1	1

Step-3: In each of the remaining positions of the rows, fill in the number of levels, as shown in the respective column headings. The modified table after performing this step is shown in Table 4.54.

Table 4.54 Modified Table with Respect to Step-3

Model component	F a i	F b j	R n k
A_i	0	b	n
B_j	a	0	n
$(AB)_{ij}$	0	0	n
$\sigma_{(ijk)}$	1	1	1

Step-4: The expected mean sum of squares (EMS) of a component of the model is computed using the following guidelines:

- Cover all the columns headed by live subscript(s) of the component under consideration.
- Identify the rows (components) that contain at least the set of subscripts of the component under consideration as their subscript(s).
- In each such identified row (component) and the row with respect to the component under consideration, find the product of the visible letters and then multiply that product with the appropriate effect (fixed effect or random effect) of that row.
- Find the sum of the results of such multiplications in the previous sub-step and treat it as the formula for the expected mean squares of the component under consideration.

The results of this step are summarized in Table 4.55.

Table 4.55 Expected Mean Squares Formulas

Model-component	F a i	F b j	R n k	Expected mean squares formula
A_i	0	b	n	$\sigma^2 + \left[bn \sum_{i=1}^a A_i^2 / (a-1) \right]$
B_j	a	0	n	$\sigma^2 + \left[an \sum_{j=1}^b B_j^2 / (b-1) \right]$
$(AB)_{ij}$	0	0	n	$\sigma^2 + \left[n \sum_{i=1}^a \sum_{j=1}^b AB_{ij}^2 / (a-1)(b-1) \right]$
$e_{(ijk)}$	1	1	1	σ^2

From Table 4.55, it is clear that the remainder of dividing expected mean squares formula of each component of the model by the expected mean squares formula of the error term (σ^2) is equal to the effect of that component.

Hence, if all the factors in a model are fixed factors, then the F ratio for each component in the model is obtained by dividing the mean sum of squares of that component by the mean sum of squares of the error component.

Note: For any, model which has mixed factors or only random factors, the F ratio of a component of consideration of that model should be formed by dividing expected mean squares formula of that component by a suitable expected mean squares formula of some other component in the model such that the remainder of the division is equal to the effect of the component under consideration.

Example 4.12 The investigator of an experiment has considered two factors, namely factor A and factor B . He has assumed that the factor A as a fixed factor and the factor B as a random factor. The number of levels for the factor A and the factor B are a and b , respectively. The number of replications under each experimental combination is n . Derive the expected mean squares formulas for different components of the model and also form the F ratios to test the significance of them.

Solution. The model of the given example is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + e_{(ijk)}$$

The components of this model are as already explained earlier.

In this model, the factor A is a fixed factor with ' a ' number of treatments and the factor B is a random factor with ' b ' number of treatments. The replicates are always random and the number of replications under each experimental combination is n . Since, the factor B is random, the component AB is also random.

The derivations of the expected mean squares formulas of this example are shown in Table 4.56. The suitable F ratios to test the significance of different components of the model are shown in Table 4.57. In this table, the F ratio to test the effect of A_i is formed by dividing the expected mean squares of A_i (EMS_A) by the expected mean squares of the

interaction component AB_{ij} (EMS_{AB}). The F ratio to test the effect of each of the remaining components (B_j and AB_{ij}) is formed by dividing the respective expected mean squares by that of the error term (EMS_{error}).

Table 4.56 Expected Mean Squares Formulas (Example 4.12)

Model component	F a i	R b j	R n k	Expected mean squares formula
A_i	0	b	n	$\sigma^2 + n\sigma_{AB}^2 + \left[bn\sum_{i=1}^a A_i^2 / (a-1)\right]$
B_j	a	1	n	$\sigma^2 + an\sigma_B^2$
$(AB)_{ij}$	0	1	n	$\sigma^2 + n\sigma_{AB}^2$
$e_{(ijk)}$	1	1	1	σ^2

Table 4.57 Suitable F Ratios to Test Components (Example 4.12)

Model component	Expected mean square formula	F ratio
A_i	$\sigma^2 + n\sigma_{AB}^2 + \left[bn\sum_{i=1}^a A_i^2 / (a-1)\right]$	EMS_A/EMS_{AB}
B_j	$\sigma^2 + an\sigma_B^2$	EMS_B/EMS_{error}
$(AB)_{ij}$	$\sigma^2 + n\sigma_{AB}^2$	EMS_{AB}/EMS_{error}
$e_{(ijk)}$	σ^2	

Through this example, one can see the difference in forming the F ratios to test the significance of different components of the model of an experiment with all fixed factors and that of another experiment with mixed factors.

Example 4.13 The investigator of an experiment has considered two factors, viz., factor A and factor B . He has assumed that the factor A as a random factor and the factor B as a fixed factor. The number of levels for the factor A and the factor B are a and b , respectively. The number of replications under each experimental combination is n . Derive the expected mean squares formulas for different components of the model and also form the F ratios to test the significance of them.

Solution. The model of the given problem is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + e_{(ijk)}$$

The components of this model are as already explained earlier.

In this model, the factor A is a random factor with ' a ' number of treatments and the factor B is a fixed factor with ' b ' number of treatments. The replicates are always random, and the number of replications under each experimental combination is n . Since, the factor A is random, the component AB is also random.

The derivations of the expected mean squares formulas of this example are shown in Table 4.58. The suitable F ratios to test the significance of different components of the model are shown in Table 4.59. In this table, the F ratio to test the effect of A_i as well as AB_{ij} is formed by dividing the respective expected mean squares by that of the error component $e_{(ij)k}$ ($\text{EMS}_{\text{error}}$). The F ratio to test the effect of the component B_j is formed by dividing its expected mean squares by the expected mean squares of the component AB_{ij} (EMS_{AB}).

Table 4.58 Expected Mean Squares Formulas (Example 4.13)

Model component	R a i	F b j	R n k	Expected mean squares formula
A_i	1	b	n	$\sigma^2 + bn\sigma_A^2$
B_j	a	0	n	$\sigma^2 + n\sigma_{AB}^2 + \left[an \sum_{j=1}^b B_j^2 / (b-1) \right]$
$(AB)_{ij}$	1	0	n	$\sigma^2 + n\sigma_{AB}^2$
$e_{(ij)k}$	1	1	1	σ^2

Table 4.59 F Ratios to Test the Components of the Model (Example 4.13)

Model component	Expected mean squares formula	F ratio
A_i	$\sigma^2 + bn\sigma_A^2$	$\text{EMS}_A / \text{EMS}_{\text{error}}$
B_j	$\sigma^2 + n\sigma_{AB}^2 + \left[an \sum_{j=1}^b B_j^2 / (b-1) \right]$	$\text{EMS}_B / \text{EMS}_{AB}$
$(AB)_{ij}$	$\sigma^2 + n\sigma_{AB}^2$	$\text{EMS}_{AB} / \text{EMS}_{\text{error}}$
$e_{(ij)k}$	σ^2	

Example 4.14 The investigator of an experiment has considered two factors, viz., factor A and factor B . He has assumed that the factor A as well as the factor B as random factors. The number of levels for the factor A and the factor B are a and b , respectively. The number of replications under each experimental combination is n . Derive the expected mean squares formulas for different components of the model and also form the F ratios to test the significance of them.

Solution. The model of the given example is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + e_{(ij)k}$$

The components of this model are as already explained earlier.

In this model, the factor A is a random factor with ' a ' number of treatments and the factor B is also a random factor with ' b ' number of treatments. The replicates are always random and the number of replications under each experimental combination

is n . Since, the factor A and the factor B are random, the component AB is also random.

The derivations of the expected mean squares formulas of this example are shown in Table 4.60. The suitable F ratios to test different components of the model are shown in Table 4.61. In this table, the F ratio to test the effect of A_i as well as B_j is formed by dividing the respective expected mean squares by that of the component AB_{ij} . The F ratio to test the effect of the component AB_{ij} is formed by dividing its expected mean squares by that of the error component $e_{(ijk)}$.

Table 4.60 Expected Mean Squares Formulas (Example 4.14)

Model component	R_a i	R_b j	R_n k	Expected mean squares formula
A_i	1	b	n	$\sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$
B_j	a	1	n	$\sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$
$(AB)_{ij}$	1	1	n	$\sigma^2 + n\sigma_{AB}^2$
$e_{(ijk)}$	1	1	1	σ^2

Table 4.61 Suitable F Ratios to Test Components (Example 4.14)

Model component	Expected mean squares formula	F ratio
A_i	$\sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$	EMS_A/EMS_{AB}
B_j	$\sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$	EMS_B/EMS_{AB}
$(AB)_{ij}$	$\sigma^2 + n\sigma_{AB}^2$	EMS_{AB}/EMS_{error}
$e_{(ijk)}$	σ^2	

Example 4.15 The investigator of an experiment has considered three factors: factor A , factor B and factor C . He has assumed that all the three factors are random. The number of levels for the factor A , factor B and factor C are a , b and c , respectively. The number of replications under each experimental combination is n . Derive the expected mean squares formulas for different components of the model and form test statistics to test the significance of different components of the model of this experiment.

Solution. The model of the given problem is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + C_k + (AC)_{ik} + (BC)_{jk} + (ABC)_{ijk} + e_{(ijk)}$$

The components of the above model are as already explained in Example 4.9.

In this model, the factor A is a random factor with ' a ' number of treatments, the factor B is a random factor with ' b ' number of treatments and the factor C is a random factor

with 'c' number of treatments. The replicates are always random and the number of replications under each experimental combination is n . Since, the factor A , factor B and factor C are random, the components AB , AC , BC and ABC are also random.

The derivations of the expected mean squares formulas of this example are shown in Table 4.62. The suitable F ratios to test the significance of interaction components of the model are shown in Table 4.63.

Table 4.62 Expected Mean Squares (Example 4.15)

Model component	R a i	R b j	R c k	R n l	Expected mean squares formula
A_i	1	b	c	n	$\sigma^2 + cn\sigma_{AB}^2 + bn\sigma_{AC}^2 + n\sigma_{ABC}^2 + bc n\sigma_A^2$
B_j	a	1	c	n	$\sigma^2 + cn\sigma_{AB}^2 + an\sigma_{BC}^2 + n\sigma_{ABC}^2 + ac n\sigma_B^2$
$(AB)_{ij}$	1	1	c	n	$\sigma^2 + n\sigma_{ABC}^2 + cn\sigma_{AB}^2$
C_k	a	b	1	n	$\sigma^2 + bn\sigma_{AC}^2 + an\sigma_{BC}^2 + n\sigma_{ABC}^2 + ab n\sigma_C^2$
$(AC)_{ik}$	1	b	1	n	$\sigma^2 + n\sigma_{ABC}^2 + bn\sigma_{AC}^2$
$(BC)_{jk}$	a	1	1	n	$\sigma^2 + n\sigma_{ABC}^2 + an\sigma_{BC}^2$
$(ABC)_{ijk}$	1	1	1	n	$\sigma^2 + n\sigma_{ABC}^2$
$\theta_{(ijk)l}$	1	1	1	1	σ^2

Table 4.63 Suitable F Ratios to Test Components (Example 4.15)

Model component	Expected mean squares formula	F ratio
A_i	$\sigma^2 + cn\sigma_{AB}^2 + bn\sigma_{AC}^2 + n\sigma_{ABC}^2 + bc n\sigma_A^2$	Directly not possible
B_j	$\sigma^2 + cn\sigma_{AB}^2 + an\sigma_{BC}^2 + n\sigma_{ABC}^2 + ac n\sigma_B^2$	Directly not possible
AB_{ij}	$\sigma^2 + n\sigma_{ABC}^2 + cn\sigma_{AB}^2$	EMS_{AB}/EMS_{ABC}
C_k	$\sigma^2 + bn\sigma_{AC}^2 + an\sigma_{BC}^2 + n\sigma_{ABC}^2 + ab n\sigma_C^2$	Directly not possible
$(AC)_{ik}$	$\sigma^2 + n\sigma_{ABC}^2 + bn\sigma_{AC}^2$	EMS_{AC}/EMS_{ABC}
$(BC)_{jk}$	$\sigma^2 + n\sigma_{ABC}^2 + an\sigma_{BC}^2$	EMS_{BC}/EMS_{ABC}
$(ABC)_{ijk}$	$\sigma^2 + n\sigma_{ABC}^2$	EMS_{ABC}/EMS_{error}
$\theta_{(ijk)l}$	σ^2	

In this table, the F ratios to test the main effects, A_i , B_j and C_k cannot be formed directly. The effects of $(AB)_{ij}$, $(AC)_{ik}$ and $(BC)_{jk}$ can be tested using the F ratio formed by dividing the respective expected mean squares by that of the component $(ABC)_{ijk}$. The effects of $(ABC)_{ijk}$ can be tested using the F ratio formed by dividing its expected mean squares by that of the error component, $e_{(ijk)}$.

Approximate tests to test the significance of main effects If the degrees of freedom for the error component is less than 6, one can pool the expected mean squares of some of the components for dividing the expected mean squares of the required component(s) such that the remainder of the division is exactly equal to the effect the required component(s).

By assuming that the degrees of freedom of the error component is less than 6, in this example, an approximate test for each of the main effects, A_i , B_j and C_k can be formed by suitably pooling the mean expected squares formulas, as shown in Table 4.64:

Table 4.64 Approximate Test Ratios to Test Main Effects, A_i , B_j and C_k

Model component	F ratio	Remainder
A_i	$\frac{\text{EMS}_{A+ABC}}{\text{EMS}_{AB+AC}} = \frac{(\sigma^2 + c\sigma_{AB}^2 + b\sigma_{AC}^2 + \sigma_{ABC}^2 + bc\sigma_A^2) + (\sigma^2 + \sigma_{ABC}^2)}{(\sigma^2 + \sigma_{ABC}^2 + c\sigma_{AB}^2) + (\sigma^2 + \sigma_{ABC}^2 + b\sigma_{AC}^2)}$	$bc\sigma_A^2$
B_j	$\frac{\text{EMS}_{B+ABC}}{\text{EMS}_{AB+BC}} = \frac{(\sigma^2 + c\sigma_{AB}^2 + a\sigma_{BC}^2 + \sigma_{ABC}^2 + ac\sigma_B^2) + (\sigma^2 + \sigma_{ABC}^2)}{(\sigma^2 + \sigma_{ABC}^2 + c\sigma_{AB}^2) + (\sigma^2 + \sigma_{ABC}^2 + a\sigma_{BC}^2)}$	$ac\sigma_B^2$
C_k	$\frac{\text{EMS}_{C+ABC}}{\text{EMS}_{AC+BC}} = \frac{(\sigma^2 + b\sigma_{AC}^2 + a\sigma_{BC}^2 + \sigma_{ABC}^2 + ab\sigma_C^2) + (\sigma^2 + \sigma_{ABC}^2)}{(\sigma^2 + \sigma_{ABC}^2 + b\sigma_{AC}^2) + (\sigma^2 + \sigma_{ABC}^2 + a\sigma_{BC}^2)}$	$ab\sigma_C^2$

The degrees of freedom of the F test in which the effect of A_i is the remainder are given as follows:

$$\text{Numerator } df = \frac{(\text{MSS}_A + \text{MSS}_{ABC})^2}{\frac{\text{MSS}_A^2}{a-1} + \frac{\text{MSS}_{ABC}^2}{(a-1)(b-1)(c-1)}}$$

$$\text{Denominator } df = \frac{(\text{MSS}_{AB} + \text{MSS}_{AC})^2}{\frac{\text{MSS}_{AB}^2}{(a-1)(b-1)} + \frac{\text{MSS}_{AC}^2}{(a-1)(c-1)}}$$

The degrees of freedom of the F test, in which the effect of B_j is the remainder, are:

$$\text{Numerator } df = \frac{(\text{MSS}_B + \text{MSS}_{ABC})^2}{\frac{\text{MSS}_B^2}{b-1} + \frac{\text{MSS}_{ABC}^2}{(a-1)(b-1)(c-1)}}$$

$$\text{Denominator } df = \frac{(MSS_{AB} + MSS_{BC})^2}{\frac{MSS_{AB}^2}{(a-1)(b-1)} + \frac{MSS_{BC}^2}{(b-1)(c-1)}}$$

The degrees of freedom of the F test, in which the effect of C_k is the remainder, are:

$$\text{Numerator } df = \frac{(MSS_C + MSS_{ABC})^2}{\frac{MSS_C^2}{c-1} + \frac{MSS_{ABC}^2}{(a-1)(b-1)(c-1)}}$$

$$\text{Denominator } df = \frac{(MSS_{AC} + MSS_{BC})^2}{\frac{MSS_{AC}^2}{(a-1)(c-1)} + \frac{MSS_{BC}^2}{(b-1)(c-1)}}$$

Example 4.16 (The Example 4.8 is reproduced here with some modifications in terms of the type of factors involved in it for quick reference. The aim of this example is to study the method of analysis of an experiment with mixed factors.)

The sales manager of Krishna Engineering Company wants to study the effect of sales regions as well as salesmen on quarterly sales revenue. So, he has designed a factorial experiment involving two factors, as shown in Table 4.65, with two replications for each experimental combination. As per this design, each salesman is assigned to each region for two different quarters. The sales region is treated as factor A and it is assumed to be a random factor. The salesman is treated as factor B and it is assumed to be a fixed factor.

Table 4.65 Quarterly Sales of Krishna Engineering Company

		Salesman (B) (Rupees in lakhs)			
		1	2	3	4
Sales region (A)	East	65	30	50	20
		70	35	65	40
	West	30	60	20	50
		35	55	25	40

Perform ANOVA and check the following at the significance level of 0.05.

- (a) (i) Whether there is significant difference in terms of sales revenue between sales regions.
- (ii) Whether there is significant difference in terms of sales revenue between salesmen.
- (iii) Whether there is significant difference in terms of sales revenue between different combinations of sales region and salesman.
- (b) If the factor A is assumed as a fixed factor and the factor B is as a random factor, comment on the results of ANOVA.

- (c) If the factor A as well as the factor B is assumed as a random factor, comment on the results of ANOVA.

Solution. (a) The model of this factorial design is as follows:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + e_{(ijk)}$$

Hypothesis with respect to treatment of the factor A:

Null hypothesis, $H_0: A_1 = A_2$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of the treatment means of the factor A .

Hypothesis with respect to treatment of the factor B:

Null hypothesis, $H_0: B_1 = B_2 = B_3 = B_4$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of the treatment means of the factor B .

Hypothesis with respect to interaction component, AB:

Null hypothesis, $H_0: A_1B_1 = A_1B_2 = A_1B_3 = \dots = A_2B_3 = A_2B_4$

Alternate hypothesis, H_1 : Interaction means are not equal for at least one pair of interaction means.

The relationship between different sum of squares of this model is:

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_{\text{error}}$$

The formulas for F ratios to test the significance of different components of this example are presented in Table 4.66. This is based on Example 4.13.

Table 4.66 F Ratios to Test the Components of the Model (Example 4.16a)

Model component	Expected mean squares formula	F ratio
A_i	$\sigma^2 + bn\sigma_A^2$	EMS_A/EMS_{error}
B_j	$\sigma^2 + n\sigma_{AB}^2 + \left[an \sum_{j=1}^b B_j^2 / (b-1) \right]$	EMS_B/EMS_{AB}
$(AB)_{ij}$	$\sigma^2 + n\sigma_{AB}^2$	$EMS_{AB}/EMS_{\text{error}}$
$e_{(ijk)}$	σ^2	

Based on Table 4.66, the results of ANOVA of this example are summarized in Table 4.67. The different sum of squares in Table 4.67 are reproduced from Table 4.35 of Example 4.8.

In Table 4.67, the value of the calculated F ratio for the factor A is 4.24, whereas its table value with the significance level of 0.05 and degrees of freedom (1, 8) is 5.32. The

calculated F ratio for the factor B is 0.12, whereas, its table value with the significance level of 0.05 and degrees of freedom 3 (3, 3) is 9.28. The calculated F ratio for the interaction AB is 19.29, whereas, its table value with the significance level of 0.05 and degrees of freedom (3, 8) is 4.07.

Table 4.67: Results of ANOVA (Example 4.16a)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
SS_A	1	225.00	$225.00/1 = 225.00$	$225.00/53.13 = 4.24$
SS_B	3	368.75	$368.75/3 = 122.92$	$122.92/1025 = 0.12$
SS_{AB}	3	3075.00	$3075/3 = 1025.00$	$1025.00/53.13 = 19.29$
SS_{error}	8	425.00	$425/8 = 53.13$	
Total	15	4093.75		

Component—factor A: For this component,

$$F_{\text{calculated}} [4.24] < F_{\alpha=0.05 \text{ and } df=(1,8)} [5.32]$$

Hence, the null hypothesis (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different sales regions.

Component—factor B: For this component,

$$F_{\text{calculated}} [0.12] < F_{\alpha=0.05 \text{ and } df=(3,3)} [9.28]$$

Hence, the null hypothesis, (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different salesmen.

Component—interaction, $A \times B$: For this component,

$$F_{\text{calculated}} [19.29] > F_{\alpha=0.05 \text{ and } df=(3,8)} [4.07]$$

Hence, the null hypothesis, (H_0) should be rejected.

Inference: This means that there is a significant difference in quarterly sales because of the interaction between the factor 'sales region' and the factor 'salesman'. Alternatively, this means that the interaction between 'sales region' and 'salesman' has significant effect on quarterly sales.

- (b) If the factor A is assumed as a fixed factor and the factor B is assumed as a random factor, the corresponding formulas for F ratios to test the significance of different components of this example are presented in Table 4.68. This is based on Example 4.12.

Table 4.68 Suitable F ratios to Test Components (Example 4.16)

Model component	Expected mean squares formula	F ratio
A_i	$\sigma^2 + n\sigma_{AB}^2 + \left[bn \sum_{i=1}^a A_i^2 / (a-1) \right]$	EMS_A / EMS_{AB}
B_j	$\sigma^2 + a\sigma_B^2$	EMS_B / EMS_{error}
$(AB)_{ij}$	$\sigma^2 + n\sigma_{Ai}^2$	EMS_{AB} / EMS_{error}
$\sigma_{(ij)k}$	σ^2	

Based on Table 4.68, the results of ANOVA of this example are summarized in Table 4.69. The different sum of squares in Table 4.69 is reproduced from Table 4.35 of Example 4.8.

Table 4.69 Results of ANOVA of Example 4.16b

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
SS_A	1	225.00	$225.00/1 = 225.00$	$225.00/1025 = 0.22$
SS_B	3	368.75	$368.75/3 = 122.92$	$122.92/53.13 = 2.31$
SS_{AB}	3	3075.00	$3075/3 = 1025.00$	$1025.00/53.13 = 19.29$
SS_{error}	8	425.00	$425/8 = 53.13$	
Total	15	4093.75		

In Table 4.69, the value of the calculated F ratio for the factor A is 0.22, whereas, its table value with the significance level of 0.05 and degrees of freedom (1, 3) is 10.13. The calculated F ratio for the factor B is 2.31, whereas, its table value with the significance level of 0.05 and degrees of freedom (3, 8) is 4.07. The calculated F ratio for the interaction AB is 19.29, whereas, its table value with the significance level of 0.05 and degrees of freedom (3, 8) is 4.07.

Component—factor A : For this component,

$$F_{\text{calculated}} [0.22] < F_{\alpha=0.05 \text{ and } df=(1,3)} [10.13]$$

Hence, the null hypothesis, (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different sales regions.

Component—factor B : For this component,

$$F_{\text{calculated}} [2.31] < F_{\alpha=0.05 \text{ and } df=(3,8)} [4.07]$$

Hence, the null hypothesis, (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different salesmen.

Component—interaction, $A \times B$: For this component,

$$F_{\text{calculated}} [19.29] > F_{\alpha=0.05 \text{ and } df=(3,8)} [4.07]$$

Hence, the null hypothesis (H_0) should be rejected.

Inference: This means that there is significant difference in quarterly sales because of the interaction between the factor 'sales region' and the factor 'salesman'. Alternatively, this means that the interaction between 'sales region' and 'salesman' has significant effect on quarterly sales.

- (c) If the factor A as well as the factor B is assumed as random factor, the corresponding formulas for F ratios to test the significance of different components of this example are presented in Table 4.70. This is based on Example 4.14.

Based on Table 4.70, the results of ANOVA of this example are summarized in Table 4.71. The different sum of squares in Table 4.71 are reproduced from Table 4.35 of Example 4.8.

Table 4.70 Suitable F ratios to Test the Components (Example 4.16c)

Model component	Expected mean squares formula	F ratio
A_i	$\sigma^2 + n\sigma_{AB}^2 + bn\sigma_A^2$	EMS_A/EMS_{AB}
B_j	$\sigma^2 + n\sigma_{AB}^2 + an\sigma_B^2$	EMS_B/EMS_{AB}
$(AB)_{ij}$	$\sigma^2 + n\sigma_{AB}^2$	$EMS_{AB}/EMS_{\text{error}}$
$e_{(ijk)}$	σ^2	

In Table 4.71, the value of the calculated F ratio for the factor A is 0.22, whereas its table value with the significance level of 0.05 and degrees of freedom (1, 3) is 10.13. The calculated F ratio for the factor B is 0.12, whereas its table value with the significance level of 0.05 and degrees of freedom (3, 3) is 9.28. The calculated F ratio for the interaction AB is 19.29, whereas, its table value with the significance level of 0.05 and degrees of freedom (3, 8) is 4.07.

Table 4.71 Results of ANOVA (Example 4.16c)

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F ratio
SS_A	1	225.00	$225.00/1 = 225.00$	$225.00/1025 = 0.22$
SS_B	3	368.75	$368.75/3 = 122.92$	$122.92/1025 = 0.12$
SS_{AB}	3	3075.00	$3075/3 = 1025.00$	$1025.00/53.13 = 19.29$
SS_{error}	8	425.00	$425/8 = 53.13$	
Total	15	4093.75		

Component—factor A: For this component,

$$F_{\text{calculated}} [0.22] < F_{\alpha=0.05 \text{ and } df=(1,3)} [10.13]$$

Hence, the null hypothesis, (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different sales regions.

Component—factor B: For this component,

$$F_{\text{calculated}} [0.12] < F_{\alpha=0.05 \text{ and } df=(3,3)} [9.28]$$

Hence, the null hypothesis (H_0) should be accepted.

Inference: This means that there is no significant difference in terms of quarterly sales between different salesmen.

Component—interaction, $A \times B$: For this component,

$$F_{\text{calculated}} [19.29] > F_{\alpha=0.05 \text{ and } df=(3,8)} [4.07]$$

Hence, the null hypothesis (H_0) should be rejected.

Inference: This means that there is significant difference in quarterly sales because of the interaction between the factor 'sales region' and the factor 'salesman'. Alternatively, this means that the interaction between 'sales region' and 'salesman' has significant effect on quarterly sales.

Note: One can notice that there is no difference in terms of inferences of Example 4.8 and that of Examples 4.16a–c though the assumptions about the type of factors in these examples are different. This is mere coincident.

One can do the analysis on his/her own constructed example to notice the difference in terms of inferences for the same problem with different assumptions about the type of factors.

QUESTIONS

1. Define ANOVA and explain the following with an example: (a) factor, (b) treatment and (c) replication.
2. Distinguish between fixed factor and random factor.
3. Prove that the total corrected sum of squares in a single factor experiment is equal to the sum of squares of treatments and the sum of squares of error.
4. Compare and contrast completely randomized design, randomized complete block design and Latin square design.
5. The personnel manager of a firm believes that a detailed experimental analysis will always help him to identify the significance of different factors, which are affecting

the effectiveness of their training programmes. Identify suitable factors and do the following:

- (a) Design an experiment using the completely randomized design. Also, explain the components of the respective model.
 - (b) Design an experiment using the randomized complete block design. Also, explain the components of the respective model.
 - (c) Design an experiment using the Latin square design. Also, explain the components of the respective model.
6. The R&D manager of a computer manufacturing firm is in a state of dilemma whether the sales revenue is affected by any one or a combination of the following factors.
- Region of sales
 - Percentage discount
 - Product warranty period
 - Rate of annual maintenance charges (AMC) after warranty period.

Design a suitable factorial experiment for the given situation.

7. What is EMS rule? Discuss its application in design of experiments.
8. Discuss the need for Duncan's multiple range test with an example.
9. Write the model of the following experiments and also explain their components:
 - (a) Factorial experiment with 2 factors
 - (b) Factorial experiment with 3 factors
 - (c) Factorial experiment with 4 factors.
10. (a) Give the generalized ANOVA table for a factorial experiment with 2 factors in which both the factors are treated to be fixed factors.
 (b) Give the generalized ANOVA table for a factorial experiment with 3 factors in which all the factors are treated to be fixed factors.
11. A production manager wishes to test the effect of five similar milling machines on the surface finish of small castings. So, he selected five such machines and conducted the experiment with four replications under each machine as per 'completely randomized design' and obtained the readings in microns as shown below:
 - (a) State H_0 and H_1 .
 - (b) Perform the required ANOVA and state the inference at the significance level of 0.05

	Machine				
	M_1	M_2	M_3	M_4	M_5
Replication	25	10	40	27	15
	30	20	30	20	8
	16	33	49	35	45
	36	42	22	48	34

12. A stock market analyst wants to study the impact of the type of company on the quarterly averages of the earnings per share (EPS). So, he collected four quarterly

averages of EPS data of four different companies during the last financial year from 'capital market', which are summarized as:

	Company			
	C ₁	C ₂	C ₃	C ₄
Replication	12	16	25	13
	8	18	15	8
	16	10	22	20
	19	11	9	5

Check whether there is a significant difference between different companies in terms of EPS at a significance level of 5%.

13. An experiment was run to determine whether four specific intervals of patrolling (in minutes) by a quality inspector affect the number of acceptable parts produced per hour in a machine. The corresponding data for four different periods with four replications are shown as:

	Period of inspection			
	30 min.	45 min.	60 min.	75 min.
Replication	50	30	40	35
	25	60	12	10
	40	20	30	48
	23	38	39	25

Perform ANOVA as per the completely randomized design and state the inference at a significance level of 5%.

14. An agriculture officer wants to study the effect of four different fertilizers on the yield (in tons) of a specific crop. Because there might be variability from one plot to another plot, he decides to use the randomized complete block design. The corresponding data are presented as in the following table:

	Fertilizer			
	A	B	C	D
Plot	100	150	120	70
	80	70	110	100
	68	90	85	78
	125	138	60	124

- Write the corresponding model.
 - Check whether each component of the model has any effect on the yield of the crop at a significance level of 5%.
15. The R&D manager of a manufacturing firm is in a state of dilemma whether the sales revenue (in crores of rupees) is affected by sales region. Because there might

be variability from one period to another period, he decides to use the randomized complete block design. The corresponding data are presented in the table:

		Sales region					
		A	B	C	D	E	F
Period	1	18	9	15	22	9	10
	2	25	7	14	18	28	13
	3	20	8	12	9	15	17
	4	11	13	30	12	20	23
	5	18	11	25	15	16	8
	6	24	30	17	16	20	30

(a) Write the corresponding model.

(b) Check whether each component of the model has any effect on the sales revenue at a significance level of 5%.

16. There are four different technological alternatives to manufacture a product. The R&D manager feels that the type of technology may have some impact on the hourly output (in units) of the product. Because there might be variability from one plant to another plant, he decides to use the randomized complete block design. The corresponding data are presented as in the following table:

		Technology			
		T ₁	T ₂	T ₃	T ₄
Plant	P ₁	73	68	74	71
	P ₂	73	57	75	52
	P ₃	45	38	68	40
	P ₄	73	41	75	75

(a) Write the model of this situation.

(b) Check whether each component of the model has any effect on the output of the product at a significance level of 5%.

17. The R&D Manager of an automobile company wishes to study the effect of the tyre brand on the tread loss (in millimetre) of tyres. Four different tyres of four different brands (A, B, C and D) are fitted at four different positions of four different cars, because there might be variability from one car to another car as well as from one position to another position, which is a Latin square design. The data as per this design are presented as:

		Car			
		1	2	3	4
Position	1	A = 8	B = 7	C = 1	D = 7
	2	B = 11	C = 2	D = 7	A = 8
	3	C = 6	D = 8	A = 6	B = 6
	4	D = 4	A = 2	B = 3	C = 8

- (a) Write the corresponding model.
- (b) Check whether each component of the model has any effect on the tread loss of tyres at a significance level of 5%.
18. The director of Alpha School of Management wants to study the effect of 'faculty' on the average performance (0–10 scale) of the participants of a training programme. Five different faculty members (A, B, C, D and E) were assigned to five different subjects of the training programmes conducted for five different batches, because there might be variability from one subject to another subject as well as from one batch to another batch, which is a Latin square design. The data as per this design are given as follows:

		Batch				
		1	2	3	4	5
Subject	1	A = 10	B = 6	C = 6	D = 6	E = 8
	2	B = 7	C = 6	D = 5	E = 1	A = 4
	3	C = 5	D = 3	E = 3	A = 2	B = 1
	4	D = 6	E = 4	A = 1	B = 2	C = 5
	5	E = 4	A = 2	B = 3	C = 8	D = 9

- (a) Write the corresponding model.
- (b) Check whether each component of the model has effect on the performance of the batches at a significance level of 5%.
19. The R&D manager of an automobile company wishes to study the effect of tyre brand on the tread loss (in millimetre) of tyres. Four tyres from each of the four different brands (A, B, C and D) are fitted to four different cars using the completely randomized design. The data as per this design are as follows:

	Tyre brand			
	A	B	C	D
6	3	8	4	
7	6	6	2	
10	2	7	1	
9	3	2	4	

- (a) Write the corresponding model.
- (b) Check whether the tyre brand has an effect on the tread loss of tyres at a significance level of 5%.
- (c) If the tyre brand has an effect on the tread loss, then perform Duncan's multiple range test to compare the significant differences between brand means and comment on the results.
20. A stock market analyst wants to study the impact of the type of company on the quarterly averages of the earnings per share (EPS). So, he collected four quarterly averages of EPS data of four different companies during the last financial year from the Financial Express which are as summarized as:

	Company			
	C ₁	C ₂	C ₃	C ₄
Replication	2	10	25	13
	8	11	15	8
	10	12	22	20
	11	8	9	5

- (a) Check whether there is any significance difference between different companies in terms of EPS at a significance level of 5%.
- (b) If the company has an effect on EPS, then perform Duncan's multiple range test to compare the significant differences between company means of EPS and comment on the results.
21. The sales manager of a leading textile showroom at Chennai wishes to study the opinion of its customers on its service quality in a 0–10 scale. In this experiment, the nature of profession as well as the monthly income level of the customers are taken as fixed factors. Two different customers have been sampled under each experimental combination and the corresponding ratings are as shown below:

		Nature of profession			
		Engineer	Doctor	Lawyer	Others
Income level	< Rs. 10,000	3 1	3 8	8 2	10 9
	> Rs. 10,000	3 7	10 4	9 7	2 8

- (a) Write the model of this factorial experiment.
- (b) Check the significance of each of the components of the model at the significance level of 0.05.
22. The sales manager of Devi Engineering Works wants to study the effect of salesman as well as sales region on its quarterly sales revenue in lakhs of rupees. He designed a factorial experiment involving two factors, as shown below, with two replications under each experimental combination. As per this design each salesman is assigned to each region for two different quarters.

		Salesman (B)			
		1	2	3	4
Sales region (A)	North	10 5	10 15	20 35	5 20
	South	9 15	40 35	5 8	30 20

Perform ANOVA and check the following at the significance level of 0.05:

- (a) Whether there is any significant difference between salesmen in terms of sales revenue.
- (b) Whether there is any significant difference between sales regions in terms of sales revenue.
- (c) Whether there is any significant difference between different combinations of salesman and sales region in terms of sales revenue.
23. A company is keen in assessing the contribution of its employees in a 0–10 scale in terms of value addition to its business operations. In this connection, the UG qualification, sex and work experience of the employees are considered to be the factors. The corresponding ratings of the employees are shown below in the table:

		UG degree			
		Engineering		Others	
		Male	Female	Male	Female
Work experience	Less than 3 years	8	2	4	2
		7	6	8	4
	3 years and above	9	4	7	5
		9	9	8	6

- (a) Write the ANOVA model of this situation.
- (b) Perform the relevant ANOVA and state the inferences at the significance level of 5%.
24. Discuss the development of contrast coefficients of 2^2 factorial experiment.
25. Explain the steps of Yates' algorithm.
26. A company is keen in assessing the contribution of its employees in a 0–10 scale in terms of value addition to its business operations. In this connection, the UG qualification, sex and work experience of the employees are considered to be the factors. The corresponding ratings of the employees are shown in the table:

		UG degree			
		Engineering		Commerce	
		Male	Female	Male	Female
Work experience	Less than 5 years	9	3	5	3
		8	7	9	5
	5 years and above	10	5	8	6
		10	10	9	7

- (a) Write the ANOVA model of this situation.
- (b) Perform the relevant ANOVA using Yates' algorithm and state the inferences at the significance level of 5%.
27. In an experiment with two factors (factors A and B), all the factors are considered to be fixed factors. The number of levels of the factor A and factor B are a and b , respectively. The number of replications in each experimental combination is n .

- (a) Write the ANOVA model.
 (b) Derive the expected mean squares formulas for the components of the model.
 (c) Give the generalized ANOVA table of this experiment.
28. In an experiment with two factors (factors A and B), the factor A is a fixed factor and factor B is a random factor. The number of levels of the factor A and factor B are a and b , respectively. The number of replications in each experimental combination is n .
- (a) Write the ANOVA model.
 (b) Derive the expected mean squares formulas for the components of the model.
 (c) Give the generalized ANOVA table of this experiment.
29. In an experiment with two factors (factors A and B), the factor A is a random factor and the factor B is a fixed factor. The number of levels of the factors A , and B are a and b , respectively. The number of replications in each experimental combination is n .
- (a) Write the ANOVA model.
 (b) Derive the expected mean squares formulas for the components of the model.
 (c) Give the generalized ANOVA table of this experiment.
30. In an experiment with two factors (factors A and B), all the factors are considered to be random factors. The number of levels of the factors A and B are a and b , respectively. The number of replications in each experimental combination is n .
- (a) Write the ANOVA model.
 (b) Derive the expected mean squares formulas for the components of the model.
 (c) Give the generalized ANOVA table of this experiment.
31. The sales manager of Devi Engineering Works wants to study the effect of salesman as well as sales region on its quarterly sales revenue (in lakhs of rupees). He designed a factorial experiment involving two factors, as shown in the following table, with two replications for each experimental combination. The factor sales region (factor A) is considered to be a fixed factor and the factor salesman (factor B) is considered to a random factor. As per this design, each salesman is assigned to each sales region for two different quarters.

		Salesman (B)			
		1	2	3	4
Sales region (A)	North	10	10	20	5
		5	15	35	20
South		9	40	5	30
		15	35	8	20

Perform ANOVA and check the following at the significance level of 0.05.

- (a) (i) Whether there is any significant difference in terms of sales revenue between salesmen.

- (ii) Whether there is any significant difference in terms of sales revenue between sales regions.
- (iii) Whether there is any significant difference in terms of sales revenue between, different combinations of salesman and sales region.
- (b) If the factor A and the factor B are assumed to be random factors, comment on the results of ANOVA.

32. The sales manager of a leading textile showroom at Chennai wishes to study the opinion of its customers on its service quality in a 0–10 scale. In this experiment, the nature of profession and the monthly income level of the customers are taken as fixed and random factors, respectively. Two different customers have been sampled under each experimental combination and the corresponding ratings are as shown below:

		Nature of profession			
		Engineer	Doctor	Lawyer	Others
Income level	< Rs. 10,000	5	9	5	10
		5	8	2	9
	> Rs. 10,000	3	10	5	8
		5	9	2	1

The factor, income level is considered to be a random factor and the other factor is considered to be a fixed factor.

- (a) Write the model of this factorial experiment.
 - (b) Check the significance of each of the components of the model at the significance level of 0.05.
33. In an experiment with three factors (factors A , B and C), all the factors are considered to be random factors. The number of levels of the factor A , factor B and factor C are a , b and c , respectively. The number of replications in each experimental combination is n .
- (a) Write the ANOVA model.
 - (b) Derive the expected mean squares formulas for the components of the model.
 - (c) Form F ratios to test the components.

CHAPTER 5

ATTITUDE MEASUREMENT AND SCALES

5.1 INTRODUCTION

In organizations, many decisions are taken in each of the functional areas of management: production, marketing, finance and personnel. Some examples of decisions in the area of production are number of shifts to be operated, amount of overtime to be operated, amount of total worth of stock of materials to be carried, etc. In marketing, there are numerous decisions which are oriented towards customers, salesmen, and products in conjunction with the other functional areas of management. Types of decisions taken in marketing are prevailing market share of a product of the company, brand loyalty, impulse buying behaviour of customers, product positioning, media planning, etc. A few examples of decisions taken in the area of finance are variety of investment plans, lending rate, credit period to its industrial customers, etc. In personnel function, decisions such as hiring/firing of labours, labour welfare measures, designing training programmes, incentives schemes, employees promotions, etc., are taken.

Among all the decisions, decisions taken in the marketing function are heavily based on attitude of the current/potential customers of a company. Hence, in this chapter a special focus is given on the measurement of attitude and related scales.

Attitude is a learned tendency of individuals to respond in a consistent manner to a given object of orientation.

Elements of measuring attitude are as follows:

- Persons, objects, events, concepts or states to be observed
- Environmental conditions of the study
- Steps to be performed in collecting the observations
- Instruments to be used to perform the steps
- Observations to be collected.

5.2 MEASUREMENT

The characteristics/attributes of objects, persons, events, concepts or states are to be represented using some notations help researchers make inferences based on statistical

analysis or comparative analysis. The way of defining such notations for the characteristics/attributes is known as *measurement*. Alternatively, measurement can be defined as the assignment of a number to an object which reflects the degree of possession of a characteristic by that object.

There are three characteristics of the real number series which are used to quantify the degree of possession of the characteristics by the objects. They are *order*, *distance* and *origin*. Order means the numbers are ordered. Distance means the difference between numbers are ordered. Origin means the real number series has unique origin, zero. The assignment of numbers to objects should be such that the relations between numbers correspond to the real-world relations between objects.

5.3 FUNDAMENTALS OF DATA COLLECTION METHODS

In this section, fundamentals of data collection methods are explained. The data collection methods can be classified into *variability* methods and *quantitative-judgment* methods. In variability methods, ordinal scaled data are collected and these are then converted into interval scaled data using some model. In quantitative-judgment methods, numerical judgments of the respondents are obtained in the form of interval scaled data.

5.3.1 Variability Methods

Different techniques under variability methods are paired comparison, ranking, ordered category sorting and rating techniques. These are explained in the following subsections.

Paired comparison technique

If there are n stimuli, then there will be nC_2 pairs of 2 stimuli. For each pair of stimuli (i, j) , if a respondent feels that the stimulus i (row i) wins over the stimulus j (column j) (the stimulus i dominates the stimulus j), then mark the respective matrix element with 1. The diagonal elements of the matrix have no values. A sample set of original data for five brands is shown in Table 5.1. Later, the matrix can be permuted such that the upper diagonal entries are with 1. Then the corresponding order of the row numbers indicates stimulus rank order. These are illustrated in Tables 5.2 and 5.3. From Table 5.3, it is clear that the stimulus rank order is 2, 1, 4, 3 and 5. The total number of 'one entry' of the rows in Table 5.3 decreases from top to bottom. The data in Table 5.3 are characterized as hypothetical data because the respondent is considered to be transitive. The meaning of the respondent being transitive is that there is no cycle in the data. Sometime, in the matrix, a cycle may be formed, shown as follows, if the entries for the cells, say (2, 5), (5, 3) and (3, 2) are assumed to be 1.

Brand 2 $\rightarrow$ Brand 5 $\rightarrow$ Brand 3 $\rightarrow$ Brand 2

Existence of such cycle signifies the fact that the respondent is non-transitive. Under such situation, some modifications to the original data should be carried out to make the matrix to the format, as shown in Table 5.3, in which all the upper diagonal cells are with 1 and all the lower diagonal cells are with 0.

In addition to the transitive property of respondent, one should check the consistency of marking of the stimulus i (row i) wins over the stimulus j (column j). This can be taken care by replicating the data using several respondents. Such replicated data can be used to arrive the proportions of preference for different pairs of stimulus i over stimulus j .

Table 5.1 Paired Comparison Responses from a Respondent

		Brand				
		1	2	3	4	5
Brand	1	—	0	1	1	1
	2	1	—	1	1	1
	3	0	0	—	0	1
	4	0	0	1	—	1
	5	0	0	0	0	—

Table 5.2 Row Permutation of Table 5.1

		Brand				
		1	2	3	4	5
Brand	2	1	—	1	1	1
	1	—	0	1	1	1
	4	0	0	1	—	1
	3	0	0	—	0	1
	5	0	0	0	0	—

Table 5.3 Column Permutation of Table 5.2

		Brand				
		2	1	4	3	5
Brand	2	—	1	1	1	1
	1	0	—	1	1	1
	4	0	0	—	1	1
	3	0	0	0	—	1
	5	0	0	0	0	—

Let us assume that this situation is replicated two more times (second and third replications) in addition to the data in Table 5.1 (first replication) and the corresponding permuted data be, as shown in Tables 5.4 and 5.5.

• By adding the data of the corresponding cells of Tables 5.3–5.5, the total frequency of preferences for the upper-diagonal cells based on the three sets of replicated data are, as shown in Table 5.6. The respective proportion of preferences are shown in Table 5.7. Each cell entry in Table 5.7 is obtained by dividing the respective frequency in Table 5.6 by 3 (no. of replications).

Table 5.4 Permuted Data of Second Replication

		Brand				
		3	1	4	5	2
Brand	3	—	1	1	1	1
	1	0	—	1	1	1
	4	0	0	—	1	1
	5	0	0	0	—	1
	2	0	0	0	0	—

Table 5.5 Permuted Data of Third Replication

		Brand				
		2	1	5	4	3
Brand	2	—	1	1	1	1
	1	0	—	1	1	1
	5	0	0	—	1	1
	4	0	0	0	—	1
	3	0	0	0	0	—

Table 5.6 Total Frequencies of Row Over Column Preferences for the Upper-diagonal Cells, Using Three Permuted Replicated Data Matrices (Tables 5.3–5.5)

		Brand				
		1	2	3	4	5
Brand	1	—	1	2	3	3
	2		—	2	2	2
	3			—	1	2
	4				—	2
	5					—

Table 5.7 Proportion of Row Over Column Preferences for the Upper-diagonal Cells

		Brand				
		1	2	3	4	5
Brand	1	—	.33	.67	1	1
	2		—	.67	.67	.67
	3			—	.33	.67
	4				—	.67
	5					—

In Table 5.7, the proportion of preference of the brand-1 over the brand-2 is 0.33. Therefore, the proportion of preference of the brand-2 over the brand-1 is 0.67 (i.e. $1 - 0.33$). Based on this logic, the entire pairwise preference proportions are summarized in Table 5.8.

Table 5.8 Proportions of Row Over Column Preferences for all Combinations of Rows and Columns

		Brand				
		1	2	3	4	5
Brand	1	—	0.33	0.67	1.00	1.00
	2	0.67	—	0.67	0.67	0.67
	3	0.33	0.33	—	0.33	0.67
	4	0.00	0.33	0.67	—	0.67
	5	0.00	0.33	0.33	0.33	—

The steps of paired comparison technique are summarized as follows:

- Step-1:* Decide on the number of respondents (subjects) which represents the number of replications of the data matrix. Let it be n .
- Step-2:* Obtain the responses for each of the respondents in a separate matrix in the format, as shown in Table 5.1.
- Step-3:* Obtain a final matrix after row and column permutations for each respondent.
- Step-4:* Obtain the matrix containing cumulative frequencies of preference for the upper-diagonal cells by adding the corresponding cell entries of all the final matrices obtained in step-3.
- Step-5:* Divide the cumulative frequency in each upper-diagonal cell of the matrix obtained in step-4 by the number of replications (n) and treat it as the proportion of preference of the row over the column of that cell.
- Step-6:* Find the proportions of preference for the lower diagonal cells. The proportion of preference of row i over column j ($i > j$), P_{ij} is obtained using the following formula.

$$P_{ij} = 1 - P_{ji}, \quad \text{where } i > j$$

- Step-7:* Treat the resultant matrix (Table 5.8) as the input to the related scale.

Ranking methods

In ranking methods, a respondent is asked to rank a set of stimuli (or brands) based on certain attribute/characteristic. The ranking methods are also useful in identifying key attributes of services/stimuli.

A suitable method may be 'k out of n' method in which the respondent ranks k items (attributes) in order of importance out of a total of n attributes of the product/service. If the ranking is with respect to stimuli, then the respondent should rank the top- k stimuli (brands) out of n stimuli (brands).

Ordered-category sorting

In ordered-category sorting method of data collection, a given set of stimuli (preferably more than 15 stimuli) is assigned to a pre-specified number of ordered category, for example, assigning 25 stimuli into, say, 5 ordered categories/piles. The number of stimuli to be assigned to each category may be unlimited (free) or limited (forced).

The interval between the boundaries of the categories may be equal (or ordered). The ordering of the category boundaries is assumed in variability method. (Equal interval between category boundaries is assumed in quantitative-judgment methods.)

If the equal interval between category boundaries is not followed, one should use stronger scale. The responses may be considered to be rating scaled, ordinal scaled or interval/ratio scaled.

Rating methods

Rating method of data collection is the easiest and widely used method. In this method, the person/object/concept which is to be rated is placed along a continuum or in any one of an ordered set of categories. Rating is nothing but the degree of an attribute or characteristic on a scale. The different forms of rating methods are numerical, graphic or verbal. In practice, two or more of these forms are combined to collect data, explained as follows:

1. Integer series along with verbal descriptions of the degree of an attribute. In this method, the respondent has to encircle the desired integer based on his opinion after reading the verbal descriptions of the attribute.
2. Only a set of categories defined verbally. Here, the respondent should check the category as per his belief and make tick mark against it.
3. A graduated thermometer scale. This has the highest reading at the top and the lowest reading at the bottom along with limited verbal descriptions of the degree of the attribute covering both extremes, median and quartile region(s).
4. Verbal descriptions in comparison with the average of the attributes/stimuli, like, the best, one of the best available, better than average, about average, below average.
5. Only a set of odd numbered categories defined verbally about the degree of importance of the attribute of an object.

5.3.2 Quantitative Judgement Methods

The techniques of quantitative-judgement methods are direct estimates, fractionation and constant sum. These are explained in the following subsections.

Direct judgement methods

In this method, the respondent marks his degree of belief/desire/judgement about a stimulus/attribute of a stimulus on a scale with unlimited response categories or limited response categories. Here, a direct numerical weight is marked by the respondent depending on his judgment about the stimulus/attribute of the stimulus which is being studied.

The unlimited response categories type contains a straight line (horizontal line)/graphic form on which the respondent has to mark about his degree of judgement about the stimulus/attribute of the stimulus, as shown in Fig. 5.1.

If multiple brands are to be judged by a respondent, then he/she should first place the index cards of those brands on the horizontal line, as shown in Fig. 5.1, and form an opinion about the distance between different brands. Then, the respondent can make the marking of the brands on the line.

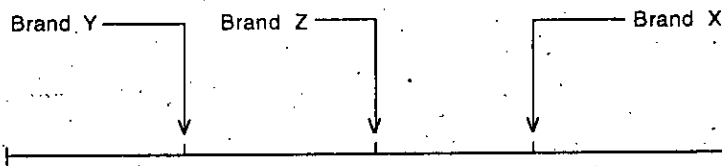


Fig. 5.1 Unlimited response categories.

The limited response categories type contains a limited number of numerical weights from high to low in continuous order as shown in Fig. 5.2. The respondent has to express his judgment by way of placing the brand name by the side of the judged numerical weight.

If multiple brands are to be judged by a respondent, then he/she should place the index card of each brand on a numerical weight in Fig. 5.2 which best fits his judgement about the brand in terms the desired attribute. Then, the respondent should make the marking of each of the brands by the side of the selected numerical weight.

5	Most liked
4	← Brand X
3	← Brand Z
2	← Brand Y
1	Most disliked

Fig. 5.2 Limited response categories.

Fractionation method

Fractionation method is another kind of quantitative judgement method in which a respondent compares each of the test brands with a standard brand at a time and expresses his/her quantitative judgement in terms of a ratio. This ratio indicates the relative degree of comparison of the performance measure of a test brand with that of the standard brand. The best example is to compare the relative mileage yield of 100-cc bikes with a standard brand bike X. A sample format of this method is shown in Table 5.9:

Table 5.9 Relative Degrees of Mileage of Test Brand Bikes with Bike X (Fraction method)

Test brand	Response
A	0.95
B	1.40
C	0.89
D	0.79
E	0.92

Constant sum method

It is the simplest among all the data collection methods. In this method, each respondent is given a number of points, say 10, 100 or 200. Then he or she allocates these points to different stimuli according to some attitudinal characteristic. The sum of all the points allocated to all the stimuli should be equal to the total number of points given to the respondent. Before allocating points to different stimuli, the respondent should compare the relative importance of the stimuli with respect to the assumed attitudinal characteristic.

A sample format of the constant sum method of data collection is shown in Table 5.10.

Table 5.10 Total Number of Points Given to Respondent = 10
(Constant sum method)

<i>Test brand</i>	<i>Response</i>
A	2
B	1
C	4
D	3

5.4 TYPES OF SCALE

Conventional scales are measurement systems to measure the length, breadth, height, volume, etc., of objects. These scales have a natural zero and constant unit of measurement.

But, in behavioural science, such clear-cut scales may not exist. The types of scales which are used in such fields are: nominal scale, ordinal scale, interval scale and ratio scale.

Nominal scale This scale has no order, distance or origin. The numbers used in this scale are used as tags or symbols to recognize or identify objects. The code numbers assigned to the customers of a telephone company, chain of stores, credit card holders of a bank, employees of a company, etc., are examples of this scale. This scale can also be used to represent yes/no, like/dislike, male/female type of responses. Only rudimentary mathematical operations can be performed on such code numbers. At the maximum, one can count the numbers and make any inference accordingly, like number of customers visiting a retail shop of a textile company during each week of a month, number of persons used their credit cards of a particular bank on a particular day. Further, contingency table summarizing the membership of persons in two categories can be formed.

Ordinal scale It is known as ranking scale which possesses the property of order. Using this scale, one can rank objects based on certain characteristic or attribute of the objects. For example, ranking the customers of a bank according to their patience while transacting with the bank, ranking the products of a company according to the satisfaction of customers, ranking operators in a shop according to their skills, ranking the products in a shop floor according to their quality, etc.

An ordinal scale has all the information of a nominal scale for the same situation which means that equivalent entities will have the same rank. The data using ordinal scale can be used to compute only the positional measures like, mode and median.

Interval scale This scale has the properties of order and distance. Using this scale, one can judge the difference between two objects. It has arbitrary zero point and constant unit of measurement. The numbers on the scale are placed at equidistance. The mathematical operations, addition, subtraction and calculation of mean are possible with any interval scaled data.

The centigrade thermometer and Fahrenheit thermometer are some examples of interval scale. In centigrade thermometer, the minimum number is 0°C and the maximum number is 100°C. Between these two numbers, the numbers are placed at equal distance. In Fahrenheit thermometer, the minimum number is 32°F and the maximum number is 212°F.

The relationship between these two systems of scales is as shown below:

$$C = \frac{5}{9}(F - 32)$$

where F is the Fahrenheit temperature and C is the centigrade temperature.

From the above equation, one can verify the fact that the minimum centigrade temperature (0°C) corresponds to the minimum Fahrenheit temperature (32°F). Similarly, the maximum centigrade temperature (100°C) corresponds to the maximum Fahrenheit temperature (212°F).

Ratio scale Ratio scale has a unique zero point. Such scales are used in physical sciences to measure length, breadth, weight, volume, etc. Equal ratio among the scale values will correspond to equal ratio among the entities, which are being measured. This means that a positive multiplicative constant will help converting feet into inches, kilogram into grams, kilogram into pounds, etc. This scale has the properties of order and distance and origin. All statistical operations can be performed on any ratio scaled data. Attitude scaling is a psychological scaling, which focusses on the study of perception, preference and motivation of individuals towards products/services.

In practice, researchers generally use ordinal scale for attitude measurement. Later, these can be converted into interval scale data to perform statistical operations.

All attitude measurement procedures are oriented in studying the responses of consumers/purchasing agents/marketing managers, etc., for a given stimuli.

The scaling methods are classified into *subject-centred* approach, *stimuli-centred* approach and *response* approach. In the subject-centred approach, the variation across the respondents is studied. The stimuli-centred approach examines the systematic variation across stimuli. In the response approach, the variation across subjects as well as the variation across stimuli are studied.

The different scales, which are used in attitude measurement, are summarized in Table 5.11 and these are explained in the following sections.

Table 5.11 Attitude Measurement Scales

Category	Purpose	Name of scale
1	Scaling stimuli	(a) Thurstone's case-V scale (b) Osgood's semantic differential scale.
2	Scaling respondents	(a) Summated scale (b) Q-sort technique
3	Scaling stimuli and respondents	(a) Cumulative scale/Scalogram analysis

5.5 SCALES FOR STIMULI

The variability method of data collection and quantitative-judgment method of data collection produce raw data responses, which can be further converted into interval-scaled data. The ordinal data produced by the variability method of data collection can be converted into interval-scaled data using Thurstone's case-V method. The raw data obtained through the quantitative-judgment method can be converted into interval scaled data using Osgood's semantic differential scale.

5.5.1 Thurstone's Case-V Scale Model

Such a model is used to construct one-dimensional interval scale based on the comparative judgment of the responses collected from the variability method. The basic principle of this scale model is that the comparative judgment between two stimuli, like brand A is better than brand B, food type I is more nutritious than food type II, etc. While collecting data, necessary number of replications are made to get the opinion of one-individual about the comparative judgment of a pair of stimuli (pair of brands) to reduce the error. The outcome of the comparative judgment is expressed in terms of proportion, like 80% of the respondent prefers the brand A in comparison with the brand B. This in turn reflects that 20% of the respondents prefers the brand B in comparison with the brand A. So, Thurstone's Case-V scale model of comparative judgment deals with the process of developing an interval scale from the stimulus-comparison proportions expressed by the respondents.

The terminologies which are associated with this scale model are explained below.

Discriminal process It is the psychological process by which a respondent reacts to a stimulus. Since, the same respondent may react differently, one should take the modal discriminial process which means that the most occurring reaction to a stimulus.

Discriminal difference It is the distance between the modal discriminial processes of two stimuli, as reacted by a respondent. It is assumed that each discriminial process is normally distributed. Hence, the discriminial difference is also normally distributed.

Based on these discussions, the generalized Thurstone's scale model is represented as:

$$D_i - D_j = Z_{ij} \sqrt{\sigma_i^2 + \sigma_j^2 - 2r_{ij}\sigma_i\sigma_j}$$

where D_i is the modal discriminial process of the stimulus i ; D_j , the modal discriminial process of the stimulus j ; $D_i - D_j$, the linear distance between modal discriminial processes

of stimulus i and the stimulus j ; Z_{ij} , the standard normal value for a given proportion; σ_i , the standard deviation of the discriminial process of the stimulus i ; σ_j , the standard deviation of the discriminial process of the stimulus j ; and r_{ij} , the correlation coefficient between the discriminial processes associated with the stimulus i and the stimulus j .

The Thurstone's case-V model is derived from the above generalized model by making the following assumptions:

- (a) The standard deviation of both the discriminial processes are the same.
- (b) All pairs of discriminial processes have the same correlation (r).

These assumptions yield the following model:

$$D_i - D_j = Z_{ij} \sqrt{2\sigma^2(1-r)}$$

In the above equation, $[2\sigma^2(1-r)]^{1/2}$ is a constant.

Since, the task is to construct an interval scale, the unit of measurement and the origin zero point can be arbitrary.

Hence, the above constant can be assumed as 1 which simplifies the formula to the following form.

$$D_i - D_j = Z_{ij}$$

Steps of construction of Thurstone's case-V scale model

The steps of construction of this model are as follows:

- Step-1:** Obtain the matrix of proportion of the comparative judgments of different pairs of stimuli. If the number of stimuli is n , then the number of pairwise combinations is $2^n C_2$.
- Step-2:** Find the matrix of standard normal values corresponding to the proportions of the comparative judgments of different pairs of stimuli.
- Step-3:** Find the total of each column of the matrix in step-2 and also find the mean of each such total.
- Step-4:** If all the column means in step-3 are positive, then find the scale values (D_j^*) by subtracting the minimum of the column means from each of the column means obtained in step-3; otherwise, find the most negative column mean. Then find the scale values (D_j^*) by adding its absolute value to each of the column means obtained in step-3.
- Step-5:** Sort the scale values in ascending order and identify the corresponding order of the stimuli.

Example 5.1 Construct Thurstone's case-V scale for the proportions of comparative judgements for different pairs of four brands, as shown in Table 5.12. Also, predict the proportion of the comparative judgement of the brand B to the brand A as well as that of the Brand A to C and compare them with actuals. It should be noted that the diagonal entries of Table 5.12 are 0.5.

Table 5.12 Proportions of Pairwise Comparative Judgements of Brands

Brand	Preferred brand			
	A	B	C	D
A	0.5	0.8	0.4	0.6
B	0.2	0.5	0.6	0.3
C	0.6	0.4	0.5	0.9
D	0.4	0.7	0.1	0.5

Solution. The standard normal values for the proportions of comparative judgements of the stimuli are shown in Table 5.13. The calculations for the step-3 and step-4 are summarized in the same table.

Table 5.13 Z_{ij} Values

Brand	Preferred brand			
	A	B	C	D
A	0.000	0.840	-0.253	0.253
B	-0.840	0.000	0.253	-0.525
C	0.253	-0.253	0.000	1.280
D	-0.253	0.525	-1.280	0.000
Column total	-0.840	1.112	-1.280	1.008
Column mean	-0.210	0.278	-0.320	0.252
D_j^*	0.110	0.598	0.000	0.572

From Table 5.13, it is clear that the ranking of the scale values of the stimuli is as:

$$B(0.598) > D(0.572) > A(0.11) > C(0)$$

The minimum of the scale value is 0 and the maximum of the scale value is 0.598.

Prediction of proportion of comparative judgements of brand B to brand A

The predicted Z value of brand B to brand A = $0.598 - 0.110 = 0.488$. The proportion for $Z = 0.488$ from the standard normal table is 0.69. This is in shortage of 0.11 when compared to the actual proportion of the brand B to the brand A.

Prediction of proportion of comparative judgements of brand A to brand C

The predicted Z value of brand A to brand C = $0.11 - 0 = 0.11$. The proportion for $Z = 0.11$ from the standard normal table is 0.544. This is in shortage of 0.056 when compared to the actual proportion of the brand A to the brand C.

Like this, one can predict for the proportion of each and every comparative judgement in Table 5.12.

5.5.2 Osgood's Semantic Differential Scale

As stated earlier, Osgood's semantic differential scale is used to convert raw data obtained using quantitative-judgment procedure into interval scaled data. This interval scaled data can be further analyzed through factor analysis.

The semantic differential scale will reflect the direction and intensity of respondents' attitude concerning corporate image, brand/service image, country image, image of professional courses, political party image, etc.

As per this scale, the respondents are expected to rate, say, an educational institution on a set of bipolar adjectives. A sample set of bipolar adjectives is shown in Fig. 5.3.

		-2	-1	0	1	2	
					X		
Students care	: Inferior	_____	_____	_____	_____	_____	Superior
				X			
Infrastructure	: Inferior	_____	_____	_____	_____	_____	Superior
					X		
Placement services	: Poor	_____	_____	_____	_____	_____	Good
						X	
Syllabus	: Old	_____	_____	_____	_____	_____	Latest
			X				
Faculty base	: Poor	_____	_____	_____	_____	_____	Good

Fig. 5.3 Institutional profile through semantic differential scale.

Each respondent will be asked to judge each institution on a five-point scale for all the attributes as mentioned in Fig. 5.3.

As an example, the different descriptive phrases of the placement services are listed as:

1. Very poor placement services
2. Poor placement services
3. Neither poor nor good placement services
4. Good placement services
5. Very good placement services.

For the five-point scale, the judgment scores of different phrases may be assumed as -2, -1, 0, 1, 2 from left to right of the scale, which are shown as in Fig. 5.3.

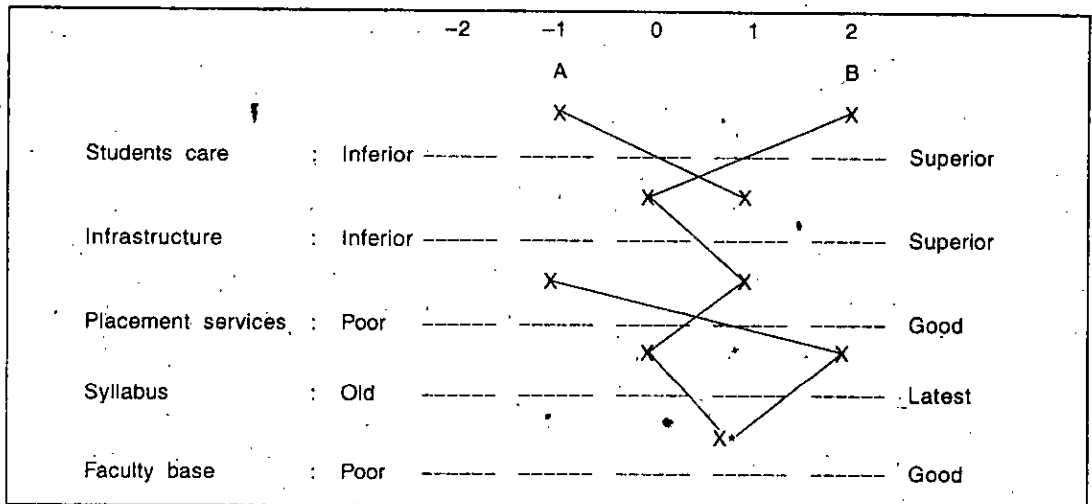
A sample set of judgements of a respondent about the profile of the institution are as follows:

1. Good students care (+1)
2. Neither good nor poor infrastructure (0)
3. Good placement services (+1)
4. Very latest syllabus (+2)
5. Poor faculty base (-1).

Steps of semantic differential scale

The steps of the semantic differential scale are presented below with reference to studying the image of organizations:

- Step-1:* Identify the attributes on which the organizations are to be judged by respondents.
- Step-2:* Decide on the number of descriptive phrases and define them. Also, decide on the numerical weights of the descriptive phrases.
- Step-4:* Design a questionnaire containing semantic differential scale for each organization.
- Step-5:* Administer the questionnaire to the respondents and seek their judgments for all the organizations.
- Step-6:* For each organization, sum the judgment scores of each attribute across respondents and find its average.
- Step-7:* For each organization, plot the average score of each attribute on the semantic differential scale as illustrated in Fig. 5.4.



*, Institution A; X, Institution B

Fig. 5.4 Semantic differential scale for institutions image.

In Fig. 5.4, the institution A is perceived as inferior students care, poor placement services, but with good infrastructure, good faculty base and very latest syllabus; the institution B is perceived as superior students care, good placement services, good faculty base but with neither inferior nor superior infrastructure and neither old nor latest syllabus.

5.6 SCALING RESPONDENTS

Sometime, the researchers may be interested in scaling respondents. There are two methods of scaling respondents: summated scale (Likert scale) and Q-sort scale.

5.6.1 Summated Scale (Likert Scale)

The summated scale was proposed by Rensis Likert. In this scale, a set of items is proposed with respect to a particular attitudinal object, like the role of educational institutions in employment market. In this case, the researcher may be interested to scale the characteristic, like public's attitude on the above-mentioned attitudinal object. The attitudinal object will have a set of items which are described in the form of statements/phrases. The items can be classified into favourable items and unfavourable items. Each favourable item is given a numerical weight like, +2, +1, 0, -1, -2. Each unfavourable item will have a numerical weight in the reverse order as -2, -1, 0, +1, +2. Each respondent is asked to assign/tick a value for each item. In the summated model, the total score of a respondent is obtained by adding the scores on individual items. A sample set of statements with respect to the role of educational institutions in employment market is given in Table 5.14. In Table 5.14, the statements 1, 3, 4 and 6 are favourable (supportive) statements with respect to educational institutions. The remaining three statements are unfavourable statements with respect to educational institutions.

Table 5.14 Sample Set of Statements (Items)

Item no.	Statement	Strongly agree	Agree	Undecided	Disagree	Strongly disagree
1	There is a hectic competition among employment institutions in providing employment opportunities.	+2	+1	0	-1	-2
2	Some educational institutions are not bothered about employment issue of their graduates.	-2	-1	0	+1	+2
3	Whatever may be the effort made by educational institutions, it will be very difficult to employ their graduates because of saturation in employment market.	+2	+1	0	-1	-2
4	Since the average salary in industries is very low, graduates of premier institutions are not willing to accept such low-paid jobs.	+2	+1	0	-1	-2

contd.

Table 5.14 (contd.)

Item no.	Statement	Strongly agree	Agree	Undecided	Disagree	Strongly disagree
5	In technical education, educational institutions are not employing qualified and experienced faculty which will affect employment opportunities adversely.	-2	-1	0	+1	+2
6	Because of higher pay in some industries, educational institutions are finding it difficult to attract talented graduates/experienced persons for faculty positions.	+2	+1	0	-1	-2
7	Some educational institutions do not plough back their surplus which badly affect institutional infrastructures development.	-2	-1	0	+1	+2

The steps of construction of the summated scale are presented below:

- Step-1:* Formulate possible number of statements about the attitudinal object which is to be studied. Ensure that the number of favourable statements is almost equal to the number of unfavourable statements.
- Step-2:* Assume the following directional intensity descriptors of the items of the attitudinal object.
- Strongly agree
 - Agree
 - Undecided
 - Disagree
 - Strongly disagree
- Assign scale values to the degree of agreement/disagreement as stated below. For favourable statements (items), the scale values from left to right are assumed as +2, +1, 0, -1, -2. For unfavourable statements (items), the scale values from left to right are assumed as -2, -1, 0, +1, +2.
- Step-3:* Administer the set of statements in the form of questionnaire to a group of respondents which consists of parents, graduates of the institutions, recruiting organizations, officials of government in UGC, state educational departments and universities.
- Step-4:* Calculate the total attitude score of each respondent by summing the scores assigned to each of the items by that respondent.
- Step-5:* 5.1: Find the highest quartile group (first quartile) and the lowest quartile group (fourth quartile) of the respondents (subjects) on the basis of the total scores. The high scores can be assumed to be with the favourable attitudes and the low scores can be assumed to be with unfavourable attitudes.

5.2: Find the mean difference between the highest quartile group and the lowest quartile group for each item. If an item (statement) is a good one, then the mean score of the favourable group will be more than that of the unfavourable group. For an item (statement), if the mean scores across the two groups are equal or near equal, then that item is to be excluded from the scale.

By comparing the differences, select only those items (statements), which appear to discriminate well between the high and low total scores.

Step-6: Among the remaining items, after dropping insignificant items in step-5.2, the favourable items and unfavourable items are mixed in random order.

Step-7: Now, administer the scale on the respondents and obtain their markings on each item. Then obtain the scores of each respondent by summing the numerical weights marked by that respondent on each item.

5.6.2 Q-sort Scale

The Q-sort scale was developed by William Stephenson. This scale has some commonality with the summated scale. In this scale, respondents are asked to sort a given set of statements (items) and classify them into a predetermined number of categories (piles), usually 11. The number of statements (items) to be included in each category is also defined in advance.

Consider again Example 5.1 of scaling the respondents in which a set of items proposed with respect to a particular attitudinal object, like the role of educational institutions in employment market. The statements in Table 5.14 are reproduced in Table 5.15 with four more statements to demonstrate the Q-sort scale.

Though the Q-sort scale is based on 11 piles, for the purpose of demonstration, only 7 piles are shown in Fig. 5.5 to which the respondents should place the statements based on their belief limited to the number of statements permitted in each pile as indicated in the same figure.

Most strongly agree	Strongly agree	Agree	Undecided	Disagree	Strongly disagree	Most strongly disagree
One item	One item	Two items	Three items	Two items	One item	One item
(+3)	(+2)	(+1)	0	(-1)	(-2)	(-3)

Fig. 5.5 Details of Q-sort scale categories (piles).

In Fig. 5.5, the degrees of agreement/disagreement and the corresponding numerical weights are as follows:

1. Most strongly agree (+3)
2. Strongly agree (+2)

3. Agree (+1)
4. Undecided (0)
5. Disagree (-1)
6. Strongly disagree (-2)
7. Most strongly disagree (-3).

The number of items to be included in each category is shown in the corresponding square of Fig. 5.5.

Each respondent will be asked to sort the given set of items and include them in different piles such that the total number of items included in each pile is equal to its prespecified number of items.

Table 5.15. Sample Set of Statements (Items) for Q-sort Scale

Item no.	Statement
1	There is hectic competition among employment institutions in providing employment opportunities.
2	Some educational institutions are not bothered about employment issue of their graduates.
3	Whatever may be the effort made by educational institutions, it will be very difficult to employ their graduates because of saturation in employment market.
4	Since the average salary in industries is very low, graduates of premier institutions are not willing to accept such low-paid jobs.
5	In technical education, educational institutions are not employing qualified and experienced faculty which will affect employment opportunities adversely.
6	Because of higher pay in some industries, educational institutions are finding it difficult to attract talented graduates/experienced persons for faculty positions.
7	Some educational institutions do not plough back their surplus which badly affect institutional infrastructures development.
8	There is a gap between job specifications in employment market and skill provided in educational institutions.
9	There is a difficulty in implementing most modern syllabus for all educational institutions because of autonomous status conferred to some colleges.
10	Insecurity of job in private colleges makes faculty members to move frequently to other institutions which affects quality of teaching greatly.
11	There is no integrated effort, like symbiotic marketing to employ graduates of different institutions in industries/service organizations.

A sample set of responses of, say, five respondents (subjects) against different items is shown in Table 5.16. Responses in Table 5.16 can be used as input to factor analysis or cluster analysis.

One can form a correlation matrix between the columns of Table 5.16 and it can be further analyzed using factor analysis to form different groups of the respondents such that

the respondents within each group are those who assigned numerical weights to the items with similar pattern.

In case, if the responses in Table 5.16 are given as input to cluster analysis, the output will be clusters of respondents (subjects) such that the respondents within each cluster are those who assigned numerical weights to the items with similar pattern.

Table 5.16 Scores of Respondents on items for Q-sort Scale

Item no.	Respondent				
	1	2	3	4	5
1	+2	-3	-1	+1	+3
2	+1	-1	+3	0	0
3	-3	+2	-1	+3	-1
4	-2	+2	-3	-1	+2
5	0	-1	0	+2	-3
6	+1	+2	+1	0	+2
7	-3	+3	+2	+1	0
8	-1	-1	+1	-3	+1
9	+2	0	+3	0	+3
10	+3	-3	-1	+2	0
11	+3	+2	-1	0	+3

5.7 SCALES FOR STIMULI AND RESPONDENTS

The previous two subsections have been devoted towards scaling stimuli and scaling respondents independently. But, it is also necessary to scale both stimuli and respondents simultaneously.

The methods used for such combined approach are classified based on the type of dimensionality of the items/characteristics/attributes of the stimuli.

The items can be one-dimensional (unidimensional) and multidimensional.

If the items are unidimensional, then cumulative scale can be used. An example of cumulative scale is the Guttman scale/scalogram analysis. If the items are multidimensional, then multidimensional scale can be used.

5.7.1 Cumulative Scaling/Guttman Scale

The cumulative scale was developed by Louis Guttman. Hence, it is called *Guttman scale*. In this scale, each respondent is provided with a set of questions. A cumulative scale exists only if unidimensionality exists among the items/questions. Then, he/she provides 'yes/no' type answer to each of the questions. If the respondent answers 'yes' to the question 2,

then he/she may answer 'yes' to the previous question 1. Similarly, if the respondent answers 'yes' to the question 3, then he/she may answer 'yes' to the preceding questions—question 1 and question 2. From these statements, one can generalize the fact that if a respondent gives his/her answer 'yes' to the n th question in the series, then the respondent may give the same answer to all the questions from 1, 2, 3, ..., up to $n-1$ questions.

Consider the case of finding the effectiveness of the advertisement for admission to an educational institution. The respondents (students) will be provided with the following questions:

1. Would you read the advertisement for admission to the institution fully?
2. Would you buy an application form from the institution after reading its advertisement for admission?
3. Would you submit the filled-in application to the institution?
4. Would you sit for the written test of the institution?
5. Would you appear for group discussion and interview, if you are called for?
6. Would you join the institution if you receive final admission letter?

• If a student answers 'yes' to the question 6, then he may give the same answer to all the preceding—questions from questions 1–5. The score of a respondent is the number of 'yes' answers given by him/her. This scale is based on the cumulative relation between items and the total scores of individuals.

In this example, the serial numbers of the questions indicate the sequential relationship among them. Let us assume that the questions are jumbled as shown below:

- (a) Would you appear for group discussion and interview, if you are called for? (5)
- (b) Would you buy an application form from the institution after reading its advertisement for admission? (2)
- (c) Would you join the institution if you receive final admission letter? (6)
- (d) Would you submit the filled-in application form to the institution? (3)
- (e) Would you read the advertisement for admission to the institution fully? (1)
- (f) Would you sit for the written test of the institution? (4)

Then, the difficulty in answering 'yes' to each question is expressed from most (6) to least (1) within parentheses by the side of that question. If a respondent has answered 'Yes' to a question with the difficulty index, say 4, then he/she may give the same answer to each other question which has the difficulty index less than 4. This fact is analyzed through scalogram analysis as shown in Table 5.17. In the Table 5.17, 'X' represents 'yes' answer and '.' represents 'No' answer. This analysis is only a post facto analysis.

Table 5.17 Scalogram Analysis with Ideal Pattern

Respondent category	Question in terms of difficulty index						Scale score
	6	5	4	3	2	1	
I	X	X	X	X	X	X	6
II		X	X	X	X	X	5
III			X	X	X	X	4
IV				X	X	X	3
V					X	X	2
VI						X	1
VII							0

5.7.2 Multidimensional Scaling

If the number of dimensions is more than one, then the related scales are called *multidimensional* scaling models. In such models, the stimuli are represented as points in the multidimensional space. The stimuli and respondents can be scaled using multidimensional scaling. Under such situation, the attributes which characterize the stimuli or respondents will form the dimensions of the multidimensional space. Some applications of multidimensional scaling are product positioning, brand positioning, market segmentation, group technology cell formation, etc. (For more details on multidimensional scaling, refer to Chapter 12.)

QUESTIONS

1. What is a measurement? What are the characteristics of the measurement, explain them?
2. Give a classification of data collection methods.
3. What is variability methods of data collection? What are the variability methods? Compare and contrast them.
4. Discuss the steps of the paired comparison technique of data collection.
5. Explain the paired comparison technique of data collection with an example.
6. What is 'k out of n' ranking method?
7. Explain the ordered category sorting method.
8. What is the rating method? What are the scales of the rating method? Explain them.

9. What is the quantitative judgment method? What are the techniques of quantitative judgement method? Compare and contrast them.
10. Distinguish between limited response categories under direct judgement methods. Explain them with examples.
11. Explain the concept of fractionation method with an example.
12. What is constant sum technique of direct judgement method? Give an example.
13. What are the types of scale? Explain them. Also, give examples under each type.
14. What are the scales for measuring stimuli? Compare and contrast them.
15. Discuss the steps to construct Thurstone's case-V scale model.
16. Illustrate Thurstone's case-V scale model with an example.
17. Discuss the steps of Osgood's semantic differential scale.
18. Explain Osgood's semantic differential scale with an example.
19. What are the scales for measuring respondents? Compare and contrast them.
20. Explain the steps of constructing summated scale/Likert scale.
21. Illustrate the construction of Likert scale/summated scale with an example.
22. Explain the construction of Q-sort scale.
23. What are the scales for measuring stimuli and respondents? Compare them.
24. What is cumulative scale/Guttman's scale? Explain its aspects.
25. What is multidimensional scaling? Discuss its areas of applications.

CHAPTER

6

PROBABILITY DISTRIBUTIONS

6.1 INTRODUCTION

Probability is defined as the chance of occurrence of an event. Many of the real-life events are subjected to probability. Whether it is the events which occur in the day-to-day life of people or in the day-to-day operation of businesses, one can visualize many examples of probability. Sometime, the probabilistic events of common people may have direct or indirect impact on the performance of business establishments.

The number of festivals/functions/marriages performed in a particular locality may not be predicted well in advance. But, the happening of these events will have some impact on the demand of consumer products sold in different shops of that locality. The demand of the items cannot be predicted accurately because of the uncertainty involved in the occurrence of the major events, viz., festivals/functions/marriages. This necessitates a careful planning of business operations in that locality with proper backward integration with their suppliers.

Like this, one can visualize several other examples of probability in other situations. So, the study of probability and related aspects are highly essential for effective use of necessary tools for data analysis.

Some more examples of probability are listed below:

- Probability of getting head while tossing a coin;
- Probability of getting 3 while rolling a die;
- Probability of rainfall on a particular day;
- Probability of winning the test series of a game by a particular team;
- Probability of getting a huge order from a customer;
- Probability of meeting the due date of a construction project;
- Probability of avoiding a battle at a crucial juncture;
- Probability of winning a battle by a country if battle takes place;
- Probability of survival of an electronic equipment in good operating condition for a specified period;
- Probability of achieving 10 per cent more over the profit of last year;

- Probability of a mother giving birth to a male child;
- Probability of a mother giving birth to a female child;
- Probability of not having strike in a company for the next 10 years;
- Probability that a lot containing 1000 electrical motors received from a supplier will be rejected during sampling inspection;
- Probability of a train reaching a particular station in time.

To understand the concept of probability, one should be clear with the terminologies, viz. experiment, sample space and event.

Experiment Tossing a coin is an example of *experiment* which has the following attributes:

- There will be two or more outcomes of an experiment.
- The list of all outcomes can be specified in advance.
- Actual occurrence of the outcomes cannot be predicted in advance.

Sample space It is the list of all possible outcomes of an experiment. For example, while tossing a coin, the possible outcomes are 'head' and 'tail'. So, the sample space of this experiment consists of head and tail.

Event In the coin-tossing experiment, the occurrence of a particular outcome or a combination of outcomes is called an *event* of the experiment. Consider the example of tossing two similar coins. The possible outcomes are listed below:

First coin	Second coin	Outcome
H	H	HH
H	T	HT
T	H	TH
T	T	TT

The sample space of the experiment is (HH, HT, TH, TT). The events of this experiment are HH, HT and TT, whose frequencies are 1, 2 and 1, respectively.

6.2 APPROACHES TO PROBABILITY

There are three approaches to probability:

- (a) Classical approach to probability.
- (b) Relative frequency approach to probability.
- (c) Subjective approach to probability.

Classical approach to probability

In this approach to probability, the probability of an event in an experiment is given by the following formula:

$$\text{Probability of an event} = \frac{\text{Number of outcomes favourable to the event}}{\text{Total number of outcomes}}$$

Consider the experiment of tossing two coins in which the sample space is (HH, HT, TH, TT), and the events are HH, HT and TT, whose frequencies are 1, 2 and 1, respectively. The probability of getting one head and one tail (HT) is:

$$\text{Probability (HT)} = \frac{\text{Number of times of occurrence of HT}}{\text{Total number of outcomes}} = \frac{2}{4} = 0.5$$

Similarly, the probability of occurrence of the other two events are:

$$\text{Probability (HH)} = \frac{1}{4} = 0.25 \quad \text{and} \quad \text{Probability (TT)} = \frac{1}{4} = 0.25$$

Relative frequency approach to probability

In this approach to probability, the probability of an event in an experiment is given by the following formula:

$$\text{Probability of an event} = \frac{\text{Number of times of occurrence of an event}}{\text{Total number of trials}}$$

Consider the case of a coaching centre giving coaching for plus-two students for seeking admission to engineering degree course in a state under single-window system.

Let the number of students admitted to the coaching class at the coaching centre be 200 and the number of students got admission to the engineering degree course of the state be 80. Then the probability of getting admission to the engineering degree course if a student is coached at the above coaching centre is:

$$P(A) = \frac{\text{Number of students got admission in engineering course}}{\text{Total number of students admitted to coaching centre}} = \frac{80}{200} = 0.4$$

where the event of getting admission in engineering course through the coaching received from the said coaching centre is denoted by A.

Subjective approach to probability

In the subjective approach to probability,

$$\text{Probability of an event} = \frac{\text{Number of successes}}{\text{Total Number of trials}}$$

Consider the case of automation in industries. Generally, there is a fear among employees that any kind of automation will lead to firing of employees. But, the recently conducted

National Productivity Congress concluded that the automation will not have such undesirable effect.

In spite of this opposition, based on the guidelines of the recently held Productivity Congress, the chief productivity engineers of 50 different companies started implementing the scheme of automation in their respective organizations. Exactly during the same time in the next year, 30 successful chief productivity engineers reported their successes to their fellow participants.

In this experiment, let the event of making successful implementation of automation be 'A'. The subjective probability is computed as shown below:

$$P(A) = \frac{\text{Number of companies successfully implemented automation}}{\text{Total number of companies tried automation}}$$

Conditional probability

Conditional probability is the probability of occurrence of an event given that another event has already occurred.

Let A and B are two events. Then the probability of occurrence of A, given that B has occurred is:

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(AB)}{P(B)}$$

Example 6.1 In an organization, out of 200 employees, 40 are having their monthly salary more than Rs. 15,000 and 120 of them are regular takers of Alpha Brand Tea. Out of those 40, who are having their monthly salary more than Rs. 15,000, 20 are regular takers of Alpha Brand Tea. If a particular employee is selected, what is the probability that he is having monthly salary more than Rs. 15,000, if he is a regular taker of Alpha Brand Tea?

Solution. Let A be the event that an employee's monthly salary is more than Rs. 15,000, B be the event that the employee is a regular taker of Alpha Brand Tea. Then

$$P(A \cap B) = P(AB) = \frac{20}{200} = 0.1$$

$$P(B) = \frac{120}{200} = 0.6$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.1}{0.6} = 0.167$$

If a particular employee is selected, and if he is a regular taker of Alpha Brand Tea, then the probability that his monthly salary is more than Rs. 15,000 is 0.167.

Extended formulas If the two events, A and B are independent events, then $P(A/B) = P(A)$. Therefore,

$$P(A \cap B) = P(A/B)P(B) = P(A)P(B)$$

If A and B are two events, then the probability of occurrence of A or B is:

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

If A, B and C are three different events, then the occurrence of A or B or C is:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Bayes' theorem In an experiment, if A and B are two mutually exclusive and collectively exhaustive events and C is another event in the same experiment, then the formulas for $P(A/C)$ and $P(B/C)$ are:

$$P(A/C) = \frac{P(C/A)P(A)}{P(C/A)P(A) + P(C/B)P(B)}$$

$$P(B/C) = \frac{P(C/B)P(B)}{P(C/A)P(A) + P(C/B)P(B)}$$

6.3 PROBABILITY DISTRIBUTIONS

In some researches, after data collection, the next stage is to present the data in the form of probability distributions which will facilitate further analysis of data in more meaningful ways.

The probability distribution can be classified into *discrete* probability distribution and *continuous* probability distribution.

Some examples of discrete probability distribution are Poisson distribution, binomial distribution, the probability distribution of tossing three coins, etc. Exponential distribution, uniform distribution, normal distribution, and *t*-distribution are some examples of continuous probability distribution.

Discrete probability distribution

As mentioned earlier, in an experiment, events can be represented in the form of frequencies which can be easily converted into the respective probabilities by dividing them with the total number of outcomes.

Consider the case of tossing three coins simultaneously. The outcomes of this experiment are shown in Fig. 6.1.

The number of outcomes of this experiment is 8. The events of this experiment are HHH, HHT, HTT and TTT. The frequencies of these events are 1, 3, 3 and 1, respectively.

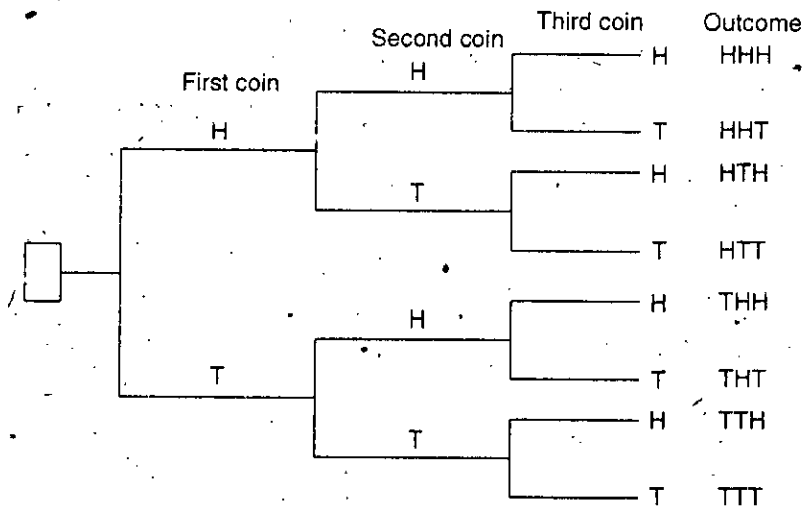


Fig. 6.1 Outcomes of tossing three coins.

The frequency distribution and probability distribution of this experiment are summarized in Table 6.1. Since, the events of this experiment are in discrete terms, the corresponding probability distribution is called *discrete probability distribution*. It should be noted that the sum of the probabilities of all the events is 1. This proves the fact that the events are mutually exclusive and collectively exhaustive.

Table 6.1 Probability Distribution of the Experiment

Event	Number of outcomes of the event	Probability of occurrence of the event
HHH	1	$1/8 = 0.125$
HHT	3	$3/8 = 0.375$
HTT	3	$3/8 = 0.375$
TTT	1	$1/8 = 0.125$

From Table 6.1, the discrete probability distribution of tossing three coins is calculated as follows:

The events of this experiment are HHH, HHT, HTT and TTT. Further, X is defined as a discrete random variable indicating the number of heads in the event occurrence, which is probabilistic.

The events HHH, HHT, HTT and TTT can be defined as follows:

- $X = 0$, when the event is TTT (number of heads is 0)
- $X = 1$, when the event is HTT (number of heads is 1)
- $X = 2$, when the event is HHT (number of heads is 2)
- $X = 3$, when the event is HHH (number of heads is 3)

Based on these assumptions, the probability distribution is:

$$P(X) = \begin{cases} 1/8 & \text{if } X = 0 \\ 3/8 & \text{if } X = 1 \\ 3/8 & \text{if } X = 2 \\ 1/8 & \text{if } X = 3 \\ 0 & \text{otherwise} \end{cases}$$

This is called *probability mass function (p.m.f.)*.

As stated earlier, the sum of the probabilities of the above discrete distribution should be equal to 1, as shown below. Such cumulative function is called *cumulative distribution function (c.d.f.)*:

$$\sum_{i=1}^n P_i(X) = 1$$

where n is the number of events which is 4 in this experiment and $P_i(X)$ is the probability of occurrence of the event i .

Continuous probability distribution

Another type of distribution, known as *continuous probability distribution*, has several practical applications.

Let us consider the following function to demonstrate the concept of the continuous distribution:

$$f(X) = 3X^2 - 3X + 1.5, \quad \text{where } 0 \leq X \leq 1 \\ = 0, \quad \text{otherwise}$$

For any probability distribution, the value of the cumulative distribution in the specified range should be 1.

The value of the cumulative distribution can be obtained by integrating $f(X)$, as shown below:

$$\int_0^1 f(X) dX = \int_0^1 (3X^2 - 3X + 1.5) dX = \left[\frac{3X^3}{3} - \frac{3X^2}{2} + 1.5X \right]_0^1 = 1$$

Since the value of the cumulative function of $f(X)$ is 1, $f(X)$ is a probability distribution $[P(X)]$ as represented below. This is called *probability density function (p.d.f.)*:

$$P(X) = 3X^2 - 3X + 1.5, \quad \text{where } 0 \leq X \leq 1 \\ = 0, \quad \text{otherwise}$$

In this distribution, the variable X is a continuous random variable because its value is continuous in the range from 0 to 1. Hence, the probability distribution $P(X)$ is a continuous probability distribution. The cumulative function of a probability density function is called *cumulative density function (c.d.f.)*.

6.3.1 Binomial Distribution

Binomial distribution is a discrete probability distribution. This distribution is based on Bernoulli process.

Consider the experiment of tossing a coin. If a coin is tossed, the outcome may be either head or tail. So, the sample space is: (Head, Tail). In this experiment, the number of events is equal to the number of outcomes which is equal to 2. The occurrence of head will prevent the occurrence of tail. Similarly, the occurrence of tail will prevent the occurrence of head. This is an example of two mutually exclusive and collectively exhaustive events. So, the probability of occurrence of head in a trial is 0.5; similarly, the probability of occurrence of tail in a trial is also 0.5.

In a game between two persons using a coin-tossing experiment, let the occurrence of head in a trial be a success to one person and the occurrence of tail be a failure to the same person in the game.

So, in a trial of tossing a coin, the probability of a success (p) to the person is 0.5 and the probability of a failure ($q = 1 - p$) to the same person is 0.5.

If n repeated trials are performed, then the objective of the game may be to find the probability of having X successes for the person. This experiment is termed as *Bernoulli process* in which n repeated trials are performed with the following assumptions:

- (a) In the experiment, there are only two mutually exclusive and collectively exhaustive events.
- (b) The probability of occurrence of the events of the experiment are the same in all the trials.
- (c) In all the n trials, the observations are independent of one another.

Based on these fundamentals, the binomial probability distribution is represented as follows:

$$P(X \text{ successes in } n \text{ trials given } p) = {}^nC_X p^X q^{n-X}$$

where $X = 0, 1, 2, 3, \dots, n$. The above distribution may be represented as shown below for convenience.

$$P(X, n, p) = {}^nC_X p^X q^{n-X}, \quad \text{where } X = 0, 1, 2, 3, \dots, n$$

where n is the number of trials; p , the probability of success in a trial; q , the probability of failure in a trial ($1 - p$); X , the random variable representing the number of successes; and $P(X, n, p)$, the probability of having X successes in n trials if the probability of success in a trial is p .

In this probability mass function, n and p are parameters.

The cumulative distribution function of the binomial distribution is:

$$F(X, n, p) = \sum_{X=0}^n P(X, n, p) = \sum_{X=0}^n ({}^nC_X p^X q^{n-X}) = 1$$

Example 6.2 If a coin is tossed for 10 times; what is the probability of having (a) 0 head, (b) 3 heads, (c) at most 2 heads, (d) at least 3 heads?

Solution. The number of trials, $n = 10$. In this coin-tossing experiment, the appearance of a head is treated as a success. Hence,

$$p = 0.5 \quad \text{and} \quad q = 1 - p = 1 - 0.5 = 0.5$$

$$(a) \quad P(X = 0, 10, 0.5) = {}^{10}C_0 p^0 q^{10-0} = {}^{10}C_0 (0.5)^0 (0.5)^{10-0} = 0.0009766$$

$$(b) \quad P(X = 3, 10, 0.5) = {}^{10}C_3 p^3 q^{10-3} = {}^{10}C_3 (0.5)^3 (0.5)^{10-3} = 0.11719$$

$$\begin{aligned} (c) \quad P(X \leq 2, 10, 0.5) &= \sum_{X=0}^2 {}^{10}C_X p^X q^{n-X} \\ &= P(X = 0, 10, 0.5) + P(X = 1, 10, 0.5) + P(X = 2, 10, 0.5) \\ &= {}^{10}C_0 (0.5)^0 (0.5)^{10-0} + {}^{10}C_1 (0.5)^1 (0.5)^{10-1} + {}^{10}C_2 (0.5)^2 (0.5)^{10-2} \\ &= 0.054684 \end{aligned}$$

$$\begin{aligned} (d) \quad P(X \geq 3, 10, 0.5) &= \sum_{X=3}^{10} {}^{10}C_X p^X q^{n-X} = 1 - \sum_{X=0}^2 {}^{10}C_X p^X q^{n-X} \\ &= 1 - \sum_{X=0}^2 {}^{10}C_X p^X q^{10-X} \\ &= 1 - 0.054684 \\ &= 0.945316 \end{aligned}$$

Example 6.3 Based on past experience, the quality control engineer of Heavy Electrical Limited has estimated that the probability of commissioning each project in time at a client site is 0.9. The company is planning to commission 5 such projects in the forthcoming year. Find the probability of commissioning (a) no project in time, (b) two projects in time, (c) at most one project in time, and (d) at least two projects in time.

Solution. The number of trials, $n = 5$.

For this example, the event of commissioning a project is treated as a success. Hence,

$$p = 0.9 \text{ and } q = 1 - p = 1 - 0.9 = 0.1$$

$$(a) P(X = 0, 5, 0.9) = {}^nC_X p^X q^{n-X} = {}^5C_0 (0.9)^0 (0.1)^{5-0} = 0.00001$$

$$(b) P(X = 2, 5, 0.9) = {}^nC_X p^X q^{n-X} = {}^5C_2 (0.9)^2 (0.1)^{5-2} = 0.0081$$

$$(c) P(X \leq 1, 5, 0.9) = \sum_{X=0}^1 {}^5C_X (0.9)^X (0.1)^{5-X} = P(X = 0, 5, 0.9) + P(X = 1, 5, 0.9) = 0.00046$$

$$(d) P(X \geq 2, 5, 0.9) = \sum_{X=2}^5 {}^nC_X p^X q^{n-X} = 1 - \sum_{X=0}^1 {}^5C_X (0.9)^X (0.1)^{5-X}$$

$$= 1 - 0.00046,$$

$$= 0.99954$$

Mean and variance of binomial distribution

The mean (μ) and variance (σ^2) of the binomial distribution are given by the following formulas:

$$\mu = np \quad \text{and} \quad \sigma^2 = npq$$

where n is the number of trials; p the probability of success in a trial, q , the probability of failure in a trial.

6.3.2 Poisson Distribution

Poisson distribution is a discrete probability distribution. This is usually used to represent the number of occurrence of an event in one unit of time.

Consider the example of vehicles arriving at a toll gate. The number of vehicles arriving per hour at the toll gate is termed as the arrival rate of vehicles. The Poisson distribution can be used to represent the arrival rate of vehicles at the toll gate. A sample arrival rate frequencies based on the observations of, say, 100 one-hour intervals is given in Table 6.2. Let the minimum and maximum arrival rate of vehicles be 20 and 27, respectively. One can find the mean (μ) and variance (σ^2) of the arrival rate from the data given in Table 6.2.

Table 6.2 Arrival Rate Distribution

Number of arrivals/hr	20	21	22	23	24	25	26	27
Number of 1-hr intervals	10	25	30	18	8	5	3	1

The shape of Poisson distribution is shown in Fig. 6.2.

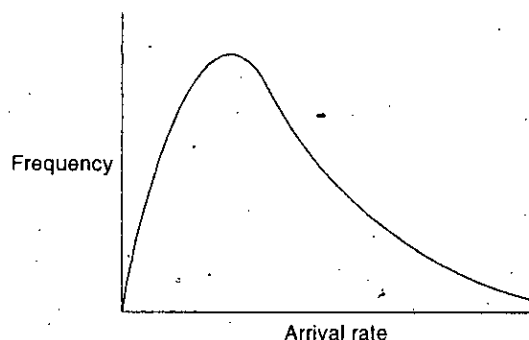


Fig. 6.2 Poisson distribution.

In this distribution, the probability of occurrence for the smaller values of the random variable X is relatively higher and it goes on increasing up to a point and then it decreases, as shown in Fig. 6.2.

The formula for the Poisson probability distribution is:

$$P(X) = \frac{e^{-\delta} \delta^X}{X!}, \quad X = 0, 1, 2, 3, \dots, \infty$$

In this probability mass function, δ is the only parameter. Alternatively, it can be represented as:

$$P(X, \delta) = \frac{e^{-\delta} \delta^X}{X!}, \quad X = 0, 1, 2, 3, \dots, \infty$$

where X is the random variable representing a specific rate of occurrence of an event (a specific arrival rate); δ the mean rate of occurrence of the event (mean arrival rate), which is the only parameter of this distribution; $P(X, \delta)$, the probability that the number of occurrences of the event (number of arrivals) per unit time is X , if the mean rate of occurrence of the event (mean arrival rate) is δ .

Mean and variance of Poisson distribution

This is a distribution for which the mean (μ) as well as the variance (σ^2) is equal to the mean rate of occurrence of the event (mean arrival rate), δ .

$$\text{Mean } (\mu) = \text{Variance } (\sigma^2) = \delta$$

Approximation of binomial distribution to Poisson distribution

The binomial distribution can be approximated to Poisson distribution under any one of the following conditions:

(i) $n \geq 20$ and $p \leq 0.05$

(ii) $n \geq 100$ and $np \leq 10$.

If any one of the above two conditions is satisfied, then the mean of the Poisson distribution is calculated using the formula: $\delta = np$.

Example 6.4 The arrival rate of customers arriving at a bank counter follows Poisson distribution with a mean arrival rate of 4 per 10 minutes interval. Find the probability that

- (a) exactly 0 customer will arrive in 10 minutes interval
- (b) exactly 2 customers will arrive in 10 minutes interval
- (c) at most 2 customers will arrive in 10 minutes interval
- (d) at least 3 customers will arrive in 10 minutes interval.

Solution. The arrival rate in 10 minutes interval, $\delta = 4$. Then

$$(a) P(X=0, 4) = \frac{e^{-\delta} \delta^X}{X!} = \frac{e^{-4} 4^0}{0!} = 0.018$$

$$(b) P(X=2, 4) = \frac{e^{-\delta} \delta^X}{X!} = \frac{e^{-4} 4^2}{2!} = 0.146$$

$$(c) P(X \leq 2, 4) = \sum_{X=0}^2 \frac{e^{-\delta} \delta^X}{X!} = \frac{e^{-4} 4^0}{0!} + \frac{e^{-4} 4^1}{1!} + \frac{e^{-4} 4^2}{2!} = 0.238$$

$$(d) P(X \geq 3, 4) = 1 - \sum_{X=0}^2 \frac{e^{-\delta} \delta^X}{X!} = 1 - \left(\frac{e^{-4} 4^0}{0!} + \frac{e^{-4} 4^1}{1!} + \frac{e^{-4} 4^2}{2!} \right) = 1 - 0.238 = 0.762$$

Example 6.5 The quality control assistant takes a sample of 25 units per hour interval at a particular workstation of a production line and inspects them one by one. Based on the past experience, he has estimated that the probability of one unit will be defective is 0.04. Find the probability that

- (a) no piece in the sample is defective.
- (b) 3 pieces in the sample will be defective.
- (c) at most 2 pieces will be defective.
- (d) at least 3 pieces will be defective.

Solution. The probability that an inspected piece will be defective (p) is 0.04. For this problem, n is 25 and p is 0.04. This follows binomial distribution. It can be approximated to Poisson distribution, because n is more than 20 and p is less than 0.05.

Therefore, mean of the Poisson distribution, δ , is computed as: $\delta = np = 25 \times 0.04 = 1$.

$$(a) P(X=0, 1) = \frac{e^{-\delta} \delta^X}{X!} = \frac{e^{-1} 1^0}{0!} = 0.368$$

$$(b) P(X=3,1) = \frac{e^{-\delta} \delta^X}{X!} = \frac{e^{-1} 1^3}{3!} = 0.061$$

$$(c) P(X \leq 2, 1) = \sum_{X=0}^2 \frac{e^{-\delta} \delta^X}{X!} = \frac{e^{-1} 1^0}{0!} + \frac{e^{-1} 1^1}{1!} + \frac{e^{-1} 1^2}{2!} = 0.92$$

$$(d) P(X \geq 3, 1) = 1 - \sum_{X=0}^2 \frac{e^{-\delta} \delta^X}{X!} = 1 - \left(\frac{e^{-1} 1^0}{0!} + \frac{e^{-1} 1^1}{1!} + \frac{e^{-1} 1^2}{2!} \right) = 0.08$$

6.3.3 Uniform Distribution

Uniform distribution is a continuous probability distribution which has wider practical applications. More specifically, it has more use in simulation.

Let us assume that the distribution of the daily demand of a product is uniformly distributed with 10 ± 2 units. The minimum daily demand is 8 and the maximum daily demand is 12. Let the minimum daily demand (8) and the maximum daily demand (12) be a and b , respectively.

The corresponding distribution is shown in Fig. 6.3. This is called a probability density function, because it is a continuous distribution.

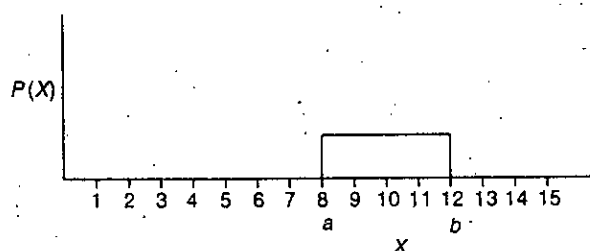


Fig. 6.3 Uniform distribution.

In Fig. 6.3; X is a continuous random variable. The probability of occurrence for the values of the random variable, X in the range a to b is constant and it is zero for all the values of the random variable outside the interval a to b . The formula for the corresponding uniform distribution is:

$$P(X) = \frac{1}{b-a}, \quad a \leq X \leq b$$

$$= 0, \quad \text{otherwise}$$

where a and b are the lower limit and upper limit for the random variable X .

As in the discrete probability distribution, here too, the sum of the probabilities should be equal to 1, as shown below:

$$f(x) = \int_a^b P(x) dx = \int_a^b \frac{1}{b-a} dx = \frac{b-a}{b-a} = 1$$

This cumulative function is called *cumulative density function*, $F(X)$.

The formulas for the mean (μ) and variance (σ^2) of the uniform distribution are:

$$\mu = \frac{a+b}{2} \quad \text{and} \quad \sigma^2 = \frac{(b-a)^2}{12}$$

Example 6.6 In a private canteen, the daily demand for packed meals follows uniform distribution as presented below:

$$P(X) = \frac{1}{450-230}, \quad 230 \leq X \leq 450$$

$$= 0, \quad \text{otherwise}$$

If the service level of satisfying the demand of the canteen is 0.8, find the highest possible demand which can be satisfied with respect to the given service level (cumulative probability).

Solution. The uniform distribution of the demand is:

$$P(X) = \frac{1}{450-230}, \quad 230 \leq X \leq 450$$

$$= 0, \quad \text{otherwise}$$

Let Q be the demand value with respect to the service level of the canteen, which is nothing but the cumulative probability of 0.8. The cumulative probability distribution of the given probability distribution in the range 230 to Q is obtained as:

$$\int_{230}^Q \frac{1}{450-230} dX = \left[\frac{X}{450-230} \right]_{230}^Q = \frac{Q-230}{220}$$

The calculation of demand (Q) for a given cumulative probability is:

$$\frac{Q-230}{220} = 0.8 \quad \text{or} \quad Q = 406 \text{ units}$$

Therefore, the demand of the product which can be satisfied with respect to the given service level is 406 units.

6.3.4 Exponential Distribution

Exponential distribution is a continuous probability distribution. The shape of this distribution is shown in Fig. 6.4.

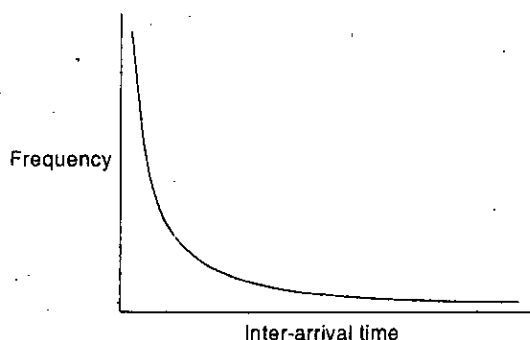


Fig. 6.4 Exponential distribution (negative exponential distribution).

Actually, it is a negative exponential distribution. For convenience, it is called as *exponential* distribution. In this distribution, the probability of occurrence for the smaller values of the random variable is very high and it quickly dampens to smaller values when the value of the random variable increases. Later, it becomes almost constant for larger values of the random variable. This distribution is used to represent the time interval between consecutive occurrence of an event, like inter-arrival time of customers, service time for customers, inter-failure time of equipments, etc.

Consider the example of the service time of customers in a queuing system. Generally, the service time in a queuing example is a random variable which follows exponential distribution. Let the number of customers served per unit time (service rate) be μ . Therefore, the service time (τ) is $1/\mu$.

The formula for the exponential distribution is shown below. This is called *probability density function*, because it is a continuous distribution.

$$P(X) = \frac{1}{\tau} e^{-X/\tau}, \quad \text{where } X > 0 \text{ and } \tau > 0$$

• or

$$P(X) = \mu e^{-\mu X}, \quad \text{where } X > 0 \text{ and } \mu > 0$$

The mean (μ) and variance (σ^2) of the exponential distribution are τ and τ^2 , respectively.

The cumulative density function, $[F(X)]$ of the exponential distribution is:

$$F(X) = 1 - e^{-\mu X}, \quad \text{where } X > 0 \text{ and } \mu > 0$$

or

$$F(X) = 1 - e^{-X/\tau}, \quad \text{where } X > 0 \text{ and } \tau > 0$$

Similarly, the inter-arrival time of customers (τ), who are entering into a queueing system will also follow exponential distribution, if α is the arrival rate of the customers ($1/\tau$). One can revise the above formulas, accordingly.

Example 6.7 In an international airport, the service time of a terminal follows exponential distribution. The service rate of a terminal serving the flights is 30 per day. Find the probability that the service time of the terminal in clearing a flight is:

- (a) less than 0.5 hr.
- (b) more than 0.75 hr.

Solution. (a) Service rate (μ) of the terminal = 30/day = 30 per 24 hr = 1.25/hr
The cumulative probability distribution is:

$$(a) F(X) = 1 - e^{-\mu X}, \text{ where } X \geq 0 \text{ and } \mu \geq 0, P(X \leq 0.5) = 1 - e^{-1.25 \times 0.5} = 0.465$$

$$(b) P(X \geq 0.75) = 1 - (1 - e^{-\mu X}) = e^{-\mu X} = e^{-1.25 \times 0.75} = 0.392$$

Example 6.8 In a mainframe computer centre, execution time of programs follows exponential distribution. The average execution time of the programs is 5 minutes. Find the probability that the execution time of programs is:

- (a) less than 4 minutes
- (b) more than 6 minutes.

Solution. (a) Execution time (τ) of the programs = 5 minutes. The cumulative probability distribution is:

$$F(X) = 1 - e^{-X/\tau}, \text{ where } X \geq 0 \text{ and } \tau \geq 0$$

Then

$$P(X \leq 4) = 1 - e^{-4/5} = 0.551$$

$$(b) P(X \geq 6) = 1 - (1 - e^{-X/\tau}) = e^{-X/\tau} = e^{-6/5} = 0.301$$

6.3.5 Normal Distribution

Normal distribution is a continuous probability distribution. The behaviour of many of the real-life situations can be modelled as normal distribution.

Some examples which follow normal distribution are as follows:

- Monthly salary of employees in a locality
- Internal diameter of bearings produced in a company
- Marks of students in an entrance test
- Height of employees in a company
- Weight of employees in a company.

The shape of the normal distribution is shown in Fig. 6.5.

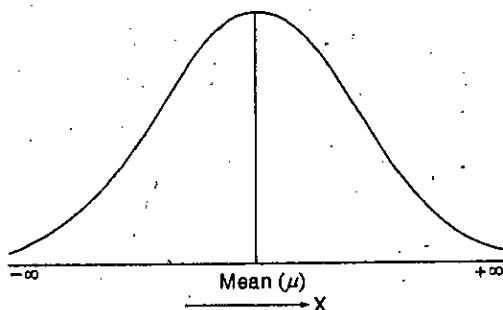


Fig. 6.5 Normal distribution.

The shape of normal distribution is bell shaped. This means that the distribution is symmetrical about its mean. The formula of this distribution is shown below. It is a probability density function, as it is a continuous distribution.

$$P(X) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(X - \mu)^2}{2\sigma^2} \right], \quad \text{where } -\infty < X < \infty$$

In this formula, μ is the mean of the distribution; σ^2 , the variance of the distribution; σ , the standard deviation of the distribution; and X , a normal random variable. Further, μ and σ are the parameters of the normal distribution.

The formula for the cumulative density distribution of the normal distribution in the range of values of the random variable from $-\infty$ to X is shown as

$$F(X) = \int_{-\infty}^X \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(X - \mu)^2}{2\sigma^2} \right] dx$$

It is impossible to provide separate table values of the normal distribution for different combinations of the parameters μ and σ , because the number of such combinations is not finite. Hence, there is a necessity for a standard distribution to use the normal distribution. Such distribution is called *standard normal distribution* which has its mean and variance as 0 and 1, respectively.

If the observations of a real-life problem follow the normal distribution with mean, μ and variance, σ^2 , then its random variable can be converted into a standard normal random variable using the following transformation:

$$Z = \frac{X - \mu}{\sigma}$$

where Z is a standard normal variable. The corresponding distribution is called standard normal distribution, whose formula is as given below:

$$P(Z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}Z^2\right), \text{ where } -\infty < Z < \infty$$

The mean and variance of this standard normal distribution are 0 and 1, respectively.

Example 6.9 In a survey with a sample of 200 respondents, the monthly income of the respondents follows normal distribution with its mean (μ) and standard deviation (σ) as Rs. 10,000 and Rs. 2,000, respectively. Assume the significance level α as 0.05 and answer the following:

- What is the probability that the monthly income is less than Rs. 11,000? Also, find the number of respondents having income less than Rs. 11,000.
- What is the probability that the monthly income is more than Rs. 12,000? Also, find the number of respondents having income more than Rs. 12,000.
- What is the probability that the monthly income is in between Rs. 7,000 and Rs. 11,200? Also, find the number of respondents having income in between Rs. 7,000 and Rs. 11,200.

Solution. Let X be a random variable representing the monthly income of respondents. It follows normal distribution with mean (μ) and standard deviation (σ) as Rs. 10,000 and Rs. 2,000, respectively. Then

$$\begin{aligned} \text{(a) } P(X \leq 11000) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{11000 - 10000}{2000}\right) \\ &= P(Z \leq 0.5) \\ &= 0.6915 \end{aligned}$$

The probability that the monthly income is less than Rs. 11,000 is 0.6915. The number of respondents having income less than Rs. 11,000 = 200×0.6915 = 138.3 $\approx$ 138.

$$\begin{aligned} \text{(b) } P(X \geq 12000) &= P\left(\frac{X - \mu}{\sigma} \geq \frac{12000 - 10000}{2000}\right) \\ &= P(Z \geq 1) \\ &= 1 - P(Z \leq 1) \\ &= 1 - 0.8413 \\ &= 0.1587 \end{aligned}$$

The probability that the monthly income is more than Rs. 12,000 is 0.1587. Therefore, The number of respondents having income more than Rs. 12,000 = 200×0.1587 = 31.74 $\approx$ 32.

$$(c) P(7000 \leq X \leq 11200) = P(X \leq 11200) - P(X \leq 7000)$$

$$= P\left(\frac{X - \mu}{\sigma} \leq \frac{11200 - 10000}{2000}\right) - P\left(\frac{X - \mu}{\sigma} \leq \frac{7000 - 10000}{2000}\right)$$

$$= P(Z \leq 0.6) - P(Z \leq -1.5)$$

$$= P(Z \leq 0.6) - [1 - P(Z \leq 1.5)]$$

$$= 0.7257 - 1 + 0.9332$$

$$= 0.659$$

The probability that the monthly income is between Rs. 7,000 and Rs. 11,200 is 0.659. The number of respondents having income in between Rs. 7,000 and Rs. 11,200 = $200 \times 0.659 = 131.8 \approx 132$.

QUESTIONS

1. Define probability and give at least ten real-life examples involving probability.
2. Define the following with examples: (a) experiment, (b) outcomes, (c) sample space and (d) event.
3. What are the approaches to probability? Explain them with suitable examples.
4. What is conditional probability? Give an example.
5. State Bayes' theorem and illustrate it with an example.
6. Define each of the following with an example.
 - (a) Probability mass function
 - (b) Cumulative distribution function
 - (c) Probability density function
 - (d) Cumulative density function.
7. Construct the probability mass functions of (a) tossing 2 coins, (b) tossing 3 coins and (c) rolling two dice.
8. What is Bernoulli process? Explain its use in binomial distribution.
9. Discuss the applications of Poisson distribution.
10. Discuss the approximation of binomial distribution to Poisson distribution.
11. Discuss the applications to exponential distribution.
12. Define and explain the following: (a) uniform distribution and (b) normal distribution.
13. What is the standard normal distribution? Explain its importance.
14. In an organization, out of 250 employees, 55 are having their monthly salary more than Rs. 20,000 and 150 of them are regular takers of Beta Brand Tea. Out of those 55 who

- are having their monthly salary more than Rs. 20,000, 30 are regular takers of Beta Brand Tea. If a particular employee is selected, what is the probability that he is having monthly salary more than Rs. 20,000, if he is a regular taker of Beta Brand Tea?
15. In an organization, out of 20 salesmen, 10 are having their monthly sales revenue more than Rs. 25 lakhs and 15 of them are regular user of computer softwares. Out of those 10 who are having their monthly sales revenue more than Rs. 25 lakhs, 6 are regular user of computer softwares. If a particular salesman is selected, what is the probability that he is having monthly sales revenue more than Rs. 25 lakhs if he is a regular user of computer softwares?
 16. If a coin is tossed for 15 times, what is the probability of having (a) 0 head, (b) 5 heads, (c) at most 3 heads, (d) at least 4 heads?
 17. If a coin is tossed 10 times, what is the probability of appearance of (a) 5 heads (b) 4 heads (c) at most 4 heads (d) at least 5 heads?
 18. Based on past experience, the chief of a consultancy organization has estimated that the probability of completing each project in time is 0.7. The company is planning to execute 12 such projects in the forthcoming quarter. Find the probability of completing (a) no project in time, (b) four projects in time, (c) at most 3 projects in time, (d) at least 4 projects in time?
 19. Based on past experience, the quality control engineer of a Heavy Electrical Limited has estimated that the probability of commissioning each project in time at a client site is 0.8. The company is planning to commission 10 such projects in the forthcoming year. Find the, probability of commissioning (a) no project in time, (b) three projects in time, (c) at most two projects in time, (d) at least three projects in time?
 20. The arrival rate of customers arriving at a petrol bunk follows Poisson distribution with a mean arrival rate of 10 per 15 minutes interval. Find the probability that (a) no customer will arrive in 15 minutes interval, (b) exactly 3 customers will arrive in 15 minutes interval, (c) at most 3 customers will arrive in 15 minutes interval, (d) at least 4 customers will arrive in 15 minutes interval.
 21. The arrival rate of vehicles arriving at a toll gate follows Poisson distribution with a mean arrival rate of 50 vehicles per hour. Find the probability that (a) no vehicle will arrive in one hour; (b) exactly 40 vehicles will arrive in one hour; (c) at most 5 vehicles will arrive in one hour; (d) at least 6 vehicles will arrive in one hour.
 22. The quality control assistant takes a sample of 20 units per one hour at a particular workstation of a production line and inspects them one-by-one. Based on the past experience, he has estimated that the probability of one unit will be defective is 0.10. Find the following:
 - (a) The probability that no piece in the sample is defective,
 - (b) The probability that exactly 4 pieces in the sample will be defective,

- (c) The probability that at most 4 pieces will be defective,
 (d) The probability that at least 5 pieces will be defective.
23. An auditing team takes a sample of 40 files per day for auditing in an organization. Based on the past experience, he has estimated the probability of one file which will deserve audit objection as 0.2. Find the following:
- (a) The probability that no file in the sample is without audit objection,
 (b) The probability that 3 files in the sample will be having some audit objections,
 (c) The probability that at most 2 files will be having some audit objections,
 (d) The probability that at least 3 files will be having some audit objections.
24. In a private canteen, the daily demand for packed meals follows uniform distribution as presented below:

$$P(X) = \frac{1}{500 - 200}, \quad 200 \leq X \leq 500$$

$$= 0, \quad \text{otherwise}$$

If the service level of satisfying the demand of the canteen is 0.95, find the highest possible demand which can be satisfied with respect to the given service level (cumulative probability).

25. In a gas agency, the daily demand for cylinders follows uniform distribution as:

$$P(X) = \frac{1}{300 - 150}, \quad 150 \leq X \leq 300$$

$$= 0, \quad \text{otherwise}$$

If the service level of satisfying the demand of the cylinders is 0.85, find the highest possible demand for the cylinders which can be satisfied with respect to the given service level (cumulative probability).

26. In a company, the service time for repairing machineries follows exponential distribution. The service rate of a maintenance crew repairing machineries is 15 per day. Find the probability that the service time of the crew in clearing a machine is (a) less than 0.5 hr, (b) more than 1.5 hrs.
27. In an international airport, the service time for serving flights by a terminal follows exponential distribution. The service rate of a terminal serving the flights is 20 per day. Find the probability that the service time of the terminal in clearing a flight is (a) less than 0.45 hr, (b) more than 1 hr.
28. At the security check counter of an international airport, checking time of passengers follows exponential distribution. The mean checking time of the passengers is 2 minutes. Find the probability that the checking time of passengers is (a) less than 1 minute, (b) more than 3 minutes.
29. In a mainframe computer centre, execution time of programs follows exponential distribution. The average execution time of the programs is 3 minutes. Find the probability that the execution time of programs is (a) less than 2 minutes, (b) more than 4 minutes.

30. In a survey with a sample of 300 respondents, the monthly income of the respondents follows normal distribution with its mean and standard deviation as Rs. 15,000 and Rs. 3,000, respectively. Answer the following:
- (a) What is the probability that the monthly income is less than Rs. 12,000? Also, find the number of respondents having income less than Rs. 12,000.
 - (b) What is the probability that the monthly income is more than Rs. 16,000? Also, find the number of respondents having income more than Rs. 16,000.
 - (c) What is the probability that the monthly income is in between Rs. 10,000 and Rs. 17,000? Also, find the number of respondents having income in between Rs. 10,000 and Rs. 17,000.
31. In a survey with a sample of 200 private clinics, the daily number of patients treated follows normal distribution with its mean and standard deviation as 75 and 15, respectively. Answer the following:
- (a) What is the probability that the daily number of patients treated is less than 80? Also, find the number of clinics having the daily number of patients treated less than 80.
 - (b) What is the probability that the daily number of patients treated is more than 60? Also, find the number of clinics having the daily number of patients treated more than 60.
 - (c) What is the probability that the daily number of patients treated is in between 65 and 85? Also, find the number of clinics having the daily number of patients treated in between 65 and 85.

CHAPTER

7

SAMPLING METHODS AND DISTRIBUTIONS

7.1 INTRODUCTION

The realities, such as residents in a region, account holders of a bank, educational institutions in a state, employees in an industrial estate, collection of machines in an industry, finished products at the final inspection stage in a factory, students in a college, etc., are considered to be different populations. Analysts may be interested in making inferences about some attribute(s) of each of the above populations. For example, the income distribution of the residents in a region may be of importance in a study. Similarly, a particular bank may be keen in assessing the satisfaction of its account holders through a survey. A state government may be interested in periodical analysis of the infrastructure facilities of different educational institutions in that state. Like this, one can imagine the purpose(s) of the respective study of the remaining systems which are mentioned above.

Here, the word *population* means the entire spectrum of a system of interest. For example, the population representing the account holders of a bank consists of individuals who are having accounts with the bank. The account holders are termed as *members/units* of the population. The members of a population need not be human beings as can be inferred from the examples of educational institutions in a state, collection of machines in a factory, etc.

Census versus sampling

The inference of each of the examples cited earlier can be drawn by analyzing the respective data. So, data collection is an important activity in this type of research studies. There are two options for data collection: *census* and *sampling*.

'Census' is the process of obtaining responses from/about each of the members of the population. The determination of the size of population of a country is an example of census. This process is also known as *population census*. While collecting data about the profile of the residents in a region to plan a marketing strategy of a product, under census method, one has to collect data from each individual of the population. If the size of the population is manageable, one can resort to census method; otherwise, this method is seldom feasible. So, under this situation, the alternate method of data collection is sampling.

'Sampling' is a process of selecting a subset of randomized number of members of the population of a study and collecting data about their attributes. The limited members of the population selected for sampling are called as *sampling units*. Based on the data of the sample, the analyst will draw inference about the population.

Following are main advantages of sampling:

- Less time taken to collect data;
- Less cost for data collection;
- It is the only option while collecting data using a destructive testing;
- Physical impossibility of complete enumeration necessitates sampling;
- More accuracy of data collected due to its limited size.

The size of the number of sampling units is relatively smaller when compared to that of the population, which results in reduced time and cost of data collection.

In some situation, data collection may be through destructive testing of units of a product of a company which will leave no units for sales. Under such situation, sampling is the only option.

As stated earlier, if the size of the population is too large, it is impossible to collect data through complete enumeration (census) method. This necessitates the use of sampling.

In complete enumeration method of data collection, since data is collected about each unit of the population, there is a greater possibility of error in data collection. Whereas, in sampling, there is more accuracy of data about the sampling units because of its limited number of sampling units.

The notations of the parameters of population and sample are summarized in Table 7.1.

Table 7.1 Notations of Parameters of Population and Sample

Measure	Population	Sample
Mean	μ	$\bar{X}$
Variance	σ^2	S^2
Sample size	N	n
Proportion	P	p

Sampling frame

The complete list of all the members/units of the population from which each sampling unit is selected is known as *sampling frame*. It should be free from error. A perfect sampling frame will contain its each unit only once. The sampling frame must be complete, accurate, adequate and up to date, as explained below:

- Presence of all legitimate units in a frame makes it complete.
- Absence of non-existing units in the frame improves its accuracy.

- The structure of the frame should make it adequate in terms of covering the entire population *unlike* the units from telephone directory of a region which has a limited subset of the population of that region.
- The units of the frame must be up to date in terms of its content.

In the following sections, sampling methods and sampling distributions are presented.

7.2 SAMPLING METHODS

Sampling methods can be classified into *probability sampling* and *non-probability sampling* methods.

In 'probability sampling', each unit of the population has a probability of being selected as an unit of the sample. But, this probability varies from one method to another method of probability sampling. This type of sampling is more rigorous and free from biases.

In 'non-probability sampling', there may be instances that certain units of the population will have zero probability of selection, because judgment, biases and convenience of the interviewers are considered to be the criteria for the selection of sample units of such sampling.

7.2.1 Probability Sampling Methods

Methods of probability sampling are listed as follows:

1. Simple random sampling
2. Systematic sampling
3. Stratified sampling
4. Cluster sampling
5. Multi-stage sampling.

Simple random sampling

Let N be the number of units of the population; n be the number units of the sample ($n < N$). There are two ways of performing simple random sampling, viz., simple random sampling with replacement of units and simple random sampling without replacement of units.

Simple random sampling with replacement In this method of sampling, each unit of the population has the equal probability of being selected as an unit of the sample given by the following formula:

$$\text{The probability of selection of each unit} = \frac{1}{N}$$

This means that the first unit of the sample will be selected from the population with a probability of $1/N$. The selection of the unit from the sample can be done using either random number table if it is done manually or using Monte-Carlo simulation, if it is done

using computer. Each of the remaining units required for the sample is selected in the same way using the same probability of selection.

The random number table contains, say, four-digit numbers in a rectangular format with rows and columns. The steps of the simple random sampling with replacement are presented as follows:

1. If N is a three-digit number, say 500, representing the size of the population, then the units of the population are numbered as 1, 2, 3, ..., 500.
2. Fix the initial position in the random number table.
3. Read the first-three digits of that number and select the unit from the population with that serial number as the first unit of the sample. Later, replace that unit in the population.
4. Then decide the direction of movement for selecting the next number either to the right or below with reference to the current position (initial position) in the random number table.
5. Accordingly select the first-three digits of the next number in the random number table.
6. Then select the unit from the population with the corresponding serial number as the second unit of the sample. Later, replace that unit in the population.
7. Repeat steps 4–6 until the required number of units of the sample are selected from the population.

(Note: In this process, the number at the extreme right of a row or at the bottom most of a column is reached, then switch over to the left of the next row or top of the next column, respectively.)

Simple random sampling without replacement In this method of sampling, each unit of the population has a varying probability of being selected as an unit of the sample given by the following formula:

$$\text{The probability of selection of the first unit} = \frac{1}{N}$$

$$\text{The probability of selection of the second unit} = \frac{1}{N-1}$$

∴

$$\text{The probability of selection of the } n\text{th unit} = \frac{1}{N-(n-1)}$$

The units of the sample are selected from the population based on the respective probabilities given above using Monte-Carlo simulation.

The steps of Monte-Carlo simulation are as follows:

1. Find the probability, $1/N$. If $N = 10$, then $1/N$ is 0.1
2. Find the cumulative probabilities of the N intervals by summing the successive $1/N$. The details of the Monte-Carlo simulation are shown in Table 7.2.

Table 7.2 Cumulative Probabilities

Unit of population	Probability	Cumulative probability	Cumulative probability interval
1	0.1	0.1	0.0–0.09
2	0.1	0.2	0.10–0.19
3	0.1	0.3	0.20–0.29
4	0.1	0.4	0.30–0.39
5	0.1	0.5	0.40–0.49
6	0.1	0.6	0.50–0.59
7	0.1	0.7	0.60–0.69
8	0.1	0.8	0.70–0.79
9	0.1	0.9	0.80–0.89
10	0.1	1.0	0.90–0.99

- Find a two digit random number from the random number table and check the range in which it lies in Table 7.2.
- Select the corresponding unit of the population and include it as the first unit of the sample.
- Find the probability $1/(N-1)$ and repeat steps 2–4 to select the second unit of the sample from the population.
- Like this, repeat steps 2–4 until all the remaining units starting from the third unit till the n th unit of the sample are selected from the population using the probabilities, $1/(N-2)$, ..., $1/[N-(n-1)]$, respectively.

Systematic sampling

This is a special kind of random sampling in which the selection of the first unit of the sample from the population is based on randomization. The remaining units of the sample are selected from the population at a fixed interval of n , where n is the sample size.

Let the size of the population (N) be 800 and the sample size (n) be 40. The units of the sampling frame are divided into n number of intervals based on the ratio N/n , as shown below:

$$\text{Sampling interval width, } I = \frac{N}{n} = \frac{800}{40} = 20$$

The sampling frame consists of units with serial numbers from 1 to 800. This range is divided into 40 intervals, viz., 1–20, 21–40, 41–60, 61–80, ..., 760–780 and 780–800, where the total number of intervals is equal to the sample size.

Then a number from the first interval 1–20 is selected randomly and the unit of the population with this serial number is treated as the first unit of the sample. Let the randomly selected unit from the first interval of the population be 12. Then, the second unit of the sample is the unit in the population with serial number 32 which is obtained by adding 20

(sampling interval width, I) to 12. Then, each of the remaining units of the sample can be obtained from the population in the same manner by adding 20 to the serial number of the previous unit selected from the population.

As per these guidelines, the units of the population with serial numbers 52, 72, 92, ..., 772 and 792 are treated as the third, fourth, fifth, ..., 39th and 40th units of the sample, respectively.

The advantage of this sampling method is that, once the first unit of the sample is randomly determined all the sampling units can be easily obtained. But, the sampling units starting from 2 to n are dependent on the first sampling unit which lacks complete randomization of the selection of the sampling units from the population.

If the order of the units of the population possesses a periodicity of I and if the first unit of the sample is selected such that all the units of the sample are from the end-points of such periodicity then the results of the sampling will be completely misleading. In such situations, suppose in a production process, while checking the quality of the products, if the first unit of the sample is good, then all other units of the sample will be good; the converse is also true. But, such occurrence is very rare.

Stratified sampling

Stratified sampling is an improvised sampling over simple random sampling and systematic sampling. This sampling will have more statistical efficiency. In this sampling method, the population is divided into a specified set of strata such that the members within each stratum have similar attributes but the members between strata have dissimilar attributes. This means that each stratum is homogeneous when compared to the population.

Consider an example of studying the educational standard of schools in a state. The location of a school will have certain influence over its standard of education. The location can be classified into rural, semi-urban and urban areas. If we use the same proportion of sampling units from each stratum, then it is known as *proportional* stratified sampling; otherwise, the sampling is known as *disproportional* stratified sampling.

In the above example, if there is no much of difference (less variance) in educational standard between schools within each area, viz., stratum (rural or semi-urban or urban), then the proportional stratified sampling is to be used.

But, due to good infrastructure facilities and other reasons, it is generally expected that the standard of education between the schools located in an urban area would be more or less same which results in less variance between them; but on the other extreme, the standard of education between the schools located in a rural area differs markedly because of varying infrastructure facilities. In this context, one should use disproportional stratified sampling.

The concept of strata in the stratified sampling is illustrated in Fig. 7.1. In Fig. 7.1, the outer rectangle represents the entire population of the study which consists of the locations of schools in rural, semi-urban and urban areas. The strata in Fig. 7.1 can be inferred as given below:

Stratum 1: Rural area which has the largest space

Stratum 2: Semi-urban area which has less space when compared to rural area, but more space when compared to urban area.

Stratum 3: Urban area which has the least space

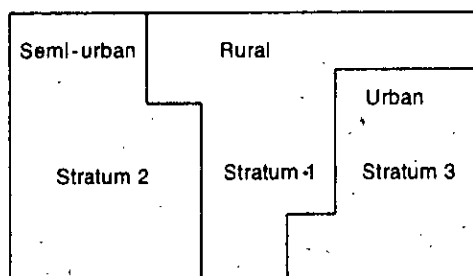


Fig. 7.1 Schema of strata.

The objective of the stratified sampling is to select a sample of size n from the population such that the following condition is satisfied. A schematic view of the sample is shown in Fig. 7.2.

$$n = n_1 + n_2 + n_3 + \dots + n_k$$

where n_i is the number of sampling-units taken from the stratum i , $i = 1, 2, 3, \dots, k$; k , the number of strata.

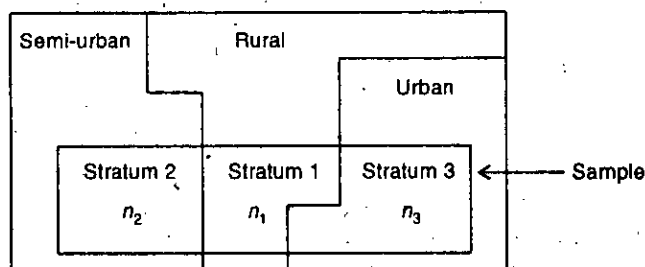


Fig. 7.2 Schematic view of sample.

Proportional stratified sampling As stated earlier, if the variance between the members within each stratum of a population is zero or very less, then one can employ proportional stratified sampling.

Let N be the size of the population; N_i be the size of the stratum i ; n_i be the size of the sub-sample to be taken from the stratum i ; k , be the number of strata in the population; and n , be the size of the total sample of the population. Then, n_i should be decided based on the following relationship:

$$\frac{n_1}{N_1} = \frac{n_2}{N_2} = \frac{n_3}{N_3} = \dots = \frac{n_k}{N_k} = \frac{n}{N}$$

and

$$N_1 + N_2 + N_3 + \dots + N_k = N$$

$$n_1 + n_2 + n_3 + \dots + n_k = n$$

Example 7.1 A survey is conducted to analyze the status of employment of the recently graduated batch of students of a premier technological university. It is planned to conduct stratified sampling for this study. So, the population which consists of different colleges is divided into three strata, viz., government colleges, aided colleges and self-financing colleges. Since, the regulations of AICTE and the respective university ensure uniform standard of infrastructure and educational standards, proportional stratified sampling is used for this study. The total number of engineering colleges in the university is 200. The number of engineering colleges in the three categories, viz., government, aided and self-financing are 20, 50 and 130, respectively. If the sample size is 20, determine the number of colleges to be sampled from each category.

Solution. We have

Total number of engineering colleges in the university, N	= 200
Number of government engineering colleges, N_1	= 20
Number of aided engineering colleges, N_2	= 50
Number of self-financing engineering colleges, N_3	= 130
Total sample size, n	= 20

The relationship of proportional stratified sampling is as follows:

$$\frac{n_1}{N_1} = \frac{n_2}{N_2} = \frac{n_3}{N_3} = \frac{n}{N}$$

where n_1 is the number of colleges to be sampled from government category; n_2 , the number of colleges to be sampled from aided category; and n_3 , the number of colleges to be sampled from self-financing category. Then

$$\frac{n}{N} = \frac{20}{200} = 0.1$$

Therefore,

$$n_1 = \frac{n}{N} \times N_1 = 0.1 \times 20 = 2$$

$$n_2 = \frac{n}{N} \times N_2 = 0.1 \times 50 = 5$$

$$n_3 = \frac{n}{N} \times N_3 = 0.1 \times 130 = 13$$

Disproportional stratified sampling Let σ_i be the variance of the stratum i , where i varies from 1 to k . If the above variances are not equal, then disproportional stratified sampling should be used. In this sampling, the stratum which has more variance will have

proportionately more sampling units when compared to another stratum which has less variance.

The formula to compute the number of sampling units of the stratum i is:

$$n_i = \frac{q_i \sigma_i n}{\sum_{i=1}^k q_i \sigma_i}, \quad i = 1, 2, 3, \dots, k$$

where n_i is the number of sampling units of the stratum i ; σ_i , the standard deviation of the stratum i ; n , the sample size (total); $q_i = N_i/N$; N_i the size of the stratum i ; N , the size of the population; q_i , the ratio of the size of the stratum i with that of the population; and k , the number of strata.

Example 7.2 The education department of a state wants to study the standard of education in schools. The analyst of the study divided the entire population of the schools into schools in rural locations, semi-urban locations and urban locations. He found more variation between schools in rural locations because of varying infrastructure than the variation between schools in semi-urban locations. Same is the case between semi-urban locations and urban locations. The analyst prefers to use disproportionate stratified sampling in which the categories of location represent strata. The total number of schools in the state is 1200. The number of schools in the rural, semi-urban and urban locations are 500, 400 and 300, respectively. The variance of educational standard of the schools in these locations on 0–10 scale are 49, 16 and 4, respectively. If the disproportionate stratified sampling is to be used with the sampling size of 90, determine the number of sampling units for each category of locations.

Solution. As per the statement of the problem, the three different strata are as defined below:

Stratum 1: Locations of schools in rural areas

Stratum 2: Locations of schools in semi-urban areas

Stratum 3: Locations of schools in urban areas

We also have

Size of the population, N = 1200

Size of the stratum 1, N_1 = 500

Size of the stratum 2, N_2 = 400

Size of the stratum 3, N_3 = 300

Variance of the stratum 1, σ_1^2 = 49

Variance of the stratum 2, σ_2^2 = 16

Variance of the stratum 3, σ_3^2 = 4

Sample size, n = 90

The details of calculations of the number of sampling units (n_1 , n_2 and n_3) from stratum 1, stratum 2 and stratum 3, respectively are summarized in Table 7.3. We find that the number of sampling units of stratum 1, stratum 2 and stratum 3 are 55, 25 and 10, respectively.

Table 7.3 Details of Calculations of Sampling Units of Strata

Stratum number i	Size of stratum N_i	$q_i = \frac{N_i}{N}$	σ_i	$q_i \sigma_i n$	$q_i \sigma_i$	$n_i = \frac{q_i \sigma_i n}{\sum_{i=1}^3 q_i \sigma_i}$
1	500	0.416667	7	262.5	2.916669	55
2	400	0.333333	4	120.0	1.333332	25
3	300	0.250000	2	45.0	0.500000	10
Total					4.750001	90

Cluster sampling

Cluster sampling is a sampling technique in which the population is divided into different clusters such that the members within each cluster are dissimilar (heterogeneous) in terms of their attributes, but different clusters are similar to each other. This leads to the inference that each cluster can be treated as a small population which possess all the attributes of the population. Hence, in cluster sampling, any one of the clusters is randomly selected and all the units of that cluster are selected (sampled) to arrive at inference about the population.

A schematic view of the cluster sampling is shown in Fig. 7.3. The population shown in the figure consists of clusters from 1 to 10. These clusters can be viewed as different geographic regions of a state. In a survey to study the culture of the population of a state and its impact on the economy of the state, one can conveniently assume that these clusters are similar to each other and the members within each cluster are heterogeneous. This kind of sampling can also be called *area sampling* as the population and clusters are defined with reference to the geographic regions.

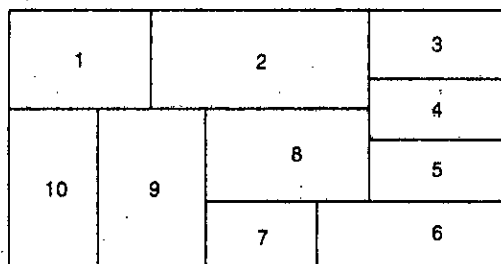


Fig. 7.3 Schematic view of cluster sampling

Multi-stage sampling

In a large scale survey covering the entire nation/subcontinent, the size of the sampling frame will be too large which leads to more time and cost of the study. In such study,

multi-stage sampling technique helps designing a smaller sampling frame which will make the study practicable in terms of cost and time.

The multi-stage sampling employs more than one stage to sample the population depending upon the reality. The combination of the types of sampling techniques to be used in the specified number of stages is unique to the reality.

Consider the case of studying the requirement of chemical laboratory equipments in all the colleges in a country by CHEMEQUIP LTD. This study can be done using multi-stage sampling, as explained below:

Stage 1: In the first stage, the different states of the country are sampled from each region using stratified sampling. The country can be divided into different regions (strata), viz., east, west, north and south. Here, it is assumed that the states (sampling units) within each region are similar and the regions are dissimilar.

Stage 2: After selecting some states from each region in stage-1 based on the particulars of the stratified sampling, again one can use cluster sampling to identify a district from each selected state by assuming different districts of each state as its clusters. Here, it is assumed that the districts of a state are similar, but the colleges in each district are dissimilar in terms of their present chemical laboratory facilities.

Stage 3: In each selected district, a random sampling may be used to select the proportionate number of colleges (sampling units) from it.

This study involves three stages of sampling. The highest level of sampling units are states and the lowest level of sampling units are colleges. From this example, one can visualize the fact that the multi-stage sampling reduces the size of the overall sampling frame.

7.2.2 Non-probability Sampling Methods

Convenience sampling

This is a non-probability sampling method in which the interviewers will decide the choice of sampling units based on their convenience.

In most of the situations, the following may be true:

- The sampling units may be distributed sparsely.
- Many respondents will refuse to fill the questionnaires.
- Some respondents will not cooperate in filling the questionnaires.
- Some of the interviewers may not be serious in selecting the sampling units as per the assumed sampling plan.

Though the probability sampling gives better accuracy in terms of confidence level of the inferences of the study, there are many practical difficulties in fully executing probability sampling because of the limitations stated earlier. So, naturally, the interviewers will be resorted to convenience sampling to overcome such difficulties. The sampling units for this type of sampling are selected from a telephone directory, newspaper subscribers list, departmental stores, etc.

Judgement sampling

The *judgement* sampling is a non-probability sampling in which the sampling units are selected on the advice of some expert or by the intuition/opinion of the researcher himself. In the first case, an expert who is familiar with the sampling frame guides the researcher in selecting the sampling units from the sampling frame. In the second case, the researcher applies his/her intuitive judgement and previous experience in selecting the sampling units from the sampling frame.

In this sampling method, there is more chance of personal biases. If the exercise is done more seriously, it would lead to better results. This sampling is also called as *purposive* sampling because the sampling units of the sample are identified from the population selectively which prevents the inclusion of other sampling units in the sample. This sampling technique is applied for sampling the population related to rare events in which the members of the population are not equally qualified to become the members of the sampling frame and hence the members of the sample.

Quota sampling

The *quota* sampling is a non-probability sampling in which the population is classified into a number of groups based on some criterion, say age of the members of population, viz., old age, middle age and young age. Let the proportion of the number of persons in the population under the old age category be 20%, and that of middle age and young age be 50% and 30%, respectively. In the quota sampling, the proportions of the number of sampling units selected from these categories are the same as in the population. If n is the sample size, then the proportions of the number of sampling units to be selected from the old-age, middle-age and young-age categories will be 20%, 50% and 30% of the sample size (n), respectively (i.e. $0.2n$, $0.5n$ and $0.3n$, respectively).

Later, while selecting the required number of sampling units from each category, one can use any one of the other non-probability sampling methods, viz., convenience sampling or judgment sampling.

Though this method comes under non-probability sampling methods, in the first phase, certain amount of proportionate selections of sampling units from different strata of the population are made which makes the sample as a representative sub-population of the main population.

Snowball sampling

The snowball sampling is a restrictive multi-stage sampling in which initially certain number of sampling units (respondents) are randomly selected. Later, additional sampling units are selected based on referral process. This means that the initially selected respondents provide addresses of additional respondents for the interviewers.

Initial respondents may be randomly selected, for example, from the information contained in telephone directories. Later, additional respondents can be included in the sample based on the references made by those initial respondents. This is an inexpensive and convenient non-probability sampling method which suits the situations where the development of sampling frame is a difficult and time-consuming task.

7.3 SAMPLING DISTRIBUTION OF MEAN WHEN NORMAL POPULATION VARIANCE IS KNOWN

In a sample, suppose there are n observations. The different means of n observations of different samples are $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots$ and so on. The distribution of above sample means is known as the *sampling distribution of the mean*.

As stated earlier, the inference about a population can be drawn based on the value of the sample mean, $\bar{X}$ with reference to the sampling distribution of the mean.

7.3.1 Sampling Distribution of Mean (when the Population is Infinite)

When the population is infinite, the mean and variance of the sampling distribution are given by the following formulas:

$$\text{Mean} = \mu \text{ (same as the population mean)}$$

$$\text{Variance} = \frac{\sigma^2}{n} \text{ (where } n \text{ is the sample size)}$$

$$\text{Standard deviation} = \frac{\sigma}{\sqrt{n}} = \text{Standard error of the mean}$$

The standard error of the mean represents the deviation of different sample means from the true mean because of sampling error. If this is lesser, then sample means are closer to the true mean.

If X is a random variable which follows normal distribution with mean μ and variance σ^2 , $\bar{X}$ is also a random variable which follows normal distribution with mean μ and variance σ^2/n . That is, If $X \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \sigma^2/n)$.

Alternatively, if the standard normal statistic (Z) for X is $(\bar{X} - \mu)/\sigma$, then the standard normal statistic for $\bar{X}$ is

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Example 7.3 The annual income of employees in an industry follows normal distribution with mean and variance as Rs. 4 lakhs and Rs. 1 lakh, respectively. A random sample of size 49 is taken from an infinite normal population. What is probability that the sample mean is greater than Rs. 4.25 lakhs?

Solution. We have

Sample size, n	= 49
Population mean, μ	= Rs. 4 lakhs
Population variance, σ^2	= Rs. 1 lakh
Population standard deviation, σ	= Rs. 1 lakh

We also know

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ or } \bar{X} \sim N\left(4, \frac{1}{49}\right)$$

Then

$$\begin{aligned}
 P(\bar{X} \geq 4.25) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{4.25 - 4}{1/\sqrt{49}}\right) \\
 &= P(Z_{\bar{X}} \geq 1.75) \\
 &= 1 - (Z_{\bar{X}} \leq 1.75) \\
 &= 1 - 0.9599 \\
 &= 0.0401
 \end{aligned}$$

Therefore, the probability that the sample mean, more than Rs. 4.25 lakhs, is 0.0401.

7.3.2 Sampling Distribution of Mean (when the Population is Finite)

If the sampling is done with infinite population then the variance of the sampling distribution of the mean is given by

$$\text{Variance} = \frac{\sigma^2}{n}$$

If the sampling is done with finite population of size N , then the variance of the sampling distribution of the mean is given by

$$\text{Variance} = \frac{\sigma^2}{n} \frac{N-n}{N-1}$$

In the above formula, $(N-n)/(N-1)$ is called *finite population multiplier*, because the variance of the sampling distribution under the finite population is same as that under the infinite population, except the additional multiplying factor namely the finite population multiplier in it.

If X is a random variable which follows normal distribution with mean μ and variance σ^2 , then $\bar{X}$ is also a random variable which follows normal distribution with the following mean and variance:

$$\text{Mean} = \mu \quad \text{and} \quad \text{Variance} = \frac{\sigma^2}{n} \frac{N-n}{N-1}$$

Also if, $X \sim N(\mu, \sigma^2)$, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right)$$

Alternatively, if the standard normal statistic (Z) for $\bar{X}$ is $(\bar{X} - \mu)\sigma$, then the standard normal statistic for $\bar{X}$ is:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

Example 7.4 The age of employees in a company follows normal distribution with its mean and variance as 40 years and 121 years, respectively. If a random sample of 36 employees is taken from a finite normal population of size 1000, what is the probability that the sample mean is: (a) lesser than 45, (b) greater than 42 and (c) in between 40 and 42?

Solution. We have

Sample size, n	= 36
Population size, N	= 1000
Population mean, μ	= 40 years
Population variance, σ^2	= 121 years
Population standard deviation, σ	= 11 years

We also get

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right)$$

(a) When $\bar{X} \leq 45$,

$$\begin{aligned} P(\bar{X} \leq 45) &= P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \leq \frac{45 - 40}{\frac{11}{\sqrt{36}} \sqrt{\frac{1000-36}{1000-1}}}\right) \\ &= P(Z_{\bar{X}} \leq 2.78) \\ &= 0.9973 \end{aligned}$$

Therefore, the probability that the sample mean will be less than 45 is 0.9973.

(b) When $\bar{X} \geq 42$,

$$P(\bar{X} \geq 42) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \geq \frac{42 - 40}{\frac{11}{\sqrt{36}} \sqrt{\frac{1000-36}{1000-1}}}\right)$$

$$= P(Z_{\bar{X}} \geq 1.11)$$

$$= 1 - P(Z_{\bar{X}} \leq 1.11)$$

$$= 1 - 0.8665$$

$$= 0.1335$$

Therefore, the probability that the sample mean will be more than 42 is 0.1335.

(c) When $40 \leq \bar{X} \leq 42$,

$$\begin{aligned} P(40 \leq \bar{X} \leq 42) &= P\left(\frac{40 - 40}{\frac{11}{\sqrt{36}} \sqrt{\frac{1000 - 36}{1000 - 1}}} \leq \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N - n}{N - 1}}} \leq \frac{42 - 40}{\frac{11}{\sqrt{36}} \sqrt{\frac{1000 - 36}{1000 - 1}}} \right) \\ &= P(0 \leq Z_{\bar{X}} \leq 1.11) \\ &= P(Z_{\bar{X}} \leq 1.11) - P(Z_{\bar{X}} \leq 0) \\ &= 0.8665 - 0.5 \\ &= 0.3665 \end{aligned}$$

Therefore, the probability that the sample mean will be in between 40 and 42 is 0.3665.

7.3.3 Central Limit Theorem

If $\bar{X}$ is the mean of a sample of size n taken from a population with mean μ and known variance σ^2 , then $Z_{\bar{X}}$ which is equal to $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ will follow standard normal distribution when n tends to ∞ .

This theorem is useful to sample a population when its distribution is unknown or non-normal.

Example 7.5 A non-normal distribution representing the number of trips performed by lorries per week in a coal field has a mean of 100 trips and variance of 121 trips. A random sample of 36 lorries is taken from the non-normal population. What is the probability that the sample mean is

- greater than 105 trips,
- less than 102 trips and
- between 101 trips and 103 trips?

Solution. We have

Sample size, n = 36

Population mean, μ = 100 trips

Population variance, σ^2 = 121 trips

Population standard deviation, σ = 11 trips

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

(a) When $\bar{X} \geq 105$

$$P(\bar{X} \geq 105) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{105 - 100}{11/\sqrt{36}}\right)$$

$$= P(Z_{\bar{X}} \geq 2.73)$$

$$= 1 - (Z_{\bar{X}} \leq 2.73)$$

$$= 1 - 0.9968$$

$$= 0.0032$$

Therefore, the probability that the sample mean will be more than 105 is 0.0032.

(b) When $\bar{X} \leq 102$,

$$P(\bar{X} \leq 102) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{102 - 100}{11/\sqrt{36}}\right) = (Z_{\bar{X}} \leq 1.09) = 0.8621$$

Therefore, the probability that the sample mean will be less than 102 trips is 0.8621.

(c) When $101 \leq \bar{X} \leq 103$,

$$P(101 \leq \bar{X} \leq 103) = P\left(\frac{101 - 100}{11/\sqrt{36}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{103 - 100}{11/\sqrt{36}}\right)$$

$$= P(0.55 \leq Z_{\bar{X}} \leq 1.64)$$

$$= P(Z_{\bar{X}} \leq 1.64) - (Z_{\bar{X}} \leq 0.55)$$

$$= 0.9495 - 0.7088$$

$$= 0.2407$$

Therefore, the probability that the sample mean will be in between 101 and 103 trips is 0.2407.

7.3.4 Sampling Distribution of Mean when Normal Population Variance is Unknown (Student's t -distribution)

If the variance of a normal population is known, then the corresponding sampling distribution is presented as in the earlier section.

If the variance of a normal population is unknown, the corresponding sampling distribution can be defined using Student's t -distribution. Here, the sample variance (S^2) is used as an estimate for the population variance σ^2 .

Let X be a random variable which follows normal distribution with mean μ and variance σ^2 ; $Z_{\bar{X}}$, the standard normal statistic of $\bar{X}$; and χ^2 be a random variable which follows chi-square distribution with τ degrees of freedom.

If the standard normal statistics of $\bar{X}$ and χ^2 are independent of each other, then the Student's t distribution is

$$t = \frac{Z_{\bar{X}}}{\sqrt{\frac{\chi^2}{\tau}}}$$

The standard normal statistic of $\bar{X}$ and the chi-square statistics χ^2 are:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

and

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \quad \text{with } (n-1) \text{ degrees of freedom}$$

By substituting the above two statistics in t , we get Student's t distribution as:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \text{with } (n-1) \text{ degrees of freedom}$$

The sketch of t distribution is shown in Fig. 7.4.

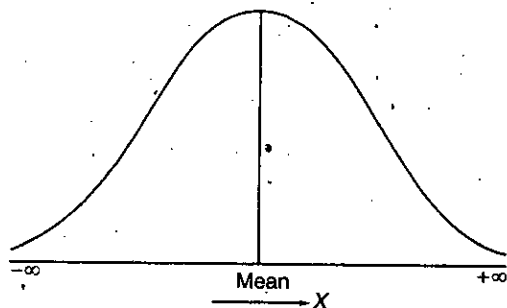


Fig. 7.4 Student's t distribution.

The t distribution can be used only when the sample size is not more than 30; when the sample size is more than 30, it can be approximated to a normal distribution.

Example 7.6 The annual sales of dealers of a company follows normal distribution with its mean as Rs. 94 lakhs. A random sample of 10 dealers of the company is taken from the normal population. The variance of the annual sales of these 10 dealers is Rs. 81 lakhs. Find the probability that the mean annual sales of the sample is

- (a) less than Rs. 98 lakhs,
- (b) more than Rs. 98 lakhs.

Solution. We have

Sample size, n	= 10. Then
Mean annual sales of the population, μ	= Rs. 94 lakhs
Variance of mean annual sales of the sample, S^2	= Rs. 81 lakhs
Standard deviation of the mean annual sales of the sample, S	= Rs. 9 lakhs

We also have

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \text{with } (n-1) \text{ degrees of freedom}$$

$$= \frac{98 - 94}{9/\sqrt{10}} = 1.41 \text{ with 9 degrees of freedom}$$

- (a) When $\bar{X} \leq 98$,

$$\begin{aligned} P(\bar{X} \leq 98) &= P\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} \leq \frac{98 - 94}{9/\sqrt{10}}\right) \\ &= P(t \leq 1.41) \\ &= 0.90 + \frac{1.41 - 1.383}{1.833 - 1.383} \times 0.05 \\ &= 0.903 \end{aligned}$$

The probability that the mean annual sales of the dealers is less than Rs. 98 lakhs is 0.903.

- (b) When $\bar{X} \geq 98$,

$$\begin{aligned} P(\bar{X} \geq 98) &= P\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} \geq \frac{98 - 94}{9/\sqrt{10}}\right) \\ &= P(t \geq 1.41) \\ &= 1 - P(t \leq 1.41) \\ &= 1 - 0.903 \\ &= 0.097 \end{aligned}$$

The probability that the mean annual sales of the dealers is more than Rs. 98 lakhs is 0.097.

7.4 SAMPLING DISTRIBUTIONS OF VARIANCE

There are two sampling distributions for the variance which are chi-square distribution and F distribution. These are explained in this section.

7.4.1 Chi-square Distribution

The variance of a sample is denoted by S^2

$$S^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$$

where X_i is the i th observation of the sample, $\bar{X}$, the mean of the sample, n , is the size of the sample.

The distribution of the variance S^2 of a random sample with size n taken from a normal population with variance σ^2 is known as chi-square (χ^2) distribution with $(n-1)$ degrees of freedom which is given as

$$\chi^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \quad \text{with } (n-1) \text{ degrees of freedom}$$

The sketch of chi-square distribution is shown in Fig. 7.5. In this distribution, degrees of freedom is the only factor. So, the distribution will have different shape for different degrees of freedom.

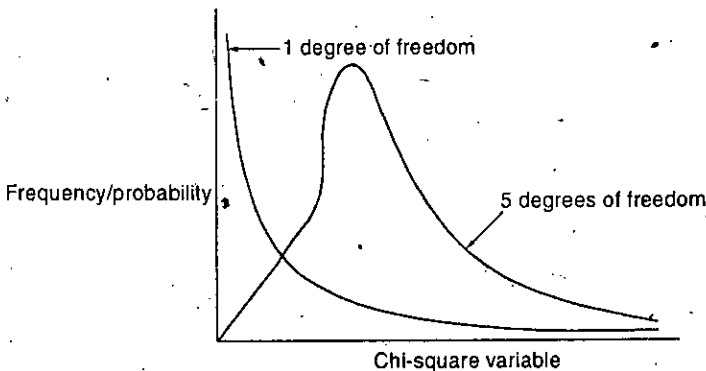


Fig. 7.5 Chi-square distribution.

Example 7.7 A random sample of 20 dealers of a company is taken from a normal population. The variance of the annual sales of dealers from the normal population and that of the random sample of 20 dealers are Rs. 81 lakhs and 125 lakhs, respectively.

Compute chi-square statistic and find the probability that the chi-square variable is more than calculated chi-square statistic.

Solution. Here, sample size, $n = 20$. Then

Variance of mean annual sales of the population, $\sigma^2 = \text{Rs. 81 lakhs}$

Variance of mean annual sales of the sample, $S^2 = \text{Rs. 125 lakhs}$

The chi-square statistic is:

$$\begin{aligned}\chi^2 &= \frac{(n-1)S^2}{\sigma^2} \quad \text{with } (n-1) \text{ degrees of freedom} \\ &= \frac{(20-1)125}{81} \\ &= 29.32 \quad \text{with 19 degrees of freedom}\end{aligned}$$

Then

$$P(\chi^2 > 29.32) \approx 0.05$$

Therefore, the probability that the chi-square variable is more than the calculated chi-square statistic is approximately 0.05.

7.4.2 F-distribution

F-distribution is the ratio of two chi-square variables.

If S_1^2 and S_2^2 are the variances of two independent random samples of size n_1 and n_2 , respectively, taken from two normal populations which are having the variances as σ_1^2 and σ_2^2 , respectively, then *F* distribution with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom is defined as shown below.

$$F = \frac{\frac{(n_1 - 1) S_1^2 / \sigma_1^2}{n_1 - 1}}{\frac{(n_2 - 1) S_2^2 / \sigma_2^2}{n_2 - 1}}$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom. If $\sigma_1 = \sigma_2$, then the formula for *F* is reduced to the form as

$$F = \frac{S_1^2}{S_2^2}$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom. The sketch of F distribution is shown in Fig. 7.6.

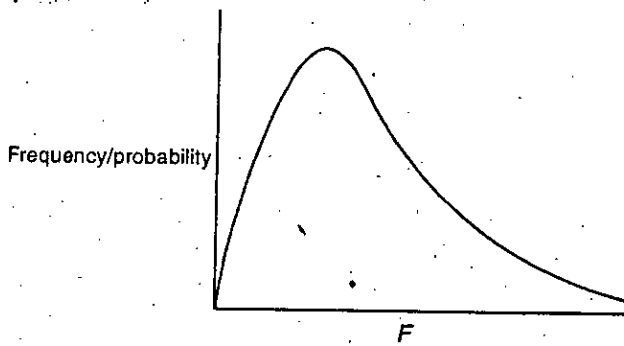


Fig. 7.6 F distribution.

In this distribution, the numerator degrees of freedom $(n_1 - 1)$ and the denominator degrees of freedom $(n_2 - 1)$ are the only two parameters affecting its shape and size.

Example 7.8 Two independent samples of students of a programme under distance education are taken from normal populations with the same variance. The size and variance of marks of the first sample are 8 and 100, respectively. The size and variance of marks of the second sample are 20 and 40, respectively.

- (a) What is the calculated F statistic?
- (b) What is the probability that the F ratio is more than the calculated F statistic?

Solution. We have

Size of the first sample, n_1	= 8
Size of the second sample, n_2	= 20
Variance of the first sample, S_1^2	= 100
Variance of the second sample, S_2^2	= 40

In this example, $\sigma_1^2 = \sigma_2^2$. Hence, the F statistic is computed as

$$F = \frac{S_1^2}{S_2^2} = \frac{100}{40} = 2.5$$

with (7, 19) degrees of freedom. Then $P(F \geq 2.5) \approx 0.05$. Therefore, the probability that the F ratio is more than the calculated F statistic of 2.5 is approximately 0.05.

Example 7.9 Two independent samples of items of vendors are taken from normal populations with the same variance. The size of the first sample and that of the second sample are 21 and 20, respectively. What is the probability that the variance of the first sample is more than 3 times that of the second sample?

Solution. We have

Size of the first sample, $n_1 = 21$

Size of the second sample, $n_2 = 20$

Then

$$F = \frac{S_1^2}{S_2^2} = 3$$

with 20 and 19 degrees of freedom. Then $P(F \geq 3) = 0.01$.

The probability that the variance of the first sample more than 3 times that of the second sample is 0.01. This is shown in Fig. 7.7.

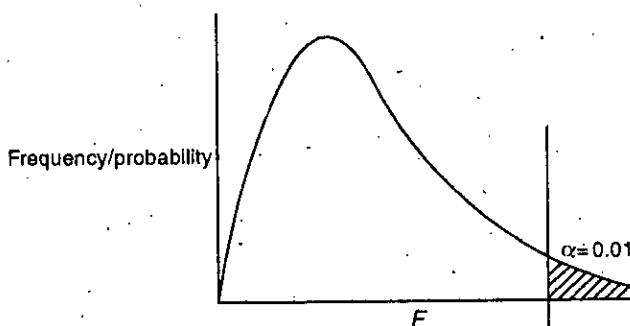


Fig. 7.7 F distribution for Example 7.9.

7.5 SAMPLING DISTRIBUTION OF PROPORTION

In a sample of n successive trials, let the probability of success in a trial be p and X be the number of successes. Then, X follows binomial distribution which has its mean and variance as:

$$\mu = np \text{ and } \sigma^2 = np(1-p) = npq$$

For large value of n , the binomial distribution can be approximated to normal distribution as

$$X \sim N(\mu, \sigma^2)$$

Let the proportion of success be $\bar{p}$ and it is given by

$$\bar{p} = \frac{X}{n}$$

Then $\bar{p}$ also approaches normal distribution with its mean and variance,

$$\text{Mean, } \bar{p} = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} np = p$$

and

$$\text{Variance, } \bar{p} = \text{var}\left(\frac{X}{n}\right) = \frac{1}{n^2} \text{var}(X) = \frac{1}{n^2} np(1-p) = \frac{p(1-p)}{n}$$

Then

$$\bar{p} \sim N\left[p, \frac{p(1-p)}{n}\right]$$

where one can approximate like this only when np as well as $n(1-p)$ is at least 5.

Example 7.10 The personnel manager of a company feels that 52% of the employees will have enhanced skill after attending the training programme if they are sponsored for an in-house training programme.

A sample of records of 49 employees of the company, who attended the training programme on skill development, reveals that only 24 of them are having enhanced skill after attending the training programme. Find the probability that the sample of employees who attended the training programme have enhanced their skill.

Solution. We have $p = 0.52$, $n = 49$, $q = 1 - p = 1 - 0.52 = 0.48$ and $\bar{p} = 24/49 = .49$. Then

$$\text{Variance, } \bar{p} = \frac{p(1-p)}{n} = \frac{0.52 \times 0.48}{49} = 0.0051$$

and $\sigma_{\bar{p}} = 0.071$. Then

$$\bar{p} \sim N\left(p, \frac{pq}{n}\right)$$

Therefore,

$$\begin{aligned} P\left(Z \geq \frac{\bar{p} - p}{\sigma_{\bar{p}}}\right) &= 1 - P\left(Z \leq \frac{\bar{p} - p}{\sigma_{\bar{p}}}\right) \\ &= 1 - P\left(Z \leq \frac{0.49 - 0.52}{0.071}\right) \\ &= 1 - P(Z \leq -0.42) \\ &= 1 - [1 - P(Z \leq 0.42)] \\ &= P(Z \leq 0.42) \\ &= 0.6628 \end{aligned}$$

The probability that the sample of employees who attended the training programme have enhanced their skill is 0.6628.

7.6 CONFIDENCE INTERVAL ESTIMATION

Based on sample characteristics, decision is taken for either accepting or rejecting the lot. There are many methods available to do the task. *Confidence interval* helps take such decision.

7.6.1 Confidence Interval (when Sample Size is Large)

Consider a population whose mean (μ) is unknown and variance (σ^2) is known. If a sample of size n ($n > 30$) is selected from the population, then the confidence interval is estimated using the sample mean at a desired significance level (α). Confidence interval is the range of the population mean (μ) calculated for a specified significance level (α).

If mean of the population = μ and mean of the sample = $\bar{X}$, then

$$\text{Variance of the population} = \frac{\sigma^2}{n}$$

and

$$\text{Standard deviation} = \frac{\sigma}{\sqrt{n}} = \text{Standard error of the mean}$$

Therefore,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Since the significance level is to be equally distributed on both the tails of the normal distribution as shown in Fig. 7.8, the rejection area at each tail of the normal distribution is $\alpha/2$.

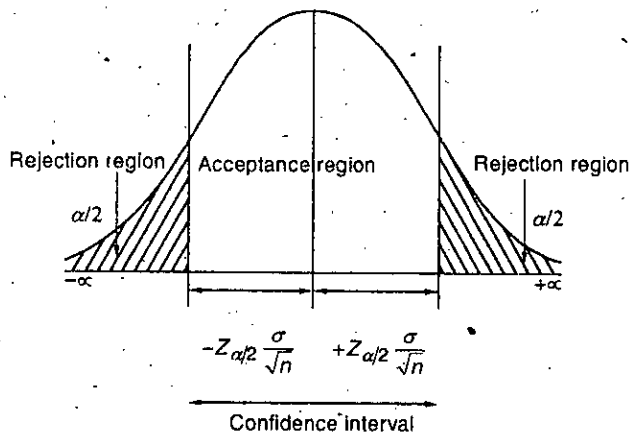


Fig. 7.8 Confidence interval with rejection regions under normal distribution.

Now, the confidence interval

$$\mu \approx \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = \left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) \text{ to } \left(\bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

The above interval gives the range for μ of the population.

Example 7.11 The diameter of shafts produced in a machine follows normal distribution with the variance of 81 mm. A random sample of 36 shafts taken from the production has its mean diameter of 30 mm. Find the confidence interval for the diameter of shafts at a significance level of 0.05.

Solution. We have $\sigma^2 = 81$ mm, $\bar{X} = 30$ mm, $n = 36$, $\alpha = 0.05$ and $Z_{\alpha/2} = 1.96$. Then

$$\text{Standard error} = \frac{\sigma}{\sqrt{n}} = \frac{9}{\sqrt{36}} = 1.5$$

and

$$\begin{aligned} \text{Confidence interval} &= \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \\ &= \left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) \text{ to } \left(\bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) \\ &= (30 - 1.96 \times 1.5) \text{ to } (30 + 1.96 \times 1.5) \\ &= 27.06 \text{ mm to } 32.94 \text{ mm} \end{aligned}$$

7.6.2 Confidence Interval (when Sample Size is Small)

Consider a normal population whose mean and variance are unknown. If a small sample of size $n (n \leq 30)$ is selected from the population, then the confidence interval is estimated using its mean and variance at a given desired significance level (α).

Let mean of the population = μ and mean of the sample = $\bar{X}$. Then

$$\text{Variance of the sample} = \frac{S^2}{n}$$

and

$$\text{Standard deviation} = \frac{S}{\sqrt{n}} = \text{Standard error of the mean}$$

When the sample size is less than or equal to 30, Student's t distribution is to be used to establish the confidence interval.

Since the significance level is to be equally distributed on both tails of the t distribution, as shown in Fig. 7.9, the rejection area at each tail of the t distribution is $\alpha/2$. The value of $t_{\alpha/2}$ with $(n-1)$ degrees of freedom is to be determined for use in the following formula:

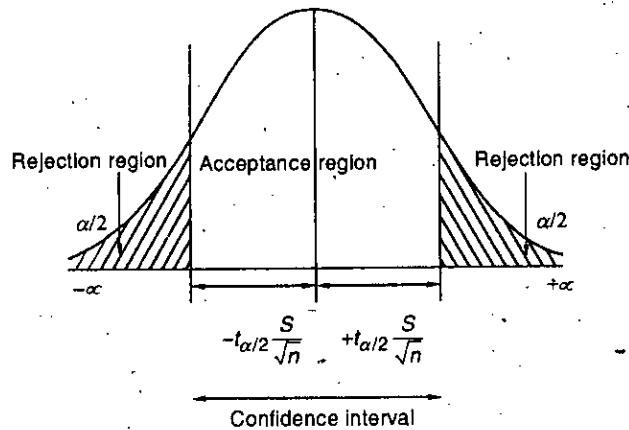


Fig. 7.9 Confidence interval with rejection regions under t distribution.

The confidence interval, $\mu = \bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}}$

$$= \left(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}} \right) \text{ to } \left(\bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}} \right)$$

This gives the range for μ of the population.

Example 7.12 The diameter of shafts produced in a machine follows normal distribution. A random sample of 16 shafts are taken from the production. The mean and variance of the diameter of the shafts in the sample are 20 mm and 9 mm, respectively. Find the confidence interval for the diameter of shafts at a significance level of 0.05.

Solution. Given $\bar{X} = 20$ mm, $S^2 = 9$ mm, $S = 3$ mm, $n = 16$ and $\alpha = 0.05$. Therefore, $\alpha/2 = 0.025$. Then

$$t_{\alpha/2} \text{ with 15 d.f.} = 2.131$$

and

$$\text{Standard error} = \frac{S}{\sqrt{n}} = \frac{3}{\sqrt{16}} = \frac{3}{4} = 0.75$$

Therefore,

$$\text{The confidence interval of } \mu = \bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}}$$

$$\begin{aligned}
 &= \left(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}} \right) \text{ to } \left(\bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}} \right) \\
 &= (20 - 2.131 \times 0.75) \text{ to } (20 + 2.131 \times 0.75) \\
 &= 18.4 \text{ mm to } 21.6 \text{ mm}
 \end{aligned}$$

7.6.3 Confidence Interval for Proportion

As stated earlier, the proportion of success, $\bar{p} = (X/n)$. Then, $\bar{p}$ approaches normal distribution with its mean and variance as (One can approximate like this only when np as well as $n(1-p)$ is at least 5):

$$\text{Mean} = p \quad \text{and} \quad \sigma_{\bar{p}}^2 = \frac{p(1-p)}{n} = \frac{pq}{n}$$

Then

$$\bar{p} \sim N\left(p, \frac{pq}{n}\right)$$

Therefore,

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{pq}{n}}$$

The confidence interval for the proportion is determined with a significance level of α . Since, the significance level is to be equally distributed on both the tails of the normal distribution, the rejection area at each tail of the normal distribution is $\alpha/2$. Then

$$\text{The confidence interval of proportion} = \bar{p} \pm Z_{\alpha/2} \sqrt{\frac{pq}{n}}$$

This interval gives the range for proportion.

Example 7.13 The marketing manager of a company feels that 40% of retailers enhance weekly sales after introducing an advertisement at a point of sales. A sample of 64 retailer shops of the company where the point of sales advertisement has been displayed reveals that only 32 of them are having enhanced sales after displaying the advertisement. Determine the confidence interval for the proportion representing enhanced sales with a confidence level of 0.9.

Solution. We have $p = 0.40$, $n = 64$, $q = 1 - p = 0.60$ and $\bar{p} = 32/64 = 0.50$. Then

$$\text{Variance} = \frac{p(1-p)}{n} = \frac{0.4 \times 0.6}{64} = 0.00375$$

and

$$\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}} = 0.061$$

We also have $1 - \alpha = 0.9$. Then $\alpha = 0.1$ and $\alpha/2 = 0.05$. Therefore, $Z_{\alpha/2} = 1.64$. Hence

$$\begin{aligned} \text{The confidence interval of proportion} &= \bar{p} \pm Z_{\alpha/2} \sqrt{\frac{pq}{n}} \\ &= 0.5 \pm 1.64 \times 0.061 \\ &= 0.5 \pm 0.1 \\ &= 0.4 \text{ to } 0.6 \end{aligned}$$

7.7 DETERMINATION OF SAMPLE SIZE

If the sample size is very large, then the error in making decision will be very less. But, it will be a costly exercise to do sampling for a large sample, which is more than the necessity. In sampling, the right size of sample minimizes the error in making decision, too. Hence, before performing sampling, it is a general practice to determine the sample size for a desired accuracy based on the population variance.

7.7.1 Sample Size for Determining Sample Mean

Consider a population whose mean (μ) is unknown and variance (σ^2) is known. As stated earlier, if a sample of size n ($n > 30$) is selected from the population, then the confidence interval is estimated using the sample mean at a desired significance level (α). Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

The confidence interval

$$\mu = \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

If the required accuracy about the mean is $\pm D$, then the confidence interval is obtained as

$$\text{Confidence interval of } \mu = \bar{X} \pm D$$

Therefore,

$$Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = D$$

From the above expression, the formula for the sample size n is

$$n = \left(\frac{Z_{\alpha/2} \sigma}{D} \right)^2$$

Example 7.14 The axial length of pistons produced follows normal distribution with the variance of 64 mm. Determine the sample size such that the mean length of pistons is within plus or minus 2 mm with a confidence level of 95%.

Solution. Here $\sigma^2 = 64$ mm, $\sigma = 8$ mm and confidence level $(1 - \alpha) = 0.95$. Therefore, $\alpha = 0.05$ and $\alpha/2 = 0.025$. Then, $Z_{\alpha/2} = 1.96$.

Since the desired accuracy about the mean $(D) = 2$ mm, we have the required sample size

$$n = \left(\frac{Z_{\alpha/2} \sigma}{D} \right)^2 = \left(\frac{1.96 \times 8}{2} \right)^2 = 61.5 \approx 62$$

7.7.2 Sample Size for Determining Sample Proportion

Consider the discussion in Section 7.6.3. If mean $= p$, we have

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{pq}{n}}$$

Then

$$\bar{p} \sim N\left(p, \frac{pq}{n}\right)$$

and

$$\text{Confidence interval of proportion} = \bar{p} \pm Z_{\alpha/2} \sqrt{\frac{pq}{n}}$$

If the required accuracy about the proportion is $\pm D$, then the confidence interval of proportion $= \bar{p} \pm D$. Therefore,

$$Z_{\alpha/2} \sqrt{\frac{pq}{n}} = D$$

From the above expression, the formula for the sample size n is

$$n = \frac{Z_{\alpha/2}^2 pq}{D^2}$$

Example 7.15 The manager of a bank feels that 35% of branches will have enhanced yearly collection of deposits after introducing a hike in interest rate. Determine the sample size such that the mean proportion is within plus or minus 0.06 at a confidence level of 90%.

Solution. Given $p = 0.35$, $q = 1 - p = 0.65$ and confidence level $(1 - \alpha) = 0.9$. Therefore, $\alpha = 0.10$ and $\alpha/2 = 0.05$. Then, $Z_{\alpha/2} = 1.64$.

For the desired accuracy $D = 0.06$, we get

$$n = \frac{Z_{\alpha/2}^2 pq}{D^2} = \frac{1.64^2 \times 0.35 \times 0.65}{0.06^2} = 170$$

Therefore, the required sample size, $n = 170$.

QUESTIONS

1. Define sampling. Distinguish it from census.
2. What are the advantages of sampling?
3. What is a sampling frame? Discuss the desirable attributes of a sampling frame.
4. Give the classification of sampling methods.
5. What is random sampling? What are the types of random sampling? Explain them with examples.
6. Explain systematic sampling.
7. What is stratified sampling? Discuss its concept with a suitable diagram?
8. Illustrate the application of proportional stratified sampling with an example.
9. Illustrate the application of disproportional stratified sampling with an example.
10. Explain cluster sampling.
11. What is multi-stage sampling? Give an example.
12. Explain convenience sampling.
13. Discuss judgment sampling.

14. What is quota sampling? Explain its advantages.
15. What is snowball sampling?
16. What is a sampling distribution? Discuss its uses.
17. What is central limit theorem? Discuss its application.
18. Explain the following:
 - (a) Student's t distribution
 - (b) Chi-square distribution
 - (c) F distribution.
19. What is a confidence interval? How will you construct a confidence interval for mean as well as for proportion?
20. Discuss the methods for determining sample size.
21. A random sample of size 85 is taken from an infinite normal population, which has its mean and variance as 100 and 225, respectively. What is the probability that the sample mean is greater than 125?
22. A random sample of size 70 is taken from an infinite normal population, which has its mean and variance as 200 and 750, respectively. What is the probability that the sample mean is greater than 205?
23. A random sample of size 44 is taken from a finite normal population of size 1200, which has its mean and variance as 85 and 64, respectively. What is the probability that the sample mean is (a) lesser than 88, (b) greater than 70 and (c) in between 83 and 87?
24. A random sample of size 66 is taken from a finite normal population of size 900, which has its mean and variance as 95 and 121, respectively. What is the probability that the sample mean is (a) lesser than 92, (b) greater than 98 and (c) in between 85 and 105?
25. A random sample of size 55 is taken from a non-normal population. The mean and variance of the population are 65 and 88, respectively. What is the probability that the sample mean is (a) greater than 70, (b) less than 64 and (c) between 55 and 65?
26. A random sample of size 49 is taken from a non-normal population. The mean and variance of the population are 30 and 49, respectively. What is the probability that the sample mean is (a) greater than 32, (b) less than 29 and (c) in between 28 and 32?
27. A random sample of 15 dealers of a company is taken from a normal population. The mean of the annual sales of the population is Rs. 50 lakhs and the variance of the annual sales of the sample is Rs. 100 lakhs, respectively. Find the probability that the mean annual sales of the sample is (a) less than Rs. 55 lakhs, (b) more than Rs. 55 lakhs.
28. A random sample of 20 customers of a company is taken from a normal population. The mean of the annual purchases made by the customers of the population is Rs. 75 lakhs. The variance of the purchases made by customers of the sample is Rs. 121 lakhs. Find the probability that the mean annual sales of the sample is (a) less than Rs. 80 lakhs, (b) more than Rs. 80 lakhs.

29. A random sample of 25 dealers of a company is taken from a normal population. The variance of the annual sales of dealers from the normal population and that of the random sample of 25 dealers are Rs. 70 lakhs and Rs. 84 lakhs, respectively. Compute chi-square statistic and find the probability that the chi-square value is more than the calculated chi-square statistic.
30. A random sample of 24 respondents is taken from a normal population. The variance of the annual income of the respondents from the normal population and that of the random sample of 24 respondents are Rs. 4 lakhs and Rs. 7 lakhs, respectively. Compute chi-square statistic and find the probability that the chi-square value is more than the calculated chi-square statistic.
31. Two independent samples of students of a programme under distance education are taken from normal populations with the same variance. The size and variance of marks of the first sample are 10 and 104, respectively. The size and variance of marks of the second sample are 25 and 45, respectively.
- (a) What is the calculated F statistic?
- (b) What is the probability that the F ratio is more than the calculated F statistic?
32. Two independent samples of diabetics are taken from normal populations with the same variance. The size and variance of blood sugar level of the first sample are 10 and 780, respectively. The size and variance of blood sugar level of the second sample are 21 and 225, respectively.
- (a) What is the calculated F statistic?
- (b) What is the probability that the F ratio is more than the calculated F statistic?
33. Two independent samples of items of vendors are taken from normal populations with the same variance. The size of the first sample and the second sample are 13 and 61, respectively. What is the probability that the variance of the first sample is more than 2.5 times that of the second sample?
34. The personnel manager of a company feels that the 45% of the employees will have enhanced skill after the training programme if they are sponsored for an in-house training programme.
- A sample of records of 36 employees of the company who attended training programme on skill development reveals that only 25 of them are having enhanced skill after attending the training programme. Find the probability that the sample of employees who attended the training programme have enhanced their skill.
35. The production manager of a company feels that 35% of the machines will have enhanced quality of its production after fixing an electronic measuring gadget to the machine.
- A sample of records of 44 machines, which are fixed with the electronic gadget, reveals that only 12 of them are producing products with enhanced quality. Find the probability that the sample of machines which are fixed with electronic gadget have enhanced their quality level.

36. The diameter of shafts produced in a machine follows normal distribution with the variance of 121 mm. A random sample of 49 shafts taken from the production has its mean diameter of 50 mm. Find the confidence interval for the diameter of shafts at a significance level of 0.05.
37. The internal diameter of shafts produced in a machine follows normal distribution with the variance of 225 mm. A random sample of 64 shafts taken from the production has its mean diameter of 100 mm. Find the confidence interval for the diameter of shafts at a significance level of 0.10.
38. The diameter of shafts produced in a machine follows normal distribution. A random sample of 20 shafts are taken from the production. The mean and variance of the diameter of the shafts in the sample are 40 mm and 11 mm, respectively. Find the confidence interval for the diameter of shafts at a significance level of 0.10.
39. The annual income of respondents follows normal distribution. A random sample of 15 respondents are taken. The mean and variance of the annual income of the respondents in the sample are Rs. 3 lakhs and Rs. 9 lakhs, respectively. Find the confidence interval for the annual income of the respondents at a significance level of 0.05.
40. The marketing manager of a company feels that 42% of retailers will have enhanced weekly sales after introducing an advertisement at the point of sales. A sample of 36 retailer shops of the company, where the point of sales advertisement has been displayed, reveals that only 18 of them are having enhanced sales after displaying the advertisement. Determine the confidence interval for the proportion representing the enhanced sales with a confidence level of 0.95.
41. The finance manager of a bank feels that 30% of its customers pay their installments against their loans in time after introducing an incentive of 1% on the installment if they pay the installment in time. A sample of 30 customers of the bank reveals that only 12 of them have availed this incentive. Determine the confidence interval for the proportion representing the number of customers making the payment in time to avail the incentive with a confidence level of 0.95.
42. The axial length of pistons produced follows normal distribution with the variance of 81 mm. Determine the sample size such that the mean length of pistons is within plus or minus 2.5 mm with a confidence level of 90%.
43. The outer length of connecting rods produced follows normal distribution with the variance of 625 mm. Determine the sample size such that the mean outer length of connecting rods is within plus or minus 1.5 mm with a confidence level of 95%.
44. The finance manager of a company feels that 55% of branches will have enhanced yearly collection of deposits after introducing a hike in interest rate. Determine the sample size such that the mean proportion is within plus or minus 0.05 at a confidence level of 90%.
45. The Director of a management school feels that 55% of students will have enhanced performance if additional input is given to them. Determine the sample size such that the mean proportion is within plus or minus 0.10 at a confidence level of 95%.

CHAPTER

8

TESTS OF HYPOTHESES

8.1 INTRODUCTION

In many realities, inferences about populations are to be drawn based on the characteristics of samples. As discussed earlier, sampling enables a researcher to draw an inference about a population. The inference may be pertaining to certain hypothesis.

Hypothesis is an assumption about a population. Consider, a study relating to buyers behaviour. A few sample hypotheses are presented as follows:

1. Mean purchases made by females (μ_1) is more than or equal to the mean purchases made by males (μ_2) in a textile store ($\mu_1 \geq \mu_2$).
2. Mean age of female shoppers (μ_1) is less than or equal to that of male shoppers (μ_2) in a book exhibition ($\mu_1 \leq \mu_2$).
3. Mean monthly income of buyers (μ) in a shop is more than or equal to Rs.10,000 ($\mu \geq 10000$).
4. The mean stay-over time of customers (μ) in a shop is at most 45 minutes ($\mu \leq 45$).

There are two types of hypothesis: namely null hypothesis and alternate hypothesis.

Null hypothesis

It is an assumption (preferred assumption) about a population. It is denoted by H_0 .

Consider a particular sub-assembly (fuel pump) which is received from a vendor for use in the assembly of motorcycle in an automobile company. Now, take the example of inspecting the number of defects found in each sub-assembly of a sample for the purpose of controlling the quality of the sub-assemblies. Let the number of defects per sub-assembly follows normal distribution with mean μ and variance σ^2 .

Let us assume that the vendor claims that the mean number of defects found in his shipments of fuel pumps is limited to 5. Then, the inspector takes a sample and inspects all the pumps in it to find the number of defects in each pump.

A critical value (cut-off value) for the mean defects found in the sample ($\bar{X}_c$) is calculated based on the given significance level (α), and population parameter (σ) and/or sample parameter(s). Based on this critical value, the decision is taken either to accept or reject the null hypothesis. A possible null hypothesis made by the inspector about the population from which he will draw a sample is H_0 : The mean number of defects per sub-assembly in the population is more than or equal to 5. That is,

$$\mu \geq 5 \quad (8.1)$$

The null hypothesis is made in the opposite spirit of the claim of the vendor. The interpretation of this null hypothesis is that the mean number of defects per sub-assembly in the population will be more than or equal to 5.

The formula for computing the critical value $\bar{X}_c$ for this type of null hypothesis is shown as (based on Fig. 8.1)

$$\bar{X}_c = \mu - Z_\alpha \sigma$$

where $\bar{X}_c$ is the critical value for the mean number of defects per sub-assembly in the sample; μ , the mean of the number of defects per sub-assembly in the population; α , the level of significance; Z_α the standard normal statistic for a given α ; and σ , the standard deviation of the number of defects per sub-assembly in the population.

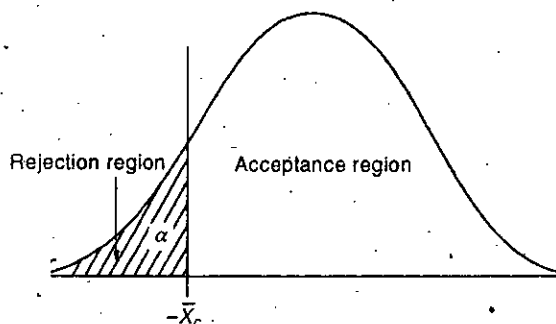


Fig. 8.1 Normal curve for $H_0 \geq 5$.

The null hypothesis is accepted if the mean number of defects found in the sample is more than the calculated critical value; otherwise, it is rejected.

Alternatively, the inspector can state the hypothesis as:

H_0 : The mean number of defects per sub-assembly in the population is at most 5. That is,

$$\mu \leq 5 \quad (8.2)$$

The null hypothesis is made in the same spirit of the claim of the vendor. The interpretation of this null hypothesis is that the mean number of defects per sub-assembly in the population is less than or equal to 5.

The formula for computing the critical value $\bar{X}_c$ for this type of null hypothesis is shown as (based on Fig. 8.2).

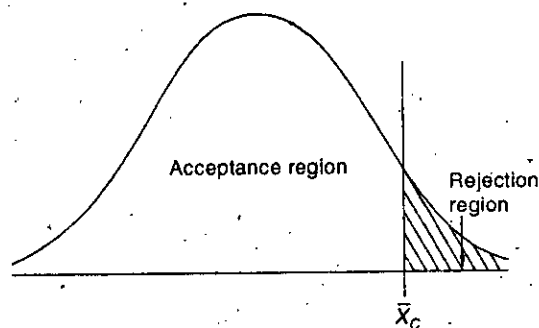


Fig. 8.2 Normal curve for $H_0 \leq 5$.

$$\bar{X}_c = \mu + Z_\alpha \sigma$$

where $\bar{X}_c$ is the critical value for the mean number of defects per sub-assembly in the sample; μ , the mean number of defects per sub-assembly in the population; α , the level of significance; Z_α is the standard normal statistic for a given α ; and σ , the standard deviation for the mean number of defects per sub-assembly in the population.

The null hypothesis is accepted if the mean number of defects per sub-assembly in the sample is less than or equal to the calculated critical value; otherwise, it is rejected.

The first type of statement for the null hypothesis is preferable.

Alternate hypothesis

If the null hypothesis (H_0) is rejected, then the opposite of the null hypothesis must be true. The hypothesis representing the opposite of the null hypothesis is called *alternate* hypothesis. It is denoted by H_1 .

If the null hypothesis is as in (8.1), then the alternate hypothesis made by the inspector about the population is stated as follows:

H_1 : The mean number of defects per sub-assembly in the sample is less than 5. That is,

$$H_1: \mu < 5$$

- (a) If the mean number of defects per sub-assembly in the sample is greater than or equal to the critical value, then the null hypothesis (H_0) is accepted and the alternate hypothesis (H_1) is rejected;
- (b) Otherwise, the null hypothesis (H_0) is rejected and the alternate hypothesis (H_1) is accepted.

If the null hypothesis (H_0) is as in (8.2), then the alternate hypothesis is as:

H_1 : The mean number of defects per sub-assembly in the sample is more than 5. That is:

$$H_1: \mu > 5$$

- (a) If the mean number of defects per sub-assembly in the sample is less than or equal

to the critical value, then the null hypothesis (H_0) is accepted and the alternate hypothesis (H_1) is rejected;

- (b) Otherwise, the null hypothesis (H_0) is rejected and the alternate hypothesis (H_1) is accepted. Based on the above discussions, the two combinations of H_0 and H_1 are as:

Case 1: $H_0: \mu \geq 5$; $H_1: \mu < 5$

Case 2: $H_0: \mu \leq 5$; $H_1: \mu > 5$

Significance level (α)

The *significance level* is the probability with which the null hypothesis (H_0) will be rejected due to sampling error, though H_0 is true.

It is a well-known fact that the inference through sampling may have certain amount of error. Based on the explanation given above, the error can be classified into Type-I error (*producer's risk*) and Type-II error (*consumer's risk*).

Type-I error It is the probability of rejecting the null hypothesis when it is true. It is denoted by α . In quality control, it is termed as producer's risk, because it is the probability of rejecting a good lot.

Type-II error It is the probability of accepting the null hypothesis when it is false. It is denoted by β . In quality control, it is termed as consumer's risk, because it is the probability of accepting a bad lot.

The above types of error are summarized in Table 8.1.

Table 8.1 Types of Error

Decision based on sample	Real state of H_0	
	True	False
Reject H_0	Type-I error	No error
Accept H_0	No error	Type-II error

The hypotheses, H_0 and H_1 , are mutually exclusive events. This means that if H_0 is accepted, then H_1 will be rejected; if H_0 is rejected, then H_1 will be accepted. Hence, defining Type-I error and Type-II error with respect to H_1 is not necessary.

Power curve

Consider the following hypotheses:

$$H_0: \mu \leq 5$$

$$H_1: \mu > 5$$

The *power curve* is a curve showing the probabilities of rejection of the null hypothesis when it is true for different values of μ with respect to a standard normal statistic. The

computation of the probability of acceptance relating to the hypothesis $H_0: \mu \leq k$ is as shown below, where k is a constant.

Let the critical value ($\bar{X}_c$) be 5.5 and the sample size (n) be 49 with the population standard deviation (σ) as 1. From these, it is clear that one should take a sample of 49 fuel pumps. If the mean number of defects found in the sample is more than 5.5, then he should reject the null hypothesis; otherwise, he should accept the null hypothesis. If the true value of the mean (μ) is 5.1, then the probability of rejecting H_0 is given by the following formula:

$$\begin{aligned} P(\bar{X}_c \geq 5.5) &= P\left(\frac{\bar{X}_c - \mu}{\sigma/\sqrt{n}} \geq \frac{5.5 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(Z \geq \frac{5.5 - 5.1}{1/\sqrt{49}}\right) \\ &= 1 - P(Z \leq 2.8) \\ &= 1 - 0.9974 \\ &= 0.0026 \end{aligned}$$

where $\bar{X}_c$ is the critical value of the mean number of defects per sub-assembly in the sample; μ , the mean number of defects per sub-assembly in the population; Z , the standard normal statistic; σ , the standard deviation of the number of defects per sub-assembly in the population; and, n , the sample size.

Like this, one can compute the probability of rejecting H_0 for different values of μ . The plot of such probabilities for different values of μ (power curve) is shown in Fig. 8.3.

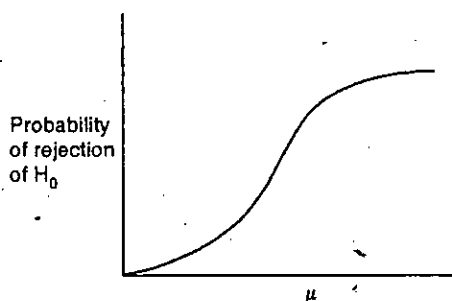


Fig. 8.3 Power curve.

Operating characteristic curve (OC curve)

OC curve is the curve showing the probability of acceptance of H_0 when it is true for different values of μ . It is the complement of the power curve. A sample OC curve is shown in Fig. 8.4.

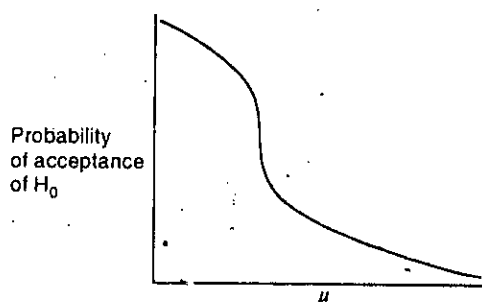


Fig. 8.4 Operating characteristics curve.

Test of hypothesis

Test of hypothesis is an experiment conducted on a population of a reality to draw inference concerning certain measure (mean or variance or proportion). This is done through sampling. The inference is drawn through a sample measure in conjunction with the defined hypothesis for the population of the reality.

A given practical situation may require any of the following tests of hypotheses:

- (a) Tests of hypotheses concerning mean(s);
- (b) Tests of hypotheses concerning variance/variances;
- (c) Tests of hypotheses concerning proportion(s).

8.2 TESTS OF HYPOTHESES CONCERNING MEAN(S)

The tests of hypotheses concerning mean(s) can be classified into tests of hypotheses concerning single mean and test of hypothesis concerning difference between two means. These are explained in the following subsections.

The inference about a population can be arrived based on its sample characteristics in terms of the following measures:

1. One-tailed tests concerning single mean when the variance of the population is known and the population is infinite.
2. Two-tailed tests concerning single-mean when the variance of the population is known and the population is infinite.
3. One-tailed tests concerning single mean when the variance of the population is known and the population is finite.
4. Two-tailed tests concerning single mean when the variance of the population is known and the population is finite.
5. One-tailed tests concerning single mean when the variance of the population is unknown and the sample size is large.
6. Two-tailed tests concerning single mean when the variance of the population is unknown and the sample size is large.

7. One-tailed tests concerning single mean when the variance of the population is unknown and the sample size is small.
 8. Two-tailed tests concerning single mean when the variance of the population is unknown and the sample size is small.
 9. One-tailed tests concerning difference between two means when the variances of the populations are known.
 10. Two-tailed tests concerning difference between two means when the variances of the populations are known.
 11. One-tailed tests concerning difference between two means when the variances of the populations are unknown and the sample sizes are large.
 12. Two-tailed tests concerning difference between two means when the variances of the populations are unknown and the sample sizes are large.
 13. One-tailed tests concerning difference between two means when the variances of the populations are unknown and the sample sizes are small.
 14. Two-tailed tests concerning difference between two means when the variances of the populations are unknown and the sample sizes are small.
- Now, they are discussed in the following sections.

8.2.1 One-tailed Tests Concerning Single Mean (when the Variance of the Population is Known and the Population is Infinite)

Consider a population in which a random variable follows normal distribution, and its mean and variance are known. A random sample of size n ($n > 30$) is drawn from the population and its mean is estimated.

One-tailed tests concerning single mean, when the variance of the population is known are classified as follows:

1. Test 1: $H_0: \mu \leq k; H_1: \mu > k$
2. Test 2: $H_0: \mu \geq k; H_1: \mu < k$

where k is a constant.

The above two tests are demonstrated with suitable examples:

Test 1: $H_0: \mu \leq k; H_1: \mu > k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing the significance level of α at the right tail of the standard normal distribution.

Example 8.1 Consider the situation in which the monthly sales revenue of salesmen in a company follows normal distribution. The targetted mean of the monthly sales of the salesmen is Rs. 10 lakhs. The variance of the monthly sales revenues of the salesmen is Rs. 16 lakhs. The regional marketing manager of the company feels that the performance of salesmen has declined in the recent past. A random sample of 36 salesmen is taken from the normal population for which the mean monthly sales revenue is found to be Rs. 10.5 lakhs. Check whether the sales revenue of the salesmen has really declined at a significance level of 0.05.

Solution. There are three approaches to solve this problem, which are explained as follows:

Let X be a random variable indicating the sales revenue of a salesman, which follows normal distribution. The mean of monthly sales revenues of salesmen in the population (μ) is Rs. 10 lakhs with variance (σ^2) of Rs. 16 lakhs. Then

$$\mu = \text{Rs. 10 lakhs}$$

$$\sigma^2 = \text{Rs. 16 lakhs}$$

$$\sigma = \text{Rs. 4 lakhs}$$

$$\text{Sample size, } n = 36$$

$$\text{Sample mean, } \bar{X} = \text{Rs. 10.5 lakhs}$$

where

$$X \sim N(\mu, \sigma^2) \quad \text{and} \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

We also have $\alpha = 0.05$. Then, $Z_\alpha = 1.64$. Now,

$$\text{Null hypothesis, } H_0: \mu \leq 10$$

$$\text{Alternate hypothesis, } H_1: \mu > 10$$

Approach-1: In this approach, a cut-off value is calculated using the following formula. If the sample mean is less than or equal to the cut-off value (C), then the null hypothesis (H_0) will be accepted; otherwise, it is rejected.

$$C = \mu + Z_\alpha \frac{\sigma}{\sqrt{n}} = 10 + 1.64 \frac{4}{\sqrt{36}} = 11.09$$

The details of this test are shown in Fig. 8.5.

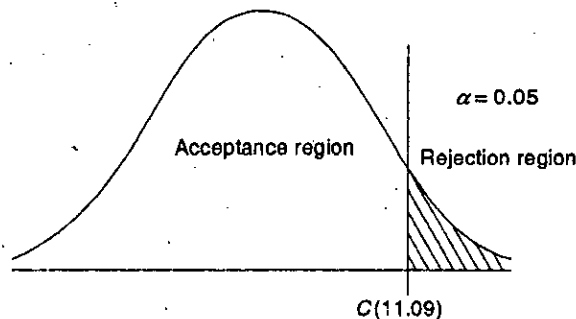


Fig. 8.5 Normal curve for Example 8.1 (approach-1).

Since the mean monthly sales revenue of the sample (10.5 lakhs) is less than the cut-off value, C (11.09 lakhs), accept the null hypothesis (H_0).

Hence the inference of the test is that the mean monthly sales revenue of the salesmen has really declined against the targetted mean of Rs. 10 lakhs.

Approach-2: In this approach, the probability of rejection of H_0 can be calculated using the following formula. If it is more than the significance level (α), then H_0 is accepted; otherwise, it is rejected.

Since sample mean, $\bar{X} = 10.5$ lakhs, we have

$$\begin{aligned} P(\bar{X} \geq 10.5) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{10.5 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(Z_{\bar{X}} \geq \frac{10.5 - 10}{4/\sqrt{36}}\right) \\ &= P(Z_{\bar{X}} \geq 0.75) \\ &= 1 - P(Z_{\bar{X}} \leq 0.75) \\ &= 1 - 0.7734 \\ &= 0.2266 \end{aligned}$$

Since the above probability is more than the given significance level of 0.05, H_0 should be accepted which amounts to the same inference given earlier. These are shown graphically in Fig. 8.6.

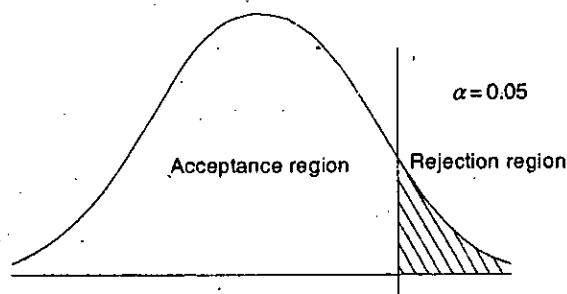


Fig. 8.6 Normal curve for Example 8.1 (approach-2).

Approach-3: In this approach, the standard normal statistic for the given problem is computed using the following formula. If this computed standard normal statistic is less than or equal to the table value (Z_{α}), then accept H_0 ; otherwise, reject it.

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{10.5 - 10}{4/\sqrt{36}} = 0.75$$

The standard normal statistic for the given significance level of 0.05, Z_{α} is 1.64, which is shown in Fig. 8.7.

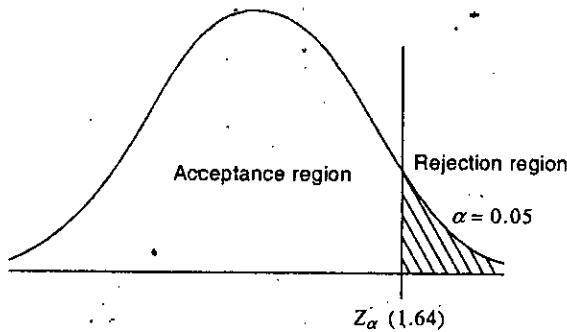


Fig. 8.7 Normal curve for Example 8.1 (approach-3).

The calculated Z -value (0.75) is less than Z_{α} (1.64) and it falls in the acceptance region. Hence, H_0 should be accepted which is same as the results of the earlier two approaches.

Test 2: $H_0: \mu \geq k$; $H_1: \mu < k$

The objective of this test to check whether the population is to be accepted based on the value of the sample mean (μ) by placing the significance level of α at the left tail of the standard normal distribution.

Example 8.2 Consider the situation in which the monthly sales revenues of salesmen follow normal distribution. The targetted mean of the monthly sales of the salesmen and their variance in the population are Rs.12 lakhs and Rs.16 lakhs, respectively. The regional marketing manager feels that the performance of salesmen has improved in the recent past. A random sample of 36 salesmen is taken from the normal population for which the mean monthly sales revenue is found to be Rs. 10.5 lakhs. Check whether the sales revenue of the salesmen has really improved at a significance level of 0.05.

Solution. Let X be the random variable indicating the sales revenue of a salesman which follows normal distribution.

The mean of monthly sales revenues of salesmen in the population (μ) is Rs.12 lakhs with variance (σ^2) of Rs. 16 lakhs. We have,

$$\mu = \text{Rs. 12 lakhs}$$

$$\sigma^2 = \text{Rs. 16 lakhs}$$

$$\sigma = \text{Rs. 4 lakhs}$$

$$\text{Sample size, } n = 36$$

$$\text{Sample mean, } \bar{X} = \text{Rs. 10.5 lakhs}$$

Hence, $X \sim N(\mu, \sigma^2)$ and $\bar{X} \sim N(\mu, \sigma^2/n)$. Then, $\alpha = 0.05$ and $Z_{\alpha} = 1.64$. Null and alternate hypotheses are:

$$H_0: \mu \geq 12$$

$$H_1: \mu < 12$$

The standard normal statistic for the given problem is computed using the following formula:

If this computed standard normal statistic is more than or equal to the table value of $-Z_{\alpha}$, then accept H_0 ; otherwise, reject it.

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{10.5 - 12}{4/\sqrt{36}} = -2.25$$

The standard normal statistic for the given significance level of 0.05, Z_{α} , is 1.64. Therefore, $-Z_{\alpha} = -1.64$.

The calculated Z value (-2.25) is less than $-Z_{\alpha}$ (-1.64) and it falls in the rejection region. Hence, H_0 should be rejected. These are shown in Fig. 8.8.

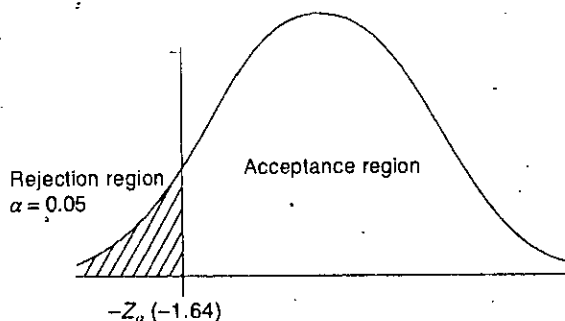


Fig. 8.8 Normal curve for Example 8.2.

Hence the performance of salesmen has not improved in the recent past.

8.2.2 Two-tailed Test Concerning Single Mean (when the Variance of the Population is Known and the Population is Infinite)

The two-tailed test is applicable to the following combination of H_0 and H_1 :

$$H_0: \mu = k, \quad H_1: \mu \neq k$$

where k is a constant.

Consider a population in which a random variable follows normal distribution, and its mean and variance are known. A random sample of size n ($n > 30$) is drawn from the population and its mean is estimated. The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing half of the given significance level ($\alpha/2$) at both the tails of the standard normal distribution.

Example 8.3 The annual sales of the dealers of a company follow normal distribution. The targetted mean of the annual sales of the dealers of the population is Rs. 50 lakhs. The variance of the annual sales of the dealers of the population is Rs. 100 lakhs. The regional marketing manager feels that the performance of the dealers has not changed from the population mean in the recent past. A random sample of 64 dealers is taken from the normal population for which the mean annual sales is found to be Rs. 52 lakhs. Check whether the sales of the dealers has not changed from Rs. 50 lakhs at a significance level of 0.01.

Solution. There are two approaches to solve this problem which are explained below:

Let X be the random variable indicating the annual sales of a dealer which follows normal distribution. The mean of annual sales of dealers in the population (μ) is Rs. 50 lakhs with variance (σ^2) of Rs. 100 lakhs. Then

$$\mu = \text{Rs. 50 lakhs}$$

$$\sigma = \text{Rs. 10 lakhs}$$

$$\text{Sample size, } n = 64$$

$$\text{Sample mean, } \bar{X} = \text{Rs. 52 lakhs}$$

We get $X \sim N(\mu, \sigma^2)$. In other words, $\bar{X} \sim N(\mu, \sigma^2/n)$. Then, $\alpha = 0.01$ and $Z_{\alpha/2} = 2.57$. Then,

$$\text{Null hypothesis, } H_0: \mu = 50$$

$$\text{Alternate hypothesis, } H_1: \mu \neq 50$$

Approach-1: In this approach, two cut-off values are calculated using the formula given below:

$$C = \mu \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The cut-off values are the lower limit of the acceptance region (C_1) and the upper limit of the acceptance region (C_2). If the sample mean is in between these two cut-off values (between C_1 and C_2), then the null hypothesis, (H_0) is accepted; otherwise, it is rejected.

Therefore,

$$C_1 = \mu - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 50 - 2.57 \frac{10}{\sqrt{64}} = 50 - 3.21 = 46.79$$

and

$$C_2 = \mu + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 50 + 2.57 \frac{10}{\sqrt{64}} = 50 + 3.21 = 53.21$$

The details of this test are shown in Fig. 8.9.

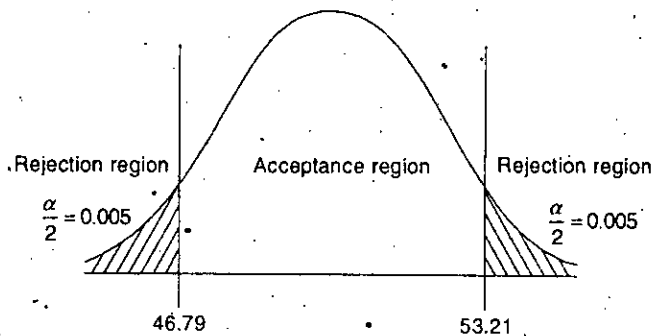


Fig. 8.9 Normal curve for Example 8.3 (approach-1).

Since the mean annual sales of the sample (52 lakhs) is in between the C_1 and C_2 , (46.79 lakhs to 53.21 lakhs), accept the null hypothesis (H_0).

Hence the inference of the test is that the annual sales of the dealers has not significantly changed from the population mean of Rs. 50 lakhs.

Approach-2: In this approach, the standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{52 - 50}{10/\sqrt{64}} = 1.6$$

If this computed standard normal statistic is in between the table values $-Z_{\alpha/2}$ and $Z_{\alpha/2}$, then accept H_0 ; otherwise, reject it.

The standard normal statistics (from the table) $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$ are -2.57 and 2.57 , respectively, which are shown in Fig. 8.10.

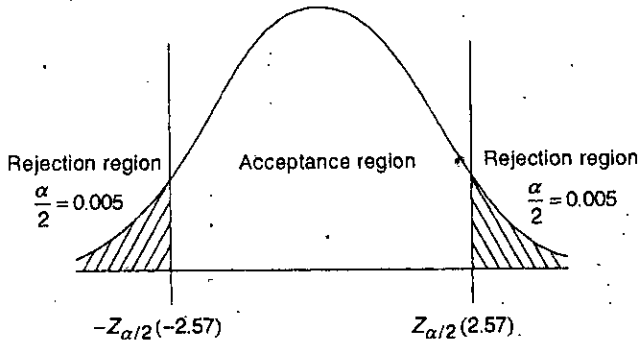


Fig. 8.10 Normal curve for Example 8.3 (approach-2).

The calculated Z value (1.6) is in between $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$ (-2.57 and 2.57) and it falls in the acceptance region. Hence, H_0 should be accepted which is same as the results of the earlier approach.

8.2.3 One-tailed Tests Concerning Single Mean (when the Variance of the Population is Known and the Population is Finite)

Consider a finite population in which a random variable follows normal distribution, and its mean and variance are known. A random sample of size n ($n > 30$) is drawn from the population and its mean is estimated.

These are same as in Section 8.2.1 except the size of the population. Here, the size of the population is finite. The one-tailed tests applicable to this situation are as follows:

Test 1: $H_0: \mu \leq k; H_1: \mu > k$

Test 2: $H_0: \mu \geq k; H_1: \mu < k$

where k is a constant.

Test 1 $H_0: \mu \leq k; H_1: \mu > k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing the significance level of α at the right tail of the standard normal distribution.

Example 8.4 The bill amount of purchases made by customers at a leading readymade garment shop in an industrial complex follows normal distribution with finite population size of 1000. The targetted mean of the bill amount of customers of the population is Rs. 10,000. The variance of the bill amount of the customers of the population is Rs. 2,50,000 lakhs. The shop manager feels that the mean bill amount of customers has declined from the targetted mean of Rs. 10,000 in the recent past. A random sample of 81 customers is taken from the normal population for which the mean bill amount is found to be Rs. 10,250. Check whether the mean bill amount has declined from the targetted mean of Rs. 10,000 at a significance level of 0.05.

Solution. Let X be a random variable indicating the bill amount of a customer which follows a normal distribution, whose population is finite. We also have

Population size, N	= 1000
Sample size, n	= 81
Population mean, μ	= Rs. 10,000
Variance of the population, σ^2	= Rs. 2,50,000
Standard deviation of the population, σ	= Rs. 500
Sample mean, $\bar{X}$	= Rs. 10,250
Significance level, α	= 0.05
Z_α	= 1.64

From, $X \sim N(\mu, \sigma^2)$, we get

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right)$$

Null and alternate hypotheses are:

$$H_0: \mu \leq 10,000$$

$$H_1: \mu > 10,000$$

Then

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{10250 - 10000}{\frac{500}{\sqrt{81}} \sqrt{\frac{1000-81}{1000-1}}} = 4.69$$

The standard normal statistic (Z_α) is 1.64, which is shown in Fig. 8.11.

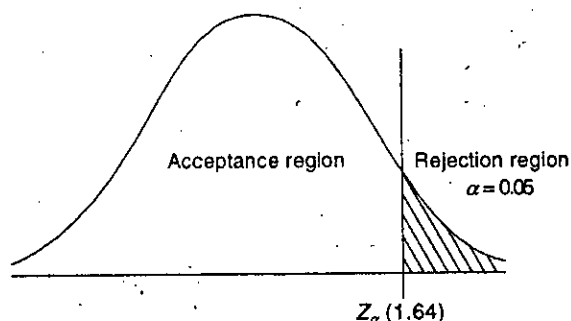


Fig. 8.11 Normal curve for Example 8.4.

The calculated Z value (4.69) is more than Z_{α} (1.64) and it falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean bill amount of the customers has not declined from the targetted mean of Rs. 10,000 in the recent past.

Test 2: $H_0: \mu \geq k$; $H_1: \mu < k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing the significance level of α at the left tail of the standard normal distribution.

Example 8.5 Consider Example 8.4 again as reproduced here below with some modifications:

The bill amount of purchases made by customers at a leading ready-made garment shop in an industrial complex follows normal distribution with finite population size of 1000. The targetted mean of the bill amount of customers of the population is Rs. 8,000. The variance of the bill amount of the customers of the population is Rs. 1,60,000 lakhs. The shop manager feels that the mean bill amount of customers has improved from the targetted mean of Rs. 8,000, in the recent past. A random sample of 64 customers is taken from the normal population for which the mean bill amount is found to be Rs. 7,900. Check whether the mean bill amount has improved from the targetted mean of Rs. 8,000 at a significance level of 0.05.

Solution. Let X be a random variable indicating the bill amount of a customer which follows a normal distribution whose population is finite. We also have

Population size, N	= 1000
Sample size, n	= 64
Population mean, μ	= Rs. 8,000
Variance of the population, σ^2	= Rs. 1,60,000
Standard deviation of the population, σ	= Rs. 400
Sample mean, $\bar{X}$	= Rs. 7,900
Significance level, α	= 0.05

Then

$$-Z_{\alpha} = -1.64$$

and

$$X \sim N(\mu, \sigma^2), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right)$$

The hypotheses are:

$$H_0: \mu \geq 8000$$

$$H_1: \mu < 8000$$

Then

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{7900 - 8000}{\frac{400}{\sqrt{64}} \sqrt{\frac{1000-64}{1000-1}}} = -2.066$$

The standard normal statistic, $-Z_{\alpha}$ is -1.64 which is shown in Fig. 8.12 along with acceptance and rejection regions.

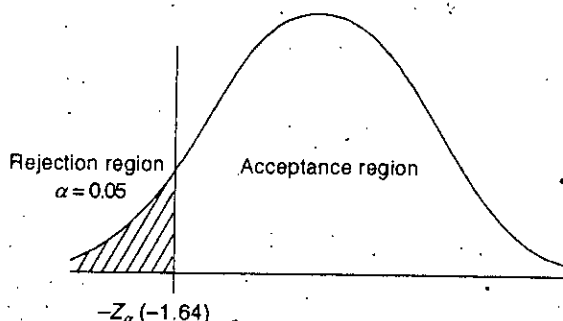


Fig. 8.12 Normal curve for Example 8.5.

The calculated Z value (-2.066) is less than $-Z_{\alpha}(-1.64)$ and it falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean bill amount of customers has not improved significantly from the targeted mean of Rs. 8,000 in the recent past.

8.2.4 Two-tailed Test Concerning Single Mean (when the Variance of the Population is Known and the Population is Finite)

As stated earlier, the two-tailed test is applicable to the following combination of H_0 and H_1 :

$$H_0: \mu = k$$

$$H_1: \mu \neq k$$

where k is a constant.

Consider a population in which a random variable follows normal distribution, and its mean and variance are known. A random sample of size n ($n > 30$) is drawn from the population and its mean is estimated. The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing half of the given significance level ($\alpha/2$) at both tails of the standard normal distribution.

The standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

If this computed standard normal statistic is in between the table values $-Z_{\alpha/2}$ and $Z_{\alpha/2}$, then accept H_0 ; otherwise, reject it.

Example 8.6 Consider Example 8.4 as reproduced below with some modifications.

The bill amount of purchases made by customers at a leading ready-made garment shop in an industrial complex follows normal distribution with finite population size of 1000. The targetted mean of the bill amount of customers of the population is Rs. 10,000. The variance of the bill amount of the customers of the population is Rs. 4,90,000 lakhs. The shop manager feels that the mean bill amount of customers has not changed from the targetted mean of Rs. 10,000, in the recent past. A random sample of 36 customers is taken from the normal population for which the mean bill amount is found to be Rs. 10,250. Check whether the mean bill amount has declined from the targetted mean of Rs. 10,000 at a significance level of 0.01.

Solution. Let X be a random variable indicating the bill amount of a customer which follows a normal distribution whose population is finite. We also have

Population size, N	= 1000
Sample size, n	= 36
Population mean, μ	= Rs. 10,000
Variance of the population, σ^2	= Rs. 4,90,000
Standard deviation of the population, σ	= Rs. 700
Sample mean, $\bar{X}$	= Rs. 10,250
Significance level, α	= 0.01

Then

$$Z_{\alpha/2} = 2.57, -Z_{\alpha/2} = -2.57$$

and

$$X \sim N(\mu, \sigma^2), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right)$$

Null and alternate hypotheses are:

$$H_0: \mu = 10000$$

$$H_1: \mu \neq 10000$$

Therefore

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{10250 - 10000}{\frac{700}{\sqrt{36}} \sqrt{\frac{1000-36}{1000-1}}} = 2.181$$

The table standard normal values, $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ are -2.57 and $+2.57$, respectively, which are shown in Fig. 8.13.

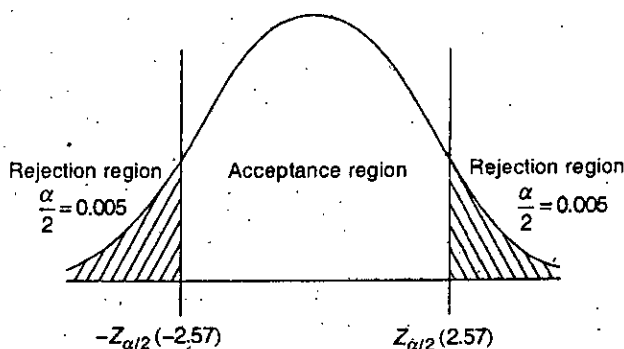


Fig. 8.13 Normal curve for Example 8.6.

The calculated Z value (2.181) is in between -2.57 and $+2.57$, and it falls in the acceptance region. Hence, H_0 should be accepted.

This means that the mean bill amount of customers has not changed from the targetted mean of Rs. 10,000 in the recent past.

8.2.5 One-tailed Tests Concerning Single Mean (when the Variance of the Population is Unknown and the Sample Size is Large)

Such one-tailed tests are applicable to the following combinations of H_0 and H_1 :

1. Test 1: $H_0: \mu \leq k$ and $H_1: \mu > k$
2. Test 2: $H_0: \mu \geq k$ and $H_1: \mu < k$

where k is a constant.

Consider a population in which a random variable follows normal distribution with a mean μ and an unknown variance, k . A large random sample of size n ($n > 30$) is drawn from the population and its mean ($\bar{X}$) and variance (S^2) are estimated. Since the size of the sample is large, one can estimate the population variance from the sample variance. The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing the given significance level (α), as follows:

- (a) At the right tail of the standard normal distribution in the case of Test 1 and
 (b) At the left tail of the standard normal distribution in the case of Test 2.

Infinite population

If the population is infinite, then the variance of the sample is treated as the variance of the population. The standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

In Test 1, if this computed standard normal statistic is less than Z_{α} , then accept H_0 ; otherwise, reject it.

In Test 2, if this computed standard normal statistic is more than Z_{α} , then accept H_0 ; otherwise, reject it.

Finite population

If the population is finite ($n/N > 0.05$), the variance of the population is estimated from the variance of the sample using the following formula:

$$\sigma^2_{\bar{X}} = \frac{\sigma^2}{n} \frac{N-n}{N-1} = \frac{S^2}{n} \frac{N-n}{N-1}$$

The standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

In Test 1, if this computed standard normal statistic is less than Z_{α} , then accept H_0 ; otherwise, reject it.

In Test 2, if this computed standard normal statistic is more than Z_{α} , then accept H_0 ; otherwise, reject it.

Example 8.7 The annual income of the respondents in a survey follows normal distribution whose population is infinite. The expected mean of the annual income of the respondents of that population is Rs. 5 lakhs and its variance is unknown. The researcher feels that the mean annual income of the respondents is significantly less than the expected mean income of the respondents. So, he has selected a random sample of 81 respondents and its mean and variance are found to be Rs. 5.25 lakhs and 1 lakh, respectively. Verify the intuition of the researcher at a significance level of 0.10.

Solution: Given that

Population mean, μ = Rs. 5 lakhs

Sample size, n = 81

Sample mean, $\bar{X}$	= Rs. 5.25 lakhs
Variance of the sample, S^2	= Rs. 1 lakh
Standard deviation of the sample, S	= Rs. 1 lakh
Significance level, α	= 0.10
Z_α	= 1.28

Since, the population is infinite, the variance of the sample is treated as the variance of the population. Hence $\sigma = \text{Rs. 1 lakh}$. Then

$$X \sim N(\mu, \sigma^2), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \sim N\left(\mu, \frac{S^2}{n}\right)$$

Null and alternate hypotheses are:

$$H_0: \mu \leq 5$$

$$H_1: \mu > 5$$

Therefore,

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{5.25 - 5}{1/\sqrt{81}} = 2.25$$

The table standard normal value (Z_α) is 1.28 which is shown in Fig. 8.14 along with acceptance and rejection regions.

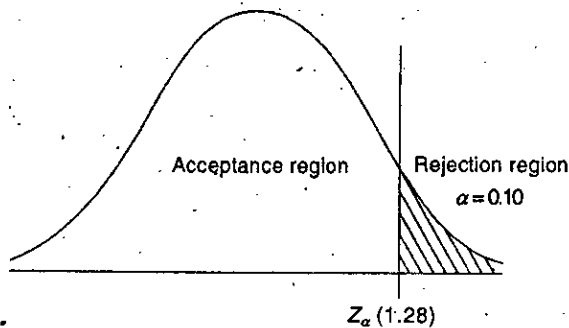


Fig. 8.14 Normal curve for Example 8.7.

The calculated Z value (2.25) is more than Z_α (1.28) and it falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean annual income of the respondents is not significantly less than the expected population mean of Rs. 5 lakhs.

Example 8.8 The weight of cement bags produced by Alpha Cement Company follows normal distribution whose population is infinite. The expected mean of the weight of the cement bags for sales of this population is 50 kg and its variance is unknown. The sales manager of the firm claims that the mean weight of the cement bags is significantly more than the expected mean weight of the population. So, he has selected a random sample of 64 bags and its mean and variance are found to be 50.4 kg and 0.64 kg, respectively. Verify the intuition of the sales manager at a significance level of 0.10.

Solution. Given

Sample size, n	= 64
Population mean, μ	= 50 kg
Sample mean, $\bar{X}$	= 50.4 kg
Variance of the sample, S^2	= 0.64 kg
Standard deviation of the sample, S	= 0.8 kg
Significance level, α	= 0.10
Z_α	= 1.28.

Since, the population is infinite, the variance of the sample is treated as the variance of the population. Therefore, $\sigma = 0.8$ kg. Then,

$$X \sim N(\mu, \sigma^2), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \sim N\left(\mu, \frac{S^2}{n}\right)$$

Null and alternate hypotheses are:

$$H_0: \mu \geq 50$$

$$H_1: \mu < 50$$

Therefore,

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{50.4 - 50}{0.8/\sqrt{64}} = 4$$

The table standard normal value, $-Z_\alpha$ is -1.28 which is shown in Fig. 8.15 along with acceptance and rejection regions.

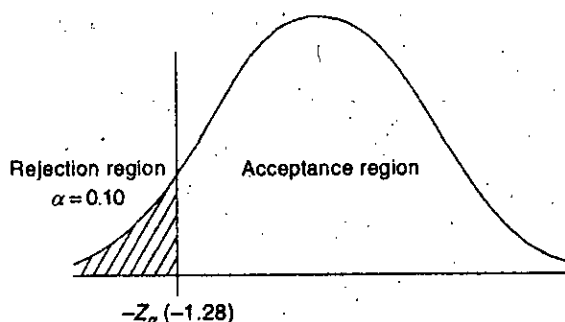


Fig. 8.15 Normal curve for Example 8.8.

The calculated Z value (4) is more than $-Z_\alpha (-1.28)$ and it falls in the acceptance region. Hence, H_0 should be accepted.

This means that the mean weight of the cement bags is significantly more than the expected population mean of 50 kg.

Example 8.9 Consider the Example 8.8, as reproduced below, with some modifications:

The weight of cement bags produced by Alpha Cement Company follows normal distribution whose population is finite and its size is 900. The expected mean of the weight of the cement bags for sales of this population is 50 kg and its variance is unknown. The sales manager of the firm claims that the mean weight of the cement bags is significantly more than the expected mean weight of the population. So, the purchase manager of Beta Construction Company who places order for cement bags with the Alpha Cement Company has selected a random sample of 49 bags and its mean and variance are found to be 49.8 kg and 1.44 kg, respectively. Verify the intuition of the sales manager of the cement company at a significance level of 0.10.

Solution. We have

Population size, N	= 900
Sample size, n	= 49
Population mean, μ	= 50 kg
Sample mean, $\bar{X}$	= 49.8 kg
Variance of the sample, S^2	= 1.44 kg
Standard deviation of the sample, S	= 1.2 kg
Significance level, α	= 0.10
$-Z_\alpha$	= -1.28

Since, the population is finite ($n/N > 0.05$), the variance of the population is computed using the following formula:

$$\sigma_{\bar{X}}^2 = \frac{S^2}{n} \frac{N-n}{N-1}$$

or

$$\sigma_{\bar{X}} = \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{1.2}{\sqrt{49}} \sqrt{\frac{900-49}{900-1}} = 0.1668$$

We also have $\bar{X} \sim N(\mu, \sigma^2)$. Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right) \sim N\left(\mu, \frac{S^2}{n} \frac{N-n}{N-1}\right)$$

Null and alternate hypotheses are:

$$H_0: \mu \geq 50$$

$$H_1: \mu < 50$$

Then,

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{49.8 - 50}{0.1668} = -1.199$$

The table standard normal value, $-Z_{\alpha}$ is -1.28 which is shown in Fig. 8.16 along with acceptance and rejection regions.

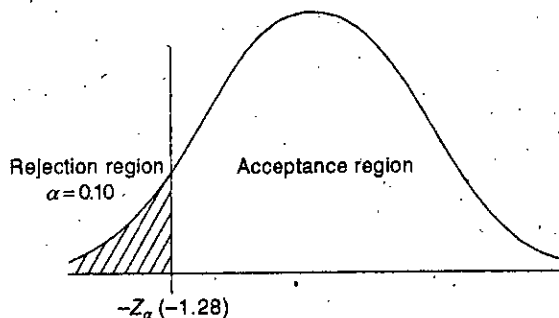


Fig. 8.16 Normal curve for Example 8.9.

The calculated Z value (-1.199) is more than $-Z_{\alpha}$ (-1.28) and it falls in the acceptance region. Hence, H_0 should be accepted.

This means that the mean weight of the cement bags is significantly more than the population mean of 50 kg.

Example 8.10 Consider Example 8.7, as reproduced below, with some modifications.

The annual income of respondents in a survey follows normal distribution whose population size is 900. The expected mean of the annual income of the respondents of this population is Rs. 6 lakhs and its variance is unknown. The researcher feels that mean annual income of the respondents is significantly more than the expected mean income of the respondents. So, he has selected a random sample of 100 respondents and its mean and variance are found to be Rs. 5.70 lakhs and 1.44 lakhs, respectively. Verify the intuition of the researcher at a significance level of 0.01.

Solution. We have

Population size, N	= 900
Sample size, n	= 100
Population mean, μ	= Rs. 6 lakhs
Sample mean, $\bar{X}$	= Rs. 5.7 lakhs
Variance of the sample, S^2	= Rs. 1.44 lakhs
Standard deviation of the sample, S	= Rs. 1.2 lakhs
Significance level α	= 0.01
$-Z_{\alpha}$	= -2.33

Since, the population is finite ($n/N > 0.05$), the variance of the population is computed using the following formula:

$$\sigma_{\bar{X}}^2 = \frac{S^2}{n} \frac{N-n}{N-1}$$

or

$$\sigma_{\bar{X}} = \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{1.2}{\sqrt{100}} \sqrt{\frac{900-100}{900-1}} = 0.1132$$

We also have $\bar{X} \sim N(\mu, \sigma^2)$. Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n} \frac{N-n}{N-1}\right) \sim N\left(\mu, \frac{S^2}{n} \frac{N-n}{N-1}\right)$$

Where null and alternate hypotheses are:

$$H_0: \mu \geq 6$$

$$H_1: \mu < 6$$

Then

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{5.7 - 6}{0.1132} = -2.65$$

The table standard normal value, $-Z_{\alpha}$ is -2.33 which is shown in Fig. 8.17 along with acceptance and rejection regions.

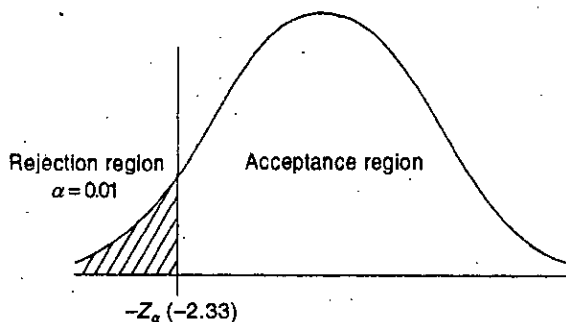


Fig. 8.17 Normal curve for Example 8.10.

The calculated Z value (-2.65) is less than $-Z_{\alpha} (-2.33)$ and it falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean annual income of the respondents is not significantly more than the expected population mean of Rs. 6 lakhs.

8.2.6 Two-tailed Test Concerning Single Mean (when the Variance of the Population is Unknown and the Sample Size is Large)

This test is applicable to the following combination of H_0 and H_1 :

$$H_0: \mu = k$$

$$H_1: \mu \neq k$$

where k is a constant.

Consider the situation as explained in Section 8.2.5 except the placement of the significance level α on the standard normal distribution.

The objective of this test is to check whether the population is to be accepted based on the value of the sample mean by placing half of the given significance level ($\alpha/2$) on both tails of the standard normal distribution.

Infinite population

If the population is infinite, then the variance of the sample may be treated as the variance of the population. The standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

If this computed standard normal statistic is in between $-Z_{\alpha/2}$ and $Z_{\alpha/2}$, then accept H_0 ; otherwise, reject it.

Finite population

If the population is finite ($n/N > 0.05$), then the variance of the population is estimated from the variance of the sample using the following formula:

$$\sigma_{\bar{X}}^2 = \frac{S^2}{n} \frac{N-n}{N-1}$$

The standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

If this computed standard normal statistic is in between $-Z_{\alpha/2}$ and $Z_{\alpha/2}$, then accept H_0 ; otherwise, reject it.

Example 8.11 The weight of a drug produced by Alpha Pharmaceutical Company follows normal distribution whose population is infinite. The specified mean weight of the drug of this population is 10 mg and its variance is unknown. The quality engineer of the firm claims that the mean weight of the drugs does not differ significantly from the specified

mean weight of the population. So, the purchase manager of Beta Hospital who places order for that drug with the Alpha Pharmaceutical Company has selected a random sample of 64 drugs. The mean and variance of the sample are found to be 9.8 mg and 0.64 mg, respectively. Verify the intuition of the quality engineer of Alpha Pharmaceutical Company at a significance level of 0.05.

Solution. Given

Sample size, n	= 64
Population mean, μ	= 10 mg
Sample mean, $\bar{X}$	= 9.8 mg
Variance of the sample, S^2	= 0.64 mg
Standard deviation of the sample, S	= 0.8 mg
Significance level, α	= 0.05
$Z_{\alpha/2}$	= 1.96

Since the population is infinite, the variance of the sample is treated as the variance of the population. Then,

$$\sigma^2 = S^2 = 0.64 \text{ mg}, \quad \sigma = 0.8 \text{ mg}$$

and

$$X \sim N(\mu, \sigma^2)$$

Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \sim N\left(\mu, \frac{S^2}{n}\right)$$

Null and alternate hypotheses are:

$$H_0: \mu = 10$$

$$H_1: \mu \neq 10$$

Therefore,

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{9.8 - 10}{0.8/\sqrt{64}} = -2$$

The table standard normal values for $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ are -1.96 and 1.96, respectively which are shown in Fig. 8.18 along with acceptance and rejection regions.

The calculated Z value (-2) is not in between $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ (-1.96 and 1.96) and it falls in the left rejection region. Hence, H_0 should be rejected.

This means that the mean weight of the drugs differs significantly from the specified mean of 10 mg.

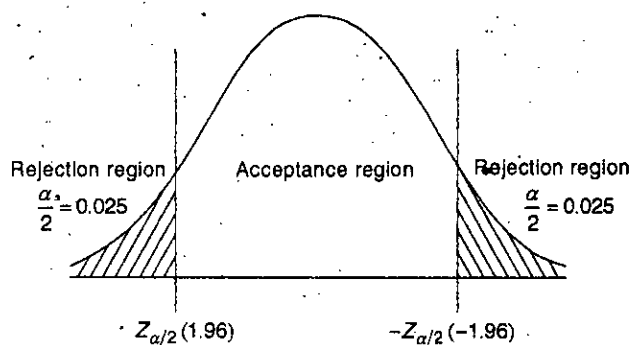


Fig. 8.18 Normal curve for Example 8.11.

Example 8.12 Consider Example 8.11 with the only difference that the population is finite and its size is 1000.

Verify the intuition of the quality engineer of Alpha Pharmaceutical Company at a significance level of 0.05.

Solution. Given

Population size, N	= 1000
Sample size, n	= 64
Population mean, μ	= 10 mg
Sample mean, $\bar{X}$	= 9.8 mg
Variance of the sample, S^2	= 0.64 mg
Standard deviation of the sample, S	= 0.8 mg
Significance level, α	= 0.05
$Z_{\alpha/2}$	= 1.96

Since, the population is finite, the variance of the population is estimated from that of the sample using the following formula:

$$\sigma_{\bar{X}}^2 = \frac{S^2}{n} \frac{N-n}{N-1} = \frac{0.64}{64} \frac{1000-64}{1000-1} = 0.00937$$

Then $\sigma_{\bar{X}} = 0.0968$ mg. Null and alternate hypotheses are:

$$H_0: \mu = 10$$

$$H_1: \mu \neq 10$$

The standard normal statistic for the given problem is computed using the following formula:

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{9.8 - 10}{0.0968} = -2.066$$

The table standard normal values for $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ are -1.96 and 1.96 , respectively, which are shown in Fig. 8.19 along with acceptance and rejection regions.

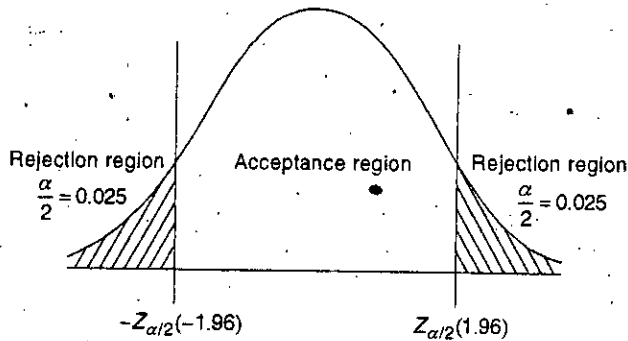


Fig. 8.19 Normal curve for Example 8.12.

The calculated Z value (-2.066) is not in between $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ (-1.96 and 1.96) and it falls in the left rejection region. Hence, H_0 should be rejected.

This means that the mean weight of the drugs differs significantly from the specified mean of 10 mg

8.2.7 One-tailed Tests Concerning Single Mean (when the Variance of the Population Is Unknown and the Sample Size is Small)

Consider a situation in which the random variable of the population follows normal distribution, its variance is not known and the size of the sample is small ($n \leq 30$). Under such situation, the standard deviation of the respective sampling distribution ($\sigma_{\bar{X}}$) cannot be estimated directly. So, the inference about the population based on a sample can be drawn using Student's t distribution whose random variable is given by the following formula, where n is the sample size:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \text{with } (n-1) \text{ degrees of freedom}$$

One-tailed tests for the above situation are shown below:

1. Test 1: $H_0: \mu \leq k$ and $H_1: \mu > k$
2. Test 2: $H_0: \mu \geq k$ and $H_1: \mu < k$

where k is a constant.

The objective of one-tailed test is to check whether the population is to be accepted based on the value of the sample mean by placing the given significance level (α):

- (a) At the right tail of t -distribution in the case of Test 1 and
- (b) At the left tail of t -distribution in the case of Test 2.

In Test 1, if the computed t statistic is less than Z_{α} then accept H_0 ; otherwise, reject it. In Test 2, if the computed t statistic is more than Z_{α} then accept H_0 ; otherwise, reject it.

Example 8.13 The supplier of electrical motors to Washtex Company—a washing machine manufacturing company—claims that the mean time between failures of the motors is at most 60 days. The quality manager of the Washtex Company wants to test the claim of the vendor. Hence, he has taken a sample of 16 motors whose mean time between failures and its variance are found to be 63 days and 25 days, respectively. Verify the claim of the vendor at a significance level of 0.01.

Solution. We have

Sample size, n	= 16
Population mean, μ	= 60 days
Sample mean, $\bar{X}$	= 63 days
Variance of the sample, S^2	= 25 days
Standard deviation of the sample, S	= 5 days
Significance level, α	= 0.01

We also have $X \sim N(\mu, \sigma^2)$. Since the sample size n is less than 30, $\bar{X} \sim t$ distribution with $(n-1)$ degrees of freedom whose statistic is given below:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \text{ with 15 degrees of freedom}$$

where null and alternate hypotheses are:

$$H_0: \mu \leq 60$$

$$H_1: \mu > 60$$

Then

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{63 - 60}{5/\sqrt{16}} = 2.4$$

The table t value with the degrees of freedom 15 at $\alpha = 0.01$ placed at right tail is 2.602 which is shown in Fig. 8.20 along with acceptance and rejection regions.

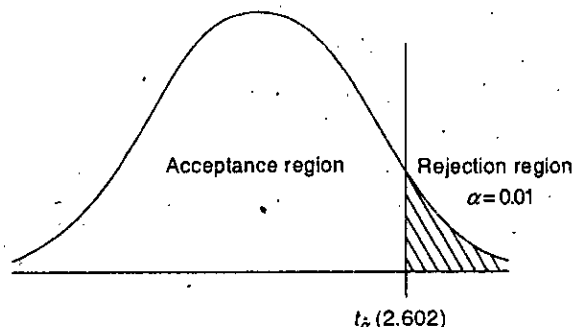


Fig. 8.20 t distribution curve for Example 8.13.

Since, the calculated t value (2.4) is less than table t_α value (2.602), the calculated t value falls in the acceptance region. Hence, H_0 should be accepted.

This means that the mean time between failures of the electrical motors is significantly lesser than 60 days (not greater than 60 days) which means that the claim of the vendor firm is true.

Example 8.14 The weight of costly electrodes purchased by a foundry follows normal distribution. The sales manager of the vendor firm claims that the mean weight of the electrodes is at least 100 gm. The quality manager of the foundry wants to verify this claim. So, he has taken a sample of 25 electrodes. The mean and the variance of the electrodes in the sample are found to be 97 gm and 64 gm, respectively. Verify the claim of the sales manager at a significance level of 0.05.

Solution. We have

Sample size, n	= 25
Population mean, μ	= 100 gm
Sample mean, $\bar{X}$	= 97 gm
Variance of the sample, S^2	= 64 gm
Standard deviation of the sample, S	= 8 gm
Significance level, α	= 0.05

So, $X \sim N(\mu, \sigma^2)$. Since the sample size (n) is less than 30, $\bar{X} \sim t$ distribution with $(n-1)$ degrees of freedom whose statistic is given below:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \text{with 24 degrees of freedom}$$

Here null and alternate hypotheses are:

$$H_0: \mu \geq 100$$

$$H_1: \mu < 100$$

Then

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{97 - 100}{8/\sqrt{25}} = -1.875$$

The table t value with the degrees of freedom 24 at $\alpha = 0.05$ placed at the left tail is -1.711 which is shown in Fig. 8.21 along with acceptance and rejection regions.

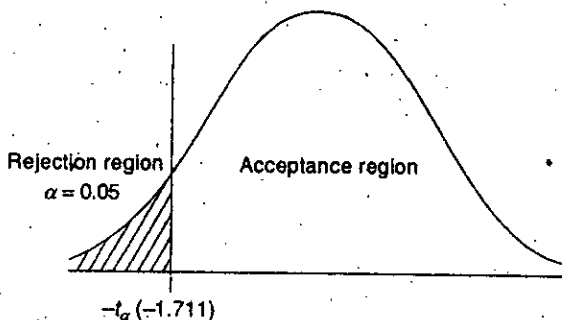


Fig. 8.21 t distribution curve for Example 8.14.

Since the calculated t value (-1.875) is less than $-t_\alpha$ value (-1.711), the calculated t value falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean weight of the electrodes is not significantly greater than 100 gm which means that the claim of the sales manager of the vendor firm is not true.

8.2.8 Two-tailed Test Concerning Single Mean (when the Variance of the Population is Unknown and the Sample Size is Small)

This is same as the one-tailed tests concerning single mean when the variance of the population is unknown and the sample size is small. But, in this test, half of the significance level ($\alpha/2$) is placed at both tails of the t distribution.

The two-tailed test is applicable to the following combination of H_0 and H_1 :

$$H_0: \mu = k$$

$$H_1: \mu \neq k$$

where k is a constant.

The objective of this test is to check whether population is to be accepted based on the value of the sample mean by placing half of the given significance level ($\alpha/2$) at both tails of t distribution.

Example 8.15 The weight of costly catalyst electrodes purchased by a chemical company follows normal distribution. The sales manager of the vendor firm claims that the mean weight of the electrodes is 150 gm. The quality manager of the chemical company wants to verify this claim. So, he has taken a sample of 25 electrodes. The mean and the variance of the electrodes in the sample are found to be 145 gm and 100 gm, respectively. Verify the claim of the sales manager at a significance level of 0.05.

Solution. Given

Sample size, n	= 25
Population mean, μ	= 150 gm
Sample mean, $\bar{X}$	= 145 gm
Variance of the sample, S^2	= 100 gm
Standard deviation of the sample, S	= 10 gm
Significance level, α	= 0.05

So, $X \sim N(\mu, \sigma^2)$. Since the sample size, n is less than 30, $\bar{X} \sim t$ distribution with $(n-1)$ degrees of freedom whose statistic is given below:

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \text{with 24 degrees of freedom.}$$

Now, null and alternate hypotheses are:

$$H_0: \mu = 150$$

$$H_1: \mu \neq 150$$

Then

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{145 - 150}{10/\sqrt{25}} = -2.5$$

The table t values with the degrees of freedom of 24 at $\alpha/2 = 0.025$ placed at left and right tails are -2.064 and $+2.064$, respectively, which are shown in Fig. 8.22 along with acceptance and rejection regions.

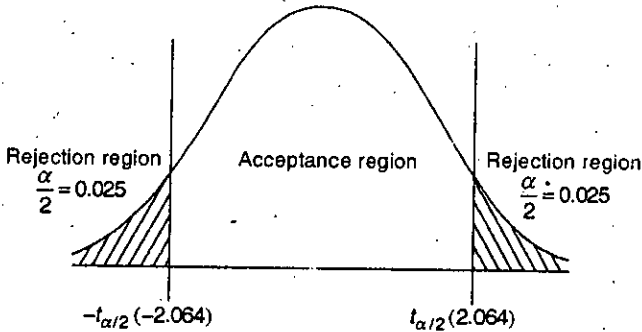


Fig. 8.22 t distribution for Example 8.15.

Since the calculated t value (-2.5) is less than $-t_{\alpha/2}$ (-2.064), the calculated t value falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean weight of the catalyst electrodes is significantly different from 150 gm which means that the claim of the sales manager of the vendor firm is not true.

8.2.9 One-tailed Tests Concerning Difference between Two Means (when the Variances of the Populations are Known)

Consider a situation in which there are two random variables, X_1 and X_2 which follow two normal distributions representing two independent populations with means μ_1 and μ_2 , respectively. Let the variances of these populations be σ_1^2 and σ_2^2 , respectively. The objective of the one-tailed test for this situation may be to test the significance of the relative difference between the population means, as presented below:

1. Test 1: $H_0: \mu_1 \leq \mu_2$ and $H_1: \mu_1 > \mu_2$

or

$$H_0: \mu_1 - \mu_2 \leq 0 \text{ and } H_1: \mu_1 - \mu_2 > 0$$

2. Test 2: $H_0: \mu_1 \geq \mu_2$ and $H_1: \mu_1 < \mu_2$

or

$$H_0: \mu_1 - \mu_2 \geq 0 \text{ and } H_1: \mu_1 - \mu_2 < 0$$

If the distributions of two independent random variables, X_1 and X_2 , have means μ_1 and μ_2 , variances σ_1^2 and σ_2^2 , respectively, then the distribution of their difference will have

the mean and variance as $(\mu_1 - \mu_2)$ and $\sigma_1^2 + \sigma_2^2$, respectively. The null hypothesis, H_0 of each test can be verified by taking one sample from each population.

Let $\bar{X}_1$ be the mean of the sample of size n_1 taken from the first population and $\bar{X}_2$ be the mean of the sample of size n_2 taken from the second population. These two samples are independent.

The expected values of $\bar{X}_1$ and $\bar{X}_2$ are μ_1 and μ_2 , respectively. The variances of the sample means X_1 and X_2 are σ_1^2/n_1 and σ_2^2/n_2 , respectively.

The mean and variance of the distribution of the difference in sample means are as given below:

$$\text{Expected value of difference in sample means} = \mu_1 - \mu_2$$

and

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, then

$$\bar{X}_1 - \bar{X}_2 \sim N \left[\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \right]$$

The objective of the one-tailed test is to check whether the null hypothesis is to be accepted based on the difference in the values of the sample means by placing the given significance level (α):

- (a) At the right tail of the standard normal distribution in the case of Test 1 and
- (b) At the left tail of the standard normal distribution in the case of Test 2.

The test statistic for each of the above hypotheses is:

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$

Note: If the distributions of the given two independent populations are non-normal, and if the sample sizes are large, one can convert them into standard normal distributions using central limit theorem. The rest of the procedure for this type of hypothesis testing is same as outlined above.

Example 8.16 The personnel manager of a firm is recruiting its graduate trainees from colleges affiliated to two different universities. The performance index (0–10 scale) of the trainees from each university follows normal distribution. The variance of the performance index of the trainees from the University-1 is 9 and that of the trainees from the University-2 is 4. The manager feels that the mean performance index of the trainees from the University-1 is less than that of the trainees from the University-2. To test his intuition, he

has selected a sample of 49 trainees from the University-1 and their mean performance index is found to be 7. Similarly, he has selected a sample of 64 trainees from the University-2 and their mean performance index is found to be 6. Test the intuition of the manager at a significance level of 0.05.

Solution. We have

Variance of the performance index of the trainees from the University-1, σ_1^2	= 9
Variance of the performance index of the trainees from the University-2, σ_2^2	= 4
Number of trainees in the sample from University-1, n_1	= 49
Number of trainees in the sample from University-2, n_2	= 64
Mean performance index of the sample-1, $\bar{X}_1$	= 7
Mean performance index of the sample-2, $\bar{X}_2$	= 6
Significance level, α	= 0.05

We also have $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$.

Therefore,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Then

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{9}{49} + \frac{4}{64} = 0.246$$

or

$$\sigma_{\bar{X}_1 - \bar{X}_2} = 0.496$$

Null and alternate hypotheses are:

$$H_0: \mu_1 \leq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \leq 0$$

$$H_1: \mu_1 > \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 > 0$$

Thus

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(7 - 6) - 0}{0.496} = 2.02$$

The standard normal statistic for the given significance level, (α) of 0.05, Z_α is 1.64, which is shown in Fig. 8.23.

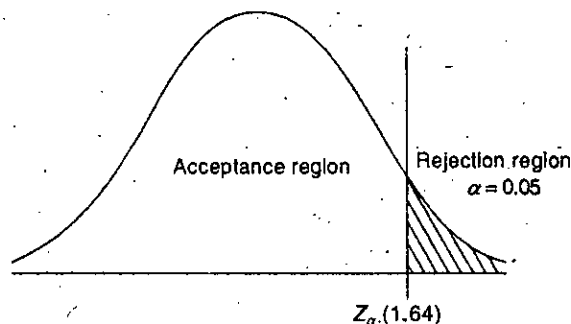


Fig. 8.23 Normal curve for Example 8.16.

Since, the calculated Z value (2.02) is more than Z_α (1.64), the calculated Z value falls in the rejection region. Hence, H_0 should be rejected. This means that the performance index of the trainees from the University-1 is not less than that of the trainees from the University-2.

Example 8.17 The sales manager of a firm is studying the sales revenues of two different categories of salesmen (Category-1 and Category-2) for their relative difference in sales performance. The variance of the yearly sales revenues of the Category-1 salesmen is Rs. 9 lakhs and that of the Category-2 salesmen is Rs. 36 lakhs. The manager feels that the sales performance of the Category-1 salesmen is better than that of the Category-2 salesmen. To test his intuition, he has selected a sample of 64 salesmen from the Category-1 and their mean annual sales is found to be Rs. 50 lakhs. Similarly, he has selected a sample of 81 salesmen from the Category-2 and their mean annual sales is found to be Rs. 51 lakhs. Test the intuition of the manager at a significance level of 0.05.

Solution. We have

Variance of the annual sales revenue of Category-1 salesmen, σ_1^2	= Rs. 9 lakhs
Variance of the annual sales revenue of Category-2 salesmen, σ_2^2	= Rs. 36 lakhs
Number of salesmen in the sample of Category-1, n_1	= 64
Number of salesmen in the sample of Category-2, n_2	= 81
Mean annual sales revenue of the sample-1, $\bar{X}_1$	= 50
Mean annual sales revenue of the sample-2, $\bar{X}_2$	= 51
Significance level, α	= 0.05

We also know $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. Therefore,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Then

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{9}{64} + \frac{36}{81} = 0.585$$

or

$$\sigma_{\bar{X}_1 - \bar{X}_2} = 0.765$$

Null and alternate hypotheses are:

$$H_0: \mu_1 \geq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \geq 0$$

$$H_1: \mu_1 < \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 < 0$$

Hence

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(50 - 51) - 0}{0.765} = -1.31$$

The standard normal statistic for the given significance level (α) of 0.05, $-Z_\alpha$ is -1.64, which is shown in Fig. 8.24.

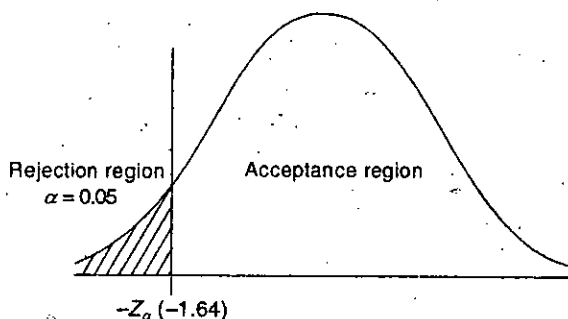


Fig. 8.24 Normal curve for Example 8.17.

Since the calculated Z value (-1.31) is more than $-Z_\alpha$ (-1.64), the calculated Z value falls in the acceptance region. Hence, H_0 should be accepted. This means that the performance of the Category-1 salesmen is better than that of the Category-2 salesmen.

8.2.10 Two-tailed Test Concerning Difference between Two Means (when the Variances of the Populations are Known)

This is same as the one-tailed test concerning difference between two means when the variances of the populations are known except the placement of rejection region in the Z distribution.

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, then

$$\bar{X}_1 - \bar{X}_2 \sim N \left[(\mu_1 - \mu_2), \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \right]$$

The objective of this two-tailed test is to check whether the null hypothesis is to be accepted based on the difference in the values of the sample means by placing half of the significance level ($\alpha/2$) at both tails.

Example 8.18 An automobile company presently buys pistons from Vendor-1. But the quality manager is desiring to consider the supply of pistons from Vendor-2 to meet the extra demand, provided the new vendor can supply the pistons with good precision on its axial length on par with that of the Vendor-1. If the axial length is more than the necessity, it will lead to knocking at the top of the cylinder head. Hence, it should be avoided. The axial length of the pistons supplied by Vendor-1 as well as Vendor-2 follow normal distribution with variances of 5 mm and 9 mm, respectively.

He has selected 64 pistons from the Vendor-1 and found that the mean axial length of the sample as 60 mm. Similarly, he selected 81 pistons from the Vendor-2 and found that the mean axial length of this sample as 61 mm. Check whether the quality manager can consider the supply from the Vendor-2 at a significance level of 0.01.

Solution. We have

Variance of axial length of pistons from Vendor-1, $\sigma_1^2 = 5$ mm

Variance of axial length of pistons from Vendor-2, $\sigma_2^2 = 9$ mm

Sample size of Vendor-1, $n_1 = 64$

Sample size of Vendor-2, $n_2 = 81$

Mean axial length of the sample-1, $\bar{X}_1 = 60$ mm

Mean axial length of the sample-2, $\bar{X}_2 = 61$ mm

Significance level, $\alpha = 0.01$

where $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. Therefore,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Then

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{5}{64} + \frac{9}{81} = 0.1892$$

or

$$\sigma_{\bar{X}_1 - \bar{X}_2} = 0.435$$

Then the null and alternate hypotheses are:

$$H_0: \mu_1 = \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \neq 0$$

Hence

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(60 - 61) - 0}{0.435} = -2.3$$

The table standard normal values, $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$ are -2.57 and $+2.57$, respectively, which are shown in Fig. 8.25.

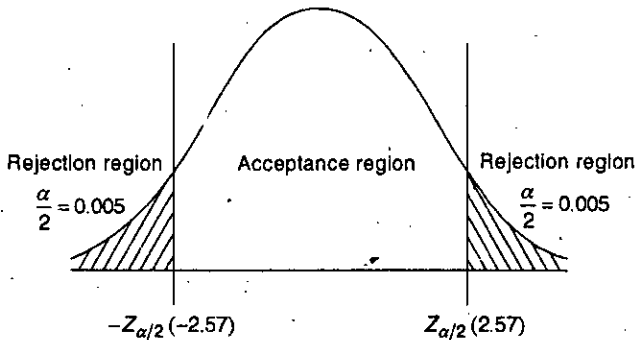


Fig. 8.25 Normal curve for Example 8.18.

Since the calculated Z value (-2.3) is in between -2.57 and $+2.57$, the calculated Z value falls in the acceptance region. Hence, H_0 should be accepted.

This means that the mean axial length of the pistons supplied by Vendor-2 does not differ from 60 mm. So, the quality manager can consider the supply from the Vendor-2, also.

8.2.11 One-tailed Tests Concerning Difference between Two Means (when the Variances of the Populations are Unknown and the Sample Sizes are Large)

Consider a situation in which there are two random variables, X_1 and X_2 which follow two normal distributions representing two independent populations with means μ_1 and μ_2 , respectively. Let the variances of these populations are not known. The objective of the one-tailed test for this situation may be to test the significance of the relative difference between the population means as presented below:

1. Test 1: $H_0: \mu_1 \leq \mu_2$ and $H_1: \mu_1 > \mu_2$

or

$$H_0: \mu_1 - \mu_2 \leq 0 \text{ and } H_1: \mu_1 - \mu_2 > 0$$

2. Test 2: $H_0: \mu_1 \geq \mu_2$ and $H_1: \mu_1 < \mu_2$

or

$$H_0: \mu_1 - \mu_2 \geq 0 \text{ and } H_1: \mu_1 - \mu_2 < 0$$

If the distributions of two independent random variables, X_1 and X_2 have means μ_1 and μ_2 , variances σ_1^2 and σ_2^2 , respectively, then the distribution of their difference will have

the mean and variance as $(\mu_1 - \mu_2)$ and $\sigma_1^2 + \sigma_2^2$, respectively. The null hypothesis (H_0) of each test can be verified by taking one sample from each population.

Let $\bar{X}_1$ be the mean of the sample of size n_1 taken from the first population and $\bar{X}_2$ be the mean of the sample of size n_2 taken from the second population. These two samples are independent and n_1 and n_2 are large and greater than 30.

The expected value of $\bar{X}_1$ and $\bar{X}_2$ are μ_1 and μ_2 , respectively. The variance of the sample means $\bar{X}_1$ and $\bar{X}_2$ are σ_1^2/n_1 and σ_2^2/n_2 , respectively.

The mean and variance of the distribution of the difference in sample means are as given below:

$$\text{Expected value of difference in sample means} = \mu_1 - \mu_2$$

and

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, then

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Since, the sample sizes, n_1 and n_2 are larger, their variances S_1^2 and S_2^2 can be substituted for σ_1^2 and σ_2^2 , respectively.

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$$

The test statistic used to verify both one-tailed tests which are presented in this section is as shown below:

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$

Example 8.19 The production rate of shafts produced in Machine-A as well as in Machine-B follows normal distribution. The variance of the production rate of the shafts produced using each machine is not known. The production manager feels that the performance of the Machine-A is not comparable with that of the Machine-B. To test his intuition, the quality assistant of the shop has selected 81 days' records of the Machine-A and found that the mean and variance of the production rate of this sample are 70 units and 9 units, respectively. Similarly, he has selected 64 days' records of the Machine-B and found that the mean and variance of the production rate of this sample are 68 units and 25 units, respectively. Check the intuition of the production manager at a significance level of 0.05.

Solution.

Sample size on Machine-A, n_1 = 81 days' records

Sample size on Machine-B, n_2 = 64 days' records

Mean production rate of the sample from Machine-A, $\bar{X}_1$ = 70 units

Variance of the production rate of the sample from Machine-A, S_1^2 = 9 units

Mean production rate of the sample from Machine-B, $\bar{X}_2$ = 68 units

Variance of the production rate of the sample from Machine-B, S_2^2 = 25 units

Significance level, α = 0.05

where $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. Therefore,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Since the sample sizes, n_1 and n_2 are more than 30, the respective variances of the samples can be substituted for the respective variances of the populations. Then

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} = \frac{9}{81} + \frac{25}{64} = 0.5017$$

or

$$\sigma_{\bar{X}_1 - \bar{X}_2} = 0.7083$$

Now, the null and alternate hypotheses are:

$$H_0: \mu_1 \leq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \leq 0$$

$$H_1: \mu_1 > \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 > 0$$

Hence

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(70 - 68) - 0}{0.7083} = 2.824$$

The table standard normal value, Z_α is 1.64, which is shown in Fig. 8.26.

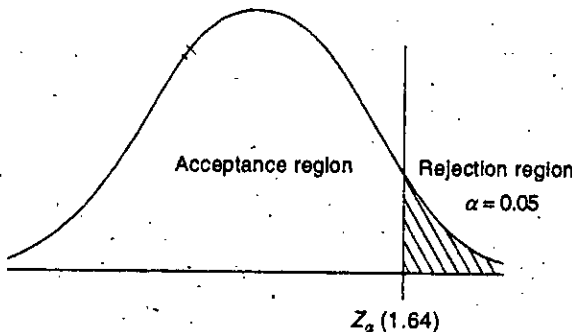


Fig. 8.26 Normal curve for Example 8.19.

Since the calculated Z value (2.824) is more than Z_α (1.64), the calculated Z value falls in the rejection region. Hence, H_0 should be rejected.

This means that the mean production rate of the Machine-A is comparable (not less than) with that of the Machine-B.

Note: Readers are directed to generate a problem to the following combination of hypotheses of Test-2 and draw inference.

$$2. \text{ Test 2: } H_0: \mu_1 \geq \mu_2 \text{ or } \mu_1 - \mu_2 \geq 0$$

$$H_1: \mu_1 < \mu_2 \text{ or } \mu_1 - \mu_2 < 0$$

8.2.12 Two-tailed Test Concerning Difference between Two Means (when the Variances of the Populations are Unknown and the Sample Sizes are Large)

This is same as one-tailed test, as presented in Section 8.2.11, except the placement of rejection region. The objective of this two-tailed test is to test the significance of the relative difference between the population means, as presented below. In this test, half of the significance level will be placed at both tails. In such a case, the null and alternate hypotheses are:

$$H_0: \mu_1 = \mu_2 \text{ or } \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2 \text{ or } \mu_1 - \mu_2 \neq 0$$

Expected value of the difference in sample means = $\mu_1 - \mu_2$. Then

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, then

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Since the sample sizes, n_1 and n_2 are larger, their variances S_1^2 and S_2^2 can be substituted for σ_1^2 and σ_2^2 , respectively.

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$$

The test statistic used to verify this two tailed test is shown below.

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$

Example 8.20 The regional manager of a bank wants to analyze the number of delayed payments of instalments of loans by farmers which were availed by them under a special loan scheme in two of its branch banks (Branch-1 and Branch-2) in rural areas. The number of delayed payments of instalments in each branch bank follows normal distribution. The manager feels that the number of delayed payments of instalments by the customers of the Branch-1 is no way different from that of the Branch-2. So, he selected the loan accounts of 50 different customers from the Branch-1 and found that the mean of the number of delayed payments of instalments and its variance were 21 and 9, respectively. Similarly, he selected the loan accounts of 80 different customers from the Branch-2 and found that the mean of the number of delayed payments of instalments and its variance were 23 and 36, respectively. Test his intuition at a significance level of 0.01.

Solution. Given that

Number of customers selected from Branch-1, n_1	= 50
Number of customers records selected from Branch-2, n_2	= 80
Mean of number of delayed payments of instalments in Branch-1, $\bar{X}_1$	= 21
Mean of number of delayed payments of instalments in Branch-2, $\bar{X}_2$	= 23
Variance of mean of number of delayed payments of instalments in Branch-1, S_1^2	= 9
Variance of mean of number of delayed payments of instalments in Branch-2, S_2^2	= 36
Significance level, α	= 0.01

For expected value of difference in sample means $= \mu_1 - \mu_2$, we have

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, then

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Since the sample sizes, n_1 and n_2 are larger, their variances S_1^2 and S_2^2 can be substituted for σ_1^2 and σ_2^2 , respectively. Then

$$\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} = \frac{9}{50} + \frac{36}{80} = 0.63$$

or

$$\sigma_{\bar{X}_1 - \bar{X}_2} = 0.794$$

Their null and alternate hypotheses are:

$$H_0: \mu_1 = \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \neq 0$$

Then

$$Z_{\bar{X}_1 - \bar{X}_2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(21 - 23) - 0}{0.794} = -2.52$$

The table standard normal values, $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ are -2.57 and $+2.57$, respectively, which are shown in Fig. 8.27.

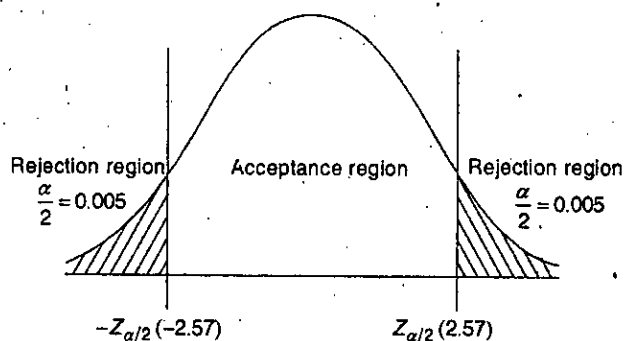


Fig. 8.27 Normal curve for Example 8.20.

Since the calculated Z value (-2.52) is in between -2.57 and $+2.57$, the calculated Z value falls in the acceptance region. Hence, H_0 should be accepted.

This means that the number of delayed payments of instalments by the customers of the Branch-1 is no way different from that of the Branch-2. Hence, the regional manager's intuition is correct.

8.2.13 One-tailed Tests Concerning Difference between Two Means (when the Variances of the Populations are Unknown and the Sample Sizes are Small)

Consider a situation in which the two random variables of two independent populations follow normal distribution, their variances are not known and the size of the sample of each population is small ($n \leq 30$). Under such situation, the standard deviation of the difference in population means ($\sigma_{\bar{X}_1 - \bar{X}_2}$) cannot be obtained directly by using the sample variances, S_1 and S_2 .

By assuming the same variance for the populations ($\sigma_1 = \sigma_2 = \sigma$), a pooled variance is computed using the following formula:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

The standard deviation for the difference between means of the populations is as given below:

$$\sigma_{\bar{X}_1 - \bar{X}_2} = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

The statistic to test the significance of the difference between population means under this situation a Student's t distribution is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{(1/n_1) + (1/n_2)}}$$

with $(n_1 + n_2 - 2)$ degrees of freedom.

The possible one-tailed tests are as presented below:

1. Test 1: $H_0: \mu_1 \leq \mu_2$ or $\mu_1 - \mu_2 \leq 0$

or

$H_1: \mu_1 > \mu_2$ or $\mu_1 - \mu_2 > 0$

2. Test 2: $H_0: \mu_1 \geq \mu_2$ or $\mu_1 - \mu_2 \geq 0$

or

$H_1: \mu_1 < \mu_2$ or $\mu_1 - \mu_2 < 0$

Example 8.21 Consider Example 8.19, as reproduced below, with some modifications:

The production rate of shafts produced in Machine-A as well as in Machine-B follows normal distribution. The variance of the production rate of the shafts produced using each machine is not known. The production manager feels that the performance of the Machine-A is far better than that of the Machine-B. To test his intuition, the quality assistant of the shop has selected 16 days' records of the Machine-A and found that the mean and variance of the production rate of this sample were 58 units and 4 units, respectively. Similarly, he has selected 25 days' records of the Machine-B and found that the mean and variance of the production rate of this sample were 60 units and 16 units, respectively. Check the intuition of the production manager at a significance level of 0.01.

Solution. We have

Sample size of Machine-A, n_1	= 16 days' records
Sample size of Machine-B, n_2	= 25 days' records
Mean production rate of the sample from Machine-A, $\bar{X}_1$	= 58 units
Variance of the production rate of the sample from Machine-A, S_1^2	= 4 units
Mean production rate of the sample from Machine-B, $\bar{X}_2$	= 60 units
Variance of the production rate of the sample from Machine-B, S_2^2	= 16 units
Significance level, α	= 0.01

Then

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} = \frac{(16 - 1) \times 4 + (25 - 1) \times 16}{16 + 25 - 2} = 11.385$$

or

$$S_p = 3.374$$

The standard deviation for the difference between means of the populations is:

$$\sigma_{\bar{X}_1 - \bar{X}_2} = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 3.374 \sqrt{\frac{1}{16} + \frac{1}{25}} = 1.08$$

The null and alternate hypotheses are:

$$H_0: \mu_1 \geq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \geq 0$$

$$H_1: \mu_1 < \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 < 0$$

Then

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{(1/n_1) + (1/n_2)}} = \frac{(58 - 60) - 0}{1.08} = -1.85$$

The table value of $-t_\alpha$ with 39 degrees of freedom is -2.326 , which is as shown in Fig. 8.28, along with rejection region and acceptance region.

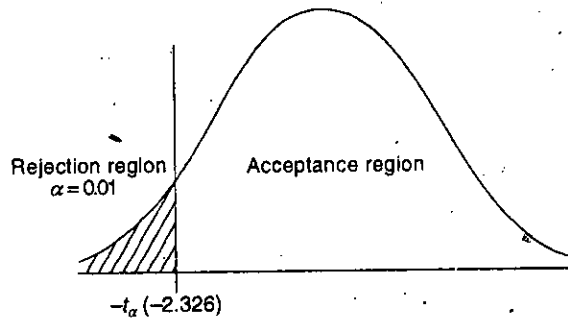


Fig. 8.28 t distribution curve for Example 8.21.

The calculated t value (-1.85) is greater $-t_\alpha$ (-2.326). Hence, accept H_0 .

This means that the performance of the Machine-A is far better than that of the Machine-B. So, the production manager's intuition is correct.

Note: The readers are directed to generate a problem to perform the following test concerning difference between two means when the variances of the populations are unknown and the sample sizes are small and draw the inference:

$$\text{Test 1: } H_0: \mu_1 \leq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \leq 0$$

$$H_1: \mu_1 > \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 > 0$$

8.2.14 Two-tailed Test Concerning Difference between Two Means (when the Variances of the Populations are Unknown and the Sample Sizes are Small)

This is same as the one-tailed tests except the placement of the rejection region for a given significance level. In this test, half of the significance level ($\alpha/2$) will be placed at both the tails of Student's t distribution and the inference will be drawn accordingly.

Example 8.22 The diameter of a particular type of bearing supplied by Vendor-1 and Vendor-2 follows normal distribution. The quality manager of the firm buying the bearings believes that the diameter of the bearings supplied by the Vendor-1 is not different from that of the bearings supplied by the Vendor-2. So, the quality assistant under him selected 16 bearings from the supply of the Vendor-1 and found that mean and variance of the diameter were 38.5 mm and 2.5 mm, respectively. Similarly, he selected 20 bearings from the supply of the Vendor-2 and found that mean and variance of the diameter were 40 mm and 4 mm, respectively. Verify the intuition of the manager at a significance level of 0.10.

Solution: We have

Sample size of Vendor-1, n_1	= 16
Sample size of Vendor-2, n_2	= 20
Mean diameter of the sample from Vendor-1, $\bar{X}_1$	= 38.5 mm
Variance of the mean diameter of the sample from Vendor-1, S_1^2	= 2.5 mm
Mean diameter of the sample from Vendor-2, $\bar{X}_2$	= 40 mm
Variance of the mean diameter of the sample from Vendor-2, S_2^2	= 4 mm
Significance level, α	= 0.10

Now

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} = \frac{(16 - 1) \times 2.5 + (20 - 1) \times 4}{16 + 20 - 2} = 3.338$$

or

$$S_p = 1.827$$

The standard deviation for the difference between the means of the populations is computed as shown below:

$$\sigma_{\bar{X}_1 - \bar{X}_2} = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 1.827 \sqrt{\frac{1}{16} + \frac{1}{20}} = 0.6128$$

The null and alternate hypotheses are:

$$H_0: \mu_1 = \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2 \quad \text{or} \quad \mu_1 - \mu_2 \neq 0$$

Then t distribution is

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{(1/n_1) + (1/n_2)}} = \frac{(38.5 - 40) - 0}{0.6128} = -2.45$$

The table value of $-t_{\alpha/2}$ and $t_{\alpha/2}$ with 34 degrees of freedom are -1.645 and $+1.645$, which are as shown in Fig. 8.29, along with rejection region and acceptance region.

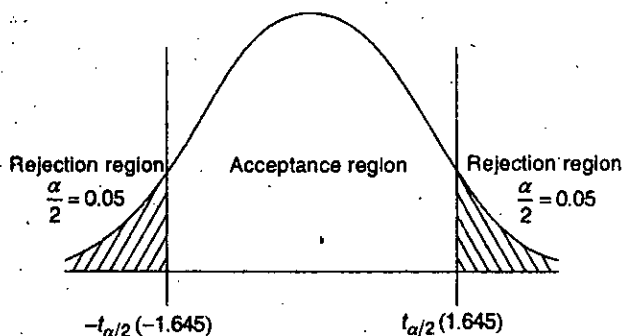


Fig. 8.29 t distribution for Example 8.22.

The calculated t value (-2.45) is not in between $-t_{\alpha/2}$ and $t_{\alpha/2}$ (-1.645 and $+1.645$). Hence, reject H_0 .

This means that diameter of the bearings supplied by the Vendor-1 is different from that of the bearings supplied by the Vendor-2. So, the quality manager's intuition is not correct.

8.3 TESTS OF HYPOTHESES CONCERNING PROPORTION(S)

The tests of hypotheses concerning proportion(s) can be classified into (a) tests of hypotheses concerning single proportion and (b) test of hypotheses concerning difference between two proportions. These are explained in the following subsections.

The inference about population(s) can be obtained based on its/their sample characteristics in terms of the following measures:

1. One-tailed tests concerning single proportion
2. Two-tailed test concerning single proportion
3. One-tailed tests concerning two proportions
4. Two-tailed test concerning two proportions.

8.3.1 One-tailed Tests Concerning Single Proportion

In n successive trials, let the probability of success in a trial be p and X be the number of successes, where n is large.

Then, X follows binomial distribution which has its mean and variance as given below:

$$\mu = np \quad \text{and} \quad \sigma^2 = np(1-p) = npq$$

For large value of n , the binomial distribution can be approximated to normal distribution, as:

$$X \sim N(\mu, \sigma^2)$$

Let the proportion of success be $\bar{p}$ which is equal to X/n . Then $\bar{p}$ also approaches normal distribution with its mean and variance, as shown below: (One can approximate like this only when np as well as $n(1-p)$ is at least 5.)

$$\text{Mean, } \bar{p} = p \quad \text{and} \quad \text{Variance, } \sigma_{\bar{p}}^2 = \frac{p(1-p)}{n}$$

Then

$$\bar{p} \sim N\left[p, \frac{p(1-p)}{n}\right] \quad \text{and} \quad \sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$$

The standard normal statistic for $\bar{p}$ is as shown below:

$$Z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}}$$

One-tailed tests concerning single proportion are classified as follows:

Test 1: $H_0: p \leq k$ and $H_1: p > k$

Test 2: $H_0: p \geq k$ and $H_1: p < k$

where k is a constant. The above two tests are demonstrated with suitable examples in this subsection.

Test 1: $H_0: p \leq k$ and $H_1: p > k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample proportion by placing the significance level of α at the right tail of the standard normal distribution.

Example 8.23 The production manager of an automobile company is convinced that the reduction in rest allowance for the employees working in its assembly line will improve the productivity of the line. But the quality manager has a contradicting view, because such reduction will decrease the proportion of the number of assemblies passing out of the assembly line which are free from missing components. Before introducing the new rest allowance, the quality manager ascertained through a study that 85% of the assemblies coming out of the assembly line will be free from missing components.

After introducing the new rest allowance for the employees in the assembly line, it is found that out of a sample of 50 assemblies which passed out of the assembly line, 44 assemblies are free from missing components. Check the view of the quality manager at a significance level of 0.05.

Solution. Here, sample size, $n = 50$; proportion, $p = 0.85$; and sample proportion, $\bar{p} = 44/50 = 0.88$. Then the proportion of the sample ($\bar{p}$), which follows normal distribution, is as shown below, because np as well as $n(1-p)$ is more than 5:

$$\bar{p} \sim N\left[p, \frac{p(1-p)}{n}\right]$$

and

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.85(1-0.85)}{50}} = 0.05$$

We also have significance level, $\alpha = 0.05$. Then the null and alternate hypotheses are:

$$H_0: p \leq 0.85$$

$$H_1: p > 0.85$$

The standard normal statistic for $\bar{p}$ is as shown below:

$$Z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}} = \frac{0.88 - 0.85}{0.05} = 0.6$$

The standard normal statistic for the given significance level of 0.05, Z_{α} is 1.64, which is shown in Fig. 8.30.

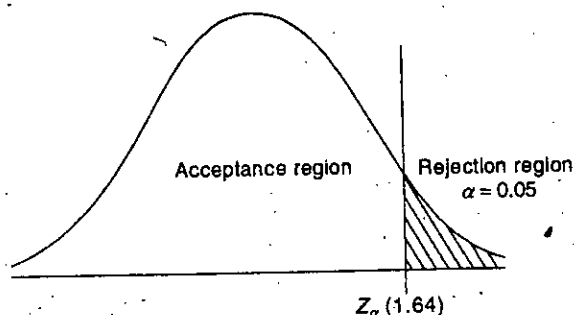


Fig. 8.30 Normal curve for Example 8.23.

Since the calculated Z value (0.6) is less than Z_{α} (1.64), the calculated Z value falls in the acceptance region. Hence, H_0 should be accepted.

This means that the quality manager's contradicting view, that "the reduction in rest allowance will decrease the proportion of the number of assemblies passing out of the assembly line which are free from missing components", is true.

Test 2: $H_0: p \geq k$ and $H_1: p < k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample proportion by placing the significance level of α at the left tail of the standard normal distribution.

Example 8.24 The personnel manager of a company is convinced that complete avoidance of overtime for the employees will improve the productivity of the company. But the production manager has a contradicting view, because such complete avoidance of the

overtime increases labour absenteeism which will have its negative consequences in planning. Before implementing complete avoidance of overtime, the production manager ascertained through a study that 15% of employees attended duty for less than the total number of working days in a month.

After implementing the complete avoidance of overtime, it is found that out of a sample of 120 employees, 17 attended duty for less than the total number of working days in a month. Check the view of the production manager at a significance level of 0.05.

Solution. Here, sample size, $n = 120$; proportion, $p = 0.15$; and sample proportion, $\bar{p} = 17/120 = 0.142$. Then, the sample proportion, $\bar{p}$, which follows normal distribution is as shown below, because np as well as $n(1-p)$ is more than 5:

$$\bar{p} \sim N\left[p, \frac{p(1-p)}{n}\right]$$

Then

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.15(1-0.15)}{120}} = 0.0326$$

We also have significance level, $\alpha = 0.05$. So, the null and alternate hypotheses are:

$$H_0: p \leq 0.15$$

$$H_1: p < 0.15$$

The standard normal statistic for $\bar{p}$ is:

$$Z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}} = \frac{0.142 - 0.15}{0.0326} = -0.245$$

The standard normal statistic for the given significance level of 0.05, $-Z_{\alpha}$ is -1.64, which is shown in Fig. 8.31.

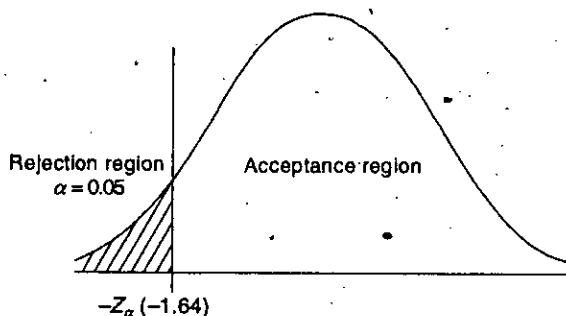


Fig. 8.31 Normal curve for Example 8.24.

Since the calculated Z value (-0.245) is more than $-Z_{\alpha}$ (-1.64), the calculated Z value falls in the acceptance region. Hence, H_0 should be accepted.

This means that the production manager's contradicting view that the complete avoidance of overtime will increase the labour absenteeism is true.

8.3.2 Two-tailed Test Concerning Single Proportion

This test is same as explained in Section 8.3.1, except the placement of the significance level on the standard normal distribution.

The combination of the hypotheses for this situation is:

$$H_0: p = k$$

$$H_1: p \neq k$$

The objective of this test is to check whether the population is to be accepted based on the value of the sample proportion by placing the significance level of $\alpha/2$ at both tails of the standard normal distribution. Here,

$$\bar{p} \sim N \left[p, \frac{p(1-p)}{n} \right]$$

Then

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$$

The standard normal statistic for $\bar{p}$ is:

$$Z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}}$$

If the calculated Z value, is in between $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$, then accept H_0 ; otherwise reject H_0 .

Example 8.25 The sales representative of a pharmaceutical company claims that a new drug cures fever in 90% of the cases. When it is administered by a doctor to 80 patients, it cured only 68 patients. Check the claim of the sales representative at a significance level of 0.05.

Solution. We have sample size, $n = 80$; proportion, $p = 0.90$; and sample proportion, $\bar{p} = 68/80 = 0.85$. The sample proportion, $\bar{p}$, which follows normal distribution is as shown below, because np as well as $n(1-p)$ is more than 5:

$$\bar{p} \sim N \left[p, \frac{p(1-p)}{n} \right]$$

Then

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.90(1-0.90)}{80}} = 0.03354$$

When the significance level, $\alpha = 0.05$, null and alternate hypotheses are:

$$H_0: p = 0.90$$

$$H_1: p \neq 0.90$$

The standard normal statistic for $\bar{p}$ is as shown below:

$$Z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}} = \frac{0.85 - 0.90}{0.03354} = -1.49$$

The standard normal statistics for $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$ are -1.96 and $+1.96$, respectively, which are shown in Fig. 8.32.

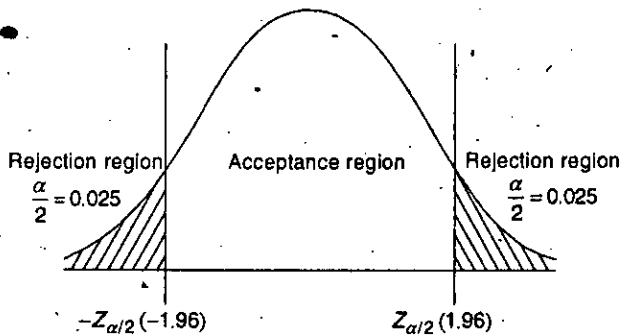


Fig. 8.32 Normal curve for Example 8.25.

Since the calculated Z value (-1.49) falls in the acceptance region, H_0 should be accepted.

This means that the sales representative's claim is true.

8.3.3 One-tailed Tests Concerning Difference between Two Proportions

In n_1 successive trials, let the probability of success in a trial be p_1 and X_1 be the number of successes, where n_1 is large.

Then, X_1 follows binomial distribution which has its mean and variance as given below:

$$\mu_1 = n_1 p_1 \quad \text{and} \quad \sigma_1^2 = n_1 p_1 (1 - p_1) = n_1 p_1 q_1$$

For large value of n_1 , the binomial distribution can be approximated to normal distribution, as shown below:

$$X_1 \sim N(\mu_1, \sigma_1^2)$$

Let the proportion of success in a sample be $\bar{p}_1$ which is equal to X_1/n_1 . Then, $\bar{p}_1$ also approaches normal distribution with its mean and variance as shown below. One can approximate like this only when $n_1 p_1$ as well as $n_1(1 - p_1)$ is at least 5. Here

Mean, $\bar{p}_1 = p_1$ and Variance, $\bar{p}_1 = \frac{p_1(1-p_1)}{n_1}$

Then

$$\bar{p}_1 \sim N\left[p_1, \frac{p_1(1-p_1)}{n_1}\right]$$

In n_2 successive trials, let the probability of success in a trial be p_2 and X_2 be the number of successes, where n_2 is large.

Then, X_2 follows binomial distribution which has its mean and variance as:

$$\mu_2 = n_2 p_2 \quad \text{and} \quad \sigma_2^2 = n_2 p_2 (1 - p_2) = n_2 p_2 q_2$$

For large value of n_2 , the binomial distribution can be approximated to normal distribution, as shown below:

$$X_2 \sim N(\mu_2, \sigma_2^2)$$

Let, the proportion of success in another sample be $\bar{p}_2$ which is equal to X_2/n_2 . Then, $\bar{p}_2$ also approaches normal distribution with its mean and variance, as shown below. One can approximate like this only when $n_2 p_2$ as well as $n_2(1-p_2)$ is at least 5. Here

Mean, $\bar{p}_2 = p_2$ and Variance, $\bar{p}_2 = \frac{p_2(1-p_2)}{n_2}$

Then

$$\bar{p}_2 \sim N\left[p_2, \frac{p_2(1-p_2)}{n_2}\right]$$

Therefore,

$$\bar{p}_1 - \bar{p}_2 \sim N\left[(p_1 - p_2), \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right]$$

Hence

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

The standard normal statistic for $\bar{p}_1 - \bar{p}_2$ is:

$$Z_{\bar{p}_1 - \bar{p}_2} = \frac{(\bar{p}_1 - \bar{p}_2) - (p_1 - p_2)}{\sigma_{\bar{p}_1 - \bar{p}_2}}$$

One-tailed tests concerning the difference between two proportions are classified as follows:

1. Test 1: $H_0: p_1 \leq p_2$ and $H_1: p_1 > p_2$

or

$$H_0: p_1 - p_2 \leq 0 \text{ and } H_1: p_1 - p_2 > 0$$

2. Test 2: $H_0: p_1 \geq p_2$ and $H_1: p_1 < p_2$

or

$$H_0: p_1 - p_2 \geq 0 \text{ and } H_1: p_1 - p_2 < 0$$

These two tests are demonstrated with suitable examples in this subsection.

Test 1: $H_0: p_1 - p_2 \leq 0$ and $H_1: p_1 - p_2 > 0$

The objective of this test is to check whether the population is to be accepted based on the value of the difference between the two sample proportions by placing the significance level of α at the right tail of the standard normal distribution.

Example 8.26 The number of defective pieces of a bought-out item supplied by Vendor-1 and Vendor-2 follows binomial distribution. The quality manager of the firm buying the item believes that the proportion of the number of defective pieces in the lot supplied by the Vendor-1 is less than the proportion of the number of defective pieces in the lot supplied by the Vendor-2. So, the quality assistant under him selected a sample of 120 pieces from the supply of the Vendor-1 and found that the number of defective pieces in it is 15. Similarly, he selected a sample of 100 pieces from the supply of the Vendor-2 and found that the number of defective pieces in it is 11. Check the intuition of the quality manager at the significance level of 0.05.

Solution. Given that

Size of the sample of the Vendor-1, $n_1 = 120$

Size of the sample of the Vendor-2, $n_2 = 100$

Number of defective pieces in the Sample-1, $x_1 = 15$

Number of defective pieces in the Sample-2, $x_2 = 11$

Significance level, $\alpha = 0.05$

Let p_1, p_2 be the proportion with respect to Vendor-1 and Vendor-2, respectively. Then

$$\bar{p}_1 = \frac{15}{120} = 0.125, \quad \bar{p}_2 = \frac{11}{100} = 0.110$$

$$\bar{p}_1 \sim N\left[p_1, \frac{p_1(1-p_1)}{n_1}\right], \quad \bar{p}_2 \sim N\left[p_2, \frac{p_2(1-p_2)}{n_2}\right]$$

Therefore,

$$\bar{p}_1 - \bar{p}_2 \sim N\left[(p_1 - p_2), \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right]$$

Then

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} = \sqrt{\frac{0.125 \times 0.875}{120} + \frac{0.11 \times 0.89}{100}} = 0.0435$$

Null and alternate hypotheses are:

$$H_0: p_1 - p_2 \leq 0$$

$$H_1: p_1 - p_2 > 0$$

The standard normal statistic for $(p_1 - p_2)$ is, as shown below:

$$Z_{\bar{p}_1 - \bar{p}_2} = \frac{(\bar{p}_1 - \bar{p}_2) - (p_1 - p_2)}{\sigma_{\bar{p}_1 - \bar{p}_2}} = \frac{(0.125 - 0.11) - 0}{0.0435} = 0.345$$

The standard normal statistic for $+Z_\alpha$ is 1.64, which is shown in Fig. 8.33.

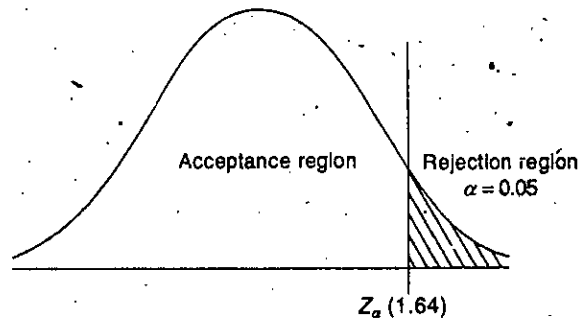


Fig. 8.33 Normal curve for Example 8.26.

Since the calculated Z value (0.345) is less than Z_α (1.64), it falls in the acceptance region. Hence, H_0 should be accepted.

Hence the intuition of the quality manager that “the proportion of the number of defective pieces in the lot supplied by the Vendor-1 is less than the proportion of the number of defective pieces in the lot supplied by the Vendor-2”, is true.

Test 2: $H_0: p_1 - p_2 \geq 0$ and $H_1: p_1 - p_2 < 0$

The objective of this test is to check whether the population is to be accepted based on the value of the difference between the two sample proportions by placing the significance level of α at the left tail of the standard normal distribution.

Example 8.27 The number of days on which the sales exceeded the targetted sales in Retail shop-1 as well as in Retail shop-2 of a company follows binomial distribution. The sales manager feels that the performance of the Retail shop-1 will exceed that of the Retail shop-2. To test his intuition, a sample of 110 days’ sales records of the Retail shop-1 is taken and it is found that on 90 days, the sales exceeded the targetted sales. Similarly, a sample of 130 days’ sales records of the Retail shop-2 is taken and it is found that on

112 days, the sales exceeded the targetted sales. Check the intuition of the sales manager at a significance level of 0.10.

Solution. Given that

Sample size of the Retail shop-1, n_1	= 110 days' records
Sample size of the Retail shop-2, n_2	= 130 days' records
Number of days on which sales exceeded the targetted sales in Retail shop-1	= 90 days
Number of days on which sales exceeded the targetted sales in Retail shop-2	= 112 days
Significance level, α	= 0.10

Let p_1 and p_2 be the proportion with respect to the Retail shop-1 and Retail shop-2, respectively. Then

$$\bar{p}_1 = \frac{90}{110} = 0.82, \quad \bar{p}_2 = \frac{112}{130} = 0.86$$

and

$$\bar{p}_1 \sim N\left[p_1, \frac{p_1(1-p_1)}{n_1}\right], \quad \bar{p}_2 \sim N\left[p_2, \frac{p_2(1-p_2)}{n_2}\right]$$

Therefore,

$$\bar{p}_1 - \bar{p}_2 \sim N\left[(p_1 - p_2), \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right]$$

Then

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} = \sqrt{\frac{0.82 \times 0.18}{110} + \frac{0.86 \times 0.14}{130}} = 0.04762$$

The null and alternate hypotheses are:

$$H_0: p_1 - p_2 \geq 0$$

$$H_1: p_1 - p_2 < 0$$

The standard normal statistic for $(\bar{p}_1 - \bar{p}_2)$ is:

$$Z_{\bar{p}_1 - \bar{p}_2} = \frac{(\bar{p}_1 - \bar{p}_2) - (p_1 - p_2)}{\sigma_{\bar{p}_1 - \bar{p}_2}} = \frac{(0.82 - 0.86) - 0}{0.04762} = -0.84$$

The standard normal statistic for $-Z_\alpha$ is -1.28 , which is shown in Fig. 8.34.

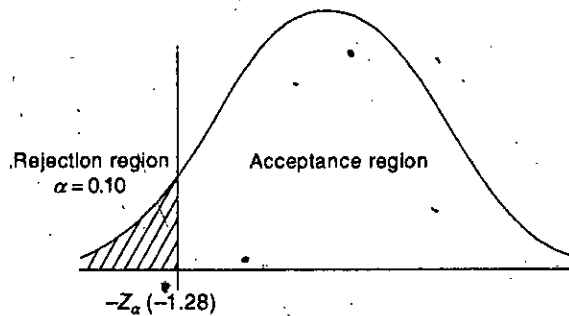


Fig. 8.34 Normal curve for Example 8.27.

Since the calculated Z value (-0.84) is more than $-Z_\alpha (-1.28)$, it falls in the acceptance region. Hence, H_0 should be accepted.

Hence the intuition of the sales manager that "the performance of the Retail shop-1 will exceed that of the Retail shop-2" is true.

8.3.4 Two-tailed Test Concerning the Difference between Two Proportions

This test is same as the tests discussed under Section 8.3.3, except the placement of the significance level on the standard normal distribution.

The combination of the hypotheses for this situation is:

$$H_0 : p_1 - p_2 = 0$$

$$H_1 : p_1 - p_2 \neq 0$$

The objective of this test is to check whether the population is to be accepted based on the value of the difference between the two sample proportions by placing the significance level of $\alpha/2$ at both tails of the standard normal distribution.

Let p_1 and p_2 be the proportion of the population-1 and population-2, respectively. Then

$$\bar{p}_1 \sim N \left[p_1, \frac{p_1(1-p_1)}{n_1} \right], \quad \bar{p}_2 \sim N \left[p_2, \frac{p_2(1-p_2)}{n_2} \right]$$

Therefore,

$$\bar{p}_1 - \bar{p}_2 \sim N \left[(p_1 - p_2), \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2} \right]$$

Then,

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

The standard normal statistic for $(p_1 - p_2)$ is:

$$Z_{\bar{p}_1 - \bar{p}_2} = \frac{(\bar{p}_1 - \bar{p}_2) - (p_1 - p_2)}{\sigma_{\bar{p}_1 - \bar{p}_2}}$$

Example 8.28 The number of credit card holders of a bank in two different cities (City-X and City-Y) settling their excess withdrawal amounts in time without attracting interest follows binomial distribution. The manager (collections) of the bank feels that the proportion of the number of such credit card holders in the City-X is not different from the proportion of the number of such credit card holders in the City-Y. To test his intuition, a sample of 150 credit card holders is taken from the City-X and it is found that 135 of them are settling their excess withdrawal amounts in time without attracting interest. Similarly, a sample of 160 credit card holders is taken from the City-Y and it is found that 133 of them are settling their excess withdrawal amounts in time without attracting interest. Check the intuition of the sales manager at a significance level of 0.10.

Solution. We have sample size with respect to City-X, $n_1 = 150$; sample size with respect to City-Y, $n_2 = 160$; number of credit card holders in the City-X settling their excess withdrawal amounts in time without attracting interest = 135; number of credit card holders in the City-Y settling their excess withdrawal amounts in time without attracting interest = 133; and the significance level, $\alpha = 0.10$.

Let p_1 and p_2 be the proportion with respect to City-X and City-Y, respectively. Then

$$\bar{p}_1 = \frac{135}{150} = 0.90, \quad \bar{p}_2 = \frac{133}{160} = 0.83$$

$$\bar{p}_1 \sim N\left[p_1, \frac{p_1(1-p_1)}{n_1}\right], \quad \bar{p}_2 \sim N\left[p_2, \frac{p_2(1-p_2)}{n_2}\right]$$

Therefore,

$$\bar{p}_1 - \bar{p}_2 \sim N\left[(p_1 - p_2), \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right]$$

Then

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} = \sqrt{\frac{0.90 \times 0.10}{150} + \frac{0.83 \times 0.17}{160}} = 0.0385$$

The null and alternate hypotheses are:

$$H_0: p_1 - p_2 = 0$$

$$H_1: p_1 - p_2 \neq 0$$

The standard normal statistic for $(\bar{p}_1 - \bar{p}_2)$ is:

$$Z_{\bar{p}_1 - \bar{p}_2} = \frac{(\bar{p}_1 - \bar{p}_2) - (p_1 - p_2)}{\sigma_{\bar{p}_1 - \bar{p}_2}} = \frac{(0.90 - 0.83) - 0}{0.0385} = 1.82$$

The standard normal statistics for $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$ are -1.64 and $+1.64$, respectively, which are shown in Fig. 8.35.

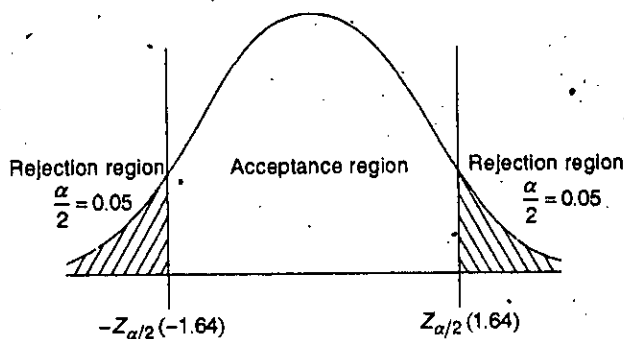


Fig. 8.35 Normal curve for Example 8.28.

Since the calculated Z value (1.82) is greater than $+Z_{\alpha/2}$ ($+1.64$), it falls in the rejection region. Hence, H_0 should be rejected.

Hence the intuition of the manager (collections) is not true.

8.4 TESTS OF HYPOTHESES CONCERNING VARIANCE(S)

As stated earlier, there are two sampling distributions for the variance which are chi-square distribution and F distribution.

For tests concerning single population variance, chi-square test should be used and for the tests concerning two-population variances, F -test should be used.

The different types of testing of hypothesis concerning variance(s) are listed below:

1. One-tailed chi-square tests concerning single-population variance
2. Two-tailed chi-square test concerning single-population variance
3. One-tailed F -tests concerning equality of two-population variances
4. Two-tailed F -test concerning equality of two-population variances.

8.4.1 One-tailed Chi-square Tests Concerning Single Population Variance

Let σ^2 be the variance of a population under study. The population may be normal or assumed to be normal. In some situations, the analyst may be keen in testing the variance of the population. Such testing of hypothesis concerning single population variance is carried out using chi-square test.

As stated in Chapter 7, the distribution of the variance S^2 of a random sample with size n taken from a normal population with variance σ^2 is known as chi-square (χ^2) distribution with $(n-1)$ degrees of freedom, which is as given below:

$$\chi^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2}$$

with $(n-1)$ degrees of freedom. In this distribution, degrees of freedom is the only factor. The two types of one-tailed chi-square tests for testing the population variance are:

Test 1: $H_0: \sigma^2 \leq k$ and $H_1: \sigma^2 > k$

Test 2: $H_0: \sigma^2 \geq k$ and $H_1: \sigma^2 < k$

where k is a constant.

Test 1: $H_0: \sigma^2 \leq k$ and $H_1: \sigma^2 > k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample variance by placing the significance level of α at the right tail of the chi-square distribution. If the calculated chi-square value is less than or equal to the table chi-square value with $(n-1)$ degrees of freedom and with a significance level of α placed at the right tail, then accept H_0 ; otherwise, reject H_0 .

Example 8.29 The diameter of crank shafts received from a vendor by Krishna Engines Limited follows normal distribution. The quality manager of the Krishna Engines Limited is given the information that the vendor has been instructed to set the production process to produce the crank shaft such that the variance of its diameter limited to at most 1.44 mm which is an essential requirement of the function of the engines. So, the quality manager has instructed his assistant to draw a sample of 12 crank shafts from the current consignment. The variance of the diameter of the crank shafts in the sample is found to be 1.96 mm. Test whether the consignment is to be accepted or rejected at a significance level of 0.05.

Solution. Here diameter of crank shafts follows normal distribution. Variance of the diameter of the population, $\sigma^2 = 1.44$ mm; sample size, $n = 12$; variance of the diameter of the sample, $S^2 = 1.96$ mm; and significance level, $\alpha = 0.05$. The two hypotheses are:

$$H_0: \sigma^2 \leq 1.44$$

$$H_1: \sigma^2 > 1.44$$

The chi-square statistic to test variance is:

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \quad [\text{with } (n-1) \text{ degrees of freedom}]$$

$$= \frac{(12-1) \times 1.96}{1.44}$$

$$= 14.972 \quad (\text{with } 11 \text{ degrees of freedom})$$

The table chi-square value with 11 degrees of freedom and at the given significance level of 0.05 placed at right tail is 19.675, which is shown in Fig. 8.36.

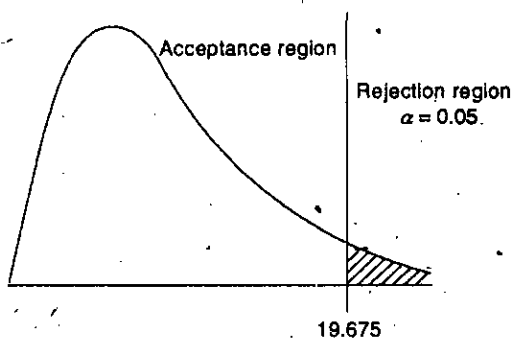


Fig. 8.36 Chi-square distribution curve for Example 8.29.

Since the calculated chi-square value (14.972) is less than the table chi-square value (19.675), it falls in the acceptance region. Hence, H_0 is to be accepted.

We, therefore, conclude that the variance of the consignment is within the permitted limit. So, the consignment is to be accepted.

Test 2: $H_0: \sigma^2 \geq k$ and $H_1: \sigma^2 < k$

The objective of this test is to check whether the population is to be accepted based on the value of the sample variance by placing the significance level of α at the left tail of the chi-square distribution. If the calculated chi-square value is more than or equal to the table chi-square value with $(n - 1)$ degrees of freedom and with a significance level of α placed at the left tail, then accept H_0 ; otherwise, reject H_0 .

Example 8.30 The weight of cement bags produced by Vignesh Cement Company follows normal distribution. The quality assistant at the final inspection section of the company feels that the variance of the weight of the cement bags has increased from a specified maximum variance of 0.64 kg which will lead to customer complaints. Hence, he has selected a sample of 8 cement bags and found that the variance of the sample is 0.36 kg. Check the intuition of the quality assistant at a significant level of 0.01.

Solution. Weight of cement bags follows normal distribution. We also have variance of the weight of cement bags of the population, $\sigma^2 = 0.64$ kg; sample size, $n = 8$; variance of the weight of the cement bags of the sample, $S^2 = 0.36$ kg; and the significance level, $\alpha = 0.01$. The null and alternate hypotheses are:

$$H_0: \sigma^2 \geq 0.64$$

$$H_1: \sigma^2 < 0.64$$

The chi-square statistic to test the variance is:

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \quad [\text{with } (n-1) \text{ degrees of freedom}]$$

$$= \frac{(8-1) \times 0.36}{0.64}$$

$$= 3.938 \quad (\text{with } 7 \text{ degrees of freedom})$$

The table chi-square value with 7-degrees of freedom and at the given significance level of 0.01 placed at the left tail (0.99 from right tail) is 1.239, which is shown in Fig. 8.37.

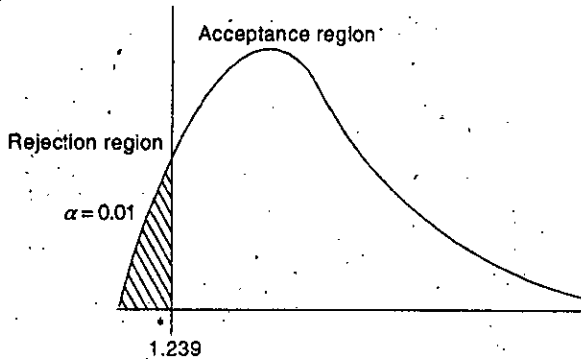


Fig. 8.37 Chi-square distribution curve for Example 8.30.

Since the calculated chi-square value (3.938) is more than the table chi-square left value (1.239) for the given significance level of 0.01, it falls in the acceptance region. Hence, H_0 is to be accepted.

Hence, we find that the variance of the weight of cement bags is more than the specified variance. Hence, the quality assistant should instruct the production supervisor to adjust the cement filling machine to avoid customer complaints in future.

8.4.2 Two-tailed Chi-square Test Concerning Single Population Variance

This test is same as the one-tailed tests presented in Section 8.4.1, except the placement of the significance level on the chi-square distribution.

The objective of this two-tailed test is to check whether the null hypothesis is to be accepted based on the sample variance by placing half of the significance level ($\alpha/2$) at both tails of the chi-square distribution.

The combination of the hypotheses for this situation is:

$$H_0: \sigma^2 = k$$

$$H_1: \sigma^2 \neq k$$

where k is a constant.

If the calculated chi-square value is in between $\chi^2_{\alpha/2}$ at left tail with $(n-1)$ degrees of freedom and $\chi^2_{\alpha/2}$ at right tail with $(n-1)$ degrees of freedom, then accept H_0 ; otherwise, reject H_0 .

Example 8.31 The weight of a drug produced by Ganga Pharmaceutical Company follows normal distribution. The specified variance of the weight of the drug of this population is 0.25 mg. The quality engineer of the firm claims that the variance of the weight of the drug does not differ significantly from the specified variance of the weight of the drug of the population. So, the purchase manager of Alpha Hospital who places order for that drug with the Ganga Pharmaceutical Company has selected a random sample of 12 drugs. The variance of the weight of the sample is found to be 0.49 mg. Verify the intuition of the quality manager of Ganga Pharmaceutical Company at a significance level of 0.10.

Solution. Weight of the drug follows normal distribution. We also have population variance of the weight of drugs $\sigma^2 = 0.25$ mg; sample size, $n = 12$; sample variance of the weight of drugs, $S^2 = 0.49$ mg; and significance level, $\alpha = 0.10$.

Null and alternate hypotheses are:

$$H_0: \sigma^2 = 0.25$$

$$H_1: \sigma^2 \neq 0.25$$

The chi-square statistic to test the variance is:

$$\begin{aligned}\chi^2 &= \frac{(n-1)S^2}{\sigma^2} \quad [\text{with } (n-1) \text{ degrees of freedom}] \\ &= \frac{(12-1) \times 0.49}{0.25} \\ &= 21.56 \quad (\text{with } 11 \text{ degrees of freedom})\end{aligned}$$

The table chi-square values with 11 degrees of freedom, $\chi^2_{\alpha/2}$ placed at left tail and $\chi^2_{\alpha/2}$ placed at right tail are 4.575 and 19.675, respectively, which are as shown in Fig. 8.38.

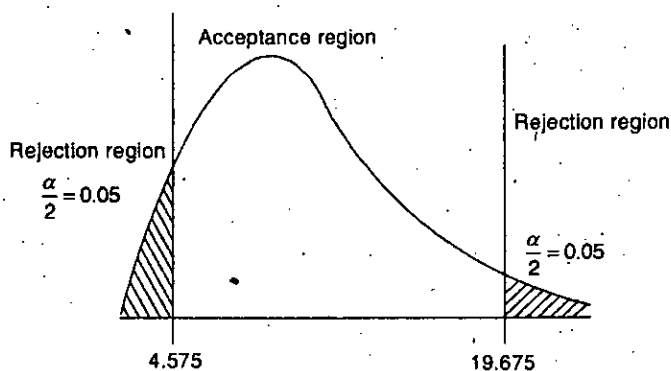


Fig. 8.38 Chi-square curve for Example 8.31.

Since the calculated chi-square value (21.56) is more than the table chi-square value at the right tail (19.675), it falls in the rejection region. Hence, H_0 is to be rejected.

Therefore, the variance of the weight of the drug is different from the specified variance. Hence, the quality engineer should take steps to adjust the process to bring the variance of the weight of the drug within the limit.

8.4.3 One-tailed F -tests Concerning Two Population Variances

Let σ_1^2 and σ_2^2 be the variances of two populations which follow normal distribution. The possible combinations of the hypotheses for one-tailed tests concerning two-population variances are:

Test 1: $H_0: \sigma_1^2 \leq \sigma_2^2$ and $H_1: \sigma_1^2 > \sigma_2^2$

Test 2: $H_0: \sigma_1^2 \geq \sigma_2^2$ and $H_1: \sigma_1^2 < \sigma_2^2$

F -statistic is a suitable statistic for testing the above combinations of hypotheses.

As stated in Chapter 7, the F distribution is the ratio of two chi-square variables. If S_1^2 and S_2^2 are the variances of two independent random samples of sizes n_1 and n_2 , respectively, taken from two normal populations which are having the variances as σ_1^2 and σ_2^2 , respectively, then F distribution is defined as:

$$F = \frac{[(n_1 - 1)S_1^2 / \sigma_1^2] / (n_1 - 1)}{[(n_2 - 1)S_2^2 / \sigma_2^2] / (n_2 - 1)}$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom.

The objective of any test is to minimize the type-1 error which will be higher at the transition point between H_0 and H_1 . Equivalent to this idea, the transition point occurs when $\sigma_1^2 = \sigma_2^2$. If $\sigma_1^2 = \sigma_2^2$, then the formula for the F distribution is reduced to the form, as shown below:

$$F = \frac{S_1^2}{S_2^2}$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom. In this distribution, the numerator degrees of freedom $(n_1 - 1)$ and the denominator degrees of freedom $(n_2 - 1)$ are the only two parameters affecting its shape and size.

Test 1: $H_0: \sigma_1^2 \leq \sigma_2^2$ and $H_1: \sigma_1^2 > \sigma_2^2$

The objective of this test is to check whether the null hypothesis, $\sigma_1^2 = \sigma_2^2$ is to be accepted based on the value of the two sample variances by placing the significance level of α at the right tail of the F distribution. If the calculated F value is less than or equal to the table F value with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom, and with a significance level of α placed at the right tail, then accept H_0 ; otherwise, reject H_0 .

Example 8.32 Two independent samples of students representing Region-1 and Region-2 of a programme under the distance education of a university are taken from normal populations with the same variance. The variance of marks of 10 students of a sample taken from the Region-1 is 64 and the variance of the marks of 15 students of another sample taken from the Region-2 is 40. Check whether the variance of the marks of the first sample is less than the variance of the marks of the second sample at a significance level of 0.05.

Solution. We have

Size of the sample from the Region-1, n_1	= 10
Size of the sample from the Region-2, n_2	= 15
Variance of the sample from the Region-1, S_1^2	= 64
Variance of the sample from the Region-2, S_2^2	= 40
Significance level, α	= 0.05

Hypotheses to be tested are:

$$H_0: \sigma_1^2 \leq \sigma_2^2$$

$$H_1: \sigma_1^2 > \sigma_2^2$$

In this example, $\sigma_1^2 = \sigma_2^2$. Hence, the F -statistic is computed as:

$$F = \frac{S_1^2}{S_2^2} = \frac{64}{40} = 1.6$$

The table F value with (9, 14) degrees of freedom at the given significance level of 0.05 is 2.65, which is shown in Fig. 8.39 along with the region of acceptance and the region of rejection.

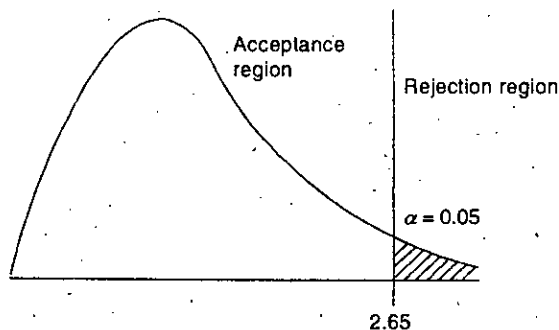


Fig. 8.39 F distribution curve for Example 8.32.

Since the calculated F ratio (1.6) is less than the table F value (2.65), it falls in the acceptance region. Hence, H_0 is to be accepted.

Therefore, the variance of the marks of the students in the Region-1 is less than the variance of the marks of the students in the Region-2.

Test.2: $H_0: \sigma_1^2 \geq \sigma_2^2$ and $H_1: \sigma_1^2 < \sigma_2^2$

The objective of this test is to check whether the null hypothesis, $\sigma_1^2 \geq \sigma_2^2$ is to be accepted based on the value of the two sample variances by placing the significance level of α at the left tail of the F -distribution. If the calculated F value is more than or equal to the table F value with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom, and with a significance level of α placed at the left tail, then accept H_0 ; otherwise, reject H_0 .

Example 8.33 The weight of cement bags packed during night shifts and day shifts in a cement company follows normal distribution. The quality assistant at the final inspection section of the company feels that the variance of the weight of the cement bags packed during night shifts is more than that of the cement bags packed during day shifts. To verify this fact, he has taken a sample of 9 bags of cement packed during night shift and found that its variance is 0.49 kg. Similarly, he has taken a sample of 13 bags of cement packed during day shift and found that its variance is 0.81 kg. Check the intuition of the quality assistant at a significant level of 0.01.

Solution. Given that

Size of the sample taken during night shift, n_1	= 9
Size of the sample taken during day shift, n_2	= 13
Variance of the sample taken during night shift, S_1^2	= 0.49 kg
Variance of the sample taken during day shift, S_2^2	= 0.81 kg
Significance level, α	= 0.01

Null and alternate hypotheses are:

$$H_0: \sigma_1^2 \geq \sigma_2^2$$

$$H_1: \sigma_1^2 < \sigma_2^2$$

The F statistic is computed as:

$$F = \frac{S_1^2}{S_2^2} = \frac{0.49}{0.81} = 0.605$$

The rejection region for the given significance level of 0.01 is placed at the left tail of F distribution, as shown in Fig. 8.40.

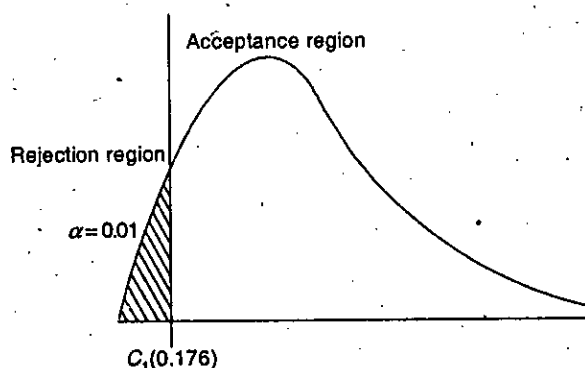


Fig. 8.40 F distribution curve for Example 8.33.

The cut-off value (C_1) at the left tail is linked with the given data, as shown in the following formula:

$$P\left(\frac{S_1^2}{S_2^2} \leq C_1\right) = 0.01$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom.

The F distribution is not a symmetrical distribution. Hence, the F value (C_1) for a given α placed at its left tail cannot be obtained directly. Let the modified form of the above equation be as follows:

$$P\left(\frac{S_2^2}{S_1^2} \geq \frac{1}{C_1}\right) = 0.01$$

with $(n_2 - 1)$ and $(n_1 - 1)$ degrees of freedom.

In the above equation, the statistic S_2^2/S_1^2 will also follow F distribution with $(n_2 - 1)$ and $(n_1 - 1)$ degrees of freedom.

The table F value with (12, 8) degrees of freedom at the given significance level of 0.01 placed at the right tail of F distribution is 5.67. Therefore, $1/C_1 = 5.67$.

From the above relation, C_1 is 0.176, which is marked in Fig. 8.40. Since the calculated F value (0.605) is more than the cut-off value, C_1 (0.176), it falls in the acceptance region. So, H_0 is to be accepted.

Thus we conclude that the variance of the weight of the cement bags packed during night shifts is more than that of the weight of the cement bags packed during day shifts.

8.4.4 Two-tailed F -test Concerning Equality of Two Population Variances

In this test, the equality of two population variances is checked using F -test. This is same as the one-tailed F -tests concerning the single population variance except in the placement of the significance level on the tails of the F -distribution.

The combination of hypotheses for this situation is:

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

The objective of this test is to check whether the null hypothesis, $\sigma_1^2 = \sigma_2^2$ is to be accepted based on the value of the two sample variances by placing half of the significance level ($\alpha/2$) at both tails of the F distribution.

Let C_1 be the F value with respect to $\alpha/2$ at the left tail with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom. C_2 be the F value with respect to $\alpha/2$ at the right tail with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom. If $\sigma_1^2 = \sigma_2^2$, then the F statistic is reduced to the following form:

$$F = \frac{S_1^2}{S_2^2}$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom.

If the calculated F value is in between C_1 and C_2 , then it falls in the acceptance region. Hence, accept H_0 ; otherwise, reject H_0 .

Example 8.34 The annual revenue of small scale industries in each of the two industrial estates (New Industrial Estate and Old Industrial Estate) situated in a city follows normal distribution. The manager of the branch bank issuing loans to the industries in these industrial estates feels that the variance of the annual revenue of the industries in the Old Industrial Estate does not differ from that of the industries in the New Industrial Estate. So, he selected a sample of 16 industries from the old industrial estate and found that the variance of the annual revenue is Rs. 225 lakhs. Similarly, he selected another sample of 13 industries from the new industrial estate and found that the variance of the annual revenue is Rs. 83 lakhs. Check the intuition of the branch manager at a significant level of 0.10.

Solution. Size of the sample taken from the old industrial estate, $n_1 = 16$. Size of the sample taken from the new industrial estate, $n_2 = 13$. Sample variance of annual revenue of the industries in the old industrial estate, $S_1^2 = \text{Rs. } 225$ lakhs. Sample variance of annual revenue of the industries in the new industrial estate, $S_2^2 = \text{Rs. } 83$ lakhs. Significance level, $\alpha = 0.10$. Hypotheses to be tested are:

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

F value is obtained as:

$$F = \frac{S_1^2}{S_2^2} = \frac{225}{83} = 2.71$$

The F value with (15, 12) degrees of freedom at half of the significance level ($\alpha/2 = 0.05$) placed at the right tail, as shown in Fig. 8.41 is 2.62. This is the cut-off value, C_2 at the right tail.

The rejection region for the other half of the significance level, ($\alpha/2$) placed at the left tail of F distribution in the same figure.

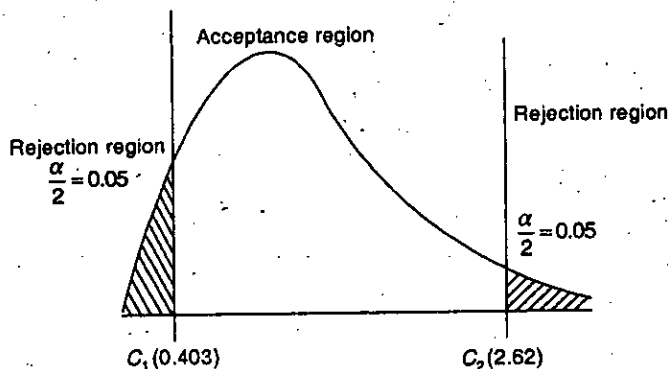


Fig. 8.41 F -distribution curve for Example 8.34.

The cut-off value (C_1) at the left tail is linked with the given data as shown in the following formula:

$$P\left(\frac{S_1^2}{S_2^2} \leq C_1\right) = 0.05$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom.

The F distribution is not a symmetrical distribution. Hence, the F ratio for a given $\alpha/2$ (0.05) placed at its left tail cannot be obtained directly. Let, the modified form of the above equation be as follows:

$$P\left(\frac{S_2^2}{S_1^2} \geq \frac{1}{C_1}\right) = 0.05$$

with $(n_1 - 1)$ and $(n_2 - 1)$ degrees of freedom. In the above equation, the statistic S_2^2/S_1^2 will also follow F distribution with $(n_2 - 1)$ and $(n_1 - 1)$ degrees of freedom.

The table F value with (12, 15) degrees of freedom at the given significance level of 0.05 placed at the right tail of F distribution is 2.48. Therefore, $1/C_1 = 2.48$. From the above relation, C_1 is 0.403 which is as marked in Fig. 8.41.

Since the calculated F value (2.71) is more than the cut-off value, C_2 (2.62) placed at the right tail, the calculated F value falls in the corresponding rejection region. So, H_0 is to be rejected.

Thus this means that the branch manager's intuition is not true. So, the variance of the annual revenue of industries in the old industrial estate differ from that of the industries in the new industrial estate.

8.5 CHI-SQUARE TEST FOR CHECKING INDEPENDENCE OF CATEGORIZED DATA

Consider two factors which may or may not have influence on the observed frequencies formed with respect to the combinations of different levels of the two factors.

Let A be one of the factors. B be another factor. a be the number of levels for the factor A. b be the number of levels for the factor B. o_{ij} be the observed frequency with respect to the i th level of the factor A and the j th level of the factor B, where $i = 1, 2, \dots, a$ and $j = 1, 2, \dots, b$.

A sample format of the above data is shown in Table 8.2. With reference to the Table 8.2, the following variables are defined:

$o_{i.}$ is the total number of the observations in the i th row. $o_{.j}$ is the total number of observations in the j th column. $o_{..}$ is the total of the observations in the entire table. $p_{i.}$ be the marginal probability of the row i , $i = 1, 2, \dots, a$. $p_{.j}$ be the marginal probability of column j , $j = 1, 2, \dots, b$. p_{ij} be the joint probability with respect to the row i and column j , where $i = 1, 2, \dots, a$ and $j = 1, 2, \dots, b$.

Table 8.2 Format of categorized Data/Contingency Table for Observed Frequency

		Factor B						Row total
		1	2	...	j	...	b	
Factor A	1	o_{11}	o_{12}	...	o_{1j}	...	o_{1b}	$o_{1.}$
	2	o_{21}	o_{22}	...	o_{2j}	...	o_{2b}	$o_{2.}$
	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
	i	o_{i1}	o_{i2}	...	o_{ij}	...	o_{ib}	$o_{i.}$
	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
	a	o_{a1}	o_{a2}	...	o_{aj}	...	o_{ab}	$o_{a.}$
Column total		$o_{.1}$	$o_{.2}$	...	$o_{.j}$	...	$o_{.b}$	$o_{..}$

Here

$$p_{i.} = \frac{o_{i.}}{o_{..}}, \quad p_{.j} = \frac{o_{.j}}{o_{..}}, \quad p_{ij} = \frac{o_{ij}}{o_{..}}$$

Expected value (expected frequency) for the row i and the column j is:

$$e_{ij} = o_{..} p_{ij} = o_{..} p_{i.} p_{.j} = o_{..} \frac{o_{i.}}{o_{..}} \frac{o_{.j}}{o_{..}} = \frac{o_{i.} o_{.j}}{o_{..}}$$

The combination of the hypotheses for this situation is:

$H_0: p_{ij} = p_{i.} p_{.j}$ (Factor A and factor B are independent)

$H_1: p_{ij} > p_{i.} p_{.j}$ (Factor A and factor B are not independent)

where $i = 1, 2, \dots, a$ and $j = 1, 2, \dots, b$.

The observed chi-square statistic

$$\chi^2 = \sum_{i=1}^a \sum_{j=1}^b \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

The objective of this test is to check if the row classification is independent of the column classification at a given significance level of α placed at the right tail of the chi-square distribution.

Let the table chi-square value with $(a-1)(b-1)$ degrees of freedom at the given significance level of α placed at the right tail of the chi-square distribution be K .

If the calculated chi-square value is less than or equal to the table chi-square value (K), then accept H_0 ; otherwise reject H_0 .

Example 8.35 A district industries centre (DIC) has collected the data summarizing the number of industries in the district under each combination of the level of technology used and the level of rate of return, as shown in Table 8.3. Check whether the rate of return is independent of the level of technology while grouping the industries, at a significance level of 0.05.

Table 8.3 Summary of Data of Industries

Rate of return	Level of technology used		
	Low	Medium	High
$R < 8\%$	20	50	10
$8\% \leq R < 10\%$	50	70	20
$10\% \leq R < 15\%$	30	75	70
$15\% \leq R$	10	35	60

Solution. The data in Table 8.3 is reproduced with row totals and column totals, as in Table 8.4.

Table 8.4 Summary of Observed Frequencies with Row Totals and Column Totals

Rate of return	Level of technology used			Row total $o_{i.}$
	Low	Medium	High	
$R < 8\%$	20	50	10	80
$8\% \leq R < 10\%$	50	70	20	140
$10\% \leq R < 15\%$	30	75	70	175
$15\% \leq R$	10	35	60	105
Column total ($o_{.j}$)	110	230	160	$o_{..} = 500$

Let p_{ij} be the joint probability with respect to the row i and column j , where $i = 1, 2, 3, 4$ and $j = 1, 2, 3$. The combination of the hypotheses is, as shown below:

$$H_0: p_{ij} = p_{i.} p_{.j}, \quad i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3.$$

(i.e. Classification of rate of return and classification of technology are independent.)

$$H_1: p_{ij} > p_{i.} p_{.j}, \text{ for at least one combination of } i \text{ and } j, \text{ where } i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3.$$

(i.e. Classification of rate of return and classification of technology are not independent.)

The expected frequencies for different combinations of rows and columns are summarized in Table 8.5 using the following formula:

$$e_{ij} = \frac{o_{i.} o_{.j}}{o_{..}}, \quad i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3$$

where $o_{i.}$ is the total number of observations in the i th row. $o_{.j}$ is the total number of observations in the j th column. $o_{..}$ is the total number of observations in the entire table.

Table 8.5 Summary of Expected Frequencies (e_{ij})

Rate of return	Level of technology used			Row total $o_{i.}$
	Low	Medium	High	
$R < 8\%$	17.6	36.8	25.6	80
$8\% \leq R < 10\%$	30.8	64.4	44.8	140
$10\% \leq R < 15\%$	38.5	80.5	56.0	175
$15\% \leq R$	23.1	48.3	33.6	105
Column total ($o_{.j}$)	110	230	160	$o_{..} = 500$

A sample calculation to obtain e_{11} is as:

$$e_{11} = \frac{o_{1.} \times o_{.1}}{o_{..}} = \frac{80 \times 110}{500} = 17.6$$

The chi-square statistic

$$\chi^2 = \sum_{i=1}^a \sum_{j=1}^b \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

The value of the chi-square statistic is computed as:

$$\begin{aligned} \chi^2 &= \frac{(20-17.6)^2}{17.6} + \frac{(50-36.8)^2}{36.8} + \frac{(10-25.6)^2}{25.6} + \frac{(50-30.8)^2}{30.8} + \frac{(70-64.4)^2}{64.4} + \frac{(20-44.8)^2}{44.8} \\ &\quad + \frac{(30-38.5)^2}{38.5} + \frac{(75-80.5)^2}{80.5} + \frac{(70-56.0)^2}{56.0} + \frac{(10-23.1)^2}{23.1} + \frac{(35-48.3)^2}{48.3} + \frac{(60-33.6)^2}{33.6} \\ &= 78.339 \end{aligned}$$

The table chi-square value with $6[(4-1)(3-1)]$ degrees of freedom at the given significance level of 0.05 placed at the right tail of the chi-square distribution is 12.592, which is shown in Fig. 8.42 along with the acceptance region and the rejection region.

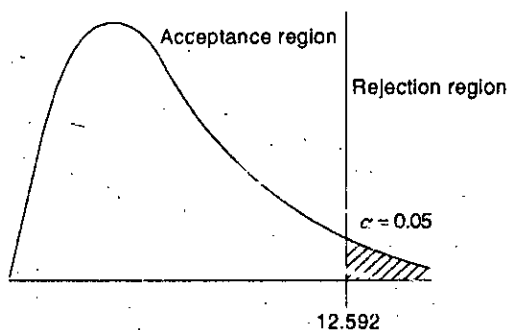


Fig. 8.42 Chi-square distribution for Example-8.35.

Since the calculated chi-square value (78.339) is more than the table chi-square value (12.592), it falls in the rejection region. Hence, reject H_0 :

Thus the classification of rate of return and classification of technology are not independent. This means that they have some influence on the grouping of industries into different combinations as indicated in the contingency table.

Example 8.36 A placement agency has conducted a competitive examination to create a potential database of applicants for the post of Quality Manager for their future recruitment. The classifications of the applicants who secured more than 80 percentile in the competitive examination are given in Table 8.6. Check whether the present monthly salary is independent of the number of years of experience while grouping the applicants at a significance level of 0.05.

Table 8.6 Summary of Data of Applicants

Present monthly salary, S (in Rs. 000)	Number of years of experience, n		
	$n < 2$	$2 \leq n < 5$	$5 \leq n$
$S < 10$	40	35	30
$10 \leq S < 14$	30	20	35
$14 \leq S < 18$	30	30	40
$18 \leq S$	35	40	35

Solution. The data in Table 8.6 is reproduced with row totals and column totals as in Table 8.7.

Let p_{ij} be the joint probability with respect to the row i and column j , where $i = 1, 2, 3, 4$ and $j = 1, 2, 3$.

Table 8.7 Summary of Observed Frequencies with Row Totals and Column Totals

Present monthly salary, S (in Rs. '000)	Number of years of experience, n			Row total $o_{i.}$
	$n < 2$	$2 \leq n < 5$	$5 \leq n$	
$S < 10$	40	35	30	105
$10 \leq S < 14$	30	20	35	85
$14 \leq S < 18$	30	30	40	100
$18 \leq S$	35	40	35	110
Column total, $(o_{.j})$	135	125	140	$o_{..} = 400$

The combination of the hypotheses is as shown below:

$$H_0: p_{ij} = p_{i.} p_{.j}, \quad i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3$$

(i.e. classification of present monthly salary and classification of number of years of experience are independent.)

$$H_1: p_{ij} > p_{i.} p_{.j}, \text{ for at least one combination of } i \text{ and } j, \text{ where } i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3.$$

(i.e. classification of present monthly salary and classification of number of years of experience are not independent.)

The expected frequencies for different combinations of rows and columns are summarized in Table 8.8 using the following formula:

$$e_{ij} = \frac{o_{i.} o_{.j}}{o_{..}}, \quad i = 1, 2, 3, 4 \text{ and } j = 1, 2, 3$$

where $o_{i.}$ is the total number of observations in the i th row, $o_{.j}$ is the total number of observations in the j th column, $o_{..}$ is the total number of the observations in the entire table.

Table 8.8 Summary of Expected Frequencies (e_{ij})

Present monthly salary, S (in Rs. '000)	Number of years of experience, n			Row total $o_{i.}$
	$n < 2$	$2 \leq n < 5$	$5 \leq n$	
$S < 10$	35.44	32.81	36.75	105
$10 \leq S < 14$	28.69	26.56	29.75	85
$14 \leq S < 18$	33.75	31.25	35.00	100
$18 \leq S$	37.13	34.28	38.50	110
Column total, $o_{.j}$	135	125	140	$o_{..} = 400$

A sample calculation to obtain e_{11} is as shown below:

$$e_{11} = \frac{o_{1.} o_{.1}}{o_{..}} = \frac{105 \times 135}{400} = 35.44$$

The chi-square statistic

$$\chi^2 = \sum_{i=1}^4 \sum_{j=1}^3 \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

The calculated value of the chi-square statistic is 7.12. The table chi-square value with $6[(4-1)(3-1)]$ degrees of freedom at the given significance level of 0.05 placed at the right tail of the chi-square distribution is 12.592, which is shown in Fig. 8.43 along with the acceptance region and the rejection region.

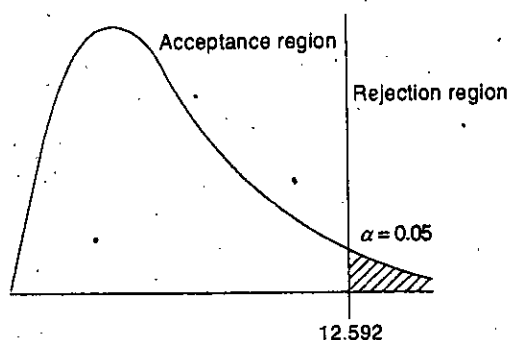


Fig. 8.43 Chi-square distribution for Example 8.36.

Since the calculated chi-square value (7.12) is less than the table chi-square value (12.592), it falls in the acceptance region. Hence, accept H_0 . Therefore, the classification of present monthly salary and classification of number of years of experience are independent. This means that they are not having influence on the grouping of potential applicants into different combinations as indicated in the contingency table.

8.6 GOODNESS OF FIT TEST

The real-world data of a system follow some distribution depending on the characteristics of the system. After collecting data from the system of interest, the essential step is to fit the data to the nearest distribution which represents the data more meaningfully for future analysis. Such fitting of data to the nearest distribution is done using the goodness of fit test. The goodness of fit of a given set of data is performed using chi-square test.

The combination of hypotheses for this situation is:

H_0 : The given data follow an assumed distribution

H_1 : The given data do not follow the assumed distribution

Steps of the goodness of fit test are as follows:

- Step-1:** Find the parameters (mean and if necessary variance) of the observed frequencies.
- Step-2:** Make a graphical plot of the observed frequencies. Based on the shape of the curve, identify the nearest distribution which can better explain the data.
- Step-3:** Find the expected frequencies for different values of the random variable using the theoretical distribution of the data by substituting the parameter(s).
- Step-4:** Compute the value of the chi-square statistic using the formula:

$$\text{The observed chi-square statistic, } \chi^2 = \sum_{i=1}^n \frac{(o_i - e_i)^2}{e_i}$$

where o_i is the observed frequency of the i th value of the random variable. e_i is the expected frequency of the i th value of the random variable. n is the number of values of the random variable.

- Step-5:** Find the table chi-square value with the applicable degrees of freedom and at the given significance level (α) placed at the right tail of the chi-square distribution.
- Step-6:** If the calculated chi-square value is less than the table chi-square value, it falls in the acceptance region. Hence, accept H_0 ; otherwise, reject H_0 .

Example 8.37 The sales manager of a showroom feels that the daily demand of a product follows uniform distribution. The observed frequencies of demand values are summarized in Table 8.9.

Table 8.9 Observed Frequencies of Daily Demand

Demand (units), (X_i)	Observed frequency, (o_i)	Demand (units), (X_i)	Observed frequency, (o_i)
30	13	35	9
31	10	36	12
32	7	37	10
33	10	38	14
34	6	39	9

Check whether the given data follow uniform distribution at a significance level of 0.05.

Solution. The combination of hypotheses for this problem is:

H_0 : The given data follow uniform distribution

H_1 : The given data do not follow uniform distribution

We also have

The total of the observed frequencies = 100

Number of demand values = 10

Expected frequency for each, X_i = $100/10 = 10$

The details of the calculations of the chi-square value are summarized in Table 8.10.

Table 8.10 Summary of Calculations of Chi-square Value

Serial number	Demand value, X_i	Observed frequency, o_i	Expected frequency, e_i	$\frac{(o_i - e_i)^2}{e_i}$
1	30	13	10	0.9
2	31	10	10	0.0
3	32	7	10	0.9
4	33	10	10	0.0
5	34	6	10	1.6
6	35	9	10	0.1
7	36	12	10	0.4
8	37	10	10	0.0
9	38	14	10	1.6
10	39	9	10	0.1

The observed chi-square statistic = $\sum_{i=1}^{10} \frac{(o_i - e_i)^2}{e_i} = 5.6$

The table chi-square value with 9 degrees of freedom at the given significance level of 0.05 placed at the right tail of the chi-square distribution is 16.919. This is shown in Fig. 8.44.

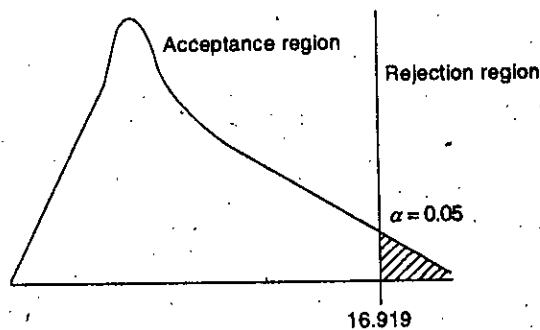


Fig. 8.44 Chi-square distribution for Example 8.37.

Since the calculated chi-square value (5.6) is less than the table chi-square value (16.919), it falls in the acceptance region. Hence, accept H_0 .

Thus the given data follow uniform distribution.

Example 8.38 The manager of a branch bank feels that the arrival rate of customers in his branch bank follows Poisson distribution. The observed frequencies of arrival rates (number of arrivals in 15 minutes intervals) are as summarized in Table 8.11.

Table 8.11 Observed Frequencies of Arrival Rates

Serial number, i	Arrival rate, X_i	Observed frequency, o_i
1	0	2
2	1	4
3	2	8
4	3	14
5	4	7
6	5	5
7	6	4
8	7	3
9	8	2
10	9	1

Check whether the given data follow Poisson distribution at a significance level of 0.05.

Solution. The combination of hypotheses for this problem is:

H_0 : The given data follow Poisson distribution

H_1 : The given data do not follow Poisson distribution

We also have

The total of the observed frequencies = 50

Number of values for the arrival rate = 10

Then

$$\text{Mean arrival rate, } \delta = \frac{\sum_{i=1}^{10} X_i o_i}{\sum_{i=1}^{10} o_i} = 3.7 \text{ arrivals per 15 min.}$$

The probability distribution for the Poisson distribution is

$$P(X) = \frac{\delta^X e^{-\delta}}{X!}, \quad X = 0, 1, 2, 3, \dots$$

The calculation of the chi-square value is summarized in Table 8.12.

Table 8.12 Summary of Calculations of Chi-square Value

Serial number	Demand value, X_i	Observed frequency, o_i	Expected probability, $P(X_i)$	Expected frequency, e_i	$\frac{(o_i - e_i)^2}{e_i}$
1	0	2	0.0247235	1.24	0.466
2	1	4	0.0914770	4.57	0.071
3	2	8	0.1692326	8.46	0.025
4	3	14	0.2087202	10.44	1.214
5	4	7	0.1930661	9.65	0.728
6	5	5	0.1428689	7.14	0.641
7	6	4	0.0881025	4.41	0.038
8	7	3	0.0465685	2.33	0.193
9	8	2	0.0215379	1.08	0.784
10	9	1	0.0088545	0.44	0.713

The observed chi-square statistic = $\sum_{i=1}^{10} \frac{(o_i - e_i)^2}{e_i} = 4.873$

The table chi-square value with 9 degrees of freedom at the given significance level of 0.05 placed at the right tail of the chi-square distribution is 16.919. This is shown in Fig. 8.45.

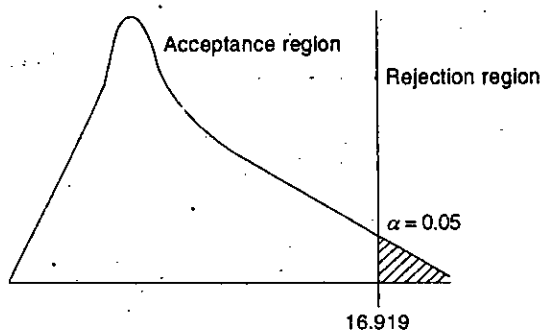


Fig. 8.45 Chi-square distribution for Example 8.38.

Since the calculated chi-square value (4.873) is less than the table chi-square value (16.919), it falls in the acceptance region. Hence, accept H_0 .

Thus the given data follow Poisson distribution.

QUESTIONS

1. Define hypothesis. What are the types of hypothesis? Explain them with suitable examples.
2. Distinguish between type-I error and type-II error.
3. What is operating characteristic (OC) curve? Compare and contrast it with power curve.
4. Give the classification of:
 - (a) tests of hypotheses concerning mean(s)
 - (b) tests of hypotheses concerning proportion(s)
 - (c) tests of hypotheses concerning variance(s).
5. The weekly sales of the salesmen of a company follows normal distribution. The targetted mean of the weekly sales of the salesmen is Rs. 2 lakhs. The variance of the weekly sales of the salesmen is Rs. 0.5 lakh. The regional marketing manager of the company feels that the performance of salesmen has declined in the recent past. A random sample of 40 salesmen is taken from the normal population for which the weekly sales is found to be Rs. 2.25 lakhs. Check whether the sales revenue of the salesmen has really declined at a significance level of 0.01.
6. The daily sales of a product in different shops follow normal distribution. The targetted mean of the daily sales of the product and their variance in the population are Rs. 10,000 and Rs. 2,500, respectively. The sales manager of the company which produces the product feels that the sales of the product has improved in the recent past. A random sample of 49 shops is taken from the normal population for which the mean daily sales is found to be Rs. 9,600. Check whether the sales of the product in different shops has really improved at a significance level of 0.05.
7. The annual sales of small scale industries in an industrial estate follow normal distribution. The targetted mean of the annual sales of the industries of the population is Rs. 75 lakhs. The variance of the annual sales of the industries of the population is Rs. 400 lakhs. The director of the industry department feels that the performance of the industries has not changed from the population mean in the recent past. A random sample of 49 industries is taken from the normal population for which the mean annual sales is found to be Rs. 70 lakhs. Check whether the sales of the industries has not changed from Rs. 75 lakhs at a significance level of 0.05.
8. The monthly income of respondents in a survey follows normal distribution with finite population size of 1000. The expected mean of the monthly income of the respondents of the population is Rs. 8,000. The variance of the monthly income of the respondents of the population is Rs. 3,00,000 lakhs. The researcher feels that the mean monthly income of the respondents has declined from the expected mean of Rs. 8,000 in the recent past. A random sample of 64 respondents is taken from the normal population for which the mean monthly income is found to be Rs. 8,750. Check whether the mean monthly income has declined from the expected mean of Rs. 8,000 at a significance level of 0.01.

9. The age of respondents in a survey follows normal distribution with finite population size of 1000. The expected mean of the age of the respondents of the population is 45 years. The variance of the age of the respondents of the population is 400 years. The researcher feels that the mean age of the respondents is more than the expected mean of 45 years in the recent past. A random sample of 81 respondents is taken from the normal population for which the mean age is found to be 42 years. Check whether the mean age has increased from the expected mean age of 45 years at a significance level of 0.10.
10. The monthly shopping amount of respondents for a particular product in a survey follows normal distribution with finite population size of 1000. The expected mean of the monthly shopping amount of the respondents of the population is Rs. 7,000. The variance of the monthly shopping amount of the respondents of the population is Rs. 2,50,000. The researcher feels that the mean monthly shopping amount of the respondents does not differ from the expected mean of Rs. 7,000 in the recent past. A random sample of 36 respondents is taken from the normal population for which the mean monthly shopping amount is found to be Rs. 6500. Check whether the mean shopping amount has changed from the expected mean of Rs. 7,000 at a significance level of 0.05.
11. The age of students undergoing postgraduate programme in distance mode follows normal distribution whose population is infinite. The expected mean of the age of the students is 45 years and its variance is unknown. The researcher feels that mean age of the students is significantly less than the expected mean age of the population. So, he has selected a random sample of 125 students and its mean and variance are found to be 50 years and 25 years, respectively. Verify the intuition of the researcher at a significance level of 0.10.
12. The weight of cement bags produced in a cement company follows normal distribution whose population is infinite. The expected mean of weight of the cement bags for sales of this population is 45 kg and its variance is unknown. The sales manager of the firm claims that the mean weight of the cement bags is significantly more than the expected mean weight of the population. So, he has selected a random sample of 36 bags and its mean and variance are found to be 44 kg and 1.25 kg, respectively. Verify the intuition of the sales manager at a significance level of 0.05.
13. The weight of cement bags produced by Beta Cement Company follows normal distribution whose population is finite and its size is 1000. The expected mean of the weight of the cement bags for sales of this population is 65 kg and its variance is unknown. The sales manager of the firm claims that the mean weight of the cement bags is significantly more than the expected mean weight of the population. So, the purchase manager of Alpha Construction Company who places order for cement bags with the Beta Cement Company has selected a random sample of 64 bags and its mean and variance are found to be 62 kg and 2.25 kg, respectively. Verify the intuition of the sales manager of the cement company at a significance level of 0.05.
14. The annual income of respondents in a survey follows normal distribution whose population size is 1000. The expected mean of the annual income of the respondents

- of this population is Rs. 4 lakhs and its variance is unknown. The researcher feels that mean annual income of the respondents is significantly more than the expected mean income of the respondents. So, he has selected a random sample of 81 respondents and its mean and variance are found to be Rs. 3.50 lakhs and 1.21 lakhs, respectively. Verify the intuition of the researcher at a significance level of 0.05.
15. The weight of a drug produced by Alpha Pharmaceutical Company follows normal distribution whose population is infinite. The specified mean of the weight of the drug of this population is 100 mg and its variance is unknown. The quality engineer of the firm claims that the mean weight of the drugs does not differ significantly from the specified mean weight of the population. So, the purchase manager of Beta hospital who places order for that drug with the Alpha Pharmaceutical Company has selected a random sample of 36 drugs. The mean and variance of the sample are found to be 96 mg and 25 mg, respectively. Verify the intuition of the quality manager of Alpha Pharmaceutical Company at a significance level of 0.01.
 16. The quality manager of a washing machine company feels that the mean time between failures of the motors received is at most 90 days. The quality manager wants to test his intuition. Hence, he has taken a sample of 25 motors whose mean time between failures and its variance are found to be 93 days and 16 days, respectively. Verify the intuition of the quality manager at a significance level of 0.05.
 17. The weight of electrodes purchased by a foundry follows normal distribution. The sales manager of the vendor firm claims that the mean weight of the electrodes is at least 125 gm. The quality manager of the foundry wants to verify this claim. So, he has taken a sample of 16 electrodes. The mean and the variance of the electrodes in the sample are found to be 120 gm and 64 gm, respectively. Verify the claim of the sales manager at a significance level of 0.01.
 18. The weight of costly catalyst electrodes purchased by a chemical company follows normal distribution. The sales manager of the vendor firm claims that the mean weight of the electrodes is 200 gm. The quality manager of the chemical company wants to verify this claim. So, he has taken a sample of 20 electrodes. The mean and the variance of the electrodes in the sample are found to be 205 gm and 81 gm, respectively. Verify the claim of the sales manager at a significance level of 0.10.
 19. The personnel manager of a firm is recruiting its graduate trainees through two consultancy agencies. The performance index (0–10 scale) of the trainees from each consultancy agency follows normal distribution. The variance of the performance index of the trainees selected through a Consultancy Agency X is 12 and that of the trainees selected through another Consultancy Agency Y is 6. The manager feels that the mean performance index of the trainees selected through the Consultancy Agency X is less than that of the trainees selected through the Consultancy Agency Y. To test his intuition, he has drawn a sample of 64 trainees selected through the Consultancy Agency X and their mean performance index is found to be 8. Similarly, he has drawn a sample of 81 trainees selected through the Consultancy Agency Y and their mean performance index is found to be 6.5. Test the intuition of the manager at a significance level of 0.01.

20. A consultant to a multi-plant organization is studying the age of employees in two different plants (Plant-1 and Plant-2) for their relative difference. The variance of the age of employees in Plant-1 is 64 and that of the employees in Plant-2 is 81. The consultant feels that the age of the employees in Plant-1 is more than that of the employees in Plant-2. To test his intuition, he has selected a sample of 81 employees from Plant-1 and their mean age is found to be 45 years. Similarly, he has selected a sample of 64 employees from Plant-2 and their mean age is found to be 48 years. Test the intuition of the consultant at a significance level of 0.05.
21. An automobile company presently buys connecting rods from Vendor-X. But the quality manager is desiring to consider the supply of connecting rods from Vendor-Y to meet the extra demand, provided the new vendor can supply the connecting rods with good precision on its length on par with that of the Vendor-X. If the length is more than the necessity, it will lead to excessive filing time while assembling. Hence, it should be avoided. The length of the connecting rods supplied by Vendor-X as well as Vendor-Y follows normal distribution with variances of 9 mm and 12 mm, respectively.
- He has selected 81 connecting rods from the Vendor-X and found that the mean length of the sample is 450 mm. Similarly, he selected 64 connecting rods from the Vendor-Y and found that the mean length of the sample is 455 mm. Check whether the quality manager can consider the supply from the Vendor-Y at a significance level of 0.05.
22. The production rate of spindles produced in Machine-A as well as in Machine-B follows normal distribution. The variance of the production rate of the spindles produced using each machine is not known. The production manager feels that the performance of Machine-A is not comparable with that of the Machine-B. To test his intuition, the quality assistant of the shop has selected 100 days' records of the Machine-A and found that the mean and variance of the production rate of this sample are 90 units and 16 units, respectively. Similarly, he has selected 120 days' records of the Machine-B and found that the mean and variance of the production rate of this sample are 85 units and 40 units, respectively. Check the intuition of the production manager at a significance level of 0.05.
23. The regional manager of a bank wants to analyze the number of delayed repayments of instalments of consumer loans in two of its branch banks (Branch-X and Branch-Y). The number of delayed payments of instalments in each branch bank follows normal distribution. The manager feels that the number of delayed payments of instalments by the consumers of the Branch-X is no way different from that of the Branch-Y. So, he selected the loan accounts of 80 different consumers from the Branch-X and found that the mean and variance of the number of delayed payments of instalments are 35 and 25, respectively. Similarly, he selected the loan accounts of 100 different consumers from the Branch-Y and found that the mean and variance of the number of delayed payments of instalments are 40 and 49, respectively. Test his intuition at a significance level of 0.01.
24. The diameter of engine cylinders supplied by Vendor-1 as well as Vendor-2 follows normal distribution. The quality manager of the firm buying the engine cylinders believes that the diameter of the cylinders supplied by the Vendor-1 is not different from

that of the cylinders supplied by the Vendor-2. So, the quality assistant under him selected 25 cylinders from the supply of the Vendor-1 and found that mean and variance of the diameter are 50 mm and 2.5 mm, respectively. Similarly, he selected 28 cylinders from the supply of the Vendor-2 and found that mean and variance of the diameter are 48 mm and 3.5 mm, respectively. Verify the intuition of the manager at a significance level of 0.01.

25. The production manager of an automobile company is convinced that the reduction in standard times for the employees working in its assembly line will improve the productivity of the line. But the quality manager has a contradicting view, because such reduction will decrease the proportion of the number of assemblies passing out of the assembly line which are free from missing components. Before introducing the new standard times, the quality manager ascertained through a study that 90% of the assemblies coming out of the assembly line will be free from missing components.

After introducing the new standard times for the employees in the assembly line, it is found that out of a sample of 60 assemblies which passed out of the assembly line, 57 assemblies are free from missing components. Check the view of the quality manager at a significance level of 0.01.

26. The personnel manager of a company is convinced that complete avoidance of overtime for the employees will improve the productivity of the company. But the production manager has a contradicting view, because such complete avoidance of the overtime will increase labour absenteeism which will have its negative consequences in planning. Before implementing complete avoidance of overtime, the production manager ascertained through a study that 20% of employees attended duty for less than the total number of working days in a month.

After implementing the complete avoidance of overtime, it is found that out of a sample of 150 employees, 24 of them attended duty for less than the total number of working days in a month. Check the view of the production manager at a significance level of 0.10.

27. The sales representative of a pharmaceutical company claims that a new drug for a disease cures in 95% of the cases. When it is administered by a doctor to 100 patients, it cured only 90 patients. Check the claim of the sales representative at a significance level of 0.05.

28. The number of defective pieces of an item produced by Machine-X as well as Machine-Y follows binomial distribution. The quality manager of the firm buying the item believes that the proportion of the number of defective pieces in the lot produced by the Machine-X is less than the proportion of the number of defective pieces in the lot produced by the Machine-Y. So, the quality assistant under him selected a sample of 100 pieces from the production of the Machine-X and found that the number of defective pieces in it is 10. Similarly, he selected a sample of 90 pieces from the production of the Machine-Y and found that the number of defective pieces in it is 6. Check the intuition of the quality manager at the significance level of 0.10.

29. The number of days on which the sales of a product exceeded the targetted sales in City-1 as well as in City-2 follows binomial distribution. A researcher feels that the

performance of the product in the City-1 will exceed that of the City-2. To test his intuition, a sample of 100 days' sales records from the City-1 is taken and it is found that on 90 days, the sales exceeded the targetted sales. Similarly, a sample of 120 days' sales records from the City-2 is taken and it is found that on 112 days, the sales exceeded the targetted sales. Check the intuition of the researcher at a significance level of 0.05.

30. The number of credit card holders of a bank in two different cities (City-X and City-Y) settling their excess withdrawal amounts in time without attracting interest, follows binomial distribution. The manager (collections) of the bank feels that the proportion of the number of such credit card holders in the City-X is not different from the proportion of the number of such credit card holders in the City-Y. To test his intuition, a sample of 200 credit card holders is taken from the City-X and it is found that 160 of them are settling their excess withdrawal amounts in time without attracting interest. Similarly, a sample of 180 credit card holders is taken from the City-Y and it is found that 150 of them are settling their excess withdrawal amounts in time without attracting interest. Check the intuition of the sales manager at a significance level of 0.05.
31. The diameter of crank shafts received from a vendor by Devi Engines Limited follows normal distribution. The quality manager of Devi Engines Limited is given the information that the vendor has been instructed to set the production process to produce the crank shaft such that the variance of its diameter is limited to at most 2.25 mm which is an essential requirement of the function of the engines. So, the quality manager has instructed his assistant to draw a sample of 15 crank shafts from the current consignment. The variance of the diameter of the crank shafts in the sample is found to be 3.15 mm. Test whether the consignment is to be accepted or rejected at a significance level of 0.05.
32. The weight of cement bags produced in a cement company follows normal distribution. The quality assistant at the final inspection section of the company feels that the variance of the weight of the cement bags has increased from a specified maximum variance of 0.81 kg which will lead to customer complaints. Hence, he has selected a sample of 10 cement bags and found that the variance of the sample is 0.64 kg. Check the intuition of the quality assistant at a significant level of 0.01.
33. The weight of a drug produced by a pharmaceutical company follows normal distribution. The specified variance of the weight of the drug of this population is 0.36 mg. The quality engineer of the firm claims that the variance of the weight of the drug does not differ significantly from the specified variance of the weight of the drug of the population. So, the purchase manager of a hospital who places order for that drug with the pharmaceutical company has selected a random sample of 15 drugs. The variance of the weight of the sample is found to be 0.56 mg. Verify the intuition of the quality engineer of the pharmaceutical company at a significance level of 0.05.
34. Two independent samples of students representing Region-1 and Region-2 of a programme under the distance education of a University are taken from normal populations with the same variance. The variance of marks of 16 students of a sample taken from the Region-1 is 225 and the variance of the marks of 12 students of another sample taken

from the Region-2 is 144. Check whether the variance of the marks of the first sample is less than the variance of the marks of the second sample at a significance level of 0.01.

35. The weight of cement bags packed during forenoon slot and afternoon slot in a cement company follows normal distribution. The quality assistant at the final inspection section of the company feels that the variance of the weight of the cement bags packed during forenoon slot is more than that of the cement bags packed during afternoon slot. To verify this fact, he has taken a sample of 8 bags of cement packed during forenoon slot and found that its variance is 0.64 kg. Similarly, he has taken a sample of 10 bags of cement packed during afternoon slot and found that its variance is 1.00 kg. Check the intuition of the quality assistant at a significant level of 0.05.
36. The monthly income of employees in each of the two Industrial estates (new industrial estate and old industrial estate) situated in a city follows normal distribution. A researcher feels that the variance of the monthly income of the employees in the old industrial estate does not differ from that of the employees in the new industrial estate. So, he selected a sample of 10 employees from the old industrial estate and found that the variance of their monthly income is Rs. 80,00,000. Similarly, he selected another sample of 16 employees from the new industrial estate and found that the variance of their monthly income is Rs. 10,50,000. Check the intuition of the researcher at a significant level of 0.10.
37. A researcher has collected the data summarizing the number of respondents in his study under each combination of the level of income and the level of qualification as shown in the following table. Check whether the income is independent of the qualification while grouping the respondents, at a significance level of 0.01.

Level of income	Level of qualification		
	Diploma	UG	PG
Low	25	55	15
Medium	60	70	25
High	50	80	75

38. A researcher has grouped the number of employees in an organization with respect to their age and income level, as shown in the following table.

Age, X	Income level		
	Low	Medium	High
$X < 40$	45	40	35
$40 \leq X < 50$	35	25	40
$50 \leq X$	35	35	45

Check whether the age is independent of the income level while grouping the employees at a significance level of 0.05.

39. The daily demand of a product seems to follow uniform distribution. The observed frequency of daily demand values are summarized in the following table:

Demand (units)	25	26	27	28	29	30	31	32	33	34
Observed frequency	18	15	12	15	11	14	17	15	19	14

Check whether the given data follow uniform distribution at a significance level of 0.01.

40. The arrival rates (number of assemblies per 10 minutes interval) of assembled product at the final inspection station of the assembly line of an automobile company are summarized below:

Serial number, i	Arrival rate, X_i	Observed frequency, o_i
1	0	3
2	1	6
3	2	12
4	3	18
5	4	12
6	5	10
7	6	4
8	7	3
9	8	2
10	9	1

Check whether the given data follow Poisson distribution at a significance level of 0.05.

CHAPTER

9

NONPARAMETRIC TESTS

9.1 INTRODUCTION

It is generally assumed that all practical data follow normal distribution. Under such situations, one can estimate the parameters such as mean, variance, etc., and use the standard tests which are presented in the previous chapter. Hence, they are known as *parametric* tests.

But, in some situations, the practical data may be non-normal and/or it may not be possible to estimate the parameter(s) of the data. The tests which are used for such situations are called *nonparametric* tests. Since, these tests are based on the data which are free from distribution and parameter, these tests are known as nonparametric tests.

The nonparametric tests can be used even for nominal data (qualitative data, like greater or less, etc.) and ordinal data, like ranked data. These tests require less calculations, because there is no need to compute parameters. Also, these tests can be applied to very small samples, more specifically during pilot studies in market research.

Inference about the population can be made by the nonparametric tests when the assumptions of the standard methods cannot be satisfied, since the nonparametric tests involve no or less restricting assumptions when compared to the parametric tests.

The nonparametric tests which are presented in this chapter are as follows:

- (a) One-sample tests
 - One-sample sign test
 - Chi-square test
 - Kolmogorov-Smirnov test
 - Run test for randomness
- (b) Two-samples tests
 - Two-samples sign test
 - Median test
 - Mann-Whitney U test (Rank-sum test)
- (c) K-samples test
 - Median test
 - Kruskal-Wallis test

9.2 ONE-SAMPLE TESTS

9.2.1 One-sample Sign Test

This one-sample sign test is applied to a situation where a sample is taken from a population which has a continuous symmetrical distribution and known to be non-normal such that the probability of having a sample value less than the mean value as well as the probability of having a sample value more than the mean value (p) is $1/2$. It will be difficult to verify the symmetry of the distribution when the sample size is small. Hence, under such situation, it is better to formulate the hypotheses using its median (μ). Here, based on the hypothesized 'population mean' (when the sample size is large), 'population median' (when the sample size is small), each sample observation will be classified into either plus (+) or minus (-) as stated below:

$$\begin{aligned}\text{Modified sample value, } Y_i &= +, \text{ if } X_i > \mu \\ &= -, \text{ if } X_i < \mu\end{aligned}$$

where X_i is the i th sample value, μ is the mean/median and Y_i is the i th modified sample observation (+ or -).

This means that if a sample value is more than μ , then it is replaced by '+' sign; if a sample value is less than μ , then it is replaced by '-' sign.

The one-sample sign test can be further classified into the following categories:

- One-tailed one-sample sign tests for small sample
- Two-tailed one-sample sign test for small sample
- One-tailed one-sample sign tests for large sample
- Two-tailed one-sample sign test for large sample.

One-tailed one-sample sign tests for small sample

One-tailed tests concerning one-sample sign test for small sample are classified as follows:

$$\text{Test 1: } H_0: p = \frac{1}{2} \text{ and } H_1: p > \frac{1}{2}$$

$$\text{Test 2: } H_0: p = \frac{1}{2} \text{ and } H_1: p < \frac{1}{2}$$

A small random sample of size n [np as well as $n(1-p)$ is less than 5] is taken and its null hypothesis is tested using binomial distribution with $p = 1/2$, where X is the random variable representing the number of plus signs.

The above two tests are demonstrated with suitable examples in this subsection.

Test 1: $H_0: p = 1/2$ and $H_1: p > 1/2$ The objective of this test is to check whether the population is to be accepted based on the value of the sample statistic by placing the significance level of α at the right tail of the distribution.

If the computed binomial value, $P(X \geq \text{number of plus signs})$ is more than α , then accept H_0 ; otherwise, reject H_0 .

Example 9.1 A medical representative visited at random 9 clinics and found that the waiting times in minutes before meeting doctors in these clinics are 16, 30, 17, 22, 31, 11, 19, 18 and 8. Check whether the waiting time is 15 minutes ($H_0: \mu = 15$) as against the alternate hypothesis $H_1: \mu > 15$ using sign test at a significance level of 0.05.

Solution. Here, sample size (n) = 9, $p = 0.5$. Then $np = n(1-p) = 4.5$. So, the sample is a small sample. We also have significance level (α) = 0.05.

On classifying each observation into '+' sign if it is more than 15 and '-' sign, if it is less than 15, the signs are as follows: +, +, +, +, +, -, +, +, -. The number of plus (+) signs = 7. The number of minus (-) signs = 2.

Let X be the random variable representing the number of plus signs. Then

$$H_0: \mu = 15 \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu > 15 \quad \text{or} \quad p > \frac{1}{2}$$

As per the statement of the problem, the significance region is 0.05 which is to be kept at the right tail. This is equal to the probability of having X or more number of plus signs. The binomial probability that the number of plus signs is more than or equal to 7 is as computed below:

$$\begin{aligned} P(X \geq 7, n=9, p=0.5) &= 1 - P(X \leq 6, n=9, p=0.5) \\ &= 1 - 0.9102 \\ &= 0.0898 \end{aligned}$$

Since $P(X \geq 7)$ is more than the significance level of 0.05, the binomial statistic falls in the acceptance region. Hence, accept H_0 .

Thus the waiting time of the medical representative at the clinics is not more than 15 minutes.

Test 2: $H_0: p = 1/2$ and $H_1: p < 1/2$ The objective of this test is to check whether the population is to be accepted based on the value of the sample statistic by placing the significance level of α at the left tail of the distribution.

If the computed binomial value, $P(X \leq \text{number of plus signs})$ is more than α , then accept H_0 ; otherwise, reject H_0 .

Example 9.2 A salesman of a company visited at random 8 cities and the number of orders obtained in different cities are 5, 6, 4, 8, 2, 4, 9, 1. Check whether the number of orders obtained in each city is 7 ($H_0: \mu = 7$) as against the alternate hypothesis $H_1: \mu < 7$ using sign test at a significance level of 0.05.

Solution. We have sample size (n) = 8, $p = 0.5$. Then $np = n(1-p) = 4$. So, the sample is a small sample. We also have significance level (α) = 0.05.

On classifying each observation into '+' sign if it is more than 7 and '-' sign, if it is less than 7, the signs are as follows: -, -, -, +, -, -, +, -. The number of plus (+) signs = 2. The number of minus (-) signs = 6.

Let X be the random variable representing the number of plus signs. Then

$$H_0: \mu = 7 \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu < 7 \quad \text{or} \quad p < \frac{1}{2}$$

As per the statement of the problem, the significance region is 0.05 which is to be kept at the left tail. This is equal to the probability of having X or less number of plus signs. The binomial probability that the number of plus signs is less than or equal to 2 is:

$$P(X \leq 2, n = 8, p = 0.5) = 0.1445$$

Since, $P(X \leq 2)$ is more than the given significance level (0.05), the binomial statistic falls in the acceptance region. Hence, accept H_0 .

Thus the number of orders obtained by the salesman from each city is not less than 7.

Two-tailed one-sample sign test for small sample

Two-tailed test concerning one-sample sign test for small sample is as follows:

$$H_0: p = \frac{1}{2}$$

$$H_1: p \neq \frac{1}{2}$$

A small random sample of size n [np as well as $n(1-p)$ is less than 5] is taken and its null hypothesis is tested using binomial distribution with $p = 1/2$, where X is the random variable representing the number of plus signs.

The objective of this test is to check whether the population is to be accepted based on the value of the sample statistic by placing half of the significance level, $\alpha/2$ at both the tails of the distribution. If the computed binomial value, $P(X \geq \text{number of plus signs})$ is more than $\alpha/2$, then accept H_0 ; otherwise, reject H_0 . The above test is demonstrated with a suitable example as shown below.

Example 9.3 The annual income in lakhs of rupees of 9 randomly selected CEOs of companies are 12, 16, 20, 15, 14, 8, 11, 17 and 13. Check whether the annual income of the CEOs is 10 lakhs ($H_0: \mu = 10$) as against the alternate hypothesis $H_1: \mu \neq 10$ using sign test at a significance level of 0.10.

Solution. For sample size (n) = 9, we get $np = n(1-p) = 4.5$. So, the sample is a small sample. We have significance level (α) = 0.10 and $\alpha/2 = 0.05$.

On classifying each observation into + sign if it is more than 10 and - sign, if it is less than 10, the signs are as follows: +, +, +, +, +, -, +, +, +. The number of plus (+) signs = 8 and the number of minus (-) signs = 1.

Let X be the random variable representing the number of plus signs. Then

$$H_0: \mu = 10 \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu \neq 10 \quad \text{or} \quad p \neq \frac{1}{2}$$

As per the statement of the problem, half of the significance level ($\alpha/2$) is to be kept at both tails of the distribution. The binomial probability that the number of plus signs is more than or equal to 8 is as computed below:

$$\begin{aligned} P(X \geq 8, n=9, p=0.5) &= 1 - P(X \leq 7, n=9, p=0.5) \\ &= 1 - 0.9805 \\ &= 0.0195 \end{aligned}$$

Since $P(X \geq 8)$ is less than one-half of the significance level ($\alpha/2 = 0.05$), the binomial statistic falls in the rejection region placed at the right tail of the distribution. Hence, reject H_0 .

Thus the annual income of the CEOs does not equal to Rs. 10 lakhs.

One-tailed one-sample sign tests for large sample

One-tailed tests concerning one-sample sign test for large sample are classified as follows:

$$\text{Test 1: } H_0: p = \frac{1}{2} \quad \text{and} \quad H_1: p > \frac{1}{2}$$

$$\text{Test 2: } H_0: p = \frac{1}{2} \quad \text{and} \quad H_1: p < \frac{1}{2}$$

A random sample of size n , where np as well as $n(1-p)$ is at least 5 is taken and its null hypothesis is tested using normal approximation to the binomial distribution with $p = 1/2$, where X is the random variable representing the number of plus signs.

If $X \sim B(X, n, p)$, then for large value of n ($n > 30$), one can approximate the binomial distribution to normal distribution as:

$$X \sim N[np, np(1-p)]$$

where np is the mean and $np(1-p)$ is the variance of the normal distribution. The formula for the Z statistic is

$$Z = \frac{X - np}{\sqrt{np(1-p)}}$$

In this section, these tests are demonstrated with suitable examples.

Test 1: $H_0: p = 1/2$ and $H_1: p > 1/2$ The objective of this test is to check whether the population is to be accepted based on the value of the sample statistic by placing the significance level of α at the right tail of the distribution. If the computed Z value is more than Z_{α} , then reject H_0 ; otherwise, accept H_0 .

Example 9.4 The weight of a drug produced by Alpha Pharmaceutical Company follows nonnormal distribution. The specified weight of the drug of this population is 100 mg. Any excess weight of the drug will lead to side reactions. The quality engineer of the firm claims that the weight of the drugs is not greater than the specified mean weight of the population. So, the purchase manager of Beta Hospital who places order for that drug with the Alpha Pharmaceutical Company has selected a random sample of 36 drugs and their weights as:

101	102	95	103	96	103	101	103	105
99	100	99	102	104	102	108	102	101
91	95	98	101	109	103	102	109	106
92	97	99	108	105	104	103	99	108

Verify the intuition of the quality manager of Alpha Pharmaceutical Company that the weight of the drugs is 100 mg ($H_0: \mu = 100$), against the alternate hypothesis ($H_1: \mu > 100$), using sign test at a significance level of 0.05.

Solution. Given sample size (n) = 36, significance level (α) = 0.05. On classifying each observation into + sign if it is more than 100 and - sign, if it is less than 100, we get the signs as follows:

+	+	-	+	-	+	+	+	+
-	0	-	+	+	+	+	+	+
-	-	-	+	+	+	+	+	+
-	-	-	+	+	+	+	-	+

The number of plus (+) signs = 24

The number of minus (-) signs = 11

The number of zeros = 1

Let X be the random variable representing the number of plus signs, then

$$H_0: \mu = 100 \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu > 100 \quad \text{or} \quad p > \frac{1}{2}$$

Since n is more than 30, the binomial distribution of this situation can be approximated to a normal distribution with the following mean and variance:

$$\text{Mean} = np = 36 \times 0.5 = 18$$

$$\text{Variance, } (\sigma^2) = np(1-p) = 36 \times 0.5 \times 0.5 = 9$$

$$\text{Standard deviation, } (\sigma) = 3$$

Then

$$X \sim N [np, np(1-p)] \sim N(18, 9)$$

The Z statistic is computed as:

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{24 - 18}{\sqrt{9}} = 2$$

The value of the standard normal statistic Z_α when α (0.05) is placed at the right tail of the standard normal distribution is 1.64.

Since the calculated Z value (2) is more than the table value of Z_α (1.64), reject the null hypothesis. Thus the weight of the drugs is more than 100 mg.

Test 2: $H_0: p = 1/2$ and $H_1: p < 1/2$ The objective of this test is to check whether the population is to be accepted based on the value of the sample statistic by placing the significance level of α at the left tail of the distribution.

If the computed Z value is more than $-Z_\alpha$ then accept H_0 ; otherwise, reject H_0 .

Example 9.5 The personnel manager of a firm is recruiting its graduate trainees through a written test. The performance index (0–10 scale) of the trainees follows non-normal distribution. The personnel manager feels that the performance index of the trainees is more than 8. To test his intuition, he has selected a sample of 40 trainees whose performance indices are as:

5	6	9	4	2	5	5	7	7	9
6	9	5	2	4	5	8	6	10	7
7	10	8	7	9	3	6	7	6	9
2	9	9	8	5	4	7	9	8	10

Test the intuition of the manager using sign test at a significance level of 0.01.

Solution. Given sample size (n) = 40, significance level (α) = 0.01.

On classifying each observation into + sign if it is more than 8 and – sign, if it is less than 8, the signs are as follows:

–	–	+	–	–	–	–	–	–	+
–	+	–	–	–	–	0	–	–	–
–	+	0	–	+	–	–	–	–	+
–	+	+	0	–	–	–	+	0	+

The number of plus (+) signs = 11

The number of minus (–) signs = 25

The number of zeros = 4

Let X be the random variable representing the number of plus signs. Then

$$H_0: \mu = 8 \text{ or } p = \frac{1}{2}$$

$$H_1: \mu < 8 \text{ or } p < \frac{1}{2}$$

Since n is more than 30, the binomial distribution of this situation can be approximated to a normal distribution with the following mean and variance:

$$\text{Mean} = np = 40 \times 0.5 = 20$$

$$\text{Variance, } (\sigma^2) = np(1-p) = 40 \times 0.5 \times 0.5 = 10$$

$$\text{Standard deviation, } (\sigma) = \sqrt{10}$$

Then

$$X \sim N[np, np(1-p)] \sim N(20, 10)$$

The Z statistic is computed as:

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{11 - 20}{\sqrt{10}} = -2.846$$

The value of the standard normal statistic $-Z_\alpha$ is -2.33 . Since the calculated Z value (-2.846) is less than the value of $-Z_\alpha$ (-2.33), reject the null hypothesis. Thus the performance index of the trainees is less than 8.

Two-tailed one-sample sign test for large sample

Two-tailed test concerning one-sample sign test for large sample is as follows:

$$H_0: p = \frac{1}{2}$$

$$H_1: p \neq \frac{1}{2}$$

A random sample of size n where np as well as $n(1-p)$ is at least 5 is taken and its null hypothesis is tested using normal approximation to the binomial distribution with $p = 1/2$, where X is the random variable representing the number of plus signs.

If $X \sim B(X, n, p)$, then for large value of n ($n > 30$), one can approximate the binomial distribution to normal distribution, as:

$$X \sim N[np, np(1-p)]$$

The formula for the Z statistic is:

$$Z = \frac{X - np}{\sqrt{np(1-p)}}$$

The objective of this test is to check whether the population is to be accepted based on the value of the sample statistic by placing one-half of the significance level ($\alpha/2$) at both the tails of the distribution. If the absolute value of the computed $Z|Z|$ is more than $Z_{\alpha/2}$, then reject the null hypothesis; otherwise, accept the null hypothesis.

The above test is demonstrated with the following example.

Example 9.6 The production rate of spindles produced in a machine follows non-normal distribution. The production manager feels that the production rate of the machine is not different from 75 spindles per hour. To test his intuition, the quality assistant of the shop has selected a random sample of 36 different hourly productions which are as shown below:

67	70	76	85	60	66	80	85	77
79	80	89	82	74	82	98	82	71
81	85	78	71	89	83	82	89	86
72	87	89	78	75	74	73	89	88

Check the intuition of the production manager using sign test at a significance level of 0.05.

Solution. We have sample size (n) = 36 and significance level (α) = 0.05.

On classifying each observation into + sign if it is more than 75 and - sign, if it is less than 75, we get the signs as follows:

-	-	+	+	-	-	+	+	+
+	+	+	+	-	+	+	+	-
+	+	+	-	+	+	+	+	+
-	+	+	+	0	-	-	+	+

The number of plus (+) signs = 25

The number of minus (-) signs = 10

The number of zeros = 1

Let X be the random variable representing the number of plus signs. Then

$$H_0: \mu = 75 \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu \neq 75 \quad \text{or} \quad p \neq \frac{1}{2}$$

Since n is more than 30, the binomial distribution of this situation can be approximated to a normal distribution with the following mean and variance:

$$\text{Mean} = np = 36 \times 0.5 = 18$$

$$\text{Variance, } \sigma^2 = np(1 - p) = 36 \times 0.5 \times 0.5 = 9$$

$$\text{Standard deviation, } \sigma = \sqrt{9} = 3$$

Then

$$X \sim N [np, np(1 - p)] \sim N(18, 9)$$

The Z statistic is

$$Z = \frac{X - np}{\sqrt{np(1 - p)}} = \frac{25 - 18}{\sqrt{9}} = 2.333$$

The value of the standard normal statistic $Z_{\alpha/2}$ is 1.96.

Since the calculated absolute value of Z (2.333) is more than $Z_{\alpha/2}$, reject the null hypothesis. Thus the production rate of the machines is different from 75 per hour.

9.2.2 Chi-square Test

Chi-square test is a nonparametric test which is used to check the goodness of fit for large samples. This test is used to compare the given observed frequencies with the corresponding expected frequencies of an assumed theoretical distribution to draw conclusion about whether or not the given data follow the assumed distribution. This test has been already discussed in Section 8.6.

9.2.3 Kolmogorov-Smirnov Test

Kolmogorov-Smirnov (K-S) test is similar to the chi-square test as discussed in Sections 8.6 and 9.2.2 to do goodness of fit of a given set of data to an assumed distribution. This test is more powerful for small samples whereas the chi-square test is suited for large samples.

The combination of hypotheses used in this test is as:

H_0 : The given set of observations follows the assumed distribution.

H_1 : The given set of observations does not follow the assumed distribution.

In K-S test, the values of the observed cumulative distribution of the random variable in a given situation are compared with the corresponding values of the theoretical cumulative distribution of the random variable and their absolute differences are calculated. Then, the maximum of these absolute differences is treated as the calculated statistic of K-S test, namely D_{cal} which is given by the following formula:

$$D_{cal} = \max |OF_i - EF_i|$$

where OF_i is the observed cumulative probability for the i th value of the random variable X , EF_i is the expected cumulative probability for the i th value of the random variable X , from the theoretical distribution.

For a given significance level (α) and the sample size, the theoretical value of D can be obtained from Annexure VI. This theoretical value of D is distribution-free, as it is a function of the significance level (α) and the sample size (n).

The Kolmogorov-Smirnov test is an one-tailed test. Hence, if the calculated value of D is more than the theoretical value of D for a given significance level, then reject the null hypothesis; otherwise, accept the null hypothesis.

Example 9.7. The sales manager of a cement company feels that the quarterly demand for cement in tons follows uniform distribution. The observed frequencies of demand values are summarized in Table 9.1.

Table 9.1 Observed Frequencies of Quarterly Demand

Demand (in tons) (X_i)	Observed frequency (o_i)	Demand (in tons) (X_i)	Observed frequency (o_i)
30	2	35	2
31	3	36	4
32	1	37	2
33	4	38	4
34	2	39	1

Check whether the given data follow uniform distribution using Kolmogorov–Smirnov test at a significance level of 0.05.

Solution. The combination of hypotheses for this problem situation is:

H_0 : The given data follow uniform distribution.

H_1 : The given data do not follow uniform distribution.

We have

The total of the observed frequencies (sample size, n) = 25

Significance level, α = 0.05

Number of values of the random variable, X_i = 10

Expected frequency for each X_i = $25/10 = 2.5$

Expected probability for each X_i = $2.5/25 = 0.10$

The details of the calculations of K–S test are summarized in Table 9.2.

Table 9.2 Summary of Calculations of K–S Test

Serial number (i)	Demand value (X_i)	Observed frequency (O_i)	Observed probability	Expected probability	Observed cumulative probability (OF_i)	Expected cumulative probability (EF_i)	$D_i = OF_i - EF_i $
1	30	2	0.08	0.10	0.08	0.10	0.02
2	31	3	0.12	0.10	0.20	0.20	0.00
3	32	1	0.04	0.10	0.24	0.30	0.06
4	33	4	0.16	0.10	0.40	0.40	0.00
5	34	2	0.08	0.10	0.48	0.50	0.02
6	35	2	0.08	0.10	0.56	0.60	0.04
7	36	4	0.16	0.10	0.72	0.70	0.02
8	37	2	0.08	0.10	0.80	0.80	0.00
9	38	4	0.16	0.10	0.96	0.90	0.06
10	39	1	0.04	0.10	1.00	1.00	0.00

In Table 9.2, the maximum value of D_{cal} is 0.06 when X_i is equal to 32 or 38. The theoretical value of D_n for the sample size (n) of 25 and the significance level of 0.05 is 0.27.

Since D_{cal} (0.06) is less than the table value of D_{25} (0.27), accept the null hypothesis, H_0 . Thus the given data follow uniform distribution.

Example 9.8 The arrival rate of customers at the loan counter of a branch bank appears to follow Poisson distribution. The observed frequencies of arrival rates (number of arrivals per hour) are as summarized in Table 9.3.

Table 9.3 Observed Frequencies of Arrival Rates

Period (i)	Arrival rate (X_i)	Observed frequency (O_i)
1	0	2
2	1	6
3	2	18
4	3	12
5	4	8
6	5	3
7	6	1

Check whether the given data follow Poisson distribution using Kolmogorov–Smirnov test at a significance level of 0.05.

Solution. The combination of hypotheses for this problem situation is:

H_0 : The given data follow Poisson distribution.

H_1 : The given data do not follow Poisson distribution.

We have

The total of the observed frequencies = 50

Number of values of the random variable, $X_i = 7$

Then

$$\text{Mean arrival rate, } \delta = \frac{\sum_{i=1}^7 X_i o_i}{\sum_{i=1}^7 o_i} = 2.62 \text{ arrivals per hour}$$

The probability distribution for Poisson distribution is:

$$P(X) = \frac{\delta^X e^{-\delta}}{X!}, \quad X = 0, 1, 2, 3, \dots$$

The calculations of D values for Kolmogorov–Smirnov test are summarized in Table 9.4.

Table 9.4 Summary of Calculations of K-S Test

Period (i)	Arrival rate (X_i)	Observed frequency (O_i)	Observed probability	Expected probability	Observed cumulative probability (OF_i)	Expected cumulative probability (EF_i)	$D_{cal} =$ $ OF_i - EF_i $ probability
1	0	2	0.04	0.073	0.04	0.073	0.033
2	1	6	0.12	0.191	0.16	0.264	0.104
3	2	18	0.36	0.250	0.52	0.514	0.006
4	3	12	0.24	0.218	0.76	0.732	0.028
5	4	8	0.16	0.143	0.92	0.875	0.045
6	5	3	0.06	0.075	0.98	0.950	0.030
7	6	1	0.02	0.033	1.00	0.983	0.017

In Table 9.4, the maximum value of D_{cal} is 0.104 when X_i is equal to 1. The theoretical value of D_n for the sample size (n) of 50 and the significance level of 0.05 is computed based on the following formula

$$D_{50} = \frac{1.36}{\sqrt{n}} = \frac{1.36}{\sqrt{50}} = 0.192$$

Since D_{cal} (0.104) is less than the table value of D_{50} (0.192), accept the null hypothesis, H_0 . Thus the given data follow Poisson distribution.

Example 9.9 The demand of a product in lakhs of units appears to follow normal distribution. The observed frequencies of the demand values are as summarized in Table 9.5.

Table 9.5 Observed Frequencies of Demand Values

Serial no. (i)	Demand (X_i)	Observed frequency (O_i)
1	0 – 5	4
2	5 – 10	9
3	10 – 15	15
4	15 – 20	20
5	20 – 25	18
6	25 – 30	7
7	30 – 35	5

Check whether the given data follow normal distribution using Kolmogorov–Smirnov test at a significance level of 0.10.

Solution. The combination of hypotheses for this problem situation is:

H_0 : The given data follow normal distribution.

H_1 : The given data do not follow normal distribution.

Now, we have

The total of the observed frequencies, $n = 78$
 Number of ranges for the random variable, $X_i = 7$

Then,

$$\text{Mean demand } \mu = \frac{\sum_{i=1}^7 X_i o_i}{\sum_{i=1}^7 o_i} = 17.6282$$

We also obtain

Variance, $\sigma^2 = 57.033$
 Standard deviation, $\sigma = 7.552$
 Significance level, $\alpha = 0.10$

The formula for the standard normal statistic is:

$$Z_i = \frac{X_i - \mu}{\sigma}, \quad i = 1, 2, 3, 4, 5, 6, 7$$

The calculations of D values for K-S test are summarized in Table 9.6.

Table 9.6 Summary of Calculation of K-S Test

Demand interval (I)	Mid-point (X_i)	Observed frequency (O_i)	Observed probability	Z-value	Observed cumulative probability up to Z(OF_i)	Expected cumulative probability (OF_i)	$D_i = OF_i - EF_i $
0-5	2.5	4	0.0513	-2.00	0.0228	0.0513	0.0285
5-10	7.5	9	0.1154	-1.34	0.0901	0.1667	0.0766
10-15	12.5	15	0.1923	-0.68	0.2483	0.3590	0.1107
15-20	17.5	20	0.2564	-0.02	0.4920	0.6154	0.1234
20-25	22.5	18	0.2308	0.65	0.7422	0.8462	0.1040
25-30	27.5	7	0.0897	1.31	0.9049	0.9359	0.0310
30-35	32.5	5	0.0641	1.97	0.9756	1.0000	0.0244

In Table 9.6, the maximum value of D_{cal} is 0.1234. The theoretical value of D_n for the sample size (n) of 78 and the significance level of 0.10 is computed based on the following formula.

$$D_{78} = \frac{1.22}{\sqrt{n}} = \frac{1.22}{\sqrt{78}} = 0.1381$$

Since D_{cal} (0.1234) is less than the table value of D_{78} (0.1381), accept the null hypothesis, H_0 . Thus the given data follow the normal distribution.

9.2.4 Run Test for Randomness

Consider the example of arrivals of customers at a branch office of the telephone department for payment of telephone bills after the due date. The senior officer of the telephone department wants to verify whether the gender of the arriving customers is random.

An example sequence of arriving customers is shown as below, in which M means male and F means female.

M M F F F M M M F F F M F M M M M F F

In this sequence of customers, the number of male customers is 10 and the number of female customers is 9.

In this context, a run is defined as a sequence which consists of repeated occurrence of a particular symbol and the immediate preceding and succeeding symbols are different from the symbol in it or no other symbol on either side.

Though the proportion of the number of male customers and that of the female customers is almost same in this example, here the objective is to test about the randomness of the occurrence of runs at a given significance level, α .

Let n_1 be the frequency of occurrence of a particular symbol in the whole stream of symbols, n_2 be the frequency of occurrence of another symbol in the whole stream of symbols and r be the number of runs.

If n_1 as well as n_2 is less than or equal to 20, then the sample is treated as a small sample. If either n_1 or n_2 , or both is/are larger than 20, then the sample is treated as a larger sample.

The combination of hypotheses for the run test of randomness is:

H_0 : The occurrence of the runs in the given stream of symbols is random.

H_1 : The occurrence of the runs in the given stream of symbols is not random.

Run test of randomness of small samples

As stated earlier, if n_1 as well as n_2 is less than or equal to 20, then the sample is treated as a small sample. In this case, the observed number of runs = r .

From Annexure VII, the smaller critical value, and the larger critical value if exists for the given combination of n_1 and n_2 at a significance level of 0.05, can be obtained. If the observed r value is less than or equal to the smaller critical value, or more than or equal to the larger critical value for the given combination of n_1 and n_2 at the given significance level (α), then reject the null hypothesis; otherwise, accept the null hypothesis.

Example 9.10 Consider the example stated earlier as reproduced below:

The customers arrive at a branch office of the telephone department for payment of telephone bills after the due date. The senior officer of the telephone department wants to verify whether the gender of the arriving customers is random. The sequence of arriving customers is as shown below, in which M means male and F means female. Check the randomness of gender of the arriving customers at a significance level of 0.05.

M M F F F M M M F F F M F M M M M F F

Solution. The given stream of arrivals of male and female customers is as shown below with underscoring to represent different runs:

M M F F F M M M F F F M F M M M M F F

Frequency of occurrence of M, $n_1 = 10$

Frequency of occurrence of F, $n_2 = 9$

Number of runs, $r_{\text{cal}} = 8$

Significance level, $\alpha = 0.05$

Since both n_1 and n_2 are less than 20, the sample of the problem is to be treated as small. Now, consider

H_0 : The occurrence of the runs of the given stream of symbols (M, F) is random.

H_1 : The occurrence of the runs of the given stream of symbols (M, F) is not random.

From Annexure VII, the smaller critical value, and the larger critical value for the given combination of n_1 (10) and n_2 (9) at a significance level of 0.05 are 5 and 16, respectively.

Since the calculated number of runs, r_{cal} (8), is not less than the smaller critical value (5) as well as not greater than the larger critical value (16), accept the null hypothesis.

Thus the occurrences of the sub-streams of M and F are random.

Run test of randomness of large samples

As stated earlier, if n_1 or n_2 or both is/are more than 20, then the sample is treated as a large sample. Under such situation, one can approximate the sampling distribution of r to normal distribution with the following mean and variance:

$$\text{Mean of } r, \mu = \frac{2n_1n_2}{n_1 + n_2} + 1$$

$$\text{Variance of } r, \sigma^2 = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}$$

The formula for the standard normal statistic to test the significance of r is as shown below:

$$Z = \frac{r - \mu}{\sigma}$$

It is a two-tailed test. Hence, the values of table $-Z_{\alpha/2}$ and $Z_{\alpha/2}$ define the left critical value and right critical value of r . If the calculated Z value is in between $-Z_{\alpha/2}$ and $Z_{\alpha/2}$, then accept the null hypothesis; otherwise reject the null hypothesis.

Example 9.11 The marketing manager of a company is keen in analyzing the outcomes of different quotations submitted to its customers. The outcome is either Winning (W) or Losing (L) the order. The sequence of outcomes of 40 different quotations are as listed below. Check whether the events of Winning or Losing the orders is random at a significance level of 0.05.

WW LL WWWWWW LL WWW L WWW LL WW LL WW LLL
W LL WWW LL WW

Solution. The given stream of Winning and Losing of orders against quotations is as shown below with underscoring to represent different runs.

WW LL WWWWWW LL WWW L WWW LL WW LL WW LLL
W LL WWW LL WW

Frequency of occurrence of W, $n_1 = 24$

Frequency of occurrence of L, $n_2 = 16$

Number of runs, $r = 17$

Significance level, $\alpha = 0.05$

Half of the significance level, $\alpha/2 = 0.025$

Since n_1 is more than 20, the sample of the problem is large. Now, consider the hypothesis:

H_0 : The occurrence of the runs of the given stream of symbols (W, L) is random.

H_1 : The occurrence of the runs of the given stream of symbols (W, L) is not random.

Then

$$\text{Mean of } r, \mu^2 = \frac{2n_1n_2}{n_1 + n_2} + 1 = \frac{2 \times 24 \times 16}{24 + 16} + 1 = 20.2$$

$$\begin{aligned} \text{Variance of } r, \sigma^2 &= \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)} \\ &= \frac{2 \times 24 \times 16(2 \times 24 \times 16 - 24 - 16)}{(24 + 16)^2(24 + 16 - 1)} \\ &= 8.96 \end{aligned}$$

Standard deviation, $\sigma = 2.993$.

The formula for the standard normal statistic to test the significance of r is as shown below:

$$Z = \frac{r - \mu}{\sigma} = \frac{17 - 20.2}{2.993} = -1.069$$

The values of $-Z_{\alpha/2}$ is -1.96 and the value of $+Z_{\alpha/2}$ is $+1.96$. These are shown in Fig. 9.1 along with acceptance region and rejection regions.

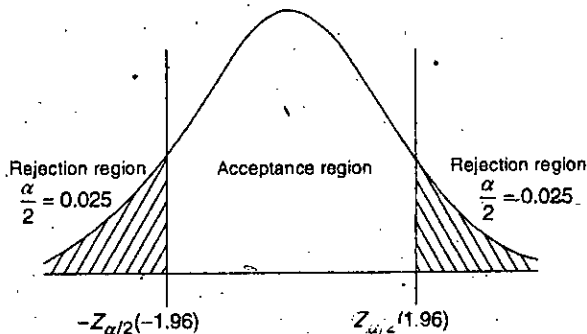


Fig. 9.1 Normal curve for Example 9.11.

Since the calculated Z value (-1.069) of r falls in the acceptance region between $-Z_{\alpha/2}$ (-1.96) and $+Z_{\alpha/2}$ ($+1.96$), accept the null hypothesis.

Thus the occurrences of the sub-streams of the symbols W and L are random.

9.3 TWO-SAMPLES TESTS

In some situation, the researchers may be taking statistical decisions based on two samples. Each of the two samples may represent a particular population under consideration. For example, one may be interested to study the effectiveness of teaching methods, like unstructured teaching and structured teaching. Suppose, colleges have changed their method of teaching from unstructured teaching to structured teaching, recently. Then, the researcher has to take a random sample of students of a particular programme from different colleges and collect their performances before introducing the structured method of teaching and after introducing the structured method of teaching. The respective observations will form two samples.

There are many more examples which are as listed below:

1. Number of defects found in the daily lots of items produced by a set of similar machines before and after installing a costly automatic inspection system in them.
2. Monthly sales revenues of a product in different regions before launching an advertisement campaign and after launching the advertisement campaign.
3. Number of daily transactions in a bank before and after introducing computers.
4. Number of outpatients treated in a government hospital before and after providing modern infrastructure.

As stated earlier, the two samples nonparametric tests can be classified into two samples sign test, median test, Mann-Whitney U test (rank-sum test) and Wilcoxon matched-pairs test (rank-sum test). These tests are explained with examples in this section.

9.3.1 Two-samples Sign Test

Two-samples sign test is applied to a situation, where two samples are taken from two populations which have continuous symmetrical distributions and known to be non-normal.

such that the probability of having the first sample value less than the corresponding second sample value as well as the probability of having the first sample value more than the corresponding second sample (p) is $1/2$.

The observations of the samples are modified as follows:

$$\begin{aligned}\text{Modified sample value, } Z_i &= +, \text{ if } X_i > Y_i \\ &= -, \text{ if } X_i < Y_i\end{aligned}$$

where X_i is the i th observation of the first sample, Y_i is the i th observation of the second sample, and Z_i is the i th combined observation of the two samples put together, in terms of '+' or '-'.

This means that if the i th observation of the first sample is more than that of the second sample, then their i th combined observation in terms of sign is '+'; if the i th observation of the first sample is less than that of the second sample, then their i th combined observation in terms of sign is '-'; if they are equal, then their i th combined observation is treated as 0.

The two-samples sign test can be further classified into the following categories:

- (a) One-tailed two-samples sign tests with binomial distribution
- (b) Two-tailed two-samples sign test with binomial distribution
- (c) One-tailed two-samples sign tests with normal approximation
- (d) Two-tailed two-samples sign test with normal approximation.

One-tailed two-samples sign tests with binomial distribution

The combinations of hypotheses for the one-tailed tests concerning two-samples sign test with binomial distribution are as given below:

$$\text{Test 1: } H_0: \mu_X = \mu_Y \text{ or } p = \frac{1}{2}$$

$$H_1: \mu_X > \mu_Y \text{ or } p > \frac{1}{2}$$

$$\text{Test 2: } H_0: \mu_X = \mu_Y \text{ or } p = \frac{1}{2}$$

$$H_1: \mu_X < \mu_Y \text{ or } p < \frac{1}{2}$$

Two random samples of size n where np , as well as $n(1-p)$ is less than 5 are taken. Their null hypothesis is tested using binomial distribution with $p = 1/2$, in which Q is the random variable representing the number of plus signs.

The objective of the first test is to check whether the null hypothesis is to be accepted based on the value of the combined samples statistic by placing the significance level (α) at the right tail of the distribution. If the computed binomial value, $P(Q \geq \text{number of plus signs})$ is more than the significance level (α), then accept H_0 ; otherwise, reject H_0 .

The objective of the second test is to check whether the null hypothesis is to be accepted based on the value of the combined samples statistic by placing the significance level (α) at the left tail of the distribution. If the computed binomial value, $P(Q \leq \text{number of plus signs})$ is more than the significance level (α), then accept H_0 ; otherwise, reject H_0 .

Example 9.12 The daily average number of outpatients treated in 9 government hospitals in a district before and after providing health care awareness programme in the surrounding regions of the hospitals are summarized in Table 9.7. Check whether the daily average inflow of the outpatients to the hospitals is decreased considerably at a significance level of 0.05.

Table 9.7 Daily Average Inflow of Outpatients of Hospitals

Hospital	Before	After	Hospital	Before	After
1	400	350	6	150	100
2	300	250	7	175	50
3	600	620	8	275	200
4	100	90	9	420	500
5	200	220			

Solution. The combination of hypotheses for this problem situation is as shown below:

$$H_0: \mu_X = \mu_Y \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu_X > \mu_Y \quad \text{or} \quad p > \frac{1}{2}$$

where μ_X is the mean of the first population (before the programme); μ_Y is the mean of the second population (after the programme).

As per the statement of the problem, the significance level, 0.05 (α) is to be placed at the right tail of the distribution.

On classifying each observation into '+' sign if the observation of the first sample is more than that of the second sample, '-' sign if the observation of the first sample is less than that of the second sample and '0', if they are equal. The sequence of signs of the observations are as shown below:

+, +, -, +, -, +, +, +, -

Number of observations of each sample, $n = 9$

The number of plus (+) signs = 6

The number of minus (-) signs = 3

Significance level, $\alpha = 0.05$

Then, $np = n(1 - p) = 4.5$. Since np as well as $np(1 - p)$ is less than 5, binomial distribution is to be used. Also, let Q be the random variable representing the number of plus signs.

The alternative hypothesis, H_1 means that the proportion of the number of plus signs is more than 0.5. The binomial probability that the number of plus signs is more than or equal to 6 is:

$$\begin{aligned} P(Q \geq 6, n=9, p=0.5) &= 1 - P(Q \leq 5, n=9, p=0.5) \\ &= 1 - 0.7461 \\ &= 0.2539 \end{aligned}$$

Since $P(Q \geq 6)$ is more than the significance level of 0.05, the binomial statistic falls in the acceptance region. Hence, accept H_0 .

Thus the inflow of outpatients to the hospitals has not decreased considerably after introducing the health care awareness programme in the surrounding regions of the hospitals.

Two-tailed two-samples sign test with binomial distribution

The combination of hypotheses for a two-tailed test concerning two-samples sign test with binomial distribution is:

$$H_0: \mu_X = \mu_Y \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu_X \neq \mu_Y \quad \text{or} \quad p \neq \frac{1}{2}$$

Two random samples of size n where np as well as $n(1-p)$ is less than 5 are taken. Their null hypothesis is tested using binomial distribution with $p = 1/2$, in which Q is the random variable representing the number of plus signs.

The objective of this test is to check whether the null hypothesis is to be accepted based on the value of the combined samples statistic by placing half of the significance level ($\alpha/2$) at both tails of the distribution. If the computed binomial value of $P(Q \geq \text{number of plus signs})$ or $P(Q \leq \text{number of plus signs})$ is more than half of the significance level ($\alpha/2$), then accept H_0 ; otherwise, reject H_0 .

Example 9.13 The number of defects found in the hourly production of the items by randomly selected 8 different similar machines before and after installing a costly tool guiding device (jig) in them are summarized in Table 9.8. The production manager wants to check whether the number of defects in the hourly production of the items before installing the tool guiding device (jig) in the machines and that of the items after installing the costly tool guiding device (jig) in them are different at a significance level of 0.05.

Table 9.8 Hourly Production of Items by Machines

Machine	Before	After	Machine	Before	After
1	10	8	5	20	12
2	8	10	6	3	6
3	12	7	7	9	8
4	15	12	8	6	4

Solution. The combination of hypotheses for this problem situation is:

$$H_0: \mu_X = \mu_Y \text{ or } p = \frac{1}{2}$$

$$H_1: \mu_X \neq \mu_Y \text{ or } p \neq \frac{1}{2}$$

where μ_X is the mean of the first population (before installing costly jig); μ_Y is the mean of the second population (after installing costly jig).

As per the statement of the problem, half of the significance level, 0.025 ($\alpha/2$), is to be placed at both tails of the distribution.

On classifying each observation into '+' sign if the observation of the first sample is more than that of the second sample, '-' sign if the observation of the first sample is less than that of the second sample and '0', if they are equal, the sequence of signs of the observations is:

+, -, +, +, +, -, +, +

Number of observations of each sample, $n = 8$

The number of plus (+) signs = 6

The number of minus (-) signs = 2

Significance level, $\alpha = 0.05$

Then $np = n(1-p) = 4$. Since np as well as $np(1-p)$ is less than 5, binomial distribution is to be used. Let Q be the random variable representing the number of plus signs.

The binomial probability that the number of plus signs is more than or equal to 6 is as computed below:

$$\begin{aligned} P(Q \geq 6, n=8, p=0.5) &= 1 - P(Q \leq 5, n=8, p=0.5) \\ &= 1 - 0.8555 \\ &= 0.1445 \end{aligned}$$

Since $P(Q \geq 6)$ is more than half of the given significance level, 0.025 ($\alpha/2s$), the calculated binomial statistic falls in the acceptance region. Hence, accept H_0 .

Thus the number of defects found in the hourly production of the items before installing the costly tool guiding device (jig) in the machines and that of the items after installing the costly tool guiding device (jig) in them are not significantly different.

One-tailed two-samples sign test with normal approximation

The combinations of hypotheses for the one-tailed tests concerning two-samples sign-test with normal approximation are:

$$\text{Test 1: } H_0: \mu_X = \mu_Y \text{ or } p = \frac{1}{2}$$

$$H_1: \mu_X > \mu_Y \text{ or } p > \frac{1}{2}$$

$$\text{Test 2: } H_0: \mu_X = \mu_Y \text{ or } p = \frac{1}{2}$$

$$H_1: \mu_X < \mu_Y \text{ or } p < \frac{1}{2}$$

Two random samples of size n where np as well as $n(1-p)$ is at least 5, are taken. Their null hypothesis is tested using normal distribution as an approximation to binomial distribution with $p = 1/2$, in which Q is the random variable representing the number of plus signs. Under this situation, the binomial distribution can be approximated to normal distribution with the following mean and variance:

$$\mu_Q = np$$

$$\sigma_Q^2 = np(1-p)$$

$$\sigma_Q = [np(1-p)]^{1/2}$$

$$Q \sim N[np, np(1-p)]$$

The formula for the standard normal statistic is:

$$Z = \frac{Q - \mu_Q}{\sigma_Q} = \frac{Q - np}{\sqrt{np(1-p)}}$$

The objective of the first test is to check whether the null hypothesis is to be accepted based on the value of the combined samples statistic by placing the significance level (α) at the right tail of the normal distribution. If the computed Z statistic is more than Z_α , then reject H_0 ; otherwise, accept H_0 .

The objective of the second test is to check whether the null hypothesis is to be accepted based on the value of the combined samples statistic by placing the significance level (α) at the left tail of the distribution. If the computed Z statistic is less than $-Z_\alpha$, then reject H_0 ; otherwise, accept H_0 .

Example 9.14 The weekly sales revenues (in lakhs of rupees) of a product in 14 randomly selected retail stores before and after releasing a competing product are summarized in Table 9.9. Check whether the weekly sales revenue of the product in these retail stores has declined after releasing the competing product at a significance level of 0.05.

Table 9.9 Weekly Sales Revenues of a Product

Retail store	Before	After	Retail store	Before	After
1	8	7	8	9	8
2	9	8	9	2	4
3	6	8	10	8	7
4	9	7	11	9	8
5	4	3	12	10	7
6	8	5	13	12	9
7	5	4	14	7	9

Solution. The combination of hypotheses for this problem situation is:

$$H_0: \mu_X = \mu_Y \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu_X > \mu_Y \quad \text{or} \quad p > \frac{1}{2}$$

where μ_X is the mean of the first population (before releasing competing product); μ_Y is the mean of the second population (after releasing competing product).

As per the statement of the problem, the significance level, 0.05 (α) is to be placed at the right tail of the distribution.

On classifying each observation into '+' sign if the observation of the first sample is more than that of the second sample, '-' sign if the observation of the first sample is less than that of the second sample and '0', if they are equal, the sequence of signs of the observations are:

+ , + , - , + , + , + , + , + , - , + , + , + , + , -

Number of observations of each sample, $n = 14$

The number of plus (+) signs = 11

The number of minus (-) signs = 3

Significance level, (α) = 0.05

Then $np = n(1-p) = 7$ which is more than 4. So, the normal distribution can be used as an approximation to the binomial distribution with np and $np(1-p)$ as its mean and variance, respectively. Let Q be the random variable representing the number of plus signs. Then

$$Q \sim N[np, np(1-p)]$$

$$\mu_Q = np = 14 \times 0.5 = 7$$

$$\sigma_Q^2 = np(1-p) = 14 \times 0.5 \times 0.5 = 3.5$$

$$\sigma_Q = 1.871$$

The formula for the standard normal statistic is:

$$Z = \frac{Q - \mu_Q}{\sigma_Q} = \frac{Q - np}{\sqrt{np(1-p)}} = \frac{11 - 7}{1.871} = 2.14$$

The value of $Z_{0.05}$ is 1.64. Since the calculated value of Z (2.14) is more than the value of Z_α (1.64), reject the null hypothesis.

Thus the weekly sales revenues of the product in different retail stores have declined after releasing its competing product.

Example 9.15 The weekly sales revenues (in lakhs of rupees) of a product in 14 randomly selected retail stores before and after introducing a price discount are summarized in Table 9.10. Check whether the weekly sales revenues of the product in these retail stores have improved after introducing the price discount at a significance level of 0.10.

Table 9.10 Weekly Sales Revenues of a Product

Retail store	Before	After	Retail store	Before	After
1	7	9	8	8	9
2	8	9	9	3	6
3	6	8	10	6	9
4	8	7	11	7	9
5	3	6	12	5	8
6	7	6	13	6	4
7	4	8	14	5	4

Solution. The combination of hypotheses for this situation is:

$$H_0: \mu_X = \mu_Y \text{ or } p = \frac{1}{2}$$

$$H_1: \mu_X < \mu_Y \text{ or } p < \frac{1}{2}$$

where μ_X is the mean of the first population (before introducing price discount); μ_Y is the mean of the second population (after introducing price discount).

As per the statement of the problem, the significance level, 0.10 (α) is to be placed at the left tail of the distribution. On classifying each observation into '+' sign if the observation of the first sample is more than that of the second sample, '-' sign if the observation of the first sample is less than that of the second sample and '0', if they are equal, the sequence of signs of the observations are as shown below:

-, -, -, +, -, +, -, -, -, -, -, -, -, +, +

Number of observations of each sample, $n = 14$

The number of plus (+) signs = 4

The number of minus, (-) signs = 10

Significance level, (α) = 0.10

Here, $np = n(1-p) = 7$ which is more than 4, So the normal distribution can be used as an approximation to the binomial distribution with np and $np(1-p)$ as its mean and variance, respectively.

Let Q be the random variable representing the number of plus signs. Then

$$Q \sim N[np, np(1-p)]$$

$$\mu_Q = np = 14 \times 0.5 = 7$$

$$\sigma_Q^2 = np(1-p) = 14 \times 0.5 \times 0.5 = 3.5$$

$$\sigma_Q = 1.871$$

The formula for the standard normal statistic is:

$$Z = \frac{Q - \mu_Q}{\sigma_Q} = \frac{Q - np}{\sqrt{np(1-p)}} = \frac{4 - 7}{1.871} = -1.60$$

The value of $-Z_{0.10}$ is -1.28 . Since the calculated value of Z (-1.60) is less than the value of $-Z_{\alpha}$ (-1.28), reject the null hypothesis.

Thus the weekly sales revenues of the product in the retail stores have improved after introducing the price discount.

Two-tailed two-samples sign test with normal approximation

The combination of hypotheses for the two-tailed test concerning two-samples sign test with normal approximation is:

$$H_0: \mu_X = \mu_Y \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu_X \neq \mu_Y \quad \text{or} \quad p \neq \frac{1}{2}$$

This is same as the one-tailed test of this category except the placement of the significance level at the end of the distribution.

The objective of this test is to check whether the null hypothesis is to be accepted based on the value of the combined samples statistic by placing half of the significance level ($\alpha/2$) at both tails of the distribution. If the computed Z statistic is in between $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$, then accept H_0 ; otherwise, reject H_0 .

Example 9.16 The performance indices of 20 randomly selected employees before and after attending a training programme are summarized in Table 9.11. Check whether the performance indices of the employees before attending the training programme are different from that of the employees after attending the training programme at a significance level of 0.10.

Table 9.11 Performance Indices of Employees

Employee	Before	After	Employee	Before	After
1	7	9	11	8	7
2	5	4	12	7	9
3	7	8	13	2	6
4	4	5	14	8	6
5	8	9	15	5	7
6	9	7	16	3	2
7	4	7	17	4	6
8	1	7	18	7	9
9	6	9	19	7	9
10	9	8	20	8	6

Solution. The combination of hypotheses for the given situation is:

$$H_0: \mu_X = \mu_Y \quad \text{or} \quad p = \frac{1}{2}$$

$$H_1: \mu_X \neq \mu_Y \quad \text{or} \quad p \neq \frac{1}{2}$$

where μ_X is the mean of the first population (before attending training programme); μ_Y is the mean of the second population (after attending training programme).

As per the statement of the problem, half of the significance level, 0.05 ($\alpha/2$) is to be placed at both tails of the distribution.

On classifying each observation into '+' sign if the observation of the first sample is more than that of the second sample, '-' sign if the observation of the first sample is less than that of the second sample and '0', if they are equal, the sequence of signs of the observations is:

-, +, -, -, -, +, -, -, -, +,
+, -, -, +, -, +, -, -, -, +

Number of observations of each sample, $n = 20$

The number of plus (+) signs = 7

The number of minus (-) signs = 13

Significance level (α) = 0.10

Here $\alpha/2 = 0.05$, and $np = n(1-p) = 10$ which is more than 4. So, the normal distribution can be used as an approximation to the binomial distribution with np and $np(1-p)$ as its mean and variance, respectively.

Let Q be the random variable representing the number of plus signs. Then

$$Q \sim N[np, np(1-p)]$$

$$\mu_Q = np = 20 \times 0.5 = 10$$

$$\sigma_Q^2 = np(1-p) = 20 \times 0.5 \times 0.5 = 5$$

$$\sigma_Q = 2.24$$

The formula for the standard normal statistic is:

$$Z = \frac{Q - \mu_Q}{\sigma_Q} = \frac{Q - np}{\sqrt{np(1-p)}} = \frac{7 - 10}{2.24} = -1.34$$

The value of $-Z_{\alpha/2}$ and $+Z_{\alpha/2}$ are -1.64 and $+1.64$, respectively. Since the calculated value of Z (-1.34) is in between $-Z_{\alpha/2}$ (-1.64) and $+Z_{\alpha/2}$ ($+1.64$), accept the null hypothesis.

Thus the performance indices of the employees before attending the training programme are not different from that of the employees after attending the training programme.

9.3.2 Two-samples Median Test

The different sign tests for two samples which are presented in the previous section are applicable if the two samples are having the same size. These sign tests cannot be applied when the two independent samples with different sizes are taken from two populations. Under this situation, one can use median test.

The objective of the median test is to check whether the two independent samples are from two different populations with the same median. The combination of hypotheses for the two samples median test is:

H_0 : The two independent samples are from the populations with the same median.

H_1 : The two independent samples are not from the populations with the same median.

Procedure for median test

Step-1: Pool the observations of the two samples.

Step-2: Find the median of the combined observations.

Step-3: Form frequency table for the pooled observations, as shown in Table 9.12. Let n_1 be the size of the first sample, n_2 be the size of the second sample; N be the size of pooled observations ($n_1 + n_2$); and μ be the median of the pooled observations.

Table 9.12 Frequency of Pooled Observations

	Sample-1	Sample-2	Row total
Above Median	p	q	$p + q$
Below Media	r	s	$r + s$
Column total	$p + r$	$q + s$	$N = n_1 + n_2$

Step-4: If the pooled sample is small go to step-5; otherwise, go to step-6.

Step-5: 5.1: Compute the probability which is to be compared with the given significance level using the following formula:

$$P = \frac{{}^{n_1}C_p {}^{n_2}C_q}{{}^{n_1+n_2}C_{p+q}}$$

5.2: If the calculated value of P as shown in the step 5.1, is more than the given significance level of α , accept the null hypothesis; otherwise, reject the null hypothesis. Go to step-7.

Step-6: 6.1: Compute chi-square statistic using the following formula:

$$\chi^2 = \frac{N [|ps - qr| - (N/2)]^2}{(p+q)(r+s)(p+r)(q+s)}$$

6.2: If the calculated chi-square statistic is less than the table chi-square value at 1 degree of freedom and the given significance level of α , accept the null hypothesis; otherwise, reject the null hypothesis.

Step-7: End.

Example 9.17 (Small samples) In a computer company, the incremental monthly sales revenues (in lakhs of rupees) in randomly selected sales regions against two different pure selling strategies (increased warranty period, increased price discount) are summarized in Table 9.13. Check whether two samples are from the populations with the same median at a significance level of 0.05.

Table 9.13 Sales Data

Sales region	Increased warranty period	Increased price discount
1	20	23
2	25	27
3	41	30
4	35	42
5	60	56
6	40	48
7	28	-

Solution. The combination of hypotheses for the two-samples median test is:

H_0 : The incremental monthly sales revenues due to the two selling strategies are with the same median.

H_1 : The incremental monthly sales revenues due to the two selling strategies are not with the same median.

Size of the first sample, n_1 = 7
 Size of the second sample, n_2 = 6
 Size of the pooled observations, $(n_1 + n_2)$ = 13
 Level of significance, α = 0.05

The pooled observations in the increasing order are:

20, 23, 25, 27, 28, 30, 35, 40, 41, 42, 48, 56, 60



Median

The median (μ) of the pooled observations, (μ) = 35. The classification of the pooled observations is as shown in Table 9.14.

Table 9.14 Frequency of Pooled Observations

	Sample-1	Sample-2	Row total
Above median	$p = 3$	$q = 3$	$p + q = 6$
Below median	$r = 4$	$s = 3$	$r + s = 7$
Column total	$p + r = 7$	$q + s = 6$	$N = n_1 + n_2 = 13$

$$P = \frac{{}^7C_3 {}^6C_3}{{}^{7+6}C_{3+3}} = 0.408$$

The above calculated value of $P(0.408)$ is more than the given significance level of 0.05. Hence, accept the null hypothesis.

Thus the incremental monthly sales revenues by the two selling strategies are with the same median.

Example 9.18 (Large samples) In a chemical company, the incremental weekly production in tons using two different catalysts are summarized in Table 9.15. Check whether two samples are from the populations with the same median at a significance level of 0.05.

Table 9.15 Production Data

Week	Catalyst X	Catalyst Y	Week	Catalyst X	Catalyst Y
1	25	28	9	41	54
2	30	32	10	51	57
3	46	42	11	64	63
4	40	47	12	66	70
5	65	62	13	77	80
6	45	53	14	82	78
7	33	29	15	96	93
8	35	34	16	68	-

Solution. The combination of hypotheses for the two-samples median test is:

H_0 : The incremental weekly productions of the chemical using the two catalysts are with the same median.

H_1 : The incremental weekly productions of the chemical using the two catalysts are not with the same median.

Size of the first sample, n_1 = 16

Size of the second sample, n_2 = 15

Size of the pooled observations, $n_1 + n_2$ = 31

Level of significance, α = 0.05

The pooled observations in the increasing order are:

25, 28, 29, 30, 32, 33, 34, 35, 40, 41, 42, 45, 46,
47, 51, 53*, 54, 57, 62, 63, 64, 65, 66, 68, 70, 77
78, 80, 82, 93, 96

The median (μ) of the pooled observations, (μ) = 53. The classification of the pooled observations is as shown in Table 9.16.

Table 9.16 Frequency of Pooled Observations

	Sample-1	Sample-2	Row total
Above median	$p = 7$	$q = 8$	$p + q = 15$
Below median	$r = 9$	$s = 7$	$r + s = 16$
Column total	$p + r = 16$	$q + s = 15$	$N = n_1 + n_2 = 31$

Since, $n_1 + n_2$ is large, the chi-square statistic is to be used for this problem:

$$\begin{aligned}\chi^2 &= \frac{N[|ps - qr| - N/2]^2}{(p+q)(r+s)(p+r)(q+s)} \\ &= \frac{31[|7 \times 7 - 8 \times 9| - 31/2]^2}{15 \times 16 \times 16 \times 15} \\ &= 0.0303\end{aligned}$$

The chi-square value for 1 degree of freedom and ($\alpha = 0.05$) = 3.841. The calculated chi-square value (0.0303) is less than the table chi-square value (3.841). Hence, accept the null hypothesis.

Thus the incremental weekly productions of the chemical using the two catalysts are with the same median.

9.3.3 Mann-Whitney U Test

Mann-Whitney U test is an alternate to the two sample t -test. This test is based on the ranks of the observations of two samples put together.¹ This test is more powerful, when compared to the Sign tests which were presented earlier. The alternate name for this test is Rank-Sum test.

Consider two independent samples taken from two different populations. Let n_1 be the size of the first sample; n_2 be the size of the second sample; and $N = n_1 + n_2$.

In this test, the observations of the two samples are pooled together and then their positional ranks are obtained from lowest (1) to highest (N) from left to right, respectively. If the same rank is assigned to more than one observation, then the average of the positions

1. For small samples, one may refer to the book by Johnson and Bhattacharyya, 1996.

of those observations is assigned as the rank for the observations having that equal rank.

Let R_1 be the sum of the ranks of the observations of the first sample; R_2 be the sum of the ranks of the observations of the second sample.

The objective of this test is to check whether the two samples are drawn from different populations having the same distribution.

If the size of the second sample (n_2) is more than 8, one can have normal approximation to the given data with the following random variable:

$$U = n_1 n_2 + \frac{n_1 (n_1 + 1)}{2} - R_1$$

$$U \sim N(\mu_U, \sigma_U^2)$$

where μ_U is the mean, σ_U^2 is the variance. The formulas for mean (μ_U) and variance (σ_U^2) of U are:

$$\mu_U = \frac{n_1 n_2}{2}$$

$$\sigma_U^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

$$Z_U = \frac{U - \mu_U}{\sigma_U}$$

The null hypothesis of this test is:

H_0 : Two samples are drawn from different populations having the same distribution.

A suitable alternate hypothesis (H_1) can be assumed as per the reality and it may be one of the following:

Case 1: Population 1 is stochastically larger than population 2 (greater than type). Under this case, if the calculated Z_U is less than $-Z_{\alpha}$, then reject H_0 ; otherwise accept H_0 .

Case 2: Population 1 is stochastically lesser than population 2 (smaller than type). Under this case, if the calculated Z_U is more than Z_{α} , then reject H_0 ; otherwise accept H_0 .

Case 3: Population 1 stochastically differs from population 2 (not equal to type). Under this case, if the calculated Z_U is less than $-Z_{\alpha/2}$ or more than $+Z_{\alpha/2}$, then reject H_0 ; otherwise accept H_0 .

In this section, Cases 1 and 3 are demonstrated with examples.

Example 9.19 In a computer company, the performance indices of a randomly selected sample of programmers of each of its two branches located in different cities are summarized in Table 9.17. Check whether the two samples are drawn from identical populations against the alternate hypothesis that the first population is stochastically larger than the second population at a significance level of 0.05.

Table 9.17 Performance Indices of Employees

Branch-1	Branch-2	Branch-1	Branch-2
87	78	91	83
76	55	68	97
57	92	73	53
50	71	79	89
43	25	59	74
73	62	50	30
40	45	35	54
90	77	82	32
75	34	73	-
85	50	66	-

Solution. The number of programmers from the Branch-1 $n_1 = 20$. The number of programmers from the Branch-2, $n_2 = 18$. Significance level, $\alpha = 0.05$. The combination of hypotheses for the problem situation is:

H_0 : Two samples are drawn from different populations having the same distribution.

H_1 : The mean performance ratings of the programmers from the population of the Branch-1 is stochastically larger than that of the programmers from the population of the Branch-2.

Rule for decision making: Under this case, if the calculated Z_U is more than Z_α , then reject H_0 ; otherwise accept H_0 .

The ranks of the observations pooled together are as shown in Table 9.18.

Table 9.18 Ranks of Pooled Observations

Sample no.	Index	Rank	Revised rank	Sample no.	Index	Rank	Revised rank
2	25	1		2	71	20	
2	30	2		1	73	21	22
2	32	3		1	73	22	22
2	34	4		1	73	23	22
1	35	5		2	74	24	
1	40	6		1	75	25	
1	43	7		1	76	26	
2	45	8		2	77	27	
1	50	9	10	2	78	28	
1	50	10	10	1	79	29	
2	50	11	10	1	82	30	
2	53	12		2	83	31	
2	54	13		1	85	32	
2	55	14		1	87	33	

contd.

Table 9.18 (Contd.)

Sample no.	Index	Rank	Revised rank	Sample no.	Index	Rank	Revised rank
1	57	15		2	89	34	
1	59	16		1	90	35	
2	62	17		1	91	36	
1	66	18		2	92	37	
1	68	19		2	97	38	

The observation 50 as well as 73 is repeated three times. The revised ranks of the observation 50 is the average of its original ranks which is equal to 10 [i.e. $(9 + 10 + 11)/3 = 10$]. Similarly, the revised ranks of the observation 73 is the average of its original ranks which is equal to 22 [i.e. $(21 + 22 + 23)/3 = 22$].

Ranks of Sample-1 = {5, 6, 7, 10, 10, 15, 16, 18, 19, 22, 22, 22, 25, 26, 29, 30, 32, 33, 35, 36}

Ranks of Sample-2 = {1, 2, 3, 4, 8, 10, 12, 13, 14, 17, 20, 24, 27, 28, 31, 34, 37, 38}

The sum of the ranks of the Sample-1, $R_1 = 418$

The sum of the ranks of the Sample-2, $R_2 = 323$

Now

$$U \sim N(\mu_U, \sigma_U^2)$$

where μ_U is the mean, σ_U^2 is the variance. Then

$$U = n_1 n_2 + \frac{n_1 (n_1 + 1)}{2} - R_1 = 20 \times 18 + \frac{20 \times (20 + 1)}{2} - 418 = 152$$

$$\mu_U = \frac{n_1 n_2}{2} = \frac{20 \times 18}{2} = 180$$

$$\sigma_U^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12} = \frac{20 \times 18 \times (20 + 18 + 1)}{12} = 1170$$

$$\sigma_U = 34.21$$

$$Z_U = \frac{U - \mu_U}{\sigma_U} = \frac{152 - 180}{34.21} = -0.819$$

The standard normal value for the given significance level of α (0.05) placed at the right tail is 1.64. Since the computed Z_U (-0.819) is less than Z_α (1.64), accept the null hypothesis.

Thus two samples are drawn from populations having identical distributions.

Example 9.20 In a company, the weekly capacity utilization (in %) of two different machines during randomly selected weeks are summarized in Table 9.19. Check whether

the two samples are drawn from identical populations against the alternate hypothesis that the populations are not identical at a significance level of 0.10.

Table 9.19 Capacity Utilization of Machines (%)

Week	Machine-1	Machine-2	Week	Machine-1	Machine-2
1	94	78	9	62	97
2	77	95	10	68	84
3	64	54	11	83	53
4	88	71	12	73	61
5	65	91	13	90	98
6	55	56	14	60	87
7	75	82	15	63	-
8	93	92			

Solution. The number of weeks of observations from the Machine-1, $n_1 = 15$. The number of weeks of observations from the Machine-2, $n_2 = 14$. Significance level $\alpha = 0.10$. Half of the significance level = $\alpha/2 = 0.05$. The combination of hypotheses for the problem situation is:

H_0 : Two samples are drawn from identical populations.

H_1 : Two samples are not drawn from identical populations.

Rule for decision making: Under this case, if the calculated Z_U is less than $-Z_{\alpha/2}$ or more than $+Z_{\alpha/2}$, then reject H_0 ; otherwise accept H_0 .

The ranks of the observations pooled together are as shown in Table 9.20.

Table 9.20 Ranks of Pooled Observations

Sample no.	Observation	Rank	Sample no.	Observation	Rank
2	53	1	2	78	16
2	54	2	2	82	17
1	55	3	1	83	18
2	56	4	2	84	19
1	60	5	2	87	20
1	62	6	1	88	21
2	61	7	1	90	22
1	63	8	2	91	23
1	64	9	2	92	24
1	65	10	1	93	25
1	68	11	1	94	26
2	71	12	2	95	27
1	73	13	2	97	28
1	75	14	2	98	29
1	77	15	-	-	-

Ranks of Sample-1 = {3, 5, 6, 8, 9, 10, 11, 13, 14, 15, 18, 21, 22, 25, 26}

Ranks of Sample-2 = {1, 2, 4, 7, 12, 16, 17, 19, 20, 23, 24, 27, 28, 29}

The sum of the ranks of the Sample-1 $R_1 = 206$

The sum of the ranks of the Sample-2 $R_2 = 229$

Now,

$$U \sim N(\mu_U, \sigma_U^2)$$

where μ_U is the mean, σ_U^2 is the variance. Then

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 = 15 \times 14 + \frac{15 \times (15 + 1)}{2} - 206 = 124$$

$$\mu_U = \frac{n_1 n_2}{2} = \frac{15 \times 14}{2} = 105$$

$$\sigma_U^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12} = \frac{15 \times 14 \times (15 + 14 + 1)}{12} = 525$$

$$\sigma_U = 22.913$$

$$Z_U = \frac{U - \mu_U}{\sigma_U} = \frac{124 - 105}{22.913} = 0.83$$

The standard normal value for half of the significance level ($\alpha/2 = 0.05$) is 1.64. Hence, $-Z_{\alpha/2} = -1.64$ and $+Z_{\alpha/2} = +1.64$.

Since the computed Z_U (0.83) is in between $-Z_{\alpha/2}$ (-1.64) and $Z_{\alpha/2}$ (1.64), accept the null hypothesis.

Thus two samples are drawn from identical populations.

9.4 K-SAMPLES TESTS

There may be situations in which simultaneously more than two samples (K samples) are to be checked whether they are from identical populations. The following tests can be used for testing such K samples. K -samples median test, Kruskal-Wallis test.

9.4.1 K-samples Median Test

As discussed in Section 9.3.2, K -samples median test can be classified into two types.

- K -samples median test for small samples
- K -samples median test for large samples

The objective of K -sample median test is to check whether the K -independent samples are from K different populations with the same median.

The combination of hypotheses for the K -samples median test is as given below:

H_0 : K independent samples are from the populations with the same median.

H_1 : K independent samples are not from the populations with the same median.

Procedure for K -samples median test

Step-1: Pool the observations of K samples.

Step-2: Find the median of the combined observations.

Step-3: Form frequency table for the pooled observations, as shown in Table 9.21. Let K be the number of samples; n_j be the size of the sample j , $j = 1, 2, 3, \dots, K$; N be the size of pooled observations ($n_1 + n_2 + \dots + n_k$); and μ be the median of the pooled observations.

Table 9.21 Frequency of Pooled Observations

	Sample					
	1	2	3	...	K	Row total
Above median	p_1	p_2	p_3	...	p_K	$p_1 + p_2 + \dots + p_K$
Below median	q_1	q_1	q_1	...	q_K	$q_1 + q_2 + \dots + q_K$
Column total	n_1	n_2	n_3	...	n_K	$N = n_1 + n_2 + \dots + n_K$

Step-4: If the pooled sample is small go to step-5; if it is large, go to step-6.

Step-5: 5.1: Compute the probability which is to be compared with the given significance level using the following formula:

$$P = \frac{{}^{n_1}C_{p_1} {}^{n_2}C_{p_2} \dots {}^{n_k}C_{p_k}}{{}^{n_1+n_2+\dots+n_K}C_{p_1+p_2+\dots+p_k}}$$

5.2: If the calculated value of P , as shown in the step-5.1, is more than the given significance level of α , accept the null hypothesis; otherwise, reject the null hypothesis. Go to step-7.

Step-6: 6.1: Compute chi-square statistic.

6.2: If the calculated chi-square statistic is less than the table chi-square value at $(K-1)$ degrees of freedom and the given significance level of α , accept the null hypothesis; otherwise, reject the null hypothesis.

Step-7: End

Example 9.21 In a company, the monthly percentage absenteeism of employees in the first quarter of the last five years are summarized in Table 9.22. Check whether there is significant difference between the monthly percentage absenteeism of the employees using median test at a significance level of 0.05.

Table 9.22 Percentage Absenteeism of Employees

January	February	March
10	18	13
15	25	21
8	16	26
20	17	14
19	11	7

Solution. The combination of hypotheses for the three samples median test is as given below:

H_0 : There is no significant difference between the percentage absenteeism of the employees in different months.

H_1 : There is significant difference between the percentage absenteeism of the employees in different months.

Size of the first sample, n_1	= 5
Size of the second sample, n_2	= 5
Size of the third sample, n_3	= 5
Size of the pooled observations, $(n_1 + n_2 + n_3)$	= 15
Level of significance, α	= 0.05

The pooled observations in the increasing order are as shown below:

7, 8, 10, 11, 13, 14, 15, 16, 17, 18, 19, 20, 21, 25, 26

↑

Median = 16

The median of the pooled observations (μ) = 16. The classification of the pooled observations is as shown in Table 9.23.

Table 9.23 Frequency of Pooled Observations

	Sample			
	1	2	3	Row total
Above median	$p_1 = 2$	$p_2 = 3$	$p_3 = 2$	7
Below median	$q_1 = 3$	$q_2 = 2$	$q_3 = 3$	8
Column total	5	5	5	$N = n_1 + n_2 + n_3 = 15$

$$P = \frac{{}^5C_2 {}^5C_3 \dots {}^5C_2}{{}^{15}C_7} = 0.155$$

The above calculated value of $P(0.155)$ is more than the given significance level of 0.05. Hence, accept the null hypothesis.

Thus there is no significant difference between the absenteeism percentages of the employees in different months.

Example 9.22 In a survey, the age of respondents in three different regions are summarized in Table 9.24. Check whether there is significant difference between the ages of the respondents in three different regions using median test at a significance level of 0.05.

Table 9.24 Age of Respondents

Region-1	Region-2	Region-3
35	45	38
42	55	36
28	32	43
31	47	30
44	53	60
50	46	39
33	41	54
52	34	57
56	49	27

Solution. The combination of hypotheses for the three samples median test is as given below:

H_0 : There is no significant difference between the ages of respondents in three regions.

H_1 : There is significant difference between the ages of respondents in three regions.

Size of the first sample, n_1 = 9

Size of the second sample, n_2 = 9

Size of the third sample, n_3 = 9

Size of the pooled observations, $(n_1 + n_2 + n_3)$ = 27

Level of significance, α = 0.05

The pooled observations in the increasing order are as shown below:

27, 28, 30, 31, 32, 33, 34, 35, 36, 38, 39, 41, 42, 43*, 44, 45,
46, 47, 49, 50, 52, 53, 54, 55, 56, 57, 60

Here, median of the pooled observations, $(\mu) = 43$. The classification of the pooled observations is as shown in Table 9.25.

Table 9.25 Frequency of Pooled Observations

	Sample			Row total
	1	2	3	
Above median	$p_1 = 4$	$p_2 = 6$	$p_3 = 3$	13
Below median	$q_1 = 5$	$q_2 = 3$	$q_3 = 6$	14
Column total	9	9	9	$N = n_1 + n_2 + n_3 = 27$

Since $(n_1 + n_2 + n_3)$ is large, the chi-square statistic is to be used for this problem. The chi-square statistic is computed as shown below:

$$\chi^2 = \sum_{i=1}^2 \sum_{j=1}^3 \frac{(o_{ij} - e_{ij})^2}{e_{ij}} = 2.0769$$

where o_{ij} is the observed frequency for the i th row and j th column; e_{ij} is the expected frequency for the i th row and the j th column; and

$$e_{ij} = \frac{\text{Row total} \times \text{Column total}}{N}$$

Chi-square value for 2 degrees of freedom and $\alpha = (0.05)$ is 5.991. The calculated chi-square value (2.0769) is less than the table chi-square value (5.991). Hence, accept the null hypothesis.

Thus there is no significant difference between the ages of respondents in three regions.

9.4.2 Kruskal-Wallis Test (H Test)

Kruskal-Wallis test is similar to the K samples median test in which the objective is to check whether the K independent samples are from K identical populations. This is called as H test. This is a nonparametric test which is an alternative to single factor (one way) ANOVA which is already presented in Chapter 4.

The combination of hypotheses for the H test is as given below:

H_0 : K independent samples are from K identical populations.

H_1 : K independent samples are not from K identical populations.

As in Mann-Whitney U test, here too, the observations of all the samples are pooled together and then their ranks are established. The smallest value of the observation will have the rank 1 and the largest value of the observation will have rank N . If there is tie, then the ranks of the observations which are having tie are to be modified by the corresponding average rank as explained in Mann-Whitney test.

H follows chi-square distribution with $(K-1)$ degrees of freedom.

The formula for H statistic of Kruskal-Wallis test is as given below, where K is the number of samples.

$$H = \frac{12}{N(N+1)} \sum_{j=1}^K \frac{R_j^2}{n_j} - 3(N+1)$$

where R_j is the sum of the ranks of the sample j ; n_j is the size of the sample j , $j = 1, 2, 3, \dots, K$; and N is the size of the pooled sample ($n_1 + n_2 + \dots + n_K$).

The calculated H value is to be compared against the chi-square value with $(K-1)$ degrees of freedom at the given significance level of α .

Rule for decision making: If the value of H is more than the table chi-square value, then reject the null hypothesis; otherwise, accept the null hypothesis.

Example 9.23 The production volume of units assembled by three different operators (1, 2 and 3) during 9 shifts are summarized in Table 9.26. Check whether there is significant difference between the production volumes of units assembled by the three operators using Kruskal-Wallis test at a significance level of 0.05.

Table 9.26 Production Volume of Units Assembled.

Shift no.	Operator-1	Operator-2	Operator-3
1	29	30	26
2	34	21	36
3	34	23	41
4	20	25	48
5	32	44	27
6	45	37	39
7	42	34	28
8	24	19	46
9	35	38	15

Solution. The number of shifts worked by Operator-1, $n_1 = 9$; the number of shifts worked by Operator-2, $n_2 = 9$; the number of shifts worked by Operator-3, $n_3 = 9$; significance level $\alpha = 0.05$.

$$N = n_1 + n_2 + n_3 = 27$$

The combination of hypotheses for the problem situation is as given below:

H_0 : There is no significant difference between the production volumes of units assembled by the three operators.

H_1 : There is significant difference between the production volumes of units assembled by the three operators.

Rule for decision making: Under this case, if the calculated H value is more than the table chi-square value with $(K-1)$ degrees of freedom at the given significance level of 0.05, then reject H_0 ; otherwise accept H_0 .

The ranks of the observations pooled together are as shown in Table 9.27.

Table 9.27 Ranks of Pooled Observations

Sample no.	Observation	Rank	Sample no.	Observation	Rank	Revised rank
3	15	1	1	34	14	15
2	19	2	1	34	15	
1	20	3	2	34	16	
2	21	4	1	35	17	
2	23	5	3	36	18	
1	24	6	2	37	19	
2	25	7	2	38	20	
3	26	8	3	39	21	
3	27	9	3	41	22	
3	28	10	1	42	23	
1	29	11	2	44	24	
2	30	12	1	45	25	
1	32	13	3	46	26	
			3	48	27	

The observation 34 is repeated three times. The revised ranks of the observation (34) is the average of the original ranks with respect to the observation 34 which is equal to 15 [i.e. $(14+15+16)/3 = 15$].

Ranks of Sample-1 = {3, 6, 11, 13, 15, 15, 17, 23, 25}

Ranks of Sample-2 = {2, 4, 5, 7, 12, 15, 19, 20, 24}

Ranks of Sample-3 = {1, 8, 9, 10, 18, 21, 22, 26, 27}

The sum of the ranks of the Sample-1, $R_1 = 128$.

The sum of the ranks of the Sample-2, $R_2 = 108$

The sum of the ranks of the Sample-3, $R_3 = 142$

Here, $H \sim$ chi-square distribution with $(K-1)$ degrees of freedom, where K is the total number of samples. Then

$$\begin{aligned}
 H &= \frac{12}{N(N+1)} \sum_{j=1}^3 \frac{R_j^2}{n_j} - 3(N+1) \\
 &= \frac{12}{27(27+1)} \left(\frac{128^2}{9} + \frac{108^2}{9} + \frac{142^2}{9} \right) - 3 \times (27+1) \\
 &= 1.03
 \end{aligned}$$

The table chi-square value with 2 degrees of freedom at the given significance level of 0.05 is 5.991.

Since the value of H (1.03) is less than the table chi-square value (5.991), accept the null hypothesis.

Thus there is no significant difference between the production volumes of units assembled by the three operators.

QUESTIONS

1. Give a classification of nonparametric tests.
2. What are the advantages and limitations of parametric tests?
3. Give brief account of one-sample tests.
4. List and explain different two-samples tests.
5. What are the tests for testing K samples? Explain them in brief.
6. A time-study engineer observed the final assembly operation of 8 randomly selected units in a product line and found that the assembly time for that operation (in minutes) are 20, 34, 28, 15, 27, 15, 26 and 12. Check whether the operation times is 20 minutes ($H_0: \mu = 20$) as against the alternate hypothesis $H_1: \mu > 20$ using sign test at a significance level of 0.10.
7. At a toll gate, 9 randomly selected interarrival times (in seconds) are found to be 40, 90, 120, 300, 250, 130, 60, 90 and 100. Check whether the interarrival time is 130 seconds ($H_0: \mu = 130$) as against the alternate hypothesis $H_1: \mu > 130$ using sign test at a significance level of 0.01.
8. The performance index (in percentage) of 8 randomly selected operators in a machine shop are 80, 65, 82, 25, 45, 89, 70 and 48. Check whether the performance index is 75% ($H_0: \mu = 75$) as against the alternate hypothesis $H_1: \mu < 75$ using sign test at a significance level of 0.10.
9. The age (in percentage) of 8 randomly selected respondents in a survey are 50, 30, 70, 65, 25, 48, 65 and 50. Check whether the age of respondents is 45 years ($H_0: \mu = 45$) as against the alternate hypothesis $H_1: \mu < 45$ using sign test at a significance level of 0.05.
10. The annual income (in lakhs) of rupees of 8 randomly selected general managers of factories are 4, 3, 6, 3, 6, 2, 8 and 5.5. Check whether the annual income of the general managers is 5 lakhs ($H_0: \mu = 5$) as against the alternate hypothesis $H_1: \mu \neq 5$ using sign test at a significance level of 0.05.
11. The yield (in tons) of a chemical process on 9 different days are 12, 15, 8, 16, 11, 20, 14, 9 and 10. Check whether the yield is 15 tons ($H_0: \mu = 15$) as against the alternate hypothesis $H_1: \mu \neq 15$ using sign test at a significance level of 0.10.

12. The volume of syrup in capsule form produced in Beta Pharmaceutical Company follows non-normal distribution. The specified volume of the syrup in the capsule of this population is 10 ml. Any excess volume of the syrup will lead to side reactions. The quality engineer of the firm claims that the volume of the syrup in the capsule is not greater than the specified mean volume of the population. So, the purchase manager of Alpha Hospital who places order for that capsule with the Beta Pharmaceutical Company has selected a random sample of 32 capsules and their volumes are as given below:

9	8	10	7	11	12	9	9.5
8.5	9.9	10	9	6	7	8	8
9	11	10	9	8	7	9	8
7	9.5	11	13	14	11	8	8

Verify the intuition of the quality manager of Beta Pharmaceutical Company that the volume of the syrup in the capsule is 10 ml ($H_0: \mu = 10$), as against the alternate hypothesis $H_1: \mu > 10$ using sign test at a significance level of 0.10.

13. The scrap (in kg) produced in a machine shop on 36 randomly selected days are as shown below:

99	80	120	130	120	90	110	120	95
80	88	65	99	110	115	125	105	85
110	99	80	77	95	88	78	95	89
83	94	85	88	85	79	89	9	77

Verify, whether the weight of the scrap produced in the machine shop is 100 kg ($H_0: \mu = 100$), against the alternate hypothesis $H_1: \mu > 100$ using sign test at a significance level of 0.05.

14. The performance of students in a competitive examination are as given below. The performance indices (0–10 scale) of 40 students follow non-normal distribution. Verify whether the performance index of students is 7 ($H_0: \mu = 7$), against the alternate hypothesis $H_1: \mu > 7$ using sign test at a significance level of 0.05.

4	5	8	3	1	5	4	6	6	8
5	8	4	1	3	4	7	5	9	6
6	9	7	6	8	2	5	6	6	8
1	8	8	7	4	3	6	8	7	9

15. The yield (in kg) of a particular variety of food grains follows non-normal distribution. The researcher feels that the yield of the food grains is not different from

750 kg. To test his intuition, he has selected a random sample of 36 plots and their yields are as shown below:

670	700	760	850	600	660	800	850	770
790	800	890	820	740	820	980	820	710
810	850	780	710	890	830	820	890	860
720	870	890	780	750	740	730	890	880

Check the intuition of the researcher using sign test at a significance level of 0.10.

16. The annual revenue (in lakhs of rupees) of small scale industries in a state follows non-normal distribution. The researcher feels that the annual revenue of the small scale industries is not different from Rs. 150 lakhs. To test his intuition, he has selected a random sample of 32 small scale industries and their annual revenues are as shown below:

170	100	160	150	100	160	100	150
190	100	190	120	140	120	180	120
110	150	180	110	190	130	120	190
120	170	190	180	150	140	130	190

Check the intuition of the researcher using sign test at a significance level of 0.05.

17. The observed frequencies of age of randomly selected 200 employees in an industrial estate are summarized below:

Age (years)	Observed frequency
25-30	10
30-35	30
35-40	20
40-45	40
45-50	30
50-55	40
55-60	30

Check whether the given data follow uniform distribution using Kolmogorov-Smirnov test at a significance level of 0.10.

18. The arrival rate of customers (no. of customers per hour) at a leading retail shop appears to follow Poisson distribution. The observed frequencies of customers are as summarized below:

Arrival rate (X_i)	Observed frequency (o_i)
0	6
1	10
2	20
3	34
4	15
5	5

Check whether the given data follow Poisson distribution using the Kolmogorov-Smirnov test at a significance level of 0.10.

19. The daily volume of a product produced by a CNC machine appears to follow normal distribution. The observed frequencies of the daily production are as summarized below:

Production volume (gm)	No. of days
100–150	20
150–200	25
200–250	40
250–300	30
300–350	15

Check whether the given data follow normal distribution using the Kolmogorov-Smirnov test at a significance level of 0.05.

20. The weight of welding rod produced in a factory appears to follow normal distribution. The observed frequencies of the weight of the welding rods are shown below:

Weight (gm)	No. of welding rods
100–102	30
102–104	35
104–106	50
106–108	40
108–110	25

Check whether the given data follow normal distribution using the Kolmogorov-Smirnov test at a significance level of 0.10.

21. The gender of arrival of applicants at a passport office is summarized below in terms of M (male) and F (female). Verify whether the gender of the arriving applicants is random at a significance level of 0.05.

M M M F F M M F F F M M M F F M M M M

22. The items inspected at a final inspection station of a production line are classified into good (G) and bad (B) as shown below:

G G B B G G G B G G G G B G G G G B

Verify whether the results of the inspection is random at a significance level of 0.01.

23. The sequence of telephone calls (STD/ISD) made at a telephone booth is as summarized below:

SSSS II SS IIII SSSS II SSS IIII SS III SSSSS I

Check whether the events of STD (S) or ISD (I) is random at a significance level of 0.10.

24. The monthly income of respondents in a survey is classified into high income (H)/low income (M). The sequence of such codes of 40 respondents is as shown below:

HHH MMMM HHHH MMMMMM HHHHH MMMMMMM HHHHHH MMM HH

Check whether the events of H/M is random at a significance level of 0.05.

25. The monthly number of accidents before and after introducing a new traffic regulation in 9 different roads are as summarized below:

Road	Before	After	Road	Before	After
1	12	10	6	16	14
2	15	13	7	9	10
3	18	20	8	11	10
4	20	17	9	17	16
5	8	6	-	-	-

Check whether the monthly number of accidents is considerably reduced using the sign test at a significance level of 0.10.

26. The daily number of rejections of items in 8 similar machines before and after introducing a quality improvement programme are as summarized below:

Machine	Before	After	Machine	Before	After
1	17	22	5	13	11
2	20	18	6	21	22
3	23	17	7	14	12
4	25	20	8	15	16

Check whether the daily number of rejections is reduced considerably using the sign test at a significance level of 0.05.

27. The number of defects found in the lots supplied by 8 randomly selected vendors before and after introducing a newly designed sampling plan is summarized as follows:

<i>Vendor</i>	<i>Before</i>	<i>After</i>	<i>Vendor</i>	<i>Before</i>	<i>After</i>
1	20	18	5	25	22
2	15	12	6	30	20
3	17	18	7	19	15
4	18	17	8	16	18

Check whether the number of defects in the lots supplied by the vendors before and after introducing the newly designed sampling plan is different, using the sign at a significance level of 0.01.

28. The number of vehicles produced per week in seven plants of Alpha Auto Company before and after introducing modern computer package for line balancing is summarized below:

<i>Week</i>	<i>Before</i>	<i>After</i>	<i>Week</i>	<i>Before</i>	<i>After</i>
1	600	625	5	575	600
2	550	600	6	630	650
3	620	575	7	575	500
4	700	720	—	—	—

Check whether the number of vehicles produced per week before and after introducing the computer package are different in the plants using sign test at a significance level of 0.05.

29. The daily consumption of water (in lakhs of gallons) in 13 different colonies before and after organizing water conservation programme in them is summarized below:

<i>Colony</i>	<i>Before</i>	<i>After</i>	<i>Colony</i>	<i>Before</i>	<i>After</i>
1	15	12	8	25	18
2	17	20	9	33	24
3	20	15	10	40	37
4	25	17	11	20	18
5	30	23	12	15	17
6	27	25	13	18	14
7	35	39	—	—	—

Check whether the consumption of water has declined after conducting the Conservation programme using the sign test at a significance level of 0.05.

30. The yearly tax paid (in thousands of rupees) by 14 randomly selected small business firms before and after introducing attractive government savings schemes is summarized as follows:

Firm	Before	After	Firm	Before	After
1	17	11	8	27	22
2	19	15	9	35	20
3	22	16	10	42	35
4	27	15	11	22	15
5	32	25	12	17	22
6	29	20	13	20	15
7	37	45	14	40	42

Check whether the tax paid by the small firms has declined after introducing attractive government savings schemes using the sign test at a significance level of 0.05.

31. The monthly revenues (in lakhs of rupees) of a product in 13 randomly selected retail stores before and after introducing point of sales advertisement is summarized below:

Retail store	Before	After	Retail store	Before	After
1	17	22	8	18	24
2	18	23	9	13	17
3	16	25	10	16	21
4	18	15	11	17	20
5	13	18	12	15	18
6	17	14	13	16	15
7	14	20	—	—	—

Check whether the monthly sales revenues of the product in these retail stores have improved after introducing the advertisement using the sign test at a significance level of 0.01.

32. The monthly production (in hundreds of tons) of cement produced in 12 randomly selected cement plants before and after introducing sales tax concession is summarized below:

Retail store	Before	After	Retail store	Before	After
1	7	9	7	12	18
2	15	18	8	16	21
3	6	5	9	10	16
4	15	19	10	11	14
5	12	20	11	13	14
6	15	16	12	15	19

Check whether the monthly production of the cement in these plants improved after introducing the tax concession using the test sign at a significance level of 0.01.

33. The monthly salary (in thousands of rupees) of 20 randomly selected part-time students before and after studying a certificate course is summarized as follows:

Student	Before	After	Student	Before	After
1	17	19	11	18	17
2	15	14	12	17	19
3	17	18	13	12	16
4	14	15	14	18	16
5	18	19	15	15	17
6	19	17	16	13	12
7	14	17	17	14	16
8	12	17	18	17	19
9	16	19	19	17	19
10	19	18	20	18	16

Check whether the monthly salaries of the students before doing the certificate course are different from that of the students after doing the certificate course using the sign test at a significance level of 0.05.

34. The percentage of employment arranged for students by 20 randomly selected management departments in a university before and after introducing a trimester system are summarized below:

Student	Before	After	Student	Before	After
1	87	79	11	88	67
2	65	54	12	67	89
3	97	98	13	42	76
4	54	65	14	38	26
5	78	89	15	75	87
6	49	37	16	63	52
7	45	67	17	94	86
8	92	99	18	87	90
9	86	80	19	27	40
10	99	88	20	28	20

Check whether the percentage of employment for students before implementing the trimester system are different from that of the students after implementing the trimester system using sign test at a significance level of 0.01.

35. In a company, the percentage incremental efficiency of different shops in a factory against two different productivity improvement strategies (increased incentive, increased welfare measure) are summarized below:

Shop	Increased incentive	Increased welfare measure
1	30	35
2	35	40
3	51	47
4	45	60
5	70	60
6	50	70
7	38	-

Check whether the two samples are from the populations with the same median at a significance level of 0.10.

36. The yield in standard bags of paddy in eight randomly selected plots against the usage of two different fertilizers (Fertilizer-A and Fertilizer-B) are summarized below:

Plot	Fertilizer-A	Fertilizer-B	Plot	Fertilizer-A	Fertilizer-B
1	15	20	5	30	40
2	20	30	6	22	27
3	25	22	7	18	15
4	15	20	8	14	16

Check whether the two samples are from the populations with the same median at a significance level of 0.01.

37. In 15 randomly selected retail shops, incremental daily sales in thousands of rupees using two different selling strategies (price discount, door delivery) are summarized below:

Retail shop	Price discount	Door delivery	Retail shop	Price discount	Door delivery
1	35	40	9	51	64
2	40	42	10	61	67
3	56	52	11	74	73
4	50	57	12	76	80
5	75	72	13	87	90
6	55	63	14	92	88
7	43	39	15	86	83
8	45	44	-	-	-

Check whether the two samples are from the populations with the same median at a significance level of 0.01.

38. In an automobile company, the quarterly sales revenues (in lakhs of rupees) of a randomly selected sample of salesmen from each of its two regions are summarized below:

Region-1	Region-2	Region-1	Region-2
57	48	61	53
46	25	38	67
27	62	43	23
20	41	49	59
13	25	29	44
43	32	20	20
10	15	15	24
60	47	52	32
45	34	43	-
55	20	36	-

Check whether the two samples are drawn from identical populations against the alternate hypothesis that the first population is stochastically larger than the second population using the Mann-Whitney U test at a significance level of 0.10.

39. In a survey, the monthly income (in thousands of rupees) of randomly selected respondents in two regions is summarized below:

Region-1	Region-2	Region-1	Region-2
17	15	22	22
16	15	28	22
7	8	23	10
20	15	29	14
13	10	19	12
33	40	14	11
10	15	12	22
20	20	11	10
15	26	9	—
17	15	16	—

Check whether the two samples are drawn from identical populations against the alternate hypothesis that the first population is stochastically larger than the second population using Mann-Whitney U test at a significance level of 0.05.

40. In a company, the daily production of units of two different operators during randomly selected days is summarized below:

Day	Operator-1	Operator-2	Day	Operator-1	Operator-2
1	74	58	9	42	77
2	57	75	10	48	64
3	44	34	11	50	40
4	68	51	12	53	41
5	45	71	13	70	78
6	35	36	14	40	67
7	55	62	15	43	—
8	73	72	—	—	—

Check whether the two samples are drawn from identical populations against the alternate hypothesis that the populations are not identical using the Mann-Whitney U test at a significance level of 0.05.

41. In a survey, the number of respondents covered by two enumerators during randomly selected days is summarized as follows:

Day	Enumerator-1	Enumerator-2	Day	Enumerator-1	Enumerator-2
1	24	30	8	13	16
2	17	20	9	22	21
3	34	15	10	28	8
4	28	22	11	27	22
5	15	31	12	15	14
6	35	24	13	13	16
7	25	12	14	17	-

Check whether the two samples are drawn from identical populations against the alternate hypothesis that the populations are not identical using the Mann-Whitney U test at a significance level of 0.01.

42. In a company, capacity utilization (in %) in each of the months in the second quarter of the last six years is summarized below:

Capacity Utilization (%)		
April	May	June
90	80	70
88	75	95
90	85	70
60	78	50
80	90	69
65	74	68

Check whether there is a significant difference between the capacity utilization (%) of different months of the quarter using the median test at a significance level of 0.05.

43. The yearly sales revenue of three competing products is shown below:

Yearly Sales Revenue (Crores of Rupees)		
Product-1	Product-2	Product-3
10	10	30
48	25	55
50	45	30
20	38	10
40	50	29
25	34	48

Check whether there is a significant difference between the yearly sales revenues of the three competing products using the median test at a significance level of 0.01.

44. The efficiency of three different algorithms applied to different problems to design assembly lines is summarized as follows:

Efficiency (%)		
Algorithm-1	Algorithm-2	Algorithm-3
75	95	78
82	85	86
68	72	63
71	87	70
84	93	90
90	76	89
73	81	84
86	59	67

Check whether there is a significant difference between the efficiencies of different algorithms using the Kruskal-Wallis test at a significance level of 0.05.

45. A stock market analyst wants to study the impact of the type of company on the quarterly averages of the earnings per share (EPS). So, he collected four quarterly averages of EPS data of four different companies during the last financial year from 'capital market', which are as summarized below:

	Company			
	C1	C2	C3	C4
Replication	12	16	25	13
	8	18	15	8
	16	10	22	20
	19	11	9	5

Check whether there is a significant difference between the quarterly averages of the earnings per share (EPS) of the four companies using the Kruskal-Wallis test at a significance level of 0.05.

46. An experiment was conducted to determine whether four specific intervals of patrolling in minutes by a quality inspector affect the number of acceptable parts produced per hour, in a machine. The corresponding data for 4 different periods with four replications are as shown below:

	Period of Inspection			
	30 min.	45 min.	60 min.	75 min.
Replication	50	30	40	35
	25	60	12	10
	40	20	30	48
	23	38	39	25

Check whether there is a significant difference between the acceptable parts produced per hour for different intervals of patrolling using the Kruskal-Wallis test at a significance level of 0.01.

47. An agricultural officer wants to study the effect of four different fertilizers on the yield (in tons) of a specific crop. The corresponding data are as shown below:

		Fertilizer			
		A	B	C	D
Plot	1	100	150	120	70
	2	80	70	110	100
	3	68	90	85	78
	4	125	138	60	124

Check whether there is a significant difference between the yields of different fertilizers using the Kruskal-Wallis test at a significance level of 0.01.

CHAPTER 10

BASIC MULTIVARIATE ANALYSIS

10.1 INTRODUCTION

Multivariate analysis is the analysis of data consisting of observations pertaining to a reality involving two or more variables. In this chapter, the following basic topics of multivariate analysis are presented:

- (a) Correlation analysis
- (b) Forecasting
- (c) Linear regression and nonlinear regression
- (d) Time series.

The advanced topics on multivariate analysis are presented in the next chapter.

Under correlation analysis, correlation coefficient for ungrouped data, correlation coefficient for grouped data, rank correlation and auto-correlation are presented. The topics on forecasting cover the concept of forecasting, types of forecasting errors, simple moving average, single exponential smoothing method and Delphi method are presented. In linear regression, simple regression, multiple regression under which regression model with two independent variables and matrix method for a generalized regression model are presented. Later, it is followed by nonlinear regression model using matrix method. At the end, the concept of time series analysis with a complete example is presented.

10.2 CORRELATION ANALYSIS

Correlation is the degree of association between two variables and it is represented in terms of a coefficient known as correlation coefficient. The range of the correlation coefficient is in between -1 and $+1$. If the correlation coefficient is negative, then the variables are inversely proportional and it is maximum when it is -1 ; if the coefficient is 0 , there is no association between the variables. If the coefficient is positive, then the variables are associated directly and it is maximum when it is $+1$.

The different types of correlation coefficient are as listed below:

- Correlation coefficient for ungrouped data (Karl Pearson's coefficient of correlation)
- Correlation coefficient for grouped data (Karl Pearson's coefficient of correlation)
- Rank correlation coefficient
- Auto-correlation coefficient.

10.2.1 Correlation Coefficient for Ungrouped Data (Karl Pearson's Coefficient of Correlation)

Consider two variables X and Y , like yearly R&D expenditures and yearly sales revenue. Here, the objective is to determine the degree of association between these variables based on a given set of ungrouped data.

The basic formula to determine the correlation coefficient (r) is:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n\sigma_X\sigma_Y}$$

where X_i is the i th observation of the variable X ; Y_i the i th observation of the variable Y ; $\bar{X}$, the mean of the observations of the variable X ; $\bar{Y}$, the mean of the observations of the variable Y ; n , the number of pairs of observations of X and Y ; σ_X , the standard deviation of the variable X ; and σ_Y is the standard deviation of the variable Y .

The above formula may be presented in the following form:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

The same may be computed using Pearson's product moment formula, as shown below:

$$r = \frac{n \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{\sqrt{n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2} \sqrt{n \sum_{i=1}^n Y_i^2 - \left(\sum_{i=1}^n Y_i \right)^2}}$$

The significance of the correlation coefficient can be tested using t statistic, whose formula is as shown as follows:

$$t_r = \frac{r}{\sqrt{(1-r^2)/(n-2)}}$$

with $(n-2)$ degrees of freedom, where r is the correlation coefficient, and n is the number of pairs of observations.

The combination of hypotheses to test the significance of r is as shown below:

$H_0: r = 0$ (The two variables are not associated.)

$H_1: r \neq 0$ (The two variables are associated.)

If the calculated value of the t statistic is more than the table t value, then reject the null hypothesis; otherwise, accept the null hypothesis.

Example 10.1 The annual advertising expenditures (in lakhs of rupees) and the corresponding annual sales (in crores of rupees) for the past ten years of a company are presented in Table 10.1. Find the correlation coefficient between annual advertising expenditure and annual sales revenue using the basic formula as well as Pearson's product moment formula. Also, test the significance of the correlation coefficient at a significance level of 0.05.

Table 10.1 Data on Annual Advertising Expenditures and Annual Sales

Year (i)	Annual advertising Expenditure (X_i)	Annual sales (Y_i)	Year (i)	Annual advertising Expenditure (X_i)	Annual sales (Y_i)
1	10	20	6	20	78
2	12	30	7	22	89
3	14	37	8	24	100
4	16	50	9	26	120
5	18	56	10	28	110

Solution. The necessary calculations for computing the correlation coefficient (r) between the annual advertising expenditure and the annual sales revenue are summarized in Table 10.2.

Table 10.2 Summary of Calculations for Correlation Coefficient

Calculations for the basic formula								Calculations for product moment formula		
<i>i</i>	X_i	Y_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$X_i Y_i$	X_i^2	Y_i^2
1	10	20	-9	-49	81	2401	441	200	100	400
2	12	30	-7	-39	49	1521	273	360	144	900
3	14	37	-5	-32	25	1024	160	518	196	1369
4	16	50	-3	-19	9	361	57	800	256	2500
5	18	56	-1	-13	1	169	13	1008	324	3136
6	20	78	1	9	1	81	9	1560	400	6084
7	22	89	3	20	9	400	60	1958	484	7921
8	24	100	5	31	25	961	155	2400	576	10000
9	26	120	7	51	49	2601	357	3120	676	14400
10	28	110	9	41	81	1681	369	3080	784	12100
Total	190	690	0	0	330	11200	1894	15004	3940	58810

We get $\bar{X} = (\Sigma X_i)/n = 19$, $\bar{Y} = (\Sigma Y_i)/n = 69$. The correlation coefficient using the basic formula is computed as shown below:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{1894}{\sqrt{330} \sqrt{11200}} = 0.985$$

The correlation coefficient between annual advertising expenditure and annual sales revenue is 0.985.

The correlation coefficient using Pearson's product moment formula is computed as shown below:

$$r = \frac{n \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{\sqrt{n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2} \sqrt{n \sum_{i=1}^n Y_i^2 - \left(\sum_{i=1}^n Y_i \right)^2}}$$

$$= \frac{10 \times 15004 - 190 \times 690}{\sqrt{10 \times 3940 - 190^2} \sqrt{10 \times 58810 - 690^2}}$$

$$= 0.985$$

The combination of the hypotheses to test the significance of r is as shown below:

$H_0 : r = 0$ (The two variables are not associated.)

$H_1 : r \neq 0$ (The two variables are associated.)

The value of the t -statistic to test the significance of r is computed, as shown below:

$$t = \frac{r}{\sqrt{(1-r^2)/(n-2)}} = \frac{0.985}{\sqrt{(1-0.985^2)/(10-2)}} = 16.146$$

The table t value at half of the significance level ($\alpha/2 = 0.025$) and 8 degrees of freedom is 2.306.

The calculated value of t statistic (16.146) is more than the table t statistic (2.306). Hence, reject the null hypothesis. This means that the correlation coefficient of the problem is significant and hence the association between the two variables is significant.

10.2.2 Correlation Coefficient for Grouped Data (Karl Pearson's Coefficient of Correlation)

If the values of X and Y are grouped data then the correlation coefficient (r_g) of such data is computed using the following formula by using the intermediate results of Table 10.3.

$$r_g = \frac{N \sum_{i=1}^m \sum_{j=1}^n f_{ij} - \sum_{i=1}^m f_i X_i \sum_{j=1}^n f_j Y_j}{\sqrt{N \sum_{i=1}^m f_i X_i^2 - \left(\sum_{i=1}^m f_i X_i \right)^2} \sqrt{N \sum_{j=1}^n f_j Y_j^2 - \left(\sum_{j=1}^n f_j Y_j \right)^2}}$$

where X_i is the midpoint of the i th class-interval of X ; Y_j , the midpoint of the j th class-interval of Y ; a_{ij} , the actual frequency of observations of the i th class-interval of X and the j th class-interval of Y ; m , the number of class-interval for X ; n , the number of class-interval for Y ; f_i , the sum of the actual frequencies in the row i ($\sum_j a_{ij}$); f_j , the sum of the actual frequencies in the column j ($\sum_i a_{ij}$); f_{ij} , the theoretical frequency of observations of the i th class-interval of X and the j th class-interval of Y and it is computed as:

$$f_{ij} = a_{ij} X_i Y_j$$

N is the grand total of all the actual frequencies ($\sum_i \sum_j a_{ij}$); and $F = \sum_i \sum_j f_{ij}$ (grand total of theoretical frequencies).

Let the given data be as shown in Table 10.3. If the midpoints of X and Y are large in magnitude, then the following codifications may be used:

Let A be the assumed mean of X , B be the assumed mean of Y . Then

Example 10.2 A district industries centre (DIC) has collected data summarizing the number of industries in the district under each combination of the equity capital and rate of return, as shown in Table 10.4. Find the correlation coefficient of this grouped data.

Table 10.4. Summary of Data of Industries

Rate of return	Equity (crores of Rs.)		
	2-4	4-6	6-8
9 to 11	2	5	1
11 to 13	5	7	2
13 to 15	3	7	8
15 to 17	1	3	6

Solution. The data in Table 10.4 is presented as in Table 10.5 to compute different terms which are required to compute r_g .

Table 10.5 Summary of Calculations of r_g for Grouped Data

Mid - Y_j	3	5	7	$f_j = \sum_j a_{ij}$	$f_j X_i$	$f_j X_i^2$	$\sum_j a_{ij} X_i Y_j$ $= \sum_j f_{ij}$
Mid - X_i							
10	60	250	70				380
	2	5	1	8	80	800	
12	180	420	168				768
	5	7	2	14	168	2016	
14	126	490	784				1400
	3	7	8	18	252	3528	
16	48	240	672				960
	1	3	6	10	160	2560	
f_j	11	22	17	$N=50$	660	8904	3508
$f_j Y_j$	33	110	119	262			
$f_j Y_j^2$	99	550	833	1482			

From Table 10.5, we have $\sum_i \sum_j f_{ij} = 3508$. Therefore

$$r_g = \frac{N \sum_{i=1}^4 \sum_{j=1}^3 f_{ij} - \sum_{i=1}^4 f_i X_i \sum_{j=1}^3 f_j Y_j}{\sqrt{N \sum_{i=1}^4 f_i X_i^2 - \left(\sum_{i=1}^4 f_i X_i \right)^2} \sqrt{N \sum_{j=1}^3 f_j Y_j^2 - \left(\sum_{j=1}^3 f_j Y_j \right)^2}}$$

$$= \frac{50 \times 3508 - 660 \times 262}{\sqrt{50 \times 8904 - 660^2} \sqrt{50 \times 1482 - 262^2}}$$

$$= 0.3427$$

To simplify the calculations, the problem can be solved using the concept of assumed mean as shown in Table 10.6.

Table 10.6 Summary of Calculations of r_g for Grouped Data

Mid- Y_j		3	5	7	$f_j = \sum_i a_{ij}$	$f_j x_j$	$f_j x_j^2$	$\sum_j a_{ij} x_j y_j$ $= \sum_j f_j y_j$
$y_j = Y_j - 5$		-2	0	2				
Mid- X	$x_i = X_i - 12$							
10	-2	8 2	0 5	-4 1	8	-16	32	4
12	0	0 5	0 7	0 2	14	0	0	0
14	2	-12 3	0 7	32 8	18	36	72	20
16	4	-8 1	0 3	48 6	10	40	160	40
$f_j = \sum_i a_{ij}$		11	22	17	$N = 50$	60	264	64
$f_j y_j$		-22	0	34	12			
$f_j y_j^2$		44	0	68	112			

We get $\sum_i \sum_j f_{ij} = 64$. Then

$$r_g = \frac{N \sum_{i=1}^4 \sum_{j=1}^3 f_{ij} - \sum_{i=1}^4 f_i x_i \sum_{j=1}^3 f_j y_j}{\sqrt{N \sum_{i=1}^4 f_i x_i^2 - \left(\sum_{i=1}^4 f_i x_i \right)^2} \sqrt{N \sum_{j=1}^3 f_j y_j^2 - \left(\sum_{j=1}^3 f_j y_j \right)^2}}$$

$$= \frac{50 \times 64 - 60 \times 12}{\sqrt{50 \times 264 - 60^2} \sqrt{50 \times 112 - 12^2}}$$

$$= 0.3427$$

10.2.3 Rank Correlation

In some situation, the data may be available in the form of ranks of variables, X and Y . Consider the situation in which the objective is to determine the coefficient of correlation between the ranks of students in two different subjects. Such coefficient of correlation is known as Spearman's rank correlation coefficient (r_s). Like this, there are many other examples as listed below:

- Ranks of selected industries based on EPS (Earnings Per Share) in two consecutive years.
- Preference ratings (ranks) of similar products by two persons.
- Ranks of employees in two different skill tests.

The formula for the Spearman's rank correlation coefficient (r_s) is

$$r_s = 1 - \frac{6 \sum_{i=1}^n (X_i - Y_i)^2}{n(n^2 - 1)}$$

where X_i is the rank of the i th observation of the variable X ; Y_i , the rank of the i th observation of the variable Y ; and n , the number of pairs of observations.

Test of significance of rank correlation coefficient

The significance of the rank correlation coefficient can be tested using t statistic, whose formula is:

$$t = r_s \sqrt{\frac{n-2}{1-r_s^2}}$$

with $(n-2)$ degrees of freedom. The combination of the hypotheses to test the significance of r_s is:

$H_0: r_s = 0$ (The ranks of the two variables are not associated.)

$H_1: r_s \neq 0$ (The ranks of the two variables are associated.)

Example 10.3 The preference ratings of 10 similar brands of television by two judges are, as shown in Table 10.7. Find the rank correlation between the preference ratings assigned by the judges for the Televisions. Also, test the significance of the coefficient at a significance level of 0.05.

Table 10.7 Data on Preference Ratings of Televisions

TV brand	Rating by judge-1	Rating by judge-2	TV brand	Rating by judge-1	Rating by judge-2
1	1	2	6	6	6
2	2	4	7	7	7
3	3	5	8	8	9
4	4	1	9	9	10
5	5	3	10	10	8

Solution. The necessary calculations for computing the rank correlation coefficient (r_s) between preference ratings given by the two judges are summarized in Table 10.8.

The rank correlation coefficient is:

$$r_s = 1 - \frac{6 \sum_{i=1}^n (X_i - Y_i)^2}{n(n^2 - 1)} = 1 - \frac{6 \times 28}{10 \times (10^2 - 1)} = 0.8303$$

Table 10.8 Calculations for Rank Correlation Coefficient

TV brand	Rating by judge-1	Rating by judge-2	$X_i - Y_i$	$(X_i - Y_i)^2$
1	1	2	-1	1
2	2	4	-2	4
3	3	5	-2	4
4	4	1	3	9
5	5	3	2	4
6	6	6	0	0
7	7	7	0	0
8	8	9	-1	1
9	9	10	-1	1
10	10	8	2	4
Total				28

The rank correlation coefficient between the preference ratings assigned by the judges for the brands of televisions is 0.8303 which is fairly high.

The combination of the hypotheses to test r_s is:

$H_0: r_s = 0$ (The ranks of the two variables are not associated.)

$H_1: r_s \neq 0$ (The ranks of the two variables are associated.)

The t statistic is:

$$t = r_s \sqrt{\frac{n-2}{1-r_s^2}} = 0.8303 \sqrt{\frac{8}{1-(0.8303)^2}} = 4.214$$

The table t value, at half of the significance level ($\alpha/2 = 0.025$) and 8 degrees of freedom, is 2.306.

The calculated value of t statistic (4.214) is more than the table t statistic (2.306). Hence, reject the null hypothesis. This means that the rank correlation coefficient of the problem is significant.

10.2.4 Auto-correlation

Consider the values of X over different periods (three periods), as shown in Table 10.9. Then, one can find the correlation coefficient between X and Y_1 (or) X and Y_2 (or) X and Y_3 and so on, where Y_1 represents the values of X with 1 period lag, Y_2 represents the values of X with 2 periods lag, Y_3 represents values of X with 3 periods lag, and so on.

Table 10.9 Auto-correlation Data

Time (t)	Variable (X_t)	One period time lag (Y_1)	Two periods time lag (Y_2)	Three periods time lag (Y_3)
1	X_1	X_2	X_3	X_4
2	X_2	X_3	X_4	X_5
3	X_3	X_4	X_5	X_6
4	X_4	X_5	X_6	X_7
5	X_5	X_6	X_7	X_8
6	X_6	X_7	X_8	$\vdots$
7	X_7	X_8	$\vdots$	X_n
8	X_8	$\vdots$	X_n	—
$\vdots$	$\vdots$	X_n	—	—
n	X_n	—	—	—

The auto-correlation coefficient (r_k) between the variable X and Y_k is given by the following formula:

$$r_k = \frac{\sum_{t=1}^n (X_t - \bar{X})(X_{t+k} - \bar{X})}{\sum_{t=1}^n (X_t - \bar{X})^2}$$

where r_k is the auto-correlation coefficient with k period lag; $\bar{X}$, the mean of X ; X_t , the observation of the period t ; X_{t+k} , the observation of X with k period lag; and n is the number of observations of X .

Example 10.4 The demand values of a product over the past 10 years are summarized in Table 10.10. Find the auto-correlation coefficients with one year lag (r_1), two year lag (r_2) and three year lag (r_3).

Table 10.10 Demand Data

Year (t)	Demand (in '000)	Year (t)	Demand (in '000)
1	20	6	90
2	30	7	100
3	50	8	120
4	60	9	140
5	80	10	160

Solution. The summary of basic data which are necessary to compute r_1 , r_2 and r_3 are summarized in Table 10.11.

Table 10.11 Basic Data to Compute r_1 , r_2 and r_3

Time (t)	Variable (X_t)	One year lag data (Y_1)	Two year lag data (Y_2)	Three year lag data (Y_3)
1	20	30	50	60
2	30	50	60	80
3	50	60	80	90
4	60	80	90	100
5	80	90	100	120
6	90	100	120	140
7	100	120	140	160
8	120	140	160	—
9	140	160	—	—
10	160	—	—	—

The summary of calculations to compute r_1 is shown in Table 10.12.

Table 10.12 Summary of Calculations to Compute r_1

Time (t)	X_t	X_{t+1}	$X_t - \bar{X}$	$X_{t+1} - \bar{X}$	$(X_t - \bar{X})(X_{t+1} - \bar{X})$	$(X_t - \bar{X})^2$
1	20	30	-65	-55	3575	4225
2	30	50	-55	-35	1925	3025
3	50	60	-35	-25	875	1225
4	60	80	-25	-5	125	625
5	80	90	-5	5	-25	25
6	90	100	5	15	75	25
7	100	120	15	35	525	225
8	120	140	35	55	1925	1225
9	140	160	55	75	4125	3025
10	160	-	75	-	-	5625
Total	850				13125	19250

Here, Mean ($\bar{X}$) = 85.

The auto-correlation coefficient r_1 is computed as:

$$r_1 = \frac{\sum_{t=1}^9 (X_t - \bar{X})(X_{t+1} - \bar{X})}{\sum_{t=1}^{10} (X_t - \bar{X})^2} = \frac{13125}{19250} = 0.682$$

The summary of calculations to compute r_2 is shown in Table 10.13.

Table 10.13 Summary of Calculations to Compute r_2

Time (t)	X_t	X_{t+1}	$X_t - \bar{X}$	$X_{t+2} - \bar{X}$	$(X_t - \bar{X})(X_{t+2} - \bar{X})$	$(X_t - \bar{X})^2$
1	20	50	-65	-35	2275	4225
2	30	60	-55	-25	1375	3025
3	50	80	-35	-5	175	1225
4	60	90	-25	5	-125	625
5	80	100	-5	15	-75	25
6	90	120	5	35	175	25
7	100	140	15	55	825	225
8	120	160	35	75	2625	1225
9	140	-	55	-	-	3025
10	160	-	75	-	-	5625
Total	850				7250	19250

Here, mean ($\bar{X}$) = 85.

The auto-correlation r_2 is computed as:

$$r_2 = \frac{\sum_{t=1}^8 (X_t - \bar{X})(X_{t+2} - \bar{X})}{\sum_{t=1}^{10} (X_t - \bar{X})^2} = \frac{7250}{19250} = 0.377$$

Similarly, one can verify that the value of r_3 is 0.134.

10.3 FORECASTING

A *forecast* is an estimate of an event which will happen in future. The event may be the demand of a product, rainfall at a particular place, population of a country, or growth of a technology. The forecast value is not a deterministic quantity. Since it is only an estimate based on the past data related to a particular event, proper care must be taken while estimating it.

In business, forecasts may be classified into technology forecasts, economic forecasts and demand forecasts.

Technology forecast Technology is a combination of hardware and software. Hardware is any physical product while software is the know-how, technique or procedure. *Technology forecast* deals with certain characteristics such as the level of technical performance, rate of technological advances. Technological forecast is a prediction of the future characteristics of useful machines, products, process, procedures or techniques.

Economic forecasts Government agencies and other organizations involve in collecting data and prediction of estimate on the general business environment. *Economic forecasts* will be useful to government agencies in predicting future tax revenues, level of business growth, level of employment, level of inflation, etc. Also, these will be useful to business circles to plan their future activities based on the level of business growth.

Demand forecast The *demand forecast* gives the expected level of demand for goods or services. This is the basic input for business planning and control. Hence, the decisions of all the functions of any corporate house are influenced by the demand forecast.

Usefulness of forecasting in decision making

The usefulness of forecasting in different functional areas of management are listed below:

Marketing Some of the applications of forecasting in marketing function are projection of demand forecasting of products, market share, forecasting the trend in prices, etc.

Production Projection of materials requirements, trends in material and labour costs, maintenance requirements, plant capacity, etc. are some of the applications of forecasting in production function.

Finance Some of the applications of forecasting in finance function are projection of cash flows, rates of expenses, revenues, etc.

Personnel Projection of number of workers in each category, labour turnover, absenteeism of employees, etc., are some of the applications of forecasting in personnel function.

Measures of forecast accuracy

Demand forecast influences most of the decisions in all the functions. Hence, it must be estimated with the highest level of precision. Some common measures are inevitable to measure the accuracy of a forecasting technique. This measure may be an aggregate error (deviation) of the forecast values from the actual demands. The different types of errors which are generally computed are:

- Mean absolute deviation (MAD)
- Mean square error (MSE)
- Mean forecast error (MFE)
- Mean absolute per cent error (MAPE).

Let D_t be the actual demand for the period t ; F_t be the forecast demand for the period t ; e_t be the forecast error for the period t ; and N be the number of time periods for which the error is computed.

The formula for error is

$$e_t = D_t - F_t$$

Mean absolute deviation *Mean absolute deviation* (MAD) is the mean of absolute deviations of forecast demands from actual demand values. The mean absolute deviation is also called as the *mean absolute error* (MAE).

$$\text{MAD} = \frac{\sum_{t=1}^N |D_t - F_t|}{N}$$

Mean square error *Mean square error* (MSE) is the mean of the squares of the deviations of the forecast demands from the actual demand values. Minimization of the sum of the mean square errors penalizes large errors more than small errors.

$$\text{MSE} = \frac{\sum_{t=1}^N (D_t - F_t)^2}{N}$$

Mean forecast error *Mean forecast error* (MFE) is the mean of the deviations of the forecast demands from the actual demands:

$$\text{MFE} = \frac{\sum_{t=1}^N (D_t - F_t)}{N}$$

Mean absolute percentage error *Mean absolute percentage error* (MAPE) is the mean of the per cent deviations of the forecast demands from the actual demands.

$$\text{MAPE} = \frac{1}{N} \sum_{t=1}^N \frac{|D_t - F_t|}{D_t} \times 100$$

Forecasting techniques

Some of the commonly used forecasting techniques are:

- Simple moving average
- Single exponential smoothing
- Regression
- Delphi type method
- Time-series analysis.

Among these methods, Delphi method is used to obtain qualitative forecast and all other methods are used to obtain quantitative forecast.¹

10.3.1 Simple Moving Average Method

A *simple moving average* is a method of computing the average of a specified number of the most recent data values in a series. The formula to compute the simple moving average (SMA) is:

$$M_t = \frac{1}{n} [D_{t-(n-1)} + D_{t-(n-2)} + \dots + D_{t-2} + D_{t-1} + D_t]$$

where M_t is the simple moving average at the end of period t (it is to be used as the forecast for period $t+1$); D_t , the actual demand in the period t ; and n is the number of periods included in each moving average.

Example 10.5 The actual demand of a product for 8 months are summarized in Table 10.14.

- (a) Find three months moving averages.
- (b) Compute mean forecast error.
- (c) Compute mean square error (MSE) and mean absolute deviation (MAD).

1. The regression method of forecasting and time-series analysis are discussed in Section 10.4 and all other methods indicated here are discussed in following sections.

Table 10.14 Actual Demand Values

Month (<i>t</i>)	Demand (<i>D_t</i>)	Month (<i>t</i>)	Demand (<i>D_t</i>)
1	24	5	39
2	30	6	45
3	27	7	42
4	24	8	51

Solution. The formula to compute moving averages is:

$$M_t = \frac{1}{n} [D_{t-(n-1)} + D_{t-(n-2)} + \cdots + D_{t-2} + D_{t-1} + D_t]$$

The calculations to obtain the moving averages are summarized in Table 10.15.

Table 10.15 Three Months Moving Averages

Month (<i>t</i>)	Demand for month <i>t</i> (<i>D_t</i>)	Moving average <i>M</i> (<i>t</i>)	Forecast (<i>F_t</i>)	Error (<i>e_t</i>)
1	24			
2	30			
3	27	27		
4	24	27	27	-3
5	39	30	27	12
6	45	36	30	15
7	42	42	36	6
8	51	46	42	9

The values printed under column *M*(*t*) are obtained as: Moving average period (*n*) = 3 months. Then

$$M(3) = \frac{24 + 30 + 27}{3} = 27, \quad M(4) = \frac{30 + 27 + 24}{3} = 27$$

The forecast for the month 9 would be: $F_9 = M_8 = 46$. Then

$$\text{Mean forecast error (MPE)} = 7.8$$

$$\text{Mean absolute deviation (MAD)} = 9.0$$

Limitations: The limitations of the simple moving average method of forecasting are summarized below:

- In most cases, this method is applied to forecast for only one period into the future.
- The forecaster must wait until demand entries are available for making the first forecast.

- This method is applicable only for horizontal demand data patterns.
- This model requires to store the first n observed values, which consumes considerable amount of computer storage space if the demands of several products are to be estimated.

10.3.2 Weighted Moving Average Method

In simple moving average method, equal weights are assigned to the data of all periods while computing moving averages. But, it may not be realistic to give such equal weight. Hence, differential weights must be given to different periods which are used for computing the moving average such that the most recent piece of data has the highest weight and then it decreases for other preceding periods. This method is called *weighted moving average* method. The formula to compute the weighted moving averages is:

$$\text{Wt. MA}_t = \frac{\sum_{i=1}^n W_i D_i}{\sum_{i=1}^n W_i}$$

where D_i is the actual demand of the period i ; n is the number of periods used for computing moving averages; and W_i is the weight of the data of the time period i in the moving average.

When i is equal to n , it corresponds to the most recent time period and when i is equal to 1, it corresponds to the oldest time period.

Example 10.6 The actual demand of a product for 6 months are summarized in Table 10.16.

Table 10.16 Actual Demand Values

Month (t)	D_t
1	80
2	90
3	70
4	100
5	70
6	90

- Find three months weighted moving averages by assuming the weights, $W_1 = 0.2$, $W_2 = 0.3$ and $W_3 = 0.5$.
- Compute forecast errors.
- Compute mean square error and mean absolute deviation.

Solution. In this example, three months weighted moving average is computed with a weight of 0.5 assigned to the most recent demand value, a weight of 0.30 assigned to the next most recent value and a weight of 0.20 assigned to the oldest of the demand values included in the average. The calculations to obtain the weighted moving averages are summarized in Table 10.17.

Table 10.17 Three Months Weighted Moving Averages

Month (<i>t</i>)	Demand (<i>D_t</i>)	Wt. moving average <i>M</i> (<i>t</i>)	Forecast (<i>F_t</i>)	Error (<i>e_t</i>)
1	80			
2	90			
3	70	78		
4	100	89	78	22
5	70	79	89	-19
6	90	86	79	11

The formula for weighted moving average is

$$\text{Wt. MA}_t = \frac{\sum_{i=1}^n W_i D_i}{\sum_{i=1}^n W_i}$$

The weighted moving average of the month 3 is:

$$\text{Weighted MA}_3 = \frac{0.2 \times 80 + 0.3 \times 90 + 0.5 \times 70}{0.2 + 0.3 + 0.5} = 78$$

Mean forecast error (MPE) = 4.67

Mean absolute deviation (MAD) = 17.33

10.3.3 Simple (Single) Exponential Smoothing Method

Another form of weighted moving average is the exponential smoothed average using exponential smoothing method. The forecast of the period *t* is computed by applying some correction over the forecast value of the immediate preceding period. The correction quantity is a portion (α) of the difference between the actual demand of the immediate preceding period and the forecast of the immediate preceding period.

The formula to compute the exponential smoothed forecast is as given below:

$$F_t = F_{t-1} + \alpha(D_{t-1} - F_{t-1})$$

where F_t is the smoothed average forecast of the period t ; F_{t-1} , the smoothed average forecast of the period $(t-1)$; D_{t-1} , the actual demand of the period $(t-1)$; and α , is the smoothing constant ($0 < \alpha < 1$).

If α is equal to 1, then the latest forecast would be equal to the previous period actual demand value. The preferred range for α is from 0.1 to 0.3.

Example 10.7 The forecast of a product for the first week of March was 600 units, whereas the actual demand turned out to be 550 units.

- Find the forecast for the second week of March by assuming the smoothing constant (α) as 0.2.
- Find the forecast for the third week of March if the actual demand of the second week is 620 units.

Solution. (a) Forecast of the first week of March, $F_1 = 600$ units. Actual demand of the first week of March, $D_1 = 550$ units. Smoothing constant, $\alpha = 0.2$. Forecast of the second week of March, F_2 is as computed as:

$$\begin{aligned} F_2 &= F_1 + \alpha(D_1 - F_1) \\ &= 600 + 0.2(550 - 600) \\ &= 600 - 10 \\ &= 590 \text{ units} \end{aligned}$$

Therefore, the forecast of the second week of March is 590 units.

(b) Forecast of the second week of March, $F_2 = 590$ units. Actual demand of the second week of March, $D_2 = 620$ units. Forecast of the third week of March, F_3 is computed as:

$$\begin{aligned} F_3 &= F_2 + \alpha(D_2 - F_2) \\ &= 590 + 0.2(620 - 590) \\ &= 590 + 6 \\ &= 596 \text{ units} \end{aligned}$$

Therefore, the forecast of the third week of March is 596 units.

10.3.4 Delphi Method

Delphi method is a forecasting technique applied to subjective nature of demand values. In view of globalization in India, Indian companies have difficulty in estimating the demand of their products mainly because of possible mixed reactions of customers towards various attributes of a specific product which is manufactured by multinational firms and indigenous firms. Under such situation, one has to resort to subjective estimates. Technology forecasting is another example where there is no quantitative data based on which the future technology can be predicted. In this situation, we will have information at various stages

of technological advancement for a particular application which can be used for predicting the future technology for that application.

In Delphi method of forecasting, several knowledgeable persons are asked to provide subjective estimates of demands or forecasts of possible advances of technology. The experts may provide several opinions. Based on the opinions of the experts, a consensus will be arrived about the demand of the product/advances of the technology.

The essential precautions to be followed in this method are as follows:

- Panel members must be unknown to each other.
- The initial questionnaire should be unambiguous and it should explain every matter about which opinion is sought.

After getting the opinions from the panel members, they are to be compared for similarity. If the variation among the opinions is too much, the summary of opinions is to be circulated again among the members without mentioning the names of persons who provided the opinions. Generally, 50 per cent of the estimate is treated as the basis for comparison. The panel members whose opinions differ significantly from the middle 50% of the estimate will be asked to reconsider their opinions. Still, if they want to stick to their original opinions, they will be asked to provide rationale for the same.

So, the Delphi method is an iterative process until the panel converges on a specific value or a range of values as defined by the required accuracy or arrives a consensus on the matter under consideration.

10.4 LINEAR REGRESSION AND TIME SERIES

Regression is the dependence of a variable on one or more variables. Consider a linear equation, as shown below, which is known as a linear regression equation.

$$Y_c = a + bX$$

where Y is the dependent variable; X , the only independent variable; a , is the intercept; and b is the slope/trend.

A practical example, where the above model can be used, may be the task of forecasting the demand of a product for a given period. In this example, the demand is the dependent variable (Y) and the time period is the independent variable (X).

This equation is shown graphically in Fig. 10.1.

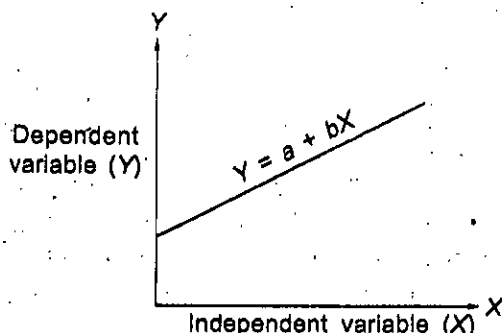


Fig. 10.1 Linear regression equation.

In Fig. 10.1, the independent variable X is represented along the X -axis and the dependent variable Y is represented along the Y -axis. The constant ' a ' is the intercept of the line, $Y = a + bX$ on the Y -axis. The constant ' b ' is the slope of the line $Y = a + bX$. The slope of the equation means the increase in Y -axis per unit increase in X -axis.

If the number of independent variables on the right-hand side of the linear equation is more than one, then the corresponding equation is called as *multiple linear regression equation*.

The model for multiple linear regression equation is:

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3 + b_iX_i + \dots + b_nX_n$$

where Y is the dependent variable; X_i , the i th independent variable; a , the intercept; and b_i is the slope of the independent variable X_i .

The equation for a sample curvilinear regression is:

$$Y = a + b_1X + b_2X^2 + b_3X^3 + \dots + b_nX^n$$

where Y is the dependent variable; $X, X^2, X^3, \dots, X^n$ are independent terms; a is the intercept; $b_1, b_2, b_3, \dots, b_n$ are the slopes of the terms $X, X^2, X^3, \dots, X^n$ respectively.

The value of the dependent variable Y may be affected by the components of time series such as trend, seasonal, cyclical and random. The integrated study of regression by taking these components into account is called *time-series analysis*.

10.4.1 Simple Regression

Consider a model for the simple linear regression as:

$$Y = a + bX$$

where

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} \quad \text{and} \quad a = \bar{Y} - b\bar{X}$$

with

$$\bar{X} = \frac{\sum X}{n} \quad \text{and} \quad \bar{Y} = \frac{\sum Y}{n}$$

and n is the number of pairs of observations made.

Note: When $\sum X = 0$, then the formulae for b and a are reduced as:

$$b = \frac{\sum XY}{\sum X^2} \quad \text{and} \quad a = \bar{Y}$$

Alternate approach for simple regression

The regression equation of Y on X is:

$$Y = a + bX$$

and

$$Y - \bar{Y} = r \frac{\sigma_Y}{\sigma_X} (X - \bar{X})$$

where $\bar{Y}$ is the mean of Y ; $\bar{X}$, the mean of X ; σ_X , the standard deviation of X ; σ_Y , the standard deviation of Y ; and r is the correlation coefficient between X and Y . (It is the degree of association between two variables and its range is from -1 to $+1$.)

By comparing the above two equations, we get

$$\text{Slope, } b = r \frac{\sigma_Y}{\sigma_X}$$

and

$$\text{Intercept, } a = \bar{Y} - r \frac{\sigma_Y}{\sigma_X} \bar{X} = \bar{Y} - b\bar{X}$$

Similarly, regression equation of X on Y is:

$$X = a + bY$$

and

$$X - \bar{X} = r \frac{\sigma_X}{\sigma_Y} (Y - \bar{Y})$$

By comparing the above equation,

$$\text{Slope, } b = r \frac{\sigma_X}{\sigma_Y}$$

and

$$\text{Intercept, } a = \bar{X} - r \frac{\sigma_X}{\sigma_Y} \bar{Y} = \bar{X} - b\bar{Y}$$

Example 10.8 The demand values (in lakhs of tons) of a product during the past 6 years are summarized in Table 10.18.

Table 10.18 Summary of Demand Values

Year (x)	Demand (Y)
1997	50
1998	60
1999	50
2000	80
2001	72
2002	90

- Fit a linear regression to estimate the demand of the product in future.
- Compute the demand of the product for the year 2008.

Solution. (a) The necessary calculations to design the regression equation to estimate the demand of the product are summarized in Table 10.19.

Table 10.19 Summary of Calculations for Regression

Year (x)	$X = x - 2000$	Y	XY	X^2
1997	-3	50	-150	9
1998	-2	60	-120	4
1999	-1	50	-50	1
2000	0	80	0	0
2001	1	72	72	1
2002	2	90	180	4
Total	-3	402	-68	19

We get

$$\bar{X} = \frac{-3}{6} = -0.5, \quad \bar{Y} = \frac{402}{6} = 67$$

with

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} = \frac{-68 - 6 \times (-0.5) \times (67)}{19 - 6 \times (-0.5) \times (-0.5)} = 7.6$$

$$a = \bar{Y} - b\bar{X} = 67 - 7.6 \times (-0.5) = 70.8$$

The regression model is

$$Y = a + bX = 70.8 + 7.6X = 70.8 + 7.6(x - 2000)$$

(b) The demand for the year 2008 is computed as

$$\begin{aligned} Y &= 70.8 + 7.6(x - 2000) \\ &= 70.8 + 7.6(2008 - 2000) \\ &= 131.6 \text{ lakhs tons} \\ &= 1,31,60,000 \text{ tons} \end{aligned}$$

Example 10.9 A company has the sales pattern (in units) during 1994 to 2002, as shown in Table 10.20.

- Fit a linear regression for the data.
- Compute the sales forecast for the year 2006.

Table 10.20 Summary of Demand Values

Year	Sales (in lakhs of units)
1994	10
1995	12
1996	15
1997	27
1998	33
1999	38
2000	44
2001	49
2002	60

Solution. (a) The calculations for the design of the regression line are summarized in Table 10.21.

Table 10.21 Summary of Calculations for Regression

Year (X)	Sales (Y) (in lakhs)	$X = x - 1998$	XY	X^2
1994	10	-4	-40	16
1995	12	-3	-36	9
1996	15	-2	-30	4
1997	27	-1	-27	1
1998	33	0	0	0
1999	38	1	38	1
2000	44	2	88	4
2001	49	3	147	9
2002	60	4	240	16
Total	288	0	380	60

We have from the table

$$\bar{X} = \frac{\sum X}{n} = \frac{0}{9} = 0, \quad \bar{Y} = \frac{\sum Y}{n} = \frac{228}{9} = 32$$

and

$$b = \frac{\sum XY}{\sum X^2} = \frac{380}{60} = 6.33, \quad a = \bar{Y} = 32$$

Then

$$y = a + bX = 32 + 6.33X = 32 + 6.33(x - 1998)$$

The forecast of sales for the year 2006 is computed by substituting $x = 2006$ in the above equation.

$$\hat{Y} = 32 + 6.33(2006 - 1998)$$

$$= 32 + 6.33 \times 8$$

$$= 82.64 \text{ lakhs units}$$

$$= 82,64,000 \text{ units}$$

Example 10.10 The annual sales (in crores of rupees) of washing machines for the past 7 years as a function of R&D expenditure (in lakhs of rupees) are summarized in Table 10.22.

Table 10.22 Sales Data of Washing Machine

Year	R&D expenditure (X)	Sales (Y)
1	10	25
2	15	35
3	20	45
4	25	55
5	30	70
6	35	65
7	40	85

- Fit a regression model to estimate the sales (Y) for a given level of R&D activity (X) and also find the mean square error if this model is used.
- Fit a regression model to estimate the R&D expenditure (X) for a desired sales (Y) and also find the mean square error if this model is used.

Solution. The calculations to fit the regression equation $Y = a + bX$ as well as $X = a + bY$ are summarized in Table 10.23.

Table 10.23 Summary of Calculations

X	$x = X - 25$	x^2	Y	$y = Y - 55$	y^2	xy
10	-15	225	25	-30	900	450
15	-10	100	35	-20	400	200
20	-5	25	45	-10	100	50
25	0	0	55	0	0	0

contd.

Table 10.23 (contd.)

X	$x = X - 25$	x^2	Y	$y = Y - 55$	y^2	xy
30	5	25	70	15	225	75
35	10	100	65	10	100	100
40	15	225	85	30	900	450
Total	175	0	700	-5	2625	1325

(a) Regression of Y on X : The desired equation is: $Y = a + bX$, where

$$\bar{x} = \frac{\sum x}{n} = \frac{0}{7} = 0, \quad \bar{y} = \frac{\sum y}{n} = \frac{-5}{7} = -0.714$$

and

$$b = \frac{\sum xy}{\sum x^2} = \frac{1325}{700} = 1.893, \quad a = \bar{y} = -0.714$$

Then

$$y = a + bx$$

$$Y - 55 = -0.714 + 1.893(X - 25)$$

$$Y = 6.961 + 1.893X$$

The calculations for estimating different values of Y and the computation of forecast errors are summarized in Table 10.24. The mean square error is 16.199.

Table 10.24 Summary of Calculations

X	Y	Forecast of Y	Error	(Error) ²
10	25	25.891	-0.891	0.794
15	35	35.356	-0.356	0.127
20	45	44.821	0.179	0.032
25	55	54.286	0.714	0.510
30	70	63.751	6.249	39.050
35	65	73.216	-8.216	67.503
40	85	82.681	2.319	5.378

Now

$$\text{Mean square error} = \frac{\sum (\text{Error})^2}{n} = \frac{113.394}{7} = 16.199$$

(b) The desired equation for regression of X on Y is: $X = a + bY$. Here

$$\bar{x} = \frac{\sum x}{n} = \frac{0}{7} = 0, \quad \bar{y} = \frac{\sum y}{n} = \frac{-5}{7} = -0.714$$

and

$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum y^2 - n\bar{y}^2} = \frac{1325 - 7 \times 0 \times (-0.714)}{2625 - 7 \times (-0.714) \times (-0.714)} = 0.5055$$

$$a = \bar{x} - b\bar{y} = 0 - 0.5055 \times (-0.714) = 0.3609$$

Therefore,

$$x = a + by$$

$$X - 25 = 0.3609 + 0.5055(Y - 55)$$

$$X = -2.4416 + 0.5055Y$$

The calculations for estimating different values of X and the computation of forecast errors are summarized in Table 10.25.

The mean square error is 4.325.

Table 10.25 Summary of Calculations

Y	X	Forecast of X	Error	(Error) ²
25	10	10.196	-0.196	0.038
35	15	15.251	-0.251	0.063
45	20	20.306	-0.306	0.094
55	25	25.361	-0.361	0.130
70	30	32.943	-2.943	8.661
65	35	30.416	4.584	21.013
85	40	40.526	-0.526	0.277

Now, we get

$$\text{Mean square error} = \frac{\sum (\text{Error})^2}{n} = \frac{30.276}{7} = 4.325$$

Example 10.11 The results of a survey on the sales of a product (Y) as a function of time period (X) are summarized below:

	X	Y
Mean	25	150
Standard deviation	2	8
Correlation coefficient, r	0.9	

- (a) Fit the regression line of Y on X and estimate the value of Y when X is 10.
 (b) Fit the regression line of X on Y and estimate the value of X when Y is 96.

Solution. (a) The regression equation of Y on X is:

$$Y - \bar{Y} = r \frac{\sigma_Y}{\sigma_X} (X - \bar{X})$$

where $\bar{Y}$ is the mean of Y ; $\bar{X}$, the mean of X ; σ_X , the standard deviation of X ; σ_Y , the standard deviation of Y ; and r is the correlation coefficient between X and Y .

We have $\bar{X} = 25$, $\bar{Y} = 150$; $\sigma_X = 2$, $\sigma_Y = 8$; and $r = 0.9$. Then from the regression equation of Y on X ,

$$Y - 150 = 0.9 \frac{8}{2} (X - 25) \quad \text{or} \quad Y = 60 + 3.6X$$

The estimated value of Y when X is 10:

$$Y = 60 + 3.6 \times 10 = 96$$

(b) The regression equation of X on Y is:

$$X - \bar{X} = r \frac{\sigma_X}{\sigma_Y} (Y - \bar{Y})$$

$$X - 25 = 0.9 \frac{2}{8} (Y - 150)$$

$$X = -8.75 + 0.225Y$$

The value of X is estimated when Y is 96, as

$$X = -8.75 + 0.225 \times 96 = 12.85$$

Testing of hypothesis on the significance of regression

After fitting a regression model for a real-world data, one should check whether the estimates from the regression model represent the real-world data.

Let Y_i be the i th observation of the dependent variable, Y ; X_i , the i th observation of the independent variable, X ; $\bar{Y}$, the mean of the dependent variable, Y ; $\hat{Y}_i$, the estimated value of the i th dependent variable Y ; n , the total number of pairs observations of the independent variable and dependent variable; a , the intercept; b , the slope; and e_i , be the random error associated with the i th pair of Y_i and $\hat{Y}_i$. Then

$$\text{Model of regression: } Y_i = a + bX_i + e_i$$

The sum of squares of the model is:

$$\text{Total sum of squares} = \text{Regression sum of squares} + \text{Errors sum of squares}$$

or

$$SS_{\text{total}} = SS_{\text{regression}} + SS_{\text{error}}$$

or

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

where

The degrees of freedom for the total = $n - 1$

The degrees of freedom for the regression = 1

The degrees of freedom for the error = $n - 2$

The related combination of hypotheses is:

H_0 : The slope, $b = 0$ (there is no linear relationship between Y and X , as shown in the model).

H_1 : The slope, $b \neq 0$ (there is linear relationship between Y and X , as shown in the model).

The ANOVA table to test the null hypothesis is as shown in Table 10.26

Table 10.26 ANOVA Table to Test the Significance of Regression

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F ratio
Due to regression	SS_R	1	$MSS_R = SS_R$	$F = MSS_R / MSS_E$
Due to error	SS_E	$n - 2$	$MSS_E = SS_E / (n - 2)$	
Total	SS_T	$n - 1$		

The coefficient of determination, $r^2 = \frac{SS_R}{SS_T}$

The coefficient of determination is the variation in Y explained by the regression model when compared to the total variation. If this is higher, higher is the degree of relationship between Y and X .

Correlation coefficient, $r = \sqrt{\frac{SS_R}{SS_T}}$

The range of r^2 is from 0 to 1 and that of r is from -1 to $+1$. The sign of r is same as that of the slope, b in the regression model.

Example 10.12 The annual sales of computer of Brand-X (in crores of rupees) are summarized in Table 10.27.

Table 10.27 Annual Sales of Computer of Brand-X (in Crores of Rupees)

Year (X)	Annual sales (Y)
1	12
2	16
3	15
4	18
5	23
6	26
7	30

- (a) Fit a regression model to estimate the annual sales.
 (b) Test the significance of regression model at a significance level of 0.01.

Solution. Let the regression model be:

$$Y = a + bX$$

The calculations to fit the regression model are summarized in Table 10.28.

Table 10.28 Summary of Calculations

	X	Y	X ²	XY
	1	12	1	12
	2	16	4	32
	3	15	9	45
	4	18	16	72
	5	23	25	115
	6	26	36	156
	7	30	49	210
Total	28	140	140	642

Then

$$\bar{X} = \frac{28}{7} = 4, \quad \bar{Y} = \frac{140}{7} = 20$$

and

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} = \frac{642 - 7 \times 4 \times 20}{140 - 7 \times 4 \times 4} = 2.929$$

$$a = \bar{Y} - b\bar{X} = 20 - 2.929 \times 4 = 8.284$$

Therefore, the regression model to estimate Y is

$$Y = a + bX \text{ or } Y = 8.284 + 2.929X$$

The calculations for estimating different sum of squares are summarized in Table 10.29.

Table 10.29 Summary of Calculations

Y_i	$\hat{Y}$	$(Y_i - \bar{Y})^2$	$(\hat{Y}_i - \bar{Y})^2$	$(Y_i - \hat{Y}_i)^2$
12	11.213	64	77.194	0.619
16	14.142	16	34.316	3.452
15	17.071	25	8.579	4.289
18	20.000	4	0.000	4.000
23	22.929	9	8.579	0.005
26	25.858	36	34.316	0.020
30	28.787	100	77.211	1.471
Total		254	240.195	13.856

We get from the table

$$SS_T = 254.00, \quad SS_R = 240.195, \quad SS_E = 13.856$$

Note: One can compute SS_E using the following formula without actually using the elaborate calculations for it

$$SS_E = SS_T - SS_R = 254 - 240.195 = 13.805$$

We also have

$$\text{The degrees of freedom for the Total} = n - 1 = 6$$

$$\text{The degrees of freedom for the Regression} = 1$$

$$\text{The degrees of freedom for the Error} = n - 2 = 5$$

The related combination of hypotheses is:

H_0 : The slope, $b=0$ (there is no linear relationship between Y and X , as shown in the model).

H_1 : The slope, $b \neq 0$ (there is linear relationship between Y and X , as shown in the model).

The ANOVA results to test the null hypothesis are, as shown in Table 10.30.

Table 10.30 ANOVA Table to Test the Significance of Regression

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F ratio
Due to regression	240.195	1	240.195	86.682
Due to error	13.856	5	2.771	
Total	254.000	6		

From Table 10.30, the calculated value of F is 86.682. The table value of F with (1, 5) degrees of freedom at the given significance level of 0.01 is 16.26.

Since the calculated value of F (86.682) is more than the table value of F (16.26), reject the null hypothesis and accept the alternate hypothesis.

Thus the slope, $b \neq 0$ (there is linear relationship between Y and X , as shown in the model).

$$\text{The coefficient of determination, } r^2 = \frac{240.195}{254} = 0.946$$

This means that the regression model, $Y = 8.284 + 2.929X$, explains 94.6% variation in Y when compared to the total variation.

$$\text{Correlation coefficient, } r = \sqrt{r^2} = (0.946)^{0.5} = 0.9726$$

10.4.2 Multiple Regression

As stated earlier, if the number of independent variables in a regression model is more than one, then the model is called as *multiple regression*. In fact, many of the real-world applications demand the use of multiple regression models.

A sample application is as stated below:

$$Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4$$

where Y is the economic growth rate of a country; X_1 , the time period; X_2 , the size of the population of the country; X_3 , the level of employment in percentage; X_4 , the percentage literacy; b_0 is the intercept; and b_1 , b_2 , b_3 and b_4 are the slopes of the variables X_1 , X_2 , X_3 and X_4 , respectively.

In this regression model, X_1 , X_2 , X_3 and X_4 are the independent variables and Y is the dependent variable.

Regression model with two independent variables using normal equations

Suppose the number of independent variables is two, as shown in the following model.

$$Y = b_0 + b_1X_1 + b_2X_2$$

One can use the following normal equations and solve for the constants to design the regression equation:

$$\sum Y = nb_0 + b_1 \sum X_1 + b_2 \sum X_2$$

$$\sum YX_1 = b_0 \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1X_2$$

$$\sum YX_2 = b_0 \sum X_2 + b_1 \sum X_1X_2 + b_2 \sum X_2^2$$

where n is the total number of combinations of observations. The solution to the above set of simultaneous equations will form the results for the coefficients (b_0 , b_1 and b_2) of the regression model.

Example 10.13 The annual sales revenue (in crores of rupees) of a product as a function of sales force (number of salesmen) and annual advertising expenditure (in lakhs of rupees) for the past 10 years are summarized in Table 10.31. Design a regression model to forecast the annual sales revenue of the product.

Table 10.31 Details of Annual Sales Revenue, Sales Force and Annual Advertising Expenditures

Annual sales revenue (Y)	Sales force (X ₁)	Annual advertising expenditures (X ₂)
20	8	28
23	13	23
25	8	38
27	18	16
21	23	20
29	16	28
22	10	23
24	12	30
27	14	26
35	20	32

Solution. Let the regression model for the given problem be:

$$Y = b_0 + b_1X_1 + b_2X_2$$

where Y is the annual sales revenue (in crores of rupees); X_1 , the sales force; and X_2 is the annual advertising expenditure (in lakhs of rupees).

The normal equations are as given below:

$$nb_0 + b_1 \sum X_1 + b_2 \sum X_2 = \sum Y$$

$$b_0 \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1X_2 = \sum YX_1$$

$$b_0 \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2 = \sum YX_2$$

where n is the total number of combinations of observations. For this example, $n = 10$.

The details of calculations to get values for forming the normal equations are summarized in Table 10.32.

Table 10.32: Details of Calculations to Get Values for Forming Normal Equations

Annual sales revenue (Y)	Sales force (X ₁)	Annual advertising expenditures (X ₂)	X ₁ ²	X ₂ ²	X ₁ X ₂	YX ₁	YX ₂
20	8	28	64	784	224	160	560
23	13	23	169	529	299	299	529
25	8	38	64	1444	304	200	950
27	18	16	324	256	288	486	432
21	23	20	529	400	460	483	420
29	16	28	256	784	448	464	812
22	10	23	100	529	230	220	506
24	12	30	144	900	360	288	720
27	14	26	196	676	364	378	702
35	20	32	400	1024	640	700	1120
Total 253	142	264	2246	7326	3617	3678	6751

Substituting the required values in the normal equations, we get the simultaneous equations as shown below:

$$10b_0 + 142b_1 + 264b_2 = 253$$

$$142b_0 + 2246b_1 + 3617b_2 = 3678$$

$$264b_0 + 3617b_1 + 7326b_2 = 6751$$

The solution to the above set of simultaneous equations is:

$$b_0 = 5.1483, \quad b_1 = 0.6190, \quad b_2 = 0.4304$$

Therefore, the regression model of the given example is:

$$Y = b_0 + b_1X_1 + b_2X_2 = 5.1483 + 0.619X_1 + 0.4304X_2$$

Matrix method for regression model (generalized method)

Consider the following model of regression:

$$Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_kX_k$$

where Y is the dependent variable; $X_1, X_2, X_3, \dots, X_k$ are the independent variables; b_0 is the intercept; $b_1, b_2, b_3, \dots, b_k$ are the slopes of the variables $X_1, X_2, X_3, \dots, X_k$, respectively; and k is the number of independent variables.

A sample set of theoretical observations of this model is presented in Table 10.33. In Table 10.33, n is the total sets of observations and X_{ji} is the i th data of the j th independent variable.

Table 10.33 Data for the Regression Model

Y	X_1	X_2	X_3	$\dots$	X_j	X_k
Y_1	X_{11}	X_{21}	X_{31}	$\dots$	X_{j1}	X_{k1}
Y_2	X_{12}	X_{22}	X_{32}	$\dots$	X_{j2}	X_{k2}
Y_3	X_{13}	X_{23}	X_{33}	$\dots$	X_{j3}	X_{k3}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Y_i	X_{1i}	X_{2i}	X_{3i}	$\dots$	X_{ji}	X_{ki}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Y_n	X_{1n}	X_{2n}	X_{3n}	$\dots$	X_{jn}	X_{kn}

The data in Table 10.33 and the coefficients in the regression model can be arranged in matrices, as presented below:

$$X = \begin{bmatrix} 1 & X_{11} & X_{21} & X_{31} & \dots & X_{j1} & \dots & X_{k1} \\ 1 & X_{12} & X_{22} & X_{32} & \dots & X_{j2} & \dots & X_{k2} \\ 1 & X_{13} & X_{23} & X_{33} & \dots & X_{j3} & \dots & X_{k3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & X_{1i} & X_{2i} & X_{3i} & \dots & X_{ji} & \dots & X_{ki} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & X_{1n} & X_{2n} & X_{3n} & \dots & X_{jn} & \dots & X_{kn} \end{bmatrix}, \quad Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_n \end{bmatrix}, \quad b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_k \end{bmatrix}$$

↑
Coefficients for b_0 term

The coefficients of the regression model can be obtained using the following equation:

$$b = [X^T X]^{-1} [X^T Y]$$

where T stands for transpose.

Example 10.14 Consider Example 10.13 again. Design a regression model to forecast the annual sales revenue of the product using matrix method.

Solution. Let the regression model for the given problem be:

$$Y = b_0 + b_1 X_1 + b_2 X_2$$

where Y is the annual sales revenue (in crores of rupees); X_1 , the sales force; and X_2 is the annual advertising expenditure (in lakhs of rupees). We also have

The number of independent variables, $k = 2$

The number of sets of observations, $n = 10$

Y is the dependent variable and X_1 and X_2 are the independent variables. Representing the given data in matrix form, we get

$$X = \begin{bmatrix} 1 & 8 & 28 \\ 1 & 13 & 23 \\ 1 & 8 & 38 \\ 1 & 18 & 16 \\ 1 & 23 & 20 \\ 1 & 16 & 28 \\ 1 & 10 & 23 \\ 1 & 12 & 30 \\ 1 & 14 & 26 \\ 1 & 20 & 32 \end{bmatrix}, \quad Y = \begin{bmatrix} 20 \\ 23 \\ 25 \\ 27 \\ 21 \\ 29 \\ 22 \\ 24 \\ 27 \\ 35 \end{bmatrix}, \quad b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$$


 Coefficients for b_0 term

Now

$$[X^T X]^{-1} = \begin{bmatrix} 5.2305160 & -0.13249480 & -0.123071600 \\ -0.1324948 & 0.00552914 & 0.002044730 \\ -0.1230716 & 0.00204473 & 0.003561987 \end{bmatrix}$$

and

$$[X^T Y] = \begin{bmatrix} 253 \\ 3678 \\ 6751 \end{bmatrix}$$

Then

$$\begin{aligned} b &= [X^T X]^{-1} [X^T Y] \\ &= \begin{bmatrix} 5.2305160 & -0.13249480 & -0.123071600 \\ -0.1324948 & 0.00552914 & 0.002044730 \\ -0.1230716 & 0.00204473 & 0.003561987 \end{bmatrix} \begin{bmatrix} 253 \\ 3678 \\ 6751 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 5.1483 \\ 0.6190 \\ 0.4304 \end{bmatrix}$$

The values of the regression coefficients are:

$$b_0 = 5.1483, \quad b_1 = 0.619, \quad b_2 = 0.4304$$

Therefore, the regression model of the given example is:

$$Y = b_0 + b_1X_1 + b_2X_2 = 5.1483 + 0.619X_1 + 0.4304X_2$$

Nonlinear regression model using matrix method

In practice, there may be situations in which the regression model will be in nonlinear forms. Hence, they should be treated as per guidelines presented below:

Consider the following nonlinear situation to estimate the dependent variable Y :

$$Y = b_0 + b_1X^{p_1} + b_2X^{p_2} + \dots + b_iX^{p_i} + \dots + b_kX^{p_k}$$

where Y is the dependent variable; X^{p_i} , the i th independent variable, $i = 1, 2, 3, \dots, k$; b_0 , the intercept; and b_i is the slope of the i th term, $i = 1, 2, 3, \dots, k$.

Based on the past data, if different sets of Y and X values are given, then one can generate the values of the other terms of the model as per the exponents given in it. Later, the matrix method which is explained in the previous section can be applied to get the coefficients of the model.

Example 10.15 The power consumption (in lakhs of units) in a region is the function of the period of the day where period is defined as four-hours intervals of the day, as shown in Table 10.34.

Table 10.34 Details of Power Consumption

Period (X)	Time interval	Power consumption (scaled units) (Y)
1	Midnight–4 a.m.	40
2	4–8 a.m.	60
3	8 a.m.–12 noon	100
4	12 noon–4 p.m.	70
5	4–8 p.m.	60
6	8 p.m.–12 midnight	90

Fit a regression model, $Y = b_0 + b_1X + b_2X^2$, for the given data. Also, find MAD and MSE of the forecasting values.

Solution. Let the regression model for the given problem be:

$$Y = b_0 + b_1 X + b_2 X^2$$

where Y is the power consumption; X , the time period; b_0 , the intercept; and b_1 and b_2 are the slopes of X and X^2 terms. We also have

The number of independent variables, $k = 2$

The number of sets of observations, $n = 6$

The sets of observations for the regression model are summarized in Table 10.35.

Table 10.35 Details of Power Consumption

Power consumption (scaled units) (Y)	Period (X)	X^2
40	1	1
60	2	4
100	3	9
70	4	16
60	5	25
90	6	36

Y is the dependent variable and X and X^2 are the independent variables. Matrix form of the given data is

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix}, \quad Y = \begin{bmatrix} 40 \\ 60 \\ 100 \\ 70 \\ 60 \\ 90 \end{bmatrix}, \quad b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$$

↑
Coefficients for b_0 term

Then

$$[X^T X]^{-1} = \begin{bmatrix} 3.20 & -1.950000 & 0.25000000 \\ -1.95 & 1.369643 & -0.18750000 \\ 0.25 & -0.187500 & 0.02678572 \end{bmatrix}$$

and

$$[X^T Y] = \begin{bmatrix} 420 \\ 1580 \\ 7040 \end{bmatrix}$$

Multiplying them, we get

$$\begin{aligned}
 b &= [X^T X]^{-1} [X^T Y] \\
 &= \begin{bmatrix} 3.20 & -1.950000 & 0.25000000 \\ -1.95 & 1.369643 & -0.18750000 \\ 0.25 & -0.187500 & 0.02678572 \end{bmatrix} \begin{bmatrix} 420 \\ 1580 \\ 7040 \end{bmatrix} \\
 &= \begin{bmatrix} 23.000000 \\ 25.035650 \\ -2.678574 \end{bmatrix}
 \end{aligned}$$

The values of the regression coefficients are:

$$b_0 = 23, \quad b_1 = 25.03565, \quad b_2 = -2.678574$$

Therefore, the regression model of the given example is:

$$Y = b_0 + b_1 X_1 + b_2 X_2 = 23 + 25.03565 X - 2.678574 X^2$$

The estimated values of Y and errors are summarized in Table 10.36. The mean absolute deviation and the mean square error of this model are 13.0714 and 240.119, respectively.

Table 10.36 Summary of Estimates and Errors

Period (X)	Power consumption (scaled units) (Y)	Estimate ($\hat{Y}$)	Error
1	40	45.35708	-5.35708
2	60	62.35701	-2.35700
3	100	73.99978	26.00022
4	70	80.28542	-10.28542
5	60	81.21390	-21.21390
6	90	76.78524	13.21476

10.4.3 Time-series Analysis

In practice, the shape of the curve representing the forecast of an event is one or more than one of the following patterns:

- Seasonal pattern
- Cyclical pattern
- Trend pattern
- Random pattern.

Consider the case of predicting the demand of a product manufactured in a company for the purpose of explaining the above patterns.

The seasonal demand exists when the demand fluctuates according to some seasonal factors. The season may be summer season or winter season, during which the sale of refrigerators, soft drinks, wool items, etc., will be more in one season and less in another season.

In the *cyclical pattern*, the length of a single cycle is longer than a year. The cycle does not repeat at constant intervals. The price of metals and gross national product (GNP) of a country are some examples which will have the cyclical pattern.

The *trend pattern* exists when the data has definite positive or negative trend, which means increase or decrease in demand values.

The *random pattern* may exist in some situations for which no reason can be given. Since it happens purely due to random process, it will be accounted under the error of the forecast if it is present.

In Section 10.3, a detailed account of the techniques to deal with the problems involving trend and random patterns is given. But, in business forecasting and economic forecasting, the other two patterns, namely seasonal pattern and cyclical pattern may be existing in addition to the trend and random patterns.

The event of our interest under such situation is said to have time-series data, and the related analysis is called *time-series analysis*. The time-series analysis is applied for short-range prediction.

The components of time series are seasonal (S), cyclical (C), trend (T) and random (R), as indicated below:

$$y = SCTR$$

where y is the time series.

The seasonal component of the time series can be removed by taking moving averages for the original time-series, as shown below. The moving average is free from seasonal effect and will have randomness to certain extent (i.e. $MA = TC$)

$$s = \frac{y}{MA}$$

where s is unadjusted seasonal index and MA is the moving average of the original time-series.

Smoothed seasonal index (S) for each season can be obtained by taking the average of the unadjusted seasonal indices of that season and further it may be adjusted by multiplying it with the following factor, X .

$$X = \frac{\text{Number of seasons}}{\text{Total of the averages of unadjusted seasonal indices of different seasons}}$$

Then deseasonalized forecast for each of the time-series values can be obtained using the following formula:

$$Y = \frac{y}{S}$$

The trend estimates ($\hat{Y}$) for different periods can be obtained using a regression model designed based on the deseasonalized forecast (Y).

The cyclical component can be obtained by dividing the deseasonalized forecast (Y) by the trend estimate ($\hat{Y}$).

Steps of time series

The steps of time-series analysis are summarized below:

Step-1: Input the original time-series data (y).

Step-2: Find moving averages (MA).

Step-3: Obtain seasonal indices (S)

Step-4: Obtain deseasonalized value of y by using the formula:

$$Y = \frac{y}{S}$$

Step-5: Fit a regression model for the deseasonalized forecast (Y):

$$Y = a + bX$$

Step-6: Obtain trend estimates ($\hat{Y}$) using $\hat{Y} = a + bX$.

Step-7: Find the cyclical component using the equation:

$$C = \frac{Y}{\hat{Y}}$$

Step-8: Stop.

Note: The cyclical component cannot be used for estimation of forecast, but it is used only for analysis.

Example 10.16 The demand (in thousands of units) of refrigerators manufactured by a company are summarized in Table 10.37.

Table 10.37 Original Time-series of Refrigerators

Year	Quarter			
	1	2	3	4
1	124	120	176	140
2	104	128	148	132
3	80	152	168	156
4	116	156	184	180

Perform the steps of time series to get the following:

- Deseasonalized forecast
- Trend forecast
- Cyclical component.

Solution. Let the original time-series be y , as given below:

$$y = SCTR$$

Step-1: The values of the original time series (y) are shown in Table 10.38.

Step-2: Since each year consists of four quarters, to even out the seasonal variations, find four quarters centred moving averages (centred MA), as shown in Table 10.38. In Table 10.38, the column, MT, consists of four quarters moving totals. Then, the centred moving total (centred MT) for each of the moving totals is obtained by taking two consecutive moving totals starting from top and placed from period 3 onwards. Theoretically, the first centred moving total is to be placed in between the second period and the third period, but it is placed in period 3.

Step-3: The last column of the Table 10.38 is obtained by dividing the original time-series data by the respective centred moving averages. Later, these indices (S) are copied onto the respective quarters, as shown in Table 10.39.

Table 10.38 Computation of Centred Moving Averages and Deseasonalized Indices

Year	Quarter	Period (X)	y	MT	Centred MT	Centred MA	S
1	1	1	124				
	2	2	120				
	3	3	176		550	137.5	1.280000
	4	4	140	560	544	136.0	1.029412
2	1	5	104	540	534	133.5	0.779026
	2	6	128	548	516	129.0	0.992248
	3	7	148	520	500	125.0	1.184000
	4	8	132	512	500	125.0	1.056000
3	1	9	80	488	522	130.5	0.613027
	2	10	152	512	544	136.0	1.117647
	3	11	168	532	574	143.5	1.170732
	4	12	156	556	594	148.5	1.050505
4	1	13	116	592	604	151.0	0.768212
	2	14	156	596	624	156.0	1.000000
	3	15	184	612			
	4	16	180	636			

In Table 10.39, the average of the indexes in each column is obtained. The total of these averages is 4.013603. Then, each such average is multiplied by the factor $4/4.013603$ to get the smoothed seasonal indices (S) for different quarters, as shown in the last row of the Table 10.39.

Table 10.39 Computation of Seasonal Indices

Year	Quarter			
	1	2	3	4
1	—	—	1.280000	1.029412
2	0.779026	0.992248	1.184000	1.056000
3	0.613027	1.117647	1.170732	1.050505
4	0.768212	1.000000	—	—
Average	0.720088	1.036632	1.211577	1.045306
Total of the averages = 4.013603				
Smoothed seasonal index (S)	0.717648	1.0331186	1.207471*	1.0417632

Step-4: In Table 10.40, the original time-series values (y) are divided by the respective smoothed seasonal indices (S) to get the corresponding deseasonalized forecast values (Y).

Step-5: The necessary calculations to fit a regression model for the deseasonalized forecast (Y) are summarized in Table 10.40. Based on these calculations, the regression model is fitted, as shown below:

$$\bar{Y} = \frac{2272.48}{16} = 142.03$$

$$b = \frac{\sum xY}{\sum x^2} = \frac{942.14}{1360} = 0.69275$$

$$a = \bar{Y} = 142.03$$

$$Y = a + bx$$

$$= 142.03 + 0.69275(x)$$

$$= 142.03 + 0.69275 \frac{X - 8.5}{0.5}$$

$$= 130.25325 + 1.3855X$$

Step-6: Using the regression model as obtained in step-5, the trend values ($\hat{Y}$) are estimated and summarized, as shown in Table 10.40.

Step-7: The values of cyclical component (C) are obtained using the following formula and these are summarized in the last column of Table 10.40.

$$C = \frac{Y}{\hat{Y}}$$

Step-8: Stop.

Table 10.40 Calculations to Fit Regression for Deseasonalized Values and Summary of Trend Values and Cyclical Component Values

PART-1							PART-2	
Period (X)	y	Seasonal index (S)	Deseasonalized value of y (Y)	$x = (X - 8.5)/0.6$	x^2	xy	$\hat{Y} = 130.25325 + 1.3855X$	$C = Y/\hat{Y}$
1	124	0.717648	172.79	-15	225	-2591.85	131.64	1.3126
2	120	1.033119	116.15	-13	169	-1509.95	133.02	0.8732
3	176	1.207471	145.76	-11	121	-1603.36	134.41	1.0844
4	140	1.041763	134.39	-9	81	-1209.51	135.80	0.9896
5	104	0.717648	144.92	-7	49	-1014.44	137.18	1.0564
6	128	1.033119	123.90	-5	25	-619.50	138.57	0.8941
7	148	1.207471	122.57	-3	9	-367.71	139.95	0.8758
8	132	1.041763	126.71	-1	1	-126.71	141.34	0.8965
9	80	0.717648	111.48	1	1	111.48	142.72	0.7811
10	152	1.033119	147.13	3	9	441.39	144.11	1.0210
11	168	1.207471	139.13	5	25	695.65	145.49	0.9563
12	156	1.041763	149.75	7	49	1048.25	146.88	1.0195
13	116	0.717648	161.64	9	81	1454.76	148.26	1.0902
14	156	1.033119	151.00	11	121	1661.00	149.65	1.0090
15	184	1.207471	152.38	13	169	1980.94	151.04	1.0089
16	180	1.041763	172.78	15	225	2591.70	152.42	1.1336
Total			2272.48	0	1360	942.14		

QUESTIONS

1. Define multivariate analysis. Give any four examples.
2. What is correlation coefficient? Discuss the role of correlation coefficient in management decision making.
3. What are rank correlation and auto-correlation?
4. The annual labour welfare fund (lakhs of rupees) and the corresponding annual production (in crores of rupees) for the past 8 years of a company are presented below:

Year (i)	Annual labour welfare fund (X_i)	Annual production (Y_i)
1	8	18
2	10	28
3	12	35
4	14	45
5	16	50
6	18	70
7	20	85
8	22	95

Find the correlation coefficient between annual labour welfare funds and annual production. Also, test the significance of the correlation coefficient at a significance level of 0.05.

5. The annual training expenditure (lakhs of rupees) and the corresponding labour productivity index (0-100) for the past 8 years of a company are presented below:

Year (i)	Annual training expenditure (X_i)	Productivity index (Y_i)
1	5	80
2	7	90
3	9	75
4	10	85
5	12	95
6	15	70
7	18	95
8	20	60

Find the correlation coefficient between annual training expenditure and labour productivity index.

6. A district industries centre (DIC) has collected data summarizing the number of industries in the district under each combination of R&D expenditure (in lakhs of rupees) and annual sales (in crores of rupees), as shown below:

Annual sales (crores of Rs.)	R&D expenditure (in lakhs of Rs.)		
	10-30	30-50	50-70
10-14	3	6	2
14-18	6	8	3
18-22	4	8	9
22-26	2	4	7

Find the correlation coefficient of this grouped data.

7. A consultant has collected data summarizing the number of projects executed by a company under each combination of project cost (in crores of rupees) and rate of return of project (in %), as shown below:

Rate of return (%)	Project cost (in crores of Rs.)		
	4-8	8-12	12-16
10-12	1	4	1
12-14	2	6	1
14-16	3	8	7
16-18	1	3	5

Find the correlation coefficient of this grouped data.

8. The preference ratings of 10 brands of washing machine by two judges are, as shown below:

Brand	Rating by judge-1	Rating by judge-2	Brand	Rating by judge-1	Rating by judge-2
1	1	3	6	6	7
2	2	5	7	7	8
3	3	6	8	8	10
4	4	2	9	9	1
5	5	4	10	10	9

Find the rank correlation between the preference ratings assigned by the judges for different brands of washing machines. Also, test the significance of the coefficient at a significance level of 0.05.

9. The preference ratings of 8 brands of soft drink by two judges are, shown as follows:

Brand	Rating by judge-1	Rating by judge-2
1	2	4
2	3	5
3	1	7
4	8	3
5	5	6
6	4	8
7	7	2
8	6	1

Find the rank correlation between the preference ratings assigned by the judges for the brands of soft drinks.

10. The demand values of a product over the past 10 years are summarized below:

Year (t)	Demand (in '000)	Year (t)	Demand (in '000)
1	40	6	75
2	45	7	87
3	52	8	90
4	60	9	95
5	68	10	110

Find the auto-correlation coefficients with one year lag (r_1), two year lag (r_2) and three year lag (r_3).

11. The demand values of a product over the past 10 years are summarized below:

Year (t)	Demand (in '000)	Year (t)	Demand (in '000)
1	100	6	700
2	150	7	890
3	250	8	950
4	350	9	1000
5	500	10	1200

Find the auto-correlation coefficients with one year lag (r_1), two year lag (r_2) and three year lag (r_3).

12. What are the types of forecast? Explain them in brief.
13. Discuss the usefulness of forecasting in decision making.
14. What are the measures of forecast accuracy? Explain them with examples.
15. Explain Delphi method of forecasting with an example.
16. What are the limitations of moving average method?
17. The actual demand of a product for 8 months are summarized as follows:

Month (t)	Demand (D_t)
1	20
2	26
3	32
4	44
5	66
6	88
7	96
8	112

- Find three months moving averages.
- Compute forecast error.
- Compute mean square error (MSE) and mean absolute deviation (MAD).

18. The actual demand (in thousands of units) of a product for 6 months are summarized below:

Month (t)	Demand (D_t)
1	16
2	24
3	40
4	60
5	85
6	98

- Find three months moving averages.
- Compute forecast error.
- Compute mean square error (MSE) and mean absolute deviation (MAD).

19. The actual demand of a product for 6 months are summarized below:

Month (t)	Demand (D_t)
1	30
2	45
3	70
4	90
5	120
6	140

- Find three months weighted moving averages by assuming the weights, $W_1 = 0.1$, $W_2 = 0.4$ and $W_3 = 0.5$.

- (b) Compute forecast errors.
- (c) Compute mean square error and mean absolute deviation.
20. The forecast of a product for the first week of July was 800 units, whereas the actual demand turned out to be 850 units.
- (a) Find the forecast for the second week of July by assuming the smoothing constant (α) as 0.25.
- (b) Find the forecast for the third week of July if the actual demand of the second week is 750 units.
21. The forecast of a product for the first week of January was 200 units, whereas the actual demand turned out to be 220 units.
- (a) Find the forecast for the second week of January by assuming the smoothing constant (α) as 0.2.
- (b) Find the forecast for the third week of January if the actual demand of the second week is 210 units.
22. The demand values (in lakhs of tons) of Alpha brand of fertilizer during the past 9 years are summarized below:

Year (X)	Demand (Y)	Year (X)	Demand (Y)
1	20	6	70
2	27	7	78
3	35	8	88
4	42	9	98
5	55	—	—

- (a) Fit a linear regression to estimate the demand of the brand in future.
- (b) Compute the demand of the brand for the year 10.
23. A company has the sales pattern (in units) during 1995 to 2002 as shown below:

Year	Sales (in lakhs of units)
1995	15
1996	19
1997	27
1998	35
1999	45
2000	56
2001	70
2002	86

- (a) Fit a linear regression for the data.
- (b) Compute the sales forecast for the year 2005.

24. The sales (in crores of rupees) of two wheeler company for the past 7 years as a function of R&D expenditure (in lakhs of rupees) are summarized below:

Year	R&D expenditure (X)	Sales (Y)
1	15	30
2	20	45
3	25	60
4	30	90
5	35	120
6	40	135
7	45	120

- (a) Fit a regression model to estimate the sales (Y) for a given level of R&D expenditure (X) and also find the mean square error if this model is used.
- (b) Fit a regression model to estimate the R&D expenditure (X) for a desired sales (Y) and also find the mean square error if this model is used.
25. The sales (in crores of rupees) of a soft drink company for the past 7 years as a function of advertising expenditure (in lakhs of rupees) are summarized below:

Year	Advertising expenditure (X)	Sales (Y)
1	25	50
2	30	60
3	40	70
4	50	90
5	55	100
6	60	120
7	90	150

- (a) Fit a regression model to estimate the sales (Y) for a given level of advertising expenditure (X) and also find the mean square error if this model is used.
- (b) Fit a regression model to estimate the advertising expenditure (X) for a desired sales (Y) and also find the mean square error if this model is used.
26. The results of a survey on the sales of a product (Y) as a function of time period (X) are summarized below:

	X	Y
Mean	40	125
Standard deviation	2.5	16
Correlation coefficient (r)		0.85

- (a) Fit the regression line of $\hat{Y}$ on X and estimate the value of Y when X is 45.
- (b) Fit the regression line of X on Y and estimate the value of X when Y is 135.

27. The results of a survey on the sales of a product (Y) as a function of time period (X) are summarized below:

	X	Y
Mean	80	150
Standard deviation	9	25
Correlation coefficient (r)		0.8

- (a) Fit the regression line of Y on X and estimate the value of Y when X is 90.
 (b) Fit the regression line of X on Y and estimate the value of X when Y is 175.
28. The annual sales of readymade garments of a company (in crores of rupees) are summarized below:

Year (X)	Annual sales (Y)
1	15
2	20
3	28
4	42
5	58
6	75
7	98

- (a) Fit a regression model to estimate the annual sales.
 (b) Test the significance of regression model at a significance level of 0.01.
29. The rate of return (in percentage) of a product as a function of R&D expenditure (in lakhs of rupees) and annual advertising expenditure (in lakhs of rupees) for the past 7 years are summarized below:

Rate of return (Y)	R&D expenditure (X_1)	Annual advertising expenditure (X_2)
12	12	30
15	16	50
14	18	65
18	20	75
17	18	80
15	25	95
20	30	105

Design a regression model to forecast the rate of return of the product.

30. The annual sales revenue (in crores of rupees) of a product as a function of time

- period and annual advertising expenditure (in lakhs of rupees) for the past 5 years are summarized below:

Annual sales revenue (X)	Year (X_1)	Annual advertising expenditures (X_2)
25	1	20
30	2	25
38	3	30
45	4	35
60	5	40

Design a regression model to forecast the annual sales revenue of the product using matrix method.

31. The sales (in crores of rupees) of a soft drink company for the past 7 years as a function of advertising expenditure (in lakhs of rupees) are summarized below:

Year	Advertising expenditure (X)	Sales (Y)
1	20	50
2	25	58
3	30	65
4	35	75
5	40	80
6	45	95
7	50	110

Fit a regression model to estimate the sales (Y) for a given level of advertising expenditure (X) using matrix method.

32. The demand of a product over 7 years are summarized below:

Year (X)	Demand (Y)
1	20
2	30
3	25
4	50
5	60
6	70
7	90

Fit a regression model, $Y = b_0 + b_1X + b_2X^2$ for the given data.

33. What are the components of time series? Explain them in brief.
 34. Discuss the steps of time-series analysis.

35. The demand (in thousands of units) of ice-cream products of a company are summarized below:

Original Time Series of Ice-cream Products

Year	Quarter			
	1	2	3	4
1	104	100	156	120
2	84	108	128	112
3	60	132	148	136
4	96	136	164	160

Perform the steps of time series to get the following:

- Deseasonalized forecasts
- Trend forecast
- Cyclical component.

36. The inflow of tourists (in thousands of units) to a state are summarized below:

Original Time Series of Tourists Inflow

Year	Quarter			
	1	2	3	4
1	54	50	106	70
2	34	59	78	62
3	10	82	98	86
4	46	86	114	110

Perform the steps of time series to get the following:

- Deseasonalized forecast
- Trend forecast
- Cyclical component.

CHAPTER

11

ADVANCED MULTIVARIATE ANALYSIS

11.1 INTRODUCTION

In this chapter, the following advanced topics of multivariate analysis are presented: discriminant analysis, factor analysis and cluster analysis.

The discriminant analysis aims at studying the effect of two or more predictor variables on certain evaluation criterion. The evaluation criterion may be good or bad, like or dislike, successful or unsuccessful, etc.

Factor analysis is intended to grouping the original input variables into factors which underlie the input variables. Each factor will account for one or more input variables. Theoretically, the total number of factors in the factor analysis is equal the total number of input variables. But, after performing factor analysis, the total number of factors in the study can be reduced by dropping the insignificant factors based on certain criterion. This is demonstrated through algorithms along with examples.

Cluster analysis aims at grouping of persons/objects/occasions into unknown number of groups such that the members of each group are having similar characteristics/attributes. The different techniques of cluster analysis form groups such that the similarity among the members within each group is maximized. The cluster algorithms are classified into *hierarchical* clustering algorithms and *non-hierarchical* clustering algorithms. The different techniques under each of these categories are presented along with examples.

11.2 DISCRIMINANT ANALYSIS

In some situation, it may be essential to study the effect of two or more predictor variables on certain evaluation criterion, as listed below:

Good or bad

- While grouping investment alternatives based on rate of return, the criterion of the rate of return will be categorized into 'good' or 'bad'. In this example, 'Investment alternatives' is an entity and each investment alternative is treated as a member of the entity.

- While grouping sales regions based on their sales potential, the criterion of sales potential will be categorized into good or bad. In this example, 'Sales regions' is an entity and each sales region is treated as a member of the entity.

Likewise one can define the entity and entity member of the following examples.

Successful or unsuccessful

- While grouping CEOs of companies based on their overall managerial skills from the point of view of organizational productivity, the criterion will be 'successful' or 'unsuccessful'.
- While grouping R&D projects based on their successful completion, the criterion will be 'successful' or 'unsuccessful'.
- While grouping salesmen in terms of meeting sales target, the criterion will be 'successful' or 'unsuccessful'.

Like or dislike

- While grouping products by consumers in terms of their flavour, the criterion will be 'like' or 'dislike'.
- While grouping executives in a company in terms of their leadership qualities, the criterion will be 'like' or 'dislike'.

Above expected level or below expected level

- While grouping employees after a training programme in terms of their enhanced skills, the criterion will be above expected level or below expected level.
- While grouping the performance of educational institutions in terms of their achievements, the criterion will be 'above expected level' or 'below expected level'.

The researcher will be keen in checking whether the predictor variables discriminate among the groups. More specifically, it is necessary to identify the predictor variable (independent variable) which is more important when compared to the other predictor variable(s). Such analysis is called *discriminant analysis*.

In this section, *two-group discriminant analysis* is presented with an illustration.

The different stages of the discriminant analysis are as listed below:

- Designing a discriminant function like the one shown below:

$$Y = aX_1 + bX_2$$

where Y is the linear composite representing the discriminant function, X_1 and X_2 are the predictor variables (independent variables) which are having effect on the evaluation criterion of the problem of interest.

- Finding the discriminant ratio (K) and determining the variables which account for intergroup differences in terms of group means.

This ratio is the maximum possible ratio between the 'variability between groups' and the 'variability within groups'.

- Finding the critical value which can be used to include a new data set (i.e. new combination of instances for the predictor variables) into its appropriate group.
- Testing the null hypothesis, H_0 : The group means are equal in importance, against the alternate hypothesis, H_1 : The group means are not equal in importance, using F test at a given significance level, α .

In this section, a detailed account of two-group discriminant analysis is given. At the end of this section, direction for multiple discriminant analysis is given.

11.2.1 Steps of Two-group Discriminant Analysis

The steps of discriminant analysis applied to two-groups are as presented below:

Step-1: Input the data. Let the predictor variables representing the two factors be, X_1 and X_2 .

Step-2: Classify the data into two mutually exclusive and collectively exhaustive groups, G_1 and G_2 .

Some sample combinations of G_1 and G_2 are shown below:

Group-1 (G_1)	Group-2 (G_2)
Dislike	Like
Bad	Good
Below	Above

Let n_1 and n_2 , be the number of sets of observations in the Group-1 and Group-2, respectively.

Step-3: Find the mean of X_1 as well as X_2 in each group.

Let $\bar{X}_{1(G1)}$ be the mean of X_1 in Group-1; $\bar{X}_{1(G2)}$ be the mean of X_1 in Group-2; $\bar{X}_{2(G1)}$ be the mean of X_2 in Group-1; and $\bar{X}_{2(G2)}$ be the mean of X_2 in Group-2.

• Also, find the grand mean of X_1 as well as X_2 .

Step-4: In each group, find: $\sum X_1^2$, $\sum X_2^2$, and $\sum X_1 X_2$.

Step-5: Define the linear composite as: $Y = aX_1 + bX_2$

Step-6: Find the values of a and b by solving the following normal equations:

$$a \sum (X_1 - \bar{X}_1)^2 + b \sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) = \bar{X}_{1(G2)} - \bar{X}_{1(G1)}$$

$$a \sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) + b \sum (X_2 - \bar{X}_2)^2 = \bar{X}_{2(G2)} - \bar{X}_{2(G1)}$$

The sum of squares in the above normal equation can be substituted with the following simple formulas:

$$\sum (X_1 - \bar{X}_1)^2 = \sum [X_1 - \bar{X}_{1(G1)}]^2 + \sum [X_1 - \bar{X}_{1(G2)}]^2$$

$$\sum (X_2 - \bar{X}_2)^2 = \sum [X_2 - \bar{X}_{2(G1)}]^2 + \sum [X_2 - \bar{X}_{2(G2)}]^2$$

$$\begin{aligned} \sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) &= \sum [X_1 - \bar{X}_{1(G1)}][X_2 - \bar{X}_{2(G1)}] \\ &\quad + \sum [X_1 - \bar{X}_{1(G2)}][X_2 - \bar{X}_{2(G2)}] \end{aligned}$$

where

$$\sum [X_1 - \bar{X}_{1(G1)}]^2 = \sum X_1^2 - n_1 \bar{X}_{1(G1)}^2$$

$$\sum [X_1 - \bar{X}_{1(G2)}]^2 = \sum X_1^2 - n_2 \bar{X}_{1(G2)}^2$$

$$\sum [X_2 - \bar{X}_{2(G1)}]^2 = \sum X_2^2 - n_1 \bar{X}_{2(G1)}^2$$

$$\sum [X_2 - \bar{X}_{2(G2)}]^2 = \sum X_2^2 - n_2 \bar{X}_{2(G2)}^2$$

$$\sum [X_1 - \bar{X}_{1(G1)}][X_2 - \bar{X}_{2(G1)}] = \sum X_1 X_2 - n_1 \bar{X}_{1(G1)} \bar{X}_{2(G1)}$$

$$\sum [X_1 - \bar{X}_{1(G2)}][X_2 - \bar{X}_{2(G2)}] = \sum X_1 X_2 - n_2 \bar{X}_{1(G2)} \bar{X}_{2(G2)}$$

Note: In each of the above equations, $\sum X_1^2$, $\sum X_2^2$, and $\sum X_1 X_2$ are computed using the data sets of the corresponding group.

Step-7: In each group, find the discriminant score for each combination of the variables X_1 and X_2 . Then, find the average of the discriminant scores of each group and also the grand mean of the discriminant scores for the entire problem.

Step-8: Find the variability between groups (V_{BG}) using the following formula:

$$V_{BG} = n_1 (\bar{S}_1 - \bar{S})^2 + n_2 (\bar{S}_2 - \bar{S})^2$$

where S_1 and S_2 are the means of the discriminant scores in the Group-1 and Group-2, respectively; and $\bar{S}$ is the grand mean of the discriminant scores for the entire problem.

Step-9: Find the variability within groups (V_{WG}) using the following formula:

$$V_{WG} = \sum_{j=1}^{n_1} (S_{1j} - \bar{S}_1)^2 + \sum_{j=1}^{n_2} (S_{2j} - \bar{S}_2)^2$$

where S_{1j} and S_{2j} are the discriminant scores for the j th set of observations in the Group-1 and Group-2, respectively; $\bar{S}_1$ and $\bar{S}_2$ are already defined in the step-8.

Step-10: Find the discriminant ratio, (K) using the following formula and identify the predictor variable(s) which has/have more importance when compared to the other predictor variables.

$$K = \frac{V_{BG}}{V_{WG}}$$

This is the maximum possible ratio between 'the variability between groups' and 'the variability within groups'.

Step-11: Validate the discriminant function using the given data sets by forming groups based on the critical discriminant score (grand mean of discriminant scores). If the discriminant score of a data set is less than the critical discriminant score, then include the member of the entity representing that data set in the 'Below' category; otherwise, include it in the 'Above' category.

Direction to classify entity member of future data set

In future, if the values of the predictor variables are known, then the discriminant score of that data set can be obtained using the discriminant function. Then, as per the guidelines stated earlier, the corresponding member of the entity representing that data set can be included in the appropriate group.

Step-12: Find F ratio using the following formula:

$$F = \frac{n_1 n_2 (n_1 + n_2 - m - 1)}{m (n_1 + n_2) (n_1 + n_2 - 2)} D^2$$

and

$$D^2 = (n_1 + n_2 - 2) \left\{ a \left[\bar{X}_{1(G2)} - \bar{X}_{1(G1)} \right] + b \left[\bar{X}_{2(G2)} - \bar{X}_{2(G1)} \right] \right\}$$

where m is the number of predictor variables (in this case, it is equal to 2); and D^2 is known as Mahalanobi's squared distance; and degrees of freedom = $[m, (n_1 + n_2 - m - 1)]$.

Step-13: Find the table F value for $[m, (n_1 + n_2 - m - 1)]$ degrees of freedom at a significance level of α .

Step-14: If the calculated F value is more than the table F value, reject the null hypothesis, H_0 ; otherwise, accept the null hypothesis (H_0), where

H_0 : The group means are equal in importance.

H_1 : The group means are not equal in importance.

(This means that one variable is more important than the other variable).

Example 11.1 The director of a management school wants to do discriminant analysis concerning the effect of two factors, namely, the yearly spending (in lakhs of rupees) on infrastructures of the school (X_1) and the yearly spending on interface events of the school (X_2) on the grading of the school by an inspection team. He has collected the data for the past 12 years and submitted to the inspection team as shown in Table 11.1. Based on the

data, the committee has awarded one of the following grades for each year, as shown in the same table.

- Design the discriminant function, $Y = aX_1 + bX_2$.
- Compute the discriminant ratio, K and identify the variable which is more important in relation to the other variable.
- Validate the discriminant function using the given data by forming groups based on the critical discriminant score.
- Test whether the group means are equal in importance at a significance level of 0.05.

Solution. The combination of the hypotheses of this example is as stated below:

H_0 : The group means are equal in importance.

H_1 : The group means are not equal in importance.

- Design of the discriminant function, $Y = aX_1 + bX_2$.

Step-1&2: Input data and the award of grades by the inspection team are as already shown in Table 11.1.

Table 11.1 Yearly Spending (in Lakhs of Rupees) on Infrastructures and Interface Events

Year	Grade	Infrastructure (X_1)	Interface events (X_2)
1	Below	3	4
2	Below	4	5
3	Above	10	7
4	Below	5	4
5	Below	6	6
6	Above	11	4
7	Below	7	4
8	Above	12	5
9	Below	8	7
10	Below	9	5
11	Above	13	6
12	Above	14	8

Step-3: The mean of X_1 as well as X_2 in each group is shown in Table 11.2. The grand means of X_1 and X_2 are also shown in the same table.

Step-4: The calculations of necessary results to solve normal equations are summarized in Table 11.3. Also, they are summarized in refined form as in Table 11.4.

Table 11.2 Summary of Means of X_1 and X_2 for Each Group

Group	Year	Standard	Infrastructure (X ₁)	Infrastructure events (X ₂)
G1	1	Below	3	4
	2	Below	4	5
	4	Below	5	4
	5	Below	6	6
	7	Below	7	4
	9	Below	8	7
	10	Below	9	5
	Total of below		42	35
	Mean of below		6	5
	G2	3	Above	10
6		Above	11	4
8		Above	12	5
11		Above	13	6
12		Above	14	8
Total of above		60	30	
Mean of above		12	6	
Grand mean			8.5	5.41666

Table 11.3 Calculations of Necessary Results to Solve Normal Equations

Group	Year	Standard	X_1	X_2	X_1^2	X_2^2	X_1X_2
G1	1	Below	3	4	9	16	12
	2	Below	4	5	16	25	20
	4	Below	5	4	25	16	20
	5	Below	6	6	36	36	36
	7	Below	7	4	49	16	28
	9	Below	8	7	64	49	56
	10	Below	9	5	81	25	45
	Total of below		42	35	280	183	217
	Mean of below		6	5			

contd.

Table 11.3 Contd.

Group	Year	Standard	X_1	X_2	X_1^2	X_2^2	X_1X_2
G2	3	Above	10	7	100	49	70
	6	Above	11	4	121	16	44
	8	Above	12	5	144	25	60
	11	Above	13	6	169	36	78
	12	Above	14	8	196	64	112
	Total of above		60	30	730	190	364
	Mean of above		12	6			
	Grand. mean		8.5	5.41666			

Table 11.4 Refined Forms of Sum of Squares

Sum of squares	Below	Above	Total
$\Sigma(X_1 - \bar{X}_1)^2 = \Sigma X_1^2 - n\bar{X}_1^2$	28	10	38
$\Sigma(X_2 - \bar{X}_2)^2 = \Sigma X_2^2 - n\bar{X}_2^2$	8	10	18
$\Sigma(X_1 - \bar{X}_1)(X_2 - \bar{X}_2) = \Sigma X_1X_2 - n\bar{X}_1\bar{X}_2$	7	4	11

Note: n is the number of sets of observations in each category (Above/Below).

Step-5: Let the linear composite discriminant function be as follows:

$$Y = aX_1 + bX_2$$

Step-6: The normal equations are as shown below:

$$a \Sigma (X_1 - \bar{X}_1)^2 + b \Sigma (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) = \bar{X}_{1(G2)} - \bar{X}_{1(G1)}$$

$$a \Sigma (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) + b \Sigma (X_2 - \bar{X}_2)^2 = \bar{X}_{2(G2)} - \bar{X}_{2(G1)}$$

Substitution of the results from the last column of Table 11.4, the normal equations are as presented below:

$$38a + 11b = 12 - 6 = 6 \quad (1)$$

$$11a + 18b = 6 - 5 = 1 \quad (2)$$

The solution to the above two simultaneous equations are: $a = 0.17229$ and $b = -0.04973$. Hence, the discriminant function is as shown below:

$$Y = 0.17229X_1 - 0.04973X_2$$

(b) *Computation of discriminant ratio, K and identification of the variable which is more important in relation to the other variable.*

Step-7: Discriminant score for each combination of X_1 and X_2 in each group, the average of discriminant scores of each group and the grand mean of the discriminant scores are summarized in Table 11.5.

Alternatively, the mean discriminant score of each group as well the grand mean discriminant score can be obtained using the discriminant function, as shown below:

$$\begin{aligned}\bar{Y}(\text{Below}) &= 0.17229\bar{X}_1 - 0.04973\bar{X}_2 \\ &= 0.17229 \times 6 - 0.04973 \times 5 \\ &= 0.78509\end{aligned}$$

Table 11.5 Summary of Discriminant Scores and their Group Averages

Discriminant function: $Y = 0.17229X_1 - 0.04973X_2$							
Below (Group-G1)				Above (Group-G2)			
Data set (j)	Year	Discriminant score (S_{1j})	$(S_{1j} - \bar{S}_1)^2$	Data set (j)	Year	Discriminant score (S_{2j})	$(S_{2j} - \bar{S}_2)^2$
1	1	0.31795	0.218220	1	6	1.69627	0.005304
2	2	0.44051	0.118735	2	8	1.81883	0.002473
3	4	0.66253	0.015021	3	11	1.94139	0.029684
4	5	0.73536	0.002473	4	12	2.01422	0.060084
5	7	1.00711	0.049293				
6	9	1.03021	0.060084				
7	10	1.30196	0.267155				
Total		5.49563	0.730981	Total		8.84550	0.253025
		Mean ($\bar{S}_1$) 0.78509				Mean ($\bar{S}_2$) 1.7691	
Grand Total of discriminant scores = 14.34113							
Grand mean of discriminant scores ($\bar{S}$) = 1.195094							

$$\begin{aligned}\bar{Y}(\text{Above}) &= 0.17229\bar{X}_1 - 0.04973\bar{X}_2 \\ &= 0.17229 \times 12 - 0.04973 \times 6 \\ &= 1.7691\end{aligned}$$

$$\begin{aligned}\bar{Y}(\text{Grand mean}) &= 0.17229\bar{X}_1 - 0.04973\bar{X}_2 \\ &= 0.17229 \times 8.5 - 0.04973 \times 5.41666 \\ &= 1.195094\end{aligned}$$

Step-8: The variability between groups (V_{BG}) is computed using the following formula :

Sum of squares between groups:

$$\begin{aligned} V_{BG} &= n_1(\bar{S}_1 - \bar{S})^2 + n_2(\bar{S}_2 - \bar{S})^2 \\ &= 7(0.78509 - 1.195094)^2 + 5(1.7691 - 1.195094)^2 \\ &= 2.824137 \end{aligned}$$

Step-9: The variability within groups (V_{WG}) is computed using the following formula:

Sum of squares within groups:

$$\begin{aligned} V_{WG} &= \sum_{j=1}^7 (S_{1j} - \bar{S}_1)^2 + \sum_{j=1}^5 (S_{2j} - \bar{S}_2)^2 \\ &= 0.730981 + 0.253025 \\ &= 0.984006 \end{aligned}$$

Step-10: The discriminant ratio, K is determined as shown below:

$$K = \frac{V_{BG}}{V_{WG}} = \frac{2.824137}{0.984006} = 2.87$$

This is the maximum possible ratio between 'the variability between groups' and 'the variability within groups'. This means that if the variable X_2 is suppressed in the discriminant function, then the revised value of the discriminant ratio will be less than the previous value.

In the discriminant function, $Y = 0.17229X_1 - 0.04973X_2$, the coefficient of X_2 has negative sign which indicates that the variable X_1 (spending on infrastructure) is more important than the variable X_2 (spending on interface events).

(c) *Validate the discriminant function using the given data by forming groups based on the critical discriminant score.*

Step-11: The discriminant function can be validated based on the grand mean of the discriminant scores which is known as the critical discriminant score. If the discriminant score of a data set is less than the critical discriminant score (1.195094), include that data set into group corresponding to 'Below' category; otherwise, include that data set into 'Above' category.

As per these guidelines, the classification of the twelve data sets of the given problem is shown in Table 11.6. One can notice that the discriminant function has grouped all the years into their original category as given in the problem except the year 10.

Table 11.6 Classification of Data Sets Based on Critical Discriminant Score
(Each Entry of the Table Represents Year)

Below	Above
1	10
2	3
4	6
5	8
7	11
9	12

Direction for including future data set: In future, say for the 11th year, if the values of the predictor variables, X_1 and X_2 are known, then its discriminant score can be obtained using the discriminant function. Then, as per the guidelines stated earlier, that year can be included in the appropriate group.

(d) *Testing whether the group means are equal in importance at a significance level of 0.05.*

Step-11: The formula to compute F is as shown below:

$$F = \frac{n_1 n_2 (n_1 + n_2 - m - 1)}{m (n_1 + n_2) (n_1 + n_2 - 2)} D^2$$

where m is the number of predictor variables, (in this case, it is equal to 2);

$$\begin{aligned} D^2 &= (n_1 + n_2 - 2) \left\{ a [\bar{X}_{1(G2)} - \bar{X}_{1(G1)}] + b [\bar{X}_{2(G2)} - \bar{X}_{2(G1)}] \right\} \\ &= (7 + 5 - 2) (0.17229 \times 6 - 0.04973 \times 1) \\ &= 9.8401 \end{aligned}$$

and

$$F = \frac{7 \times 5 \times (7 + 5 - 2 - 1)}{2 \times (7 + 5) \times (7 + 5 - 2)} \times 9.8401 = 12.915$$

Step-12: The degrees of freedom for the F ratio is given by the formula $[m, (n_1 + n_2 - m - 1)]$, where m is the number of factors (2).

The table F value for (2, 9) degrees of freedom and at the given significance level of 0.05 is 4.26.

Step-13: The calculated F value (12.915) is more than the table F value (4.26). Hence, reject the null hypothesis, H_0 and accept H_1 . Thus, we get the alternate hypothesis,

H_1 : The group means are not equal in importance.

Based on H_1 and the discriminant function, it is clear that the variable X_1 (annual spending on infrastructures) is more important than the other variable X_2 (annual spending in interface events) of the business school.

11.2.2 Directions for Multiple Discriminant Analysis

In a discriminant analysis, let the number of groups be N and the number of predictor variables be m (which should be at least 2). Consider a situation wherein the entity of interest

is salesmen of a company. The profile of the entity members in terms of predictor variables are defined as shown below:

Let X_1 be the age of the salesman; X_2 be the marketing work experience of the salesman.

The salesmen can be classified, say, into three groups in terms of meeting sales target, as listed below:

- Good performer (Good)
- Normal performer (Normal)
- Poor performer (Poor)

If the number of groups in a discriminant analysis is more than 2, then the number of non-redundant discriminant functions to be proposed can be the minimum of $(N - 1)$ and m , where N is the number of groups for classification of entity members and m is the number of predictor variables.

Let $N = 3$ (Good, Normal and Poor); and $m = 2$ (X_1 and X_2). For this problem, the maximum number of discriminant functions which can be proposed is 2 as $\min(N - 1, m) = \min(3 - 1, 2) = 2$.

If we plan to propose two non-redundant discriminant function, then a sample combination of the discriminant functions is:

$$Y_1 = aX_1 + bX_2, \quad Y_2 = cX_1 + dX_2$$

The DISCRIMINANT module of SPSS package or DISCRIM module of SAS package can be used to do multiple discriminant analysis.

11.3 FACTOR ANALYSIS

In many real-life applications, the number of independent variables used in predicting a response variable will be too many. The difficulties in having too many independent variables in such exercise are as follows:

- Increased computational time to get solution
- Increased time in data collection
- Too much expenditure in data collection
- Presence of redundant independent variables
- Difficulty in making inferences.

These can be avoided using *factor analysis*. Factor analysis aims at grouping the original input variables into factors which underlie the input variables. Each factor will account for one or more input variables. Theoretically, the total number of factors in the factor analysis is equal to the total number of input variables. But, after performing factor analysis, the total number of factors in the study can be reduced by dropping the insignificant factors based on certain criterion.

To demonstrate these aspects, consider the problem of studying the customer feedback about a two-wheeler produced by a company, as explained below.

The marketing manager of a two-wheeler company designed a questionnaire to study the customers feedback about its two-wheeler and in turn he is keen in identifying the factors of his study. He has identified six variables which are:

1. Fuel efficiency (X_1)
2. Life of the two-wheeler (X_2)
3. Handling convenience (X_3)
4. Quality of original spares (X_4)
5. Breakdown rate (X_5)
6. Price (X_6).

After administering a questionnaire among respondents, the opinion of the customers can be obtained on the above variables. Let the range of the score for each of the above variables is assumed to be between 1 and 10. The score 1 means the lowest rating and 10 means the highest rating. Let us assume that the application of factor analysis groups these variables, as follows:

- X_1, X_2, X_4 and X_5 into factor-1;
- X_6 into factor-2;
- X_3 into factor-3.

If all the three factors are significant, then they are retained for future analysis. A careful examination of these groupings of variables into factors reveals that the factor-1, factor-2 and factor-3 can be named as technical factor, price factor and personal factor, respectively.

So, in future while conducting a detailed study, it is sufficient to get opinion of the customers on the three factors which are obtained through factor analysis.

As explained earlier, if there are n variables, there will be n factors. Each factor, say factor k is represented by a linear composite. Let F_k be the linear composite of the factor k which is given by

$$F_k = w_{1k}X_1 + w_{2k}X_2 + w_{3k}X_3 + \cdots + w_{ik}X_i + \cdots + w_{nk}X_n$$

where w_{ik} is the weight of the original variable X_i in the linear composite of the factor k .

$$F_k = \sum_{i=1}^n w_{ik}X_i, \quad \text{for } k = 1, 2, 3, \dots, n$$

The above formula for F_k can be used to find the scores of each set of observations (each row) for the factor k by substituting the values of $X_i, i = 1, 2, 3, \dots, n$ in it.

Consider a situation in which there are only two input variables, X_1 and X_2 . Let, F_1 and F_2 be the two factors which underlie the input variables X_1 and X_2 , respectively.

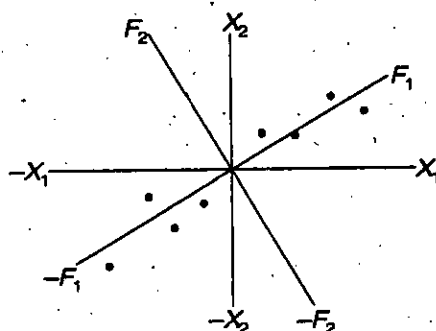


Fig. 11.1 Plot of original variables and factors.

The geometric interpretation of the factors in relation to the input variables is shown in Fig. 11.1.

The plot of the input variables can be covered by the line F_1 which is known as factor-1. This passes through the inter-distances of points. It is clear that all the points are not lying on F_1 . To capture the remaining variation, another factor, F_2 is introduced which is orthogonal to F_1 . Now, with the new linear composites, F_1 and F_2 , the original set of input variables can be studied. After analyzing the factor loadings of the variables on F_1 and F_2 which can be obtained through factor analysis, one can take decision about the number of factors to be retained for future study.

11.4 TERMINOLOGIES OF FACTOR ANALYSIS

Factor analysis involves many terminologies which are presented in this subsection for better understanding of the related techniques.

Correlation coefficient matrix

It is the matrix of correlation coefficients of the original observations between different pairs of input variables.

Factor loadings, $L_i(j)$

It is a matrix representing the correlation between different combinations of variables and factors. $L_i(j)$ is the factor loading of the variable j on the factor i , where $i = 1, 2, 3, \dots, n$ and $j = 1, 2, 3, \dots, n$.

Communality, h_i^2

It is the sum of squares of the factor loadings of the variable i on all factors:

$$h_i^2 = \sum_{j=1}^n L_{ij}^2$$

Eigenvalue

It is the sum of squares of the factor loadings of all variables on a factor.

$$\text{Eigenvalue of the factor } j = \sum_{i=1}^n L_{ij}^2$$

Note: The sum of the eigenvalues of all factors (if no factor is dropped) is equal to the sum of the communalities of all variables.

Rotation

After obtaining factor loadings, one should examine whether the factor loading matrix possesses simple structure. If a factor loading matrix has a simple structure, it is easy to make interpretations about the factors. If there is no simple structure, then the n -dimensional

space of the factors should be rotated by an angle such that the factor loadings are revised to have a simple structure which will simplify the process of interpretation of the factors. Such rotation is called *rotation of factors*.

A simple structure means that each variable has very high factor loading (as high as 1) on one of the factors and very low factor loading (as low as 0) on other factors. The communalities of each variable before and after factor rotation will be the same.

The popular methods of rotation of factors are *varimax* method and *promax* method. Varimax method of factor rotation employs orthogonality between different pairs of factor axes. This means that the angles between different pairs of factors axes are 90° even after rotation. The promax method employs oblique rotation. This means that the angles between different pairs of factors axes are not 90° after rotation. Both the techniques aim at better interpretations.

Significant number of factors

As explained earlier, the main objective of the factor analysis is to group the given set of input variables into minimal number factors with the maximum capability of extracting information with the reduced set of factors.

There are certain criteria to determine the number of factors to be retained for future study which are, as explained below:

Minimum eigenvalue criterion If the eigenvalue (sum of squares of the factor loadings of all variables on a factor) of a factor is more than or equal to 1, then that factor is to be retained; otherwise, that factor is to be dropped.

Scree plot criterion As per this criterion, plot the eigenvalues of the factors by taking the factor number on X-axis and the eigenvalues on Y-axis. Then, identify the factor number at which the slope of the line connecting the points changes from steep to a gradual trailing off towards the right of the identified factor number. Such change in slope in the graph is known as *scree* and the point is known as *scree point*. The factors which are marked up to the scree point from the origin are to be retained for future study and all the factors to the right of the scree point are to be dropped from the study.

Factor scores

Though a factor is not visible like an original input variable, it is still a variable which can be used to find the scores for respondents. At the initial stage, the respondents assign scores for the variables. After performing factor analysis, each factor assigns a score for each respondent. Such scores are called *factor scores*.

The expression to compute the factor score of a respondent by the factor k is shown below. By substituting the standardized values of the input variables assigned by a respondent in this expression, the factor score of that respondent can be obtained:

$$F_k = w_{1k}X_1 + w_{2k}X_2 + w_{3k}X_3 + \dots + w_{ik}X_i + \dots + w_{nk}X_n = \sum_{i=1}^n w_{ik}X_i$$

where w_{ik} is the weight of the input variable X_i in the linear composite of the factor, k , for $k = 1, 2, 3, \dots, n$.

11.5 METHODS OF FACTOR ANALYSIS

The popular methods of factor analysis are *centroid* method, *principle components* method. Each of the methods is explained in the following section. For the principle component method, there are two methods, the *conventional* method and an *alternate* method. For the conventional method, a clear algorithm is given. The alternate method of principle component analysis is demonstrated with a clear algorithm and a numerical example. Also, at the end of this section, the steps of varimax factor rotation method is demonstrated with a hypothetical ill-structured problem.

11.5.1 Centroid Method

This method maximizes the sum of absolute loadings of each factor. In this method, the coefficients of the terms in the linear composite of each factor will be either +1 or -1. This method is less cumbersome when compared to other methods. The steps of this method are explained below. Later, this method is demonstrated using a problem with 6 variables and 10 sets of observations.

Steps of centroid method

The steps of this method are presented below:

Step-1: Input the data matrix of $m \times n$ size, where m is the number of sets of observations and n is the number of characteristics (variables).

Input the total number of factors (F) to be considered. Initialize $f = 1$, where f is the current factor number.

Step-2: Find $n \times n$ correlation coefficient matrix (R_f , i.e. R_1) summarizing the correlation coefficient for each pair of the variables. Each diagonal value of the correlation coefficient matrix is assumed as 1, as the respective variable is correlated to itself.

Step-3: Perform reflections if necessary:

For each column j in the matrix R_1 , do the following:

3.1: Find the total of the column j in the matrix R_1 excluding the diagonal entry in it.

3.2: If the total of each column is positive, then the matrix is positive manifold and assign a weight (w_j) of +1 to each column j (variable j) in the matrix; then, treat the matrix R_1 as the matrix R_1' (the matrix of R_1 with reflections) without any modification in R_1 and go to step-4; otherwise, go to step-3.3.

3.3: Perform reflections: *Reflection* means the determination of the minimum number columns in which the negative values are to be converted into positive values, as per following guidelines:

(a) In a column j , if $R_1(i, j) < 0$ as well as $R_1(j, i) < 0$, then the column j should be reflected. In this case, for each negative value in the row i of the column j do the following updation:

$R_1(i, j) = \text{Absolute value of } R_1(i, j)$

$R_1(j, i) = \text{Absolute value of } R_1(j, i)$

- (b) In the remaining columns, if $R_1(j, j) < 0$, then the column j should be reflected. In this case, do the following updation.

$R_1(j, j) = \text{Absolute value of } R_1(j, j)$

- (c) Treat this revised matrix as the matrix R'_1 , which is known as the matrix with reflections.
- (d) Assignment of weight for the columns $[w_j]$: Assign -1 as the weight for the columns which are reflected and $+1$ as the weight for columns which are not reflected.

Step-4: Determination of first centroid factor:

4.1: For each column j in the matrix R'_1 , find the sum of the entries in it $[S_j]$

4.2: Find the grand total (T) of S_j , $j = 1, 2, 3, \dots, n$.

$$T = \sum_{j=1}^n S_j$$

4.3: Find the root of T , $(\sqrt{T})$.

4.4: Divide each column sum, S_j by $\sqrt{T}$ to get unweighted factor loading $UL_1(j)$ [i.e. $UL_1(j)$] for that column j .

4.5: For each column j , find its weighted factor loading $L_1(j)$ [i.e. $L_1(j)$] by multiplying $UL_1(j)$ with its weight w_j .

Step-5: Increment the current factor number by 1 ($f = f + 1$)

Step-6: Determination of f th centroid factor:

6.1: Determination of cross-product matrix $[P_f]$.

Obtain the product of each pair of factor loadings of $(f-1)$ th centroid factor and store it in the respective row and column of the cross-product matrix, (P_f) .

Let the factor loadings of $(f-1)$ th centroid factor be as: $l_1, l_2, l_3, \dots, l_n$.

The format of cross-product matrix is shown in Table 11.7:

Table 11.7 Format of Cross-product Matrix $[P_f]$

Row (i)	Column (j)				
	1	2	3	...	n
1	$l_1 \times l_1$	$l_1 \times l_2$	$l_1 \times l_3$	...	$l_1 \times l_n$
2	$l_2 \times l_1$	$l_2 \times l_2$	$l_2 \times l_3$	...	$l_2 \times l_n$
3	$l_3 \times l_1$	$l_3 \times l_2$	$l_3 \times l_3$	...	$l_3 \times l_n$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	$l_n \times l_1$	$l_n \times l_2$	$l_n \times l_3$	...	$l_n \times l_n$

6.2: Find the residual matrix R_f by element by element subtraction of the cross-product matrix P_f from the matrix R_{f-1} .

6.3: Perform reflections, if necessary:

- (a) In a column j , if $R_f(i, j) < 0$ as well as $R_f(j, i) < 0$, then the column j should be reflected.

In this case, for each negative value in the row i of the column j do the following updation:

$$R_f(i, j) = \text{Absolute value of } R_f(i, j)$$

$$R_f(j, i) = \text{Absolute value of } R_f(j, i)$$

- (b) In the remaining columns, if $R_f(j, j) < 0$, then the column j should be reflected. In this case, do the following updation.

$$R_f(j, j) = \text{Absolute value of } R_f(j, j)$$

- (c) Treat this revised matrix as the matrix R'_f which is known as the *matrix with reflections*; (If there is no negative weight for any of the columns, then treat the matrix R_f as the matrix R'_f without any modification in it.)

- (d) Assignment of weight for the columns $[w_j]$:

Assign '-1' as the weight for the columns which are reflected and '+1' as the weight for columns which are not reflected.

6.4: For each column j in the matrix R'_f , find the sum of the entries in it $[S_j]$.

6.5: Find the grand total (T) of S_j , $j = 1, 2, 3, \dots, n$.

$$T = \sum_{j=1}^n S_j$$

6.6: Find the root of T , that is $\sqrt{T}$.

6.7: Divide each column sum, S_j by $\sqrt{T}$ to get unweighted factor loading $UL_f(j)$ for that column j .

6.8: For each column j , find its weighted factor loading $L_f(j)$ by multiplying $UL_f(j)$ with its weight w_j .

Step-7: If the current factor number, (f) is less than the total number of factors to be considered (F), then go to step-5; otherwise go to step-8.

Step-8: Arrange the loadings of F factors in columns as shown in Table 11.8.

Table 11.8 Factor Loadings

Variable	Factor							Communality (sum of squares of row)
	1	2	3	...	f	...	F	
X_1	$L_1(1)$	$L_2(1)$	$L_3(1)$	...	$L_f(1)$	...	$L_F(1)$	$\sum L_i(1)^2$
X_2	$L_1(2)$	$L_2(2)$	$L_3(2)$	...	$L_f(2)$	...	$L_F(2)$	$\sum L_i(2)^2$
X_3	$L_1(3)$	$L_2(3)$	$L_3(3)$	...	$L_f(3)$	...	$L_F(3)$	$\sum L_i(3)^2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	
X_i	$L_1(i)$	$L_2(i)$	$L_3(i)$	...	$L_f(i)$	...	$L_F(i)$	$\sum L_i(i)^2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	
X_n	$L_1(n)$	$L_2(n)$	$L_3(n)$	...	$L_f(n)$	...	$L_F(n)$	$\sum L_i(n)^2$
Eigen value	$\sum L_i(i)^2$	$\sum L_2(i)^2$	$\sum L_3(i)^2$	...	$\sum L_f(i)^2$	...	$\sum L_F(i)^2$	$\sum \sum L_i(i)^2$

Step-9: Assign each variable to a factor with which it has the maximum absolute loading (irrespective of sign).

Step-10: Find the sum of squares of the loadings in each row which is known *communality* (h^2) and sum of squares of the loadings in each column (eigenvalue of the factor), as shown in Table 11.8. The grand total of the sum of squares of the loadings in Table 11.8 is shown at its bottom-right corner (common variance).

For each factor (f) find the following and state inferences.

$$\text{Proportion of total variance of the factor } f = \frac{\text{Eigenvalue of factor } f}{\text{Total variance}}$$

where, the total variance is equal to the number of variables, n . And

$$\begin{aligned} \text{Proportion of common variance of the factor } f &= \frac{\text{Eigenvalue of factor } f}{\text{Common variance}} \\ &= \frac{\text{Eigenvalue of factor } f}{\sum h^2} \end{aligned}$$

Step-11: End.

Example 11.2 The marketing manager of a two-wheeler company designed a questionnaire to study the customers feedback about its two-wheeler and in turn he is keen in identifying the factors of his study. He has identified six variables which are as listed below:

1. Fuel efficiency (X_1)
2. Life of the two-wheeler (X_2)
3. Handling convenience (X_3)

4. Quality of original spares (X_4)
5. Breakdown rate (X_5)
6. Price (X_6).

So, the company administered a questionnaire among 10 customers to obtain their opinion on the above variables. The range of the score for each of the above variables is assumed to be between 0 and 10. The score 0 means low rating and 10 means high rating. The survey data are summarized in Table 11.9. Perform factor analysis using centroid method to identify three factors which can represent the variables of the study.

Table 11.9 Survey Data

Respondent	Variable					
	X_1	X_2	X_3	X_4	X_5	X_6
1	6	8	9	9	1	2
2	4	4	6	8	2	1
3	4	1	6	5	1	2
4	1	2	6	3	2	3
5	4	3	5	5	2	3
6	4	4	6	8	2	3
7	3	3	0	9	1	3
8	7	7	6	9	9	2
9	5	3	1	8	1	2
10	5	4	2	3	1	0

Solution.

Step-1: The survey data are as already shown in Table 11.9.

Number of variables (n) = 6

Number of sets of observations (respondents), m = 10

Number of factors to be identified, F = 3

Initialize, current factor number, f = 1

Step-2: The correlation coefficient between each pair of the variables is as shown in Table 11.10. This matrix is known as R_1 .

Table 11.10 Correlation Coefficient Matrix (R_1)

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	1.000	0.742	0.168	0.496	0.484	-0.430
X_2	0.742	1.000	0.424	0.568	0.474	-0.204
X_3	0.168	0.424	1.000	0.050	0.238	0.092
X_4	0.496	0.568	0.050	1.000	0.290	0.196
X_5	0.484	0.474	0.238	0.290	1.000	0.037
X_6	-0.430	-0.204	0.092	0.196	0.037	1.000

Step-3: Since the matrix R_1 is not a positive manifold, the variable X_6 is reflected, as shown in Table 11.11.

The weight of the variables (columns) from 1 to 6 are 1, 1, 1, 1, 1, -1.

Table 11.11 Correlation Coefficient Matrix with Reflections on the Variable 6 [R'_1] along with Column Totals

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	1.000	0.742	0.168	0.496	0.484	0.430
X_2	0.742	1.000	0.424	0.568	0.474	0.204
X_3	0.168	0.424	1.000	0.050	0.238	0.092
X_4	0.496	0.568	0.050	1.000	0.290	0.196
X_5	0.484	0.474	0.238	0.290	1.000	0.037
X_6	0.430	0.204	0.092	0.196	0.037	1.000
Total (S_j)	3.320	3.412	1.972	2.600	2.523	1.959

Step-4: The first centroid factor is determined, as shown below:

The sum (S_j) of each column of the matrix R'_1 is shown in Table 11.11 itself.

$$\text{Grand total of column totals, } T = \sum_{j=1}^6 S_j = 15.786$$

or

$$\sqrt{T} = 3.973159$$

The calculations to determine the loadings of factor 1 [$L_1(j)$] are summarized in Table 11.12.

Table 11.12 Calculations to Determine Loading Values of Factor 1 [$L_1(j)$]

j	S_j	$UL_1(j) = S_j/\sqrt{T}$	$L_1(j)$
1	3.320	0.8356	0.8356
2	3.412	0.85876	0.85876
3	1.972	0.49633	0.49633
4	2.600	0.65439	0.65439
5	2.523	0.6350	0.6350
6	1.959	0.49306	-0.49306

Step-5: Increment the factor number by 1; i.e. $f = 1 + 1 = 2$.

Step-6: The second centroid factor is determined in this step.

The cross-product matrix $[P_2]$ which consists of the product of each pair of coefficients of the loadings $L_1(j)$, $j = 1, 2, 3, 4, 5$ and 6 is shown in Table 11.13.

Table 11.13 Cross-product Matrix $[P_2]$

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	0.698	0.718	0.415	0.547	0.531	-0.412
X_2	0.718	0.737	0.426	0.562	0.545	-0.423
X_3	0.415	0.426	0.246	0.325	0.315	-0.245
X_4	0.547	0.562	0.325	0.428	0.416	-0.323
X_5	0.531	0.545	0.315	0.416	0.403	-0.313
X_6	-0.412	-0.423	-0.245	-0.323	-0.313	0.243

The residual matrix (R_2) is obtained by subtracting the cross-product matrix (P_2) from R_1 and it is as shown in Table 11.14.

Table 11.14 Residual Matrix $[R_2]$

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	0.302	0.024	-0.247	-0.051	-0.047	-0.018
X_2	0.024	0.263	-0.002	0.006	-0.071	0.219
X_3	-0.247	-0.002	0.754	-0.275	-0.077	0.337
X_4	-0.051	0.006	-0.275	0.572	-0.126	0.519
X_5	-0.047	-0.071	-0.077	-0.126	0.597	0.350
X_6	-0.018	0.219	0.337	0.519	0.350	0.757

From Table 11.14, it is clear that the columns (variables) 3, 4, 5 and 6 must be reflected. The reflected matrix R'_2 is as shown in Table 11.15. The weights of the columns (variables) from 1 to 6 are +1, +1, -1, -1, -1 and -1, respectively.

Table 11.15 Residual Matrix $[R'_2]$ with Reflections on Variables, 3, 4, 5 and 6

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	0.302	0.024	0.247	0.051	0.047	0.018
X_2	0.024	0.263	0.002	0.006	0.071	0.219
X_3	0.247	0.002	0.754	0.275	0.077	0.337
X_4	0.051	0.006	0.275	0.572	0.126	0.519
X_5	0.047	0.071	0.077	0.126	0.597	0.350
X_6	0.018	0.219	0.337	0.519	0.350	0.757
Total (S_j)	0.689	0.585	1.692	1.549	1.268	2.200

$$\text{Grand total of column totals } T = \sum_{j=1}^6 S_j = 7.983$$

or

$$\sqrt{T} = 2.825421$$

The second centroid factor loadings, $L_2(j)$, are summarized in Table 11.16 after multiplying with the respective column weights.

Table 11.16 Calculations to Determine Loading Values of Factor 2 [$L_2(j)$]

j	S_j	$UL_2(j) = S_j/\sqrt{T}$	$L_2(j)$
1	0.689	0.24386	0.24386
2	0.585	0.20705	0.20705
3	1.692	0.59885	-0.59885
4	1.549	0.54824	-0.54824
5	1.268	0.44878	-0.44878
6	2.200	0.77865	-0.77865

Step-7: The current factor number (f) is 2 which is less than the total number of factors to be considered ($F = 3$). So, go to step-5.

Step-5: Increment the factor number by 1; i.e. $f = 2 + 1 = 3$.

Step-6: The third centroid factor is determined in this step.

The cross-product matrix [P_3] which consists of the product of each pair of coefficients of the loadings $L_2(j)$, $j = 1, 2, 3, 4, 5$ and 6 is shown in Table 11.17.

The residual matrix R_3 is obtained by subtracting the cross-product matrix P_3 from R_2 and it is as shown in Table 11.18.

Table 11.17 Cross-product Matrix [P_3]

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	0.059	0.050	-0.146	-0.134	-0.109	-0.190
X_2	0.050	0.043	-0.124	-0.114	-0.093	-0.161
X_3	-0.146	-0.124	0.359	0.328	0.269	0.466
X_4	-0.134	-0.114	0.328	0.301	0.246	0.427
X_5	-0.109	-0.093	0.269	0.246	0.201	0.349
X_6	-0.190	-0.161	0.466	0.427	0.349	0.606

Table 11.18 Residual Matrix [R_3]

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	0.243	-0.026	-0.101	0.083	0.062	0.172
X_2	-0.026	0.220	0.122	0.120	0.022	0.380
X_3	-0.101	0.122	0.395	-0.603	-0.346	-0.129
X_4	0.083	0.120	-0.603	0.271	-0.372	0.092
X_5	0.062	0.022	-0.346	-0.372	0.396	0.001
X_6	0.172	0.380	-0.129	0.092	0.001	0.151

From Table 11.18, it is clear that the columns (variables) 2, 3, 4, 5 and 6 must be reflected. The reflected matrix R'_3 is as shown in Table 11.19. The weights of the columns (variables) from 1 to 6 are +1, -1, -1, -1, -1 and -1, respectively.

Table 11.19 Residual Matrix [R'_3] with Reflections on Variables 2, 3, 4, 5 and 6

Variable	X_1	X_2	X_3	X_4	X_5	X_6
X_1	0.243	0.026	0.101	0.083	0.062	0.172
X_2	0.026	0.220	0.122	0.120	0.022	0.380
X_3	0.101	0.122	0.395	0.603	0.346	0.129
X_4	0.083	0.120	0.603	0.271	0.372	0.092
X_5	0.062	0.022	0.346	0.372	0.396	0.001
X_6	0.172	0.380	0.129	0.092	0.001	0.151
Total (S_j)	0.687	0.890	1.696	1.541	1.199	0.925

$$\text{Grand total of column totals, } T = \sum_{j=1}^6 S_j = 6.938$$

or

$$\sqrt{T} = 2.634$$

The third centroid factor loadings, $L_3(j)$, are summarized in Table 11.20 after multiplying with the respective column weights.

Table 11.20 Calculations to Determine Loadings of Factor 3 [$L_3(j)$]

j	S_j	$UL_3(j) = S_j/\sqrt{T}$	$L_3(j)$
1	0.687	0.26081	0.26081
2	0.890	0.33789	-0.33789
3	1.696	0.64389	-0.64389
4	1.541	0.53504	-0.53504
5	1.199	0.45520	-0.45520
6	0.925	0.35118	-0.35118

Step-7: The current factor number (f) is 3 which is equal to the total number of factors to be considered ($F=3$). So, go to step-8.

Step-8: The loadings of all the three factors are shown in three columns, as in Table 11.21.

Table 11.21 Complete Three Factor Loading

Variable	Factor-1	Factor-2	Factor-3	Communality (h^2)
X_1	0.83560	0.24386	0.26081	0.82572
X_2	0.85876	0.20705	-0.33789	0.89451
X_3	0.49633	-0.59885	-0.64389	1.01956
X_4	0.65439	-0.54824	-0.58504	1.07107
X_5	0.63500	-0.44878	-0.45520	0.81184
X_6	-0.49306	-0.77865	-0.35118	0.97273
Eigenvalue	2.7566	1.56923	1.26959	$\Sigma h^2 = 5.59543$
Proportion of total variance	0.46	0.26	0.21	
Proportion of common variance	0.49	0.28	0.23	

Note: Σh^2 is the common variance which is equal to 5.59543 Total variance = number of variables (6).

Step-9: Assign each variable to the factor with which it has the maximum absolute loading as shown in Table 11.21.

As per this guidelines, the assignment of variables to different factors is shown in Table 11.22.

Table 11.22 Assignment of Variables to Factors

Factor number	Name of factor	Variable	Description
1	Technical factor	X_1	Fuel efficient
		X_2	Life of the two-wheeler
		X_4	Quality of original spares
		X_5	Breakdown rate
2	Price factor	X_6	Price
3	Personal factor	X_3	Handling convenience

Step-10: The eigenvalues of the factors, communalities (h^2), common variance (Σh^2), proportion of total variance of each factor, proportion of common variance of each factor are summarized in Table 11.21.

The proportion of total variance of the factor-1 = 0.46 (46%).

The proportion of total variance of the factor-2 = 0.26 (26%).

The proportion of total variance of the factor-3 = 0.21 (21%).

The proportion of common variance of the factor-1 = 0.49 (49%).

The proportion of common variance of the factor-2 = 0.28 (28%).

The proportion of common variance of the factor-3 = 0.23 (23%).

11.5.2 Principal Components Method

This method maximizes the sum of squares of loadings of each identified factor.

This is a popular technique which determines loadings of variables on different factors by using the standard normal values of the observations of the original (input) variables.

Let n be the total number of input variables; m , the total number of sets of observations for the variables (total number of respondents); n , the total number of factors which is same as the total number of variables; X_j , the input variable j , $j = 1, 2, 3, \dots, n$; a_{ij} , the i th original observation/opinion (marked by the i th respondent) for the j th input variable; $\bar{X}_j$, the mean of the original observations of the variable j ; σ_j , the standard deviation of the original observations of the variable j ; and z_{ij} be the standard normal value of the i th observation of the j th variable given by the following formula:

$$z_{ij} = \frac{a_{ij} - \bar{X}_j}{\sigma_j}$$

A generalized data format of the original observations and the standardized observations are shown in Tables 11.23 and 11.24, respectively.

Table 11.23 Generalized Data Format of Original Observations

Observation (i)	Input variable						
	X_1	X_2	X_3	...	X_j	...	X_n
1	a_{11}	a_{12}	a_{13}	...	a_{1j}	...	a_{1n}
2	a_{21}	a_{22}	a_{23}	...	a_{2j}	...	a_{2n}
3	a_{31}	a_{32}	a_{33}	...	a_{3j}	...	a_{3n}
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
i	a_{i1}	a_{i2}	a_{i3}	...	a_{ij}	...	a_{in}
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
m	a_{m1}	a_{m2}	a_{m3}	...	a_{mj}	...	a_{mn}

Table 11.24 Generalized Data Format of Standardized Values of Original Observations

Observation (<i>l</i>)	Variable						
	X_1	X_2	X_3	...	X_j	...	X_n
1	z_{11}	z_{12}	z_{13}	...	z_{1j}	...	z_{1n}
2	z_{21}	z_{22}	z_{23}	...	z_{2j}	...	z_{2n}
3	z_{31}	z_{32}	z_{33}	...	z_{3j}	...	z_{3n}
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
<i>l</i>	z_{l1}	z_{l2}	z_{l3}	...	z_{lj}	...	z_{ln}
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
<i>m</i>	z_{m1}	z_{m2}	z_{m3}	...	z_{mj}	...	z_{mn}

Let F_k be the linear composite of the principal component k , known as *factor k* , which is given by the following expression:

$$F_k = w_{1k}X_1 + w_{2k}X_2 + w_{3k}X_3 + \cdots + w_{ik}X_i + \cdots + w_{nk}X_n = \sum_{i=1}^n w_{ik}X_i$$

for $k = 1, 2, 3, \dots, n$, where w_{ik} is the weight of the variable X_i in the linear composite of the factor k .

The above formula for F_k can be used to find the standardized score of each set of observations (each row) for the factor k in Table 11.24 by substituting the values of X_i , $i = 1, 2, 3, \dots, n$ in it. The format of the above weights is shown in Table 11.25.

Table 11.25 Format of Weights [w_{ik}]

Variable (<i>l</i>)	Factor- k principal component- k						
	1	2	3	...	k	...	n
1	w_{11}	w_{12}	w_{13}	...	w_{1k}	...	w_{1n}
2	w_{21}	w_{22}	w_{23}	...	w_{2k}	...	w_{2n}
3	w_{31}	w_{32}	w_{33}	...	w_{3k}	...	w_{3n}
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
<i>l</i>	w_{l1}	w_{l2}	w_{l3}	...	w_{lk}	...	w_{ln}
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
<i>n</i>	w_{n1}	w_{n2}	w_{n3}	...	w_{nk}	...	w_{nn}

w_{ik} is the unique weight determined by the application of principal-components procedure. These weights are used to determine the unstandardized factor scores. A combination of w_{ik} values for $i = 1, 2, 3, \dots, n$ and $k = 1, 2, 3, \dots, n$ should yield the unstandardized factor scores of the entire set of observations in Table 11.24 whose variance is maximal.

The weights in the principal components are chosen such that the following requirements are satisfied:

- The variances of the principal components (eigenvalues) are in the decreasing order from the principal component-1 to the principal component- n .
- The values of the factor loadings of the principal components are uncorrelated.
- The sum of the squares of the weights used in principal components is equal to 1.

The scores of different sets of observations in Table 11.24 on different factors can be obtained by using the following formula:

$$[f_{ij}]_{m \times n} = [z_{ij}]_{m \times n} \times [w_{ij}]_{n \times n}$$

where, $[f_{ij}]$ is the matrix of unstandardized factor scores (f_{ij} is the unstandardized factor score of the i th set (i th row) of standardized observations in Table 11.24 on the factor j). $[z_{ij}]$ is the standardized values of the original sets of observations of the variables (z_{ij} is the i th standardized value under the j th variable). $[w_{ij}]$ is the matrix of weights of variables in different factors (w_{ij} is the weight of the i th variable in the linear composite of factor j).

The format of the unstandardized factor scores matrix, $[f_{ij}]$, is shown in Table 11.26.

Table 11.26. Generalized Data Format of Unstandardized Factor Scores $[f_{ij}]$

Observation (i)	Factor (j)						
	1	2	3	...	j	...	n
1	f_{11}	f_{12}	f_{13}	...	f_{1j}	...	f_{1n}
2	f_{21}	f_{22}	f_{23}	...	f_{2j}	...	f_{2n}
3	f_{31}	f_{32}	f_{33}	...	f_{3j}	...	f_{3n}
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
l	f_{l1}	f_{l2}	f_{l3}	...	f_{lj}	...	f_{ln}
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
m	f_{m1}	f_{m2}	f_{m3}	...	f_{mj}	...	f_{mn}

The loadings of the variables on the factors are obtained by finding the correlation coefficients between the values in different pairs of columns of matrix $[z_{ij}]$ and $[f_{ij}]$. The generalized format of the factor loading matrix $[L_{ij}]$ is shown in Table 11.27.

Table 11.27 Format of Factor Loadings Matrix [L_{ij}]

Variable (i)	Factor (j)						
	1	2	3	...	k	...	n
1	L_{11}	L_{12}	L_{13}	...	L_{1k}	...	L_{1n}
2	L_{21}	L_{22}	L_{23}	...	L_{2k}	...	L_{2n}
3	L_{31}	L_{32}	L_{33}	...	L_{3k}	...	L_{3n}
...	...	...	...	...	...	...	...
i	L_{i1}	L_{i2}	L_{i3}	...	L_{ik}	...	L_{in}
...	...	...	...	...	...	...	...
n	L_{n1}	L_{n2}	L_{n3}	...	L_{nk}	...	L_{nn}

The standardized factor scores can be obtained using the following formula:

$$S_{ij} = \frac{f_{ij} - \bar{M}_j}{s_j}; \quad i = 1, 2, 3, \dots, m, \quad j = 1, 2, 3, \dots, n$$

where S_{ij} is the standardized factor score of the i th set of observations on the factor j ; $\bar{M}_j$, the mean of the unstandardized factor scores of the factor j ; s_j the standard deviation of the unstandardized factor scores of the factor j .

The prediction of the original standardized observations (Z_{ij}) can be obtained by using the following formula. The generalized format of [Z_{ij}] matrix is shown in Table 11.28, where

$$[Z_{ij}]_{m \times n} = [S_{ij}]_{m \times n} \times [L_{ij}]_{n \times n}$$

Table 11.28 Generalized Format of Predictions of Original Standardized Observations [Z_{ij}]

Observation (i)	Factor (j)						
	1	2	3	...	j	...	n
1	Z_{11}	Z_{12}	Z_{13}	...	Z_{1j}	...	Z_{1n}
2	Z_{21}	Z_{22}	Z_{23}	...	Z_{2j}	...	Z_{2n}
3	Z_{31}	Z_{32}	Z_{33}	...	Z_{3j}	...	Z_{3n}
...	...	...	...	...	...	...	...
i	Z_{i1}	Z_{i2}	Z_{i3}	...	Z_{ij}	...	Z_{in}
...	...	...	...	...	...	...	...
m	Z_{m1}	Z_{m2}	Z_{m3}	...	Z_{mj}	...	Z_{mn}

Steps of principal component method

The steps of the principal components analysis are summarized as follows:

Step-1: Input the original sets of observation $[a_{ij}]$; $i = 1, 2, 3, \dots, m$ and $j = 1, 2, 3, \dots, n$.

Step-2: Find the standardized sets of observations $[z_{ij}]$ from $[a_{ij}]$ using the following formula:

$$z_{ij} = \frac{a_{ij} - \bar{X}_j}{\sigma_j}; \quad i = 1, 2, 3, \dots, m, \quad j = 1, 2, 3, \dots, n$$

where z_{ij} is the standardized observation of the i th original observation under the variable j ; a_{ij} the i th original observation of the j th variable; $\bar{X}_j$, the mean of the original observations of the variable j ; and σ_j is the standard deviation of the original observations of the variable j .

Step-3: Determine the weights of variables in different linear composites of factors $[w_{ij}]$ such that the variance of the unstandardized factor scores of the entire set of observations is maximal. These weights are nothing but directional cosines of the respective vectors.

Step-4: Find the unstandardized factor scores using the following formula:

$$[f_{ij}]_{m \times n} = [z_{ij}]_{m \times n} \times [w_{ij}]_{n \times n}$$

Step-5: Find the loadings of the variables on the factors $[L_{ij}]$. The value of L_{ij} is the correlation coefficient between the values in column i of matrix $[z_{ij}]$ and that of column j of the matrix $[f_{ij}]$ for $i = 1, 2, 3, \dots, n$ and $j = 1, 2, 3, \dots, n$.

Step-6: Find the standardized factor scores using the following formula:

$$S_{ij} = \frac{f_{ij} - \bar{M}_j}{s_j}; \quad i = 1, 2, 3, \dots, m, \quad j = 1, 2, 3, \dots, n$$

where S_{ij} is the standardized factor score of the i th set of observations on the factor j ; $\bar{M}_j$, the mean of the unstandardized factor scores of the factor j ; and s_j is the standard deviation of the unstandardized factor scores of the factor j .

Step-7: Find the prediction of the standardized original observations using the following formula:

$$[Z_{ij}]_{m \times n} = [S_{ij}]_{m \times n} \times [L_{ij}]_{n \times n}$$

Step-8: Find the sum of squares of loadings of each column- j (principal component- j /factor- j) which is known as eigenvalue of that column j .

Step-9: Drop insignificant principal components which have eigenvalues less than 1. Let the number of principal components retained be X .

Step-10: Perform the rotation of the retained principal components for better interpretation. The rotation can be done by 'varimax rotation method' of 'promax rotation method'.

Step-11: Assign each variable to the principal component (factor), with which it has the maximum absolute loading (irrespective of sign).

Step-12: Find the sum of squares of loadings for each variable i (row i). It is denoted by h_i^2 . Also, find the common variance, whose formula is given below:

$$\text{Common variance} = \sum_{i=1}^X h_i^2$$

where X is the number of retained principal components.

Step-13: For each retained principal component, k ($k = 1, 2, 3, \dots, X$), find the following and state inferences:

Proportion of total variance of the principal component k

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Total variance}}$$

where the total variance is equal to the number of variables, n . Also,

Proportion of common variance of the principal component k

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Common variance}}$$

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Sum of the eigenvalues of the retained principal components}}$$

11.5.3 Principal Component Method (Alternate Method)

The steps of this method are easy to understand and simple to implement:

Step-1: Input the data matrix of $m \times n$ size, where m is the number of sets of observations and n is the number of characteristics (input variables). Input the total number of principal components (k) to be identified.

Initialize $k = 1$, where k is the current principal component number.

Step-2: Find $n \times n$ correlation coefficient matrix R_1 summarizing the correlation coefficient for each pair of the variables. Each diagonal value of the correlation coefficient matrix is assumed as 1, since the respective variable is correlated to itself.

Step-3: Perform reflections if necessary:

3.1: If the matrix is positive manifold, then assign a weight (w_j) of +1 to each column j (variable j) in the matrix; then, treat the matrix R_k as the matrix R_k without any modification in R_k and go to step-4; otherwise, go to step-3.2.

3.2: Perform reflections:

(a) Initially assign a weight (w_j) of +1 for all the columns.

(b) If there is a negative entry in column j , then it should be reflected; assign -1 as the weight for that column j .

In this case, for each negative value in the row i of the column j , do the following updation:

$$R_k(i, j) = \text{Absolute value of } R_k(i, j)$$

$$R_k(j, i) = \text{Absolute value of } R_k(j, i)$$

- (c) Treat this revised matrix as the matrix R'_k which is known as the matrix with reflections;

Step-4: Determination of k th principal component:

- 4.1: For each column j in the matrix R'_k , find the sum of the entries in it $[S_j]$.

- 4.2: Find the grand total (T_1), which is the sum of squares of the column sums:

$$T_1 = \sum_{j=1}^n S_j^2$$

- 4.3: Find the normalizing factor, W which is given by the formula: $W = \sqrt{T_1}$

- 4.4: Divide each column sum, S_j by W to get unweighted normalized loading UL_j for that column j .

- 4.5: For each column j , find its weighted factor loading L_j by multiplying UL_j with its weight w_j .

- 4.6: Set row number i of the matrix R_k to 1 ($i = 1$).

- 4.7: Multiply the value of L_j with the corresponding value in the i th row of R_k matrix for each j and sum such products and treat it as M_i , where

$$M_i = \sum_{j=1}^n R_k(i, j) L_j$$

- 4.8: Set $i = i + 1$.

- 4.9: If $i \leq n$, then go to step-4.7; otherwise go to step 4.10.

- 4.10: Find the grand total (T_2), which is the sum of squares of M_j , $j = 1, 2, 3, \dots, n$ as

$$T_2 = \sum_{j=1}^n M_j^2$$

- 4.11: Find the normalizing factor N which is given by the formula: $N = \sqrt{T_2}$.

- 4.12: Divide each M_j by N to get normalized loadings Q_j , where

$$Q_j = \frac{M_j}{N}$$

- 4.13: For each j , compare L_j and Q_j towards convergence (manual checking may be used by displaying L_j and Q_j for all j . If all L_j and Q_j are more or less closer, then go to step-5; otherwise go to step-4.14.

4.14: Update $L_j = Q_j$, for all $j = 1, 2, 3, \dots, n$. Go to step-4.6.

Step-5: Store the loadings of the k th principal components P_{ik} , as shown below. Latest normalizing factor for Q in step-4.11 of the last iteration of convergence = N . Then,

$$P_{ik} = L_i \sqrt{N}, \quad i = 1, 2, 3, \dots, n$$

Step-6: Increment k by 1 ($k = k + 1$).

Step-7: If $k > K$ then go to step-10.

Step-8: Determination of R_k :

8.1: Determination of cross-product matrix $[C_k]$: Obtain the product of each pair of factor loadings of $(k - 1)$ th principal component and store it in the respective row and column of the cross-product matrix (C_k), as explained in the centroid method.

8.2: Find the residual matrix R_k by element-by-element subtraction of the cross-product matrix C_k from the matrix R_{k-1} .

Step-9: Go to step-3.

Step-10: Arrange the loadings of K principal components by keeping the principal components on columns and the variables in rows, as shown in centroid method.

Step-11: Find the sum of squares of loadings of each column j (principal component j) which is known as eigenvalue of that column j .

Step-12: Drop insignificant principal components which have eigenvalues less than 1. Let the number of principal components retained be X .

Step-13: Perform the rotation of the retained principal components for better interpretation. The rotation can be done by 'varimax rotation method' or 'promax rotation method'.

Step-14: Assign each variable to the principal component (factor) with which it has the maximum absolute loading (irrespective of sign).

Step-15: Find the sum of squares of loadings for variable i (row i). It is denoted by h_i^2 . Also, find the common variance whose formula is given below:

$$\text{Common variance} = \sum_{i=1}^X h_i^2$$

where X is the number of retained principal components.

Step-16: For each principal component, k ($k = 1, 2, 3, \dots, X$), find the following and state inferences:

Proportion of total variance of the principal component k

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Total variance}}$$

where the total variance is equal to number of variables, n . Also,

Proportion of common variance of the principal component k

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Common variance}}$$

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Sum of the eigenvalues of the retained principal components}}$$

Example 11.3 A telephone industry wanted to study the operations of telephone booths with a view to establish norms for better customer service. A consultant for this task has identified the following variables:

1. Need for deep differential rates (0, no need; 10, highly essential) (X_1).
2. Telephone unit rates, X_2 (0, nominal; 10, very high).
3. Secrecy of discussion, X_3 (0, low; 10, high).
4. Influence of external sound on conversation, X_4 (0, nil; 10, unable to do conversation)

In a pilot study, the questionnaires containing above questions were administered to 10 respondents in different telephone booths and their responses are summarized as in Table 11.29. Perform factor analysis using principal component analysis and

- (a) Find factor loadings for four principal components (factors).
- (b) Identify the significant number of principal components.
- (c) Check whether factor rotation is required. If necessary, perform it using varimax rotation method and make inferences.
- (d) Group the variables into factors.
- (e) Find the proportion of total variance as well as common variance for each of the factors.

Note: The number of respondents is restricted to 10 mainly to reduce computational difficulties at textbook level.

Table 11.29 Survey Data

Respondent	Variable			
	X_1	X_2	X_3	X_4
1	9	4	6	6
2	7	2	8	3
3	7	2	8	10
4	5	8	7	8
5	10	6	9	5
6	5	5	1	1
7	9	2	6	8
8	9	2	8	2
9	8	2	5	5
10	1	3	10	5

Solution.

Step-1: The data matrix which contains the scores given by the respondents against four variables are already shown in Table 11.29.

Required number of principal components, $K = 4$

Set the current principal competent number, $k = 1$

Step-2: The correlation coefficient matrix between the four variables is, as given in Table 11.30:

Table 11.30 Correlation Coefficient Matrix (R_1)

Variable (i)	Variable (j)			
	1	2	3	4
1	1.000	-0.155	-0.065	0.058
2	-0.155	1.000	-0.120	0.059
3	-0.065	-0.120	1.000	0.320
4	0.058	0.059	0.320	1.000

Step-3: Since there are negative entries in Table 11.30, the matrix with reflections of columns 1 and 2 (R'_1) is shown in Table 11.31.

Table 11.31 Correlation Coefficient Matrix with Reflection on Columns 1 and 2 [R'_1]

Variable (i)	Variable (j)			
	1	2	3	4
1	1.000	0.155	0.065	0.058
2	0.155	1.000	0.120	0.059
3	0.065	0.120	1.000	0.320
4	0.058	0.059	0.320	1.000
S_j	1.278	1.334	1.505	1.437

Step-4: Determination of the first principal component:

4.1: The sum of each column of the matrix R'_1 is shown in Table 11.31 itself.

4.2: The grand total (T_1) is computed, as shown below:

$$T_1 = \sum_{j=1}^4 S_j^2 = 7.742834$$

4.3: The normalizing factor (W) is computed, as shown below:

$$W = \sqrt{T_1} = (7.742834)^{0.5} = 2.782595$$

4.4: Divide each column sum (S_j) by W to get unweighted loading UL_j for each column j , as shown in Table 11.32.

$$UL_j = \frac{S_j}{W}, \quad j = 1, 2, 3 \text{ and } 4$$

Table 11.32 Unweighted Loadings

Variable	UL_j
1	0.4592835
2	0.4794086
3	0.5408620
4	0.5164244

4.5: For each column- j , its normalized weighted factor loading L_j is obtained by multiplying UL_j with its weight w_j determined in step-3 (+1 if it is not reflected; -1 if it is reflected). These are summarized in Table 11.33.

Table 11.33 Normalized Loadings

Variable	L_j
1	-0.4592835
2	-0.4794086
3	0.5408620
4	0.5164244

4.6 to 4.9: Multiply the value of L_j with the corresponding value in the i th row of R_i matrix shown in Table 11.30 for all j and sum such products and treat it as M_i , as given by the following formula. The results of M_i for $i = 1, 2, 3$ and 4 are shown in Table 11.34.

$$M_i = \sum_{j=1}^4 R_k(i, j) L_j, \quad i = 1, 2, 3 \text{ and } 4$$

Table 11.34 M_i Values

Variable (i)	$M(i)$
1	-0.3901787
2	-0.4426541
3	0.7935004
4	0.635768

4.10: The grand total (T_2) which is the sum of squares of M_j , $j = 1, 2, 3$ and 4 is computed as shown below:

$$T_2 = \sum_{j=1}^4 M_j^2 = 1.380513$$

4.11: The normalizing factor (N) is computed using the following formula:

$$N = \sqrt{T_2} = \sqrt{1.380513} = 1.174952$$

4.12: Each M_j is divided by N to get normalized loadings Q_j , as shown in Table 11.35.

Table 11.35 Q_j Values

Variable (j)	$M(j)$	$Q(j)$
1	-0.3901787	-0.3320805
2	-0.4426541	-0.3767422
3	0.7935004	0.6753470
4	0.635768	0.5411012

4.13: For each j , L_j and Q_j are compared for convergence interactively with computer, as shown in Table 11.36. The first-two rows of Table 11.36, clearly shows that there is no convergence. Hence, go to step-4.14

4.14: The current Q_j is stored in L_j for all $j = 1, 2, 3$ and 4.

4.6 to 4.13: The steps from step-4.6 to step-4.13 via step-4.14 are repeated until convergence occurs.

As shown in Table 11.36, the convergence occurs after 10 iterations. So, go to step-5.

Table 11.36 Convergence of First Principal Component Loadings at the End of 10th Iteration

Iteration number	$L(j)$ and $Q(j)$	Variable j			
		1	2	3	4
1	$L(j)$	-0.4592836	-0.4794087	0.5408621	0.5164245
	$Q(j)$	-0.3320805	-0.3767422	0.6753470	0.5411012
2	$L(j)$	-0.3320805	-0.3767422	0.6753470	0.5400873
	$Q(j)$	-0.2284490	-0.2988281	0.7301942	0.5703766
3	$L(j)$	-0.2284490	-0.2988281	0.7301942	0.5703766
	$Q(j)$	-0.1522791	-0.2459490	0.7465683	0.5991293
4	$L(j)$	-0.1522791	-0.2459490	0.7465683	0.5991293
	$Q(j)$	-0.0977629	-0.2113560	0.7471249	0.6225556
5	$L(j)$	-0.0977629	-0.2113560	0.7471249	0.6225556
	$Q(j)$	-0.0587854	-0.1890720	0.7422871	0.6401608

(contd.)

Table 11.36 (Contd.)

6	$L(j)$	-0.0587854	-0.1890724	0.7422871	0.6401608
	$Q(j)$	-0.0307062	-0.1749147	0.7364061	0.6528154
7	$L(j)$	-0.0307062	-0.1749147	0.7364061	0.6528154
	$Q(j)$	-0.0102664	-0.1661114	0.7310851	0.6616769
8	$L(j)$	-0.0102664	-0.1661114	0.7310850	0.6616769
	$Q(j)$	0.0047807	-0.1608371	0.7267620	0.6677762
9	$L(j)$	0.0047807	-0.1608371	0.7267624	0.6677762
	$Q(j)$	0.0159864	-0.1578823	0.7234255	0.6719175
10	$L(j)$	0.0159864	-0.1578823	0.7234255	0.6719175
	$Q(j)$	0.0244289	-0.1564397	0.7209166	0.6746921

Step-5: Compute the loadings of the first principal component P_{i1} for all variables, $i = 1, 2, 3$ and 4, using the following formula and store them in Table 11.37. Normalizing factor of the latest $Q(j)$ in step-4.11 of iteration 10 (N) = $\sqrt{T_2}$ = 1.326569. Then

$$P_{ik} = (\text{Latest } L_i) \sqrt{N}, \quad i = 1, 2, 3 \text{ and } 4$$

Table 11.37 Loadings of the First Principal Component

Variable (i)	Latest $L(i)$	Loading (P_{i1}) = $L(i) \times \sqrt{N}$
1	0.0159864	0.0184126
2	-0.1578823	-0.1818437
3	0.7234255	0.8332182
4	0.6719175	0.7738929

Step-6: Increment k by 1 ($k = 1 + 1 = 2$).

Step-7: Since k is not greater than 4, go to step-8.

Step-8: Determine R_2 :

Table 11.38 Cross-product Coefficient Matrix [C_2]

Variable (i)	Variable (j)			
	1	2	3	4
1	0.000	-0.003	0.015	0.014
2	-0.003	0.033	-0.152	-0.141
3	0.015	-0.152	0.694	0.645
4	0.014	-0.141	0.645	0.599

8.1: Determination of cross-product matrix $[C_2]$: The product of each pair of factor loadings of first principal component are stored in the respective row and column of the cross-product matrix (C_2), as shown in Table 11.38.

8.2: The residual matrix (R_2) is obtained by element-by-element subtraction of the cross-product matrix C_2 from the matrix R_1 and it is shown in Table 11.39.

Table 11.39 Residual Matrix $[R_2]$

Variable (i)	Variable (j)			
	1	2	3	4
1	1.000	-0.152	-0.080	0.044
2	-0.152	0.967	0.032	0.200
3	-0.080	0.032	0.306	-0.325
4	0.044	0.200	-0.325	0.401

Step-3: The residual matrix R_2 with reflection on columns 1 and 3 is shown in Table 11.40. The reflected matrix is known as R'_2 .

Table 11.40 Residual Matrix with Reflection on Columns 1 and 3 $[R'_2]$

Variable (i)	Variable (j)			
	1	2	3	4
1	1.000	0.152	0.080	0.044
2	0.152	0.967	0.032	0.200
3	0.080	0.032	0.306	0.325
4	0.044	0.200	0.325	0.401
S_j	1.276	1.351	0.743	0.970

Step-4: Determination of the second principal component:

4.1: The sum of each column of the matrix R'_2 is shown in Table 11.40 itself.

4.2: The grand total (T_1) is computed, as shown below:

$$T_1 = \sum_{j=1}^4 S_j^2 = 4.946326$$

4.3: The normalizing factor, (W) is computed as shown below:

$$W = \sqrt{T_1} = (4.946326)^{0.5} = 2.224034$$

4.4: Divide each column sum (S_j) by W to get unweighted loading UL_j for each column j , as shown in Table 11.41.

Table 11.41 Unweighted Loadings

Variable	UL_j
1	0.5737324
2	0.6074548
3	0.3340777
4	0.4361445

4.5: For each column j , its normalized weighted factor loading (L_j) is obtained by multiplying UL_j with its weight w_j determined in step-3 (+1 if it is not reflected; -1 if it is reflected). These are shown in Table 11.42.

Table 11.42 Normalized Loadings

Variable	L_j
1	-0.5737324
2	0.6074548
3	-0.3340777
4	0.4361445

4.6 to 4.13: As explained for the first principal component, the steps from 4.6 to 4.13 via step-4.14 are carried out and the values of L_j and Q_j for 15 iterations are shown in Table 11.43. One can see the fact that the values of $L(j)$ and $Q(j)$ converge at the 15th iteration.

Table 11.43 Convergence of Second Principal Component Loadings

Iteration number	$L(j)$ and $Q(j)$	Variable (j)			
		1	2	3	4
1	$L(j)$	-0.5737324	0.6074548	-0.3340777	0.4361445
	$Q(j)$	-0.5847045	0.7082226	-0.1684276	0.3580134
2	$L(j)$	-0.5847045	0.7082226	-0.1684276	0.3580134
	$Q(j)$	-0.5922520	0.7501661	-0.0879309	0.2806359
3	$L(j)$	-0.5922520	0.7501661	-0.0879309	0.2806359
	$Q(j)$	-0.6026786	0.7622345	-0.0409989	0.2325859
4	$L(j)$	-0.6026786	0.7622345	-0.0409989	0.2325859
	$Q(j)$	-0.6147976	0.7620549	-0.0135428	0.2027634
5	$L(j)$	-0.6147976	0.7620549	-0.0135428	0.2027634
	$Q(j)$	-0.6268459	0.7571958	0.0030683	0.1836018

(contd.)

Table 11.43 (Contd.)

6	$L(j)$	-0.6268459	0.7571958	0.0030683	0.1836018
	$Q(j)$	-0.6378014	0.7509215	0.0135936	0.1707088
7	$L(j)$	-0.6378014	0.7509215	0.0135936	0.1707088
	$Q(j)$	-0.6472735	0.7446407	0.0206053	0.1616255
8	$L(j)$	-0.6472735	0.7446407	0.0206053	0.1616255
	$Q(j)$	-0.6552279	0.7389256	0.0255039	0.1549672
9	$L(j)$	-0.6552279	0.7389256	0.0255039	0.1549672
	$Q(j)$	-0.6617953	0.7339635	0.0290686	0.1499327
10	$L(j)$	-0.6617953	0.7339635	0.0290686	0.1499327
	$Q(j)$	-0.6671634	0.7297656	0.0317472	0.1460390
11	$L(j)$	-0.6671634	0.7297656	0.0317472	0.1460390
	$Q(j)$	-0.6715256	0.7262694	0.0338083	0.1429801
12	$L(j)$	-0.6715256	0.7262694	0.0338083	0.1429801
	$Q(j)$	-0.6750585	0.7233855	0.0354206	0.1405518
13	$L(j)$	-0.6750585	0.7233855	0.0354206	0.1405518
	$Q(j)$	-0.6779148	0.7210216	0.0366964	0.1386111
14	$L(j)$	-0.6779148	0.7210216	0.0366964	0.1386111
	$Q(j)$	-0.6802214	0.7190918	0.0377133	0.1370533
15	$L(j)$	-0.6802214	0.7190918	0.0377133	0.1370533
	$Q(j)$	-0.6820836	0.7175208	0.0385280	0.1357993

Step-5: Compute the loadings of the second principal component P_{12} for all variables, $i = 1, 2, 3$ and 4 using the following formula and store them in Table 11.44. Normalizing factor of latest $Q(j)$ in step-4.11 of iteration 15 = $N = \sqrt{T_2} = 1.153099$. Then

$$P_{ik} = (\text{Latest } L_i) \sqrt{N}, \quad i = 1, 2, 3 \text{ and } 4$$

Table 11.44 Loadings of the Second Principal Component

Variable (i)	Latest $L(i)$	Loadings (P_{12}) = $L(i)\sqrt{N}$
1	-0.6802214	-0.7304385
2	0.7190918	0.7721785
3	0.0377133	0.0404975
4	0.1370533	0.1471712

Step-6: Increment k by 1 ($k = 2 + 1 = 3$).

Step-7: Since, k is not greater than 4, go to step-8.

Step-8: Like this, the steps are repeated from step-3 to step-7 through necessary steps in between, until k is greater than 4 and then the control is transferred to step-10. The number of iterations for convergence of the principal component 3 and the principal component 4 are 12 and 2, respectively.

Step-10: Arrange the loadings of the four principal components by keeping the principal components on columns and the variables in rows, as shown in Table 11.45.

Step-11: The sum of squares of loadings of each column k which is known as eigenvalue of that column k is shown in Table 11.45 itself.

Step-12: In Table 11.45, only the first-two principal components are having their eigenvalues more than one. So, they are retained and the remaining two principal components (third and fourth) are dropped.

Table 11.45 Loadings of the Principal Components

Variable (I)	Principal component (k)				Communalities (h_i^2)
	Retained		Dropped		
	1	2	3	4	
1	0.0184126	-0.7304385*	0.6306556	-0.2615855	1.000033
2	-0.1818437	0.7721785*	0.5258854	-0.3071488	1.000035
3	0.8332182*	0.0404975	-0.2770974	-0.4770050	1.000055
4	0.7738929*	0.1471712	0.4243926	0.4463172	0.999877
Eigenvalue	1.3266	1.1531	0.9312	0.5895	$\Sigma h_i^2 = 4.000$

Step-13: Since the loadings of this problem are having simple structure, there is no need to do factor rotation.

Step-14: The assignment of variables to different factors (principal components) is shown by a '*' in each row of Table 11.45. Further, the groupings of variables with meaningful names for the factors are shown in Table 11.46.

Table 11.46 Assignment of Variables to Factors

Principal component	Name of factor	Variable	Description
1	Quality of conversation facilities	X_3	Secrecy of discussion
		X_4	Influence of external sound on conversation
2	Telephone charges	X_1	Need for deep differential rates
		X_2	Telephone unit rates

Step-15: Common variance = $\sum_{i=1}^4 h_i^2 = 2.4797$

Step-16: For each principal component (k), the proportion of total variance and the proportion of common variance are computed using the following formulas. These are summarized in Table 11.47.

Table 11.47 Proportion of Total Variances and Common Variances of Principal Components

	Factor (principal component)	
	1	2
Eigenvalue	1.3268	1.1531
Proportion of total variance	0.33(33%)	0.29(29%)
Proportion of common variance	0.54(54%)	0.46(46%)

Proportion of total variance of the principal component k

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Total variance}}$$

where the total variance is equal to number of variables (n), which is 4 for this example. Also,

Proportion of common variance of the principal component k

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Common variance}}$$

$$= \frac{\text{Eigenvalue of principal component } k}{\text{Sum of the eigenvalues of the retained principal components}}$$

11.5.4 Varimax Method of Factor Rotation

After obtaining factor loadings using any one of the methods, one should examine whether the factor loading matrix possesses simple structure. If a factor loading matrix has a simple structure, it is easy to make interpretations about the factors. If there is no simple structure, then the factor axes should be rotated by an angle such that the factor loadings are revised to have a simple structure which will simplify the process of interpretation of factors. The communalities of each variable before and after factor rotation will be the same.

The process of factor rotation is explained through a sample factor loading matrix, as given in Example 11.4.

Steps of varimax factor rotation method

The steps of varimax factor rotation method for two factors are presented below with reference to Figs. 11.2–11.4.

Step-1: Input the factor loading matrix. Number of variables = n . Number of principal components (factors) = 2. Angle of rotation = θ .

- Step-2:** Plot the factor loadings on a two-dimensional space, $F_1 - F_2$ plane, where F_1 and F_2 are Factor-1 and Factor-2, respectively. Let ϕ be the angle between the nearest axis (F_1 or F_2) and each vector of factor loadings.
- Step-3:** Rotate the $F_1 - F_2$ plane by an angle such that the factor loadings are revised to have a simple structure. A simple structure means that each variable has very high factor loading (as high as 1) on one of the factors and very low factor loading (as low as 0) on other factor(s). Let the rotated plane be $F'_1 - F'_2$.
- Step-4:** Set variable Number, $i = 1$.
- Step-5:** Let a_i and b_i be the factor loadings of the variable i on F_1 and F_2 , respectively. Find the magnitude of the vector C_i of the factor loadings of the variable i , using the following formula:
- $$c_i = (a_i^2 + b_i^2)^{0.5}$$
- Let the angle of the vector of factor loadings with the nearest part (positive side or negative side) of F_1 axis be α .
- Step-6:** Find $\cos \phi$ and treat it as the factor loading on the nearest axis (a'_i if the nearest axis as explained in step-1 is F'_1 -axis, b'_i if the nearest axis is F'_2 -axis). Fix the sign of the revised loading depending on the side of the factor (plus for positive side, and minus for negative side).
- Step-7:** Find $\cos (90 - \phi)$ and treat it as the factor loading on the other axis (b'_i if the nearest axis as explained in step-1 is F'_1 -axis, a'_i if the nearest axis is F'_2 -axis). Fix the sign of the revised loading depending on the side of the factor (plus for positive side, and minus for negative side).
- Step-8:** $i = i + 1$.
- Step-9:** If $i \leq n$, then go to step-5; otherwise go to step-10.
- Step-10:** Print the revised factor loading matrix.
- Step-11:** Group variables into factors and make inferences.

Example 11.4 The factor loading matrix in a factor analysis with 6 variables and 2 factors is, as shown in Table 11.48.

Table 11.48 Loadings of the Principal Components

Variable (i)	Factor		Communality h^2
	1	2	
1	-0.900	-0.400	0.9700
2	-0.850	-0.450	0.9250
3	0.600	-0.750	0.9225
4	0.650	-0.700	0.9125
5	0.600	-0.450	0.5625
6	-0.875	-0.400	0.9256
Eigenvalue	3.4410	1.7771	$\Sigma h^2 = 5.2181$

Plot the factor loadings on a two-dimensional plane (F_1 - F_2 plane) and find the revised factors loadings using varimax factor rotation method.

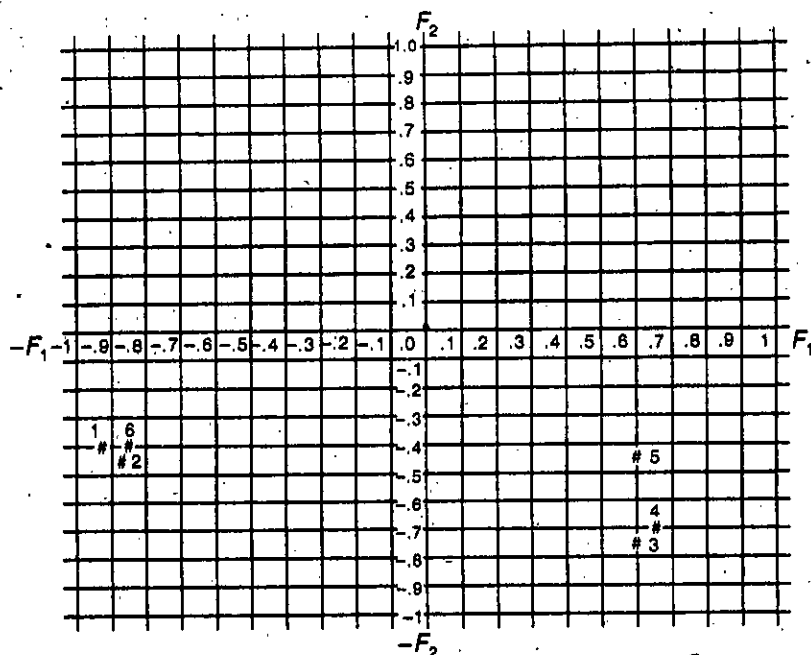


Fig. 11.2 Plot of factor loadings of Example 11.4.

Solution. The plot of the loadings is shown in Fig. 11.2. By carefully observing the plots in Fig. 11.2, the angle of rotation (θ) is fixed at 35° such that each variable has very high factor loading (as high as 1) on one of the factors and very low factor loading (as low as 0) on other factor.

For the factor loading points of the variables 1, 2 and 6, the axis F'_1 is closer. So, Fig. 11.3 is to be used to compute the respective revised factor loadings.

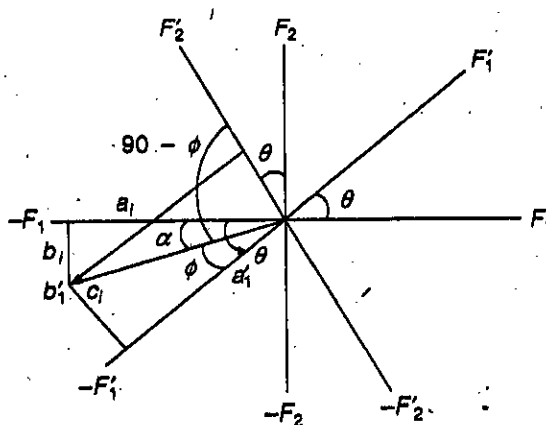


Fig. 11.3 Varimax rotation of factor loadings for variables 1, 2 and 6.

For the factor loading points of the variables 3, 4 and 5, the axis F'_2 is closer. So, Fig. 11.4 is to be used to compute the respective revised factor loadings.

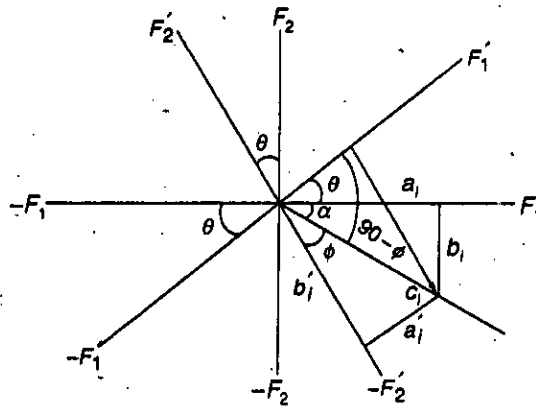


Fig. 11.4 Varimax rotation of factor loadings for variables 3, 4, and 5.

Calculation of factor loadings of variable-1: Referring to Fig. 11.3, we have, $a_1 = -0.9$, $b_1 = -0.4$, $\theta = 35^\circ$. Then

$$c_1 = (a_1^2 + b_1^2)^{0.5} = 0.985$$

$$\alpha = \tan^{-1}(b_1/a_1) = 23.96^\circ$$

$$\phi = \theta - \alpha = 35^\circ - 23.96^\circ = 11.04^\circ$$

$$a'_1 = -c_1 \cos \phi = -0.985 \times \cos 11.04^\circ = -0.967$$

$$b'_1 = c_1 \cos (90 - \phi) = -0.985 \times \cos (90 - 11.04^\circ) = 0.189$$

The value of a'_1 is fixed with minus sign because the horizontal component of c_1 falls on the negative side of F'_1 . Similarly, the value of b'_1 is fixed with plus sign because the vertical component of c_1 falls on the positive side of F'_2 . Similar logic should be followed to fix the sign of each revised loading of other variables.

Calculation of factor loadings of variable-2: Referring to Fig. 11.3, we have $a_2 = -0.85$, $b_2 = -0.45$, $\theta = 35^\circ$. Then

$$c_2 = (a_2^2 + b_2^2)^{0.5} = 0.962$$

$$\alpha = \tan^{-1}(b_2/a_2) = 27.9^\circ$$

$$\phi = \theta - \alpha = 35^\circ - 27.9^\circ = 7.1^\circ$$

$$a'_2 = -c_2 \cos \phi = -0.962 \times \cos 7.1^\circ = -0.955$$

$$b'_2 = c_2 \cos (90 - \phi) = 0.962 \times \cos (90 - 7.1^\circ) = 0.119$$

Calculation of factor loadings of variable-3: Referring to Fig. 11.4, we have $a_3 = 0.6$, $b_3 = -0.75$, $\theta = 35^\circ$. Then

$$c_3 = (a_3^2 + b_3^2)^{0.5} = 0.96$$

$$\alpha = \tan^{-1}(|b_3|/a_3) = 51.34^\circ$$

$$\phi = 90 - (\theta + \alpha) = 90^\circ - (35^\circ + 51.34^\circ) = 3.66^\circ$$

$$b'_3 = -c_3 \cos \phi = -0.96 \times \cos 3.66 = -0.958$$

$$a'_3 = c_3 \cos (90 - \phi) = 0.96 \times \cos (90 - 3.66) = 0.061$$

Calculation of factor loadings of variable-4: Referring to Fig. 11.4, we have $a_4 = 0.65$, $b_4 = -0.7$, $\theta = 35^\circ$. Then

$$c_4 = (a_4^2 + b_4^2)^{0.5} = 0.955$$

$$\alpha = \tan^{-1}(|b_4|/a_4) = 47.12^\circ$$

$$\phi = 90 - (\theta + \alpha) = 90^\circ - (35^\circ + 47.12^\circ) = 7.88^\circ$$

$$b'_4 = -c_4 \cos \phi = -0.955 \times \cos 7.88 = -0.946$$

$$a'_4 = c_4 \cos (90 - \phi) = 0.955 \times \cos (90 - 7.88) = 0.131$$

Calculation of factor loadings of variable-5: Referring to Fig. 11.4, we have $a_5 = 0.6$, $b_5 = -0.45$, $\theta = 35^\circ$. Then

$$c_5 = (a_5^2 + b_5^2)^{0.5} = 0.75$$

$$\alpha = \tan^{-1}(b_5/a_5) = 36.87^\circ$$

$$\phi = 90 - (\theta + \alpha) = 90^\circ - (35^\circ + 36.87^\circ) = 18.13^\circ$$

$$b'_5 = -c_5 \cos \phi = -0.75 \times \cos 18.13 = -0.713$$

$$a'_5 = c_5 \cos (90 - \phi) = 0.75 \times \cos (90 - 18.13) = 0.233$$

Calculation of factor loadings of variable-6: Referring to Fig. 11.3, we have $a_6 = -0.875$, $b_6 = -0.4$, $\theta = 35^\circ$. Then

$$c_6 = (a_6^2 + b_6^2)^{0.5} = 0.962$$

$$\alpha = \tan^{-1}(b_6/a_6) = 24.56^\circ$$

$$\phi = \theta - \alpha = 35^\circ - 24.56^\circ = 10.44^\circ$$

$$b'_6 = -c_6 \cos \phi = -0.962 \times \cos 10.44 = -0.946$$

$$a'_6 = c_6 \cos (90 - \phi) = 0.962 \times \cos (90 - 10.44) = 0.174$$

The rotated factor loadings are summarized in Table 11.49. From the table, it is clear that the communality for each variable is same as the communality before rotation. The minor differences in them are due to rounding-off errors. One can compare Table 11.48 and Table 11.49 in terms of simple structure. It exists only in Table 11.49 because of rotation.

The variables are grouped into factors, as shown by '*' marks in Table 11.49. The variables 1, 2 and 6 are grouped into Factor-1 and variables 3, 4 and 5 are grouped into Factor-2. The proportion of total variances and common variances of the principal components (factors) are shown in Table 11.50.

Table 11.49 Rotated Factor Loadings of the Principal Components

Variable (I)	Factor		Communalities (h^2)
	1	2	
1	-0.967*	0.189	0.9708
2	-0.955*	0.119	0.9262
3	0.061	-0.958*	0.9215
4	0.131	-0.946*	0.9121
5	0.233	-0.713*	0.5626
6	-0.946*	0.174	0.9252
Eigenvalue	2.8172	2.4012	$\Sigma h^2 = 5.2184$

Table 11.50 Proportion of Total Variances and Common Variances of Principal Components

Eigenvalue		*Factor (principal component)	
		1	2
		2.8172	2.4012
	Proportion of total variance	0.47(47%)	0.40(40%)
	Proportion of common variance	0.54(54%)	0.46(46%)

Note: Total variance = 6.

11.6 CLUSTER ANALYSIS

Cluster analysis is a technique which groups persons/objects/occasions into unknown number of groups such that the members of each group are having similar characteristics/attributes. The different techniques of cluster analysis form groups such that the similarity among the members of the groups is maximized. The main difference of the cluster analysis from the discriminant analysis is that the number of groups and the characteristics of the groups formed in clustering analysis are not known in advance.

Some examples of cluster analysis are presented in Table 11.51. The Table 11.51 summarizes the objects/persons to be clustered into groups based on certain attributes/variables.

Table 11.51 Examples of Cluster Analysis

System	Objects/persons	Variables/attributes
Health system	● Patients	● Diagnostic categories
Production system	● Components ● Machines	● Machines ● Components
Marketing system	● Customers ● Customers ● Sales regions ● Products	● Buying behaviour ● Demographic data ● Buying patterns ● Features
Finance system	● Companies ● Branch branches	● Financial performance variables ● Services rendered
Personnel system	● Employees ● Employees	● Skills ● Demographic factors
Educational System	● Students ● Faculty	● Electives taken ● Specializations

11.6.1 Concept of Clustering

Consider the data given in Table 11.52 which summarizes the highest sales per order and monthly sales of 8 salesmen of a computer company.

Table 11.52 Sales Data (in Lakhs of Rupees)

Salesman	Highest sales per order (X_1)	Monthly sales (X_2)
1	2	8
2	6	10
3	3	6
4	8	12
5	7	13
6	8	13
7	4	7
8	6	9

The observations in Table 11.52 are marked in Fig. 11.5 by taking the highest sales per order on X_1 -axis and the monthly sales on X_2 -axis. This graph is known as *scattered diagram*. A careful examination of the points in the graph reveals the existence of the following groups. Since, this is a constructed problem, the salesmen are clustered into three distinct groups. But, in reality, such distinct groups may not always exist.

Group-1: 1, 3, 7

Group-2: 2, 8

Group-3: 4, 5, 6

The same may be inferred through a profile diagram if the number of observations is small. A *profile graph* of the given example is shown in Fig. 11.6. In this figure, the variables X_1 and X_2 are taken on X -axis and the magnitude of these variables are marked on the Y -axis for each salesman. The groupings from the profile diagram are also same as that obtained from the scattered diagram, which are reproduced below:

Group-1: 1, 3, 7

Group-2: 2, 8

Group-3: 4, 5, 6

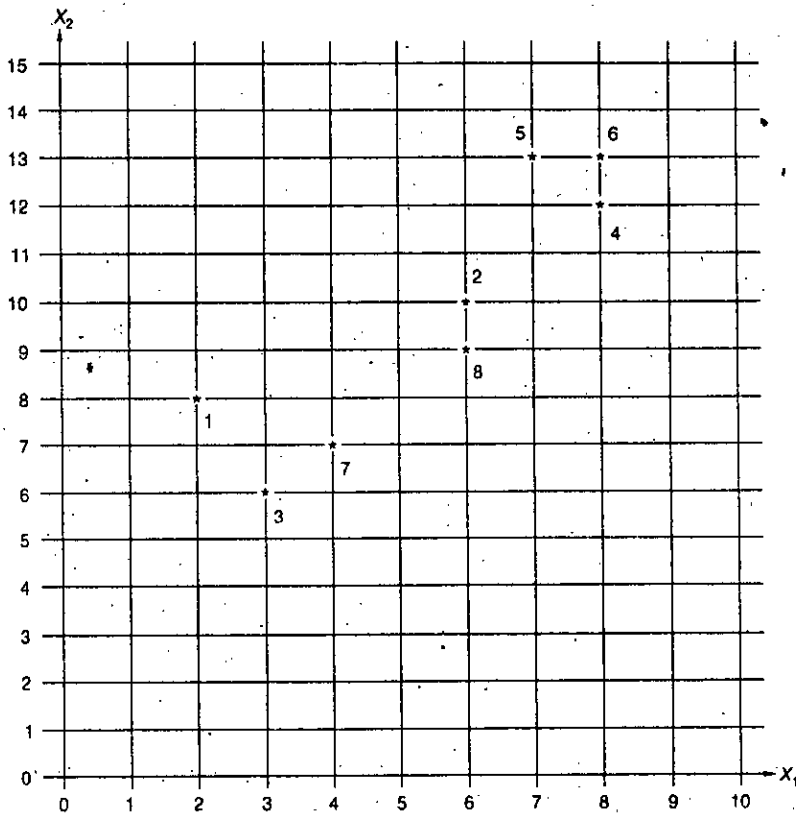


Fig. 11.5 Scattered plot of X_1 and X_2 .

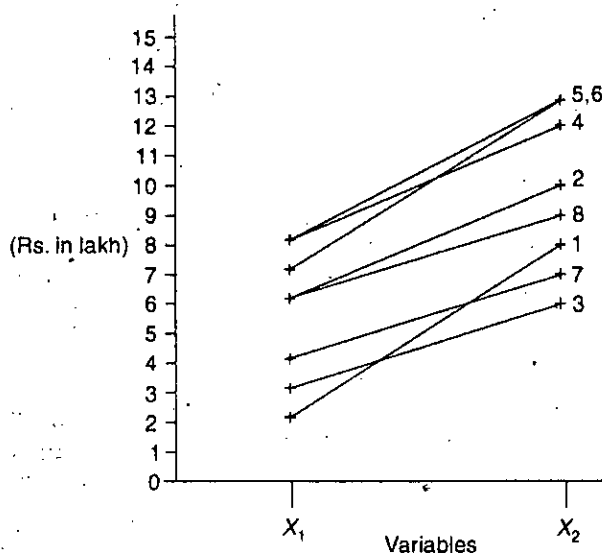


Fig. 11.6 Profile diagram.

11.6.2 Similarity Measures

The different similarity measures which are used in clustering analysis are listed below:

- Distance measure which is known as *dissimilarity measure*.
- Correlation coefficients
- Association coefficients.

Distance measure

The widely used similarity measure is the *Euclidean distance* measure. Actually, any distance measure is known as dissimilarity measure. The formula for the Euclidean distance between any two points is as shown below:

If the coordinates of point-1 and point-2 are (X_{11}, X_{12}) and (X_{21}, X_{22}) , respectively, then the distance between the point-1 and the point-2 is:

$$d_{12} = \left[(X_{11} - X_{21})^2 + (X_{12} - X_{22})^2 \right]^{1/2}$$

The generalized formula for n variables under the above situation is:

$$d_{ij} = \left[\sum_{k=1}^n (X_{ik} - X_{jk})^2 \right]^{1/2}$$

where d_{ij} is the distance between the points i and j ; X_{ik} , the coordinate of the point i along the axis k ; and X_{jk} , the coordinate of the point j along the axis k ; and n is the total number of variables (axes).

The representative point for all the objects within a cluster may be the *centroid* of that cluster. The formula for the coordinates of the centroid of a cluster is:

$$Y_k = \frac{\sum_{i=1}^a X_{ik}}{a}$$

where Y_k is the coordinate of the centroid of the cluster along the axis k ; X_{ik} , the coordinate of the point i along the axis k ; and a is the total number of points (objects) in the cluster.

If the distance between a point (object)/partial cluster and another point (object)/partial cluster is the least among all such values, then they can be grouped together.

Correlation coefficient Correlation coefficient between variables can act as a measure in cluster analysis. But, it is not widely used. The correlation coefficient is a kind of similarity coefficient. So, grouping of objects is done based on the maximum value of such coefficient.

Association coefficient It is a measure defined in terms of attributes of the objects. If a particular attribute is present in an object, then the respective association coefficient is assumed as 1; otherwise, it is assumed as 0. The association coefficient is a kind of similarity coefficient. So, grouping of objects is done based on the maximum value of such coefficient.

11.6.3 Clustering Techniques

The clustering techniques can be classified into *hierarchical* clustering techniques and *non-hierarchical* clustering techniques.

The hierarchical clustering techniques can be classified into agglomerative method and divisive method. An *agglomerative method* is a bottom-up approach in which each object is assumed as a separate cluster and then they will be clustered in succession until a single cluster which consists of all the objects is formed. A *divisive method* is a top-down approach in which initially all the objects are included in a single cluster. Then, the single cluster will be divided into sub-clusters until each object constitutes a cluster.

In non-hierarchical clustering technique, an object of one cluster will be permitted to join another cluster while the clusters are formed using either agglomerative method or divisive method. This means that lateral shifting of objects from one cluster to another cluster by violating the hierarchy is permitted. In this approach, some initial seeds of the objects will be assumed based on certain threshold value of the similarity coefficient. Each seed represents a cluster which contains certain objects. Each object should be present in any one of the initial seeds.

Hierarchical clustering techniques

Some of the popular hierarchical clustering techniques are:

- (a) Single linkage clustering method
- (b) Complete linkage clustering method

- (c) Average linkage clustering method
- (d) Ward's method
- (e) Centroid method.

In *single linkage* clustering method, initially, two objects which are having the shortest distance will be clustered together. Then, again, if another object which has the least distance with any one of the existing clusters is found, it will be clustered with that cluster; or if another two unclustered objects which are having the least distance are found, they will be clustered together. This process continues until all the objects are included in a single cluster. The identification of the final set of clusters will be done based on certain clustering criterion, like sudden jump in the distance while merging a cluster with another cluster.

The *complete linkage* clustering method is same as the single linkage clustering method except the clustering criterion. In this method, the longest distance is the clustering criterion.

The *average linkage* clustering method is same as that of single linkage clustering method or complete linkage clustering method with the clustering criterion as the average distance. The average distance between two clusters is the average of the distances from the objects of one cluster to the objects of another cluster.

The *Ward's method* uses a different clustering criterion which signifies the loss of information by way of grouping objects into clusters. This measure is known as *total sum of squares of deviations (or error sum of squares)* of the objects coordinates from the respective mean coordinates of the cluster to which the objects are assigned.

In *centroid method*, the clustering criterion is the distance between the centroids of two clusters.

11.6.4 Hierarchical Clustering Algorithm

In this section, the agglomerative hierarchical clustering method is explained first. Later, it is demonstrated with an example problem.

The steps of this method are presented below:

- Step-1:* Initialize i to the total number of objects, $i = m$.
- Step-2:* Imagine the points in n -dimensional plane, where n is the number of variables (plotting will be easy only with two-dimensional plane).
- Step-3:* Find the distance between each pair of points.
- Step-4:* Identify the two points (p, q) which are having the least distance between them.
- Step-5:* Find the centroid of the two points (p, q) . Let them be c_i and the distance between them be d_i .
- Step-6:* Identify the next two nearest clusters of point(s) and group them together. Then, find the centroid of the newly formed cluster.
- Step-7:* Set $i = i - 1$
- Step-8:* If $i > 1$, then go to step-6; otherwise go to step-9.
- Step-9:* Draw dendrogram of the sequence of cluster formation.

Step-10: As per the clustering criterion, determine the number of clusters and the objects of the clusters. An example of clustering criterion may be the sudden jump in the distance between clusters while adding a cluster to another cluster. Draw a line in the dendrogram where there is a sudden jump in the distance between any two clusters and identify the final set of clusters accordingly.

Example 11.5 In a survey, the number of years of experience of employees in two different skills are summarized in Table 11.53.

Table 11.53 Survey Data about Number of Years of Experience on Skills

Employee	Skill-X	Skill-Y
1	2	8
2	8	15
3	3	6
4	6	9
5	8	7
6	10	10

Cluster the employees using agglomerative hierarchical clustering method.

Solution. Let the axis- X_1 be the Skill-X and the axis- X_2 be the Skill-Y.

Plot the points on a X_1 - X_2 plane, as shown in Fig. 11.7.

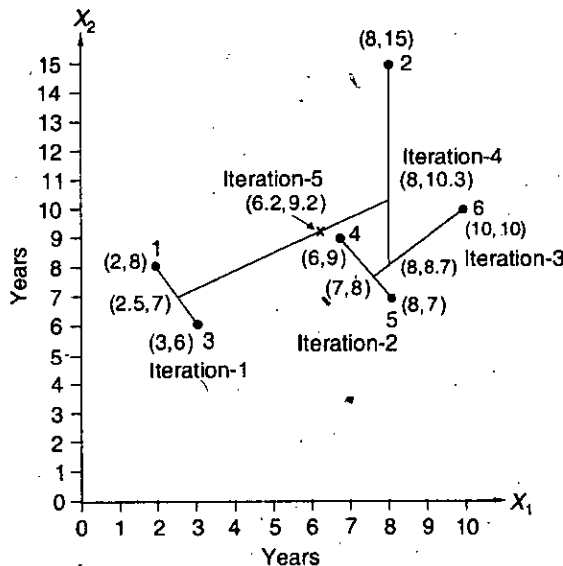


Fig. 11.7 Hierarchical clustering of Example 11.5.

Initially, the total number of clusters is equal to the total number of points (objects), 6. Let the clusters be C_1 , C_2 , C_3 , C_4 , C_5 and C_6 representing the point-1, point-2, point-3, point-4, point-5 and point-6, respectively.

From Fig. 11.7, it is clear that the points 1 and 3 are having the least distance which is equal to 2.236. So, cluster the points 1 and 3 and represent it as C_{13} . The coordinates of the centroid of the points 1 and 3 are (2.5, 7). Now, the number of clusters is reduced to 5.

The next closest clusters are the points 4 and 5 and the distance in between is 2.828. So, cluster these two points and represent it as C_{45} . The centroid of these two points is (7, 8). Now, the number of clusters is reduced to 4.

The next closest clusters are C_{45} and the point 6 and the distance between these two clusters is 3.6. So, join these two clusters and represent it as C_{456} . The centroid of these two clusters is (8, 8.7). Now, the number of clusters is reduced to 3.

The next closest clusters are C_{456} and the point 2 and the distance between these two clusters is 6.3. So, join these two clusters and represent it as C_{4562} . The centroid of these two clusters is (8, 10.3). Now, the number of clusters is reduced to 2.

The next closest clusters are C_{4562} and C_{13} and the distance between these two clusters is 6.414. So, join these two clusters and represent it as C_{4562} . The centroid of these two clusters is (6.2, 9.2). Now, the number of clusters is reduced to 1.

The above steps are summarized in Table 11.54.

Table 11.54 Results of Hierarchical Clustering of Example 11.5.

Iteration number (i)	No. of clusters	Nearest clusters	Centroid of of nearest clusters	Distance between nearest clusters	Set of cluster after merging nearest clusters
1	6	C_1, C_3	(2.5, 7)	2.236	$C_{13}, C_2, C_4, C_5, C_6$
2	5	C_4, C_5	(7, 8)	2.828	C_{13}, C_2, C_{45}, C_6
3	4	C_{45}, C_6	(8, 8.7)	3.600	C_{13}, C_2, C_{456}
4	3	C_{456}, C_2	(8, 10.3)	6.300	C_{13}, C_{4562}
5	2	C_{4562}, C_{13}	(6.2, 9.2)	6.414	C_{134562}

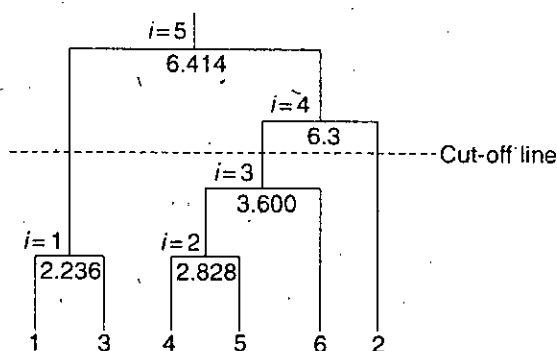


Fig. 11.8 Dendrogram of cluster formation.

Determination of number of clusters: The distance in the last but one column of Table 11.54 increases suddenly from iteration-3 to iteration-4. Hence, the number of clusters for implementation is 3 and they are summarized as follows:

Cluster 1: Employees 1, 3

Cluster 2: Employee 2

Cluster 3: Employees 4, 5, 6.

The results in Table 11.54 are represented in the form of a dendrogram as in Fig. 11.8. A cut-off line is drawn at a point where there is steep hike in the distance while merging clusters C_{456} , C_2 . The clusters below this critical line are treated as the final set of clusters for the given problem.

11.6.5 P Mean Clustering Algorithm Based on Variables

In this algorithm (Macqueen, 1967), an initial set of P clusters is formed intuitively or through arbitrary method. This initial set of P clusters is known as *initial seeds*.

The P mean clustering algorithm is a non-hierarchical clustering technique designed to rearrange the original members (objects) of initial P clusters for better similarity on attributes within each cluster. The rearrangement of an object from a cluster to another clusters is done based on a distance measure calculated between the centroids/means of the cluster to which the object belong and the potential cluster to which it can be shifted. In a remote situation, the final number of clusters may be less than the specified number of clusters, P . So, it can be generalized that the final number of clusters may be at most P .

Steps of P mean clustering algorithm

- Step-1: Input the data matrix, $[a_{ij}]$, where $i = 1, 2, 3, \dots, m$ and $j = 1, 2, 3, \dots, n$.
Where a_{ij} is the observation of the j th variable of the object i ; m , the total number of objects; and n is the total number of variables.
Number of clusters (at most number of clusters) to be formed, P .
- Step-2: Form the initial seeds consisting of P clusters based on intuition or using arbitrary method.
- Step-3: Find the centroid of each cluster.
- Step-4: Set the object number, $i = 1$ and Index = 0.
- Step-5: Identify the cluster to which the object i belongs and let it be Q .
- Step-6: Find the distance of the object i to the centroid of each of the clusters, d_{ij} , $j = 1, 2, 3, \dots, P$.
- Step-7: Find the minimum of these distances and identify the corresponding cluster:

$$X = \min \{d_{ij}, j = 1, 2, 3, \dots, P\}$$

Let the corresponding cluster be R .

- Step-8: If R is same as Q , then retain the object i in the present cluster Q itself and go to step-11; otherwise go to step-9.
- Step-9: Reassign the object i to the cluster R .
Set index = 1.

Step-10: Check whether the cluster Q has atleast one object after shifting the object to the cluster R in step-9. If yes, go to step-11; otherwise, reduce the number of clusters to be formed by 1, $p = P - 1$.¹

Step-11: $i = i + 1$

Step-12: If $i \leq m$, then go to step-5; otherwise, go to step-13.

Step-13: Update $P = p$ if the value of P is changed in step-10.¹

Step-14: If index = 1 then go to step-4; otherwise go to step-15.

Step-15: Print the objects of each clusters.

11.6.6 Rank Order Clustering Algorithm (ROC) for Groupings Based on Attributes (Association/Similarity Coefficients)

In this section, rank order clustering (ROC) applied to clustering is presented.

This algorithm was developed by J.R. King (1980) for the formation of machine-component cells in group technology. This can be used in its present form without any modification in cluster analysis for the formation of object groups.

This algorithm considers the following data:

1. Number of objects
2. Number of attributes
3. The association/similarity coefficient matrix.

This algorithm is used to obtain groupings of the objects based on attributes. Based on the attributes of the objects, an object-attribute incidence matrix is developed. The rows of the object-attribute incidence matrix represent the objects. The columns of the matrix represent the attributes. Let us treat this matrix as $[C_{ij}]$. This matrix contains 0-1 entries. If the object i has the attribute j , then the respective matrix element C_{ij} is assumed as 1; otherwise, it is assumed as 0.

Each row as well as each column of the matrix is considered as a binary number. For example, in a row, if we have numbers (1 0 1 0 1). Then the decimal equivalent is computed as follows:

$$\begin{aligned}\text{Row decimal equivalent} &= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 16 + 0 + 4 + 0 + 1 \\ &= 21\end{aligned}$$

Similarly, if a column has the following entries from top to bottom, the decimal equivalent is computed as explained below:

$$\begin{aligned}\text{Column entries} &= (1 \ 1 \ 0 \ 1 \ 0) \\ \text{Decimal equivalent} &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ &= 16 + 8 + 0 + 2 + 0 \\ &= 26\end{aligned}$$

1. The chance of updating $p = P - 1$ in step-10 and $P = p$ in step-13 is very remote.

The row with the largest decimal equivalent is considered to have the highest rank 1 among the rows. Similarly, the column with the largest decimal equivalent is considered to have the highest rank among the columns. The procedure to obtain final object-attribute incidence matrix is summarized below.

Steps of ROC

The steps of ROC algorithm are given below:

- Step-1:* Input: total number of objects, attributes.
- Step-2:* Input the object-attributes incidence matrix.
- Step-3:* Compute binary equivalent of each row.
- Step-4:* Rearrange the rows of the matrix in rank-wise (high to low from top to bottom).
- Step-5:* Compute binary equivalent of each column and check whether the columns of the matrix are in rank-wise (high to low from left to right). If not, go to step-6; otherwise, go to step-8.
- Step-6:* Rearrange the columns of the matrix in rank-wise and compute the binary equivalent of each row.
- Step-7:* Whether the rows of the matrix are in rank-wise. If not go to step-4; otherwise, go to step-8.
- Step-8:* Print the final object-attribute incidence matrix.

Example 11.6 Consider the object-attribute incidence matrix shown in Fig. 11.9. Find the groupings of the objects using ROC algorithm.

	Attribute (j)			
	1	2	3	4
Object (i)	1	0	1	0
	2	1	0	1
	3	1	0	0
	4	0	1	0
	5	0	0	1

Fig. 11.9 Object-attribute incidence matrix.

Solution. The results of different iterations are shown from Figs. 11.10–11.13.

Iteration-1:

	Attribute (j)				Decimal value	Rank
	1	2	3	4		
	1	2	3	4		
Object (i)	1	0	1	0	10	1
	2	1	0	1	5	3
	3	1	0	0	10	2
	4	0	0	0	4	4
	5	0	1	0	2	5

Fig. 11.10 Result of iteration-1.

Iteration-2:

		Attribute (j)			
		1	2	3	4
16	1	1	0	1	0
8	3	1	0	1	0
4	Object (i)	2	0	1	0
2	4	0	1	0	0
1	5	0	0	1	0
Decimal value		24	6	25	4
Rank		2	3	-1	4

Fig. 11.11 Result of iteration-2.

Iteration-3:

		Attribute (j)					
		3	1	2	4	Decimal value	Rank
1	1	1	1	0	0	12	1
3	1	1	1	0	0	12	2
Object (i)	2	0	0	1	1	3	4
4	0	0	1	0	0	2	5
5	1	0	0	0	0	8	3

Fig. 11.12 Result of iteration-3.

Iteration 4:

		Attribute (j)			
		3	1	2	4
16	1	1	1	0	0
8	3	1	1	0	0
4	5	1	0	0	0
Object (i)	2	0	0	1	1
2	2	0	0	1	1
1	4	0	0	1	0
Decimal value		28	24	3	2
Rank		1	2	3	4

Fig. 11.13 Result of iteration-4.

Based on Fig. 11.13, the groupings of the objects are as presented below:

Group-1: 1, 3, 5

Group-2: 2, 4

11.6.7 Mathematical Model for Clustering of Objects into P Groups Based on Attributes (Association/Similarity Coefficients)

In this section, a mathematical model for cluster of objects based on attributes into P groups is presented..

Consider a set of 6 objects each with one or more of the five attributes as shown in Fig. 11.14. The object-attribute incidence matrix which is shown in Fig. 11.14 has 0-1 entries.

	Object (<i>j</i>)					
	1	2	3	4	5	6
Attribute (<i>i</i>)	1	0	1	0	0	1
	2	1	0	0	0	1
	3	0	0	1	0	0
	4	1	0	1	0	0
	5	0	1	0	1	0

Fig. 11.14 Object-attribute incidence matrix.

Let n be the total number of objects (total number of columns, 6); m , the total number of attributes (total number of rows which is equal to 5); and $a_{ij} = 1$ (if the object j has the attribute i) and $= 0$, otherwise.

Similarity coefficient

A simple and meaningful similarity coefficient is defined as follows:

Consider an arrangement of attributes which is set up according to the attributes of an object. This arrangement is assumed to be a part of an object-attribute cell and the respective object is assumed to be a main object (Panneerselvam et al. 1985, 1990 and Balasubramanian and Panneerselvam, 1993) based on which the object-attribute cell is mainly formed. Suppose a group of objects which have either the same or partially the same attributes sequences as that of the main object is identified.

The similarity coefficient S_{ij} , the similarity of the object- i with the object- j (main object) is defined as follows:

$$S_{ij} = \frac{\text{Number of common attributes between the object-}i \text{ and the object-}j}{\text{Number of attributes in the object-}i} = \frac{\sum_{x=1}^m a_{xi}a_{xj}}{\sum_{x=1}^m a_{xi}}$$

Based on this formula, the similarity coefficient matrix $[S_{ij}]$ for the example problem is shown in Fig. 11.15.

	Object (<i>j</i>)					
	1	2	3	4	5	6
Object (<i>i</i>)	1	1.0	0.0	0.5	0.0	0.5
	2	0.0	1.0	0.0	0.5	1.0
	3	0.5	0.0	1.0	0.0	0.5
	4	0.0	1.0	0.0	1.0	0.0
	5	0.0	1.0	0.0	0.5	1.0
	6	0.5	0.0	0.5	0.0	1.0

Fig. 11.15 Similarity coefficient matrix.

Model to Identify object-attribute cells

Let n be the total number of objects (total number of rows or columns of the similarity matrix); P , the utmost number of object-attribute cells desired; q , the maximum number of objects to be grouped in any object-attribute cell; and $y_{ij} = 1$ (if the object i is grouped with the object j) and $= 0$, otherwise.

Then, a 0-1 programming model for the object-attribute cell formation is as follows:

Model Maximize

$$Z = \sum_{i=1}^n \sum_{j=1}^n S_{ij} y_{ij}$$

Subject to

$$\sum_{j=1}^n y_{ij} \leq P \quad (11.1)$$

$$\sum_{j=1}^n y_{ij} = 1, \quad \text{for } i = 1, 2, 3, \dots, n \quad (11.2)$$

$$\sum_{i=1}^n y_{ij} \geq y_{jj}, \quad \text{for } j = 1, 2, 3, \dots, n \quad (11.3a)$$

$$\sum_{i=1}^n y_{ij} \leq q y_{jj}, \quad \text{for } j = 1, 2, 3, \dots, n \quad (11.3b)$$

where $y_{ij} = 0$ or 1 , for $i = 1$ to n , $j = 1$ to n .

In the above model, the objective function maximizes the sum of the similarity coefficients of the object-attribute cells.

The first constraint restricts the maximum number of the final object-attribute cells to utmost p . The constraints in the second set make sure that each object is assigned to exactly one object-attribute cell.

The constraints in the third set guarantee that:

- if a main object from $1, 2, \dots, n$ is not selected in any of the final object-attribute cells then no object is assigned to that main object (all the y_{ij} values in the column corresponding to that main object are set to 0),
- if a main object from $1, 2, 3, \dots, n$ is selected in any of the final object-attribute cells, then at least one object (at least the object itself, if $y_{jj} = 1$) is assigned to that main object, and
- the total number of objects assigned to any object-attribute cell does not exceed q .

If we assume that there is no upper limit for the value of q , then the maximum value of q can be equal to the total number of objects, n . Let the value of k be 2 and q be n . Under this situation, the model for the example problem gives the results, as shown in Table 11.55. The corresponding final object-attribute cells are shown in Fig. 11.16. LINDO package may be used to solve this model. From Table 11.55, it is clear that the total number of objects groups is 2, which are as listed below:

- Objects 2, 4 and 5 are clustered together into one group based on the attributes 1 and 5.
- Objects 1, 3 and 6 are clustered together into another group based on the attributes 2, 3 and 4.

Table 11.55 Results for the Example Problem Using Model with $k = 2$ and $q = n = 6$

Object number for which $y_{ij} = 1$	Attribute-object cell	
	Object groups	Attribute groups
2	2,4,5	1,5
3	1,3,6	2,3,4

Here, objective value = 5 and $y_{13} = y_{22} = y_{33} = y_{42} = y_{52} = y_{63} = 1$.

		Object (<i>j</i>)					
		1	3	6	2	4	5
Attribute (<i>i</i>)	2	1	0	1	0	0	0
	3	0	1	1	0	0	0
	4	1	1	0	0	0	0
	1	0	0	0	1	0	1
	5	0	0	0	1	1	1

Fig. 11.16 Object-attribute cells.

QUESTIONS

1. Define discriminant analysis. What are the types of evaluation criterion used in discriminant analysis? Give examples of each type. Explain the steps of two-group discriminant analysis.
2. The performance standard of employees as a function of their age (X_1) and family size (X_2) is classified into 'Above average' and 'Below average'. The data on 10 different employees in a company are presented in the following table:

Employee	Standard	X_1	X_2
1	Below	43	3
2	Below	24	4
3	Above	30	6
4	Below	55	3
5	Below	56	5
6	Above	41	3
7	Below	37	3
8	Above	22	4
9	Below	38	6
10	Below	59	4

- Design the discriminant function, $Y = aX_1 + bX_2$.
 - Compute discriminant ratio, K and identify the variable which is more important in relation to the other variable.
 - Validate the discriminant function using the given data by forming groups based on the critical discriminant score.
 - Test whether the group means are equal in importance at a significance level of 0.05.
3. The potential customers of a computer company rate the product of the company into good or bad based on the time to respond to breakdown calls (X_1) and the percentage discount on product price (X_2). The ratings by customers are presented in the following table.

Customer	Rating	X_1 (in hours)	X_2 (%)
1	Good	24	5
2	Good	12	8
3	Bad	36	4
4	Bad	12	0
5	Bad	36	3
6	Good	36	10
7	Good	24	3
8	Bad	48	4
9	Bad	96	5
10	Good	36	12

- Design the discriminant function, $Y = aX_1 + bX_2$.
- Compute discriminant ratio, K and identify the variable which is more important in relation to the other variable.
- Validate the discriminant function using the given data by forming groups based on the critical discriminant score.

- (d) Test whether the group means are equal in importance at a significance level of 0.05.
4. Define factor analysis. Discuss the application areas of factor analysis.
 5. Explain the following:
 - (a) Correlation coefficient matrix
 - (b) Factor loadings
 - (c) Communality [A_i^2]
 - (d) Eigenvalue
 - (e) Factor rotation
 - (f) Minimum eigenvalue criterion
 - (g) Scree plot criterion
 - (h) Factor scores.
 6. Explain the steps of centroid method.
 7. Explain the steps of principal components method.
 8. In a survey on a product, data on three variables have been collected, as shown in the following table:

Survey Data

Respondent	Variable		
	X_1	X_2	X_3
1	5	7	8
2	3	3	5
3	3	1	5
4	1	1	5
5	3	2	4
6	3	3	5
7	2	2	1
8	6	6	5
9	4	2	1
10	4	3	2

Perform factor analysis using centroid method to identify two factors which can represent the variables of the study.

9. Discuss the steps of the alternative method of principal component analysis.
10. A television company wanted to study the quality of its TV with a view to improve its product quality. A consultant for this task has identified the following variables:
 - (i) Picture quality, X_1 (0, very poor; 10, excellent)
 - (ii) Reliability of TV, X_2 (0, very poor; 10, very good)
 - (iii) Response to breakdown calls, X_3 (0, low; 10, high)

In a pilot study, the questionnaires containing above questions were administered to 10 randomly selected respondents and their responses are summarized in the following table. Perform factor analysis using the alternate method of principal component analysis to obtain the following:

- Find factor loadings for three principal components (factors).
- Identify the significant number of principal components.
- Check whether factor rotation is required. If necessary, perform it using varimax rotation method.
- Group the variables into factors.
- Find the proportion of total variance as well as common variance for each of the factors.

Survey Data			
Respondent	Variable		
	X_1	X_2	X_3
1	8	3	5
2	6	1	7
3	6	1	7
4	4	7	6
5	9	5	8
6	4	4	1
7	8	1	5
8	8	2	7
9	7	2	4
10	1	3	9

11. The factor loading matrix in a factor analysis with 6 variables and two factors, is as shown in the following table:

Variable (I)	Factor	
	1	2
1	-0.875	-0.350
2	-0.840	-0.400
3	0.610	-0.700
4	0.600	-0.650
5	0.625	-0.475
6	-0.850	-0.450

Plot the factor loadings on a two-dimensional plane (F_1 - F_2 plane) and find the revised factors loadings using the varimax factor rotation method.

12. Define cluster analysis. What are the areas of applications of cluster analysis? Illustrate the concept of cluster analysis using scattered diagram.

13. Illustrate the concept of profile diagram used in cluster analysis with an example.
14. What are the measures of similarity used in cluster analysis? Explain them.
15. Distinguish between hierarchical clustering algorithms and non-hierarchical algorithms.
16. Distinguish between agglomerative hierarchical clustering algorithm and divisive hierarchical clustering algorithm.
17. List and explain the following clustering methods:
 - (a) Single linkage clustering method
 - (b) Complete linkage clustering method
 - (c) Average linkage clustering method
 - (d) Ward's method
 - (e) Centroid method.
18. Explain the steps of agglomerative hierarchical clustering method.
19. In a survey, the monthly income (in thousands of rupees) and monthly purchases of health care products (in hundreds of rupees) of respondents are summarized in the following table:

<i>Respondent</i>	<i>Monthly income</i>	<i>Monthly purchase</i>
1	12	5
2	19	8
3	14	6
4	15	9
5	18	7
6	10	2

- (a) Cluster the employees using agglomerative hierarchical clustering method with distance measure.
- (b) Draw a dendrogram and find the clusters using distance measure.
20. Explain the steps of P mean clustering algorithm based on variables.
21. List the steps of rank order clustering algorithm applied to cluster analysis.
22. Consider the following object-attribute incidence matrix involving association coefficients. Find the groupings of the objects using rank order clustering algorithm

Object-attribute Incidence Matrix

		Attribute (j)				
		1	2	3	4	5
Object (i)	1	0	0	1	0	0
	2	1	0	1	0	0
	3	0	1	0	1	0
	4	0	0	0	1	1
	5	0	1	0	1	1
	6	1	0	1	0	0

23. Write a generalized 0-1 programming model for clustering of objects into P groups based on attributes (association/similarity coefficients).

MULTIDIMENSIONAL SCALING AND CONJOINT ANALYSIS

12.1 INTRODUCTION

Multidimensional scaling (MDS) and conjoint analysis are the extensions of multivariate techniques for measuring human perceptions and preferences. Multidimensional scaling deals with the judgments of respondents about the degree of similarity of pairs of stimuli. The similarity may be defined as the degree of resemblance between pairs of stimuli on similarity basis or distance basis.

Conjoint analysis is similar to the multidimensional scaling in measuring customers preferences except that the stimuli are designed in advance using some factorial design/structure.

12.2 MULTIDIMENSIONAL SCALING

As mentioned earlier, it is used to measure human perceptions and preferences towards some stimuli (objects), like products, organizations, places, events, brands, etc., and position them in a perceptual space.

The scale value of each stimulus may be in the form of either metric or non-metric. The metric data are interval-scaled or ratio-scaled, whereas the non-metric data are ordinal-scaled. The input as well as output of fully metric form of MDS is ratio-scaled. This means that the output data are either multiple or linear functions of input data. Similarly, the input as well as output of fully non-metric form of MDS is ordinal-scaled. This means that the input as well as output are rank ordered. The input and output of non-metric form of MDS are ordinal-scaled and interval-scaled, respectively.

Multidimensional scaling involves three basic steps:

- Identification of the dimensions of the perceptual space on which customers perceive objects
- Input the respondents' judged values of perception/preference of the objects into MDS package
- Positioning of the objects on the perceptual space. The output of multidimensional scaling is a perceptual map which consists of location of the objects in it.

j to the location i , distances are shown only in the upper diagonal cells of Table 12.1. In this example, the number of dimensions is assumed as two. Multidimensional scaling procedure will take these distances which are known as *proximities as input* and simultaneously obtain the number of dimensions and the *derived distances* as output which will help construct a configuration of points in that dimensionality such that their derived distances (interpoint distances) are best matched with the input data (proximities).

Hypothetical plots of the coordinates of derived distances (output) from multidimensional scaling are shown by 'o' in Fig. 12.2 along with the original locations which are based on proximities as indicated by '*'. Sometime, the pattern of the plots of the recovered points based on the derived distances from multidimensional scaling may not be like that of the original plots at the first instance. After suitably rotating the dimensions, one can find the right type of derived dimension for each of the original dimensions of the example.

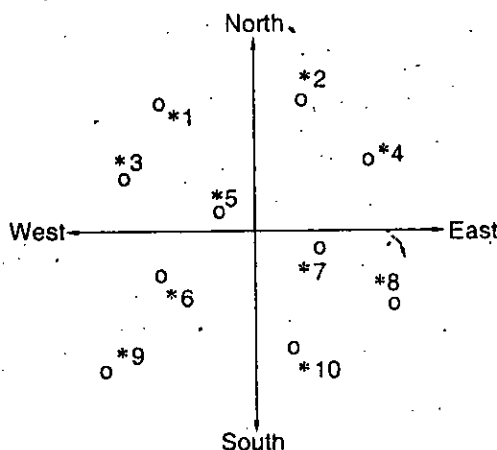


Fig. 12.2 Original locations (*) and locations obtained using MDS (o).

But, in general, the number of dimensions is not known in advance. Multidimensional scaling procedure will determine the minimum number of dimensions to represent the output, such that the recovered plots are closer to the respective original plots. This means that the derived distances (interpoint distances) of the recovered points are best matched with the input data (proximities).

If the number of dimensions is n , then the distance between any two points (positions) in multidimensional scaling is given by

$$d_{ij} = \sqrt{\sum_{k=1}^n (X_{ik} - X_{jk})^2}$$

where d_{ij} is the distance (proximity) between the point i and the point j ; X_{ik} , the coordinate of the point i along the dimension k ; X_{jk} , the coordinate of the point j along the dimension k ; and n is the total number of dimensions.

One can reconsider the data shown in Table 12.1 and convert them into ranks such that the smallest distance corresponds to the rank 1 and the largest distance corresponds to the rank 45. This type of problem is called non-metric MDS problem. If such data are given to MDS, then the required number of dimensions (in this case, it is 2) and the configuration of points in that dimensionality can be recovered, such that their inter-point distances are best matched with the input ranks generated from Table 12.1.

The above example is just an illustration with physical distances. But, the main purpose of multidimensional scaling is to scale psychological distances. These psychological distances are called as *dissimilarities*.

Psychological distance

In the previous example, we assumed physical distances which are known as actual distances, in reality as proximities. But, in practice, the judgments of a set of respondents about the proximities (estimated distances) between different pairs of locations of tourist spots will be obtained based on their knowledge and awareness about them. These distances are called *psychological distances*. Then, the average proximity for each pair of locations is to be obtained. By taking these average proximities data, MDS settles with minimum number of dimensions and gives derived distances based on which the configuration of points in that dimensionality can be developed such that their derived distances are best matched with the input data (proximities).

Later, one can plot the derived data in the conceptual space and compare them with the actual locations of different tourist spots. This exercise helps us compare the mental picture of respondents about tourist locations with the actual picture of tourist locations.

Determination of number of dimensions

As stated earlier, the important task of MDS is to determine the number of dimensions of the perceptual space in which the configuration of plots of the derived data of objects can be made. The difficulty of making inference of the results of MDS directly increases with the number of dimensions. If it is two, one can have the visual comparison of the positioning of the objects in the perceptual map. But, for a given problem, MDS will select the number of dimensions such that the interpoint distances of the objects in the derived data best match with the corresponding input data. MDS will assume some dimensions initially and the corresponding 'stress' will be calculated. The value of stress decreases with an increase in the number of dimensions. So, MDS will increase the number of dimensions one by one and the corresponding stress value will be computed. Like the scree plot in factor analysis, plot of 'stress' versus 'number of dimensions' will be made in MDS. The desired number of dimensions for the given problem corresponds to the point on the line graph of 'stress' versus 'dimensionality' plot where there is a steep change in the slope. At this point, the interpoint distances of the objects in the derived data will be best matched with the corresponding input data.

A sample plot of stress versus dimensionality graph is shown in Fig. 12.3.

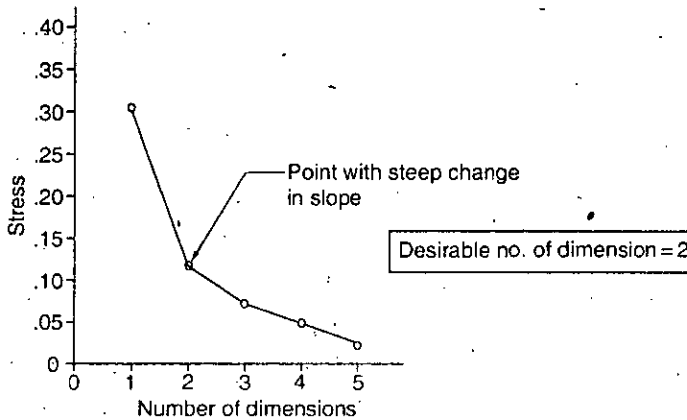


Fig. 12.3 Stress versus number of dimensions.

Naming dimensions An important task of MDS is to name each and every dimension meaningfully. If the attribute ratings of the objects form the basic data for MDS, then the respective MDS is called attribute MDS which is explained below. If the perceptual values between different pairs of objects are directly obtained from the respondents without mentioning any attribute to them, then the MDS which uses this type of data is called non-attribute MDS. In this situation, the perceptual values are like similarity/dissimilarity values signifying the interpoint distances on the perceptual map.

The correlation between each of the dimensions and attribute ratings of the objects can be studied and accordingly the dimensions can be named. One can use multiple regression model for such purpose. In the regression model, the attribute ratings are assumed as predictor variables (independent variables) and dimensional coordinates are assumed as the criterion variables (dependent variables).

Attribute of multidimensional scaling

Let us assume that a set of educational institutions (A, B, C, D, E, F, G, H, I and J) as objects and their attributes be as listed below:

1. Good academic input
2. Employment focus
3. Parental care
4. Institutional image
5. Nominal fees structure
6. Infrastructures.

Application of MDS to this problem begins with obtaining objects-attributes data in the form of object-attribute matrix. Later, this should be converted into object-object correlation data matrix. The object-object correlation for a pair of objects is the correlation

between the pair of rows of the corresponding objects in the object-attribute matrix. This correlation matrix gives values equivalent to the pairwise perception values of the educational institutions assigned by respondents. The average pairwise perception values of the institutions are the input to MDS. These are treated as the original interpoint distances of the institutions in the perceptual space/map. The output of MDS for this input data will reflect the positioning of the institutions on the perceptual map involving required number of dimensions. Let the number of dimensions decided by MDS be 2.

Assume that the plot of the institutions on the perceptual map as shown in Fig. 12.4. Now, the task is to name the dimensions. Based on the following discussion, the dimensions are named as *Overall success* and *Students centred*.

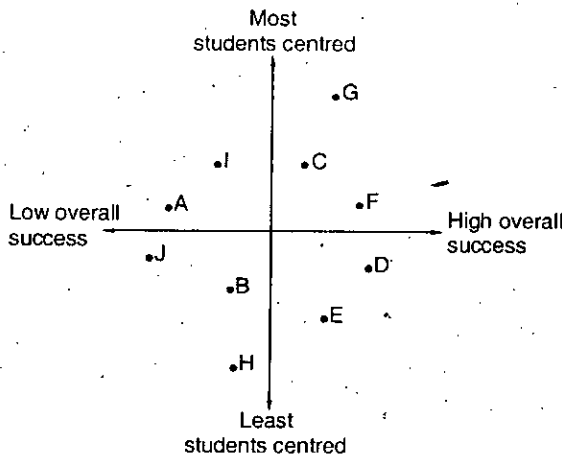


Fig. 12.4 Positioning of educational institutions.

At this stage, a multiple regression model can be carried out between the attribute ratings (predictor variables) and dimensional coordinates (criterion variables). For the purpose of illustration, let the regression coefficients of the attributes for each of the criterion variables are as shown in Table 12.2.

Table 12.2 Hypothetical Regression Coefficients

Attribute	Dimension 1	Dimension 2
Good academic input	0.80	0.10
Employment focus	0.95	-0.10
Parental care	-0.20	0.70
Institutional image	0.90	-0.40
Nominal fees structure	-0.30	0.75
Infrastructures	0.78	-0.22

From Table 12.2, it is clear that the attributes, Good academic input, Employment focus, Institutional image and Infrastructures are highly associated with the dimension 1.

These attributes jointly aim to place the graduates of institutions successfully in the employment market. Hence, this dimension can be named as *overall success* of the institution. The remaining attributes have high association with the dimension 2. These attributes jointly aim to facilitate students to pursue their studies. Hence, this dimension can be named as *students centred*. One should remember the fact that the intuition of researchers while naming dimensions will vary from person to person. But, by and large, the names of dimensions will convey the facts that underlie them.

From Fig. 12.4, it is clear that many institutions are clustered around the horizontal dimension, *overall success*.

The interpretations of a few institutions are as listed below:

- The institution D has high overall success and it is moderate students centred.
- The institution H is least students centred and has moderate overall success.

12.2.2 Applications Areas of MDS

As mentioned earlier, MDS can be applied for the following problems:

- Product positioning
- Organization positioning (as explained earlier)
- Group technology machine groups formation
- Technology positioning.

These are explained in the following sections.

Product positioning

Product positioning is the task of subdividing the given set of products in a study. This study will help the researching organization have better insight into existing product groups which will aid to find a relevant product group in which it can concentrate to bring out a new product with relevant marketing strategy in conjunction with the competing products. If this study is carried out by a company whose product is a part of the study, then, this study will help that organization formulate a suitable marketing strategies in relation to the competing products.

Non-attribute MDS for product positioning

Assume that the consultant of a company producing 50-cc, 60-cc and 100-cc two-wheelers has collected data on pairwise perceptual values of different pairs of products from respondents without mentioning any attribute to them. The list of products are $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3, \beta_4, \phi_1, \phi_2$ and ϕ_3 . In this list, α_1, α_2 and α_3 are 50-cc two-wheelers; $\beta_1, \beta_2, \beta_3$ and β_4 are 60-cc two-wheelers; ϕ_1, ϕ_2 and ϕ_3 are 100-cc two-wheelers. The consultant has employed non-attribute MDS technique for positioning the company's products. The matrix of average pairwise perceptual values of two-wheelers is the input to MDS. Let us assume that the output of MDS is mapped onto two-dimensional perceptual map, as shown in Fig. 12.5.

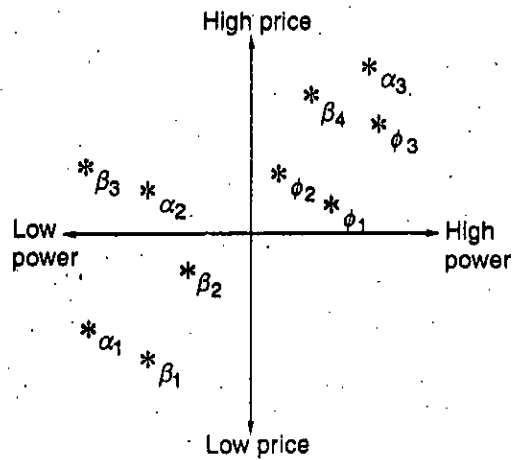


Fig. 12.5 Non-attribute MDS perceptual map of two-wheelers.

The horizontal dimension is named as 'power' in terms of cc which is the volume of combustion chamber of two wheelers in cubic centimetre. The vertical dimension is named as 'price'.

There are five groups of vehicles in Fig. 12.5, which are summarized in Table 12.3. The 50-cc two-wheelers are categorized as follows:

- α_1 : Low power and low price
- α_2 : Low power and medium price
- α_3 : High power and high price.

Hence, the revised marketing strategies of these three two-wheelers (α_1 , α_2 and α_3) can be formulated based on the competing two-wheelers within the respective groups, as shown in Table 12.3.

Table 12.3 Two-wheelers Groups

Group no.	Attributes of groups	Product names
1	High power and high price	α_3 , β_4 and ϕ_3
2	More than medium power and medium price	ϕ_1 and ϕ_2
3	Low power and medium price	α_2 and β_3
4	Medium power and medium price	β_2
5	Low power and low price	α_1 and β_1

Group technology machine groups formation—MDS approach

In production management, *group technology* is a system of layout which combines the benefits of process layout and product layout. Assume that the number of components to

be manufactured in a shop as n and the total number of unique machines which are required to process them be m . The objective of group technology layout is to subdivide the total number of components and machines into subgroups of machine-component cells such that the required machines of all the components which are assigned to each cell are maximally available within that cell itself. In this problem, the machines are assumed as objects and the components which are processed on each machine are treated as the attributes of that machine. The processing requirements of the components can be represented in the form of machine-component (object-attribute) incidence matrix. The rows of the machine-component incidence matrix represent the machines (objects). The columns of the matrix represent the attributes. Let us treat this matrix as $[a_{ij}]$. This matrix contains 0-1 entries. If the machine i is used in the component j , then the respective matrix element a_{ij} is assumed as 1; otherwise, it is assumed as 0.

Consider a set of 6 components each requiring one or more of the five machines, as shown in Fig. 12.6. (Panneerselvam et. al., 1990). The machine-component incidence matrix which is shown in Fig. 12.6 has 0-1 entries.

Let n the total number of components (total number of columns, 6); m the total number of machines (total number of rows which is equal to 5); and $a_{ij} = 1$ (if the machine i is used in the component j) and $= 0$, otherwise.

		Component (j)					
		1	2	3	4	5	6
Machine (i)	1	0	1	0	0	1	0
	2	1	0	0	0	0	1
	3	0	0	1	0	0	1
	4	1	0	1	0	0	0
	5	0	1	0	1	1	0

Fig. 12.6 Machine-component incidence matrix.

Similarity index A simple and meaningful similarity index between machine i and machine j (S_{ij}) is defined as follows (Srinivasan, Narendran and Mahadevan, 1990).

$$S_{ij} = \sum_{k=1}^n s_k, \text{ for } i \neq j, \text{ where } s_k = 1, \text{ if } a_{ik} = a_{jk};$$

$$= 0, \quad \text{if } i = j$$

Further, the value of S_{ij} is equal to that of S_{ji} . Based on this formula, the similarity matrix $[S_{ij}]$ for the example problem is shown in Fig. 12.7.

	Machine (j)				
	1	2	3	4	5
Machine (i)	1	0	2	2	5
	2		0	4	1
	3			0	1
	4				0
	5				

Fig. 12.7 Similarity index matrix.

Multidimensional scaling of machine-component cell formation The machine-component cell formation can be done using multidimensional scaling. In this problem, the similarity index between different pairwise combinations of machines (objects) are shown in Fig. 12.7. This similarity index matrix is used as the input to MDS. Then, MDS limits the number of dimensions to 2 and the corresponding results of machine groupings are shown in Table 12.4. In this example, naming of the dimensions is not necessary.

Table 12.4 Machine Groups Using MDS

Machine group number	Machines included
1	1, 5
2	2, 3, 4

The complete results of the given problem with component groupings in different machine groups are shown in Table 12.5. The corresponding block-diagonal form of the machine-component cells is shown in Fig. 12.8. It is clear that the total number of components groups, is 2, which are as listed below:

- Components 2, 4 and 5 are clustered together into one group based on the machines 1 and 5.
- Components 1, 3 and 6 are clustered together into another group based on the machines 2, 3 and 4.

Table 12.5 Results for the Example Problem using MDS

Machine group number	Machine group	Component group
2	1, 5	2, 4, 5
3	2, 3, 4	1, 3, 6

		Component (j)					
		1	3	6	2	4	5
Machine (i)	2	1	0	1	0	0	0
	3	0	1	1	0	0	0
	4	1	1	0	0	0	0
	1	0	0	0	1	0	1
	5	0	0	0	1	1	1

Fig. 12.8 Block-diagonal form of machine-component cells.

Technology positioning In view of depletion of energy sources, customers are very keen in choosing energy alternative for home appliances with appropriate technology subject to price, handling convenience, ecological issues of energy sources, service life of gadgets, etc.

In the case of water heater, there are following energy alternatives:

1. Electric current
2. Petroleum gas
3. Gobar gas/biogas
4. Solar energy
5. Kerosene
6. Fire wood
7. Rice husk.

The type of energy alternative will affect the technology of water heater. Let us assume that a company has designed a new technology water heater which can be operated with three different energy sources. Then, the construction of a perceptual map of the water heaters with existing technologies (energy technologies) along with the company's product may help the company to identify the dimensions of these technologies and positioning its water heater with other water heaters of different technologies.

Application of non-attribute MDS to this problem begins with obtaining non-attribute data from respondents for 21 $[n(n-1)/2]$ upper diagonal cells of the object-object perceptual matrix, where n is the number of objects (water heaters) which is 7. Then the average object-object perception value matrix which consists of average values across the respondents is to be given as the input to MDS. This object-object perception value matrix gives average values of the pairwise perception values of water heaters from respondents. The output of MDS for this input data will reflect the positioning of the water heaters on the perceptual map involving required number of dimensions. Careful examination of the relative positioning of the objects in the conceptual map will help the analyst name the dimensions.

Based on these details, the company can decide about the marketing strategy for its water heater in comparison with its competing products.

12.3 CONJOINT ANALYSIS

Conjoint analysis is technique which deals with the measurement of customer preferences. Let the number of factors affecting the preference of a product be m and n_i be the number of levels of the i th factor, where i varies from 1 to m . Then, the total number of product profiles or product concepts will be as follows: $n_1 \times n_2 \times n_3 \cdots \times n_m$. This is demonstrated with an example.

Consider a state-of-the-art computer in PC range. Assume that the customer preference of this product is affected by the following factors:

- Price
- Clock speed
- Disk capacity

Let the number of levels of each of the above factors be as given below:

- Price (Rs. 35,000, Rs. 40,000 and Rs. 50,000)
- Clock Speed (1 GHz, 2 GHz, 3 GHz)
- Hard disk capacity (20 GB, 40 GB).

The complete factorial combinations of the product profiles in terms of the factors and their levels are shown in Fig. 12.9. The interpretations of some of the factorial blocks which are known as product profiles are as presented below:

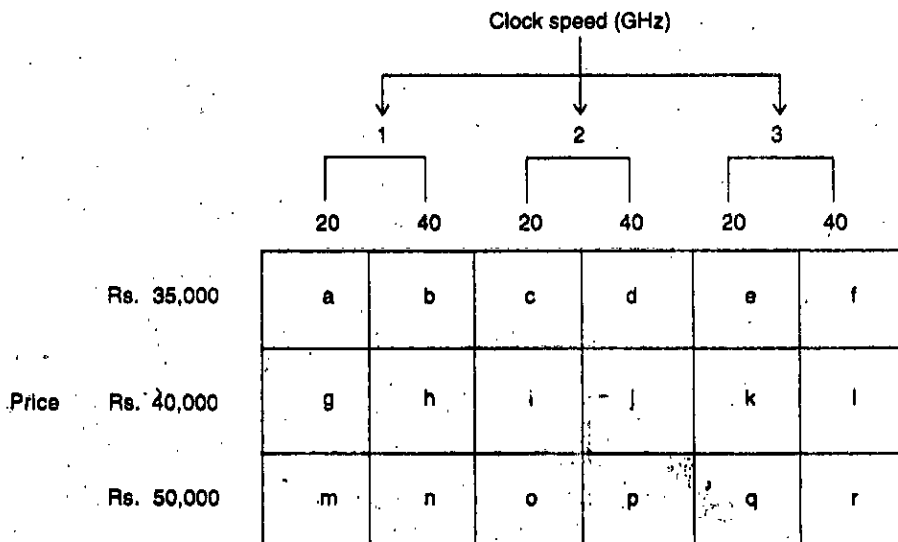


Fig. 12.9 Complete factorial design of product profiles.

Profile *a*: Price = Rs. 35,000, Speed = 1 GHz, Disk capacity = 20 GB

Profile *b*: Price = Rs. 35,000, Speed = 2 GHz, Disk capacity = 40 GB

Profile *c*: Price = Rs. 35,000, Speed = 2 GHz, Disk capacity = 20 GB

Profile *d*: Price = Rs. 35,000, Speed = 2 GHz, Disk capacity = 40 GB

:

Profile *r*: Price = Rs. 50,000, Speed = 3 GHz, Disk capacity = 40 GB

After collecting data about customer preferences, the utility value for each of the levels of different factors can be calculated using conjoint analysis. The most appealing product profile is the combination of levels by selecting one from each of the factors such that each level included in the product profile has the maximum utility within the respective factor.

The objective of conjoint analysis is to identify the most appealing product profile and also the most profitable product profile based on customer preferences. If they are one and the same, the respective product profile can be further developed and marketed. Sometime, the most appealing product profile may not be the most profitable product profile. Under such situation, a compromise can be arrived about the final desired product profile.

Approaches of conjoint analysis

There are two approaches for conjoint analysis: *two-factor* evaluation approach and *multi-factor* evaluation approach

The multi-factor approach is similar to the complete factorial design, which is shown in Fig. 12.9. It is the most ideal approach in terms of integrity in thinking while customers express their preferences. But, if the number of factors increases, then the customers will have difficulty in integrating the levels of different factors. Under such situation, one can resort to the two-factor evaluation approach in which customers are expected to express their preferences for different combinations of levels of each pair of the factors taken at a time.

12.3.1 Multi-factor Evaluation Approach for Conjoint Analysis

As explained earlier, the multi-factor evaluation approach defines different product profiles by taking all the factors together. Each product profile is written on a card and it is shown to a customer to seek his overall preference value/rating about the product.

Theoretically, the total number of product profiles is the multiplication of the number of levels of each of the factors. In Fig. 12.9, the total number product profiles is 18 ($3 \times 3 \times 2$).

Orthogonal design If the total number of product profiles (combinations) is too large, then one can select a limited number of product profiles using *orthogonal design*. Orthogonal design/arrays is a special type of factorial design used to study the effects of main factors through the smallest number of experimental combinations (product profiles) provided the effect of the second- and higher-order interaction terms are assumed to be insignificant. If the number of levels of four different factors are 3, 3, 3 and 2, then the total number of combinations is 54. But, as per orthogonal design, it is sufficient to select only 18 combinations from the complete factorial design provided the effects due to the interaction terms (second and higher order interaction terms) are insignificant. [For details, refer to the articles by Green (1974) as well as Box and Hunter (1957), and the textbook by Montgomery (1984).]

Application of multi-factor approach to example problem -

Reconsider the state-of-the-art computer in PC range. As per the assumption stated earlier, the customer preference of this product is affected by the following factors:

- Price
- Clock speed
- Disk capacity.

The number of levels of each of the above factors be as reproduced below:

- Price (Rs. 35,000, Rs. 40,000 and Rs. 50,000)
- Clock speed (1 GHz, 2 GHz, 3 GHz)
- Hard disk capacity (20 GB, 40 GB).

The complete factorial combinations of the product in terms of the factors and their levels are already shown in Fig. 12.9. In this study, the total number of product profiles is only 18. Hence, there is no need for orthogonal design. This means that all the 18 product profiles shown in Fig. 12.9 are considered in this study. A sample card defining the product profile with respect to the combination 'a' is shown in Fig. 12.10. One can prepare remaining 17 cards in the format as shown in Fig. 12.10.

Card 1	
Price:	Rs. 35,000
Speed:	1 GHz
Disk Capacity:	20 GB
Respondent Rating:	

Fig. 12.10 Product profile card with respect to the combination 'a'.

Types of preference value By showing these cards to a respondent, the researcher can obtain his/her preference value in terms of either *interval* scaled data or *ordinal* scaled data.

In the case of interval scaled data, each respondent is expected to give an overall preference value in the scale from 0 to 10. Here, 0 means that the customer has absolutely no interest to buy the product with the profile shown in the respective card and 10 means that the customer has extremely high interest to buy that product.

In the case of ordinal scaled data, each respondent is expected to rank all the product profiles shown to him/her from the most desirable product profile (rank 1) to the least desirable product profile (rank 18). Here, rank 1 means that the customer has extremely high interest to buy the product with the profile shown in the respective card and the rank 18 means that customer has absolutely no interest to buy that product.

For the example problem, if we assume interval scaled data from 0 to 10, then a sample set of preferences values given by a respondent for all the 18 product profiles is shown in Table 12.6.

Table 12.6 Sample Preference Values

Card no.	Price (Rs.)	Speed (GHz)	Disk capacity (GB)	Preference value
1	35,000	1	20	5.0
2	35,000	1	40	6.0
3	35,000	2	20	7.0
4	35,000	2	40	9.5
5	35,000	3	20	8.5
6	35,000	3	40	9.0
7	40,000	1	20	4.0
8	40,000	1	40	5.5
9	40,000	2	20	6.5
10	40,000	2	40	8.0
11	40,000	3	20	7.5
12	40,000	3	40	9.5
13	50,000	1	20	3.0
14	50,000	1	40	4.5
15	50,000	2	20	5.0
16	50,000	2	40	6.5
17	50,000	3	20	7.0
18	50,000	3	40	9.0

Now, the utility (part worth) of each level under each factor of the respondent can be obtained based on the data in Table 12.6 through a multiple regression program using dummy variable coding. For the selected respondent, a sample set of hypothetical utility graphs for the factors 'price', 'speed' and 'disk capacity' are shown in Figs. 12.11a, 12.11b and 12.11c, respectively. The preference value is treated as the criterion variable and the levels of the factors are treated as predictor variables in the regression model. The utilities are selected such that the sum of the errors between the actual preference values and the corresponding estimated preference values is minimized.

The estimation of the preference value for any product profile is obtained by adding the utility (part worth) values of the respective levels of the factors of that product profile. Based on the values shown in Fig. 12.11, the preference value of the first product profile shown by 'a' in Fig. 12.9 is estimated as shown below:

Utility when price is Rs. 35,000	= 3.00
Utility when speed is 1 GHz	= 1.50
Utility when disk capacity is 20 GB	= 0.75
Estimated preference value	= 5.25

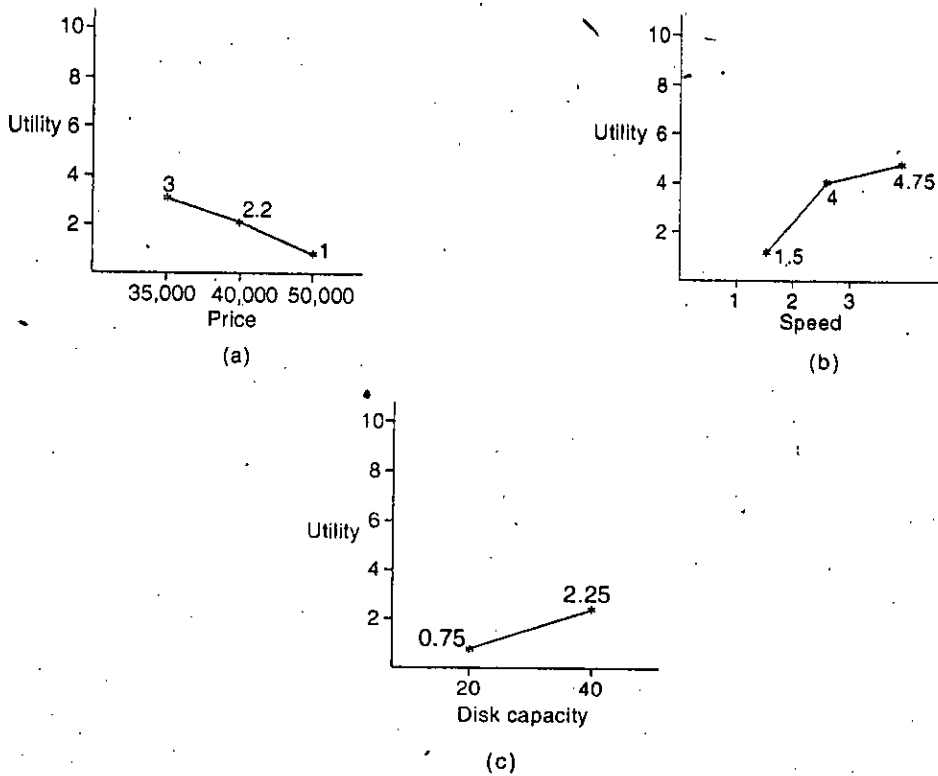


Fig. 12.11 Sample utility graphs of factors.

The comparison of the estimated preference values with the actual preference values assigned by the respondent are presented in Table 12.7. From this table, it is clear that the estimated preference values are more or less closer to the actual preference values assigned by the respondent. The respondent can select the product profile whose estimated preference value (sum of part worth value/utility values of the respective levels of the factors) is the maximum. Similar exercise should be done for the remaining respondents.

From Fig. 12.11, the most appealing product profile is the combination of the levels of the factors in which the utility of each level is maximum. From the figure, the first level (Rs. 35,000) of the factor, price has the maximum utility of 3; the third level (3 GHz) of the factor, speed has the maximum utility of 4.75; the second level (40 GB) of the factor disk capacity has the maximum utility of 2.25. So, the most appealing product profile is formed with a price of Rs. 35,000, speed of 3 GHz and disk capacity of 40 GB.

But, comparison of the utility graphs of different respondents will help the researcher form different groups of respondents such that the utility functions of the respondents within each group are similar. This will help the researcher identify the underlying market segments.

Table 12.7 Comparison of the Estimated Preference Values with the Actual Preference Values

Card no.	Price (Rs.)	Speed (GHz)	Disk capacity (GB)	Preference value	
				Actual	Estimated
1	35,000	1	20	5.0	5.25
2	35,000	1	40	6.0	6.75
3	35,000	2	20	7.0	7.75
4	35,000	2	40	9.5	9.25
5	35,000	3	20	8.5	8.50
6	35,000	3	40	9.0	10.00
7	40,000	1	20	4.0	4.45
8	40,000	1	40	5.5	5.95
9	40,000	2	20	6.5	6.95
10	40,000	2	40	8.0	8.45
11	40,000	3	20	7.5	7.70
12	40,000	3	40	9.5	9.20
13	50,000	1	20	3.0	3.25
14	50,000	1	40	4.5	4.75
15	50,000	2	20	5.0	5.75
16	50,000	2	40	6.5	7.25
17	50,000	3	20	7.0	6.50
18	50,000	3	40	9.0	8.00

In practice, the total number of factors with different levels in them will make the total number of product profiles a large number. Under such situation, one should select a limited number of product profiles using orthogonal design as mentioned earlier. Then the regular conjoint analysis is to be carried out for necessary number of respondents. The utility functions of different factors will help us estimate the preference values of the assumed number of product profiles. If one is interested, the preference values of the unassumed product profiles of the analysis can also be estimated using the utility graphs of the factors. This helps identify some unconventional product profile which will be more profitable to the organization.

The steps of the multi-factor approach with interval scaled data are summarized below:

- Step-1:* Select the required number of product profiles using orthogonal design.
- Step-2:* Administer these product profiles to a respondent and obtain his/her preference values in 0–10 scale.
- Step-3:* Obtain utility weights (part worths) of the levels of each factor such that the sum of the errors between the estimated preference values of the product profiles and the corresponding original preference values is minimized.
- Step-4:* Obtain the estimated preference values of each of the product profiles using the part-worths. The estimated preference value of a product profile is the sum of the part-worths/utility values of the levels of the factors in that product profile.

Step-5: Repeat step-2 through step-4 for necessary number of respondents.

Step-6: By comparing the estimated preference values of the product profiles of different respondents, identify possible market segments.

Directions to analyze ordinal scaled data

If the preference values are in terms of ranks, then the following procedure is to be used:

Step-1: Select the required number of product profiles using orthogonal design.

Step-2: Administer these product profiles to a respondent and obtain his/her ranks. Rank 1 is the highest and the rank 18 is the lowest for the example problem.

Step-3: Obtain the utility weights (part worths) of the levels of each factor such that the estimated ranks of the product profile using these part worths highly correlate with the original ranks of the product profiles.

Step-4: Obtain the estimated ranks of the product profiles using the part-worths. The estimated rank of a product profile is the sum of the part-worths/utility values of the levels of the factors in that product profile.

Step-5: Repeat step-2 through step-4 for necessary number of respondents.

Step-6: By comparing the estimated ranks of the product profiles of different respondents, identify possible market segments.

12.3.2 Two-factor Evaluation Approach for Conjoint Analysis

If the number of factors is very high, then it will be difficult for the respondents to have integrated idea about a product profile and express his/her rating in terms of preference value or rank of that product profile. Under such situation, it is desirable to use two-factor evaluation approach.

Again, consider the example with 'Computer' as the product which was presented earlier with three factors as reproduced below:

- Price
- Clock speed
- Disk capacity

The number of levels of each of the above factors be as given below:

- Price (Rs. 35,000, Rs. 40,000 and Rs. 50,000)
- Clock speed (1 GHz, 2 GHz, 3 GHz)
- Hard disk capacity (20 GB, 40 GB)

The total number of two-factor tables is $n(n-1)/2$, where n is the total number of factors of the study. The number of factors of this problem is three. By taking two at a time, there will be 3 $[3 \times 2/2]$ pairwise combinations of the factors as shown below:

- (a) Price $\times$ speed
- (b) Price $\times$ Disk capacity
- (c) Speed $\times$ Disk capacity.

Consider the two-factor table for the first pairwise combination of factors as shown in Table 12.8 with their corresponding levels.

Table 12.8 Two-factor Table for 'Price $\times$ Speed' combination

Speed (GHz)	Price (Rs.)		
	35,000	40,000	50,000
3			
2			
1			

Now, a selected respondent has to rank all the nine combinations of the levels of the two factors, 'Price and Speed' based on trade-off between them. Rank 1 represents the highest rank and rank 9 represents the lowest rank.

Let the hypothetical ranks assigned by the respondent be, as shown in Table 12.9. Like this table, the two-factor table for the remaining two combinations of the factors are shown in Tables 12.10 and 12.11. In Table 12.10, there are six combinations of Price and Disk capacity and the corresponding ranks given by the respondent be, as shown in that table. In Table 12.11, there are six combinations of 'speed and disk capacity' and the corresponding ranks given by the respondent be as shown in that table.

Table 12.9 Two-factor Table for 'Price $\times$ Speed'

Speed (GHz)	Price (Rs.)		
	35,000	40,000	50,000
3	1	3	4
2	2	6	8
1	5	7	9

Table 12.10 Two-factor Table for 'Price $\times$ Disk capacity'

Disk capacity (GB)	Price (Rs.)		
	35,000	40,000	50,000
40	1	2	4
20	3	5	6

Table 12.11 Two-factor Table for 'Speed × Disk Capacity'

Disk capacity (GB)	Speed (GHz)		
	3	2	1
40	1	3	5
20	2	4	6

After collecting the trade-off values/ranks for all the two factor tables, the utility value/part-worth of each of the levels in each of the factors will be determined.

Then, the trade-off value/rank in each cell of the each of the two-factor tables can be estimated by adding the corresponding row and column part-worths to that cell such that there is greater correlation between the original ranks and the estimated ranks of the cells across all the two-factor tables.

The steps of two-factor approach of conjoint analysis are summarized below:

- Step-1:* Determine the number of two-factor tables for the given number of factors.
- Step-2:* Administer all $n(n-1)/2$ two-factor tables one by one to the respondent and obtain his/her ranks for different cells in that table. In each two-factor table, rank 1 is the highest and the lowest rank is equal to the total number of cells in it.
- Step-3:* Obtain utility weights (part-worths) of the levels of each factor such that the estimated ranks of the cells of each two-factor table using these part worths correlate highly with the original ranks of the cells of the respective two-factor table.
- Step-4:* Obtain the estimated ranks of each cell in each of the two-factor tables using the respective row and column part-worths. The estimated rank of a cell in a two-factor table is the sum of the part-worths/utility values of the corresponding levels (row and column) of that cell.
- Step-5:* Repeat step-2 through step-4 for necessary number of respondents.
- Step-6:* By comparing the estimated ranks of all the two-factor tables of different respondents, identify possible market segments.

QUESTIONS

1. Distinguish between multidimensional scaling and conjoint analysis.
2. Distinguish between physical distance and psychological distance.
3. Illustrate the concept of multidimensional scaling through its application to geographical locations of cities in a country.

4. How will you determine the number of dimensions of the conceptual space of multidimensional scale?
5. Discuss the importance of naming the dimensions of the conceptual space of MDS.
6. What are the application areas of multidimensional scale?
7. Illustrate the application of MDS to product positioning.
8. Illustrate the application of MDS to machine groupings to form machine-component cells in group technology.
9. What is technology positioning? Illustrate the application of MDS to technology positioning.
10. What are the approaches of conjoint analysis? Explain them in brief.
11. Discuss the steps of multi-factor approach for conjoint analysis with interval-scaled data.
12. Illustrate the multi-factor approach for conjoint analysis applied to an example with interval-scaled data.
13. Discuss the steps of multi-factor approach for conjoint analysis with ordinal-scaled data.
14. Illustrate the multi-factor approach for conjoint analysis applied to an example with ordinal-scaled data.
15. Discuss the steps of two-factor approach for conjoint analysis.
16. Illustrate the application of two-factor approach of conjoint analysis with an example.

ALGORITHMIC RESEARCH

13.1 INTRODUCTION

Algorithm is a well defined sequence of steps to solve a problem of interest. It is nothing but a procedure for solving a problem, with special focus on solving problems using computer. Some examples of standard algorithms are as follows:

- Search algorithms (binary search, Fibonacci search, golden section search, etc.)
- Sorting algorithms (straight insertion sort, bubble sort, quick sort, heap sort, Parsort, etc.)
- Shortest path algorithms (Dijkstra's algorithm, Floyd's algorithm, etc.)
- Minimum spanning tree algorithms (PRIM algorithm, Kruskal's algorithm, etc.)
- Branch and bound algorithm (minimization type, maximization type.)
- Hungarian method for assignment method
- Simplex algorithm for linear programming
- Zero-one implicit enumeration algorithm for 0-1 programming
- Simulated annealing algorithm
- Genetic algorithm
- Tabu search algorithm
- Greedy algorithm for Travelling Salesman Problem.

This is only a sample list of standard algorithms. The algorithm of a reality may consist of steps which are unique to it and if necessary may consist one or more than one of the above such algorithms as its subroutine(s).

13.2 ALGORITHMIC RESEARCH PROBLEMS

A sample list of research problems in different functional areas of management (Production management, Marketing management, Financial management and Personnel management) and in some other areas which requires algorithms is as presented as follows:

Production management

- Minimizing the mean tardiness in single machine scheduling problem
- Minimizing the weighted mean tardiness in single machine scheduling problem
- Minimizing the sum of the tardiness and earliness in single machine scheduling problem
- Minimizing the sum of the weighted earliness and tardiness in single machine scheduling problem
- Minimizing the makespan in parallel machines scheduling problem
- Minimizing the makespan in flowshop scheduling problem
- Minimizing the mean flow time in flowshop scheduling problem
- Minimizing the makespan in jobshop scheduling problem
- Maximizing the balancing efficiency of an assembly line
- Process layout design which results in minimum cost of materials handling
- Resource levelling in project network
- Resource allocation in project network
- Plant location problems (single facility location problem, multi-facility location problem)
- Line balancing
- Aggregate planning problem
- Master production planning problem.

Marketing management

- Dealer location problem
- Warehouse location problem
- Salesmen assignment problem
- Product pricing problem
- Media planning problem
- Location of after sales service centres
- Travelling Salesman Problem
- Location of marketing branch offices.

Financial management

- Determination of rate of return
- Capital budgeting (allocation of fund to different plants and each plant with different alternatives)
- Selection of mutually exclusive and collectively exhaustive projects
- Portfolio management.

Personnel management

- Employees in-house training problem
- Manpower planning problem
- Allocation of fund to personnel activities for improved labour productivity
- Assignment problem (matching of employees with tasks for better productivity).

Other areas

- Fire stations location problem
- Schools location problem
- Branch banks location problem
- Scheduling processors in parallel computers
- Encryption and decryption problems
- Organization of B-tree.

13.3 TYPES OF ALGORITHMIC RESEARCH PROBLEMS

The types of algorithmic research problems can be classified into polynomial problem and combinatorial/NP-hard problems. The algorithms for solving these real-life problems are classified into polynomial algorithms and exponential algorithms, respectively.

13.3.1 Polynomial Algorithm for Polynomial Problem

If the complexity of an algorithm for a problem is in polynomial form, then that algorithm is known as *polynomial algorithm*. Such, algorithms will take finite time to solve a problem optimally.

Here, the complexity of an algorithm means a function representing the number of steps (time) required to solve a problem under the worst case behaviour. An algorithm consists of a set of steps. For some problems, the algorithm will be executed for all its steps or for its subset of steps. The worst case behaviour of an algorithm means the maximum number of steps executed (or time taken) for a problem.

Let the expression representing the complexity of an algorithm is as shown below:

$$f(n) = 5n^4 + 6n^2 + 25$$

where n is the problem size, $f(n)$ is the time complexity function.

The function is a polynomial of order 4. Hence, the algorithm which has this function as its complexity function is known as polynomial algorithm.

The worst case behaviour of the algorithm is known as the *order of the algorithm*. The highest polynomial of the given function is treated to be the worst case. Hence, the order of the given algorithm is $O(n^4)$. Here, the big O means 'order'.

The problems which will have polynomial time complexity function are called *polynomial problems* and these problems are solved in finite time.

Some examples of polynomial problems are listed below:

- Shortest path problem (Dijkstra's algorithm, Floyd's algorithm) (Panneerselvam, 2002)
- Finding the roots of a quadratic equation
- Minimizing mean flow time in single machine scheduling (Panneerselvam, 1999, 2002). The algorithm used for this problem is the Shortest path time (SPT) rule.
- Minimizing the maximum tardiness in single machine scheduling problem (Panneerselvam, 1999). Earliest due date (EDD) rule is used for this problem.
- Minimizing the number of tardy jobs in single machine scheduling problem (Panneerselvam, 1999). The algorithm used for this problem is Hodgson's algorithm.
- Minimizing the makespan in 2 machines and n jobs case of flowshop scheduling problem (Panneerselvam 1999). Johnson's algorithm is used for this problem.
- Greedy heuristic for travelling salesman problem (Goodman and Hedetniemi, 1977). All heuristics will have polynomial time complexity function.

Example of computing time complexity function of polynomial problem

Consider the shortest path problem in which the objective is to find the shortest distance as well as the corresponding path for any given pair of nodes in a distance network.

This type of problem can be solved using Floyd's algorithm (Panneerselvam, 2002).

This algorithm takes the initial distance network $[D^0]$ and the initial precedence matrix $[P^0]$ as input. Then it performs n iterations (n is the number of nodes in the distance network) and generates the final distance matrix $[D^n]$ and the final precedence matrix $[P^n]$. One can find the shortest distance between any two nodes from the final distance matrix $[D^n]$ and can trace the corresponding path from the final precedence matrix $[P^n]$.

Floyd's algorithm Following are the steps:

Step-1: Set $k = 0$

Step-2: From the initial distance matrix $[D^0]$ and the initial precedence matrix $[P^0]$ from the distance network.

The distance matrix as well as the precedence matrix are with n rows and n columns. In the distance matrix, between any two nodes, if there is no direct arc, then the corresponding distance is assumed as ∞ . The diagonal values of the initial distance matrix are assumed as 0. In the initial precedence matrix, the diagonal values are assumed as '-' and the other entries in the i th row are assumed as i , for $i = 1, 2, 3, \dots, n$ and i is not equal to j .

Step-3: Set $k = k + 1$

Step-4: Obtain the values of the distance matrix, $[D^k]$ for all the cells, where i is not equal to j using the following formula:

$$D_{ij}^k = \min \left[D_{ij}^{k-1}, \left(D_{ik}^{k-1} + D_{kj}^{k-1} \right) \right]$$

Step-5: Obtain the values of the precedence matrix, $[P^k]$, for all the cells, where i is not equal to j using the following formula:

$$P_{ij}^k = P_{ij}^{k-1}, \quad \text{is } D_{ij}^k \text{ not equal to } D_{ij}^{k-1} \\ = P_{ij}^{k-1}, \quad \text{otherwise}$$

Step-6: If $k = n$, go to step-7, otherwise, set $k = k + 1$ and go to step-4.

Step-7: For each source-destination nodes combination, as required in reality, find the shortest distance from the final distance matrix, $[D^n]$ and trace the corresponding shortest path from the final precedence matrix $[P^n]$. Guidelines for tracing the shortest path for a given combination of source node and destination node are presented below:

7.1: Let the source node be X and the destination node be Y .

7.2: Fix node Y as the last node in the partially formed shortest path.

7.3: Find the value from the final precedence matrix $[P^n]$ for the row corresponding to node X and the column corresponding to Y . Let it be Q . Prefix node Q in the partially formed shortest path.

7.4: Check whether Q is equal to X . If not so, set $Y = Q$ and go to step-7.3; otherwise go to step-7.5.

7.5: The path constructed is the required shortest path from the source node X to the destination node Y .

Complexity function In this algorithm, the critical analysis of execution of steps are presented below:

- (a) The step-1 is executed only once.
- (b) The step-2 will be executed n^2 times to read the distance matrix, $[D^0]$ as well as the precedence matrix, $[P^0]$.
- (c) The step-3 is repeated for one time..
- (d) In each of the step-4 and the step-5, calculations for n^2 cells in the distance matrix, $[D_{ij}^k]$ as well as in the precedence matrix, $[P_{ij}^k]$ are done for n times.
- (e) The step-6 is repeated for n times.
- (f) In step-7, for a given source-destination combination, a maximum of n set of calculations comparisons will be done if the shortest path involves all the nodes of the network.

Based on the explanations from (a) to (f), the time complexity function of Floyd's algorithm is presented below:

$$f(n) = 1 + 2a_1n^2 + 1 + n(2a_2)n^2 + n + a_3n \\ = 1 + 2a_1n^2 + 1 + 2a_2n^3 + (1 + a_3)n$$

where a_1 , a_2 and a_3 are some constants. The highest degree of the above polynomial is 3. Hence, the worst case time complexity function of Floyd's algorithm is $O(n^3)$.

Since the highest power of the polynomial is a constant (3), Floyd's algorithm is a polynomial algorithm. Further, this algorithm gives the optimal solution in terms of the shortest distance and the corresponding path for a given pair of source and destination node.

Note: If the power of the worst case time complexity function of an algorithm is a constant, then the function is called polynomial function and the corresponding problem is called as polynomial problem.

13.3.2 Exponential Algorithm for Combinatorial/NP-hard Problem

If the time complexity function of an algorithm for a problem is in exponential form, then that algorithm is known as *exponential algorithm*. Such algorithm takes a long time to solve a problem. For example, even the branch and bound technique for travelling salesman problem takes a century to solve a problem involving about 20 nodes in its distance network (Goodman and Hedetniemi, 1977).

Consider the following time complexity functions of some algorithms:

1. $f(n) = n!$
2. $f(n) = 15(2^n + 3n^2 + 20)$
3. $f(n) = e^n + 7n + 2$

In all these functions, n is the size of the problem and $f(n)$ is the time complexity function of the respective algorithm.

If a graphical plot is made for each of these functions, one can verify the fact that the value of $f(n)$ grows exponentially. This means that even for a small increment in n will cause a very large unproportionate positive increase in $f(n)$.

The algorithms which are having such functions are known as 'exponential algorithms'. The order of complexity of the three functions are $O(n!)$, $O(2^n)$ and $O(e^n)$, respectively which are in exponential order.

The problem which is having exponential time complexity function is called *combinatorial problem* or *NP-hard problem*. Np-hard problem means non-polynomial hard problem.

Note: If the power of the time complexity function of an algorithm is the size of the problem (n) or if the time complexity function is in factorial form of the size of the problem ($n!$), then the order of the function is exponential and the corresponding problem is called 'combinatorial problem/NP-hard problem'.

Some examples of combinatorial problems are listed below:

- Travelling salesman problem (TSP) aiming at optimal solution: The branch and bound method can be used to solve this problem optimally (Goodman and Hedetniemi, 1977).
- Minimizing mean tardiness in single machine scheduling: The branch and bound method can be used for this problem (Panneerselvam, 1999, 2002).

- Minimizing weighted mean tardiness in single machine scheduling problem. One can use complete enumeration technique to solve this problem optimally.
- Minimizing sum of the weighted number of early and tardy jobs in single machine scheduling problem (Panneerselvam et al., 1991). One can use a mathematical model or complete enumeration technique to solve this problem optimally.
- Machine-component cell formation in group technology. A mathematical model can be used to solve this problem optimally (Panneerselvam, 1999).
- Minimizing the makespan in M machines and N jobs flowshop problem (Panneerselvam, 1999). The branch and bound technique can be used to solve this problem optimally.
- Minimizing the makespan in M machines and N jobs jobshop problem (Panneerselvam, 1999, 2002). The branch and bound technique can be used to solve this problem optimally.
- Minimizing the total number of facilities to serve a maximum number of customers under total covering problem (Panneerselvam, 1990). One can use either a mathematical model or complete enumeration technique to solve this problem optimally.

Example of computing time complexity function of combinatorial problem

Consider the case of minimizing the mean tardiness in single machine scheduling problem (Panneerselvam, 1999).

A single machine scheduling problem contains n different single operation jobs whose processing times and due dates are known. All the jobs are available with the shop at the time of generating the schedule.

The generalized data of this problem is shown in Table 13.1.

Table 13.1 Generalized Data of Single Machine Scheduling Problem

Job (i)	Processing time (t_i)	Due date (d_i)
1	t_1	d_1
2	t_2	d_2
3	t_3	d_3
$\vdots$	$\vdots$	$\vdots$
i	t_i	d_i
$\vdots$	$\vdots$	$\vdots$
n	t_n	d_n

The objective of this problem is to find the sequence of jobs which minimizes the mean tardiness.

The tardiness of the job i is as defined below:

$$T_i = C_i - d_i, \quad \text{if } C_i > d_i$$

$$= 0, \quad \text{otherwise.}$$

where T_i is the tardiness of the job i ; C_i , the completion time of the job i ; d_i is the due date of the job i .

A sample problem is shown in Table 13.2.

Table 13.2 Sample Single-machine Scheduling Problem

Job (i)	Processing time (t_i)	Due date (d_i)
1	7	12
2	5	8
3	9	17

The number of jobs (n) in this problem is 3. So, the number of possible sequences of the jobs is $3!$, and the corresponding sequences are listed in Table 13.3.

Table 13.3 List of Sequences for Three-jobs Problem

Sequence number	Sequence
1	1-2-3
2	1-3-2
3	2-1-3
4	2-3-1
5	3-1-2
6	3-2-1

The details of computation of the total tardiness of the fourth sequence are summarized in Table 13.4.

Table 13.4 Details of Computation of Total Tardiness for 2-3-1

Job (i)	t_i	C_i	d_i	T_i
2	5	5	8	0
3	9	14	17	0
1	7	21	12	9

$$\text{Mean tardiness} = \frac{\sum_{i=1}^n T_i}{n} = \frac{0+0+9}{3} = 3$$

Like this, one can determine the mean tardiness of each of all other sequences of this problem. Finally, the sequence which has the least mean tardiness is to be identified as the sequence giving the optimal mean tardiness. This approach gives the optimal solution for the example problem in finite time, because the number of sequences in the example problem is only 6 ($3!$).

In this approach, all the possible solution combinations are enumerated and then each one is solved optimally. Finally the solution combination which has the best objective function value is selected as the solution for implementation. Such approach is known as *complete enumeration* method.

The number of sequences of this problem will grow exponentially with respect to the problem size (n), as shown in Table 13.5.

Table 13.5 Growth of Number of Sequences versus Problem Size (n)

Problem size (n)	No. of sequences ($n!$)
1	1
2	2
3	6
4	24
5	120
6	720
7	5040
8	40320
9	362880
10	3628800

From Table 13.5, it is clear that even for a small increment in n , the value of $n!$ increases unproportionately, even for moderate values of n .

From Tables 13.3 and 13.5, it is clear that the order of the time complexity function to find the sequence which minimizes the mean tardiness is $O(n!)$. Hence, this problem of minimizing the mean tardiness in the single machine scheduling problem is known as a *combinatorial* problem or NP-hard problem.

13.4 TYPES OF SOLUTION PROCEDURE/ALGORITHM

The solution procedures/algorithms can be classified into three types:

1. Complete enumeration method
2. Branch and bound method
3. Heuristic method.

If a given problem is polynomial in nature, then one can use either complete enumeration method or branch and bound method or some other method which is unique to the problem to solve it optimally.

If the given problem is combinatorial/NP-hard and its size is very large, then obtaining the optimal results in finite time is impossible. Hence, one should use some efficient heuristic to solve such problem near-optimally or optimally. The complete enumeration method or the branch and bound method can be used for solving small/moderate size combinatorial problems optimally.

13.4.1. Complete Enumeration Method

In this approach, all the possible solution combinations are enumerated and each one is solved optimally. Finally, the solution combination which has the best objective function value is selected as the solution for implementation. Such an approach is known as *complete enumeration* method. An example for the complete enumeration method is already given in Section 13.3.2.

The complete enumeration method can be applied to polynomial problems or small-size combinatorial problems. This cannot be applied to large-size combinatorial problems because the number of possible solution combinations will grow exponentially with respect to the problem size.

13.4.2. Branch and Bound Method

This is a curtailed enumeration method. Unlike the complete enumeration method, in this method, the creation of certain subproblems will be curtailed by the usage of a strong lower bound in the case of minimization problem or a strong upper bound in the case of maximization problem.

The branching procedure replaces an original problem by a set of new subproblems that are:

- Mutually exclusive and collectively exhaustive subproblems of the original problem
- Partially solved version of the original problem
- Smaller problems than the original problem.

The branch and bound method employs any one of the following approaches for branching to create subproblems:

- Breath first search method (BFS method)
- Depth first search method (DFS method)

The BFS method can find the solution quickly in comparison with DFS method, but it requires lot of memory space to store the subproblems. The DFS method requires less memory space to store the subproblems. Hence, depending on the problem, one should select the right type of branching procedure.

Caution: The worst case time complexity function of the branch and bound method is exponential in nature. Hence, it should be applied to only small size or moderate size problems. The usage of more and more stronger bound (lower bound or upper bound) can enhance the size of the problem that can be solved using the branch and bound method.

Hence, any problem specific research may be focussed in this direction.

13.4.3 Heuristics

It is known that the number of subproblems of a given combinatorial problem will grow exponentially with respect to its problem size. Hence, it will be difficult to solve any

combinatorial problem optimally using exact algorithms like, complete enumeration method, branch and bound method, etc.

If one can solve a combinatorial problem optimally in finite time using some method, then similar method can be extended to solve all other combinatorial problems in literature. Alternatively, if one can lift Himalayan mountain by some technique, the same technique can be followed to lift all other mountains on the earth. But, it is impossible to do so.

Solving a combinatorial problem optimally by any method in finite time is similar to the above example. Hence, it is impossible to solve all the combinatorial problems in literature, optimally in finite time. But, the practitioners need to solve such problems in finite time to carry out their businesses. Hence, they prefer to solve the combinatorial problems near-optimally in finite time using some suitable method.

Any method which is used to solve a combinatorial problem near-optimally in finite time is called *heuristic*. A heuristic can also give the optimal solution on some trials.

Let us assume that a given combinatorial problem is solved using a method for 1,00,000 times. Again, let us assume that the method applied to this combinatorial problem gives the optimal solution for 99,999 times, but it fails to give the optimal solution for one time.

Though the method gives the optimal solution for 99,999 times, the method should be termed as a heuristic because it fails to give the optimal solution for one time.

In general, a method is called heuristic until there is a proof that it will give an optimal solution for all the problems and for all trials on each problem.

While designing a heuristic, one should always aim to achieve the objective of the problem of interest as much as possible in terms of solution accuracy and add steps minimally and effectively such that the heuristic takes least time to solve the problem.

Example of heuristic

Consider the travelling salesman problem in which a salesman has to initiate his tour from his base city and visit each city exactly once and return to his base city. In this process, his organization's objective may be to minimize the total distance of his travel or total time of his travel.

It is like the single machine scheduling problem in which the mean tardiness is to be minimized. If n is the number of nodes in the distance network pertaining to the salesman, then the number of such tours is $n!$. This shows that this problem is combinatorial. Hence, some heuristic should be used to have a quick and practical solution.

There are many heuristics available for this purpose. Just to have a feel of a heuristic, the 'greedy' heuristic applied to the travelling salesman problem is presented below which has the order of complexity as $O(n^2)$. The name 'greedy' means that the problem solver is greedy of saving the time for solving the problem and also greedy of obtaining better solution.

Greedy heuristic for travelling salesman problem Following are the steps:

Step-1: Input the distance matrix, $[D_{ij}]$, $i = 1, 2, 3, \dots, n$ and $j = 1, 2, 3, \dots, n$, where n is the number of nodes in the distance network.

If there is no direct arc between two nodes in the distance network, then assume the distance between them as ∞ . Also, the diagonal values of the matrix should be assumed as ∞ .

- Step-2:* Randomly select a base city. Let it be X and delete the column X of the distance matrix.
- Step-3:* Include X as the first city in the tour.
- Step-4:* In the row X, find the least matrix cell entry and identify the corresponding column. Let this column be Y.
- Step-5:* Include Y as the next city to be visited in the tour.
- Step-6:* Delete column Y of the distance matrix
- Step-7:* Check whether all the columns of the distance matrix are deleted. If yes, go to step-9; otherwise go to step-8.
- Step-8:* Set $X = Y$ and go to step-4.
- Step-9:* Include the first city as the last city in the tour.
- Step-10:* List the cities in the tour along with the corresponding total distance of travel of the tour.

13.5 STEPS FOR DEVELOPMENT OF ALGORITHM

The basic steps of complete development of an algorithm to solve a problem are presented below:

1. Writing the statement of the problem
2. Developing a model
3. Designing the algorithm
4. Checking the correctness of the algorithm
5. Implementing the algorithm
6. Analyzing the complexity of the algorithm
7. Testing the program
8. Documenting the algorithm.

Statement of the problem

A proper statement of the problem is very important because in many situation it will be very difficult to clearly state the problem because of the confusing nature of the problem. If the problem is not clearly stated, it will result in garbage-in and garbage-out (GIGO).

So, while defining the problem, one should write down the *problem statement* in macro term. Then on repeated trials by superimposing the requirements of the reality, the macro statement can be refined to a clear problem statement.

Development of model

Model is an abstraction of reality. A model will help people represent a reality in an understandable form. A model may be in the form of a graph, network, matrix, equation, sequence of logic, mathematical problem with an objective function and a set of constraints, etc.

Design of algorithm

As stated earlier, algorithm is a sequence of well-defined steps. For many problems, algorithm design is an iterative process which continues until a clear-cut set of steps are obtained. One can use the concept of top-down programming concept to design an algorithm. As per this concept, the main problem is subdivided into meaningful number of subproblems, then again each subproblem should be subdivided like this until a well-defined set of steps are obtained to solve the entire main problem.

Correctness of the algorithm

The correctness of the algorithm means the correctness of the logic/steps of the algorithm. Generally, algorithms will have many steps, branching and loops included in different modules. One has to verify whether each and every module is working as per the expected results by giving a set of predetermined input. This is possible by doing hand calculation for a few iterations and comparing the intermediate results with that of known results.

Implementation

Once the correctness of the algorithm is verified, then the algorithm is to be converted into a set of codes which can be executed in a computer. At this stage, the variables and their types, arrays and their sizes, data structures and subroutines which are required by the program should be decided depending on the reality. Above all, the language in which the algorithm to be implemented is an important decision which will have great impact on the programming productivity. With all these preparatory work, finally, the given algorithm should be converted into a computer program.

Analysis and complexity of algorithm

As explained in Section 13.3, the complexity of algorithm should be analyzed. There are two types of complexity of an algorithm, namely time complexity and volume complexity.

Time complexity The *time complexity* function of an algorithm gives the worst case estimate in terms of the number of steps to be executed for the algorithm. The order of the algorithm is defined like, $O(n^2)$, $O(n!)$, $O(2^n)$, etc. The big 'O' means the order of the algorithm. $O(n^2)$ means that order of the algorithm is n^2 which indicates the maximum number of steps required to solve any problem by that algorithm is n^2 , where n is the problem size.

Again, as explained in Section 13.3, the time complexity function may be either polynomial or exponential.

Volume complexity The *volume complexity* function of an algorithm represents the amount of prime memory space required while executing the algorithm. Again, this may be either polynomial or exponential. In the case of branch and bound technique, if breadth first search (BFS) is used, the function representing the memory space required to store the subproblems will be in exponential form; if depth first search (DFS) is used, the function representing the memory space required to store the subproblems will be in polynomial form. The volume complexity is mainly affected by the type of algorithm used and also by the type

of data structure used. This analysis will be helpful in deciding the type of computer to be used for implementing the algorithm.

Program testing

Once the algorithm complexity is analyzed, the program should be tested for its correctness. Though, the correctness of the algorithm has been verified earlier, while converting the algorithm into a program, there may be some error in terms of implementation. Such errors should be corrected by carefully going through different modules of the program with known input and comparing the intermediate results of the program with predetermined results. Also, the program should be tested for extreme values. Also, the performance of the program for different inputs in terms of time taken to solve the problem can be studied well in advance which will act a guideline to use the program in future, accordingly.

Documentation

The final step of the algorithm development is the documentation. The documentation can be classified into *internal* documentation and *external* documentation.

Internal documentation This means the documentation in the program in terms of comment statements to distinguish each and every module. Over-documentation will increase the program length which is not desirable. Under-documentation will result in poor understanding of the program while modifying the program in future by someone else or by the same person for its enhanced features.

External documentation This means the documentation of the program in the form of manuals. There are different types of manual which are tutorial manual, user manual and operation manual. The tutorial manual will take the user through all the features of the program with sample data. The user manual will contain complete guidelines of the working of the algorithm. The operations manual will contain instructions about installation and maintenance of the program.

13.6 STEPS OF ALGORITHMIC RESEARCH

The steps of algorithmic research are as follows:

- Step-1:* Perform preliminary review of literature and select the problem of interest and state its objective clearly.
- Step-2:* Check the nature of the selected research problem.
 - (a) If the selected research problem is a problem with polynomial time complexity function, then the researcher should develop the following:
 - (i) A mathematical model to solve the problem optimally
 - (ii) An efficient exact algorithm which will give the optimal solution for the selected problem.
 - (b) If the selected research problem is a combinatorial problem, then the researcher should develop the following:

- (i) A mathematical model to solve the problem optimally (this model can be used to solve only small or moderate size problems).
- (ii) An efficient heuristic to find the near optimal solution for the selected problem.

Step-3: Conduct review of literature and identify the most efficient existing algorithm for the problem of interest. Also, study its limitations.

Step-4: Develop a new algorithm.

Step-5: Design an experiment to compare the new algorithm with the most efficient existing algorithm.

Step-6: Conduct the experiment and compare the results of the new algorithm with that of the model as well as that of the most efficient existing algorithm.

(a) If the selected problem is with polynomial time complexity function, then do the following comparisons:

- (i) Compare the results of the new algorithm with that of the newly developed mathematical model for few sample problems to check the correctness of the results.
- (ii) Compare the optimal results of the new algorithm with that of the most efficient exact algorithm.

(b) If the selected problem is a combinatorial problem, then do the following comparisons:

- (i) Compare the results of the new algorithm with that of the newly developed mathematical model for small size problem to study the extent of deviation of the results of the new algorithm from the optimal results of the model.
- (ii) Compare the results of the new algorithm (heuristic) with that of the most efficient existing algorithm (heuristic).

Step-7: Apply the new model and/or the new exact algorithm/new heuristic to a case problem.

All the above steps are self-explanatory. However, steps 5 and 6 are demonstrated with an example in the next section.

13.7 DESIGN OF EXPERIMENTS AND COMPARISON OF ALGORITHMS

In the mass production system, the production is achieved through line layout (Product layout). In the line layout, the product will be assembled in different stages manually. If there is no system of grouping the activities of the assembly line into different stations, it will be difficult to control the rate of production as per the production plan.

Hence, the objective of assembly line balancing is to group the activities of the assembly line into different workstations by meeting the precedence constraints and cycle time constraints such that the balancing efficiency is maximized. The *cycle time* is the time between consecutive releases of fully-assembled product from the last workstation of the assembly line or the total time allotted to each and every workstation. (For more details, readers are directed to refer the book by Panneerselvam, 1999.)

In this direction, Narayanan and Panneerselvam (2001) have compared their heuristic titled 'New global search heuristic for assembly line balancing, (NGSH)' with an existing heuristic titled 'Heuristics for assembly line balancing, (HAL)' which was developed by Panneerselvam et al. (1993).

Design of experiment

The design of ANOVA experiment to compare NGSH with HAL is presented below. In research, it is assumed that the response variable 'Balancing Efficiency' is affected by two factors, namely, 'algorithm' and 'problem', which are denoted by A and P , respectively.

The number of levels of the factor, algorithm (A) is 2 and that of the factor, problem (P) is 8. For each experimental combination of the two factors, A and P , 3 replications have been carried out by varying the cycle time (CT). The balancing efficiencies (BE) as per the above experimental design are summarized in Table 13.6.

Table 13.6 Balancing Efficiencies (%)

Problem (P)	No. of work elements	Algorithm (A)	
		HAL	NGSH
1	45	86.63	92.05
		87.66	96.43
		81.82	98.18
2	60	88.59	91.76
		88.83	96.91
		86.38	98.72
3	48	87.38	91.35
		87.66	97.40
		82.73	99.28
4	87	89.48	91.96
		90.68	96.33
		90.01	98.19
5	64	88.96	96.38
		87.07	99.51
		83.06	99.67
6	52	88.23	97.05
		87.31	91.46
		85.53	99.78
7	55	88.16	91.32
		89.75	95.36
		84.99	99.16
8	45	90.35	95.11
		88.77	95.11
		86.94	97.80

The model of ANOVA of the experiment is:

$$Y_{ijk} = \mu + A_i + P_j + AP_{ij} + e_{(ij)k}$$

where $i = 1, 2, 3, \dots, a$, $j = 1, 2, 3, \dots, b$, and $k = 1, 2, 3, \dots, n$; μ , the overall mean; A_i , the effect of the i th algorithm on the response variable; and P_j is the effect of the j th problem on the response variable. AP_{ij} , the effect of the i th algorithm and j th problem on the response variable; $e_{(ij)k}$, the error for the k th replication under the i th algorithm and j th problem; a , the number of levels of the factor A ($a = 2$); b , the number of levels of the factor P ($b = 8$); and n is the number of replications under each experimental combination ($n = 3$).

Here, the factor A (algorithm) is assumed as a *fixed* factor and the factor P (problem) is assumed as a *random* factor. Hence, the interaction action component AP_{ij} is a random component.

Comparison of algorithm

The significance of the differences between the treatments of each of the components of the model is tested using the following hypotheses.

Hypothesis with respect to treatment of the factor A:

Null hypothesis, $H_0: A_1 = A_2$

Alternate hypothesis, H_1 : Treatment means are not equal for the only one pair of the treatment means of the factor A .

Hypothesis with respect to treatment of the factor P:

Null hypothesis, $H_0: P_1 = P_2 = P_3 = \dots = P_8$

Alternate hypothesis, H_1 : Treatment means are not equal for at least one pair of the treatment means of the factor P .

Hypothesis with respect to interaction component, AP:

Null hypothesis, $H_0: A_1P_1 = A_1P_2 = \dots = A_2P_7 = A_2P_8$

Alternate hypothesis, H_1 : Interaction means are not equal for at least one pair of interaction means.

The expected mean square formulas (EMS) and suitable F -ratios to test the significance of different components of the model are summarized in Table 13.7. As per Tables 13.6 and 13.7, the results of ANOVA are summarized in Table 13.8.

Component: Factor A (algorithm): For this component,

$$F_{(\text{calculated})}[135.902] > F_{\alpha=0.05 \text{ and } df=(1,7)}[5.59]$$

Hence, the null hypothesis H_0 should be rejected. This means that there is a significant difference in terms of balancing efficiencies between different algorithms.

Table 13.7 EMS Formulas and F-ratios

Model component	Expected mean squares formula	F-ratio
A_i	$\sigma^2 + n\sigma_{AP}^2 + \left[bn \sum_{i=1}^a A_i^2 / (a-1) \right]$	EMS_A / EMS_{AP}
P_j	$\sigma^2 + a\sigma_P^2$	EMS_P / EMS_{Error}
$(AP)_{ij}$	$\sigma^2 + n\sigma_{AP}^2$	EMS_{AP} / EMS_{Error}
$\alpha_{(ij)k}$	σ^2	

Table 13.8 Results of ANOVA

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares (MSS)	F-ratio ($\alpha = 0.05$)	
				Calculated	Table
SS_A	1	910.00	910.000	135.902	5.590
SS_P	7	26.48	3.780	0.470	2.314
SS_{AP}	7	46.87	6.696	0.832	2.314
SS_{Error}	32	257.65	8.052		
Total	47	1241.00			

Component: Factor P (Problem): For this component,

$$F_{(calculated)}[0.470] < F_{\alpha=0.05 \text{ and } df=(7,32)}[2.314]$$

Hence, the null hypothesis, H_0 should be accepted. Therefore, this means that there is no significant difference in terms of balancing efficiency between different problems.

Component: Interaction, AXP: For this component,

$$F_{(calculated)}[0.832] < F_{\alpha=0.05 \text{ and } df=(7,32)}[2.314]$$

Hence, the null hypothesis (H_0) should be accepted. Therefore, this means that there is no significant difference in balancing efficiencies because of the interaction between the factor 'algorithm' and the factor 'problem'. Alternatively, this means that the interaction between 'algorithm' and 'problem' has no significant effect on the balancing efficiency.

Second-level test

The null hypothesis for the component, A_i alone in the model is rejected. Since there is a significant difference between different algorithms, one should go for a second-level test, namely, *Duncan multiple range test*, to further discriminate the differences between algorithms.

Since the number of levels for the factor A (algorithm) is 2, the number of pairs in the Duncan multiple range test will be only one. So, mere comparison of the average of the responses under each algorithm will give the inference that which algorithm is better instead of seeking the same inference through the Duncan multiple range test.

One can verify the fact that the average balancing efficiency of NGSH is more than that of HAL. So, the algorithm NGSH performs better than HAL.

Note: If the number of levels for any significant component is more than 2, then one has to test the significance of differences between different pairs of treatment means of that component using Duncan multiple range test.

13.8 META HEURISTICS FOR COMBINATORIAL PROBLEMS

As stated earlier, the medium and large-size combinatorial problems cannot be solved in polynomial time. So, a researcher is left with the following options for solving such problems:

- (a) Developing an efficient single-pass heuristic.
- (b) Developing any one of the following meta heuristics:
 - Simulated annealing algorithm.
 - Genetic algorithm
 - Tabu search.

The first option of developing an efficient single-pass heuristic will be unique to the problem situation and will give the solution after performing a polynomial number of steps. Hence, the solution through such heuristic will be local optimum.

The second option of developing a meta heuristic aims at a global optimum which will be almost closer to the optimum solution.

Areas of applications The areas of applications of the meta heuristics are production scheduling, travelling salesman problem, vehicle routing problem, just-in-time manufacturing, group technology, line balancing, media mix, dealer location, plant location, etc.

13.8.1 Simulated Annealing Algorithm

Metropolis et al. (1953) have developed an algorithm to find an equilibrium of a system containing large number of atoms for a known temperature (T). This algorithm forms the basis for the development of many simulated annealing algorithms to solve combinatorial problems. Kirkpatrick et al. (1983) established an analogy between minimizing the cost function of a combinatorial optimization problem and slow cooling of a solid metal, by utilizing the optimization process presented by Metropolis et al. (1953). By executing the Metropolis algorithm at a sequence of slowly decreasing temperature values, Kirkpatrick and his co-researchers obtained a combinatorial optimization algorithm which they called *simulated annealing algorithm*. Simulated annealing is a computational process which attempts to solve hard combinatorial optimization problems through controlled randomization. Simulated annealing emulates the process of annealing which attempts to force a system to its lowest energy state through controlled cooling.

In practice, the atoms of the metallurgical structure of some work piece will not be aligned in the desired direction because of the following reasons. Such non-alignment of atoms will reduce the workability/machinability of the work piece:

- due to work done on the work piece by machine tools
- due to internal trapping of gases in castings while cooling hot metal in the moulds.

The process of annealing is used to release the internal stress and strain of such metal piece which will improve its workability/machinability.

In general, the annealing process undergoes the following steps:

1. The temperature of the system is raised to a sufficient level.
2. While cooling, the temperature of the system is maintained for prescribed amount of time at each of the predetermined temperatures.
3. The system is allowed to cool under controlled conditions until the desired energy state is attained.

The initial temperature, the amount of time the system remains at this temperature and the rate at which the system is cooled are referred to as the *annealing schedule*. If the system is cooled too fast, it may freeze at an undesirable, high energy level. With respect to optimization problems, the state of the system corresponds to the objective function. Similarly, the freeze of a system at an undesirable energy state corresponds to an optimization problem which is frozen at a local optimum.

In simulated annealing, the problem starts at some initial solution, and a series of moves (changes in the values of decision variables) are made according to a user-defined annealing schedule until either the optimal solution is attained or the problem becomes frozen at a local optimum from which it cannot be improved. To avoid freezing at a local optimum, the algorithm moves slowly (with respect to the objective function value) through the solution space. This controlled improvement of the objective function value is accomplished by accepting non-improving moves (non-improving objective function values) with certain probability (based on the resulting change in the objective function value and the current temperature) which decreases as the algorithm progresses.

The generalized procedure for implementing a simulated annealing algorithm in the case of maximization objective function applied to a combinatorial problem is presented below.

In this algorithm, a parameter r in the range 0 to 1 is used to lower the temperature at each computational step and s is a small positive number provided by the user.

Steps of simulated annealing algorithm

- Step-1:* Get an initial feasible solution s_0 and compute the related value of the objective function $f(s_0)$.
- Step-2:* Get an initial temperature, $T > 0$.
- Step-3:* Generate a feasible solution s_1 in the neighbourhood of s_0 and compute the related value of the objective function $f(s_1)$.
- Step-4:* Compute $d = f(s_1) - f(s_0)$
- Step-5:* If $d \geq 0$, set $s_0 = s_1$; otherwise set $s_0 = s_1$ with the probability $e^{(d/T)}$.
- Step-6:* Set $T = r \times T$.

Step-7: If $T > s$, then go to step-3; otherwise, go to step-8.

Step-8: Use local optimization to reach a local optimum starting from the last s_0 value.

Finally, keep the best solution among all those that have been obtained. The real difficulty in applying simulated annealing algorithm is usually the definition of the initial feasible solution, the initial temperature, the parameter s and the coefficient r ($0 < r < 1$). T , r and s are known as *control parameters* which are unique to a given combinatorial problem and hence these are to be defined for each such problem.

The neighbourhood of a given solution s_0 is the set of solutions obtained by applying an elementary perturbation to s_0 . The concept of perturbation depends on the problem of our interest. A sample set of perturbation schemes is: transfer, trade and singleton.

Transfer In the *transfer* scheme, there is transfer of data from one set to another set in the initial solution or in the intermediate solution.

Trade In the *trade* scheme, there is exchange of data from one set to another in the initial solution or in the intermediate solution.

Singleton In the *singleton* scheme, there is creation of a new set with single member which is transferred from some other set in the initial solution or in the intermediate solution.

13.8.2 Genetic Algorithm

Genetic algorithm (GA) is also another meta heuristic which is used to find the solution to any combinatorial problem. It mimics the mechanism of selection and evaluation. In order to achieve the objective, GA generates successive population of alternate solutions until a solution is obtained that yields acceptable results. Within the generation of each successive operation, the improvements in the quality of the individual solutions are gained. Hence, GA can quickly move to a successful outcome without need to determine every possible solution to the problem. The procedure used in the genetic algorithm is based on the fundamental processes that control the evolution of biological organisms, namely, natural selection and reproduction. The organism's ability to survive within its environment is improved by these two processes as explained below:

- (a) Natural selection determines which organism has the opportunity for reproduction and survival within a population.
- (b) Reproduction involves genes from two separate individuals combining to form offspring that inherits the survival characteristics of their parents.

Terminologies

Chromosome *Chromosome* is a complete description of an individual in the population. In GA, it is the collection of primitive features called *genes*. It is a string of symbols.

Population A set of chromosome forms a single *population*. In GA, a set of strings each represents a potential solution to the given problem.

Gene *Gene* is a single feature within a chromosome..

Allele *Allele* is a particular value that may be taken on by a gene. Different genes will have different alleles. For example, the alleles of the gene that determine hair colour may be red, black, brown, etc.

Generation Successive evolution of chromosome is called *generation*.

Offspring The next generation of a chromosome is called its *offspring*.

Schema *Schema* is a collection of genes in a chromosome which are having certain specified values.

Encoding of chromosome In GA, initially, the chromosomes have to be encoded. *Encoding* of chromosome depends on the problem. Binary encoding is most popular.

Crossover *Crossover* selects genes from the parent chromosomes and create a new offspring.

There are many type of crossover. Some important types of crossover are discussed now.

(a) *Single-point crossover*: Let the chromosomes of parents be as follows:

Chromosome-1: 110 101

Chromosome-2: 111 001

In this type of crossover, the crossover point is randomly selected. Let the point of crossover be the third position from left. Then create an offspring by copying all the genes which are on or before this crossover point from one parent and all the genes which are after this crossover point from another parent.

Based on these guidelines, the possible offspring from chromosome-1 and chromosome-2 are as listed below:

Offspring-1: 110 001

Offspring-2: 111 101

(b) *Two-point crossover*: Let the chromosomes of parents be as follows:

Chromosome-1: 11 01 01

Chromosome-2: 11 10 01

In this type of crossover, the two crossover points are randomly selected. Let the points for the crossovers be the second position and fourth position from left. Then an offspring is created as explained below:

- Copy all the genes which are on or before the first crossover point from one parent.
- Copy all the genes which are after the first crossover point and on or before the second crossover point from another parent.
- Copy all the genes which are after the second crossover point from the first parent.

Based on these guidelines, the possible offspring from chromosome-1 and chromosome-2 are as listed as follows:

Offspring-1: 11 10 01

Offspring-2: 11 01 01

(c) *Uniform crossover*: In this type of crossover, an offspring is formed by copying genes randomly from the chromosomes of both parents.

(d) *Arithmetic crossover*: In this type of crossover, some arithmetic operation is performed over the genes of the chromosomes of the parents to form an offspring. Let the chromosomes of parents be as follows:

Chromosome-1: 110 101

Chromosome-2: 111 001

If AND operation is performed between the genes of the corresponding positions of the chromosomes, the following offspring is created:

Offspring: 110 001

Mutation *Mutation* is performed on an offspring after the crossover. This is done on the offspring mainly to avoid the solution ending with local optimum. Mutation performs the changing of bits (alleles) in the offspring from 1 to 0 or from 0 to 1, on some randomly selected bits just to randomize the bits in the offspring.

Selection *Selection* is the process of picking some chromosome from the population for crossover. There are many methods of selection, but the most popular *Roulette wheel selection* is explained below.

In 'Roulette wheel selection', the parents are selected which have better fitness values. Assume that all the chromosomes are placed on a Roulette wheel. When a marble is thrown onto the wheel, the chromosome which has more fitness will have more chance of selection. Based on this logic, a procedure to simulate this selection is presented below:

- Step-1:** Compute the sum of the fitness values of all the chromosomes in the population. Let it be S .
- Step-2:** Generate uniformly distributed random number from the interval 0 to S . Let it be K .
- Step-3:** Starting from the first chromosome in the population, obtain the cumulative fitness values of the chromosomes until the cumulative fitness value just exceeds K . Then identify the corresponding chromosome for which the cumulative fitness value up to that chromosome just exceeds K and treat it as one of the parents for crossover.

Steps of genetic algorithm

The general structure of the genetic algorithm is given below:

- Step-1:** Input the number of iterations to be carried out.
Let the generation count (GC) be 1.
- Step-2:** Generate a random population of n chromosomes (suitable solutions for the problem).

- Step-3:** Evaluate the fitness function, $f(x)$ of each chromosome x in the population.
- Step-4:** Create a new population of n chromosomes by repeating the following steps until the new population is complete.
- 4.1: Selection:** Sort the population by the objective function value and copy a given percentage (say 20%) of the population for remainder of the population.
Then, randomly select two parent chromosomes from the above population according to their fitness (better the fitness, there is a higher probability to be selected).
 - 4.2: Crossover:** Perform the crossover of the parents for a crossover probability to form new offspring. If no crossover is performed, then the offspring are exact copies of the parents.
 - 4.3: Mutation:** Perform the mutation of offspring at each locust (position in the chromosome) for a mutation probability. Also, evaluate the fitness function value, $f(x)$ of the offspring.
 - 4.4: Acceptance:** Place the new offspring in a new population.
 - 4.5:** Increment the generation count by one. ($GC = GC + 1$).
- Step-5:** Use the newly generated population for the next run of the algorithm.
- Step-6:** If the end condition is satisfied (If $GC > N$), then go to step-7; otherwise go to step-4.
- Step-7:** Return the best solution in the current population and stop.

13.8.3 Tabu Search

Tabu search is another heuristic under meta heuristic aiming at near-optimal solution to a combinatorial problem. This tabu search avoids becoming trapped in local optima by exploiting memory and data structures that prevent immediately moving back to a previously examined solution, and more generally, prevent moving to solutions that share certain attributes with previous solutions. Tabu search proceeds by defining a current solution. A neighbourhood structure is then imposed that enables the algorithm to develop other solutions from the current solution. A candidate list of the neighbours is examined to determine the best move available in terms of the selected objective function. The current solution is updated to the best of the neighbours that is not tabu. Common forms of memory include recency-based and frequency-based memory. A defined relative to the recency-based memory, a neighbour is *tabu* if it has an 'attribute' (which may be a combination or a function of elementary attributes) that has appeared in a solution within a designated number of previous algorithm steps. The primary parameter required for this type of memory is the 'tabu length', which is usually the number of algorithm steps for which the designated attribute is declared tabu. Often, tabu search algorithms incorporate 'aspiration criteria' to allow a tabu move to be accepted whenever such a move is deemed advantageous to the search. Other procedures can be designed to encourage the algorithm to intensify or diversify the search. Those incorporating frequency-based memory keep track of how often solution attributes occur in different classes of solutions visited. A strict diversification approach, for

example, keeps track of attribute frequencies over all solutions visited during the history of the search (or of the search within a given region). Then diversification is initiated by seeking to discourage moves to solutions that embody higher-frequency attributes or to encourage moves to solutions that embody lower-frequency attributes. A strict intensification approach instead keeps track of frequencies over sets of elite solutions (which may be differentiated by clustering). During an intensification phase, high-frequency attributes from an elite domain are encouraged or even 'locked in'. Ideally, intensification and diversification should not be independent but allowed to interact so that each makes reference to the relevant influences from the other. Within a framework that varies the emphasis on such influences, the tabu search method terminates after a designated number of iterations or a stated amount of computation time.

Tabu search may be described as a feedback-driven approach, where the reliance on adaptive memory determines both the form and the use of this feedback.

Let s be a feasible solution; $z(s)$, the objective function value of s ; $N(s)$, the neighbourhood which is the set of all feasible solutions that we can determine with some predefined transformation (or move) of the solution s ; $z(\bar{s})$, the minimum of the objective function values of $z(s')$, $s' \in N(s)$ and s' is not in Tabu_List in the case of minimization type problem; $\bar{s}$, the feasible solution with respect to $z(\bar{s})$; $\bar{z}$, the objective function value with respect to $\bar{s}$; k , the iteration number; and N be the maximum number of iterations to be carried out.

Based on the above definitions, the steps of a simple tabu search algorithm applied to a minimization problem is presented below:

- Step-1: Initialize the Tabu_List to a null set: $\text{Tabu_List} = \emptyset$.
- Step-2: Find a feasible solution s and its $z(s)$. Set, $z^* = Z(s)$, $s^* = s$.
- Step-3: $k = 1$
- Step-4: Find $\min\{z(s')\}$ such that $s' \in N(s)$ and s' is not in Tabu_List. Let it be $z(\bar{s})$, and the corresponding feasible solution be $\bar{s}$.
- Step-5: If $z(\bar{s}) < z^*$, then update $z^* = z(\bar{s})$, $s^* = \bar{s}$.
- Step-6: $\text{Tabu_List} = \text{Tabu_List} \cup \{\bar{s}\}$ and $s = \bar{s}$.
- Step-7: $k = k + 1$
- Step-8: If $k \leq N$, then go to step-4; otherwise go to step-9.
- Step-9: Print s^* and z^* and stop.

For a combinatorial problem, if a feasible solution s and the corresponding $Z(s)$ are given, we define the neighbourhood $N(s)$ as the set of all feasible solutions that we can determine with some predefined transformation (or move) of the solutions. Let us consider a problem in minimization form. A local search algorithm determines the minimum value $Z(\bar{s})$, where $s' \in N(s)$ and s' is not in the tabu list, then if $z(\bar{s}) < z^*$, it sets $s = \bar{s}$ and repeats the search; otherwise it stops. When the search is terminated, the current solution s is a local minimum.

The Tabu Search technique tries to escape from the local minimum allowing the movement from solution s to $\bar{s}$ even if $z(\bar{s}) > z^*$. In order to avoid returning in a previously examined solution, a tabu list is defined which stores these solutions. The local search is then restricted to the solutions in $N(s)$ which do not belong to the tabu list. [For more details

on tabu search, readers are directed to refer to the articles by Glover (1986, 1989, 1990) and Glover et al.(1993).]

QUESTIONS

1. Define algorithm and give some examples of standard algorithms.
2. Give brief account of algorithmic research problems in different areas of management.
3. What is polynomial algorithm? Give some examples of polynomial algorithm.
4. Write the steps of Floyd's algorithm and derive its worst case time complexity function.
5. What is exponential algorithm? Give some examples of exponential algorithm.
6. Derive the worst case time complexity function of the complete enumeration technique applied to the minimization of mean tardiness of single machine scheduling problem.
7. Distinguish between polynomial problem and NP-hard problem.
8. What are the types of solution procedure? Explain them in brief.
9. What is complete enumeration method? Illustrate it with an example.
10. What is branch and bound method? Explain its working.
11. What is heuristic? Explain it with an example.
12. Explain the steps of complete development of an algorithm.
13. Explain the steps of algorithmic research.
14. A researcher feels that the response variable of a problem (in percentage) will be affected by algorithm. So, he has designed a single-factor experiment with two algorithms. Under each algorithm, he has taken six replications. As per this design, the values of the response variable in percentages of the problem are summarized in the following table:
 - (a) Write the ANOVA model.
 - (b) Perform ANOVA and check the significance of the effect of the factor 'algorithm' on the response variable at $\alpha = 0.05$.

Algorithm-1	Algorithm-2
80	70
90	85
75	80
90	95
80	75
75	70

15. A researcher feels that the response variable of a problem (in percentage) will be affected by two factors namely, algorithm and problem. So, he has designed a factorial experiment with two algorithms and three problems. For each experimental combination, he has taken three replications. As per this design, the values of the response variable in percentages of the problem are summarized in the following table:

Problem	Algorithm	
	Algorithm-1	Algorithm-2
1	80	70
	90	85
	75	80
2	90	95
	80	75
	75	70
3	90	80
	70	65
	85	90

- Write the ANOVA model.
 - Perform ANOVA and check the significance of the effect of each component of the model on the response variable at $\alpha = 0.05$.
16. List the heuristics under meta heuristics, and compare and contrast them.
17. What is simulated annealing algorithm? Explain its steps.
18. What is genetic algorithm? Discuss the terminologies of genetic algorithms. Explain the steps of genetic algorithm.
19. What is tabu search? Explain its steps.

CHAPTER 14

MODELS FOR INDUSTRIES AND PUBLIC SYSTEMS

14.1 INTRODUCTION

Model is an abstraction of reality. Some examples of models are road map of a city to trace the shortest route from a given source to a given destination, three-dimensional view of a factory to plan the material movements in its shop floor, electrical network to compute the current flow in a particular arc, linear equation to forecast the demand of a product, etc.

Model of a system will help the analyst have better insight into the system. In an emerging area, where the number of solution procedures is very much limited, the optimal results of a model can be used as the benchmark to compare the results of new algorithms in that area. Modelling is an art which cannot be taught in a definite way. But, one can improve his/her modelling skill by going through more and more models in diverse areas.

In this chapter, a preliminary exposure to model building in different areas of management, interface areas and public systems is provided for the benefit of researchers.

14.2 STEPS OF MODELLING

The steps of modelling are listed below with reference to Fig. 14.1.

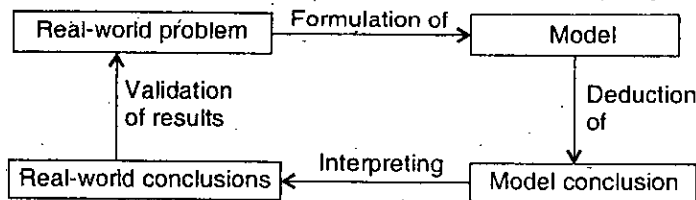


Fig. 14.1 Schematic representation of modelling steps.

Step-1: Identify the management decision problem of real world.

Step-2: Formulate a model for the real-world problem which in turn has the following steps:

- 2.1: Identify the parameters and variables which are involved in the management decision problem. Define them verbally and then introduce symbols to represent each one of them.
 - 2.2: Select the variables that appear to be the most influential so that the model may be kept as simple as possible. Classify the variables into controllable and non-controllable variables.
 - 2.3: State verbal relationship among the variables based on known principles, specially, gathered data and intuition. Make assumptions of predictions concerning the behaviour of the non-controllable variables.
 - 2.4: Construct a model by combining all the real-world relationships into a system of symbolic relationships.
- Step-3:* Perform symbolic manipulations such as solving a system of equations, iterating set of steps or making statistical analysis to optimize certain measure of performance and draw model conclusions.
- Step-4:* Interpret the model conclusions in terms of the characteristics of the real world.
- Step-5:* Test and validate the results.
- Step-6:* Implement the results.
- Step-7:* Revise the model as and when necessary.

14.3 OPERATIONS RESEARCH MODELS

Operations research is a scientific approach to problem solving for executive decision making.

Operations Research Society of America (ORSA) has defined operations research as a tool which is concerned with the design and operation of the man-machine system scientifically, usually under conditions requiring the allocation of scarce resources.

As per the Operations Research Society of the Great Britain, operations research is the application of the methods of science to complex problems arising in the direction and management of large systems of men, machines, materials and money in industry, business and government.

14.3.1 Important Topics in Operations Research

Operations research mainly discusses different optimization techniques which are as listed below:

Linear programming It involves linear objective function with a set of linear constraints.

Integer programming It is an extension of linear programming with only integer values for the decision variables of the problem.

Distance related network techniques There are problems like, transportation problem, shortest-path problem, minimum spanning-tree problem and travelling salesman problem.

The objective of the transportation problem is to find the optimal shipping schedule of a commodity from a set of sources to a set of destinations such that the total cost of shipment is minimized.

The objective of the shortest path problem is to find the shortest path from a given source node to a given destination node in a distance network such that the total distance of travel is minimized.

The objective of the minimum spanning-tree problem is to find the optimal connection of all the nodes using some edges in a given distance network such that the total length of the edges is minimized. The objective of the travelling salesman problem is to find a tour for the salesman such that the total distance/cost of the tour is minimized.

Project management It is a technique to schedule the activities of a project (e.g. construction of a bridge) such that the total project completion time is minimized. Also, it has several other control techniques.

Inventory control It is a technique to optimally plan and procure/produce raw materials/semi-finished products, respectively, such that the total cost of the inventory system is minimized.

Dynamic programming It is systematic complete enumeration technique to solve a problem optimally in an emerging area by integrating the solutions of its subproblems.

Queueing theory It is a technique based on probability to study the waiting behaviour of some real-life queueing systems.

Replacement analysis It is a technique to determine the economic life of an asset with the minimum total cost. Also, it discusses the method of selecting the best option between individual replacement policy and group replacement policy such that the total cost of the system is minimized.

Game theory It is a technique to deal with uncertainty situations related to management decisions, like bidding for tenders.

Goal programming It is an extension of linear programming with multiple objectives (goals) in the objective function.

Simulation It is a technique to deal with probabilistic situation where empirical/mathematical models fail to provide solutions to real-life problems.

Scheduling It is the process of preparing calendar for executing a set of jobs mostly in shop floors.

Non-linear programming It is an extended version of linear programming problems with non-linear objective function and linear constraints or with non-linear objective function and non-linear constraints which will enable analysts to incorporate realistic assumptions while solving problems.

14.3.2 Scope of Operations Research

The techniques of operations research can be applied to several real-world problems. A sample set of applications in different sectors is presented below:

Defence applications Many of defence operations involve scientific decision making. So, the methods, viz., network techniques for shortest path problems, scheduling algorithms for

vehicle routing, allocation techniques for shipping foodgrains and ammunition can be applied in defence.

Industrial applications Industrial management consists of four functional areas: production, marketing, personnel and finance. In addition to these functions, there are many other support services to carry out the business of any industrial enterprise.

A sample set of techniques which are used in the *production* function is presented below:

- Linear programming for aggregate planning
- Integer programming for shop floor production scheduling
- Network-based techniques for line balancing and project management
- Inventory control techniques for planning and procuring raw materials
- Replacement analysis for equipment replacement decision
- Queueing theory for designing in-process buffer stock.

A sample set of techniques which are used in the *marketing* function is presented below:

- Linear programming for product-mix problem
- Game theory for order bidding decision
- Distance network related techniques for shipping finished goods.

A sample set of techniques which are used in the *personnel* function is presented below:

- Linear programming for man power planning
- Queueing theory for determining the size of maintenance crews
- Scheduling techniques for manpower scheduling.

A sample set of techniques which are used in the *finance* function is presented below:

- Integer programming for capital budgeting
- Linear programming for break-even analysis
- Integer programming or dynamic programming for portfolio selection.

Applications in public system Some of the operations research techniques can be used to plan, design and operate different public systems like, government and government institutions/ departments, postal system, banks, highways, railways, airways, hospitals, educational institutions, etc.

In *government*, the techniques like goal programming for policy decisions, integer programming for budgeting, etc., can be used.

In *postal systems*, techniques which are related to vehicle scheduling, manpower planning and transfer policy can be used.

In *banks*, the techniques like, queueing theory for determining the number counters, portfolio models for effective deployment of funds subject to government regulations, etc. can be used.

In *highways*, the techniques like integer programming for project selection and vendor selection, simulation for traffic system design, etc. can be used.

For *railways*, the techniques like, linear programming/integer programming for cargo loading, scheduling techniques for railway traffic control, queueing theory to determine the number of platforms, etc., can be used.

In *airways*, the techniques like, simulation for simulating air traffic, and runway design, linear/integer programming for cargo loading, etc., can be used.

A sample set of techniques which are used in *hospitals* is presented below:

- Queueing theory for outpatient system design
- Linear programming/integer programming for nurse scheduling/duty doctor scheduling
- Inventory control for procurement of medicines
- Algorithms and models for operation theatre scheduling.

14.4 EXAMPLES OF APPLICATIONS OF MODELS

In this section some examples of applications of models in different functional areas of management, interface areas and in public systems are presented.

14.4.1 Application of Modelling in Production System

Model to minimize total tardiness

In this section, a mathematical model to minimize the total tardiness in the single machine scheduling is presented (Panneerselvam, 1991).

Consider the problem involving n independent jobs which can be processed on a single machine.

One of the measures of this problem is minimizing the total tardiness of the jobs. The basic data for the single machine scheduling problem is presented in Table 14.1.

Table 14.1 Generalized Data of Single Machine Scheduling Problem

Job (i)	Processing time (t_i)	Due date (d_i)
1	t_1	d_1
2	t_2	d_2
3	t_3	d_3
$\vdots$	$\vdots$	$\vdots$
i	t_i	d_i
$\vdots$	$\vdots$	$\vdots$
n	t_n	d_n

Let C_i be the completion time of the job i in a given schedule; T_j , the tardiness of the job assigned to the j th position in a given schedule; and $Y_{ij} = 1$, if job i is assigned to the j th position and $= 0$, otherwise.

The objective of the problem is to sequence the jobs such that the total tardiness is minimized. Assume that all the jobs are available at time zero, i.e. ready time, $r_i = 0$, $i = 1, 2, 3, \dots, n$. Under this situation:

C_i = Completion time of the job in the sequence immediately preceding to the job $i + t_i$

and

$$T_j = \max \{0, (\text{Completion time of the job at the position } j - \text{Due date of the job at the position } j)\}$$

A descriptive model to minimize the total tardiness is presented below:

$$\text{Minimize } Z = \sum_{j=1}^n \max \{0, (\text{Completion time of the job at the position } j - \text{Due date of the job at the position } j)\}$$

$$= \sum_{j=1}^n \max \left\{ 0, \sum_{i=1}^n \sum_{q=1}^j t_i Y_{iq} - \sum_{i=1}^n d_i Y_{ij} \right\}$$

subject to

$$\sum_{j=1}^n Y_{ij} = 1, \quad i = 1, 2, 3, \dots, n$$

$$\sum_{i=1}^n Y_{ij} = 1, \quad j = 1, 2, 3, \dots, n$$

where

$$Y_{ij} = 0 \text{ or } 1, \quad i = 1, 2, 3, \dots, n, \quad j = 1, 2, 3, \dots, n$$

For easy understanding of the above problem, the different possible variables of Y_{ij} are presented in Table 14.2.

Table 14.2 Combinations of 0–1 Variables

	Position in the sequence						
	1	2	3	...	j	...	n
1	Y_{11}	Y_{12}	Y_{13}	...	Y_{1j}	...	Y_{1n}
2	Y_{21}	Y_{22}	Y_{23}	...	Y_{2j}	...	Y_{2n}
3	Y_{31}	Y_{32}	Y_{33}	...	Y_{3j}	...	Y_{3n}
Job	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	Y_{i1}	Y_{i2}	Y_{i3}	...	Y_{ij}	...	Y_{in}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	Y_{n1}	Y_{n2}	Y_{n3}	...	Y_{nj}	...	Y_{nn}

Linear Integer and 0-1 programming model

The non-linear objective function of the descriptive model is converted into a linear function by replacing the objective term by T_j (tardiness of the job assigned to the j th position in a given schedule), as shown below, by adding constraints set-3 and set-5.

$$\text{Minimize } Z = \sum_{j=1}^n T_j$$

subject to

$$\sum_{j=1}^n Y_{ij} = 1, \quad i = 1, 2, 3, \dots, n \quad (\text{Set-1})$$

$$\sum_{i=1}^n Y_{ij} = 1, \quad j = 1, 2, 3, \dots, n \quad (\text{Set-2})$$

$$T_j \geq \sum_{i=1}^n \sum_{q=1}^i t_i Y_{iq} - \sum_{i=1}^n d_i Y_{ij}, \quad j = 1, 2, 3, \dots, n \quad (\text{Set-3})$$

where

$$Y_{ij} = 0 \text{ or } 1, \quad i = 1, 2, 3, \dots, n, \quad j = 1, 2, 3, \dots, n \quad (\text{Set-4})$$

$$T_j \geq 0, \quad j = 1, 2, 3, \dots, n \quad (\text{Set-5})$$

In this model,

1. The objective function minimizes the total tardiness.
2. The constraints in the set-1 guarantee that each job is assigned to exactly one position in the schedule.
3. The constraints in the set-2 guarantee that each position in the schedule is assigned exactly one job.
4. The constraints in the sets-3 and 5, jointly guarantee that T_j is more than zero if the job in the position j is late; otherwise it is zero.
5. So, the constraint sets 3 and 5 are either or constraint sets.
6. In this model, for a general problem of size n , the number of variables is $(n^2 + n)$ and the number of constraints is $3n$.

Example 14.1 Consider the data of a single machine scheduling problem with 3 jobs, as shown in Table 14.3.

Table 14.3 Data

Job (<i>i</i>)	Processing time (<i>t_i</i>)	Due date (<i>d</i>)
1	6	11
2	4	6
3	9	16

Find the optimal sequence of jobs and the corresponding total tardiness using the linear integer 0–1 programming model.

Solution. The development of the model for the given problem is left as an exercise. Application of the model to this problem gives the following results: $Y_{12} = Y_{21} = Y_{33} = 1$ and other variables are zero. From this result, one can infer the best sequence as 2–1–3. The corresponding total tardiness is 3.

Model to minimize makespan in parallel processors under single-machine scheduling

In single-machine scheduling, if the number of machines is more than one, then it becomes parallel processors situation in which each machine can be used for processing any job. Under this situation, the minimization of makespan becomes an important criterion.

An integer programming formulation of this problem is presented below:

Let y be the makespan of the schedule; x_{ij} , a zero-one variable; $x_{ij} = 1$, if the job i is assigned to the machine j and $= 0$, otherwise; t_i , the processing time of the job i ; n , the number of jobs; and m be the number of parallel processors.

The model to minimize the makespan of this problem is as given below:

$$\text{Minimize } Z = y$$

subject to

$$y - \sum_{i=1}^n t_i x_{ij} \geq 0, \quad j = 1, 2, 3, \dots, m$$

$$\sum_{j=1}^m x_{ij} = 1, \quad i = 1, 2, 3, \dots, n$$

$$x_{ij} = 0 \text{ or } 1, \quad i = 1, 2, 3, \dots, n \text{ and } j = 1, 2, 3, \dots, m$$

Here, the first set of constraints ensures that the makespan is at least as large as the processing time assigned to any machine. The second set of constraints ensures that each job is assigned to only one machine. This formulation contains $(m + n)$ constraints and $(mn + 1)$ variables.

14.4.2 Application of Modelling in Marketing System

In this section, some models for application in marketing are presented.

Model for locating dealers

Consider the problem of locating dealers of a computer company in a state.

The marketing chief of the firm is of the opinion that the customers are very important to the company. Hence, he is desiring to locate the optimal number (minimum number) of dealers to cover all the customer regions in the state such that no customer from any customer region is travelling more than a specified maximum distance (say 10 km) to seek details about computers or to buy computers. Incidentally, this objective is same as that of total covering problem (Panneerselvam, 1990a), which has the objective of locating the optimal number of facilities such that no customer is travelling more than a specified maximum distance to avail the service from any of the facilities. The specified maximum distance depends on the convenience of the customers and the practical difficulties involved in reaching the facility by the customers.

Let d_{ij} be the distance from the customer region i to the facility location (dealer location) j ; n , the number of customer regions; q , the maximum distance that a customer from a customer region can travel to avail services from any dealer; C_{ij} , the covering coefficient with respect to the i th customer region and j th facility location (dealer location) such that

$$C_{ij} = 1, \quad \text{if } d_{ij} \leq q;$$

$$= 0, \quad \text{otherwise}$$

and

$$Y_j = 1, \quad \text{if a facility (dealer) is assigned to the potential location } j;$$

$$= 0, \quad \text{otherwise}$$

A model to determine the minimum number of dealers to be located such that all the customer regions are covered is presented below.

In this model, each customer region is considered a potential site for locating a dealer. If the number of customer regions is n , then the number of potential sites for locating dealers is also n .

$$\text{Minimize } Z = \sum_{j=1}^n Y_j$$

subject to

$$\sum_{j=1}^n C_{ij} Y_j \geq 1, \quad i = 1, 2, 3, \dots, n$$

where $Y_j = 0$ or 1 , $j = 1, 2, 3, \dots, n$.

In this model, the objective function minimizes the total number of sites selected for locating dealers from the given set of potential sites. The constraints ensure that each customer from a customer region is covered by at least one dealer located in any of the selected potential sites.

Example 14.2 The number of customer regions under consideration for locating dealers is 6. The distance matrix of these 6 regions is shown in Table 14.4. The maximum distance a customer from any customer region can travel to reach a dealer is 8 km. Find the optimal number of dealers to be located to cover all the customer regions:

Table 14.4 Distance Matrix

		Region (potential site)					
		1	2	3	4	5	6
Region	1	20	7	15	12	6	16
	2	12	9	12	15	18	5
	3	4	5	11	15	18	12
	4	11	12	16	4	6	12
	5	6	15	6	14	12	8
	6	17	23	17	7	15	26

Solution. Based on the specified maximum distance of travel of 8 km by the customers in reaching the dealers, the covering coefficient matrix is shown in Table 14.5.

Table 14.5 Covering Coefficient Matrix

		Potential Site					
		1	2	3	4	5	6
Region	1	0	1	0	0	1	0
	2	0	0	0	0	0	1
	3	1	1	0	0	0	0
	4	0	0	0	1	1	0
	5	1	0	1	0	0	1
	6	0	0	0	1	0	0

The model to determine the minimum number of sites to be selected for locating dealers to cover all the customer regions is presented below:

$$\text{Minimize } Z = Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Y_6$$

subject to

$$Y_2 + Y_3 \geq 1$$

$$Y_6 \geq 1$$

$$Y_1 + Y_2 \geq 1$$

$$Y_4 + Y_5 \geq 1$$

$$Y_1 + Y_3 + Y_6 \geq 1$$

$$Y_4 \geq 1$$

where $Y_j = 0$ or 1 , $j = 1, 2, 3, 4, 5, 6$.

The results of this model using STORM package are as: Minimum number of dealers to be located = 3. The sites at which dealers to be located are 2, 4 and 6. The allocations of customer regions to dealers are indicated by squares, as shown in Table 14.6.

Table 14.6 Allocation of Customer Regions to Dealers

	Potential dealer location		
	2	4	6
1	1	0	0
2	0	0	1
3	1	0	0
4	0	1	0
5	0	0	1
6	0	1	0

Model for media planning

Determining optimal media-mix is a productivity improvement technique which will enhance the sales revenue of the products of concern. The objective of media-mix planning is to maximize the effective exposure (reach) of the advertisements in different media subject to relevant constraints. In practice, there are media alternatives such as magazines, radio, television, newspaper, internet.

Example 14.3 An automobile company has almost completed its R&D activities to bring out a new fuel-efficient car into the market. It is in the process of deciding media mix to advertise the car.

The company has identified its target customers in the following categories with their relative importance indicated in parentheses:

1. Male (employed) (0.3)
2. Female (employed) (0.2)
3. Age in the range 21–50 years (0.4)
4. Entrepreneur (0.1).

The total advertising budget allocation for the forthcoming year is Rs. 50 lakhs. The company has collected additional data for this exercise, as shown in Table 14.7.

Table 14.7. Additional Data for Media Planning

Data item	Media (% of reading/watching)		
	Magazine	Television	Newspaper
Readers/viewers characteristics			
Male (employed)	70	60	80
Female (employed)	65	85	70
Age in the range 21–50 years	90	70	75
Entrepreneur	90	50	70
Cost per advertisement (Rs. in lakhs)	0.5	5	1
Minimum no. of ads allowed	20	4	6
Maximum no. of ads allowed	30	16	20
Audience size per ad.	5,00,000	75,00,000	2,00,00,000

Develop an integer linear programming model to determine the optimal media mix and also obtain the corresponding results.

Solution. The effectiveness coefficient for each media is obtained using the following formula:

$$\text{Effectiveness coefficient for a media} = \sum_{i=1}^4 W_i R_i$$

where W_i is the weight of the reader characteristic i ; R_i is the reach in percentage of the reader characteristic i .

Magazine:

$$\begin{aligned} \text{Effectiveness coefficient} &= 0.7 \times 0.3 + 0.65 \times 0.2 + 0.9 \times 0.4 + 0.9 \times 0.1 \\ &= 0.79 \end{aligned}$$

$$\begin{aligned} \text{Reach of the magazine/ads} &= (\text{Effectiveness coefficient of magazine}) \\ &\quad \times (\text{Audience size per advertisement}) \\ &= 0.79 \times 5,00,000 \\ &= 3,95,000 \end{aligned}$$

Television:

$$\begin{aligned} \text{Effectiveness coefficient} &= 0.6 \times 0.3 + 0.85 \times 0.2 + 0.7 \times 0.4 + 0.5 \times 0.1 \\ &= 0.68 \end{aligned}$$

$$\begin{aligned} \text{Reach of TV/advertisement} &= 0.68 \times 75,00,000 \\ &= 51,00,000 \end{aligned}$$

Newspaper:

$$\begin{aligned}\text{Effectiveness coefficient} &= 0.8 \times 0.3 + 0.7 \times 0.2 + 0.75 \times 0.4 + 0.7 \times 0.1 \\ &= 0.75\end{aligned}$$

$$\begin{aligned}\text{Reach of newspaper/advertisement} &= 0.75 \times 200,00,000 \\ &= 1,50,00,000\end{aligned}$$

Let Y_j be the number of times of advertisement to be made in the j th media. The integer programming model for this situation is presented below:

$$\text{Maximize } Z = 3,95,000Y_1 + 5,10,000Y_2 + 1,50,00,000Y_3$$

subject to

$$0.5Y_1 + 5Y_2 + Y_3 \leq 50$$

$$Y_1 \geq 20$$

$$Y_2 \geq 4$$

$$Y_3 \geq 6$$

$$Y_1 \leq 30$$

$$Y_2 \leq 16$$

$$Y_3 \leq 20$$

where $Y_j \geq 0$, $j=1, 2, 3$, if the solutions to LP problem are integers; or, $Y_j \geq 0$ and integers, $j=1, 2, 3$, if the solutions to LP problem are not integers.

The solutions to this problem obtained through LP model are presented below. (One can notice that the values of the decision variables are integers even by using LP model.)

$$Y_1 = 20, \quad Y_2 = 4, \quad Y_3 = 20$$

The corresponding objective function value = 32,83,00,000.

We get the following inference of the solution:

The number of times of advertisement to be made in magazine = 20

The number of times of advertisement to be made in TV = 4

The number of times of advertisement to be made in newspaper = 20

The total reach of the entire media-mix = 32,83,00,000

14.4.3 Application of Modelling in Finance System

In this section, a capital budgeting problem and portfolio selection problem are presented.

Model for selecting projects for maximum return

Example 14.4 Consider the capital budgeting problem where five projects are being considered for execution over the next 3 years. The expected returns for each project and the yearly expenditures (in lakhs of rupees) are shown in Table 14.8. Assume that each approved project will be executed over the 3-year period. The objective is to select a combination of projects that will maximize the total returns.

Table 14.8 Data

Project	Expenditures for			Returns
	Year-1	Year-2	Year-3	
1	7	2	7	40
2	5	6	9	45
3	2	10	3	35
4	6	5	2	30
5	9	7	8	50
Maximum fund	25	25	25	-

Formulate the problem as a 0–1 integer programming problem and solve it.

Solution. Let

$$Y_j = 1, \quad \text{if the } j\text{th project is selected}$$

$$= 0, \quad \text{Otherwise}$$

Therefore, a 0–1 programming model is, as given below:

$$\text{Maximize } Y_0 = 40Y_1 + 45Y_2 + 35Y_3 + 30Y_4 + 50Y_5$$

subject to

$$7Y_1 + 5Y_2 + 2Y_3 + 6Y_4 + 9Y_5 \leq 25$$

$$2Y_1 + 6Y_2 + 10Y_3 + 5Y_4 + 7Y_5 \leq 25$$

$$7Y_1 + 9Y_2 + 3Y_3 + 2Y_4 + 8Y_5 \leq 25$$

where $Y_j = 0$ or 1 , $j = 1, 2, 3, 4$ and 5 .

The results of this problem are as follows:

$$Y_1 = 1, \quad Y_2 = 0, \quad Y_3 = 1, \quad Y_4 = 1, \quad Y_5 = 1$$

Interpretation: All the projects except the project 2 are to be selected which will result with a total maximum return of Rs. 155 lakhs.

Portfolio selection problem

Portfolio selection problem is an important one for many business establishments to deploy

their funds for productive purposes in different investment opportunities. A mathematical model for portfolio selection is presented below:

Example 14.5 Portfolio selection problem considers the factors such as risk, working capital liquidity, growth potential, rate of return, and diversification requirements.

A leading insurance company has identified seven investment alternatives for a 15 year investment period to invest its 10,000 crores rupees. The details of the investment alternatives are presented in Table 14.9. In Table 14.9, length of investment means the expected number of years required for the annual rate of return to be realized taking into account the possibility of reinvestment. The objective of the insurance company is to maximize its returns subject to the following constraints:

- The average length of investment should be less than or equal to 5.
- The average risk for the portfolio should not exceed 4.
- The average growth potential for the portfolio should be at least 12%.
- At least 15% of all available funds must be retained in the form of cash at all times to maintain working capital liquidity.
- Sum of the proportions of investments should be 1.

Table 14.9 Details of Investment Alternatives

<i>Investment alternatives</i>	<i>Length of investment</i>	<i>Annual rate of return (%)</i>	<i>Risk coefficient</i>	<i>Growth potential</i>
Treasury bills	5	5	1	0
Equity	8	14	6	15
Company bonds	10	10	5	9
Real estate	7	15	9	30
Mutual fund	12	12	7	18
Saving and loans	6	5	4	8
Cash	0	0	0	0

Develop a linear programming model for the above situation and solve.

Solution. Let Y_j be the proportion of the fund invested in the j th alternative, $j = 1, 2, 3, 4, 5, 6, 7$. Then

$$\text{Maximize } Z = 0.05Y_1 + 0.14Y_2 + 0.1Y_3 + 0.15Y_4 + 0.12Y_5 + 0.05Y_6 + 0Y_7$$

subject to

$$5Y_1 + 8Y_2 + 10Y_3 + 7Y_4 + 12Y_5 + 6Y_6 + 0Y_7 \leq 5$$

$$Y_1 + 6Y_2 + 5Y_3 + 9Y_4 + 7Y_5 + 4Y_6 + 0Y_7 \leq 4$$

$$0Y_1 + 15Y_2 + 9Y_3 + 30Y_4 + 18Y_5 + 8Y_6 + 0Y_7 \geq 12$$

$$Y_7 \geq 0.15$$

$$Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Y_6 + Y_7 = 1$$

where $Y_j \geq 0$, $j = 1, 2, 3, 4, 5, 6, 7$.

The solution of the above model using STORM package is presented below:

$$Y_1 = 0.4, \quad Y_4 = 0.4, \quad Y_7 = 0.2$$

The objective function value = 0.08.

For the investment of Rs. 10,000 crores, the investment pattern should be as shown below:

Investment in treasury bills, $Y_1 = 4000$ crores

Investment in real estate, $Y_4 = 4000$ crores

Cash, $Y_7 = 2000$ crores

The above pattern of investment will give the annual return of Rs. 800 crores.

14.4.4 Application of Modelling in Personnel System

In this section, modelling of a problem in personnel function is presented.

Manpower planning

Example 14.6 In an international airport, security persons report to duty at the beginning of every 4-hour interval as shown in Table 14.10. Each person, after reporting to duty works for 8 continuous hours. The minimum required number of security persons in each period is also furnished in the same table. Develop a mathematical model to determine the number of security persons to report for duty at the beginning of each period such that the total number of security persons required for the day is minimized.

Table 14.10 Details of Manpower Planning Problem

Interval	Time period		Minimum number of security persons required
	From	To	
1	12 midnight	4 a.m.	25
2	4 a.m.	8 a.m.	40
3	8 a.m.	12 noon	65
4	12 noon	4 p.m.	55
5	4 p.m.	8 p.m.	60
6	8 p.m.	12 midnight	50

Solution. Let, X_j be the number of security persons to be reported for duty at the beginning of the period j ; $j = 1, 2, 3, 4, 5, 6$. The required model is shown below:

$$\text{Minimize } Z = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$$

subject to

$$X_6 + X_1 \geq 25$$

$$X_1 + X_2 \geq 40$$

$$X_2 + X_3 \geq 65$$

$$X_3 + X_4 \geq 55$$

$$X_4 + X_5 \geq 60$$

$$X_5 + X_6 \geq 50$$

where $X_j \geq 0$, $j=1, 2, 3, 4, 5, 6$, if the solutions are integer, or $X_j \geq 0$, and integer for $j=1, 2, 3, 4, 5, 6$, if the solutions are not integers for linear programming model.

For this problem, as a linear programming model, the solutions are integers: $X_1 = 25$, $X_2 = 20$, $X_3 = 45$, $X_4 = 10$ and $X_5 = 50$. Therefore, the number of security persons to be reported for duty in a day is 150.

As per the above reporting pattern, there is a surplus of 5 security persons working during the second period and there is no surplus security persons working in the rest of the periods.

14.4.5 Application of Modelling in Interface Area

The problem of transportation is an example of interface area. In this section, the modelling of the transportation problem with quantity discounts is presented.

Modelling the transportation problem with quantity discounts

In the classical transportation problem, the unit cost of shipment from a given source to a given destination is assumed to be a constant irrespective of the quantity shipped. However, in reality, the normal practice in most of the transport agencies is to offer a discount on the unit cost for the increased volume of shipment.

Further, the quantity discount can be offered in two ways: *all* quantity discounts and *incremental* quantity discounts.

Let, C_{ij} be the unit cost of shipment and X_{ij} be the quantity shipped from the i th source to the j th destination. The cost structure (price breaks) under quantity discount scheme is as follows:

$$\begin{aligned} C_{ij} &= \text{Rs. } 10, & \text{if } 0 \leq X_{ij} < \text{Rs. } 10 \\ &= \text{Rs. } 8, & \text{if } 10 \leq X_{ij} < \text{Rs. } 25 \\ &= \text{Rs. } 5, & \text{if } 25 \leq X_{ij} < \infty \end{aligned}$$

Consider a shipment of 20 units from the source i to the destination j under the above cost structure. If the unit cost is taken as Rs. 8 for all the 20 units, then the total cost of shipping these 20 units is Rs. 160. This scheme is termed as 'all quantity discounts scheme (AQDS)'. This concept is graphically shown in Fig. 14.2.

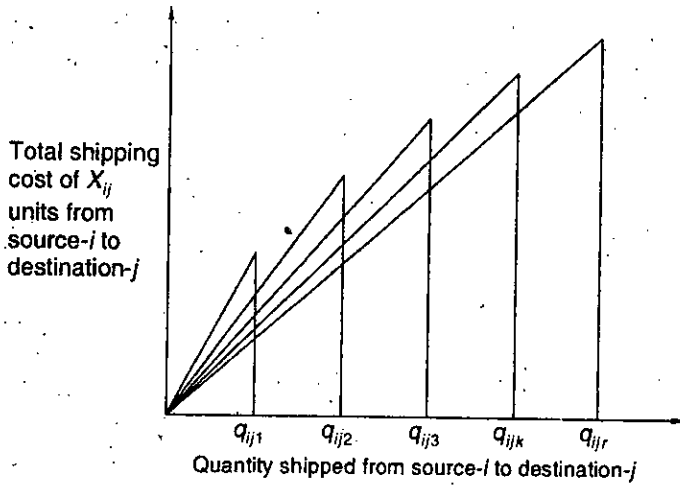


Fig. 14.2 All quantity discount scheme (AQDS).

On the other hand, if the unit cost is taken as Rs. 10 for the first nine units of the shipment and Rs. 8 for the remaining 11 units of shipment, then the total cost of shipping these 20 units is Rs. 178. This scheme is termed as 'incremental quantity discount scheme (IQDS)'. This concept is graphically shown in Fig. 14.3.

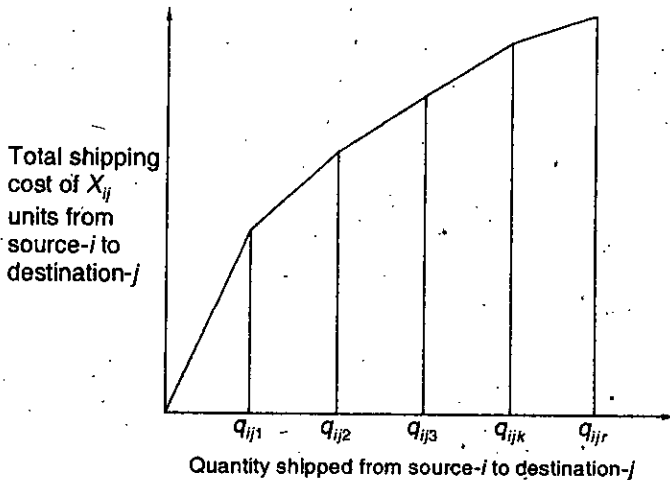


Fig. 14.3 Incremental quantity discount scheme (IQDS).

A generalized representation of the intervals of quantity discounts of the transportation problem in a tabular form is shown in Table 14.11.

Table 14.11 Generalized Format

	Destination				
	1	2	j	n	
Source	$0 \leq X_{111} < q_{111} : p_{111}$ $q_{111} \leq X_{112} < q_{112} : p_{112}$ $\vdots$ $q_{11(r-1)} \leq X_{11r} < \infty : p_{11r}$	$0 \leq X_{121} < q_{121} : p_{121}$ $q_{121} \leq X_{122} < q_{122} : p_{122}$ $\vdots$ $q_{12(r-1)} \leq X_{12r} < \infty : p_{12r}$	...	$0 \leq X_{1n1} < q_{1n1} : p_{1n1}$ $q_{1n1} \leq X_{1n2} < q_{1n2} : p_{1n2}$ $\vdots$ $q_{1n(r-1)} \leq X_{1nr} < \infty : p_{1nr}$	a_1
	$0 \leq X_{211} < q_{211} : p_{211}$ $q_{211} \leq X_{212} < q_{212} : p_{212}$ $\vdots$ $q_{21(r-1)} \leq X_{21r} < \infty : p_{21r}$	$0 \leq X_{221} < q_{221} : p_{221}$ $q_{221} \leq X_{222} < q_{222} : p_{222}$ $\vdots$ $q_{22(r-1)} \leq X_{22r} < \infty : p_{22r}$	...	$0 \leq X_{2n1} < q_{2n1} : p_{2n1}$ $q_{2n1} \leq X_{2n2} < q_{2n2} : p_{2n2}$ $\vdots$ $q_{2n(r-1)} \leq X_{2nr} < \infty : p_{2nr}$	a_2
	$\vdots$	$\vdots$	...	$\vdots$	a_i
	$0 \leq X_{m11} < q_{m11} : p_{m11}$ $q_{m11} \leq X_{m12} < q_{m12} : p_{m12}$ $\vdots$ $q_{m1(r-1)} \leq X_{m1r} < \infty : p_{m1r}$	$0 \leq X_{m21} < q_{m21} : p_{m21}$ $q_{m21} \leq X_{m22} < q_{m22} : p_{m22}$ $\vdots$ $q_{m2(r-1)} \leq X_{m2r} < \infty : p_{m2r}$	...	$0 \leq X_{mn1} < q_{mn1} : p_{mn1}$ $q_{mn1} \leq X_{mn2} < q_{mn2} : p_{mn2}$ $\vdots$ $q_{mn(r-1)} \leq X_{mnr} < \infty : p_{mnr}$	a_m
	b_1	b_2	b_j	b_n	

Here, a_i be the capacity of source i , $i = 1, 2, 3, \dots, m$; b_j , the demand of the destination j , $j = 1, 2, 3, \dots, n$; r , the total number of price breaks in any given combination of source and destination; c_{ijk} , the cost/unit of shipping from the source- i to destination- j under the k th price break, $k = 1, 2, 3, \dots, r$; q_{ijk} , the upper bound on the quantity shipped from the source- i to the destination- j under the k th price break, $k = 1, 2, 3, \dots, r$; q_{ijr} , the upper bound in the last price break in any given cell can be either finite or infinite; X_{ijk} , the number of units to be shipped from the source- i to the destination- j under the k th price break; and, p_{ijk} is the price per unit of transportation for the k th price break from the source- i to the destination j .

Model for AQDS

As per these guidelines, a model to minimize the total cost of shipment under AQDS, developed by Panneerselvam et al. (1991), is presented below:

Let

$$Y_{ijk} = \begin{cases} 1, & \text{if } X_{ijk} > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Minimize } Z = \sum_{i=1}^m \sum_{j=1}^n \sum_{k=1}^r c_{ijk} X_{ijk}$$

subject to

$$\sum_{j=1}^n \sum_{k=1}^r X_{ijk} = a_i, \quad i = 1, 2, \dots, m \quad (1)$$

$$\sum_{i=1}^m \sum_{k=1}^r X_{ijk} = b_j, \quad j = 1, 2, \dots, n \quad (2)$$

$$\sum_{k=1}^r Y_{ijk} \leq 1, \quad i = 1, 2, \dots, m, \quad j = 1, 2, 3, \dots, n \quad (3)$$

$$X_{ijk} \leq (q_{ijk} - 1)Y_{ijk}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n \quad \text{and} \quad k = 1, 2, \dots, r \quad (4)$$

$$X_{ijk} \geq q_{ij(k-1)} Y_{ijk}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n \quad \text{and} \quad k = 2, 3, \dots, r \quad (5)$$

where $X_{ijk} \geq 0$ and $Y_{ijk} = 0$ or 1 , $i = 1, 2, \dots, m$; $j = 1, 2, \dots, n$; $k = 1, 2, \dots, r$.

In this model, the objective is to minimize the total cost of shipping under AQDS.

The constraint set-1 and constraint set-2 are same as the supply and demand constraint sets in the classical transportation model.

The constraint set-3 makes sure that the allocation is made under one and only price break within any given combination of the source- i and the destination- j .

The constraint set-4 limits the allocation of units under any price break to the respective upper bound within any given combination of the source- i and the destination- j .

The constraint set-5 assures that the units allocated under any price break is more than or equal to the respective lower bound within any given combination of the source- i and the destination- j .

Model for (IQDS)

The model for IQDS is presented in section.

$$\text{Minimize } Z = \sum_{i=1}^m \sum_{j=1}^n \sum_{k=1}^r C_{ijk} X_{ijk}$$

subject to

$$\sum_{j=1}^n \sum_{k=1}^r X_{ijk} = a_i, \quad i = 1, 2, \dots, m \quad (1)$$

$$\sum_{i=1}^m \sum_{k=1}^r X_{ijk} = b_j, \quad j = 1, 2, \dots, n \quad (2)$$

$$X_{ij1} \leq (q_{ij1} - 1)Y_{ij1}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n \quad (3a)$$

$$X_{ijk} \leq [q_{ijk} - q_{ij(k-1)}]Y_{ijk}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n, \quad k = 2, 3, \dots, r \quad (3b)$$

$$X_{ij1} \geq (q_{ij1} - 1)Y_{ij2}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n \quad (4a)$$

$$X_{ijk} \geq [q_{ijk} - q_{ij(k-1)}]Y_{ij(k+1)}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n, \quad k = 2, 3, \dots, (r-1) \quad (4b)$$

$$X_{ijk} \geq 0$$

$$Y_{ijk} = 0 \text{ or } 1, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n, \quad k = 1, 2, \dots, r \quad (5)$$

In this model, the objective is to minimize the total cost of shipping under IQDS.

The constraint set-1 and constraint set-2 are same as the supply and demand constraints in the classical transportation model.

The constraint-sets (3a) and (3b) limit the allocation of units in any cell to its upper bound on the incremental quantity within any given combination of the source- i and destination- j .

The constraint sets (4a) and (4b) assure the following:

For a given source- i and destination- j , if the allocation is made with respect to the $(k+1)$ th price break ($k \geq 2$), then the allocation with respect to the k th price break must be equal to the respective upper bound on the incremental quantity, i.e. $[q_{ijk} - q_{ij(k-1)}]$.

Example 14.7 Consider a problem with two sources and two destinations. Assume that the number of price breaks is equal to three for each combination of the sources and destinations. The corresponding data is shown in Table 14.12.

Table 14.12 Data

		Destination		Supply
		1	2	
Source	1	$0 \leq X_{111} < 10:5$ $10 \leq X_{112} < 20:4$ $20 \leq X_{113} < \infty:3$	$0 \leq X_{121} < 5:8$ $5 \leq X_{122} < 10:7$ $10 \leq X_{123} < \infty:6$	45
	2	$0 \leq X_{211} < 20:8$ $20 \leq X_{212} < 65:6$ $65 \leq X_{213} < \infty:5$	$0 \leq X_{221} < 5:10$ $5 \leq X_{222} < 10:9$ $10 \leq X_{223} < \infty:8$	
	Demand	60	30	90/90

Solution. (a) *Solution for AQDS:*

The model for AQDS is presented below and the corresponding optimal result is shown in Table 14.13. The total cost of the optimal shipping plan is Rs. 475.

Model formulation for AQDS:

$$\begin{aligned} \text{Minimize } Z = & 5X_{111} + 4X_{112} + 3X_{113} + 8X_{121} + 7X_{122} + 6X_{123} + 8X_{211} \\ & + 6X_{212} + 5X_{213} + 10X_{221} + 9X_{222} + 8X_{223} \end{aligned}$$

subject to

$$X_{111} + X_{112} + X_{113} + X_{121} + X_{122} + X_{123} = 45$$

$$X_{211} + X_{212} + X_{213} + X_{221} + X_{222} + X_{223} = 45$$

$$X_{111} + X_{112} + X_{113} + X_{211} + X_{212} + X_{213} = 60$$

$$X_{121} + X_{122} + X_{123} + X_{221} + X_{222} + X_{223} = 30$$

$$Y_{111} + Y_{112} + Y_{113} \leq 1$$

$$Y_{121} + Y_{122} + Y_{123} \leq 1$$

$$Y_{211} + Y_{212} + Y_{223} \leq 1$$

$$Y_{221} + Y_{222} + Y_{223} \leq 1$$

$$-9Y_{111} + X_{111} \leq 0$$

$$-19Y_{112} + X_{112} \leq 0$$

$$-1000Y_{113} + X_{113} \leq 0$$

$$-4Y_{121} + X_{121} \leq 0$$

$$-9Y_{122} + X_{122} \leq 0$$

$$-1000Y_{123} + X_{123} \leq 0$$

$$-19Y_{211} + X_{211} \leq 0$$

$$-64Y_{212} + X_{212} \leq 0$$

$$-1000Y_{213} + X_{213} \leq 0$$

$$-4Y_{221} + X_{221} \leq 0$$

$$-9Y_{222} + X_{222} \leq 0$$

$$-1000Y_{223} + X_{223} \leq 0$$

$$-10Y_{112} + X_{112} \geq 0$$

$$-20Y_{113} + X_{113} \geq 0$$

$$-5Y_{122} + X_{122} \geq 0$$

$$-10Y_{123} + X_{123} \geq 0$$

$$-20Y_{212} + X_{212} \geq 0$$

$$-65Y_{213} + X_{213} \geq 0$$

$$-5Y_{222} + X_{222} \geq 0$$

$$-10Y_{223} + X_{223} \geq 0$$

where $X_{ijk} \geq 0$ and $Y_{ijk} = 0$ or 1 for $i = 1, 2$, $j = 1, 2$ and $k = 1, 2, 3$.

Table 14.13 Optimal Shipping Strategy under AQDS Destination

		Destination		Supply
		1	2	
Source	1	40	5	45
	2	20	25	45
Demand		60	30	90

(b) *Solution for IQDS:*

The model for IQDS is presented below and its optimal solution is presented in Table 14.14. The total cost of the optimal shipping plan is Rs. 536.

Model formulation for IQDS:

$$\text{Minimize } Z = 5X_{111} + 4X_{112} + 3X_{113} + 8X_{121} + 7X_{122} + 6X_{123} + 8X_{211} \\ + 6X_{212} + 5X_{213} + 10X_{221} + 9X_{222} + 8X_{223}$$

subject to

$$X_{111} + X_{112} + X_{113} + X_{121} + X_{122} + X_{123} = 45$$

$$X_{211} + X_{212} + X_{213} + X_{221} + X_{222} + X_{223} = 45$$

$$X_{111} + X_{112} + X_{113} + X_{211} + X_{212} + X_{213} = 60$$

$$X_{121} + X_{122} + X_{123} + X_{221} + X_{222} + X_{223} = 30$$

$$-9Y_{111} + X_{111} \leq 0$$

$$-10Y_{112} + X_{112} \leq 0$$

$$-1000Y_{113} + X_{113} \leq 0$$

$$-4Y_{121} + X_{121} \leq 0$$

$$-5Y_{122} + X_{122} \leq 0$$

$$-1000Y_{123} + X_{123} \leq 0$$

$$-19Y_{211} + X_{211} \leq 0$$

$$-45Y_{212} + X_{212} \leq 0$$

$$-1000Y_{213} + X_{213} \leq 0$$

$$-4Y_{221} + X_{221} \leq 0$$

$$-5Y_{222} + X_{222} \leq 0$$

$$-1000Y_{223} + X_{223} \leq 0$$

$$-9Y_{112} + X_{111} \geq 0$$

$$-10Y_{113} + X_{112} \geq 0$$

$$-4Y_{122} + X_{121} \geq 0$$

$$-5Y_{123} + X_{122} \geq 0$$

$$-19Y_{212} + X_{211} \geq 0$$

$$-45Y_{213} + X_{212} \geq 0$$

$$-4Y_{222} + X_{221} \geq 0$$

$$-5Y_{223} + X_{222} \geq 0$$

where $X_{ijk} \geq 0$ and $Y_{ijk} = 0$ or 1 for $i = 1, 2$, $j = 1, 2$, and $k = 1, 2, 3$.

Table 14.14 Optimal Shipping Strategy under IQDS

		Destination		Supply
		1	2	
Source	1	45	-	45
	2	15	30	45
Demand		60	30	90

14.4.6 Application of Modelling in Public System

In this section, a model of application in public system is presented.

Model for locating fire stations

Consider the problem of locating fire stations in a district. The district authority is of the opinion that the timely help in the event of fire to residential regions is very important. Hence, the authority is desiring to locate the optimal number (minimum number) of fire stations to cover all the residential regions in the district such that no residential region has the distance from the serviceable fire station(s) more than a specified maximum distance (say 20 km). This is called *total covering problem*. The objective of this total covering problem is to determine the optimal number of sites for locating fire stations such that no residential region has the actual distance from the serviceable fire station(s) more than the specified maximum distance (say 20 km). The specified distance depends on the road and traffic conditions.

Let d_{ij} be the distance from the residential region i to the fire station location j ; n , the number of residential regions; q , the specified maximum distance of a residential region from the serviceable fire station(s); C_{ij} , the covering coefficient with respect to the i th residential region and location of the j th fire station.

$$C_{ij} = 1; \quad \text{if } d_{ij} \leq q;$$

$$= 0, \quad \text{otherwise}$$

$$Y_j = 1, \text{ if a fire station is assigned to the potential location } j$$

A model to determine the minimum number of fire stations which are to be located such that all the residential regions are covered is presented below.

In this model, each residential region is considered as a potential sites for locating a fire station. If the number of residential regions is n , then the number of potential sites for locating fire stations is also n .

$$\text{Minimize } Z = \sum_{j=1}^n Y_j$$

subject to

$$\sum_{j=1}^n C_{ij} Y_j \geq 1, \quad i = 1, 2, 3, \dots, n$$

where $Y_j = 0$ or 1 , $j = 1, 2, 3, \dots, n$.

In this model, the objective function minimizes the total number of sites selected for locating fire stations from the given set of potential sites. The constraints ensure that each residential region is covered by at least one fire station located in any of the select potential sites.

Example 14.8 The number of residential regions under consideration for locating fire stations is 5. The distance matrix of these 5 regions is shown in Table 14.15. The maximum distance a fire engine should travel to any of the residential region is 20 km. Find the optimal number of fire stations to be located to cover all the residential regions.

Table 14.15 Distance Matrix

		Potential site for fire station				
		1	2	3	4	5
Region	1	40	15	55	22	16
	2	60	65	72	8	68
	3	10	35	8	55	48
	4	51	17	36	14	5
	5	36	44	9	34	25

Solution. Based on the specified maximum distance of travel of 20 km by the fire engines in reaching the residential regions, the corresponding covering coefficient matrix is shown in Table 14.16.

Table 14.16 Covering Coefficient Matrix

		Potential site for fire station				
		1	2	3	4	5
Region	1	0	1	0	0	1
	2	0	0	0	1	0
	3	1	0	1	0	0
	4	0	1	0	1	1
	5	0	0	1	0	0

The model to determine the minimum number of sites to be selected for locating fire stations such that they cover all the residential regions is presented below:

$$\text{Minimize } Z = Y_1 + Y_2 + Y_3 + Y_4 + Y_5$$

subject to

$$Y_2 + Y_5 \geq 1$$

$$Y_4 \geq 1$$

$$Y_1 + Y_3 \geq 1$$

$$Y_2 + Y_4 + Y_5 \geq 1$$

$$Y_3 \geq 1$$

where $Y_j = 0$ or 1 , $j = 1, 2, 3, 4, 5$.

The results of this model using STORM package are: Minimum number of fire stations to be located = 3. The sites at which fire stations to be located are 2, 3 and 4.

The allocations of residential regions to fire stations are indicated by squares in Table 14.17. The results show that there is a possibility of backup coverage for the residential region 4 because it is covered by more than one fire station (2 and 4).

Table 14.17 Allocation of Residential Regions to Fire Stations

		Potential fire station location		
		2	3	4
Region	1	1	0	0
	2	0	0	1
	3	0	1	0
	4	1	0	1
	5	0	1	0

QUESTIONS

1. Define Operations Research.
2. Define model. Discuss the steps of modelling.
3. Discuss the scope of Operations Research.
4. Discuss the formulation of the mathematical model to minimize the mean tardiness in single machine scheduling problem.
5. Consider the data of a single machine scheduling problem with 3 jobs as shown below:

Job (<i>i</i>)	Processing time (<i>t</i>)	Due date (<i>d</i>)
1	7	12
2	5	9
3	12	20

- Develop a 0–1 programming model to find the sequence which minimizes the total tardiness.
6. Discuss the formulation to minimize makespan of scheduling n single operations jobs under n parallel processors.
 7. Discuss the formulation of location of dealers to cover all the customer regions such that the total number of dealers located is minimized.
 8. The number of customer regions under consideration for locating dealers is 5. The distance matrix of these 5 regions is shown below:

		Distance Matrix Potential sites				
		1	2	3	4	5
Region	1	7	17	25	8	16
	2	22	9	32	7	28
	3	6	15	21	45	38
	4	4	32	26	24	5
	5	36	15	26	3	22

The Maximum distance a customer from any customer region can travel to reach a dealer is 10 km.

(a) Develop a 0-1 programming model to find the optimal number of dealers to be located to cover all the customer regions.

(b) *Only for assignment:* Solve the model in part (a) and give your recommendations.

9. A garment company has almost completed the R&D activities of bringing out a new advertisement for its products. The company has identified its target customers in the following categories with their relative importance indicated in parentheses.

1. Male (employed) (0.2)
2. Female (employed) (0.3)
3. Age in the range 18-50 years (0.5).

The total advertising budget allocation for the forthcoming year is Rs. 45 lakhs. The company has collected additional data for this exercise, as shown below:

Additional Data for Media Planning

Data Item	Media (% of reading/watching)		
	Magazine	Television	Newspaper
Readers/viewers characteristics			
Male (employed)	80	70	75
Female (employed)	85	95	80
Age in the range 18 - 50 years	70	75	73
Cost per advertisement (Rs. in lakhs)	0.1	2	0.5
Minimum no. of ads	10	4	6
Maximum no. of ads	20	10	25
Audience size per ad.	7,00,000	90,00,000	3,00,00,000

Develop an integer linear programming model to determine the optimal media-mix.

10. Consider the capital budgeting problem where five projects are being considered for execution over the next 3 years. The expected returns for each project and the yearly expenditures (in lakhs of rupees) are shown below. Assume that each approved project will be executed over the 3 year period. The objective is to select a combination of projects that will maximize the total returns.

Project	Expenditures for			Returns
	Year-1	Year-2	Year-3	
1	4	2	9	25
2	13	8	12	50
3	2	10	8	35
4	6	3	7	25
5	6	9	10	40
Maximum fund	25	25	25	-

(a) Formulate the problem as a 0–1 integer programming problem.

(b) *Only for assignment:* Solve the model of part (a) and give your recommendations.

11. Portfolio selection problem considers the factors such as risk, working capital liquidity, growth potential, rate of return, and diversification requirements.

A leading insurance company has identified seven investment alternatives for a 15-year investment period to invest its 500 crores rupees. The details of the investment alternatives are presented in the following table:

Details of Investment Alternatives

Investment alternatives	Length of investment	Annual rate of return (%)	Risk coefficient	Growth potential
Treasury bills	4	4	3	0
Equity	7	12	8	12
Company bonds	8	8	4	8
Real estate	6	14	8	20
Mutual fund	10	10	6	15
Saving and loans	5	4	3	5
Cash	0	0	0	0

In the table, length of investment means the expected number of years required for the annual rate of return to be realized taking into account the possibility of reinvestment. The objective of the insurance company is to maximize its returns subject to the following constraints:

- The average length of investment should be less than or equal to 6.
- The average risk for the portfolio should not exceed 7.
- The average growth potential for the portfolio should be at least 10%.
- At least 12% of all available funds must be retained in the form of cash at all times to maintain working capital liquidity.
- Sum of the proportions of the investments should be 1.

- (a) Develop a linear programming model for the above situation.
 (b) *Only for assignment:* Solve the model of part (a).
12. In a company, maintenance persons report to duty at the beginning every 4-hour interval, as shown in the following table:

Details of Manpower Planning Problem

Interval	Time period		Minimum number of maintenance persons required
	From	To	
1	12 midnight	4 a.m.	3
2	4 a.m.	8 a.m.	4
3	8 a.m.	12 noon	8
4	12 noon	4 p.m.	10
5	4 p.m.	8 p.m.	7
6	8 p.m.	12 midnight	4

Each person, after reporting to duty will work for 8 continuous hours. The minimum required number of maintenance persons in each period is also furnished in the same table.

- (a) Develop a mathematical model to determine the number of maintenance persons to report to duty at the beginning of each period such that the total number of maintenance persons required for the day is minimized.
 (b) *Only for assignment:* Solve the model of part(a).
13. Discuss the modelling of AQDS problem of transportation problem.
 14. Discuss the modelling of IQDS problem of transportation problem.
 15. Discuss the modelling of locating fire stations to cover all the residential regions such that the total number of fire stations is minimized.
 16. The number of residential regions under consideration for locating district libraries is 5. The distance matrix of these 5 regions is shown in the following distance matrix:

		Potential site for library				
		1	2	3	4	5
Region	1	40	55	20	45	25
	2	12	85	30	18	60
	3	28	14	38	20	38
	4	10	27	36	34	15
	5	46	8	29	54	45

The maximum distance a person from any of these regions should travel to a district library is 25 km.

(a) Develop a model to find the optimal number of district libraries to be located to cover all the residential regions.

◦ (b) *Only for assignment:* Solve the model of part (a).

17. Assume a problem of each of the following areas and develop mathematical models to solve them:

- (a) production
- (b) marketing
- (c) personnel
- (d) finance
- (e) interface
- (f) public system.

CHAPTER

15

SIMULATION

15.1 INTRODUCTION

Simulation is an experiment conducted on a model of some system of interest. Some of the systems on which the simulation experiment can be carried out are listed below:

- Simulation of waiting behaviour of customers at reservation counters of a transport corporation;
- Simulation of customers' waiting at bank counters;
- Simulation of jobs' waiting at the inspection station of an assembly line;
- Simulation of a kanban system of an automobile assembly line;
- Simulation of data-traffic in a computer network;
- Simulation of queueing behaviour of vehicles at a toll gate;
- Simulation of utilization of berths in a harbour;
- Simulation of sales revenue of a company;
- Simulation of cash flows with uncertainty of different investment alternatives;
- Simulation of labour absenteeism;
- Simulation of labour turnover;
- Simulation of stock exchange index movement;
- Simulation of portfolio mix;
- Simulation of petrol bunks for effective space planning.

In general, the queueing system consists of one or more queues and one or more servers. The queueing system operates under a set of procedures. Consider the inspection station of an assembly line producing television sets. Television sets come for inspection from the preceding station of the inspection station. The inspection time of the television sets will follow some probability distribution. The actual inspection time taken will vary from one set to another set. If the inspection times of the television sets do not match with the inter-arrival times of the television sets, there may be queueing of television sets in front of the inspection station which will introduce the problem of waiting space management. One can simulate the functioning of the inspection station with different number of inspectors to arrive the desired number of waiting television sets in front of it. If an incoming set finds

a free inspector at the inspection station, then it will be immediately taken for inspection by that free inspector; otherwise, it will be put into the queue in front of the inspection station. If an inspector completes the inspection on a television set and if the queue has some waiting television sets, then he will take the first set from the queue for inspection.

In this process, over a period of time, the system may experience queueing of television sets and/or idleness of inspector(s). In any service system/manufacturing system involving queueing situation, the objective is to design the system such that the average waiting time of the customers as well as the average waiting number of customers is minimized and the percentage utilization of the server is maintained above a desired level.

Most of the practical problems to which simulation can be applied belong to queueing systems.

The queueing system exists in many service/manufacturing situations as described in the preceding paragraphs.

In addition to these examples, many more subsystems of production, finance, personnel and marketing functions of an organization can be modelled as queueing systems for effective management decisions.

15.2 NEED FOR SIMULATION

Consider an example of the queueing system, namely the reservation system of a transport corporation. The elements of the system are booking counters (servers) and waiting customers (queues). Generally, the arrival rate of the customers follows Poisson distribution and the service time follows exponential distribution. If the reservation system has the above combination of distributions, then the queueing model, namely $(M/M/1): (GD/\infty/\infty)$ can be used to find the standard results. But, in reality, the following combinations of distributions may exist:

- (a) Arrival rate does not follow Poisson distribution but the service time follows exponential distribution.
- (b) Arrival rate follows Poisson distribution but the service time does not follow exponential distribution.
- (c) Arrival rate does not follow Poisson distribution and the service time does not follow exponential distribution.

In each of the above cases, the non-standard distribution for the queueing system may be discrete distribution, uniform distribution, normal distribution, etc.

Under such situation, the standard model $(M/M/1): (GD/\infty/\infty)$ cannot be used. The last resort to find the solution for the non-standard queueing model is to use simulation.

Like this, if there is a deviation from other standard models of queueing systems, one can use simulation. In practice, a system with any kind of assumptions can be tackled using simulation.

To quote a few of them, the following list of variations of the queueing system is provided. Under these variations, simulation is the only method to find solution for a given problem.

Bulk arrival Generally, it is assumed that the customers arrive at the system one by one. But, in reality, customers may arrive in groups. Such arrival is called *bulk arrival*.

Jockeying If there is more than one queue, the customers from one queue will be tempted to join another queue because of its smaller size. This behaviour of the customers is known as *queue jockeying*.

Balking On arrival of a customer, if the queue length is very large from his/her view point, he/she may not join the queue. This property is known as *balking* of customers.

Reneging Sometime, a customer who is already in the queue will leave the queue in anticipation of longer waiting time. This kind of departure from the queue without receiving the service is known as *reneging*.

Priority This refers to the differential importance given to customers mainly to distinguish them from one another while selecting them for providing service.

Model *Model* is an abstraction of a reality and it is used for experimentation. For example, consider the maintenance shop of a manufacturing industry. The shop can be modelled as a queueing system with 'maintenance mechanics' as 'servers' and the 'breakdown machines' as 'customers'. Later, this model can be experimented to obtain the standard results, as listed below:

- Percentage utilization of servers
- Average waiting number of breakdown machines
- Average waiting time per breakdown machine.

15.3 TYPES OF SIMULATION

There are following four types of simulation:

Identity simulation Observing the behaviour of the real world under as many operational configurations as possible to get an insight into system behaviour is called *identity simulation*.

But, the identity simulation is very expensive, seldom feasible, time consuming and permits little control over parameters that affect the response of the system.

Quasi-identity simulation In *quasi-identity* simulation, aspects of the real-world system are preserved while some elements whose presence would make the identity simulation impossible are excluded.

Simulation of an air attack over a country by its enemy aircraft is an example of *quasi-identity simulation*.

Laboratory simulation This is a cheaper method than the earlier two methods. Consider the example of operational gaming in which the business system is represented by its entities like production, personnel, finance, marketing, logistics and its players (executives) within its competing environment. In this type of simulation, transactions will be carried out over a specified period and then the operating performance of the business will be reviewed.

Computer simulation If we remove people from the laboratory simulation with a well-designed computer logic and retain other aspects, then it is called 'computer simulation'.

Advantages of simulation

The following are the advantages of simulation:

- The time for experimentation can be compressed.
- The system performance can be studied under all possible conditions.
- The success or failure of a system can be tested using a conceptual system.

Caution: Because, the use of simulation is a costly exercise, it should be used as the last resort.

Major steps of simulation

The major steps of simulation are presented in Fig. 15.1 in the form of a flow diagram.

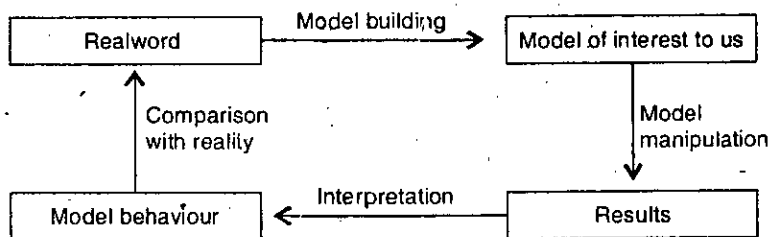


Fig. 15.1 Major steps of simulation experiment.

15.4 SIMULATION LANGUAGES

There are many specialized simulation languages like GPSS, SIMSCRIPT, GASP, etc. In this section, guidelines for simulation using high-level languages (like Fortran, C, Basic, etc.) are presented which will help the readers have better insight of the simulation concept. For details on GPSS, readers are directed to refer to the books by Panneerselvam (2002) and Scriber (1974).

Simulation using high-level languages

Simulation method can be classified into *discrete-event* simulation and *continuous* simulation.

The following terminologies of the single server model are explained first to understand the concept of discrete-event simulation:

- Customer arrival time (CAT)
- Service completion time (SCT)

- End of simulation (ES). (This is known as the run length of simulation)
- Current time/clock time (CT)
- Next-most-imminent event (NMIE).

At any point of time, the CAT, SCT and ES will have some specific values. In the discrete-event simulation, the progress of simulation experiment is done by updating CT to the time of next-most-imminent event which is the minimum of values of CAT, SCT and ES.

In continuous simulation, the current time (CT) is incremented by an equal amount of very small value. The equal increment in the CT is decided based on the sensitivity of the performance measures of the system.

Single server queueing system

A logical flowchart for the single server queueing system is presented in Fig. 15.2.

The terminologies used in Fig. 15.2 are listed as follows:

CT	Current time
CAT	Customer arrival time
IAT	Inter-arrival time
N	Total number of customers entered into the system
SCT	Service completion time
R	Uniformly distributed random number between 0 and 1
ST	Service time
CST	Cumulative service time
ES	End of simulation
NMIE	Next-most-imminent-event (the earliest of CAT, SCT and ES)
CTU	Customer time unit
TLCQ	Time of last change in the queue
QL	Queue length
BIG	A very large value
PU	Percentage utilization of server
AWT	Average waiting time per customer
AWN	Average waiting number of customer
$A \pm B$	Uniform distribution of inter-arrival time (range is from $A-B$ to $A+B$)
$C \pm D$	Uniform distribution of service time (Range is from $C-D$ to $C+D$).

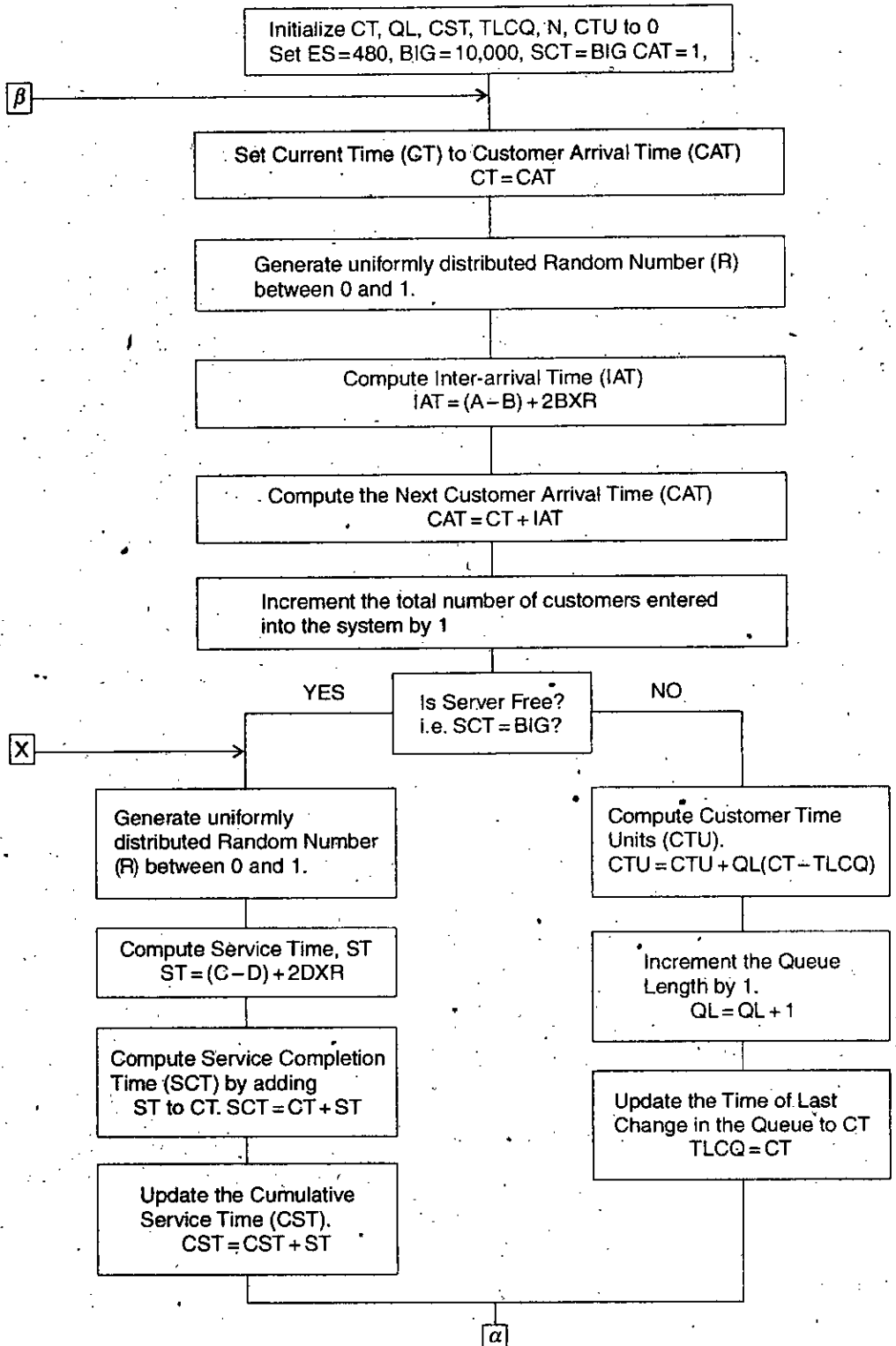


Fig. 15.2 Contd.

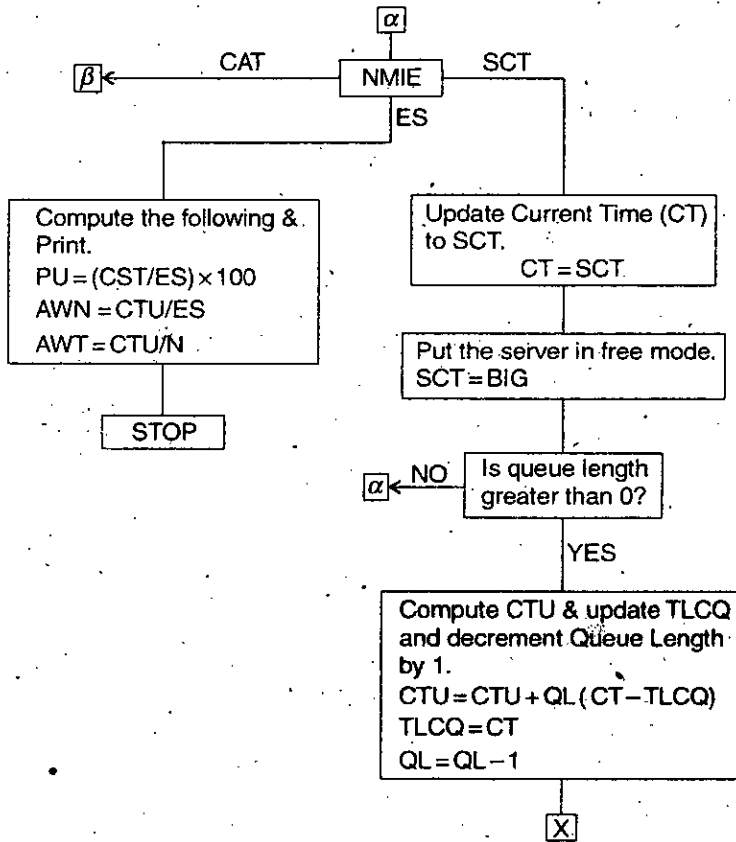


Fig. 15.2 Logical flowchart for single server model.

Sample run

The simulation logic is demonstrated for a few iterations with the following data: Inter-arrival time follows uniform distribution with 24 ± 6 minutes. Service time follows uniform distribution with 18 ± 8 minutes.

Set $CT = QL = CST = TLCQ = N = CTU = 0$

$ES = 480$; $BIG = 10,000$, $SCT = BIG$, $CAT = 1$.

Update CT:

$CT = CAT = 1$

Compute IAT:

Let the random number, R be 0.5.

$IAT = (A - B) + 2 \times B \times R = (24 - 6) + 2 \times 6 \times 0.5 = 24$

Compute CAT:

$$CAT = CT + IAT = 1 + 24 = 25$$

$$N = N + 1 = 0 + 1 = 1$$

Check Server Status:

Since, $SCT = BIG$, ST is to be computed.

Compute ST and SCT:

Let the uniformly distributed random number (R) be 0.95.

$$ST = (C - D) + 2 \times D \times R = (18 - 8) + 2 \times 8 \times 0.95 = 25$$

$$SCT = CT + ST = 1 + 25 = 26$$

Compute CST:

$$CST = CST + ST = 0 + 25 = 25$$

Determine NMIE:

$CAT = 25$, $SCT = 26$ and $ES = 480$. The minimum of the three values is 25 which is for the event, CAT . Hence, the NMIE is CAT .

Update CAT:

$$CT = CAT = 25$$

Compute CAT:

Let, $R = 0.85$

$$IAT = (A - B) + 2 \times B \times R = (24 - 6) + 2 \times 6 \times 0.85 = 28$$

$$CAT = CT + IAT = 25 + 28 = 53$$

$$N = N + 1 = 1 + 1 = 2$$

Check Server Status:

Because SCT is not equal to BIG , add the new customer in the queue.

Add Customer into Queue:

$$CTU = CTU + QL (CT - TLCQ) = 0 + 0 (25 - 0) = 0$$

$$TLCQ = CT = 25$$

$$QL = QL + 1 = 0 + 1 = 1$$

Determine NMIE:

CAT = 53, SCT = 26 & ES = 480. The minimum of the three values is 26 which is for the event, SCT. Hence, the NMIE is SCT.

Update SCT:

$$CT = SCT = 26$$

Set Server in Free Mode:

$$SCT = BIG$$

Check Queue Content:

Because $QL > 0$, fetch the first customer (at present, the only customer) from the queue for service.

Updating Queue Details and Picking Customer from Queue:

$$CTU = CTU + QL (CT - TLCQ) = 0 + 1(26 - 25) = 1$$

$$TLCQ = 26$$

$$QL = QL - 1 = 1 - 1 = 0$$

Compute SCT:

$$\text{Let, } R = 0.75$$

$$ST = (C - D) + 2 \times D \times R = (18 - 8) + 2 \times 8 \times 0.75 = 22$$

$$SCT = CT + ST = 26 + 22 = 48$$

Compute CST:

$$CST = CST + ST = 25 + 22 = 47$$

Determine NMIE:

CAT = 53, SCT = 48 and ES = 480. The minimum of the three values is 48, which is for the event, SCT. Hence, the NMIE is SCT.

The above results along with some more calculations are summarized in Table 15.1. Like this, the steps are to be carried out till the NMIE is ES. Finally, the results, PU, AWT and AWC can be computed using the given formulas.

Now, the concept of CTU calculation is demonstrated in Fig. 15.3 with the help of Table 15.1. The area of each rectangle in Fig. 15.3 is given within itself. The sum of the rectangles in Fig. 15.3 gives the total value for CTU. From this value, the AWT and AWC can be calculated by using the following formulas:

$$AWT = \frac{CTU}{N}, \quad AWC = \frac{CTU}{ES}$$

Table 15.1 Summary of Calculation for Single Server Model

Iteration number	CT	R	IAT	CAT	N	SCT?	R	ST	SCT	CST	CTU	TLCQ	QL	NMIE	CT	SCT	QL>0 ?
1	1	0.5	24	25	1	BIG	0.95	25	26	25	-	-	0	CAT = 25* SCT = 26 ES = 480	-	-	-
2	25	0.85	28	53	2	26	-	-	-	-	0	25	1	CAT = 53* SCT = 28* ES = 480	26	BIG YES	
											1	26	0				
3	26	-	-	-	-	BIG	0.75	22	48	47	-	-	0	CAT = 53* SCT = 48* ES = 480	48	BIG NO	
4	48	-	-	-	-	-	-	-	-	-	-	-	0	CAT = 53* SCT = BIG ES = 480	-	-	-
5	53	0.20	20	73	3	BIG	0.90	24	77	71	-	-	0	CAT = 73* SCT = 77 ES = 480	-	-	-
6	73	0.40	23	96	4	77	-	-	-	-	1	73	1	CAT = 96* SCT = 77* ES = 480	77	BIG YES	
											5	77	0				
7	77	-	-	-	-	BIG	0.80	23	100	94	-	-	0	CAT = 96* SCT = 100 ES = 480	-	-	-

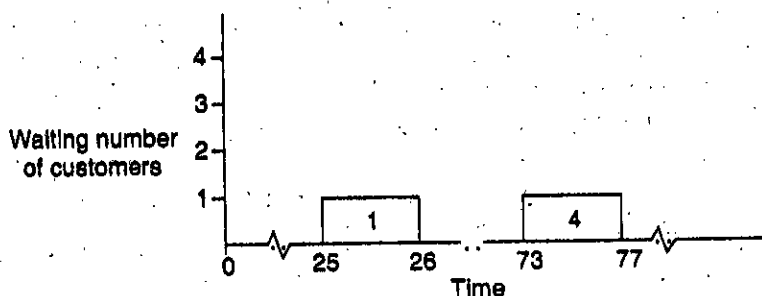


Fig 15.3 Schematic view of customer-time unit (CTU) based on Table 15.1.

15.5 SIMULATION OF A SEASON TICKET ISSUING COUNTER OF A TRANSPORT CORPORATION—A CASE STUDY

In this section, a study on simulation of a season ticket issue counter of a transport corporation (Panneerselvam and Veerapandian, 1990) is presented.

15.5.1 Problem Statement

A transport corporation in a city gives monthly season tickets to students at various depots. A counter in one of the depots of the corporation was selected for this study.

Every month, the tickets are issued to the students at a single counter in two queues: one for male students and the other for female students. Alternate service is provided to the male and the female queues and the service discipline is first-in-first-out within each queue. The students arrive at the system in bulk. Also, they may buy more than one ticket, that is they may buy for their friends who have not come.

The counter functions during the forenoon and from 3 p.m. to 8 p.m. everyday (except government holidays) from 1st to 20th of every month. Normally, a few conductors (who report to duty, but not on trip) are assigned the counter operation. Since the counter is specially assigned for the students, during the forenoon period, the queues are found to be manageable in size. But, the observations during 3 p.m. to 8 p.m. warranted this study. Again, based on the observations, the time period from 3 p.m. to 8 p.m. was divided into 5 one-hour intervals to study the variations between periods.

15.5.2 Objectives of the Study

The objectives of this study are to determine the following items during the various time periods from 3 p.m. to 8 p.m. with 5 one-hour intervals to check the needs for one more additional counter:

- (a) The average number of students waiting in each of the queues.
- (b) The average waiting time of the students in each of the queues.
- (c) The percentage utilization of the server (clerk) at the counter.

15.5.3 Data Collection

The following data were collected:

- (a) The inter-arrival time of the male students
- (b) The inter-arrival time of the female students
- (c) The bulk arrival pattern of the male students
- (d) The bulk arrival pattern of the female students
- (e) The distribution of the number of season tickets purchased per student.
- (f) The service time distribution per ticket.

15.5.4 Data Analysis

The inter-arrival time in all the periods for both the queues follows exponential distribution (arrival rates follow Poisson distribution). The mean arrival rate for each period is given in Table 15.2.

Table 15.2 Arrival Rate

Time period	Arrival rate/hr	
	Male queue	Female queue
3 p.m. to 4 p.m.	20	16
4 p.m. to 5 p.m.	33	24
5 p.m. to 6 p.m.	45	36
6 p.m. to 7 p.m.	40	21
7 p.m. to 8 p.m.	23	14

The service time follows exponential distribution and the mean service time is 40.45 sec (mean service rate is 89/hr). The probability distribution of the number of tickets purchased per student and the bulk arrival pattern of the male students and the female students are given in Table 15.3.

Table 15.3 Probability Distributions of Bulk Arrival, Number of Tickets Purchased by Students

	Value for random variable			
	1	2	3	4
Bulk arrival (male)	0.595	0.290	0.115	—
Bulk arrival (female)	0.591	0.268	0.141	—
No. of tickets/student	0.789	0.126	0.063	0.022

15.5.5 Fitting the Problem to Simulation Study

The arrival rates of both the queues follow Poisson distribution and the service time follows exponential distribution. Students arrive at the system in bulk and they come to buy more than one ticket. The coexistence of these attributes of the queueing system vindicated the use of simulation approach.

15.5.6 Simulation Model

A simplified logic of the simulation model for the queueing situation described is shown in Fig. 15.4. The logic is based on the variable increment discrete event simulation.

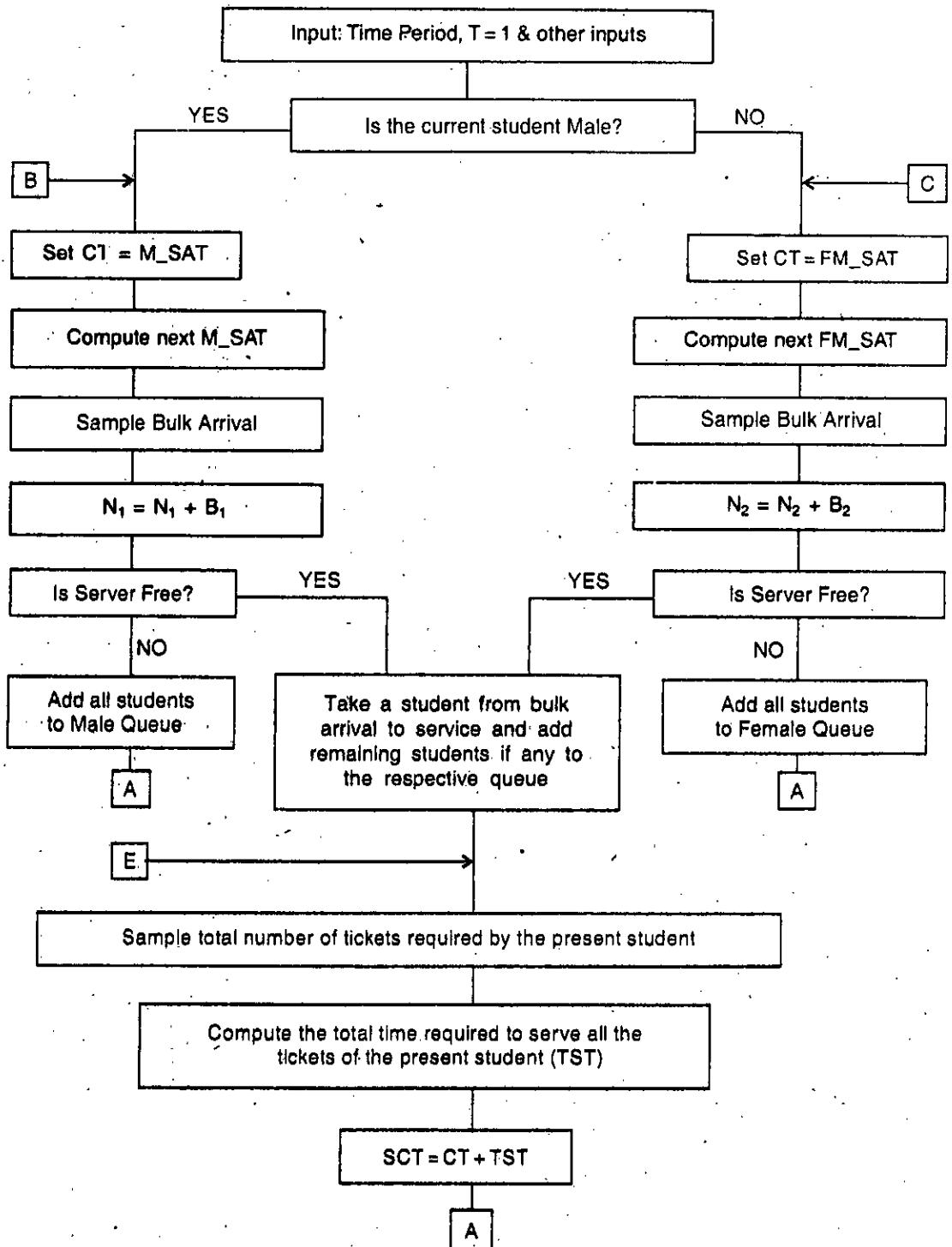


Fig. 15.4 Contd.

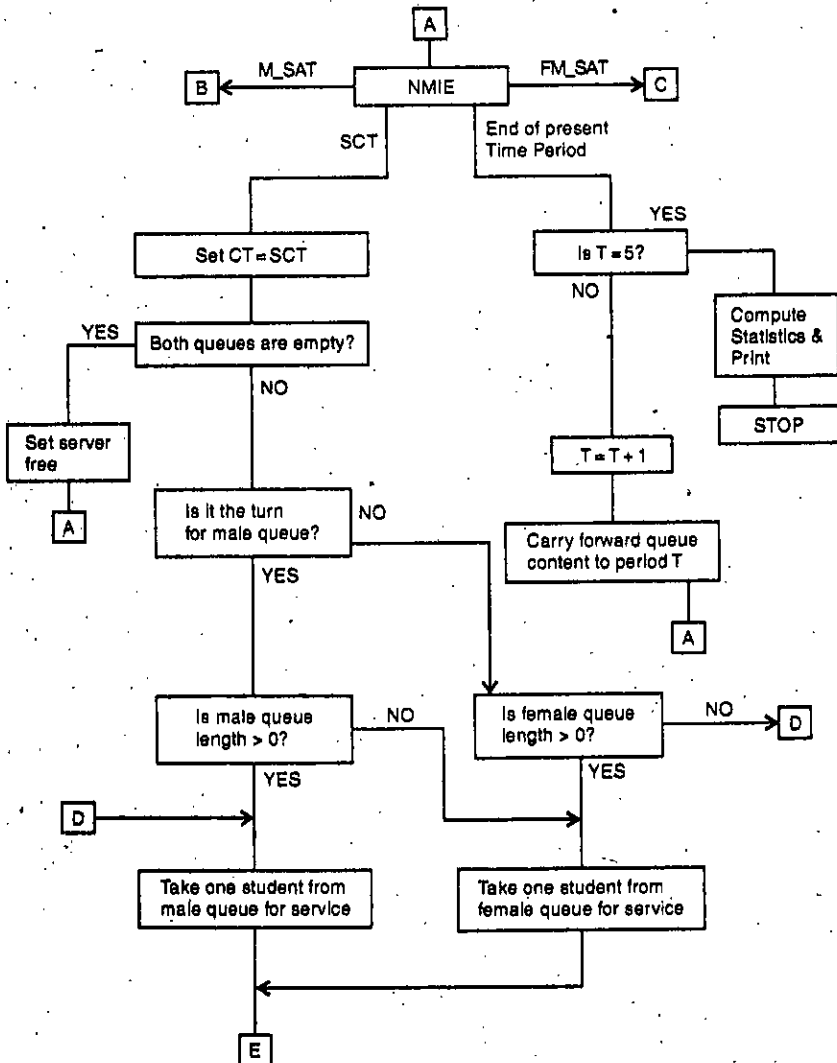


Fig. 15.4 Logical flowchart for simulating season ticket issue counter.

The various abbreviations used in the logic are as follows:

NMIE	Next-most-imminent event
CT	Current time (clock time)
M_SAT	Male student arrival time
FM_SAT	Female student arrival time
TST	Total service time for the present student
SCT	Service completion time
N1	Number of male students entered into the system
N2	Number of female students entered into the system
T	Time period (time interval number which varies from 1 to 5)
B ₁	Size of the bulk arrival in the male queue
B ₂	Size of the bulk arrival in the female queue

15.5.7 Output Analysis

The output of the simulation is shown in Table 15.4. From the table, it is clear that an additional counter from 6 p.m. onwards is justified because of the increased average number of waiting students, average waiting time per student in both the queues and the full utilization of the server. It is suggested to allot one more person to operate the counter as it is not preferable to open a separate counter from 6 p.m. onwards due to some operational difficulties. As there is no extra cost involved in assigning one extra person to the counter, the suggestion was well received and implemented by the transport corporation.

Table 15.4 Output of Simulation

Time period	Average no. of waiting students		Average waiting time/student (sec)		Server utilization (%)
	Male queue	Female queue	Male queue	Female queue	
3 p.m. to 4 p.m.	1.074	0.915	138	156	83.3
4 p.m. to 5 p.m.	0.792	2.644	86	396	77.4
5 p.m. to 6 p.m.	5.285	9.084	358	760	95.6
6 p.m. to 7 p.m.	22.459	10.840	1138	868	100.0
7 p.m. to 8 p.m.	23.195	4.170	1415	577	100.0

QUESTIONS

1. Define simulation. Also, discuss various application areas of simulation.
2. Discuss the advantages of simulation.
3. What are types of simulation? Explain them in brief.
4. Discuss the steps of simulation.
5. Give a logical flowchart for the single-server simulation model.
6. Discuss the applications of simulation in production management.
7. Discuss the applications of simulation in:
 - (a) Marketing management
 - (b) Financial management
 - (c) Personnel management
 - (d) System management
 - (e) Public systems management.
8. 'Simulation should be used as the last resort'. Explain.

9. A transport corporation in a city gives monthly season tickets to students at various depots. A counter in one of the depots of the corporation was selected for this study. Every month, the tickets are issued to the students at a single counter in two queues: one for male students and another for female students. Alternate service is followed between the male and the female queues and the service discipline is first-in-first-out within each queue. The students arrive at the system in bulk. Also, they may buy it more than one ticket, that is, they may buy it for their friends who have not come.

The counter functions during the forenoon and from 3 p.m to 8 p.m. everyday (except government holidays) from 1st to 20th of every month. Normally, a few conductors (who report to duty, but not on trip) are assigned the counter operation. Because the counter is specially assigned for the students, during the forenoon period, the queues are found to be manageable in size. But, the observations during 3 p.m. to 8 p.m. warranted this study. Again, based on the observations, the time period from 3 p.m. to 8 p.m. is divided into 5 one-hour intervals to study the variations between periods.

The objectives of this study are to determine the following items during the various time periods from 3 p.m. to 8 p.m. with 5 one-hour intervals to justify the needs for one more additional counter:

- The average number of students waiting in each of the queues
- The average waiting time of the students in each of the queues
- The percentage utilization of the server (clerk) at the counter.

The following data were collected:

- The inter-arrival time of the male students
- The inter-arrival time of the female students
- The bulk arrival pattern of the male students
- The bulk arrival pattern of the female students
- The distribution of the number of season tickets purchased per student
- The service time distribution per ticket.

The inter-arrival time in all the periods for both the queues follows exponential distribution (arrival rates follow Poisson distribution). The mean arrival rate for each period is given in the following table.

Time period	Arrival Rate	
	Arrival Rate/hr	
	Male queue	Female queue
3 p.m. to 4 p.m.	22	19
4 p.m. to 5 p.m.	35	27
5 p.m. to 6 p.m.	47	39
6 p.m. to 7 p.m.	42	24
7 p.m. to 8 p.m.	25	17

The service time follows exponential distribution and the mean service time is 50 sec. The probability distribution of the number of tickets purchased per student and the bulk arrival pattern of the male students and the female students are given in the following table:

Probability Distributions of Bulk Arrival, Number of Tickets Purchased by Students

	Value for random variable			
	1	2	3	4
Bulk arrival (male)	0.495	0.340	0.165	—
Bulk arrival (female)	0.491	0.318	0.191	—
No. of tickets/student	0.689	0.176	0.113	0.022

Develop a logical flowchart to simulate this system.

CHAPTER 16

REPORT WRITING AND PRESENTATION

16.1 INTRODUCTION

The purpose of conducting a research is to come out with inferences and suggestions. Generally, a research involves utilization of different types of resources and results in significant cost of conducting it. Hence, the outcome of the research should be well documented in the form of a research report for implementation and future use. The design and presentation of the research report involves a pre-specified sequence of steps which are presented in this chapter.

16.2 TYPES OF REPORT

A research report can be classified into *decision-oriented (technical)* report and *research-oriented* report. Further, the research-oriented report can be classified into *survey-based* research report and *algorithmic* research report.

16.2.1 Decision-oriented (Technical) Report

The steps of preparing decision-oriented report are presented below:

- i. Identification of the problem
- ii. Establishment objectives
- iii. Generation of decision alternatives
- iv. Evaluation of decision alternatives
- v. Selection of the best decision alternative
- vi. Development of action plan
- vii. Provision for correction plan after implementation of the decision.

Problem identification is the process selecting a problem which will give very good pay-off in terms of value addition to the organization. The objectives or criteria of the decision problem should be listed and finally a meaningful combination of the objectives is to be selected by taking the ground realities of the decision environment into account. The

objectives/criteria of the decision problem can be fulfilled/achieved by many ways. The researcher should list all the meaningful decision alternatives of achieving the set of criteria for the decision problem. Then, the best combination of the decision alternatives is to be selected such that the value/pay-off of the decision is maximized. The next step is to draw an action plan to implement the selected set of decision alternatives to achieve the objectives of the decision problem. Sometimes, there is deviation from the action plan while implementing the decision alternatives. So, the decision maker should be provided with contingency plan of tackling such a situation.

16.2.2 Survey-based Research Report

The main body of the report for the survey-based research contains the following:

- i. Problem definition
- ii. Objectives of the research
- iii. Research methodology
- iv. Data analysis
- v. Interpretation of results and suggestion
- vi. Conclusions.

16.2.3 Algorithmic Research Report

There are problems, viz., production scheduling, JIT, supply chain management, line balancing, layout design, portfolio management, etc., exist in reality. The solution for each of the above problems can be obtained through algorithms. So, the researchers should come out with newer algorithms or improved algorithms for such problems. For a combinatorial problem, the researcher should attempt to develop an efficient heuristic. The algorithmic research report can be classified into the following categories:

- Algorithmic research report for combinatorial problem
- Exact algorithmic research report for polynomial problem.

Algorithmic research report with modelling for combinatorial problem

The main body of this type of research report will contain the following:

- i. Problem identification
- ii. Literature review
- iii. Objectives of the research
- iv. Development of mathematical model
- v. Design of algorithm (heuristic)
- vi. Experimentation and comparison of the algorithm with the model in terms of solution accuracy
- vii. Experimentation and comparison of the algorithm with the best existing algorithm (heuristic) in terms of solution accuracy

- viii. Case study
- ix. Conclusions.

In this type of research, the results of the algorithm will be compared with the optimal results of the mathematical model as well as with the results of the best existing algorithm to check its solution accuracy through a carefully designed experiment (i.e. ANOVA). This type of report is mainly applicable for research on combinatorial problems.

Note: In a research related to combinatorial problems in new and complex area, development of a mathematical model to obtain the optimal solution may not be easy. Under such situation, the results of the algorithm (heuristic) should be compared with that of the best existing heuristic alone for checking its solution accuracy through a carefully designed experiment (ANOVA).

Exact algorithmic research report for polynomial problem

The main body of this type of research report will contain the following:

- i. Problem identification
- ii. Literature review
- iii. Objectives of the research
- iv. Design of exact algorithm
- v. Experimentation and comparison of the exact algorithm with the best existing exact algorithm in terms of computational time
- vi. Case study
- vii. Conclusions.

In a research related to polynomial problem, the researcher will have to develop an efficient exact algorithm in terms of computational time and compare it with the best existing exact algorithm for that problem through a carefully designed experiment. The comparison in terms of solution accuracy does not apply here because all exact algorithms will give optimal solution.

16.3 GUIDELINES FOR REVIEWING DRAFT

The draft of the report should be reviewed for an appropriate number of times so that the errors are completely avoided. While reviewing the draft, certain guidelines are to be followed, as indicated below:

- The text of the report should be free from ambiguity.
- The text should convey the intended message.
- Because the readers are with different profiles, the style and presentation of the text of the report should suit the profile of the targetted group of readers; otherwise, the purpose of the report will be lost.
- The content of the report should fully reveal the scope of the research in logical sequence without omitting any item and at the same time it should be crisp and clear.

- The report should be organized in hierarchical form with chapters, main sections, subsections within main sections, etc.
- There should be continuity between chapters and also between sections as well subsections.
- The abstract at the beginning should reveal the essence of the entire report which gives the overview of the report.
- The chapter on conclusions and suggestions is again enlarged version of the abstract with more detailed elaboration on the inferences and suggestions.
- A reading of abstract and conclusion of a report should give the clear picture of the report content to the readers.
- Avoid using lengthy sentences unless warranted.
- Each and every table as well as figure should be numbered and it must be referred in the main text.
- The presentation of the text should be lucid so that every reader is able to understand and comprehend the report content without any difficulty.
- The report should have appropriate length. The research report can be from 300 to 400 pages, but the technical reports should be restricted to 50 to 75 pages.

16.4 REPORT FORMAT

The outcomes of research works are to be conveyed to end-users or recorded for future use. The research efforts will be useful to the end-users only when they are documented in the form of reports. So, a standardized format for the report will help researchers present their contributions and findings more systematically.

The different items of a research report are presented in Fig. 16.1.

1. Cover page
2. Introductory pages
- Vita
- Acknowledgement
- Abstract
- Table of contents
- List of tables
- List of figures
- List of variables
3. Text
- Chapters
- main sections
- subsections
- Footnotes
- Conclusions
4. Bibliography
5. Appendices

Fig. 16.1 Items of research report.

16.4.1 Cover Page

The cover page gives the title of the research, department/faculty to which the thesis/report is submitted, degree for which it is submitted, name of the researcher, name of the supervisor (if applicable), university/organizational emblem, name of the university/organization and its address, month and year of submission. A sample cover page is shown in Exhibit 16.1.

EXHIBIT 16.1 COVER PAGE

Development of New Efficient Heuristic for Deterministic
Assembly Line Balancing Problem

A thesis submitted to the
Faculty of Management, Pondicherry University
in partial fulfilment of the requirements
for the award of the degree of

Doctor of Philosophy
by
S. Narayanan, B.E., M.S.

Research Supervisor
R. Panneerselvam, M.E., Ph.D.
Professor

University Seal

School of Management
Pondicherry University
Pondicherry 605 014. (India)
November 2000

16.4.2 Introductory Pages

As mentioned in Exhibit 16.1, the introductory pages are vita, acknowledgments, abstract, table of contents, list of tables, list of figures and list of variables.

Vita *Vita* gives a brief bio-data of the researcher. This should contain the following:

- Name of the researcher
- Date of birth and place of birth
- Degrees already holding and the respective years of pass and institutions of study
- Details of experience in chronological sequence
- List of publications by the researcher which are classified as follows:
 - Publications by the researcher from the current research
 - Publications by the researcher in other areas

Acknowledgements In the process of carrying out any research, the researcher would have received help from different persons and organizations. So, all those helps should be *acknowledged* under this heading. In an academic research, a sample list of people who are to be acknowledged is as follows:

- Research supervisor
- Organizational head
- Colleagues of the department who rendered their valuable suggestions and criticism during the research period
- The executives in different organizations who provided data and literature for the research
- The typesetting office and personnel.

Abstract *Abstract* gives the overview of the research report outlining the title of the research, objective(s), shortcomings of exiting literatures, need for further research, proposed measure of performance and research methods, results of comparison and highlights of inferences and a brief mention about a case study.

Table of contents *Table of contents* is the catalogue of the research report which summarizes every aspect of the report. The table of contents of an algorithmic research report is shown in Exhibit 16.2. The cover page and table of contents of a survey based research are shown in Exhibit 16.3 and Exhibit 16.4, respectively.

List of tables In research reports, the data and results are presented in the form of tables for quick grasp. Each and every table must be numbered. The numbering should be such that it is internal to each chapter. For example, if the total number of tables in Chapter 4 is 20, then the numbering should be from Table 4.1 to Table 4.20.

List of figures In research reports, the results are presented in the form of figures for better understanding and comparison. Each and every figure must be numbered. The numbering should be internal to each chapter. For example, if the total number of figures in Chapter 5 is 30, then the numbering should be from Fig. 5.1 to Fig. 5.30. Figures and charts help infer the relative comparison of results of a research study. The geographic locations and conceptual foundations of the research are also presented in the form of figures. Further, the decision making logic, algorithmic processing logic, etc., are presented in the form of figures to augment their stepwise presentation.

EXHIBIT 16.2 TABLE OF CONTENTS OF ALGORITHMIC RESEARCH REPORTS

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<i>Acknowledgements</i>		v
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1.4.2 Single Model Stochastic Problems		6
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1.6.1 Technical Benefits		8
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EXHIBIT 16.3 COVER PAGE AND TABLE OF CONTENTS OF SURVEY-BASED RESEARCH

A Study on Usage of Operations Research Techniques in Indian Industries

This thesis submitted to Alpha University
in partial fulfillment of the requirements
for the award of the degree of
Master of Business Administration.

by
D. Janakiraman

Research Supervisor
G. Kumaraguru, Ph.D.
Professor

University Seal

School of Management
Alpha University
Moon City, 100 876,
January 2002

EXHIBIT 16.4 TABLE OF CONTENTS

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<i>Abstract</i>	iii
<i>List of Tables</i>	iv
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List of variables The researcher will be forced to use different variables and abbreviations in different chapters of the research report to discuss about the models, equations and results. They must be defined under the heading 'List of Variables'

16.4.3 Text

As mentioned in Exhibit 16.1, the different items under 'Text' are Chapters, Main Sections, Subsections, Footnotes and Conclusions.

Chapters The text of the report should be divided into different *chapters* depending on the type of research. In the case of survey based research, the different chapters are as listed below:

1. Introduction
2. Literature Review
3. Objectives and Hypotheses
4. Research Methodology
5. Data Analysis
6. Conclusions.

Introduction The chapter on *introduction* should begin with the background of the research topic and state-of-the-art scenario in brief. The origin and the development of the research concept along with the most significant contributions in the past must be summarized in this chapter. At the end of this chapter, the plan of research in the form of a flowchart should be presented. Further, this chapter should give a lead for literature search.

Literature review Literature review is the crucial part of any research. This helps the researcher avoid reinventing past contributions and results. This focusses on the past literature in the field of research and its classification under different angles. The critique comparison of the past literatures will narrow down the research topic and provide a base for selecting a research topic which will have more pay-off. The outcome of the literature review gives a direction for feature research.

Objectives and hypotheses Based on the direction of the literature review, the researcher should clearly spell out the *objectives* of the research and related hypotheses. As mentioned earlier, *hypothesis* is an assumption about the population where the truth of the assumption is not sure. The researcher should list a meaningful number of hypotheses which will be tested under the chapter 'Data Analysis'

Research methodology *Research methodology* aims to finalize the activities of research design and data collection procedure which are as listed below:

- Selection of research approach
- Design of sampling plan
- Design of experiment
- Design of questionnaire
- Determination of sample size

- Design of sampling frame
- Execution of survey.

After designing the questionnaire, the researcher should organize the data collection activity with the help of interviewers as per the assumed sampling plan and experimental design. Any one or a combination of primary data collection methods, viz., observation method, personal interview, telephone interview and mail survey is to be used in the study.

Data analysis The *data analysis* can be classified into *preliminary* analysis and *hypotheses* testing. This chapter deals with the application of statistical tools for the analysis of data and making inferences.

In preliminary analysis, the data are presented in the form of charts and graphs to have insight into the research problem. This helps the researcher come out with some inferences, but these are based on the observations from the charts and graphs.

The second type of analysis is the testing of hypotheses. The different hypotheses which have been formulated earlier are tested at this stage. This exercise will reveal certain underlying facts of the research problem which will help the researcher formulate strategies for the operations of any business system. This chapter will have a major subsection on results and discussions. In some cases, this section deals with mathematical models and regression models, in addition to the above items.

Conclusions The chapter on conclusions summarizes research findings, contribution of the study, limitations of the study, and scope for the future research.

16.4.4 Bibliography

Each and every research work is based on a set of literatures which forms the foundation for the research theme. This includes identification of the broader area of research and then narrow-downing the specific research issues which are not yet tried by other researchers. Further, the literatures are helpful for using appropriate tools of analysis. So, bibliography is an alphabetical catalogue of different literatures which are used in the research.

The references can be classified into journals, books, magazines, newspapers, research studies, etc. These are illustrated below.

Journal articles The articles which are taken from journals should be included in the bibliography as per the following format.

Author(s), Title of the article, Name of the journal, Volume of the journal, Issue number of the volume, Year of publication, From-to numbers of the pages of the article.

A few examples of journal publications are presented below. If the number of authors is more than two, it is preferable to use et al. after the first author.

1. Panneerselvam, R. and C. Oudaya Sankar, 'New heuristics for assembly line balancing problems', *International Journal of Management and Systems*, Vol. 9, No. 1, 1993, pp. 25-36.

2. Panneerselvam, R. et al., 'Models for warehouse location problem', *International Journal of Management and Systems*, Vol. 6, 1990, pp. 1-8.
3. Shah, A.M., 'Environmental factors for strategy formulation', *Productivity*, Vol. 36, No. 3, 1996, pp. 594-599.

Books If some of the items from a book are referred in a research, the format to include that book in the research report is as given below:

Author(s), Title of the book, Name of the publisher, Place of publication, Year of publication.

Some examples of such references are as shown below:

1. Gopalakrishnan, P. and M. Sundaresan, *Materials Management: An Integrated Approach*, Prentice-Hall of India, New Delhi, 1979.
2. Green, P.E. et al., *Research for Marketing Decisions*, Prentice-Hall of India, New Delhi, 1988.
3. Panneerselvam, R., *Operations Research*, Prentice-Hall of India, New Delhi, 2002.

Articles in a book The format of an article in a book is as shown below and it is demonstrated with some sample articles.

Name of the author(s), Title of the article, Title of the book, Name of the Editor, Name of the publisher, Pages, Year of publication.

1. Panneerselvam, R., 'Integrated business logistic—a tool for effective supply chain management', *Supply Chain Management*, Sahay, B.S. (Ed.), Macmillan India Limited, pp. 336-346, 1999.
2. Prem Vrat, 'Supply chain management in India: problems and challenges', *Supply Chain Management*, Sahay, B.S. (Ed.), Macmillan India Limited, pp. 10-24, 1999.

Government publications

1. Ministry of Law, Government of India, The Copyright Act, 14 of 1957, Delhi, The Manager of Publications, 1960, p. 10.

Conference articles The format of an article in a conference proceedings is, as shown below, it is demonstrated with some sample articles:

Name of the author(s), Title of the article, Name of conference proceedings, Place of the conference, Dates-month-year of the conference, Pages if applicable.

In some conferences, only abstracts of the articles will be published in which case pages are not applicable.

1. Pires, S.R.I. et al., 'Measuring supply chain performance', *Proceedings of the Twelfth Annual Conference of the Production and Operations Management Society*, POM-2001, Orlando, March 30-April 2, 2001.

2. Senthilkumar, P. et al., 'Algorithm for AGVS scheduling', *Proceedings of 43rd National Convention of Indian Institution of Industrial Engineering*, Chennai, September, 7-8, 2001, pp. 42-46.

Articles in newspapers

1. Preeti Mishra, 'Super computing and drug discovery', *The Hindu*, Vol. 125, No. 298, p. 17, December 19, 2002.
2. Shankar, T.S., 'Aircraft maintenance: anticipate and manage', *The Hindu*, Vol. 125, No. 298, p. 17, December 19, 2002.

More than one item of author(s)

1. Panneerselvam, R., 'Efficient heuristics for total covering problem', *Productivity*, Vol. 36, No. 4, 1996, pp. 649-657.
2. —*Production and Operations Management*, Prentice-Hall of India, New Delhi, 1999.

Research studies

1. Narayanan, S., 'Development of new efficient heuristic for deterministic assembly line balancing problem', *Unpublished Ph. D. Thesis*, School of Management, Pondicherry University, Pondicherry, November, 2000.
2. Panneerselvam, R., 'Algorithmic grouping of operation sequences as an aid to cellular production system', *Unpublished Ph. D. Thesis*, Industrial Engineering Division, Anna University, Chennai, August 1986.

When the contribution of author(s) is referred in the main text of the report, the respective authors' names and year of publication separated by comma are enclosed within parentheses immediately after the point of reference (Panneerselvam, 2002). If the name of the author of a book/article is a part of a sentence, then the year of publication of the article alone should be enclosed within parentheses immediately after the name of the author(s), Panneerselvam (2002).

If the article corresponding to the location indicated by '*' in the main text of a report is to be shown as *footnotes*, then the same is presented as given below at the bottom of the respective page.

* Panneerselvam, 2002

or

* Panneerselvam, R., *Operations Research*, Prentice-Hall of India, New Delhi, 2002.

16.4.5 Appendices

Each and every chapter of the text in a research report should be crisp so as to enable the readers to have quick grasp of its content. Appendices help achieve such quick grasp of the content of each chapter of the report. Appendices are placed at the end of the report.

In a chapter, if there is any material whose absence within that chapter itself does not affect the flow of reading and grasp of the content, then it can be presented at the end of the report under the heading 'Appendices'. The types of materials which are to be placed under appendices are as listed below:

- The work of any past research which is used in the current report as a module;
- The format of the questionnaire;
- Original data;
- Intermediate tables of any algorithm which are used in the report;
- Some micro level charts and figures;
- Long tables;
- Organizational details.

Each of the above items should be referred in the respective chapter(s).

16.5 TYPING INSTRUCTIONS

After thoroughly checking the draft of the research report, it should be given for final typing. This section deals with important instructions for typing the report. The different items which are to be concentrated are paper, margins, indention, spacing between lines and spacing within a sentence.

Paper The quarto-size white thick unruled paper is used for the final printing of the report. The printing should be done on one side only.

Margins The left margin and the right margin of the report should be 1.5 inch and 1 inch, respectively. The top as well as bottom margin should be 1 inch. But, in the pages starting with a chapter, the top margin should be 3 inch.

Indention The paragraph can be classified into *general* and *indented*. The first line of general paragraph is not indented, whereas in indented paragraph, the first line of the paragraph is indented by 10 spaces from the left margin.

Spacing between lines The entire report should be typed with double spacing except indented paragraphs, tables and footnotes. The indented paragraphs, tables and footnotes are to be typed with single spacing. But there should be double spacing between each neighbouring pair of these single spaced items, that is, between adjacent indented paragraphs as well as adjacent tables.

Triple spacing should be given for the following cases:

- Before each paragraph heading.
- Before and after each centre heading, centre subheading, side heading and indented paragraph.

Spacing within a sentence The details of spacing within a sentence are summarized in Table 16.1.

Table 16.1 Details of Spacing within a Sentence

<i>Description of item</i>	<i>Number of spaces</i>
Between words	1
After a semicolon	1
After a colon	1
After a comma	1
After a full stop	1
Before the first parenthesis or bracket	1
Between the last letter of a word in a sentence and each of comma, semicolon, colon, exclamation mark, question mark, last parenthesis and last bracket.	0
Between the first quotation mark and the very next word.	0
Between the end quotation mark and the immediate preceding word.	0
Before and after a hyphen (-).	0
Before and after a slash (/)	0

16.6 ORAL PRESENTATION

After the completion of any research project, the concept, techniques and results of the research project should be presented to the audience of interested group. The presenter should take all necessary steps to make his/her presentation more effective so that the audience receive the essence of the report without any omission.

The presenter should consider the following aspects of communication to make his presentation more effective:

- Purpose of the presentation should be identified. The purpose may be to inform the audience, selling the idea, persuading, changing the attitude, etc.
- Knowledge about the mix of audience.
- The breadth and depth of the message should be decided to suit the audience.
- If the presenter has the option of selecting presentation timing, then it should be either at the beginning of the day or after a break.
- The size of the room should be decided based on the size of the audience.
- The audience should be provided with pre-presentation handout if necessary.
- Best fitting audio-visual aids should be selected and used. Some of the audio-visual aids are tape recorder, overhead projector, epidiascope, slide projector, video cassette, etc.
- It is better to augment the audio-visual aids with writings on conventional black-board.
- Computer-aided presentation can be made if there is required infrastructure at the place of presentation.
- The oral presentation should be augmented with minimal number of transparencies.
- The presenter should avoid simply reading the transparencies without explanation.

- The beginning of the presentation must be impressive.
- The presenter should avoid distracting the audience.
- The presenter should take continuous effort to involve the audience.
- It is preferable to allow questions even before the completion of the presentation.
- The presenter should maintain eye contact with the audience.
- The presenter should continuously sense the response of the audience through their facial expression and symbolic actions and tune the presentation accordingly.

QUESTIONS

1. What are the types of report? Explain them in brief.
2. Discuss the guidelines for reviewing the draft of a report.
3. What are items in a research report? Explain them in brief.
4. Give a sample cover page of a research report.
5. Discuss the items of the introductory pages in detail.
6. Give a sample table of contents of a survey based research report.
7. Give a sample table of contents of an algorithmic research report.
8. What are the items under the text of a research report? Explain them in brief.
9. Discuss the guidelines for preparing bibliography.
10. Give a brief account of typing/printing instructions while preparing a research report.
11. Discuss the guidelines for oral presentation of a research report.
12. Assume a research topic of your choice and give the complete format of its research report.

Annexure I: F Tables

Percentage Points of F-Distribution ($\alpha = 0.10$)

Denominator degrees of freedom (n_2)	Numerator degrees of freedom (n_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19	60.71	61.22	61.74	62.00	62.26	62.53	62.79	63.06	63.33
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.20	5.18	5.18	5.17	5.16	5.15	5.14	5.13
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.79	3.78	3.76
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.14	3.12	3.10
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.90	2.87	2.84	2.82	2.80	2.78	2.76	2.74	2.72
7	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70	2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.47
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.29
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.16
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.28	2.24	2.20	2.18	2.16	2.13	2.11	2.08	2.06
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.03	2.00	1.97
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.15	2.10	2.06	2.04	2.01	1.99	1.96	1.93	1.90
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.90	1.88	1.85
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.05	2.01	1.96	1.94	1.91	1.89	1.86	1.83	1.80
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.02	1.97	1.92	1.90	1.87	1.85	1.82	1.79	1.76
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	1.99	1.94	1.89	1.87	1.84	1.81	1.78	1.75	1.72
17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.96	1.91	1.86	1.84	1.81	1.78	1.75	1.72	1.69
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.93	1.89	1.84	1.81	1.78	1.75	1.72	1.69	1.66
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.91	1.86	1.81	1.79	1.76	1.73	1.70	1.67	1.63
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.89	1.84	1.79	1.77	1.74	1.71	1.68	1.64	1.61

contd.

Annexure I: Contd.

Denominator degrees of freedom (n_2)	f	Numerator degrees of freedom (n_1)																	
		2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.87	1.83	1.78	1.75	1.72	1.69	1.66	1.62	1.59
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.86	1.81	1.76	1.73	1.70	1.67	1.64	1.60	1.57
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89	1.84	1.80	1.74	1.72	1.69	1.66	1.62	1.59	1.55
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.83	1.78	1.73	1.70	1.67	1.64	1.61	1.57	1.53
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.82	1.77	1.72	1.69	1.66	1.63	1.59	1.56	1.52
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.81	1.76	1.71	1.68	1.65	1.61	1.58	1.54	1.50
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.80	1.75	1.70	1.67	1.64	1.60	1.57	1.53	1.49
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.79	1.74	1.69	1.66	1.63	1.59	1.56	1.52	1.48
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.78	1.73	1.68	1.65	1.62	1.58	1.55	1.51	1.47
30	2.88	2.49	2.28	2.14	2.03	1.98	1.93	1.88	1.85	1.82	1.77	1.72	1.67	1.64	1.61	1.57	1.54	1.50	1.46
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.71	1.66	1.61	1.57	1.54	1.51	1.47	1.42	1.38
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.66	1.60	1.54	1.51	1.48	1.44	1.40	1.35	1.29
120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65	1.60	1.55	1.48	1.45	1.41	1.37	1.32	1.26	1.19
∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.55	1.49	1.42	1.38	1.34	1.30	1.24	1.17	1.00

Annexure II: F Tables

Percentage Points of F-Distribution ($\alpha = 0.05$)

Denominator degrees of freedom (n_2)	Numerator degrees of freedom (n_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84

contd.

Annexure II: Contd.

Denominator degrees of freedom (n_2)	Numerator degrees of freedom (n_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.55	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Annexure III : F Tables

Percentage Points of F-Distribution ($\alpha = 0.01$)

Denominator degrees of freedom (n_2)	Numerator degrees of freedom (n_1)																			∞
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120		
1	4052	4999.5	5403	5625	5764	5859	5928	5982	6022	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366	
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50	
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.00	26.50	26.41	26.32	26.22	26.13	
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46	
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02	
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88	
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65	
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86	
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31	
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91	
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60	
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36	
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17	
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00	
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87	
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75	
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65	
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57	
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49	
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42	

contd.

Annexure III : Contd.

Denominator degrees of freedom (n_2)	Numerator degrees of freedom (n_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

Annexure IV: Significant Ranges of Duncan's Multiple Range Test

Significance Level (α) = 0.01

Degrees of freedom (f)	P													
	2	3	4	5	6	7	8	9	10	20	50	100	50	100
1	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00	90.00
2	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00	14.00
3	18.26	8.50	8.60	8.70	8.80	8.90	8.90	9.00	9.00	9.00	9.30	9.30	9.30	9.30
4	6.51	6.80	6.90	7.00	7.10	7.10	7.20	7.20	7.30	7.30	7.50	7.50	7.50	7.50
5	5.70	5.96	6.11	6.18	6.26	6.33	6.40	6.44	6.50	6.50	6.80	6.80	6.80	6.80
6	5.24	5.51	5.65	5.73	5.81	5.88	5.95	6.00	6.00	6.00	6.30	6.30	6.30	6.30
7	4.95	5.22	5.37	5.45	5.53	5.61	5.69	5.73	5.80	5.80	6.00	6.00	6.00	6.00
8	4.74	5.00	5.14	5.23	5.32	5.40	5.47	5.51	5.50	5.50	5.80	5.80	5.80	5.80
9	4.60	4.86	4.99	5.08	5.17	5.25	5.32	5.36	5.40	5.40	5.70	5.70	5.70	5.70
10	4.48	4.73	4.88	4.96	5.06	5.13	5.20	5.24	5.28	5.28	5.55	5.55	5.55	5.55
11	4.39	4.63	4.77	4.86	4.94	5.01	5.06	5.12	5.15	5.15	5.39	5.39	5.39	5.39
12	4.32	4.55	4.68	4.76	4.84	4.92	4.96	5.02	5.07	5.07	5.26	5.26	5.26	5.26
13	4.26	4.48	4.62	4.69	4.74	4.84	4.88	4.94	4.98	4.98	5.15	5.15	5.15	5.15
14	4.21	4.42	4.55	4.63	4.70	4.78	4.83	4.87	4.91	4.91	5.07	5.07	5.07	5.07
15	4.17	4.37	4.50	4.58	4.64	4.72	4.77	4.81	4.84	4.84	5.00	5.00	5.00	5.00
16	4.13	4.34	4.45	4.54	4.60	4.67	4.72	4.76	4.79	4.79	4.94	4.94	4.94	4.94
17	4.10	4.30	4.41	4.50	4.56	4.63	4.68	4.73	4.75	4.75	4.89	4.89	4.89	4.89
18	4.07	4.27	4.38	4.46	4.53	4.59	4.64	4.68	4.71	4.71	4.85	4.85	4.85	4.85
19	4.05	4.24	4.35	4.43	4.50	4.56	4.61	4.64	4.67	4.67	4.82	4.82	4.82	4.82
20	4.02	4.22	4.33	4.40	4.47	4.53	4.58	4.61	4.65	4.65	4.79	4.79	4.79	4.79
30	3.89	4.06	4.16	4.22	4.32	4.36	4.41	4.45	4.48	4.48	4.65	4.65	4.65	4.65
40	3.82	3.99	4.10	4.17	4.24	4.30	4.34	4.37	4.41	4.41	4.59	4.59	4.59	4.59
60	3.76	3.92	4.03	4.12	4.17	4.23	4.27	4.31	4.34	4.34	4.53	4.53	4.53	4.53
100	3.71	3.86	3.98	4.06	4.11	4.17	4.21	4.25	4.29	4.29	4.48	4.48	4.48	4.48
∞	3.64	3.80	3.90	3.98	4.04	4.09	4.14	4.17	4.20	4.20	4.41	4.41	4.41	4.41

Annexure V: Significant Ranges of Duncan's Multiple Range Test

Significance Level (α) = 0.05

Degrees of freedom (f)	P												
	2	3	4	5	6	7	8	9	10	20	50	100	
1	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	
2	6.09	6.09	6.09	6.09	6.09	6.09	6.09	6.09	6.09	6.09	6.09	6.09	
3	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50	
4	3.93	4.01	4.02	4.02	4.02	4.02	4.02	4.02	4.02	4.02	4.02	4.02	
5	3.64	3.74	3.79	3.83	3.83	3.83	3.83	3.83	3.83	3.83	3.83	3.83	
6	3.46	3.58	3.64	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	
7	3.35	3.47	3.54	3.58	3.60	3.61	3.61	3.61	3.61	3.61	3.61	3.61	
8	3.26	3.39	3.47	3.52	3.55	3.56	3.56	3.56	3.56	3.56	3.56	3.56	
9	3.20	3.34	3.41	3.47	3.50	3.52	3.52	3.52	3.52	3.52	3.52	3.52	
10	3.15	3.30	3.37	3.43	3.46	3.47	3.47	3.47	3.47	3.48	3.48	3.48	
11	3.11	3.27	3.35	3.39	3.43	3.44	3.45	3.46	3.46	3.48	3.48	3.48	
12	3.08	3.23	3.33	3.36	3.40	3.42	3.44	3.44	3.46	3.48	3.48	3.48	
13	3.06	3.21	3.30	3.35	3.38	3.41	3.42	3.44	3.45	3.47	3.47	3.47	
14	3.03	3.18	3.27	3.33	3.37	3.39	3.41	3.42	3.44	3.47	3.47	3.47	
15	3.01	3.16	3.25	3.31	3.36	3.38	3.40	3.42	3.43	3.47	3.47	3.47	
16	3.00	3.15	3.23	3.30	3.34	3.37	3.39	3.41	3.43	3.47	3.47	3.47	
17	2.98	3.13	3.22	3.28	3.33	3.36	3.38	3.40	3.42	3.47	3.47	3.47	
18	2.97	3.12	3.21	3.27	3.32	3.35	3.37	3.39	3.41	3.47	3.47	3.47	
19	2.96	3.11	3.19	3.26	3.31	3.35	3.37	3.39	3.41	3.47	3.47	3.47	
20	2.95	3.10	3.18	3.25	3.30	3.34	3.36	3.38	3.40	3.47	3.47	3.47	
30	2.89	3.04	3.12	3.20	3.25	3.29	3.32	3.35	3.37	3.47	3.47	3.47	
40	2.86	3.01	3.10	3.17	3.22	3.27	3.30	3.33	3.35	3.47	3.47	3.47	
60	2.83	2.98	3.08	3.14	3.20	3.24	3.28	3.31	3.33	3.47	3.48	3.48	
100	2.80	2.95	3.05	3.12	3.18	3.22	3.26	3.29	3.32	3.47	3.53	3.53	
∞	2.77	2.92	3.02	3.09	3.15	3.19	3.23	3.26	3.29	3.47	3.61	3.67	

Annexure VI: Area under Standard Normal Distribution from its Mean

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2703	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990

Annexure VII: Values of Student's t Statistic (t_α)

Degrees of freedom	α									
	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0025	0.001	
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	127.320	318.310	
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	
6	0.265	0.727	1.440	1.943	2.447	3.143	3.707	4.317	5.208	
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.019	4.785	
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	

contd.

Annexure VII: Contd.

Degrees of freedom	<i>a</i>									
	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0025	0.001	
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	
40	0.255	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	
60	0.254	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	
120	0.254	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	

Annexure VIII: Value of Chi-square Statistic for a Given Significance Level and Degrees of Freedom

a

Degrees of freedom	0.995	0.99	0.975	0.95	0.05	0.025	0.01	0.005
1	0.0000393	0.000157	0.000982	0.00393	3.841	5.024	6.635	7.879
2	0.0100	0.020	0.051	0.103	5.991	7.378	9.210	10.597
3	0.0717	0.115	0.216	0.352	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	11.070	12.832	15.086	16.750
6	0.676	0.872	1.237	1.635	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	31.410	34.170	37.566	39.997

contd.

Annexure VIII: Contd.

Degrees of freedom	0.995	0.99	0.975	0.95	0.05	0.025	0.01	0.005
21	8.034	8.897	10.283	11.591	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	33.924	36.781	40.289	42.796
23	9.250	10.196	11.689	13.091	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.484	36.415	39.364	42.980	45.558
25	10.520	11.524	13.120	14.611	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	42.557	45.772	49.588	52.336
30	13.787	14.953	16.791	18.493	43.773	46.979	50.892	53.672
40	20.706	22.164	24.433	26.509	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	67.505	71.420	76.154	79.490
60	35.535	37.485	40.482	43.118	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	101.879	106.629	112.329	116.321
90	59.196	61.754	65.646	69.126	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	124.342	129.561	135.807	140.169

Annexure IX: Critical Values of d for Kolmogorov-Smirnov One-sample Test

Sample size (n)	Level of significance (α)		
	0.10	0.05	0.01
1	0.950	0.975	0.995
2	0.776	0.842	0.929
3	0.642	0.708	0.828
4	0.564	0.624	0.733
5	0.510	0.565	0.669
6	0.470	0.521	0.618
7	0.438	0.486	0.577
8	0.411	0.457	0.543
9	0.388	0.432	0.514
10	0.368	0.410	0.490
11	0.352	0.391	0.468
12	0.338	0.375	0.450
13	0.325	0.361	0.433
14	0.314	0.349	0.418
15	0.304	0.338	0.404
16	0.295	0.328	0.392
17	0.286	0.318	0.381
18	0.278	0.309	0.371
19	0.272	0.301	0.363
20	0.264	0.294	0.356
25	0.240	0.270	0.320
30	0.220	0.240	0.290
35	0.210	0.230	0.270
Above 35	$1.22/\sqrt{n}$	$1.36/\sqrt{n}$	$1.63/\sqrt{n}$

Annexure X: Critical Values of r of Run Test

Any Observed Value of r which is Less than or Equal to the Smaller Value or Greater than or Equal to the Large Value for Given n_1 and n_2 is Significant at $\alpha = 0.05$

n_2

n_1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2											2	2	2	2	2	2	2	2	2
3					2	2	2	2	2	2	2	2	2	3	3	3	3	3	3
4				2	2	2	3	3	3	3	3	3	3	3	4	4	4	4	4
5			2	2	3	3	3	3	3	4	4	4	4	4	4	4	5	5	5
6		2	2	3	3	3	3	4	4	4	4	5	5	5	5	5	5	6	6
7			2	2	3	3	4	4	4	5	5	5	5	6	6	6	6	6	6
8				2	3	3	4	4	4	5	5	5	5	6	6	6	7	7	7
9					2	3	4	4	4	5	5	5	5	6	6	6	7	7	7
10						2	3	4	4	5	5	5	5	6	6	6	7	7	7
11							2	3	4	4	5	5	5	6	6	6	7	7	7
12								2	3	4	4	5	5	5	6	6	7	7	7
13									2	3	4	4	5	5	5	6	6	7	7

contd.

Annexure X: Contd.

n _i	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
14	2	2	3	4	5	5	6	7	7	8	8	9	9	9	10	10	10	11	11
15	-	-	-	-	-	15	16	17	18	19	20	20	21	22	22	23	23	23	24
	2	3	3	4	5	6	6	7	7	8	8	9	9	10	10	11	11	11	
16	-	-	-	-	-	15	16	18	18	19	20	21	22	22	23	23	24	24	25
	2	3	4	4	5	6	6	7	8	8	9	9	10	10	11	11	11	12	12
17	-	-	-	-	-	-	17	18	19	20	21	21	22	23	23	24	25	25	25
	2	3	4	4	5	6	7	7	8	9	9	10	10	11	11	11	12	12	13
18	-	-	-	-	-	-	17	18	19	20	21	22	23	23	24	25	25	26	26
	2	3	4	5	5	6	7	8	8	9	9	10	10	11	11	12	12	13	13
19	-	-	-	-	-	-	17	18	19	20	21	22	23	24	25	25	26	26	27
	2	3	4	5	6	6	7	8	8	9	10	10	11	11	12	12	13	13	13
20	-	-	-	-	-	-	17	18	20	21	22	23	23	24	25	26	26	27	27
	2	3	4	5	6	6	7	8	9	9	10	10	11	12	12	13	13	13	14
	-	-	-	-	-	-	17	18	20	21	22	23	24	25	25	26	27	27	28

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Answers

CHAPTER 3

1. Rs. 14,357.14
2. Rs. 143.57 crores
3. Rs. 17,548.39
4. No. of large scale industries = 50
No. of medium scale industries = 100
5. 39.74 years
6. Rs. 15.4 thousands
7. 350 units
8. Overall weighted rating = 7.42
9. Rs. 14 thousands
10. 7.8 years
11. Rs. 10.27 crores
12. 13.273 months
13. Rs. 20.225 crores
14. 6.765 years
15. 22.37%
16. 53.34 kilometre per litre
17. Rs. 7.018 lakhs
18. Rs. 7.01 lakhs
19. Average deviation = 90.39 shares, coefficient of average deviation = 0.374
20. Average deviation = Rs. 9.722 lakhs, coefficient of average deviation = 0.365
21. Mean = Rs. 74.16; Standard deviation = Rs. 12.17
22. Mean = Rs. 9.4 thousands, standard deviation = Rs. 3.2 thousands
23. Business A is more consistent
24. Salesman Y is more consistent
25. 0.013
26. -0.05405

27. Coefficient of skewness = 0, the distribution is symmetrical
 28. Coefficient of skewness = -0.0187, the distribution is negatively skewed.

CHAPTER 4

11. (a) H_0 : There is no significant difference in terms of surface finish between milling machines
 H_1 : There is significant difference in terms of surface finish between milling machines.
 (b) Calculated F -value = 0.4502; accept H_0 . There is no significant difference in terms of surface finish between milling machines.
12. Calculated F -value = 0.8175; accept H_0 . There is no significant difference in terms of EPS between different companies.
13. Calculated F -value = 0.23; accept H_0 . There is no significant difference in terms of number of acceptable parts produced per hour between different intervals of patrolling.
14. (a) Model: $Y_{ij} = \mu + F_j + P_i + e_{ij}$
 (b) Fertilizer [F_j]: Calculated F -value = 0.429; accept H_0 . There is no significant difference in terms of yield between different fertilizers.
 Plot [P_i]: Calculated F -value = 1.1676; accept H_0 . There is no significant difference in terms of yield between different plots.
15. (a) Model: $Y_{ij} = \mu + R_j + P_i + e_{ij}$
 (b) Sales region [R_j]: Calculated F -value = 0.82; accept H_0 . There is no significant difference in terms of sales revenue between different sales regions.
 Period [P_i]: Calculated F -value = 1.727; accept H_0 . There is no significant difference in terms of sales revenue between different periods.
16. (a) Model: $Y_{ij} = \mu + T_j + P_i + e_{ijk}$
 (b) Technology [T_j]: Calculated F -value = 4.11; reject H_0 . There is significant difference in terms of hourly output of the product between different technology.
 Plant [P_i]: Calculated F -value = 4.89; reject H_0 . There is significant difference in terms of hourly output of the product between different plants.
17. (a) Model: $Y_{ijk} = \mu + C_j + P_i + T_k + e_{ijk}$
 (b) Car [C_j]: Calculated F -value = 1.217; accept H_0 . There is no significant difference in terms of tread loss between different cars.
 Position [P_i]: Calculated F -value = 0.683; accept H_0 . There is no significant difference in terms of tread loss between different positions.
 Tires [T_k]: Calculated F -value = 0.603; accept H_0 . There is no significant difference in terms of tread loss between different tires.
18. (a) Model: $Y_{ijk} = \mu + B_j + S_i + T_k + e_{ijk}$
 (b) Batch [B_j]: Calculated F -value = 1.86; accept H_0 . There is no significant difference in terms of average performance of the participants of the training programme between different batches.

Subject [S_j]: Calculated F -value = 3.74; reject H_0 . There is significant difference in terms of average performance of the participants of the training programme between different subjects.

Faculty [T_k]: Calculated F -value = 1.65; accept H_0 . There is no significant difference in terms of average performance of the participants of the training programme between different faculties.

19. (a) $Y_{ij} = \mu + T_j + e_{ij}$
 (b) Tire Brand [T_j]: Calculated F -value = 5.807; reject H_0 . There is significant difference in terms of tread loss between different tire brands.
 (c) Inference through Duncan multiple range test:
 – Tire brands, D and A significantly differ;
 – Tire brands, B and A significantly differ;
 – All other pairs of tire brands are insignificant.
20. (a) $Y_{ij} = \mu + C_j + e_{ij}$
 (b) Company [T_j]: Calculated F -value = 2.546; accept H_0 . There is no significant difference in terms of quarterly averages of EPS between different companies.
 (c) There is no need of Duncan multiple range test.
21. (a) $Y_{ijk} = \mu + P_j + I_i + PI_{ij} + e_{ijk}$
 (b) Nature of profession [P_j]: Calculated F -value = 1.089; accept H_0 . There is no significant difference in terms of service quality between different nature of professions.
 Income level [I_i]: Calculated F -value = 0.228; accept H_0 . There is no significant difference in terms of service quality between different income levels.
 PI_{ij} : Calculated F -value = 1.291; accept H_0 . There is no significant difference in terms of service quality between different combinations of Nature of Professions and income levels (interaction).
22. (i) Salesman [S_j]: Calculated F -value = 3.76; accept H_0 . There is no significant difference in terms of sales revenue between different salesmen.
 (ii) Sales region [R_i]: Calculated F -value = 2.633; accept H_0 . There is no significant difference in terms of sales revenue between different sales regions.
 (iii) SR_{ij} : Calculated F -value = 9.013; reject H_0 . There is significant difference in terms of sales revenue between different combinations of salesman and sales region (Interaction).
23. (a) $Y_{ijkl} = \mu + E_i + D_j + ED_{ij} + S_k + ES_{ik} + DS_{jk} + EDS_{ijk} + e_{ijkl}$
 (b) Sex [S_k]: Calculated F -value = 7.563; reject H_0 . There is significant difference in terms of value addition between different sexes.
 All other components of the model are insignificant.
26. (a) $Y_{ijkl} = \mu + E_i + D_j + ED_{ij} + S_k + ES_{ik} + DS_{jk} + EDS_{ijk} + e_{ijkl}$
 (b) Sex [S_k]: Calculated F -value = 7.563; reject H_0 . There is significant difference in terms of value addition between different sexes.
 All other components of the model are insignificant.

$$31. \quad F\text{-ratio for sales region} = \frac{\text{Mean sum of squares of sales region}}{\text{Mean sum of squares of interaction}}$$

$$F\text{-ratio for salesman} = \frac{\text{Mean sum of squares of salesman}}{\text{Mean sum of squares of error}}$$

$$F\text{-ratio of interaction} = \frac{\text{Mean sum of squares of interaction}}{\text{Mean sum of squares of error}}$$

- (a) (i) Sales region: Calculated F -value = 0.292; accept H_0 . There is no significant difference in terms of sales revenue between sales regions.
 (ii) Salesman: Calculated F -value = 3.76; accept H_0 . There is no significant difference in terms of sales revenue between salesmen.
 (iii) Interaction: Calculated F -value = 9.013; reject H_0 . There is significant difference in terms of sales revenue for the interaction between sales regions and salesmen.

$$(b) \quad F\text{-ratio for Sales Region} = \frac{\text{Mean sum of squares of sales region}}{\text{Mean sum of squares of interaction}}$$

$$F\text{-ratio for Salesman} = \frac{\text{Mean sum of squares of salesman}}{\text{Mean sum of squares of interaction}}$$

$$F\text{-ratio of interaction} = \frac{\text{Mean sum of squares of interaction}}{\text{Mean sum of squares of error}}$$

- (i) Sales region: Calculated F -value = 0.292; accept H_0 . There is no significant difference in terms of sales revenue between sales regions.
 (ii) Salesman: Calculated F -value = 0.417; accept H_0 . There is no significant difference in terms of sales revenue between salesmen.
 (iii) Interaction: Calculated F -value = 9.013; reject H_0 . There is significant difference in terms of sales revenue for the interaction between sales regions and salesmen.

$$32. \quad (a) \quad Y_{ijk} = \mu + P_j + I_i + PI_{ij} + e_{ijk}$$

$$(b) \quad F\text{-ratio for profession} = \frac{\text{Mean sum of squares of profession}}{\text{Mean sum of squares of interaction}}$$

$$F\text{-ratio for income level} = \frac{\text{Mean sum of squares of income level}}{\text{Mean sum of squares of error}}$$

$$F\text{-ratio of interaction} = \frac{\text{Mean sum of squares of interaction}}{\text{Mean sum of squares of error}}$$

Nature of profession [P_j]: Calculated F -value = 3.567; accept H_0 . There is no significant difference in terms of service quality between different nature of professions.

Income level [I_i]: Calculated F -value = 1.351; accept H_0 . There is no significant difference in terms of service quality between different income levels.

PI_{ij} : Calculated F -value = 1.5; accept H_0 . There is no significant difference in terms of service quality between different combinations of Nature of professions and income levels (interaction).

CHAPTER 6

7. (a) $P(X) = 1/4$ if $X = 0$;
 $= 1/2$ if $X = 1$;
 $= 1/4$ if $X = 2$,

where X is the number of the number of heads.

- (b) $P(X) = 1/8$ if $X = 0$;
 $= 3/8$ if $X = 1$;
 $= 3/8$ if $X = 2$;
 $= 1/8$ if $X = 3$,

where X is the number of heads.

- (c) $P(X) = 1/36$ if $X = 2$;
 $= 2/36$ if $X = 3$;
 $= 3/36$ if $X = 4$;
 $= 4/36$ if $X = 5$;
 $= 5/36$ if $X = 6$;
 $= 6/36$ if $X = 7$;
 $= 5/36$ if $X = 8$;
 $= 4/36$ if $X = 9$;
 $= 3/36$ if $X = 10$;
 $= 2/36$ if $X = 11$;
 $= 1/36$ if $X = 12$;

where X is total of the numbers on both dies.

14. 0.2

15. 0.4

- | | | | |
|-------------------|--------------|---------------|---------------|
| 16. (a) 0.00003 | (b) 0.0916 | (c) 0.0218 | (d) 0.9782 |
| 17. (a) 0.246 | (b) 0.205 | (c) 0.377 | (d) 0.623 |
| 18. (a) 0 | (b) 0.0078 | (c) 0.0019 | (d) 0.9981 |
| 19. (a) 0 | (b) 0.000786 | (c) 0.0000737 | (d) 0.9999 |
| 20. (a) 0.0000454 | (b) 0.0076 | (c) 0.00833 | (d) 0.9917 |
| 21. (a) 0 | (b) 0.0215 | (c) 0 | (d) 1 |
| 22. (a) 0.1216 | (b) 0.0898 | (c) 0.956 | (d) 0.044 |
| 23. (a) 0 | (b) 0.2054 | (c) 0.0079421 | (d) 0.9920579 |
| 24. 485 units | | | |
| 25. 278 units | | | |
| 26. (a) 0.2684 | (b) 0.3916 | | |

27. (a) 0.3127 (b) 0.4346
 28. (a) 0.3935 (b) 0.2231
 29. (a) 0.4866 (b) 0.2636
 30. (a) Probability = 0.1587, 48 respondents
 (b) Probability = 0.3707, 111 respondents
 (c) Probability = 0.7011, 210 respondents
 31. (a) Probability = 0.6293, 126 clinics
 (b) Probability = 0.8413, 168 clinics
 (c) Probability = 0.4972, 99 clinics

CHAPTER 7

21. 0.001
 22. 0.063
 23. (a) 0.9943 (b) 0.9999 (c) 0.909
 24. (a) 0.0107 (b) 0.0107 (c) 0.998
 25. (a) 0.001 (b) 0.2148 (c) 0.499
 26. (a) 0.0228 (b) 0.1587 (c) 0.9544
 27. (a) 0.962 (b) 0.038
 28. (a) 0.971 (b) 0.029
 29. 0.349
 30. 0.0159
 31. (a) Calculated F -value = 2.311 (b) 0.05
 32. (a) Calculated F -value = 3.47 (b) 0.01
 33. 0.01
 34. 0.0016
 35. 0.8599
 36. 46.921 mm to 53.079 mm
 37. 96.925 mm to 103.075 mm
 38. 38.716 mm to 41.284 mm
 39. Rs. 1.338 lakhs to 4.662 lakhs
 40. 33.873% to 66.127%
 41. 23.6% to 56.4%
 42. 35
 43. 1067
 44. 267
 45. 95

CHAPTER 8

5. Z of sample mean = 2.236; accept H_0 . The sales revenue has declined.
6. Z of sample mean = -56; reject H_0 . The sales of the product has not improved.
7. Z of sample mean = -1.75; accept H_0 . The sales of industries has not changed from Rs. 75 lakhs.
8. Z of sample mean = 11.317; reject H_0 . Mean monthly income has not declined from the expected mean of Rs. 8000.
9. Z of sample mean = -1.41; reject H_0 . The mean age has not increased from 45 years.
10. Z of sample mean = -6.11; reject H_0 . Mean shopping amount has changed from the expected mean of Rs. 7,000.
11. Z of sample mean = 11.18; reject H_0 . Mean age of the students is not significantly less than the expected mean age of the population.
12. Z of sample mean = -21.47; reject H_0 . Mean weight of the cement bags is not significantly more than the expected mean weight of the population.
13. Z of sample mean = -16.53; reject H_0 . Mean weight of the cement bags is not significantly more than the expected mean weight of the population.
14. Z of sample mean = 4.265; accept H_0 . Mean annual income of the respondents is significantly more than Rs. 4 lakhs.
15. Z of sample mean = -4.8; reject H_0 . Mean weight of drugs differs significantly from the specified mean of 100 mg.
16. $t = 3.75$; reject H_0 . Mean time between failures of motors is not less than 90 days.
17. $t = -2.5$; accept H_0 . The sales manager's claim is true [the mean weight is at least 125 gm].
18. $H_0: \mu = 200$ gm; $H_1: \mu \neq 200$ gm; $t_{cal} = 2.485$; reject H_0
19. $H_0: \mu_X - \mu_Y \leq 0$; $H_1: \mu_X - \mu_Y > 0$; $Z_{cal} = 2.933$; reject H_0
20. $H_0: \mu_1 - \mu_2 \geq 0$; $H_1: \mu_1 - \mu_2 < 0$; $Z_{cal} = -2.0923$; reject H_0
21. $H_0: \mu_X - \mu_Y = 0$; $H_1: \mu_X - \mu_Y \neq 0$; $Z_{cal} = -9.15$; reject H_0
22. $H_0: \mu_A - \mu_B = 0$; $H_1: \mu_A - \mu_B \neq 0$; $Z_{cal} = 7.123$; reject H_0 . The claim is true.
23. $H_0: \mu_X - \mu_Y = 0$; $H_1: \mu_X - \mu_Y \neq 0$; $Z_{cal} = -5.58$; reject H_0 . The feeling of the manager is not true.
24. $H_0: \mu_1 - \mu_2 = 0$; $H_1: \mu_1 - \mu_2 \neq 0$; $t_{cal} = 4.177$; reject H_0 . The quality manager's belief is not true.
25. $H_0: p \leq 0.9$; $H_1: p > 0.9$; $Z_{cal} = 1.292$; accept H_0 . The view of the quality manager is correct.
26. $H_0: p \geq 0.2$; $H_1: p < 0.2$; $Z_{cal} = -1.2247$; accept H_0 . The view of the production manager is correct.
27. $H_0: p = 0.95$; $H_1: p \neq 0.95$; $Z_{cal} = -2.29$; reject H_0 . The claim of the sales representative is not true.
28. $H_0: p_X - p_Y < 0$; $H_1: p_X - p_Y > 0$; $Z_{cal} = 0.827$; accept H_0 . The belief of the quality manager is true.

29. $H_0: p_1 - p_2 \geq 0$; $H_1: p_1 - p_2 < 0$; $Z_{\text{cal}} = -0.875$; accept H_0 . The feelings of the researcher is true.
30. $H_0: p_X - p_Y = 0$; $H_1: p_X - p_Y \neq 0$; $Z_{\text{cal}} = -0.831$; accept H_0 . The intuition of the bank manager is correct.
31. $H_0: \sigma^2 \leq 2.25$; $H_1: \sigma^2 > 2.25$; $C^2_{\text{cal}} = 19.6$; accept H_0 . The consignment is to be accepted.
32. $H_0: \sigma^2 \geq 0.81$ kg; $H_1: \sigma^2 < 0.81$ kg; $C^2_{\text{cal}} = 7.11$; accept H_0 . The intuition of the quality manager is correct.
33. $H_0: \sigma^2 = 0.36$ mg; $H_1: \sigma^2 \neq 0.36$ mg; $C^2_{\text{cal}} = 21.78$; accept H_0 . The intuition of the quality engineer is true.
34. $H_0: \sigma_1^2 - \sigma_2^2 \leq 0$; $H_1: \sigma_1^2 - \sigma_2^2 > 0$; $F_{\text{cal}} = 1.563$; accept H_0 . Variance of the marks of the first sample is less than that of the second sample.
35. $H_0: \sigma_1^2 - \sigma_2^2 \geq 0$; $H_1: \sigma_1^2 - \sigma_2^2 < 0$; $F_{\text{cal}} = 0.64$; $F_{\text{left}} = 0.272$; accept H_0 ; The intuition of the quality assistant is true.
36. $H_0: \sigma_1^2 - \sigma_2^2 = 0$; $H_1: \sigma_1^2 - \sigma_2^2 \neq 0$; $F_{\text{cal}} = 7.619$; $C_2 = 2.59$, reject H_0 . The intuition of the researcher is not correct.
37. $C^2_{\text{cal}} = 30.5739$; reject H_0 . Income is not independent on qualification.
38. $C^2_{\text{cal}} = 4.484$; accept H_0 . Age is independent of income level.
39. $C^2_{\text{cal}} = 3.735$; accept H_0 . Demand distribution follows uniform distribution.
40. $C^2_{\text{cal}} = 2.819$; accept H_0 . Arrival rate follows poisson distribution.

CHAPTER 9

6. $H_0: \mu = 20$; $H_1: \mu > 20$; $P(X \geq 4) = 0.64$; accept H_0 . Operation time is not more than 20 minutes.
7. $H_0: \mu = 130$; $H_1: \mu > 130$; $P(X \geq 2) = 0.98$; accept H_0 . The inter-arrival time is not more than 130 seconds.
8. $H_0: \mu = 75$; $H_1: \mu < 75$; $P(X \leq 3) = 0.3633$; accept H_0 . The performance index is not less than 75.
9. $H_0: \mu = 45$; $H_1: \mu < 45$; $P(X \leq 6) = 0.96485$; accept H_0 . The age of respondents is not less than 45 years.
10. $H_0: \mu = 5$; $H_1: \mu \neq 5$; $P(X \geq 4) = 0.6367$; accept H_0 . The annual income of general managers is equal to Rs. 5 lakhs.
11. $H_0: \mu = 15$; $H_1: \mu = 15$; $P(X \geq 2) = 0.98$; accept H_0 . The yield is 15 tons.
12. $H_0: \mu = 10$; $H_1: \mu > 10$; $Z_{\text{cal}} = -3.182$; accept H_0 . The intuition of the quality manager is correct.
13. $H_0: \mu = 100$; $H_1: \mu > 100$; $Z_{\text{cal}} = -2.67$; accept H_0 . The weight of the scrap produced in the machine shop is 100 kg.
14. $H_0: \mu = 7$; $H_1: \mu > 7$; $Z_{\text{cal}} = -2.846$; accept H_0 . The performance index of students is 7.
15. $H_0: \mu = 750$ kg; $H_1: \mu \neq 750$ kg; $Z_{\text{cal}} = 2.333$; reject H_0 . The intuition of the researcher is not correct.

16. $H_0: \mu = 150$ lakhs; $H_1: \mu = 150$ lakhs; $Z_{\text{cal}} = -1.06$; accept H_0 . The intuition of the researcher is correct.
17. $D_{\text{cal}} = 0.21855$; reject H_0 . The age distribution of employees does not follow uniform distribution.
18. $D_{\text{cal}} = 0.1101$; accept H_0 . The inter-arrival rate distribution follows Poisson distribution.
19. $D_{\text{cal}} = 0.3015$; reject H_0 . The daily volume of product produced does not follow normal distribution.
20. $D_{\text{cal}} = 0.524$; reject H_0 . The weight of welding rods does not follow normal distribution.
21. The gender of arriving applicants is random.
22. The results of the inspection are random.
23. The occurrences of STD (S) or ISTD (I) are not random.
24. The occurrences of the events H/M are not random.
25. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} > \mu_{\text{after}}$ or $p > 1/2$; $P(Q \geq 7) = 0.0898$; reject H_0 .
26. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} > \mu_{\text{after}}$ or $p > 1/2$; $P(Q \geq 5) = 0.3633$; accept H_0 .
27. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} > \mu_{\text{after}}$ or $p > 1/2$; $P(Q \geq 6) = 0.1445$; accept H_0 .
28. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} \neq \mu_{\text{after}}$ or $p \neq 1/2$; $P(Q \geq 2) = 0.9375$; accept H_0 .
29. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} > \mu_{\text{after}}$ or $p > 1/2$; $Z_{\text{cal}} = 1.9415$; reject H_0 .
30. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} > \mu_{\text{after}}$ or $p > 1/2$; $Z_{\text{cal}} = 2.1318$; reject H_0 .
31. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} < \mu_{\text{after}}$ or $p < 1/2$; $Z_{\text{cal}} = -1.9415$; accept H_0 .
32. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} < \mu_{\text{after}}$ or $p < 1/2$; $Z_{\text{cal}} = -2.89$; reject H_0 .
33. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} \neq \mu_{\text{after}}$ or $p \neq 1/2$; $Z_{\text{cal}} = -1.3416$; accept H_0 .
34. $H_0: \mu_{\text{before}} = \mu_{\text{after}}$ or $p = 1/2$; $\mu_{\text{before}} \neq \mu_{\text{after}}$ or $p \neq 1/2$; $Z_{\text{cal}} = 0$; accept H_0 .
35. $p_{\text{cal}} = 0.408$; accept H_0 . The percentage incremental efficiency of the shops by the two productivity improvement strategies are with the same median.
36. $p_{\text{cal}} = 0.3427$; accept H_0 . Yield of paddy using two different fertilizers are with the same median.
37. Calculated chi-square value = 0.13393; accept H_0 . Incremental daily sales using two different selling strategies are with the same median.
38. $Z_{\text{cal}} = -0.16077$; accept H_0 . Two samples are drawn from populations, which are having identical distributions.
39. $Z_{\text{cal}} = -0.643$; accept H_0 . Two samples are drawn from populations, which are having identical distributions.
40. $Z_{\text{cal}} = 0.9821$; accept H_0 . Two samples are drawn from populations, which are having identical distributions.
41. $Z_{\text{cal}} = -1.0432$; accept H_0 . Two samples are drawn from populations, which are having identical distributions.
42. $p_{\text{cal}} = 0.0411$; reject H_0 . There is significant difference between the capacity utilization of different months of the quarter.

43. $p_{\text{cal}} = 0.137$; accept H_0 . There is no significant difference between the yearly sales revenues of the three competing products.
44. $H = 0.54125$; accept H_0 . There is no significant difference between the efficiencies of different algorithms.
45. $H = 1.9136$; accept H_0 . There is no significant difference between the EPS of different companies.
46. $H = 0.4025$; accept H_0 . There is no significant difference between the number of acceptable parts produced for different intervals of patrolling.
47. $H = 1.0643$; accept H_0 . There is no significant difference between the yields of using different fertilizers.

CHAPTER 10

4. $r = 0.98559$, $t_{\text{cal}} = 14.2728$, reject H_0 . The correlation coefficient is significant.
5. $r = -0.314069$
6. $r_s = 0.2648$
7. $r_s = 0.253$
8. $r_s = 0.406061$, $t_{\text{cal}} = 1.2569$, accept H_0 . Rank correlation coefficient is not significant.
9. $r_s = -0.61905$
10. $r_1 = 0.677959$; $r_2 = 0.4193$; $r_3 = 0.168982$
11. $r_1 = 0.719668$; $r_2 = 0.454697$; $r_3 = 0.174148$
17. (a) and (b)

t	MA_t	F_t	Error
3	26		
4	34	26	18
5	47.3	34	32
6	66	47.3	40.7
7	83.3	66	30
8	98.67	83.3	28.7

(c) $MSE = 945.636$, $MAD = 29.88$

18. (a) and (b)

t	MA_t	F_t	Error
3	26.67		
4	41.33	26.67	33.33
5	61.67	41.33	43.67
6	81	61.67	36.33

(c) $MSE = 1445.94$, $MAD = 37.78$

19. (a) and (b)

t	MA_t	F_t	Error
3	56		
4	77.5	56	34
5	103	77.5	42.5
6	127	103	37

(c) $MSE = 1443.75$, $MAD = 37.83$

20. (a) $F_2 = 813$ (b) $F_3 = 797$
21. (a) $F_2 = 204$ (b) $F_3 = 205.2$
22. (a) $Y = 6.25 + 10.15X$ (b) $Y(X = 10) = 107.75$
23. (a) $Y = 39.0715 + 10.107(X - 1998)$
(b) $Y(X = 2005) = 109.8205$
24. (a) $Y = -23.55 + 3.642X$; $MSE = 133.164$
(b) $X = 8.63 + 0.24935Y$; $MSE = 9.119$
25. (a) $Y = 11.615 + 1.5963X$; $MSE = 31.65$
(b) $X = -5.58 + 0.6079Y$; $MSE = 12.056$
26. (a) $Y = -92.6 + 5.44X$; $Y(X = 45) = 152.2$
(b) $X = 23.4 + 0.1328Y$; $X(Y = 135) = 41.328$
27. (a) $Y = -27.6 + 2.22X$; $Y(X = 90) = 172.2$
(b) $X = 36.8 + 0.288Y$; $X(Y = 175) = 87.2$
28. (a) $Y = -7.56 + 13.89X$
(b) $F_{cal} = 114.5$; reject H_0 . There is a linear relationship.
29. $Y = 9.6721 + 0.0774X_1 + 0.0651X_2$
30. $Y = 14.1 + 8.5X_1$
31. $Y = 41.1429 + X$
32. $Y = 16.4286 + 2.5595X + 1.131X^2$

35. Period X	Desseasonalized forecast	$x = (X - 8.5)/0.5$	Trend forecast = (122.1944 + 0.662681x)	Cyclical component
1	155.4094	-15	112.2542	1.384442
2	96.28346	-13	113.5795	0.847718
3	125.4725	-11	114.9049	1.091968
4	114.4056	-9	116.2303	0.984301
5	125.523	-7	117.5556	1.067775
6	103.9861	-5	118.881	0.874708
7	102.9518	-3	120.2064	0.856459

Continued

Period X	Deseasonalized forecast	$x = (X - 8.5)/0.5$	Trend forecast = (122.1944 + 0.662681x)	Cyclical component
8	106.7785	-1	121.5317	0.878606
9	89.65929	1	122.8571	0.729785
10	127.0942	3	124.1824	1.023447
11	119.038	5	125.5078	0.948451
12	129.6596	7	126.8332	1.022285
13	143.4549	9	128.1585	1.119355
14	130.9455	11	129.4839	1.011288
15	131.907	13	130.8093	1.008392
16	152.5408	15	132.1346	1.154434

36. Period X	Deseasonalized value of Y	$x = (X - 8.5)/0.5$	Trend forecast = 73.91997 + 0.427044 x	Cyclical component
1	128.93980	-15	67.51431	1.909815
2	46.70278	-13	68.36840	0.683105
3	74.29733	-11	69.22249	1.073312
4	64.57565	-9	70.07657	0.921501
5	81.18434	-7	70.93066	1.144559
6	55.10928	-5	71.78475	0.767702
7	54.67162	-3	72.63884	0.752650
8	57.19557	-1	73.49293	0.778246
9	23.87775	1	74.34701	0.321166
10	76.59256	3	75.20110	1.018503
11	68.68998	5	76.05519	0.903160
12	79.33579	7	76.90928	1.031550
13	109.83760	9	77.76337	1.412460
14	80.32879	11	78.61745	1.021768
15	79.90468	13	79.47154	1.005450
16	101.47600	15	80.32563	1.263308

CHAPTER 13

14. (a) $Y_{ij} = \mu + A_j + e_{ij}$
 (b) Algorithm $[A_j]$: Calculated F -value = 0.266; accept H_0 . There is no significant difference in terms of values of the response variable of the problem between different algorithms.
15. (a) $Y_{ijk} = \mu + A_j + P_i + AP_{ij} + e_{ijk}$
 (b) Algorithm $[A_j]$: Calculated F -value = 0.3378; accept H_0 . There is no significant difference in terms of the values of the response variable of the problem between different algorithms.

Problem $[P_i]$: Calculated F -value = 0.01353; accept H_0 . There is no significant difference in terms of the values of the response variable of the problem between different problems.

AP_{ij} : Calculated F -value = 0.01353; accept H_0 . There is no significant difference in terms of the values of the response variable of the problem between different combinations of algorithm and problem (interaction).

CHAPTER 14

8. (a) Minimize $Z = Y_1 + Y_2 + Y_3 + Y_4 + Y_5$
 Subject to
 $Y_1 + Y_4 \geq 1$
 $Y_2 + Y_4 \geq 1$
 $Y_1 \geq 1$
 $Y_1 + Y_5 \geq 1$
 $Y_4 \geq 1$
 $Y_j = 0 \text{ or } 1, j = 1, 2, 3, 4, 5$
- (b) $Y_1 = Y_4 = 1$ and $Z_{\text{optimum}} = 2$
9. (a) Maximize $Z = 535500Y_1 + 72,00,000Y_2 + 2,26,50,000Y_3$
 Subject to
 $0.1Y_1 + 2Y_2 + 0.5Y_3 \leq 45$
 $Y_1 \geq 10; Y_2 \geq 4; Y_3 \geq 6; Y_1 \leq 20; Y_2 \leq 10; Y_3 \leq 25$
 $Y_j \geq 0, j = 1, 2, 3$
- (b) $Y_1 = 20, Y_2 = 10, Y_3 = 25, Z_{\text{optimum}} = 64,89,60,000$
10. (a) Maximize $Z = 25Y_1 + 50Y_2 + 35Y_2 + 35Y_3 + 25Y_4 + 40Y_5$
 Subject to
 $4Y_1 + 13Y_2 + 2Y_3 + 6Y_4 + 6Y_5 \leq 25$
 $2Y_1 + 8Y_2 + 10Y_3 + 3Y_4 + 9Y_5 \leq 25$
 $9Y_1 + 12Y_2 + 8Y_3 + 7Y_4 + 10Y_5 \leq 25$
 $Y_j = 0 \text{ or } 1, j = 1, 2, 3, 4, 5$
- (b) $Y_3 = Y_4 = Y_5 = 1$ and $Z_{\text{optimum}} = 100$
11. (a) Maximize $Z = 0.04X_1 + 0.12X_2 + 0.08X_3 + 0.14X_4 + 0.1X_5 + 0.04X_6$
 Subject to
 $4X_1 + 7X_2 + 8X_3 + 6X_4 + 10X_5 + 5X_6 \leq 6$
 $3X_1 + 8X_2 + 4X_3 + 8X_4 + 6X_5 + 3X_6 \leq 7$
 $12X_2 + 8X_3 + 20X_4 + 15X_5 + 5X_6 \geq 10$
 $X_7 \geq 0.12$
 $X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 = 1$
 $X_j \geq 0, j = 1, 2, 3, 4, 5, 6, 7$ and it is the proportion of investment in the j th alternative.

- (b) $X_3 = 0.01$, $X_4 = 0.87$, $X_7 = 0.12$ and $Z_{\text{optimum}} = 0.1226$

Investment in: Company bonds = Rs. 5 crores, real estate = Rs. 435 crores and cash = Rs. 60 crores. Total return = Rs. 61.3 crores.

12. (a) Minimize $Z = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$

Subject to

$$X_1 + X_6 \geq 3$$

$$X_1 + X_2 \geq 4$$

$$X_2 + X_3 \geq 8$$

$$X_3 + X_4 \geq 10$$

$$X_4 + X_5 \geq 7$$

$$X_5 + X_6 \geq 4$$

$$\text{and } X_j \geq 0, j = 1, 2, 3, 4, 5, 6$$

- (b) $X_1 = 3$; $X_2 = 1$; $X_3 = 7$; $X_4 = 3$; $X_5 = 4$ and $Z_{\text{optimum}} = 18$

16. (a) Minimize $Z = Y_1 + Y_2 + Y_3 + Y_4 + Y_5$

Subject to

$$Y_3 + Y_5 \geq 1$$

$$Y_1 + Y_4 \geq 1$$

$$Y_2 + Y_4 \geq 1$$

$$Y_1 + Y_5 \geq 1$$

$$Y_2 \geq 1$$

$$\text{and } Y_j = 0 \text{ or } 1, j = 1, 2, 3, 4, 5.$$

- (b) $Y_1 = Y_2 = Y_3 = 1$ and $Z_{\text{optimum}} = 3$

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